NOTE: The cover page of this standard has been changed for administrative reasons. There are no other changes to this document.

MIL-STD-188-200
28 JUNE 1983

SUPERSEDING:
SEE FOREWORD

DEPARTMENT OF DEFENSE INTERFACE STANDARD

SYSTEM DESIGN AND ENGINEERING STANDARDS FOR TACTICAL COMMUNICATIONS

TCTS

NO DELIVERABLE DATA REQUIRED BY THIS DOCUMENT

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
Washington, DC 20301

System Design and Engineering Standards
for Tactical Communications
MIL-STD-188-200

1. This Military Standard is approved and mandatory for use by all Departments and Agencies of the Department of Defense, in accordance with the OASD (CCCI) Memorandum, dated 10 May 1977. (See Appendix A.)

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:

Commander
US Army Communications-Electronics Command
ATTN: DRSEL-SEI-I
Fort Monmouth, NJ 07703

by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

FOREWORD

1. In the past, Military Standard 188 (MIL-STD-188) covering military communications system technical standards, has evolved from a document applicable to all military communications (MIL-STD-188, MIL-STD-188A and MIL-STD-188B) to one applicable to tactical communications only (MIL-STD-188C).

2. The Defense Communications Agency (DCA) has published DCA Circulars (DCAC) promulgating standards and criteria applicable to the Defense Communications System (DCS) and to the technical support of the National Military Command System (NMCS).

3. Standards for all military communications are now being published as part of a MIL-STD-188 series of documents. Military communications system technical standards are subdivided into common long haul/tactical standards (MIL-STD-188-100 series), tactical standards (MIL-STD-188-200 series) and long haul standards (MIL-STD-188-300 series). Appendix B contains a list of MIL-STD-188-100 and MIL-STD-188-200 series documents that are currently in preparation.

4. This document contains technical standards and design objectives for tactical communications equipment, subsystems and systems. These standards and design objectives have been based on corresponding parameter values contained in MIL-STD-188-100, but this document differs in its outline from MIL-STD-188-100 and covers in more detail those areas applicable to tactical communications.

5. This document supersedes MIL-STD-188C, including Change Notices 1 and 2, with the exception of paragraphs 6.2 and 6.3 and subparagraphs 4.5.6 through 4.5.11. Standards and design objectives for analog telephone sets and analog facsimile equipment contained in paragraphs 6.2 and 6.3, respectively, and standards and design objectives for radio equipment and radio subsystems contained in 4.5.6 through 4.5.11 of MIL-STD-188C are being updated and will be published in other documents of the MIL-STD-188 series. (See Appendix B.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

IDENTIFICATION OF
INTERNATIONAL STANDARDIZATION AGREEMENT

Certain provisions of this document are the subject of international standardization agreements. When a change notice, revision, or cancellation of this document is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph	Title	Page
	Promulgation Sheet	ii
	Foreword	iii
	Identification of International Standardization Agreement	iv
1	SCOPE	1
1.1	Purpose	1
1.2	Application	2
1.3	Objectives	2
1.4	System standards and design objectives	3
2	REFERENCED DOCUMENTS	5
2.1	Issues of documents	5
2.1.1	Federal specifications	5
2.1.2	Military specifications	5
2.1.3	Federal standards	5
2.1.3.1	Federal telecommunications standards	5
2.1.3.2	Federal information processing standards publications (FIPS PUB)	6
2.1.4	Military standards	7
2.1.4.1	DoD standards	7
2.1.4.2	North Atlantic Treaty Organization (NATO) Standardization Agreements (STANAG) and other publications	8
2.1.4.3	Quadripartite Standardization Agreements (QSTAG)	8
2.1.5	Military handbooks	8
2.2	Other publications	9
2.2.1	DoD publications	9
2.2.2	Allied Communication Publications (ACP)	10
2.2.3	Standards of the American National Standards Institute (ANSI)	10
2.2.4	Standard of the Institute of Electrical and Electronics Engineers (IEEE)	11
2.2.5	International regulations	11
3	DEFINITIONS	12
3.1	Definition of terms	12
3.2	Abbreviations and acronyms	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
4	GENERAL DESCRIPTION OF TACTICAL COMMUNICATIONS	13
4.1	Tactical communications systems	13
4.1.1	Tactical subsystems type I and type II	13
4.1.2	Tactical subsystem type III	15
4.1.3	Tactical subsystem type IV	15
4.1.4	Single channel radio subsystems	15
4.1.5	Interconnection of communications subsystems	16
4.1.6	Tactical communications equipment	16
4.1.7	Design, installation and operation	16
4.1.8	Types of traffic	17
4.1.9	Communications security	17
4.1.10	Precedence and preemption	17
4.2	General design considerations	17
4.2.1	Compromising emanations (TEMPEST) requirements	18
4.2.2	Electromagnetic compatibility (EMC) requirements	18
4.2.3	Electronic warfare (EW) vulnerability and electronic counter-countermeasure (ECCM) capabilities	18
4.2.4	Human engineering design	18
4.2.5	Reliability	19
4.2.6	Maintainability	19
4.2.7	Survivability	19
4.2.8	Climatic conditions	20
4.2.9	Environmental test methods	20
4.2.10	Electrical measurement and test methods	20
4.2.11	Grounding, bonding and shielding	20
4.3	Functions of transmission facilities	20
4.3.1	Loop	20
4.3.2	Link	22
4.3.3	Trunk	22
4.3.4	Node	22
4.3.5	Channel	23
4.3.6	Circuit	23
4.3.7	Network	23
4.4	Types of signals	23
4.4.1	Analog signals	23
4.4.2	Digital signals	23
4.4.3	Quasi-analog signals	24
4.5	Multiplexing subsystems	24
4.5.1	Non-multiplexed subsystems	24
4.5.2	FDM subsystems	24
4.5.3	TDM subsystems	24
4.6	A-D conversion and modulation techniques	25
4.6.1	Analog (or non-discrete) pulse modulation	25
4.6.2	Digital (or discrete) pulse modulation	25
4.6.2.1	Pulse code modulation (PCM)	25
4.6.2.2	Delta modulation	26

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
4.6.2.3	Linear predictive coding (LPC)	27
4.7	Transmission media	27
4.7.1	Metallic lines	27
4.7.2	Radio relay subsystems	28
4.7.3	Radio subsystems operating in medium frequency (MF) and lower bands	28
4.7.4	HF radio subsystems	28
4.7.5	VHF radio subsystems	29
4.7.6	UHF radio subsystems	29
4.7.7	SHF radio subsystems	29
4.7.8	Fiber optic communications subsystems	30
4.8	Basic functions of tactical communications systems	30
4.9	Basic configurations of tactical communications systems	31
4.9.1	Basic configurations of tactical data communications systems	31
4.9.1.1	Data terminal equipment (DTE)	33
4.9.1.2	Digital interface adapters	33
4.9.1.3	Data circuit terminating equipment (DCE)	33
4.9.1.4	Modes of operation	33
4.9.1.4.1	Synchronous mode	33
4.9.1.4.2	Asynchronous mode	35
4.9.1.4.3	Start-stop mode	35
4.9.1.5	Signal conversion	35
5	REQUIREMENTS FOR TACTICAL COMMUNICATIONS SYSTEMS	37
5.1	General interconnection and design requirements	37
5.1.1	Interconnection of communications subsystems and systems	37
5.1.1.1	Interconnection of tactical subsystems type I or III with tactical subsystem type II	37
5.1.1.2	Interconnection of tactical subsystem type IV with tactical subsystem type I, II or III	37
5.1.1.3	Interconnection of single channel radio subsystems with switched multichannel subsystems	37
5.1.1.4	Interconnection with tactical NATO systems	38
5.1.1.5	Interconnection of long haul and tactical systems	38
5.1.2	General design requirements	38
5.1.2.1	Compromising emanations (TEMPEST) requirements	38
5.1.2.2	EMC requirements and guidance	38
5.1.2.2.1	Equipment requirements	38
5.1.2.2.2	Subsystem and system requirements	39
5.1.2.2.3	Equipment and subsystem EMC design guidance	39
5.1.2.2.4	rf spectrum characteristics	39
5.1.2.3	EW vulnerability and ECCM capabilities	39
5.1.2.3.1	Determination of EW technical threat	39

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.1.2.3.2	Analysis of EW vulnerability	39
5.1.2.3.3	Development of ECCM techniques	40
5.1.2.3.4	EMP vulnerability	40
5.1.2.4	Human engineering design	40
5.1.2.5	Reliability	40
5.1.2.6	Maintainability	40
5.1.2.7	Survivability	41
5.1.2.8	Climatic conditions	41
5.1.2.9	Environmental test methods	41
5.1.2.10	Electrical measurement and test methods	41
5.1.2.11	Grounding, bonding and shielding	41
5.2	Parameters for communications services (user-to-user)	42
5.2.1	General	42
5.2.2	Parameters for voice service (user-to-user)	42
5.2.2.1	Parameters for analog voice service (user-to-user)	42
5.2.2.1.1	Transmitted speech volume	42
5.2.2.1.2	Received speech volume	42
5.2.2.1.3	Received noise power	44
5.2.2.1.4	Insertion loss versus frequency characteristic	44
5.2.2.1.5	Envelope delay distortion	47
5.2.2.1.6	Single tone interference	47
5.2.2.1.7	Crosstalk	47
5.2.2.1.8	Echo	47
5.2.2.1.9	Frequency displacement	48
5.2.2.1.10	Net loss variations	48
5.2.2.2	Parameters for digitized voice service (user-to-user)	48
5.2.2.2.1	Bit error rate (BER)	48
5.2.2.2.2	Character-count and bit-count integrity	48
5.2.2.2.3	CVSD modulation	49
5.2.3	Parameters for telegraph and data service (user-to-user)	49
5.2.3.1	Parameters of VF circuits for telegraph and data service (user-to-user)	49
5.2.3.1.1	Bit error rate (BER)	49
5.2.3.1.2	Error control	50
5.2.3.1.3	Character-count and bit-count integrity	51
5.2.3.1.4	Transmitted quasi-analog signal level	51
5.2.3.1.5	Output parameters	52
5.2.3.1.6	Transfer function parameters	54
5.2.3.1.7	Signal discontinuities	59
5.2.3.1.8	Equalization	60
5.2.3.1.9	Regeneration	60
5.2.3.2	Parameters for telegraph and data service over 48 kHz FDM group circuits (user-to-user)	60

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.2.3.3	Parameters for telegraph and data service over digital circuits (user-to-user)	60
5.2.3.3.1	Bit error rate (BER)	60
5.2.3.3.2	Character-count and bit-count integrity	61
5.2.4	Parameters for facsimile service (user-to-user)	61
5.2.4.1	Parameters for analog facsimile service (user-to-user)	61
5.2.4.1.1	Transmitted signal level	61
5.2.4.1.2	Received signal level	61
5.2.4.1.3	Received noise power	61
5.2.4.1.4	Insertion loss versus frequency characteristic	61
5.2.4.1.5	Envelope delay distortion	61
5.2.4.1.6	Single tone interference	61
5.2.4.1.7	Frequency displacement	62
5.2.4.1.8	Net loss variations	62
5.2.4.2	Parameters for digital facsimile service (user-to-user)	62
5.2.5	Parameters for tactical digital information links (TADIL)	62
5.2.5.1	Parameters for TADIL A	62
5.2.5.2	Parameters for TADIL B	62
5.2.5.3	Parameters for TADIL C	62
5.3	Terminal subsystems	63
5.3.1	Analog terminal equipment	63
5.3.1.1	Analog telephone sets	63
5.3.1.2	Analog facsimile equipment	63
5.3.2	Data terminal equipment (DTE)	63
5.3.2.1	Digital facsimile equipment	65
5.3.3	Analog interface characteristics	65
5.3.3.1	Nominal 3 kHz VF channel interface	65
5.3.3.1.1	Impedance	65
5.3.3.1.2	Transmitted analog signal level	65
5.3.3.1.3	Received analog signal level	65
5.3.3.2	Nominal 4 kHz VF channel interface	65
5.3.3.2.1	Impedance	65
5.3.3.2.2	Transmitted analog signal level	65
5.3.3.2.3	Received analog signal level	65
5.3.4	Quasi-analog interface characteristics	65
5.3.4.1	Nominal 3 kHz VF channel interface	65
5.3.4.1.1	Impedance	66
5.3.4.1.2	Transmitted quasi-analog signal level	66
5.3.4.1.3	Received quasi-analog signal level	66
5.3.4.2	Nominal 4 kHz VF channel interface	66
5.3.4.2.1	Impedance	66
5.3.4.2.2	Transmitted quasi-analog signal level	67
5.3.4.2.3	Received quasi-analog signal level	67
5.3.4.3	48 kHz FDM group channel interface	67

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.3.4.3.1	Impedance	67
5.3.4.3.2	Transmitted quasi-analog signal level	67
5.3.4.3.3	Received quasi-analog signal level	67
5.3.5	Telegraph and data interface characteristics	67
5.3.5.1	Modulation and data signaling rates	67
5.3.5.2	Logic and signal sense for binary signals	69
5.3.5.3	Electrical characteristics of digital interface	71
5.3.5.4	Data signal connections	72
5.3.5.5	Input/output signal characteristics	72
5.3.5.6	AUTODIN interface characteristics	72
5.3.5.7	Data interface characteristics with tactical NATO systems	72
5.3.6	Clock equipment, control and timing	72
5.3.6.1	Transmission modes of station clocks	72
5.3.6.1.1	Bit synchronous mode	73
5.3.6.1.2	Character interval synchronous mode	74
5.3.6.1.3	Character interval control (stepping) mode	74
5.3.6.1.4	Bit-by-bit asynchronous mode	75
5.3.6.2	Clock characteristics	75
5.3.6.2.1	Clock modulation rates	75
5.3.6.2.2	Clock modulation rate stability	75
5.3.6.2.3	Modulation rate phase adjustment	76
5.3.6.2.4	Clock output signal	77
5.3.6.2.5	Clock signal period	77
5.3.6.3	Phase relationship between clock and data signals	78
5.3.6.3.1	Direct clock control	78
5.3.6.3.2	Indirect clock control	78
5.3.6.3.3	Sampling of data signals	81
5.3.6.3.4	Intermittent data transmission	81
5.3.6.3.5	Start-stop signals	81
5.3.6.4	Standard arrangements for supplying clock signals	84
5.3.6.4.1	Case one	84
5.3.6.4.2	Case two	84
5.3.6.4.3	Case three	86
5.3.6.4.4	Case four	86
5.3.6.4.5	Case five	87
5.3.7	Codes for information interchange	87
5.3.7.1	Scope and application	87
5.3.7.2	Character substitutions	88
5.3.7.3	Character parity	88
5.3.7.4	Basic ASCII character set	88
5.3.7.4.1	Character representation	90
5.3.7.4.2	Control characters	90
5.3.7.4.3	Graphic representation of control characters	90
5.3.7.4.4	Graphic characters	90

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.3.7.5	Serial-by-bit data communications	93
5.3.7.5.1	Bit sequencing	93
5.3.7.5.2	Character structure for synchronous data communications	94
5.3.7.5.3	Character structure for asynchronous data communications	94
5.3.7.5.4	Parity sense for synchronous data communications	95
5.3.7.5.5	Parity sense for asynchronous data communications	96
5.3.7.6	Parallel-by-bit data communications	96
5.3.7.6.1	Bit-to-channel relationship	96
5.3.7.6.2	Character structure	97
5.3.7.6.3	Character parity sense	97
5.3.7.7	ASCII subsets	97
5.3.7.7.1	95-Character ASCII graphic subset	97
5.3.7.7.2	64-Character ASCII graphic subset	99
5.3.7.7.3	16-Character ASCII graphic numeric subset	99
5.3.7.8	Code extension techniques	99
5.3.7.9	NATO 7-bit code for information interchange	102
5.3.8	Data modems	102
5.3.9	Keyboards	102
5.4	Transmission subsystems	103
5.4.1	Telegraph and data transmission subsystems	103
5.4.1.1	Bit error rate (BER)	103
5.4.1.2	Interface characteristics	103
5.4.1.3	Clock equipment, control and timing	103
5.4.1.4	Signal conditioning, diphase and conditioned diphase modulation	103
5.4.1.4.1	Signal conditioning	103
5.4.1.4.2	Diphase modulation	103
5.4.1.4.3	Conditioned diphase modulation	105
5.4.1.5	Speech plus half-duplex operation	105
5.4.1.5.1	Telephone channel characteristics	105
5.4.1.5.2	Data modems	106
5.4.1.5.3	Telegraph channel characteristics	106
5.4.1.6	Speech plus full-duplex operation	106
5.4.1.7	Monitoring, signal connections and path characteristics	106
5.4.1.7.1	Monitoring	106
5.4.1.7.2	Signal paths	106
5.4.1.7.3	Data signal connections	106
5.4.1.7.4	Transmission path performance	106
5.4.1.8	Recording media	107
5.4.1.8.1	Teletypewriter paper	107
5.4.1.8.2	Perforated tape	107
5.4.1.8.3	Magnetic tape	108

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.4.2	FDM telegraph subsystems	109
5.4.2.1	Twelve-channel FDM telegraph subsystem	109
5.4.2.1.1	Digital interface characteristics	109
5.4.2.1.2	Quasi-analog interface characteristics	109
5.4.2.1.3	Telegraph distortion	109
5.4.2.1.4	Crosstalk coupling loss	110
5.4.2.1.5	Center frequencies	110
5.4.2.1.6	Signal type and signal sense	111
5.4.2.1.7	Frequency tolerance	111
5.4.2.1.8	Frequency shift	111
5.4.2.2	Sixteen-channel FDM telegraph subsystem	111
5.4.2.2.1	Digital interface characteristics	111
5.4.2.2.2	Quasi-analog interface characteristics	111
5.4.2.2.3	Telegraph distortion	111
5.4.2.2.4	Crosstalk coupling loss	111
5.4.2.2.5	Center frequencies	111
5.4.2.2.6	Signal type and signal sense	111
5.4.2.2.7	Frequency tolerance	111
5.4.2.2.8	Frequency shift	111
5.4.3	Telegraph and data equipment	111
5.4.3.1	Common parameters	111
5.4.3.2	Data modems	111
5.4.3.3	Telegraph multiplexer equipment	113
5.4.3.4	Modulation rate converters	113
5.4.3.4.1	General	113
5.4.3.4.2	Clocking methods	113
5.4.3.5	Regenerative repeaters	114
5.4.3.5.1	Synchronous repeaters	114
5.4.3.5.2	Synchronous repeaters controlled by station clocks	116
5.4.3.5.3	Synchronous repeaters not controlled by station clocks	116
5.4.3.5.4	Start-stop regenerative repeaters	116
5.4.3.6	Telegraph code converters	117
5.4.3.7	Error control devices	117
5.4.3.8	Alphabet translators	118
5.4.3.9	Message composers	118
5.4.4	Radio subsystems	119
5.4.4.1	Radio regulations	119
5.4.4.2	Parameters for radio subsystems according to functional employment	119
5.4.4.2.1	Single channel point-to-point radio subsystems	119
5.4.4.2.2	Signal channel net radio subsystems	119
5.4.4.2.3	Telegraph and data transmission over single channel radio subsystems	120

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.4.4.2.4	Multichannel radio subsystems	121
5.4.4.2.5	Satellite subsystems	122
5.4.4.2.6	Manpack and handheld radio sets	122
5.4.4.3	Parameters for radio subsystems according to frequency bands	122
5.4.4.3.1	ELF band (below 300 Hz) and ILF band (300 Hz to 3 kHz)	122
5.4.4.3.2	VLF band (3 kHz to 30 kHz)	122
5.4.4.3.3	LF band (30 kHz to 300 kHz)	123
5.4.4.3.4	MF band (300 kHz to 3000 kHz)	123
5.4.4.3.5	HF band (3000 kHz to 30 MHz)	123
5.4.4.3.6	VHF band (30 MHz to 300 MHz)	123
5.4.4.3.7	UHF band (300 MHz to 3000 MHz)	124
5.4.4.3.8	SHF band (3000 MHz to 30 GHz)	124
5.4.4.3.9	Extremely high frequency (EHF) band (30 GHz to 300 GHz)	124
5.4.5	Interconnection of single channel radio subsystems and telephone facilities	124
5.4.5.1	Radiotelephone procedures	125
5.4.5.2	Analog NRI	125
5.4.5.2.1	Impedance	125
5.4.5.2.2	Insertion loss versus frequency characteristic	125
5.4.5.2.3	Total harmonic distortion	125
5.4.5.2.4	dc closure control	125
5.4.5.2.5	Ringdown signaling and dc closure control	127
5.4.5.2.6	Modulating signal	127
5.4.5.2.7	Dual-tone multifrequency (DTMF) signaling control	127
5.4.5.3	Digital NRI	129
5.4.6	Wire and cable transmission subsystems	129
5.4.6.1	Wire and cable loops	129
5.4.6.2	Wire and cable trunks	130
5.4.6.2.1	Crosstalk coupling loss	130
5.4.6.3	Digital transmission over wires and cables	130
5.4.6.4	Signal conditioning, diphase and conditioned diphase modulation	130
5.4.7	Fiber optic communications subsystems	130
5.5	Loop subsystems	131
5.5.1	General	131
5.5.2	Parameters of wire or cable loops for analog and quasi-analog signals	131
5.5.2.1	Insertion loss of wire or cable loops for tactical subsystems type I and type III	131
5.5.2.1.1	Loops for voice service and telegraph and data service	131
5.5.2.1.2	Loops for telegraph and data service	132

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.5.2.2	Insertion loss of wire or cable loops for tactical subsystem type II	132
5.5.2.3	Insertion loss of wire or cable loops for tactical subsystem type IV	132
5.5.2.4	Insertion loss versus frequency characteristics of wire or cable loops	132
5.5.2.4.1	Loops for analog voice and analog facsimile service	132
5.5.2.4.2	Loops for telegraph and data service with modulation rates of up to 1200 Bd	135
5.5.2.4.3	Loops for telegraph and data service with modulation rates of up to 2400 Bd	137
5.5.2.5	Envelope delay distortion of wire or cable loops	139
5.5.2.5.1	Loops for analog voice and analog facsimile service	139
5.5.2.5.2	Loops for telegraph and data service with modulation rates of up to 1200 Bd	139
5.5.2.5.3	Loops for telegraph and data service with modulation rates of up to 2400 Bd	140
5.5.2.6	Noise power	141
5.5.2.7	Impedance	142
5.5.2.8	Net loss variations	142
5.5.2.9	Test signal characteristics	142
5.5.3	Parameters for two-wire radio loops	142
5.5.3.1	Loop insertion loss	142
5.5.3.2	Noise power	142
5.5.3.3	Terminal impedance	143
5.5.3.4	Test signal characteristics	143
5.5.4	Parameters for four-wire radio loops	143
5.5.5	Parameters of wire or cable loops for data service over 48 kHz FDM group circuits	143
5.5.6	Parameters of wire or cable loops for digital signals	143
5.6	Nodal subsystems	144
5.6.1	General	144
5.6.2	Switching equipment	144
5.6.2.1	Analog automatic circuit switches	144
5.6.2.1.1	Interconnection requirements	144
5.6.2.1.2	Transmission characteristics	144
5.6.2.1.3	Control and signaling supervision procedures	146
5.6.2.1.4	Confirmation signaling	148
5.6.2.1.5	Signaling and supervisory signals	149
5.6.2.1.6	Tone detectors	149
5.6.2.1.7	Release signal protection (voice guard)	152
5.6.2.2	Analog semiautomatic circuit switches	152
5.6.2.2.1	Interconnection requirements	152
5.6.2.2.2	Transmission characteristics	152

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.6.2.2.3	Control and signaling supervision procedures	152
5.6.2.2.4	Confirmation signaling	152
5.6.2.2.5	Signaling and supervisory signals	152
5.6.2.2.6	Tone detectors	152
5.6.2.2.7	Release signal protection (voice guard)	152
5.6.2.3	Analog manual switchboards	152
5.6.2.3.1	Interconnection with tactical NATO circuit switches	152
5.6.2.3.2	Trunk signaling	154
5.6.2.3.3	Loop signaling	155
5.6.2.3.4	Insertion loss of signaling equipment	155
5.6.2.4	Digital automatic circuit switches	155
5.6.2.5	Message switches	155
5.6.3	Telephone directory numbering plan	155
5.6.3.1	Switch identification	156
5.6.3.2	Routing	156
5.6.3.3	Primary area (PR) codes	156
5.6.3.4	Switchboard locator (SL) codes	157
5.6.3.5	Subscriber number codes	157
5.6.3.6	Abbreviated dialing codes	157
5.6.3.7	Manual switchboard codes	157
5.6.3.8	Fixed directory codes	157
5.6.4	Control facilities	157
5.6.5	Patch and test facilities	157
5.6.6	Analog signal characteristics for testing, measuring and monitoring	157
5.6.6.1	Standard test signal	157
5.6.6.2	Standard test tone	158
5.6.6.3	Monitoring test tone	159
5.6.6.4	Levels for intermodulation distortion (IMD) measurements	159
5.6.6.5	Summary of test levels	159
5.6.7	Digital signal characteristics for testing, measuring and monitoring	159
5.7	Trunk subsystems	161
5.7.1	General	161
5.7.2	Multiplexing equipment	161
5.7.3	Modulation links	161
5.7.3.1	Parameters for FDM modulation links	162
5.7.3.1.1	FDM channel loading	162
5.7.3.1.2	Test signal levels	167
5.7.3.1.3	Noise power parameters	167
5.7.3.1.4	Link input/output impedances	172
5.7.3.1.5	Insertion loss versus frequency characteristics	174
5.7.3.1.6	Envelope delay distortion	180
5.7.3.1.7	Distortion parameters	183

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.7.3.1.8	Frequency displacement	183
5.7.3.1.9	Net loss variations	183
5.7.3.2	Parameters for TDM/PCM VF link	185
5.7.3.2.1	Characteristics of TDM/PCM signal	185
5.7.3.2.2	Link input/output parameters	186
5.7.3.2.3	Noise power parameters	188
5.7.3.2.4	Transfer function parameters	189
5.7.3.2.5	Crosstalk	192
5.7.3.2.6	Data transmission over TDM/PCM links	192
5.7.4	Trunk parameters	192
5.7.4.1	Trunk input parameters	193
5.7.4.1.1	FDM channel loading	193
5.7.4.1.2	TDM/PCM channel loading	193
5.7.4.1.3	Speech volume	193
5.7.4.1.4	Quasi-analog signal level	193
5.7.4.1.5	Standard test signal	193
5.7.4.1.6	Standard test tone	193
5.7.4.1.7	Monitoring test tone	193
5.7.4.1.8	Levels for IMD measurements	193
5.7.4.2	Trunk output parameters	193
5.7.4.2.1	Speech volume	193
5.7.4.2.2	Quasi-analog signal level	193
5.7.4.2.3	Noise power for FDM trunks of tactical subsystem type I	193
5.7.4.2.4	Noise power for TDM/PCM trunks of tactical subsystem type II	194
5.7.4.2.5	Noise power for FDM trunks of tactical subsystem type III	194
5.7.4.2.6	Noise power of trunks derived from SSB HF radio links	195
5.7.4.2.7	Noise power of wire and cable trunks	195
5.7.4.2.8	Single tone interference	195
5.7.4.3	Transfer function parameters	195
5.7.4.3.1	Insertion loss versus frequency characteristics	195
5.7.4.3.2	Envelope delay distortion	199
5.7.4.3.3	Total harmonic distortion	202
5.7.4.3.4	Intermodulation distortion (IMD)	203
5.7.4.3.5	Frequency displacement	203
5.7.4.3.6	Insertion gain	203
5.7.4.3.7	Limiting	204
5.7.4.3.8	Net loss variations	204
5.7.4.4	Crosstalk coupling loss	205
5.7.4.5	Trunk input/output impedances	205
5.7.4.6	Transmission level points (TLP) and test signal levels	205
5.7.4.7	Signal discontinuities	206

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.7.4.8	Grade of service for trunk groups	206
5.8	Transmission channel subsystems	207
5.8.1	Parameters for nominal 3 kHz VF transmission channels	207
5.8.2	Parameters for nominal 4 kHz VF transmission channels	207
5.8.2.1	Transmission channel input parameters	207
5.8.2.1.1	Speech volume	207
5.8.2.1.2	Quasi-analog signal level	207
5.8.2.1.3	Standard test signal	207
5.8.2.1.4	Standard test tone	207
5.8.2.1.5	Monitoring test tone	208
5.8.2.1.6	Levels for IMD measurements	208
5.8.2.2	Transmission channel output parameters	208
5.8.2.2.1	Speech volume	208
5.8.2.2.2	Quasi-analog signal level	208
5.8.2.2.3	Noise power for FDM transmission channels	208
5.8.2.2.4	Equivalent PCM noise power for TDM/PCM transmission channels	209
5.8.2.2.5	Idle channel noise power for TDM/PCM transmission channels	210
5.8.2.2.6	Single tone interference	210
5.8.2.3	Transmission channel input/output impedances	210
5.8.2.4	Insertion loss versus frequency characteristic	210
5.8.2.5	Distortion parameters	211
5.8.2.5.1	Envelope delay distortion	211
5.8.2.5.2	Total harmonic distortion	211
5.8.2.5.3	Intermodulation distortion (IMD)	212
5.8.2.6	Signal discontinuities	212
5.8.2.7	Frequency displacement	212
5.8.2.8	Net loss variations	212
5.8.3	Parameters for 48 kHz FDM group transmission channels	214
5.8.3.1	Transmission channel input parameters	214
5.8.3.1.1	Quasi-analog signal level	214
5.8.3.1.2	Test signal level	214
5.8.3.2	Transmission channel output parameters	214
5.8.3.2.1	Quasi-analog signal level	214
5.8.3.2.2	Test signal level	214
5.8.3.2.3	Noise power	214
5.8.3.2.4	Impulse noise power	215
5.8.3.3	Transmission channel input/output impedances	215
5.8.3.4	Insertion loss versus frequency characteristic	215
5.8.3.5	Envelope delay distortion	215
5.8.3.6	Net loss variations	217
5.8.4	Parameters for digital transmission channels	217

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
5.8.5	Methods for interconnecting VF channels of tactical subsystems type I, II, III and IV	217
5.8.5.1	Special method for interconnecting VF channels of tactical subsystems type I or III with type II	217
5.8.5.2	Method for interconnecting VF channels of tactical subsystems type I with type III	219
5.8.5.3	Method for interconnecting channels of tactical subsystems type I, II or III with type IV	220
5.9	Hypothetical reference circuits	222
5.9.1	General	222
5.9.2	Reference trunks	222
5.9.2.1	Reference trunks of tactical subsystem type I	222
5.9.2.2	Reference trunks of tactical subsystem type II	225
5.9.2.3	Reference trunks of tactical subsystem type III	230
5.9.2.4	Reference trunks of tactical subsystem type IV	230
5.9.3	Transfer characteristics of tandem connections	234
5.9.3.1	General	234
5.9.3.2	Noise power	234
5.9.3.3	Insertion loss versus frequency characteristic	235
5.9.3.4	Envelope delay distortion	236
5.9.3.5	Net loss variation	236
5.9.4	Examples of hypothetical reference circuits	236
5.9.4.1	General	236
5.9.4.2	Hypothetical reference circuits of tactical subsystem type I	236
5.9.4.3	Hypothetical reference circuits of tactical subsystem type II	243
5.9.4.4	Hypothetical reference circuits of tactical subsystem type III	250
5.9.4.5	Hypothetical reference circuits of tactical subsystem type IV	255
5.9.4.6	Common hypothetical reference circuit of tactical subsystems type I and type III	257
5.9.4.7	Common hypothetical reference circuit of tactical subsystems type II and type III	259

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the currentversion before use.

FIGURES

Number	Title	Page
1	Basic circuit configuration of multichannel trunking system	21
2	Basic configurations of tactical communications systems	32
3	Basic configurations of tactical data communications systems	34
4	Distribution of transmitted speech volume of telephones based on talker volume distribution	43
5	Subjective ratings of received noise power	45
6	VF circuit parameters for analog voice and analog facsimile service (user-to-user)	46
7	VF circuit parameters for telegraph and data service with modulation rates of up to 1200 Bd (user-to-user)	55
8	VF circuit parameters for telegraph and data service with modulation rates of up to 2400 Bd (user-to-user)	58
9	Example of block diagram of data terminal subsystem	64
10	Standard interfaces between data terminal equipment and data circuit terminating equipment	68
11	Phase relationship between clock and data signals for direct clock control	79
12	Phase relationship between clock and data signals for indirect clock control	80
13	Phase relationship between clock and data signals for sampling of data (direct clock control)	82
14	Phase relationship between clock and data signals for intermittent data transmission (gated clock)	83
15	Standard arrangement case 1 for supplying clock signals	85

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Number	Title	Page
16	Standard arrangement case 2 for supplying clock signals	85
17	Standard arrangement case 3 for supplying clock signals	86
18	Standard arrangement case 4 for supplying clock signals	87
19	ASCII character structure for synchronous serial-by-bit data communications	93
20	ASCII character structure for asynchronous serial-by-bit data communications	95
21	Signal conditioning, diphase and conditioned diphase modulation	104
22	Block diagram of net radio interface	126
23	Assumed insertion loss distribution for wire or cable loops of tactical subsystem type II	133
24	Parameters of wire or cable loops for analog voice and analog facsimile service	134
25	Parameters of wire or cable loops for telegraph and data service with modulation rates of up to 1200 Bd	136
26	Parameters of wire or cable loops for telegraph and data service with modulation rates of up to 2400 Bd	138
27	Basic configuration of FDM modulation link without through-equipment	163
28	Block diagram of FDM equipment	164
29	Basic configuration of FDM modulation link with through-equipment	165
30	FDM channel loading curves	166
31	Interface and test levels for FDM VF links	169

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Number	Title	Page
32	Parameters for FDM VF links of tactical subsystem type I	176
33	Parameters for FDM VF link without through-equipment of tactical subsystem type III	177
34	Parameters for FDM VF link with through-equipment of tactical subsystem type III	178
35	Block diagram of TDM/PCM VF link of tactical subsystem type II	185
36	Parameters for TDM/PCM VF links of tactical subsystem type II	190
37	Parameters for FDM trunks of tactical subsystem type I	197
38	Parameters for TDM/PCM trunks of tactical subsystem type II	198
39	Parameters for FDM trunks of tactical subsystem type III	200
40	Parameters for nominal 4 kHz transmission channels	213
41	Parameters for unconditioned 48 kHz FDM group transmission channels	216
42	Signal level and interface diagram for connecting VF channels of tactical subsystems type I or III with type II	218
43	Singal level and interface diagram for connecting VF channels of tactical subsystems type I with type III	221
44	FDM VF reference trunks of tactical subsystem type I	223
45	TDM/PCM VF reference trunks with two-wire terminations of tactical subsystem type II	226
46	TDM/PCM VF reference trunks with two-wire/four-wire terminations of tactical subsystem type II	227
47	TDM/PCM VF reference trunks with four-wire terminations of tactical subsystem type II	228

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Number	Title	Page
48	FDM VF reference trunk with LOS radio links of tactical subsystem type III	231
49	FDM VF reference trunk with LOS and tropospheric scatter radio links of tactical subsystem type III	232
50	Hypothetical 300 km (162 nmi) switched VF reference circuit for voice service of tactical subsystem type I	238
51	Hypothetical 300 km (162 nmi) switched VF reference circuit for voice service and telegraph and data service of tactical subsystem type I	239
52	Hypothetical 300 km (162 nmi) nonswitched (point-to-point) VF reference circuit for telegraph and data service of tactical subsystem type I	241
53	Hypothetical 300 km (162 nmi) switched VF reference circuit for voice service of tactical subsystem type II	244
54	Hypothetical 300 km (162 nmi) switched VF reference circuit for voice service and telegraph and data service of tactical subsystem type II	246
55	Hypothetical 300 km (162 nmi) nonswitched (point-to-point) VF reference circuit for telegraph and data service of tactical subsystem type II	247
56	Hypothetical 1800 km (972 nmi) switched VF reference circuit for voice service of tactical subsystem type III	251
57	Hypothetical 1800 km (972 nmi) switched VF reference circuit for voice service and telegraph and data service of tactical subsystem type III	252
58	Hypothetical 1800 km (972 nmi) nonswitched (point-to-point) VF reference circuit for telegraph and data service of tactical subsystem type III	254
59	Common hypothetical 1150 km (621 nmi) switched VF reference circuit for voice service and telegraph and data service of tactical subsystems type I and type III	258

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

Number	Title	Page
60	Common hypothetical 1150 km (621 nmi) switched VF reference circuit for voice service of tactical subsystems type II and type III	262
61	Block diagrams of CVSD encoder/decoder	D-285
62	Signal level and interface diagram for CVSD encoder/decoder	D-289
63	Block diagram of linear predictive encoder/decoder	D-291
64	Examples of binary state symmetric coded waveforms	E-297
65	Examples of binary state unsymmetric coded waveforms	E-300
66	Examples of binary differential symmetric coded waveforms	E-303
67	Examples of binary differential unsymmetric coded waveforms	E-306
68	Phase delay versus frequency characteristic	G-318
69	Envelope delay versus frequency characteristic	G-319
70	Absolute and relative envelope delay versus frequency characteristic	G-320
71	Noise weighting curves	L-366
72	Sum of two dB values	L-372

Source: https://assist.dla.mil Downloaded: 202640-26T07:28Z
Check the source to verify that this is the current version before use.

TABLES

Number	Title	Page
I	Switched multichannel communications subsystems	14
II	Subjective ratings of received speech volume	44
III	Insertion loss versus frequency characteristic of VF circuits for analog voice and analog facsimile service (user-to-user)	45
IV	Insertion loss versus frequency characteristic of VF circuits for telegraph and data service with modulation rates of up to 1200 Bd (user-to-user)	54
V	Insertion loss versus frequency characteristic of VF circuits for telegraph and data service with modulation rates of up to 2400 Bd (user-to-user)	56
VI	Envelope delay distortion of VF circuits for telegraph and data service with modulation rates of up to 1200 Bd (user-to-user)	57
VII	Envelope delay distortion of VF circuits for telegraph and data service with modulation rates of up to 2400 Bd (user-to-user)	57
VIII	Logic and signal sense for binary signals	71
IX	Basic ASCII character set	89
X	Control characters of basic ASCII character set	91
XI	Graphic characters of basic ASCII character set	92
XII	ASCII bit-to-channel relationship	96
XIII	95-Character ASCII graphic subset	98
XIV	64-Character ASCII graphic subset	100
XV	16-Character ASCII graphic numeric subset	101

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the currentversion before use.

Number	Title	Page
XVI	Frequency characteristics for 12-channel FDM telegraph subsystem	110
XVII	Frequency characteristics for 16-channel FDM telegraph subsystem	112
XVIII	Input signal distortion for synchronous and start-stop regenerative repeaters	115
XIX	DTMF and SF signals for NRI control unit	128
XX	Insertion loss versus frequency characteristic of wire or cable loops for analog voice and analog facsimile service	135
XXI	Insertion loss versus frequency characteristic of wire or cable loops for telegraph and data service with modulation rates of up to 1200 Bd	137
XXII	Insertion loss versus frequency characteristic of wire or cable loops for telegraph and data service with modulation rates of up to 2400 Bd	139
XXIII	Envelope delay distortion of wire or cable loops for telegraph and data service with modulation rates up to 1200 Bd	140
XXIV	Envelope delay distortion of wire or cable loops for telegraph and data service with modulation rates of up to 2400 Bd	141
XXV	Transmission levels for analog circuit switches	146
XXVI	Confirmation signaling for analog automatic circuit switches	149
XXVII	Signaling and supervisory signals for analog automatic circuit switches	150
XXVIII	Signaling and supervisory signals for analog semiautomatic circuit switches	153
XXIX	Telephone directory numbering plan	156
XXX	Summary of test levels	160
XXXI	Test signal levels for FDM modulation links	168

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Number	Title	Page
XXXII	Multiplex noise power for FDM equipment and VF links of tactical subsystem type III	170
XXXIII	Multiplex noise power for FDM equipment and group links of tactical subsystem type III	171
XXXIV	Transmission media noise power for FDM modulation links of tactical subsystem type III	173
XXXV	Insertion loss versus frequency characteristic for FDM VF links of tactical subsystem type I	174
XXXVI	Insertion loss versus frequency characteristics for FDM equipment and VF links of tactical subsystem type III	175
XXXVII	Insertion loss versus frequency characteristics for FDM equipment and group links of tactical subsystem type III	179
XXXVIII	Envelope delay distortion for FDM VF links of tactical subsystem type I	180
XXXIX	Envelope delay distortion for FDM equipment and VF links of tactical subsystem type III	181
XL	Envelope delay distortion for FDM equipment and group links of tactical subsystem type III	182
XLI	Net loss variations for FDM equipment and modulation links of tactical subsystem type III	184
XLII	Idle channel noise power for TDM/PCM VF links of tactical subsystem type II	188
XLIII	Quantizing noise ratio for TDM/PCM VF links of tactical subsystem type II	189
XLIV	Insertion loss versus frequency characteristic for TDM/PCM VF links of tactical subsystem type II	189
XLV	Envelope delay distortion for TDM/PCM VF links of tactical subsystem type II	191
XLVI	Insertion loss versus frequency characteristic for FDM trunks of tactical subsystem type I	196

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

Number	Title	Page
XLVII	Insertion loss versus frequency characteristic for TDM/PCM trunks of tactical subsystem type II	196
XLVIII	Insertion loss versus frequency characteristic for FDM trunks of tactical subsystem type III	199
XLIX	Envelope delay distortion for FDM trunks of tactical subsystem type I	201
L	Envelope delay distortion for TDM/PCM trunks of tactical subsystem type II	202
LI	Envelope delay distortion for FDM trunks of tactical subsystem type III	202
LII	Limiting characteristic for trunks	204
LIII	Insertion loss versus frequency characteristic for nominal 4 kHz transmission channels	211
LIV	Envelope delay distortion for nominal 4 kHz transmission channels	212
LV	Envelope delay distortion for 48 kHz FDM group transmission channels	215
LVI	Parameters for FDM VF reference trunks of tactical subsystem type I	224
LVII	Parameters for TDM/PCM VF reference trunks of tactical subsystem type II	229
LVIII	Parameters for FDM VF reference trunks of tactical subsystem type III	233
LIX	Parameters for hypothetical 300 km (162 nmi) VF reference circuits of tactical subsystem type I	242
LX	Parameters for hypothetical 300 km (162 nmi) VF reference circuits of tactical subsystem type II	249
LXI	Parameters for hypothetical 1800 km (972 nmi) VF reference circuits of tactical subsystem type III	256
LXII	Parameters for common hypothetical VF reference circuit of tactical subsystems type I and type III	260

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Number	Title	Page
LXIII	Calculations of circuit noise power for reference circuit of Figure 60	263
LXIV	Parameters for common hypothetical VF reference circuit of tactical subsystems type II and type III	265
LXV	Quantizing noise ratio for CVSD converter	D-290
LXVI	American version of ITA No. 2	H-325
LXVII	Spectrum designation of frequency	J-341
LXVIII	Amplitude modulated emission designators	J-347
LXIX	Frequency and phase modulated emission designators	J-353
LXX	Pulse modulated emission designators	J-356
LXXI	Characteristics of noise weighting networks	L-368
LXXII	Conversion formulas for noise power units	L-370
LXXIII	Addition factors for calculating the sum of two dB values	L-371

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDICES

Appendix	Title	Page
A	Memorandum from the Assistant Secretary of Defense for Communications, Command, Control and Intelligence, 10 May 1977, Subject: Mandatory Use of Military Standards in the 188 Series	A-267
B	List of Tactical and Common Long Haul/ Tactical Standardization Projects of the MIL-STD-188 Series	B-271
C	List of Abbreviations and Acronyms used in MIL-STD-188-200	C-275
D	Modulation Characteristics for Continuously Variable Slope Delta (CVSD) and Linear Predictive Coding (LPC)	D-281
E	Coding and Modulation Techniques for Binary Digital Signals	E-293
F	General Guidance for the Engineering of Telegraph and Data Circuits	F-307
G	Phase and Envelope Delays, and Delay Distortions	G-315
H	Codes for Information Interchange	H-321
I	Timing and Synchronization of Communications Networks	I-331
J	Spectrum Designation, Emissions and Necessary Bandwidths	J-339
K	FDM Channel Loading	K-359
L	Hypothetical Reference Circuits and Related Information	L-363
INDEX		375

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Purpose. The purpose of this document is to promulgate technical parameters in the form of mandatory system standards and optional design objectives that are considered necessary to ensure interoperability and to promote compatibility and commonality among tactical communications equipment, subsystems and systems. It is also the purpose of this document to establish a level of performance of tactical communications equipment, subsystems and systems considered necessary to satisfy the requirements of a majority of users. The technical parameters promulgated by this document represent, in general, minimum interoperability and performance characteristics which may be exceeded in order to satisfy specific requirements. For example, other codes may be added to the minimum mandatory capability of the American Standard Code for Information Interchange (ASCII), as stated in the note of 5.3.7.1.

Standards contained in this document have been based on measured performance of existing communications equipment according to the definition of system standards as stated in Federal Standard (FED-STD) 1037. Consequently, it is not the purpose of this document to establish all standards considered necessary to ensure interoperability of future communications equipment that is currently being designed and developed. However, this document provides guidance to the designers of future equipment by standardizing the technical characteristics of existing communications assets to facilitate the necessary design decisions required to achieve interoperability between existing and future communications facilities. In addition to addressing interoperability parameters, emphasis has been placed on establishing acceptable overall system performance with maximum flexibility of system configuration to satisfy diverse user requirements.

It is not the purpose of this document to serve as a stand-alone, comprehensive reference containing all technical parameters and other details required for the design of new equipment and facilities or the preparation of specifications. Therefore, parameters for such items as size and weight limitations, connectors, cable assemblies, or power supplies are not contained in this document. These parameters and other design details have to be established, based on specific requirements, and have to be carefully tailored in accordance with the policies of DoD Directive (DoDD) 4120.21.

This document is not intended to be a communications engineering textbook or a reference handbook. It is assumed that users of this document have a basic technical background in communications system engineering. However, this document contains considerable technical background information to explain why certain parameters were standardized, how some of the standards were derived, where these standards apply and how the standards are to be used. The purpose of this tutorial information is to assist the user of this document in determining

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

how certain standards and design objectives are to be applied to particular situations.

It is not the purpose of this document to inhibit advances in communications technology. Such advances are encouraged by including design objectives which should be achieved, if economically feasible. Additionally, such advances are facilitated by standardizing upper or lower limits rather than fixed parameter values and by avoiding to specify the technology that should be used to meet the required standards.

1.2 Application. This document applies to the design and development of new equipment, assemblages and subsystems used in tactical communications systems. This document applies to switched and non-switched multichannel subsystems and to single channel radio subsystems. Therefore, this document applies whenever there is an interface between a weapons platform, such as ships, aircraft and vehicles, and a land-based single channel radio net or multichannel area network.

When standards are selected from this document, it should be ascertained that only those standards are chosen that apply to the particular type of equipment, subsystem or system under consideration.

This document applies also to the engineering and installation of tactical communications equipment, subsystems and systems. It is not intended that existing communications facilities be immediately converted to comply with the standards contained in this document. New facilities and those undergoing major modification or rehabilitation shall comply with the standards contained in this document subject to the applicable requirements of current procurement regulations.

When tactical communications circuits are interconnected with long haul circuits, there are additional standards in MIL-STD-188-100 which apply. (See 5.1.1.5.)

Unless otherwise stated, performance parameters contained in this document do not apply to operation in an electronic warfare environment.

1.3 Objectives. The main objectives of this document are to ensure interoperation of tactical equipment, subsystems and systems consistent with military requirements, to provide a degree of system performance acceptable to a majority of users of tactical communications systems, and to achieve the necessary degree of interoperation, performance and compatibility in the most economical way. These objectives will be accomplished by continuing efforts in the following areas:

a. Standardizing user-to-user performance characteristics and hypothetical reference circuits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

b. Specifying the probable maximum transmission distance of communications circuits.

c. Standardizing the characteristics of signals at various interface points in a communications system.

d. Specifying the maximum permissible degradation of a signal in the process of transmission, and allocating the permissible degradation among various parts of a system.

e. Establishing performance parameters and operating features of equipment, not of detail design, but of other factors which govern the interface characteristics with subsystems and systems in which the equipment will be used.

f. Defining performance parameters without specifying the technology that should be used to obtain the required performance.

An additional objective of communications system standards is to prevent proliferation of equipment serving the same or similar function. The variety of equipment shall be the minimum necessary to effectively support the missions of the tactical forces in accordance with DoDD 4630.5.

1.4 System standards and design objectives. The parameters and other requirements specified in this document are mandatory system standards (see Appendix A) if the word "shall" is used in connection with the parameter value or requirement under consideration. Nonmandatory design objectives are indicated by parentheses after a standardized parameter value or by the word "should" in connection with the parameter value or requirement under consideration. For a definition of the terms "System Standard" and "Design Objective" see FED-STD-1037.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

2.1.1 Federal specifications.

UU-P-547	Paper, Teletypewriter Roll
UU-T-120	Tape, Teletypewriter, Perforator

2.1.2 Military specifications.

MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-P-40023	Paper, Teletypewriter, Continuous Flatfold

2.1.3 Federal standards.

2.1.3.1 Federal telecommunications standards.

FED-STD-1005	Coding and Modulation Requirements for Non-diversity 2400 Bit/Second Modems
FED-STD-1006	Coding and Modulation Requirements for 4800 Bit/Second Modems
FED-STD-1010	Bit Sequencing of the American National Standard Code for Information Interchange in Serial-by-Bit Data Transmission
FED-STD-1011	Character Structure and Character Parity Sense for Serial-by-Bit Data Communication in the American National Standard Code for Information Interchange
FED-STD-1012	Character Structure and Character Parity Sense for Parallel-by-Bit Data Communication in the American National Standard Code for Information Interchange
FED-STD-1013	Synchronous Signaling Rates Between Data Terminal Equipment and Data Circuit-Terminating Equipment Utilizing 4 kHz Circuits
FED-STD-1037	Glossary of Telecommunication Terms

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

2.1.3.2 Federal information processing standards publications (FIPS PUB).

FIPS PUB 1-1	Code for Information Interchange
FIPS PUB 2	Perforated Tape Code for Information Interchange
FIPS PUB 3-1	Recorded Magnetic Tape for Information Interchange (800 cpi, NRZI)
FIPS PUB 7	Implementation of the Code for Information Interchange and Related Media Standards
FIPS PUB 15	Subsets of the Standard Code for Information Interchange
FIPS PUB 16-1	Bit Sequencing of the Code for Information Interchange in Serial-by-Bit Data Transmission
FIPS PUB 17-1	Character Structure and Character Parity Sense for Serial-by-Bit Data Communication in the Code for Information Interchange
FIPS PUB 18-1	Character Structure and Character Parity Sense for Parallel-by-Bit Data Communication in the Code for Information Interchange
FIPS PUB 22-1	Synchronous Signaling Rates Between Data Terminal and Data Communication Equipment
FIPS PUB 25	Recorded Magnetic Tape for Information Interchange (1600 cpi, Phase Encoded)
FIPS PUB 26	One-Inch Perforated Tape for Information Interchange
FIPS PUB 35	Code Extension Techniques in 7 or 8 Bits
FIPS PUB 36	Graphic Representation of the Control Characters of ASCII (FIPS 1)
FIPS PUB 50	Recorded Magnetic Tape for Information Interchange, 6250 cpi (246 cpmm), Group Coded Recording

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

2.1.4 Military standards.

2.1.4.1 DoD standards.

MIL-STD-188C	Military Communication System Technical Standards
MIL-STD-188-100	Common Long Haul and Tactical Communication System Technical Standards
MIL-STD-188-110	Equipment Technical Design Standards for Common Long Haul/Tactical Data Modems
MIL-STD-188-114	Electrical Characteristics of Digital Interface Circuits
MIL-STD-188-124	Grounding, Bonding and Shielding for Common Long Haul/Tactical Communication Systems
MIL-STD-188-140	Equipment Technical Design Standards for Common Long Haul/Tactical Radio Communications in the Low Frequency Band and Lower Frequency Bands
MIL-STD-188-161	Design Standards for Common Long Haul and Tactical Facsimile Equipment
MIL-STD-188-203-1	Subsystem Design and Engineering Standards for Tactical Digital Information Link (TADIL) A
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-449	Radio Frequency Spectrum Characteristics, Measurement of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definition and System of Units, Electromagnetic Interference and Electromagnetic Compatibility Technology
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation
MIL-STD-781	Reliability Tests: Exponential Distribution

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-1280	Keyboard Arrangements
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

2.1.4.2 North Atlantic Treaty Organization (NATO) Standardization Agreements (STANAG) and other publications.

STANAG 5018	NATO Manual Interface Between the Manual Switched Telecommunications Systems of the Combat Zone
STANAG 5036	Parameters and Practices for the Use of the NATO 7-Bit Code
STANAG 5040	NATO Automatic and Semi-Automatic Interfaces Between the National Switched Telecommunications Systems of the Combat Zone and Between these Systems and the NATO Integrated Communications System (NICS) - Period from 1975 to the 1990's
STANAG 5046	The NATO Military Communications Directory System
ACP 127 NATO SUPP-3	Message Relay Procedures

2.1.4.3 Quadripartite Standardization Agreements (QSTAG).

QSTAG-263A	Standards to Achieve Interoperability of ABCA Armies High Frequency Combat Net Radio Equipments
QSTAG-263B	Standards to Achieve Interoperability of ABCA Armies Very High Frequency Combat Net Radio Equipments
QSTAG-263C	Standards to Achieve Interoperability of ABCA Armies Ultra High Frequency Combat Net Radio Equipments

2.1.5 Military handbooks.

MIL-HDBK-232	RED/BLACK Engineering-Installation Guidelines (U)

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

MIL-HDBK-235	Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment, Subsystems and Systems
MIL-HDBK-237	Electromagnetic Compatibility/Interference Program Requirements
MIL-HDBK-241	Design Guide for EMI Reduction in Power Supplies
MIL-HDBK-253	Guidance for the Design and Test of Systems Protected Against the Effects of Electromagnetic Energy

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or requests for proposal shall apply.

2.2.1 DoD publications.

DCAC 300-175-9, DCS Operating-Maintenance Electrical Performance Standards

DCAC 370-D175-1, DCS AUTODIN Interface and Control Criteria

DCAC 370-V175-6, System Interface Criteria

Joint Army Navy Air Force Publication (JANAP) 128, Automatic Digital Network (AUTODIN) Operating Procedures

Joint Chiefs of Staff Publication (JCS PUB) 10, Tactical Command and Control, and Communications Systems Standards (U)

NACSIM 5100, Compromising Emanations Laboratory Test Requirements, Electromagnetics (U)

NACSEM 5201, TEMPEST Guidelines for Equipment/System Design (U)

DoDD 4120.21, Specifications and Standards Application

DoDD C-4600.3, Electronic Counter-Countermeasures (ECCM) Policy (U)

DoDD 4630.5, Compatibility and Commonality of Equipment for Tactical Command and Control, and Communications

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

2.2.2 Allied Communication Publications (ACP).

ACP 125, Communication Instructions, Radiotelephone Procedure

ACP 127, Communication Instructions, Tape Relay Procedures (U)

(Copies of specifications, standards, and publications listed in 2.1.1 through 2.2.2, above, and required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2.3 Standards of the American National Standards Institute (ANSI).

ANSI X3.1	Synchronous Signaling Rates for Data Transmission
ANSI X3.4	Code for Information Interchange
ANSI X3.6	Perforated Tape Code for Information Interchange
ANSI X3.15	Bit Sequencing of the American National Standard Code for Information Interchange in Serial-by-Bit Data Transmission
ANSI X3.16	Character Structure and Character Parity Sense for Serial-by-Bit Data Communication in the American National Standard Code for Information Interchange
ANSI X3.18	One-Inch Perforated Paper Tape for Information Interchange
ANSI X3.22	Recorded Magnetic Tape for Information Interchange (800 cpi, NRZI)
ANSI X3.25	Character Structure and Character Parity Sense for Parallel-by-Bit Data Communication in the American National Standard Code for Information Interchange
ANSI X3.32	Graphic Representation of the Control Characters of American National Standard Code for Information Interchange
ANSI X3.39	Recorded Magnetic Tape for Information Interchange (1600 cpi, PE)
ANSI X3.41	Code Extension Techniques for Use with the 7-Bit Coded Character Set of American National Standard Code for Information Interchange

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

ANSI X3.54	Recorded Magnetic Tape for Information Interchange (6250 cpi, Group-Coded Recording)

(Copies of the above standards may be purchased from the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

2.2.4 Standard of the Institute of Electrical and Electronics Engineers (IEEE).

IEEE 152	Volume Measurements of Electrical Speech and Program Waves, Recommended Practice for

(Copies of the above standard may be purchased from the Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

2.2.5 International regulations.

International Telecommunication Union (ITU), Radio Regulations

(Copies of the above regulations may be purchased from the International Telecommunication Union, Place des Nations, CH-1211 Geneva 20, Switzerland.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

3. DEFINITIONS

3.1 Definition of terms. Definition of terms used in this document shall be as specified in FED-STD-1037 which incorporates and supersedes MIL-STD-188-120. Definitions of electromagnetic compatibility terms shall be as specified in MIL-STD-463.

3.2 Abbreviations and acronyms. The abbreviations and acronyms used in this document are listed in Appendix C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

4. GENERAL DESCRIPTION OF TACTICAL COMMUNICATIONS

This section describes tactical communications in a general way consistent with the definitions as stated in FED-STD-1037. This section is intended to supplement, not repeat, definitions contained in FED-STD-1037.

4.1 Tactical communications systems. These systems are normally configured by various types of mobile equipment and assemblages. The assemblages are designed to allow swift installation and reconfiguration of communications facilities and subsystems in order to meet the frequently changing requirements for command, control and support of tactical forces. Normally, tactical communications equipment and assemblages are organic to the tactical forces and are installed and operated by military personnel. Tactical communications systems provide securable voice and data communications to fixed and mobile users. These systems may range from single channel radio nets to large transportable multichannel communications complexes. Based on various requirements for multichannel trunking networks, different tactical switched multichannel subsystems have been designed. In this document, these multichannel subsystems are classified as tactical subsystems type I, II, III and IV. Table I summarizes the basic differences among these types of tactical multichannel subsystems.

4.1.1 Tactical subsystems type I and type II. These types are switched multichannel subsystems that have been designed for distances up to several hundred kilometers (km). The probable maximum transmission distance over which communications are required has been taken as 300 km or 162 nautical miles (nmi) which is also used as the nominal distance of hypothetical reference circuits of the tactical subsystems type I and II. (See 5.9.4.) Tactical subsystems type I and II require extremely short communications facility installation times (on the order of hours) caused by the frequent relocation requirements of these facilities. The selection of radio sites is usually dictated more by tactical requirements than by radio propagation requirements, often resulting in less than optimum radio path profiles. The conditions under which tactical subsystems type I and II have to be installed, reconfigured and operated make it difficult to control and maintain required performance characteristics of communications circuits. The tactical subsystem type I uses frequency division multiplexing (FDM) equipment which has been designed to operate with a test signal level of 0 dBm at the four-wire input and output terminals of a trunk and with a zero transmission level point (OTLP) at these terminals. The tactical subsystem type II uses time division multiplexing/pulse code modulation (TDM/PCM) equipment which has been designed to operate with a test signal level of -4 dBm at the four-wire input and output terminals of a trunk and with a -4TLP at these terminals. (See Table I.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE I. Switched multichannel communications subsystems.

MIL-STD-188-200 Terms	Probable Maximum Distance	Multiplexer Equipment	TLP (4-Wire Trunk)	Test Signal Level	Test Tone Level	MIL-STD-188-100 Terms
Tactical Subsystem Type I	300 km	FDM	0TLP	0 dBm0	-10 dBm0	Tactical Highly Maneuverable System
Tactical Subsystem Type II	300 km	TDM/PCM	-4TLP	0 dBm0	-3 dBm0	
		FDM*	-4TLP	0 dBm0	-3 dBm0	
Tactical Subsystem Type III	1800 km	FDM	0TLP	0 dBm0	-10 dBm0	Tactical Less Maneuverable System
Tactical Subsystem Type IV	1200 km	TDM/CVSD	Not Appl.	Not Appl.	Not Appl.	Not applicable

*There is older FDM equipment still in the inventory. This equipment has been classified as belonging to the tactical subsystem type II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

There is older FDM equipment still in the inventory that has been designed to operate with the same test signal level of -4 dBm, as is used for TDM/PCM equipment. Subsystems employing this FDM equipment are also classified as tactical subsystem type II and must follow the same interconnection requirements as specified in 5.8.5.1.

4.1.2 Tactical subsystem type III. This type is a switched multichannel subsystem that has been designed for much longer distances than tactical subsystems type I and II. The probable maximum transmission distance over which communications are required has been taken as 1800 km (972 nmi) which is also used as the nominal distance of hypothetical reference circuits of the tactical subsystem type III. (See 5.9.4.) The tactical subsystem type III is normally required to cover much longer distances than the tactical subsystems type I and II. Consequently, the selection of radio sites has to be carefully planned and more time is needed to install communications facilities and to align transmission links than is usually available for the tactical subsystems type I and II. The tactical subsystem type III uses FDM equipment which has been designed to operate with a test signal level of 0 dBm at the four-wire input and output terminals of a trunk and with a OTLP at these terminals. (See Table I.)

4.1.3 Tactical subsystem type IV. This type is a switched multichannel subsystem employing automatic message and circuit switches with end-to-end security. The type IV subsystem uses continuously variable slope delta (CVSD) modulation as the analog-to-digital (A-D) conversion method. (See Table I.) Store-and-forward (S-F) message switches and automatic circuit switches connected by TDM transmission facilities form the backbone of this subsystem. A full range of terminal equipment (e.g., secure and non-secure voice, message, facsimile and data equipment) will be provided for combat, combat support and combat service support missions. The subsystem is being designed with a mix of analog and digital switching, transmission and terminal equipment. The analog equipment will consist of equipment presently in the inventory which may be used in its current form or be modified to assure integrity of the mixed analog/digital network. The digital equipment, which is being introduced, has an integrated communications security capability. The major types of equipment include hybrid analog/digital circuit switches, secure and non-secure digital telephones, a family of unit level switches, digital multiplexing and transmission equipment, nodal control elements and a family of modular record traffic terminals. The tactical subsystem type IV has short communications facility installation times (in most cases less than an hour) which meet the frequent relocation requirements of combat units. The probable maximum transmission distance has been taken as 1200 km (648 nmi) which will be used as the nominal distance of hypothetical reference circuits of the tactical subsystem type IV.

4.1.4 Single channel radio subsystems. These subsystems consist of manpack and vehicular mounted radio sets used primarily in radio nets for

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

command and control of combat, combat support, and combat service support operations between and among ground, naval and airborne forces. These subsystems provide a single wideband or narrowband channel for securable voice, record traffic (teletype and facsimile) and other types of data. Single channel radio equipment is also used in point-to-point communications, as radio loops (see 5.5.4) in multichannel switched networks, and with satellites in single-access and multiple-access modes of operations. Interconnections of single channel radio equipment with multichannel switched subsystems is possible by means of net radio interface (NRI) equipment. (See 5.4.5.) NRI is a special use of radio and wire integration (RWI) equipment which connects radio facilities with wire circuits.

4.1.5 Interconnection of communications subsystems. Different requirements for communications subsystems have led to the development of equipment, such as multiplexers and wideband radios, with significantly different signal levels (voice, data and test tone) and overload characteristics. Therefore, special methods have to be used for interconnecting channels of different types of existing communications subsystems. These special methods are stated in 5.1.1.

4.1.6 Tactical communications equipment. Tactical communications equipment is that equipment organic to the tactical forces required to convey information, by electrical or electromagnetic means, between persons or places within the tactical forces. The term includes transmission (multichannel or single channel), switching, communications security, general purpose terminal devices (such as teletypewriter, data, facsimile and telephones), technical control and system control, interface devices and satellite terminal equipment. The term does not include sensory devices and specialized equipment (e.g., radar, identification friend or foe (IFF) and navigation aids), related communications support ancillary items, such as tools, test equipment and installer and maintenance devices.

4.1.7 Design, installation and operation. The conditions under which tactical equipment will operate vary widely. Tactical communications systems range from manpack equipment used in tactical single channel radio nets to large transportable multichannel communications facilities and even some fixed installations used by tactical forces for support of military operations. In general, for tactical systems, it is not known beforehand precisely what sites, distances, terrain and other conditions will be encountered. Thus, no particular system configuration can be specified. Instead, emphasis is placed on weight, size, portability, ruggedness, quick installation time due to frequent relocation requirements, ability to operate over extreme ranges of temperature and humidity, and operation within the capability of mobile power sources or batteries. Furthermore, human engineering (see 4.2.4) is an extremely important aspect of equipment design to facilitate operation in diverse environments and to compensate for the minimal technical training of users

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

and operators. On the other hand, for the tactical subsystem type III, the constraints (weight, size, power consumption, etc.) on equipment design due to mobility requirements can be relaxed to some extent. In general, however, the equipment assemblages are required to be air transportable. Usually, more time is available for installation and frequent relocation is not required. These installations may have larger antenna sizes, more power, and frequently have a reliable source of primary power. Human engineering of equipment design remains an important requirement. (See 4.2.4.)

4.1.8 Types of traffic. Tactical communications systems provide communications service in every essential functional area, such as command and control, logistics, intelligence, weather and administration. Traffic in the systems may be in the form of voice, teletypewriter, data and facsimile, and may be transmitted either as analog, quasi-analog or digital signals. The systems accept traffic from, and deliver traffic to individual users and other communications subsystems and systems.

4.1.9 Communications security. Most tactical communications systems have requirements for handling traffic on a secure basis. This requirement has caused a continuing increase in the use of digital transmission in tactical communications. Both end-to-end and link encryption methods are being used. Character-count and bit-count integrity must be ensured. (See 5.2.2.2.2.) Many of the standards contained in this document have been established to facilitate encryption. However, due to the security classification that is involved, no standards are contained herein for security devices. Future system planning for voice transmission will be based on CVSD A-D conversion techniques for digital transmission over wideband circuits and linear predictive coding (LPC) A-D conversion techniques for narrowband circuits. End-to-end security will be provided by on-line encryption devices.

4.1.10 Precedence and preemption. A tactical system is often required to observe degrees of importance (precedence) associated with communications traffic. The higher the precedence level, the more operationally important it is for that traffic to reach its destination. Recognition of precedence levels and appropriate actions, such as preemption of trunks, are accomplished at the switching facilities. Most communications subsystems use five levels of precedence and allow the preemption of only lower precedence levels which result in four levels of preemption. Precedence recognition may be automatic in software controlled switching facilities or manually controlled in both automatic and manual switches. The number of precedence levels available in a subsystem using automatic switching equipment is a function of the switch design.

4.2 General design considerations. The following items and factors stated in 4.2.1 through 4.2.11 must be taken into consideration, where applicable, in the design of communications equipment, subsystems and systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

4.2.1 Compromising emanations (TEMPEST) requirements. These requirements are addressed under the TEMPEST program and are concerned with preventing the inadvertent disclosure of any classified information being processed by communications systems. TEMPEST is the unclassified short name referring to investigations and studies of compromising emanations and is sometimes used synonymously for the term "compromising emanations", e.g., TEMPEST tests and TEMPEST inspections. (See 5.1.2.1.)

4.2.2 Electromagnetic compatibility (EMC) requirements. EMC is the capability of communications-electronics equipment, subsystems and systems, together with electromechanical devices, to operate without suffering or causing unacceptable degradation from unwanted electromagnetic radiation or response. (See 5.1.2.2.)

NOTE: Radio frequency interference (RFI) and electromagnetic interference (EMI) are older terms designating lack of EMC. Use of the new term EMC is preferred.

4.2.3 Electronic warfare (EW) vulnerability and electronic counter-countermeasure (ECCM) capabilities. EW is the military action involving the use of electromagnetic energy to determine, exploit, reduce or prevent hostile use of the electromagnetic spectrum while retaining friendly use of that same spectrum. ECCM is that portion of EW involving actions to insure friendly effective use of the electromagnetic spectrum despite the enemy's use of EW. (See 5.1.2.3.)

4.2.4 Human engineering design. Human engineering design for communications equipment, subsystems and systems involves the application of scientific principles concerning human physical and psychological characteristics. (See 5.1.2.4.) The objectives of human engineering design are to:

a. Provide a man-machine interface which facilitates the operation, control and maintenance of the equipment in the specified environments.

b. Provide for safe operation and maintenance.

c. Minimize the level of skill and the number of personnel required to operate, control and maintain the equipment.

d. Achieve the required reliability of personnel-equipment combinations; and

e. Achieve the required performance of communications equipment, subsystems and systems with a minimum of training time for operator, control and maintenance personnel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

4.2.5 Reliability. Reliability is the characteristic, expressed as a probability, that equipment, subsystems and systems successfully complete their mission under stated operational conditions, given that the equipment, subsystem or system was in proper working condition at the beginning of the mission. Reliability is a quantitative probabilistic factor which must be predictable in design, measurable in tests, assurable in production and maintainable during operation of the equipment, subsystem or system. Reliability must be controlled so that the failure probability does not exceed a permissible upper limit throughout the life cycle of the equipment, subsystem or system. For establishing the upper limit of the failure probability, a trade-off must be made among reliability parameters, such as the mean time between failures (MTBF) and other factors, such as cost, weight, size and schedules. Reliability, when considered early in the development cycle of equipment, subsystems and systems, will result in considerable operating cost savings and in substantial improvement in equipment, subsystem and system effectiveness. (See 5.1.2.5.)

4.2.6 Maintainability. Maintainability is the characteristic of design and installation which imparts to equipment, subsystems and systems a greater inherent ability to be maintained, resulting in lower manhours, skill levels and cost for maintenance, less complicated tools and procedures, and greater mission availability. Maintainability is expressed as a probability that equipment, subsystems or systems in need of maintenance will be retained in, or restored to, specified conditions within a given period of time. Figures of merit used in determining this probability are mean time to repair (MTTR) and mean time between preventive maintenance (MTBPM). Maintainability is also concerned with specific features for fault detection, isolation, correction and verification, such as built-in test equipment (BITE). Maintainability, when considered early in the development cycle of equipment, subsystems and systems will result in considerable maintenance cost savings and substantial improvement in equipment, subsystems and systems effectiveness. (See 5.1.2.6.)

4.2.7 Survivability. Survivability is the characteristic of equipment, subsystems and systems to withstand or avoid a variety of damage mechanisms, such as blast fragments, bullets, explosive and incendiary devices, as well as the effects of natural phenomena, such as lightning strikes, without causing a malfunction. Enhancement of survivability can be accomplished by such measures as adding armor plating, duplicating and separating critical components and simplifying the design to reduce the number of critical components. Survivability measures, when considered early in the development cycle of equipment, subsystems and systems, will result in a considerable increase in survivability for the least possible cost. (See 5.1.2.7.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

4.2.8 Climatic conditions. Many equipment failures are caused by inadequate design and testing to withstand the extremes of environmental conditions and induced stresses representative of global climates during transportation, storage and operation of the equipment. Environmental conditions include thermal and humidity stress, precipitation, wind, penetration and abrasion (such as blowing sand, dust and snow) and atmospheric pressure. The conditions and stresses under which any equipment or assemblage must be capable of operating and withstanding during storage and transit must be carefully evaluated and selected. (See 5.1.2.8.)

4.2.9 Environmental test methods. The wide range of environmental conditions that tactical communications equipment may experience requires uniform test methods for determining the resistance of equipment to the effects of natural and induced environments peculiar to military operations. The test methods include such conditions as low pressure (altitude), temperature, shock, vibration, humidity, fungus, salt, fog, rain, and solar radiation. (See 5.1.2.9.)

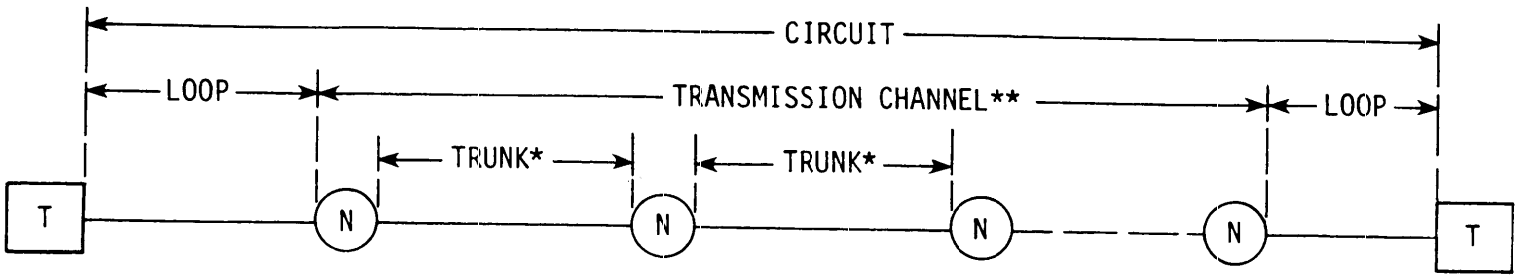
4.2.10 Electrical measurement and test methods. Uniform electrical measurement and test methods, periodically performed, are essential to an effective quality monitoring and control program to achieve reliable and efficient communications. Measurement and test methods are adopted for various conditions, such as those performed on active communications circuits with or without interrupting service. Standards and design objectives for electrical measurement and test methods are under development. (See 5.1.2.10.)

4.2.11 Grounding, bonding and shielding. The objectives of grounding, bonding and shielding are to protect personnel and communications equipment from hazardous voltages caused by lightning strikes, electrical power faults and high-level electromagnetic radiation, and to keep circuit noise levels within acceptable limits under normal operating conditions. Proper grounding, bonding and shielding practices are required to meet applicable EMC and TEMPEST requirements for designing, constructing, installing and operating communications equipment, power generators and communications facilities, such as shelters and buildings. (See 5.1.2.11.)

4.3 Functions of transmission facilities. A transmission facility may be classified as a loop, a link, a trunk, a node, a transmission channel or a circuit, depending on the function the facility performs in the system. Figure 1 shows the relationships among these system parts.

4.3.1 Loop. A loop (also called a local line or a user's line) is a single permanent connection between a node and a terminal equipment or a terminal subsystem. A loop may be a metallic connection, a radio path, or it may employ any other suitable transmission medium, such as fiber optics. A loop may be a two-wire or a four-wire connection and, in general, it provides for transmission of analog, quasi-analog or digital signals.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



LEGEND: T = Terminal Equipment or Terminal Subsystem.

N = Node. A node may include circuit and message switching, patching, signal processing, service supervision and technical control facilities (see 4.3.4).

* A trunk may consist of one link or of two or more links in tandem (see 4.3.3).

** A transmission channel consists of all transmission facilities from the input of the originating node to the output of the terminating node, that is, a transmission channel plus two loops constitutes a circuit. The transmission channel shown may be a nominal 3 kHz transmission channel, a nominal 4 kHz FDM or TDM/PCM transmission channel, or a 48 kHz FDM group transmission channel. The term "transmission channel" is not to be confused with the more general term "channel" (see 4.3.5 and FED-STD-1037).

FIGURE 1. Basic circuit configuration of multichannel trunking system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

4.3.2 Link. A link is a portion of a communications circuit which generally is designed to be connected in tandem with other links to form a connection between two distant points. A link may be a metallic connection, a radio path, or other suitable transmission media, such as fiber optics or laser. A link may be uni-directional, half-duplex or duplex and it may carry analog signals, quasi-analog signals or digital signals. In general, a link is considered to provide duplex transmission on a four-wire basis. The essential feature of a link is that signals are accepted and delivered in the same mode or form at both ends of the link. A link is not considered to be equipped with trunk or loop terminal equipment per se. The term "link" should be defined or qualified when used, in order to avoid ambiguity; e.g., line-of-sight (LOS) radio link or FDM voice frequency (VF) link. For the purpose of this standard, a link is understood to include only four-wire, or equivalent four-wire, portions of a connection and does not include two-wire/four-wire hybrid transformers.

4.3.3 Trunk. A trunk is a connection between two nodes, both of which are switching centers or single distribution points. The connection may be a metallic, fiber optic or radio transmission path that is usually shared by many users or terminal subsystems on a switched basis. A trunk is provided at its ends with equipment, such as multiplexers, suited to the operation and design of the nodes. A trunk will accept and deliver control and supervisory signals which indicate occupancy and idle conditions, etc. The trunk terminal equipment is to be considered as distinct from transmission terminal equipment, although transmission bridges, repeating coils, etc., may be included on the same mounting with the trunk terminal equipment, particularly in VF transmission facilities. A trunk may consist of one link or of two or more links in tandem. A trunk handles one call or one message at a time. When two or more parallel connections are provided between the same two nodes to serve as trunks, these connections constitute a trunk group. For the purpose of this standard, a trunk is understood to include only four-wire, or equivalent four-wire, portions of a connection and does not include two-wire/four-wire hybrid transformers.

4.3.4 Node. A node is a distribution point of loops or trunks, or both, in a communications network. The configuration of a node may range from only two-wire/four-wire hybrid transformers (to connect, for instance, two-wire loops with four-wire multiplexer equipment) to very complex facilities which may include circuit and message switching, patching, signal processing, service supervision and technical control. When a node contains circuit or message switching, it may also provide functions for terminal equipments, such as central battery talking current for telephones, acceptance and translation of the user's control or ordering signals, special signals to attract the user's attention to incoming calls, supervision of the state of the loop or of the terminal equipment and routing of calls or messages.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

4.3.5 Channel. A channel may signify either a one-way connection providing transmission in one direction only, or a two-way connection providing transmission in both directions. The term "channel" is generally used to indicate separation by FDM or TDM. In this standard, the term "transmission channel" is also used to classify that portion of a circuit which connects the originating node with the terminating node. (See Figure 1.) In this context, a transmission channel consists of all nodes and trunks between the terminal equipment or terminal subsystems.

4.3.6 Circuit. A circuit is the complete connection between terminal equipment or terminal subsystems over which one-way or two-way communications may be provided. In general, a circuit consists of a transmission channel and two loops extending to the terminal equipment or terminal subsystems. (See Figure 1.) A circuit may be classified by the type of signals it is designed to handle.

4.3.7 Network. A communications network may consist of a number of users equipped with various terminal equipment interconnected by transmission facilities. The transmission facilities may interconnect the terminal equipment either direct or through one or more nodes where such functions as switching, patching, processing, service supervision and technical control may be provided. A transmission system consists of all the transmission facilities between terminal equipment, between terminal equipment and nodes, and between nodes. Extensive use is made of multiplexing to obtain various numbers of channels by providing parallel, simultaneous paths of various bandwith (FDM), or by successively assigning the entire bandwidth of a facility to each of a number of channels in rapid time sequence to give the effect of parallel continuous channels (TDM). Transmission facilities may be used in a variety of ways, and may be considered with respect to their functions (see 4.3), the type of signal handled (see 4.4), the kind of multiplexing used (see 4.5), and the type of transmission medium over which connections are provided (see 4.7).

4.4 Types of signals. Signals may be continuous or discrete. Continuous signals are also called analog signals. Discrete signals may be digital or quasi-analog signals. In general, the transmission requirements for analog, digital and quasi-analog signals vary widely.

4.4.1 Analog signals. Analog signals may take any value within specified ranges of amplitude, frequency and phase. Circuits transmitting analog signals cannot be provided with signal regeneration, since there is no physical way to distinguish the desired signal from noise and distortion at intermediate or receiving points. However, the amplitude of the signals may be restored by amplification, but the noise will also be amplified; thus, the signal-to-noise ratio (SNR) will not be improved.

4.4.2 Digital signals. Digital signals are constrained to two or more defined and discrete states. For example, a binary signal alternates only between two states, such as "Mark" and "Space", "1" and "0", or "on" and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

"off". The signal which is transmitted between transitions from one state to another is a signal element. In a binary system, a signal element represents one bit, that is either "1" or "0". In a quarternary digital system, i.e., a four-phase system, four discrete states may be transmitted; therefore, each signal element represents two bits. Due to the discrete nature of signals, digital circuits can be provided with signal regeneration because, in general, the receiving device selects the output state which most closely corresponds with the signal state received if the signal is above threshold.

4.4.3 Quasi-analog signals. Quasi-analog signals may take only two or more discrete values within specified ranges of amplitude, frequency and phase. These discrete values represent the states of digital signals but in a form that is compatible with the characteristics of an analog circuit. Quasi-analog signals allow the transmission of data over analog circuits. Data modems are widely used to convert digital signals to quasi-analog signals for transmission over VF channels. Quasi-analog signals cannot be regenerated but may be reconverted to digital signals which can then be reshaped and retimed.

4.5 Multiplexing subsystems. Transmission links may be provided by non-multiplexed subsystems, by FDM subsystems and by TDM subsystems. Each method has distinctive characteristics affecting the objectives and use of circuits so derived.

4.5.1 Non-multiplexed subsystems. Multi-pair cables provide parallel pairs for simultaneous transmission of signals along the route. This can be referred to as space division multiplexing.

4.5.2 FDM subsystems. FDM is a method of deriving two or more, simultaneous and continuous, channels from a transmission facility by assigning separate portions of the available frequency spectrum to each of the individual channels. The most extensive exploitation of this principle is the transmission plant which provides message quality telephone circuits occupying adjacent nominal 4 kHz slots in wideband transmission facilities, both metallic and radio, and provides an effective bandwidth from about 300 Hz to 3400 Hz. The procedures for placing many telephone channels in adjacent positions in multiplex basebands extending to 12 MHz and higher, have become highly standardized.

4.5.3 TDM subsystems. TDM is a method of deriving two or more, apparently simultaneous, channels from a given frequency spectrum of a transmission facility by assigning the entire spectrum to the different channels at different times (usually at regular intervals and by automatic distribution). In general, pulse transmission is used in TDM. The multiplexed pulse train may be considered to be the interleaved pulse trains of the individual channels. This multiplexed pulse train constitutes the mission bit stream. The individual channel pulses may be modulated either in an analog or a digital manner. Digital information

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

transmitted over the TDM subsystem on an element for element basis requires that either the contributing individual channels be held in bit synchronism with the TDM subsystem or that some form of buffering be employed.

4.6 A-D conversion and modulation techniques.

4.6.1 Analog (or non-discrete) pulse modulation. A stream of pulses may be modulated to carry analog information by changing a selected parameter of each pulse over a continuous range of values within specified limits. A sampling rate at least twice as high as the highest frequency in the analog channel (Nyquist criteria) must be used to preserve all essential analog information. The selected modulation parameter may be pulse amplitude, pulse duration or pulse timing. Transmissions with analog pulse modulation are not digital; that is, the signal cannot be regenerated since the receiver has no way to distinguish the desired signal modulation from the effects of distortion and noise, although the unmodulated characteristics of pulses can be restored. An analog pulse modulation scheme may handle both analog signals and quasi-analog signals.

4.6.2 Digital (or discrete) pulse modulation. With digital pulse modulation, a specific pulse parameter is varied in a discrete manner. That is, each pulse is constrained to assume one state out of a set of two or more defined and discrete states. Such a signal may be regenerated because the receiving device can be designed so that the device is limited to assume a set of output states which correspond to the states of the transmitted signal. If noise or distortion, or both, induce the receiver to select an erroneous output state, the analog causes of the digital error are removed; the analog noise is significantly reduced but the error in the digital signal remains. Therefore, the noise and distortion do not have cumulative effects as such; only the errors add up. The same parameters used in analog pulse modulation may also be used for digital pulse modulation. The pulse amplitude may be varied in steps (this includes reversal in polarity or phase), the pulse duration may be varied in steps or the pulse position may be discretely varied. Analog information may be transmitted over a digital subsystem if it is first placed in digital form. This requires that the analog values be quantized and in some cases coded, as in PCM (see 4.6.2.1) and delta modulation (see 4.6.2.2). Analog signals transmitted in coded digital form are not degraded directly by analog noise and distortion of the transmission medium. The errors in the received code will have a random noise effect on the recovered analog signal, as opposed to the progressive degradation in bandwidth and SNR which occurs in very long circuits transmitting analog signals.

4.6.2.1 Pulse code modulation (PCM). PCM is an A-D conversion technique using a two-step pulse modulation process. In PCM, a band-limited analog signal is periodically sampled (normally at synchronized time intervals) by a narrow sampling pulse, resulting in a pulse amplitude modulation (PAM) signal. The PAM signal contains all the amplitude and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

frequency variations of the original analog signal, provided the Nyquist criteria (see 4.6.1) is met. Each PAM signal is then quantized and coded into an n-bit code word, resulting in a PCM signal. Usually, the sampling process is performed together with TDM to allow many analog channels to share the same PAM sampler and PCM coder, thus reducing the complexity and cost of PCM equipment. Basically, the PCM signal is a stream of time synchronized pulses which have constant positions on the time axis and have the same shape and amplitude. The logic value (such as 0 or 1, presence or absence) of a pulse in each time slot of a code word indicates the information derived from the original analog signal and carried by the PCM signal. Framing and synchronization pulses are added to the PCM pulse stream to synchronize the receiving equipment with the transmitting equipment for correctly demodulating each code word in each analog channel. At the receiving end, the PCM signal is decoded, demultiplexed (if TDM is used) and the resulting PAM samples are demodulated to analog baseband. The coding process of PCM introduces quantizing noise which is an important parameter for describing and measuring the quality of digitized voice transmission. Once the analog information is converted into a quantized digital signal, the information can be transmitted over tactical and global distances by using regenerative repeaters with little or no additional degradation if the signal is above threshold. However, excessive media noise in digital transmission links will cause bit errors which regeneration, even with error control, may not be able to correct.

4.6.2.2 Delta modulation. Delta modulation is an A-D conversion technique resulting in a form of digital pulse modulation. In delta modulation, the amplitude of a band-limited analog signal is periodically sampled (normally at synchronized time intervals) and the amplitude differences of two adjacent samples are coded into n-bit code words. For simplicity, a one-bit code is generally used. At the receiving end, an integrating network converts the delta modulated bit stream to analog baseband.

Basically, a delta modulator consists of a pulse generator, a feedback network and a comparator. The pulse generator provides a sequence of binary pulses with a constant pulse repetition rate. The binary value of each pulse is determined by the output of the comparator. The sequence of binary pulses is transmitted as the delta modulated signal and is also fed into the feedback network (which, in its simplest form, is an integrator) that converts the binary pulses to an approximation of the original analog signal. The approximated analog signal at the output of the feedback network is compared in the comparator with the original analog signal and a control signal is derived. The control signal at the comparator output instructs the pulse generator to emit a pulse with a positive or negative binary value depending on whether the approximation signal is smaller than, equal to, or larger than the original analog signal. The same or a similar integrator as used in the feedback network, is used at the receiving end to demodulate the delta modulated bit stream to analog baseband. Over the years, several versions of delta modulators

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

have been developed with the objective to optimize the A-D conversion of one type of analog signal, such as voice or television. One version of delta modulation for voice is CVSD which will be used as the A-D conversion process for the next generation of tactical communications equipment. More detailed information on CVSD is contained in Appendix D.

4.6.2.3 Linear predictive coding (LPC). LPC is the A-D conversion technique to be used in tactical narrowband digital voice equipment. The LPC approach is an analysis method of digital voice coding. The encoder consists of a low-pass pre-filter, A-D converter, analyzer and multiplexer. The A-D conversion changes the speech signal into a digital format suitable for processing by the analyzer. The analyzer is a digital processor which extracts vocal source information, such as the fundamental frequency or pitch, the voiced/unvoiced decision, and a gain which is a measure of the input speech energy from the speech signal. The analyzer also contains the n-th order linear predictor (non-recursive filter) whose coefficients are optimized to produce a minimum mean-square difference between the actual and predicted values of the speech signal over finite analysis intervals. The output of the analyzer consists of time quantized parameters. These parameters are predictor coefficients, root-mean-square (rms) value of speech energy, pitch and the voiced/unvoiced decision. The multiplexer formats the speech signal parameters into successive frames for transmission over a channel. More detailed information on LPC is contained in Appendix D.

4.7 Transmission media. Transmission media currently used are primarily metallic lines and radio. The latter may be divided into relatively stable media, such as millimeter LOS, microwave LOS and satellites; varying media, such as tropospheric scatter; and those media subject to wide and often prolonged propagation variations, such as in the high frequency (HF) band. An extremely stable transmission medium has been achieved by a new technology utilizing glass or plastic fibers as an optical waveguide. The use of fiber optic communications is expected to increase in the future due to the many advantages that optical fibers have over metallic lines and radio subsystems. (See 4.7.8.)

4.7.1 Metallic lines. The term metallic lines includes open wire, twisted pair cable, multipair cable, coaxial cable and waveguides. Characteristics of these lines include attenuation, noise and mutual coupling. These characteristics affect the parameters of transmission systems designed to operate over metallic lines. Attenuation increases continuously and smoothly with increase of frequency and length of line and is also sensitive to temperature changes. Noise on metallic lines can be induced by electric power systems and crosstalk from other similar parallel circuits, the latter tending to rise with the volume of traffic. Impulse noise may also be introduced by natural static, transients from signaling circuits and switching transients from power systems. In some areas, electric railways, local rail services, radar equipment and

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

industrial sources may introduce noise. In the case of baseband digital transmission over metallic lines, the cable shunt capacitance is the principal factor which determines the upper bound of the data signaling rate.

4.7.2 Radio relay subsystems. The term radio relay is generally applied to radio subsystems which are suitable for the application of either FDM or TDM and which operate in radio frequency (rf) bands where transmission, while varying continually within a restricted range, is relatively stable. The bands of the radio spectrum classified as very high frequency (VHF), ultra high frequency (UHF), and super high frequency (SHF) are currently used by radio relay subsystems of various types. The channel capacity of these subsystems varies from about twelve to several hundred VF channels. The loosely defined term microwave is also commonly applied to such radio relay subsystems. Although several types of radio propagation may be present at the same time in a specific direction, radio paths for radio relay subsystems generally may be classified as paths within the radio horizon, such as terrestrial LOS and satellite, and paths beyond the radio horizon, such as tropospheric scatter, diffraction and meteor burst propagation. In general, an important characteristic of radio relay subsystems is the use of highly directional antenna configurations which may allow extensive duplication of rf channel assignments in a relatively small geographical area.

4.7.3 Radio subsystems operating in medium frequency (MF) and lower bands. Radio subsystems using frequencies classified as MF (300 kHz to 3000 kHz), low frequency (LF, 30 kHz to 300 kHz), very low frequency (VLF, 3000 Hz to 30 kHz), infra low frequency (ILF, 300 Hz to 3000 Hz) and extremely low frequency (ELF, below 300 Hz) have propagation characteristics which are relatively stable and which may cover long distances. These radio frequencies are used for special purposes, such as reaching mobile and fixed elements located anywhere in the world. These radio subsystems generally have very limited traffic handling capability due to limited available bandwidth.

4.7.4 HF radio subsystems. Radio subsystems using frequencies between about 3000 kHz and 30 MHz are subject to wide changes of received signal levels due to slow and rapid fading. The fading may affect various frequencies differently even in the same VF channel. The fading handicap is offset by the long range of communications that can be obtained by utilizing the reflection or refraction properties of the ionosphere. This requires the selection of frequencies which depend on the time of day, season and transmission distance. The range of such HF subsystems make them useful for maintaining contact with tactical forces with wide operating radii, although such circuits are frequently of degraded performance. Due to congested spectrum conditions, telephone channels are limited to a nominal 3 kHz bandwidth, and single sideband (SSB) and independent sideband (ISB) techniques are widely used for spectrum efficiency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

4.7.5 VHF radio subsystems. Radio subsystems using frequencies between about 30 MHz and 300 MHz have relatively stable propagation characteristics over LOS distances in open terrain, particularly in the lower VHF range. Signal attenuation may occur due to interference between the free-space wave and the ground reflected wave, and due to shadowing by natural and man-made obstructions, such as hills and large buildings. This type of attenuation is strongly dependent on the direction of polarization, on earth surface features and on the type of obstruction. Attenuation due to vegetation is low in the temperate zone but can be substantial in tropical jungles. Attenuation due to atmospheric conditions (rain, hail, snow, fog, etc.) is negligible for all practical purposes. The VHF band is widely used for tactical surface-to-surface, air-to-surface and air-to-air communications over distances ranging from several miles to path lengths extending slightly beyond the horizon. VHF radio subsystems also find application in diffraction links, meteor burst and ionospheric scatter communications.

4.7.6 UHF radio subsystems. Radio subsystems using frequencies between about 300 MHz and 3000 MHz have similar propagation characteristics as those subsystems operating in the VHF band. Multipath fading and bending of the beam path (caused by abnormal refractive index profiles of the atmosphere) affect LOS propagation. These effects may be reduced by employing diversity techniques. Propagation through vegetation leads to appreciable attenuation that generally increases with frequency and, to a certain degree, depends on the direction of polarization. Attenuation due to atmospheric conditions (rain, hail, snow, fog, etc.) is negligible in the lower and middle portion of the UHF band. However, heavy rains may cause noticeable attenuation in the upper portion of the UHF band. The UHF band is extensively used for tactical communications. Applications include tactical surface-to-surface, air-to-surface and air-to-air communications, LOS multichannel radio relay links, tropospheric scatter links, knife-edge diffraction links and single channel satellite links. Furthermore, the middle portion of the UHF band appears to be an attractive frequency range for communications in built-up urban areas.

4.7.7 SHF radio subsystems. Radio subsystems using frequencies between about 3000 MHz and 30 GHz have fading effects due to multipath and atmospheric beam bending that are similar in nature to, but more pronounced than, those in the UHF band. At SHF, high-gain antennas of moderate physical dimensions are feasible, making this band particularly suited for efficient point-to-point communications over LOS paths. However, large obstructions in the LOS path effectively block SHF beams. Propagation through vegetation is subject to rather high attenuation due to scattering and absorption. Depending on vegetation density and path length, this attenuation may be prohibitive. Atmospheric effects, in particular scattering and absorption by rain, contribute significantly to path attenuation. Rain attenuation is dependent on frequency and precipitation rates. The SHF band is employed in tactical communications for LOS multichannel radio relay and point-to-point transmission links,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

satellite communications and navigation, and for tropospheric scatter and knife-edge diffraction links operating in the lower SHF range.

4.7.8 Fiber optic communications subsystems. Certain fibers made of special glasses or plastics are used as a light transmission medium in the rapidly developing technology of fiber optic communications. In a typical fiber optic transmission subsystem, electrical signals modulate a light source which may be a light emitting diode (LED) or an injection laser diode (ILD). The light signal is then conducted through an optical fiber, received by a photo detector device and reconverted to electrical signals. Voice, data and other information can be transmitted by such a subsystem.

Fiber optic cables have many advantages over metallic wires and cables. For the same information carrying capacity, a fiber optic cable has a much smaller diameter and less weight than a metallic wire or a coaxial cable. Increased bandwidth capability with information rates exceeding 1000 Mb/s has been obtained by fiber optic cables. Fiber optic cables are immune to EMI effects. These cables neither accept nor produce electromagnetic fields, thus providing a transmission medium with excellent EMC characteristics. Low loss fiber optic cables transmitting signals in excess of 100 Mb/s can extend from 5 km to 10 km without repeaters.

Fiber optic cables also possess certain disadvantages. Without the addition of metallic conductors, fiber optic cables cannot carry power for remote terminal equipment or repeaters. Adding metallic conductors, however, introduces electromagnetic pulse (EMP) vulnerability. Tactical applications will expose cables to mechanical abuse from laying and retrieving as well as possible damage from vehicular traffic. The development of simple and reliable fiber optic connectors and field splicing techniques will be a challenge.

4.8 Basic functions of tactical communications systems. Tactical communications systems may provide voice, data, and facsimile service to the users, indicating the type of signal which is received from a user or terminal equipment and delivered by a system to another user or terminal equipment. In general, the type of signal generated by a terminal equipment is converted for transmission over the different media, such as radio, metallic lines or fiber optics. The conversion process depends on the type of transmission medium and on other factors and may include multiplexing/demultiplexing, modulation/demodulation, analog-to-digital (A-D) and digital-to-analog (D-A) conversion of analog signals with or without coding/decoding. An analog signal from a terminal equipment may be transmitted through an FDM channel as an analog signal or the signal may be digitized, time division multiplexed and transmitted as a digital signal. Similarly, a digital signal from a terminal equipment may be converted to a quasi-analog signal for transmission through an FDM channel or the digital signal may be time division multiplexed and transmitted together with other digital signals. The user, generally, will not be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

interested in and will have no knowledge of, the signal conversions at transmission facilities. From a communications system point of view, analog service to a user may be provided by analog transmission or by digital transmission and, likewise, data service to a user may also be provided by digital transmission or by analog transmission. However, more and more emphasis is being placed on digital transmission which offers significant advantages over analog transmission in such areas as cost, reliability and encryption.

4.9 Basic configurations of tactical communications systems. The basic configuration of tactical communications systems consists of user or terminal equipment that is interconnected by a circuit as shown in Figure 2A. The circuit may consist of a non-multiplexed single channel radio or wire connection, switched multichannel connections (derived from FDM, TDM or a mix of both) with loops extending to the terminal equipment, or a single channel radio set connected to a switched multichannel network by means of RWI. (See 5.4.5.) The terminal equipment may be an analog device or a digital device. The circuit may be designed to handle analog or digital signals and may consist of a portion of the tactical subsystem type I or type II and a portion of the tactical subsystem type III with an interface at a node as shown in Figure 2B. Figure 2C shows analog terminal equipment interconnected by a digital transmission channel through the use of A-D and D-A communications equipment. (See 4.6.) Figure 2D shows data terminal equipment (DTE) that is interconnected by a digital circuit. Figure 2E shows DTE interconnected by an analog or digital transmission channel by means of data circuit terminating equipment (DCE). The DCE is required to interface the DTE with the transmission channel. (See 4.9.1.3.) Where it is required to interface with an analog circuit, the DCE converts the digital signals into quasi-analog signals. (See 4.4.3.) The basic system configurations as represented by Figure 2 are not inclusive of all possible variations of terminal equipment, transmission channels and circuits. However, these system configurations are considered as basic system layouts.

4.9.1 Basic configurations of tactical data communications systems. Figure 3 shows simplified configurations of data communications systems based on the configurations depicted in Figure 2. The primary function of a data communications system is the transmission of record traffic and digitized voice or the transmission of data related to information processing. The data communications system may consist of DTE, also called terminal subsystem, and a data circuit. The DTE may consist of digital end instrument(s) and intermediate equipment, if required. The data circuit may include DCE which interfaces with the DTE. The DCE may also be located in the terminal subsystem as shown in Figure 3A.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the currentversion before use.

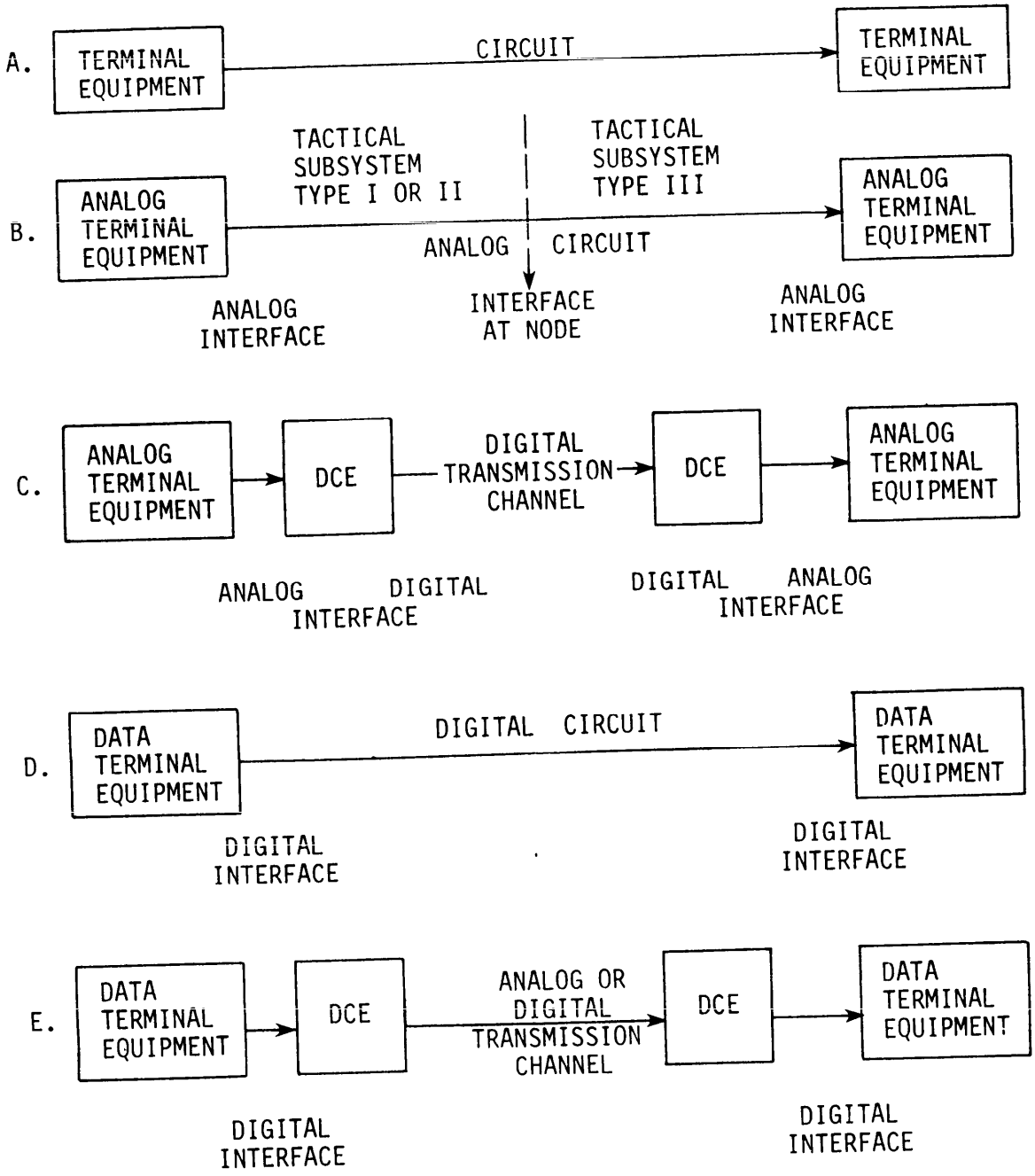


FIGURE 2. Basic configurations of tactical communications systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

4.9.1.1 Data terminal equipment (DTE). This equipment consists of digital end instruments that convert the user information into digital signals for transmission or reconverts the received digital signals into user information. The DTE may range from a single equipment (which provides all the required functions necessary to permit the user to communicate) to an interconnected subsystem of multiple equipments.

4.9.1.2 Digital interface adapters. Digital interface adapters may also be required to properly interface the DTE and the DCE. These adapters, if required, may be an integral part of the DTE, or an integral part of the DCE, or may be one or more separate equipment. If separate equipment is used, it is called intermediate equipment, as shown in Figures 3C and 3D. Part of the intermediate equipment may be the digital end instrument ancillary devices, such as clock equipment, modulation rate converters, code converters, error control devices, alphabet translators and message composers. Transmission of bits (or characters) between digital end instruments may be in serial (serial-by-bit, serial-by-character) form or in parallel (parallel-by-bit, serial-by-character) form. Ultimately, the bit stream must be serialized before transmission between terminal subsystems. (See 5.3.5.4.)

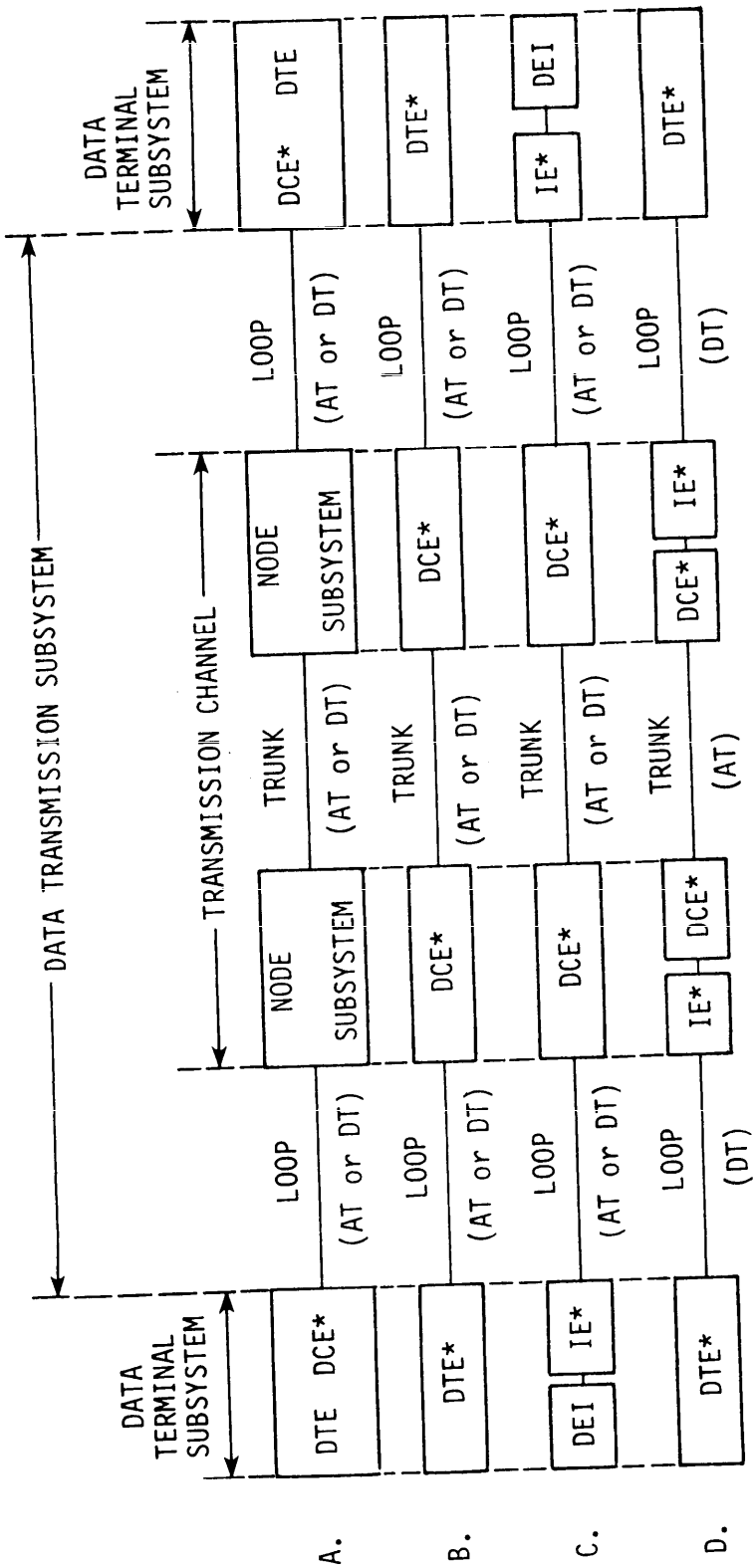
4.9.1.3 Data circuit terminating equipment (DCE). The DCE may consist of interfacing equipment required to couple the DTE output into a data circuit (see Figure 3A) or into a transmission channel (see Figure 3B) and from a data circuit or transmission channel into the DTE. The DCE, in its simplest form, may be a passive electrical device (transformer, attenuator, etc.) or it may be a complex device which performs multiple functions, such as modulation, multiplexing and signal conversion.

NOTE: The term data communications equipment is no longer used and has been replaced by data circuit terminating equipment (DCE).

4.9.1.4 Modes of operation. Generally, a data communications system will be operated either in a synchronous mode or in an asynchronous mode described in 4.9.1.4.1 through 4.9.1.4.3. Clock control and timing for synchronous and asynchronous transmission modes are stated in 5.3.6. Terminal equipment may be operated in serial-by-bit or parallel-by-bit mode. (See Note of 5.3.5.4.)

4.9.1.4.1 Synchronous mode. In a synchronous mode of operation, the DTE, the DCE and the data transmission equipment are all operated in bit synchronism by using an accurate clocking system. In a synchronized communications system, there will always be an integral number of unit intervals between any two significant instants in a data stream. In a binary synchronous serial system, the signal at the digital interface (see Figure 2) is serial and isochronous, (i.e., a two-state signal with all signal elements nominally equal to their unit interval or multiple thereof). In this case, the modulation rate in bauds (Bd) and the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



LEGEND: * May include digital interface adapters, such as modems, error control devices, control units and other equipment as required.

AT = Analog Signal Transmission
DT = Digital Signal Transmission
IE = Intermediate Equipment
DCE = Data Circuit Terminating Equipment
DEI = Digital End Instrument
DTE = Data Terminal Equipment

FIGURE 3. Basic configurations of tactical data communications systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

information rate in bits per second (b/s) are numerically equal and either unit is correct for a binary isochronous signal.

4.9.1.4.2 Asynchronous mode. In an asynchronous mode of operation, all or part of a communications system is operated such that both of the following conditions are met:

a. Between any two significant instants in the same group (in data transmission this group is a block or a character, in telegraphy this group is a character), there is always an integral number of unit intervals. An example of this is the character-release operation between the end instrument and the intermediate equipment.

b. Between any two significant instants located in different groups there is not always an integral number of unit intervals. An example of this is the free-running start-stop operation of a teletypewriter.

4.9.1.4.3 Start-stop mode. In a start-stop mode of operation, the stop element may not be a multiple of the unit interval (as in 7.42-unit per character interval). Also, the interval between characters in direct keyboard operation is not necessarily a multiple of the unit interval. To express the "instantaneous" modulation rate at which signal elements are transmitted during a character interval (and the required bandwidth is proportional to this), it is necessary to use the unit Bd. (See also 4.9.1.5.)

4.9.1.5 Signal conversion. Analog channels, such as FDM VF channels, can be utilized for data transmission provided the digital signals are converted into a form compatible with the characteristics of the analog channel. Characteristics of a number of conversion devices (modems) are standardized (see MIL-STD-188-110) in order to accommodate the variety of data to be transmitted over different types of transmission media. A quasi-analog signal does not have to be in binary form since multilevel (m-ary) modulation schemes also are being utilized for converting the binary dc signal into a quasi-analog form. If the quasi-analog signals are binary in a synchronous system, the nominal time structure of the digital signal is not affected by the modem, and the rate of the quasi-analog signal may be expressed in either Bd or b/s. But, in more sophisticated systems, this is generally not true. For instance, a 4-phase modem handling 2400 b/s at the digital interface emits a line signal modulated at 1200 Bd. The modulation rate in Bd should be used at the quasi-analog side of the modem and at the transmission equipment as a more meaningful term. Parenthetical additions should be used with the modulation rate where this will help to clarify the meaning, e.g., 2400 Bd (binary) or 1200 Bd (4-level). For wire and cable transmission subsystems, signals may be transmitted in digital form without conversion to quasi-analog form and without regeneration over relatively long distances.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

The distance depends on the data signaling rate and the wire or cable characteristics. (See MIL-STD-188-114 for more details.) There are a number of ways by which digital signals can be represented by electrical signals, resulting in different waveforms for the same sequence of digital symbols. In Appendix E, examples of various types of waveforms of digital signals are listed that are used in military and commercial data communications systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5. REQUIREMENTS FOR TACTICAL COMMUNICATIONS SYSTEMS

The design of the various parts of a communications system, such as terminal equipment, transmission facilities and nodal subsystems, must be based on equipment and subsystem performance requirements and constraints, such as cost, size and weight, power consumption and mobility, as well as on required overall system performance. The subsystems, generally consisting of equipment assemblages, are arranged and interconnected to provide optimum performance under the most extreme operational conditions.

The standards of this section are considered to be the minimum essential technical requirements necessary to ensure interoperability and acceptable overall system performance, as well as maximum flexibility of system lay-out free of restrictions caused by equipment interoperability problems and problems of incompatibility among systems, subsystems and equipment.

Some of the parameters of section 5 are stated as Design Objectives rather than System Standards (see 1.4) due to a lack of measured and verified data available at the present time or a lack of general consensus in the interpretation of the data. Tactical standards which are identical to long haul standards and are published as common long haul/tactical standards in MIL-STD-188-100 are identified in this section by "(MIL-STD-188-100)" entered after each subparagraph title.

5.1 General interconnection and design requirements. The following general requirements of 5.1.1 through 5.1.2.11 shall apply to all communications equipment, assemblages, subsystems and systems regardless of unique requirements that may have been established for highly specialized communications facilities.

5.1.1 Interconnection of communications subsystems and systems.

5.1.1.1 Interconnection of tactical subsystems type I or III with tactical subsystem type II. VF channels of the tactical subsystems type I or type III shall be interconnected with VF channels of the tactical subsystem type II on the basis of the standard test tone as specified in 5.8.5.

5.1.1.2 Interconnection of tactical subsystem type IV with tactical subsystems type I, II or III. Under consideration.

5.1.1.3 Interconnection of single channel radio subsystems with switched multichannel subsystems. VF channels of single channel radio subsystems shall be interconnected with VF channels of the tactical subsystems type I, II or III in accordance with the applicable requirements of 5.4.5.

NOTE: Requirements for interconnecting single channel radio subsystems with the tactical subsystem type IV are under consideration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.1.1.4 Interconnection with tactical NATO systems. VF channels of US switched trunking subsystems shall be interconnected with VF channels of a tactical switched trunking system of another NATO member nation in accordance with the applicable requirements of the current editions of STANAG 5018 and STANAG 5040. Telegraph and data channels of US subsystems shall be interconnected with telegraph and data channels of a tactical system of another NATO member nation in accordance with the applicable requirements of the current edition of STANAG 5040.

NOTE 1: STANAG 5018 covers the interface between two manual switched subsystems or between a manual switched subsystem and the manual position of an automatic switched subsystem of the combat zone. STANAG 5040 covers the interface between two automatic or semi-automatic switched subsystems of the combat zone.

NOTE 2: Additional requirements for a NATO data interface are stated in 5.3.5.7.

5.1.1.5 Interconnection of long haul and tactical systems. (MIL-STD-188-100.) VF channels of the tactical subsystems type I or type III shall be interconnected with VF channels of a long haul system on the basis of the standard test tone as specified in MIL-STD-188-100, Figure 4.7-10 or Figure 4.7-11, as applicable. VF channels of the tactical subsystem type II shall be interconnected with VF channels of a long haul system on the basis of the standard test tone as specified in MIL-STD-188-100, Figure 4.7-8 or Figure 4.7-9, as applicable.

NOTE: Requirements for interconnecting the tactical subsystem type IV with a long haul system are under consideration.

5.1.2 General design requirements. The general design requirements of 5.1.2.1 through 5.1.2.9 invoke documents outside of the mandatory MIL-STD-188 series. (See Appendix A.) Extreme care must be used to ensure that the provisions, selected from these documents as applicable to a given design task, are tailored in accordance with the policies of DoDD 4120.21.

5.1.2.1 Compromising emanations (TEMPEST) requirements. All communications equipment, subsystems and systems shall comply with the applicable TEMPEST criteria of the current edition of the NACSIM 5100 series.

NOTE: NACSEM 5201 provides design guidance and MIL-HDBK-232 provides installation guidelines for compromising emanations.

5.1.2.2 EMC requirements and guidance.

5.1.2.2.1 Equipment requirements. Any item including subassemblies and parts, serving functionally in an electromagnetic environment in the broadest sense, shall comply with the applicable requirements of the current edition of MIL-STD-461. Techniques used for the measurement and

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

determination of EMC characteristics shall comply with the applicable requirements of the current edition of MIL-STD-462.

5.1.2.2.2 Subsystem and system requirements. Communications subsystems and associated systems shall be designed to achieve intra-system and inter-system EMC. There shall be neither unacceptable responses nor malfunctions of any item of the system or subsystem beyond the tolerances established by the applicable requirements of the current edition of MIL-STD-461.

5.1.2.2.3 Equipment and subsystem EMC design guidance. Equipment and subsystems should be designed in accordance with the applicable EMC guidance contained in MIL-HDBK-235, MIL-HDBK-237, MIL-HDBK-241 and MIL-HDBK-253.

NOTE: MIL-HDBK-237 provides guidance for implementing an EMC program and MIL-HDBK-241 provides guidance for EMC enhancement (EMI reduction) in equipment power supplies.

5.1.2.2.4 rf spectrum characteristics. The spectral characteristics of all rf transmitters, receivers and antennas shall be measured in accordance with the applicable requirements of the current edition of MIL-STD-449.

5.1.2.3 EW vulnerability and ECCM capabilities. It is imperative that EW vulnerability analyses be performed on all communications systems beginning with the concept formulation stage, and that appropriate ECCM capabilities be developed to protect these systems from the applicable EW threat. The development of communications systems should include, but not necessarily be limited to, the following:

a. A determination of the potential EW technical threat for each phase of development and production;

b. An evaluation of candidate system design approaches in the face of the EW technical threat as part of the tradeoff studies to determine the system design approach; and

c. An EW vulnerability and ECCM assessment during each phase of development and production.

5.1.2.3.1 Determination of EW technical threat. Appropriate EW intelligence sources should be utilized to provide an EW technical threat model during the concept formulation stage of communications system development.

5.1.2.3.2 Analysis of EW vulnerability. Simulation techniques should be utilized to assess the effects of EW on the radio links of a communications system. A first analysis should be made of EW effects on candidate system communications techniques as an aid in eliminating

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

unacceptable approaches. Subsequent analyses should be made of emerging candidate communications techniques or equipment at several stages of development. EW vulnerability analyses should be performed in accordance with applicable Department/Agency directives.

5.1.2.3.3 Development of ECCM techniques. During each phase of development and production, available ECCM technology should be reviewed for applicability to a system and should be assessed during EW vulnerability analyses of system communications throughout the system development cycle. Where necessary, ECCM capability should be made integral to the system design. It should not be assumed that ECCM remedies can be applied cost-effectively at later stages of system development. EW/ECCM test requirements will be stated in applicable specifications, based on the system under consideration, and tailored in accordance with DoDD 4120.21.

NOTE: DoDD C-4600.3 provides specific guidance for incorporating ECCM capabilities in the development of communications systems.

5.1.2.3.4 EMP vulnerability. EMP vulnerability should be considered during the concept formulation stage. The development of communications systems should include, but not necessarily be limited to, an evaluation of:

a. filters between EMP collectors, such as antennas, and sensitive components;

b. amplitude limiters in wireline systems to reduce transient power pulses and; and

c. circuit layout procedures to reduce internal collection of induced power.

5.1.2.4 Human engineering design. All communications equipment, facilities, subsystems and systems shall be designed in accordance with the applicable requirements of the current edition of MIL-STD-1472 and MIL-H-46855.

5.1.2.5 Reliability. All communications equipment, subsystems and systems shall be designed to meet quantitative reliability requirements. The reliability program shall be established in accordance with the applicable requirements of the current edition of MIL-STD-785. Reliability acceptance tests shall be performed in accordance with the applicable requirements of the current edition of MIL-STD-781.

5.1.2.6 Maintainability. All communications equipment, subsystems and systems shall be designed to meet quantitative maintainability requirements. The maintainability program shall be established in accordance with the applicable requirements of the current edition of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

MIL-STD-470. Maintainability acceptance tests shall be performed in accordance with the applicable requirements of the current edition of MIL-STD-471.

5.1.2.7 Survivability. The survivability of all communications equipment, subsystems and systems should be assessed by performing vulnerability reduction studies in accordance with applicable Department/Agency directives, regulations and instructions.

5.1.2.8 Climatic conditions. All communications equipment shall be designed to meet the applicable climatic conditions specified in the current edition of MIL-STD-210. The climatic condition and induced stress requirements for an equipment or an assemblage shall be consistent with the degree of exposure that is anticipated for the intended field applications. Where the intended applications involve different environmental requirements for two or more mission profiles, the equipment shall be designed to satisfy the most stringent mission profile.

5.1.2.9 Environmental test methods. All communications equipment shall be designed to comply with the applicable environmental test methods specified in the current edition of MIL-STD-810.

5.1.2.10 Electrical measurement and test methods. Under consideration.

NOTE: Electrical measurement and test methods applicable to the entire MIL-STD-188 series are being developed and will be published in MIL-STD-188-190. (See Appendix B.) Until this document is available, Appendix D of MIL-STD-188-100 may be used.

5.1.2.11 Grounding, bonding and shielding. Methods and practices for grounding, bonding and shielding of ground-based telecommunications equipment and facilities, including buildings and structures supporting tactical and long haul communications, shall comply with the applicable requirements of the current edition of MIL-STD-188-124.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the currentversion before use.

5.2 Parameters for communications services (user-to-user).

5.2.1 General. Parameters for communications services are subdivided into parameters for voice service (see 5.2.2), for telegraph and data service (see 5.2.3) and for facsimile service (see 5.2.4), indicating the type of service provided to the users by a communications system.

5.2.2 Parameters for voice service (user-to-user). Parameters for voice service on a user-to-user basis are subdivided into parameters for analog voice service (see 5.2.2.1) and for digitized voice service (see 5.2.2.2). The parameter values are based on measured performance and on experience and are considered to be the minimum essential values necessary to provide acceptable service to the users. The parameters apply to all single channel and multichannel VF circuits regardless of transmission distance, transmission medium and multiplexing scheme employed for multichannel transmission, except where stated otherwise in the applicable subparagraph.

5.2.2.1 Parameters for analog voice service (user-to-user).

5.2.2.1.1 Transmitted speech volume. (MIL-STD-188-100.) The talker volume distribution at the line terminals of a telephone set has been found to approximate a normal or Gaussian distribution when expressed in Volume Units (VU). Based on the talker volume distribution, the transmitted speech volume of a two-wire as well as a four-wire telephone set has a mean value of -10 VU with a standard deviation (sigma) of 5 VU (see Figure 4) when measured in accordance with IEEE Standard 152.

5.2.2.1.2 Received speech volume. (MIL-STD-188-100.) Subjective tests indicate that received speech volume should range from -21.5 VU to -33.5 VU, measured at the line terminals of the listener telephone set. The mean received speech volume at the line terminals of the listener telephone set should be nominally -28 VU.

NOTE 1: Table II shows the relationship between received speech volume and the percentage of calls rated good or better, when the associated noise level is so low that its influence on subjective ratings can be neglected.

NOTE 2: The mean received speech volume of -28 VU and the values given in Table II are based on subjective tests made with a commercial type of telephone in a typical office environment. Subjective ratings for military telephones in a field environment are not available at the present time. Experience has shown that mean received speech volumes for single channel and multichannel transmission in a tactical field environment have to be higher than -28 VU for satisfactory service due to a generally higher background noise level and to higher received noise power.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

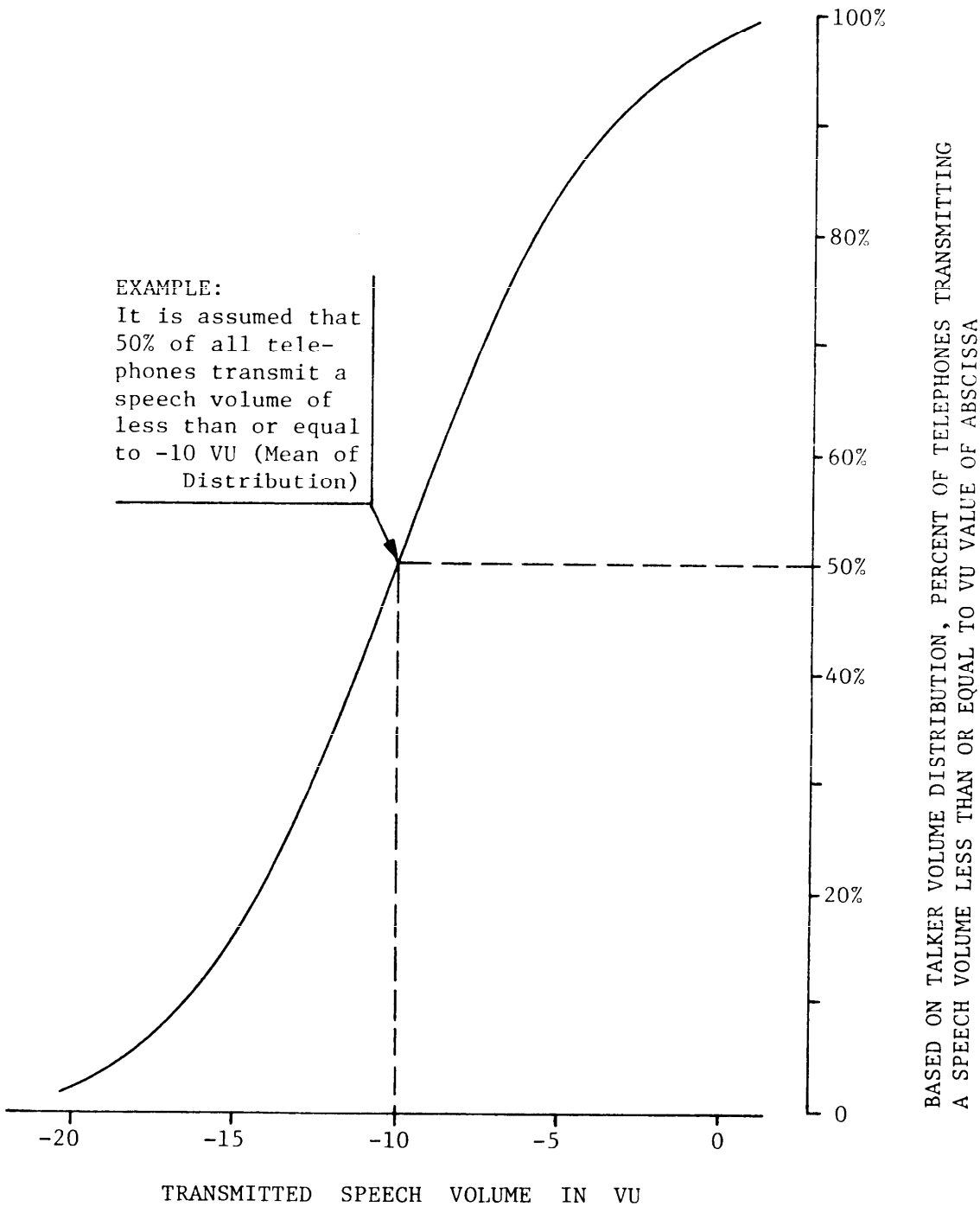


FIGURE 4. Distribution of transmitted speech volume of telephones based on talker volume distribution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE II. Subjective ratings of received speech volume.

Received Speech Volume	VU	Percentage of Calls Rated Good or Better
Maximum	-21.5	95
Mean	-28	100
Minimum	-33.5	95

5.2.2.1.3 Received noise power. For acceptable voice service over FDM or metallic circuits with a mean received speech volume of -28 VU (see 5.2.2.1.2), the received noise power shall not exceed 25,000 pWp (44.0 dBrnc) with a Design Objective of 6300 pWp (38.0 dBrnc), measured at the line terminals of a listener telephone set. (See Figure 5.) For acceptable voice service over TDM/PCM circuits, the PCM idle channel noise power shall not exceed 25,000 pWp (44.0 dBrnc) and the equivalent PCM noise power should not exceed 25,000 pWp (44.0 dBrnc), measured at the line terminals of a listener telephone set.

NOTE 1: For a definition of equivalent PCM noise see FED-STD-1037.

NOTE 2: A higher received noise power value than stated in 5.2.2.1.3 is tolerated for single channel radio subsystems due to the conditions under which these systems must operate.

5.2.2.1.4 Insertion loss versus frequency characteristic. (MIL-STD-188-100.) The insertion loss versus frequency characteristic, referenced to 1000 Hz and measured between the line terminals of talker and listener telephone sets, shall not exceed the values given in Table III over the frequency bands stated. Loss is indicated by a (+) and gain by a (-) sign. (See Figure 6A.)

NOTE: The parameter values listed in Table III are identical to the values for the V1 circuit parameters of the DCS technical schedule published in DCAC 300-175-9.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE III. Insertion loss versus frequency characteristic of VF circuits for analog voice and analog facsimile service (user-to-user).

Frequency in Hz	Insertion Loss in dB (Referenced to 1000 Hz)
$f < 400$	≥ -8
$400 \leq f < 600$	-8 to +20
$600 \leq f < 2400$	-7 to +12
$2400 \leq f < 2800$	-8 to +20
$2800 \leq f$	≥ -8

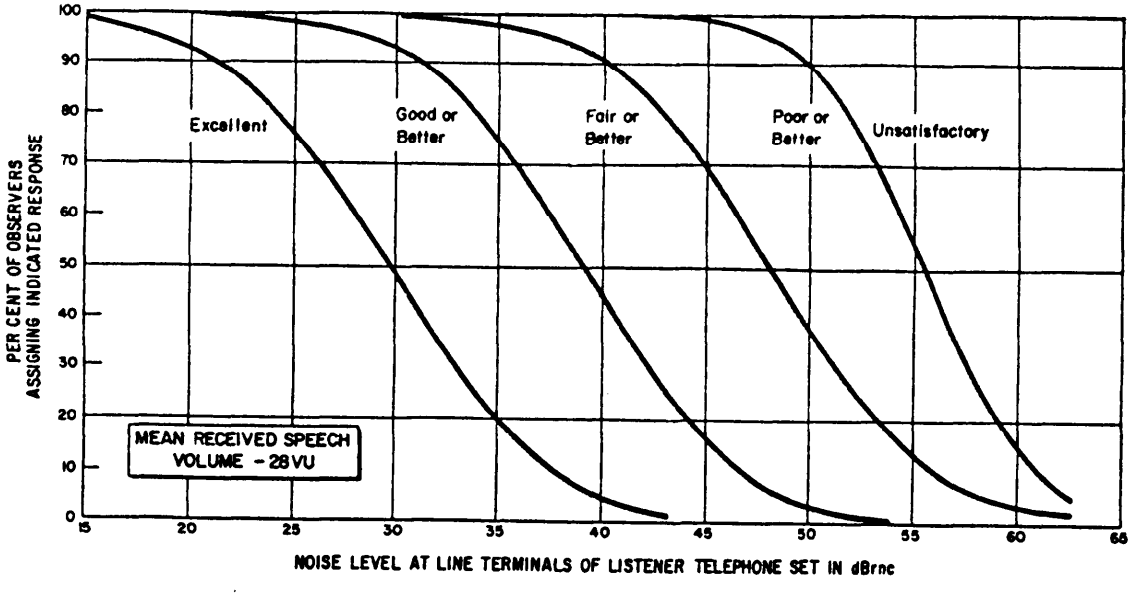
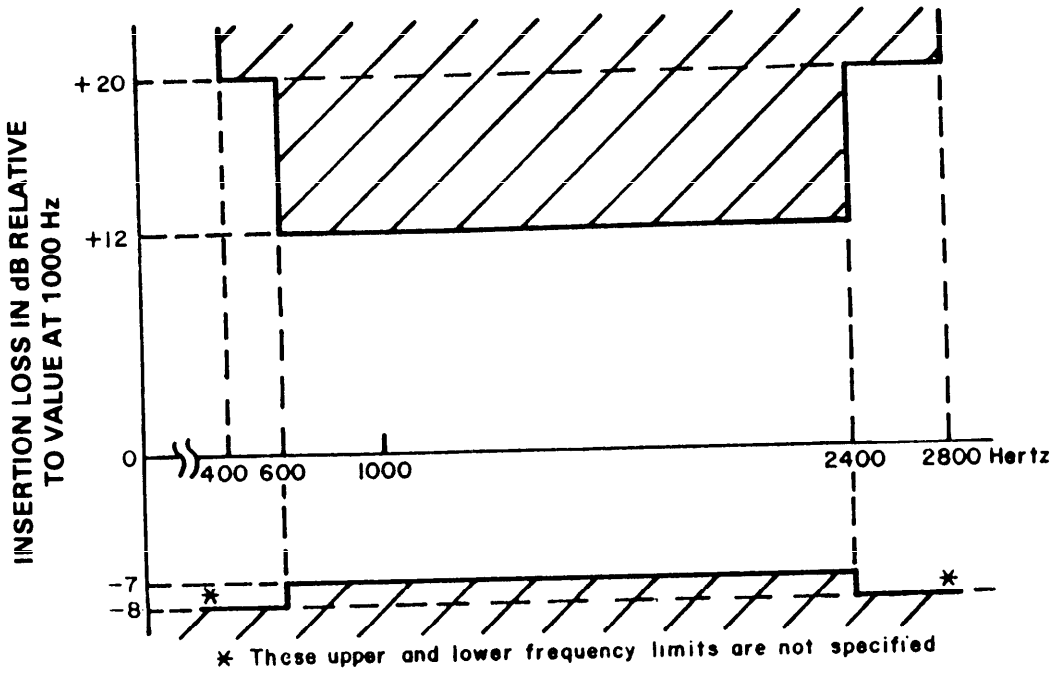
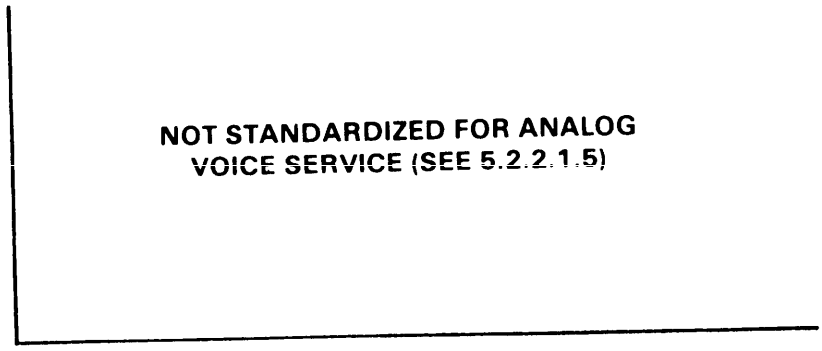


FIGURE 5. Subjective ratings of received noise power.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 6. VF circuit parameters for analog voice and analog facsimile service (user-to-user).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.2.2.1.5 Envelope delay distortion. Not standardized for analog voice service.

NOTE: No standard or design objective is given for the envelope delay distortion of VF circuits for voice service since the human ear is not sensitive to this type of distortion. However, envelope delay distortion plays a significant role in telegraph and data transmission through VF channels. Envelope delay distortion for telegraph and data service over a nominal 4 kHz VF circuit is given in 5.2.3.1.6.2, and over a 48 kHz FDM group transmission channel is given in 5.8.3.5.

5.2.2.1.6 Single tone interference. (MIL-STD-188-100.) No interfering single-frequency tone shall exceed 30 dBrnc (Design Objective: 24 dBrnc), measured at the line terminals of the listener telephone set.

5.2.2.1.7 Crosstalk. (MIL-STD-188-100.) For multichannel voice transmission, both the near-end and the far-end intelligible crosstalk shall each be at a level which is at least 55 dB below reference level (-55 dBm0) in the frequency band of interest.

NOTE: Subjective listener tests have indicated that intelligible crosstalk is more disturbing than unintelligible crosstalk of the same level. Therefore, a distinction is made between intelligible and unintelligible crosstalk even if no objective measurement technique exists to separate those two crosstalk components. Unintelligible crosstalk is considered noise for which the accepted value is stated in 5.2.2.1.3.

5.2.2.1.8 Echo. Not standardized for analog voice service.

NOTE 1: The effect of echo on a user is a function of both the magnitude and the time delay of the echo. Time delays in transmission over two-wire and return loss in four-wire tactical circuits are usually so small that echoes resulting from two-wire terminations and coupling from the receiving to the transmitting transducer are not disturbing to the near-end users. However, if channels of a tactical system are connected to channels of a long haul system then subparagraph 4.2.2.8 of MIL-STD-188-100 applies.

NOTE 2: No standard or design objective for round trip echo loss over channels containing satellite links can be provided at the present time. An objective of 30 dB is used by commercial carriers.

NOTE 3: In a telephone connection which is completely four-wire from user to user, including the telephone sets, but excluding any sidetone circuits, electrical echo need not be considered. However, if there are two-wire sections or elements at either or both ends of the connection, echo will exist. This is due to the lack of perfect electrical balance between the impedances of the line and the balancing network associated

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

with the hybrid transformer at the points where the two-wire and four-wire sections or elements join. While both parties may hear echoes, the talker is more disturbed by these echoes, so that talker echo considerations should govern overall circuit design.

5.2.2.1.9 Frequency displacement. Not standardized for analog voice service.

NOTE: No standard or design objective is given for voice service since the frequency displacement generally encountered over VF circuits has a negligible effect on intelligibility.

5.2.2.1.10 Net loss variations. (MIL-STD-188-100.) The net loss variations measured between the line terminals of talker and listener telephone sets shall not exceed ±5 dB over any 30 consecutive days and ±1.5 dB over any 15 consecutive minutes.

NOTE: The values given in 5.2.2.1.10 for the net loss variations of a circuit do not account for the net loss variations that occur in unprotected field wire loops due to weather conditions. (See Note of 5.5.2.8.)

5.2.2.2 Parameters for digitized voice service (user-to-user). Current multiplexer equipment provides digitized voice service employing TDM/PCM (see 5.7.3.2). Voice transmission for future tactical systems will be based on CVSD and LPC modulation techniques (see Appendix D). Standards for this type of service will be established when sufficient data based on measured performance are available for installed circuits operating under various conditions.

5.2.2.2.1 Bit error rate (BER). As a Design Objective for the tactical subsystem type IV (see Table I), the BER for digitized voice service on a user-to-user basis should not exceed 1 erroneous bit in 1000 information bits for 98 percent or more of the time (see Notes 1 and 2) at the applicable data signaling rate, exclusive of equipment outages. This requirement should be achieved by the tactical subsystem without using error detection and correction (EDC) measures.

NOTE 1: The 98 percent time period may occur in a continuous uninterrupted interval or may occur accumulatively.

NOTE 2: The BER requirements are applicable to system designs utilizing a mathematical model of error probability distributions. For other applications, such as acceptance tests, the unit of time (specified as a percentage of time) has to be determined, using the special requirements of the system under consideration as a basis.

5.2.2.2.2 Character-count and bit-count integrity. No extraneous characters or bits (see Note) shall be inserted or deleted in bit streams

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

representing digitized voice or message texts. This requirement shall apply to all modulation and data signaling rates. As a Design Objective, the mean-time-between-losses of character-count and bit-count integrity should be 24 hours or longer, measured from the transmitting terminal or data source to the receiving terminal or data sink.

NOTE: Extraneous characters or bits include time differential blanks associated with asynchronous/synchronous transmission equipment. These characters or bits are permissible in the transmission subsystem if they can be recovered at the receiving terminal equipment prior to forwarding the signal to the user.

5.2.2.2.3 CVSD modulation. Under consideration.

NOTE: Common long haul/tactical standards and design objectives for A-D conversion techniques including CVSD are being developed and will be published in MIL-STD-188-113 (see Appendix B). Appendix D contains characteristics for the CVSD modulation scheme that will be used in the tactical subsystem type IV (see Table I).

5.2.3 Parameters for telegraph and data service (user-to-user).

5.2.3.1 Parameters of VF circuits for telegraph and data service (user-to-user). The parameters of 5.2.3.1.1 through 5.2.3.1.7.5 apply to VF circuits connecting DTE or terminal subsystems for the transmission of quasi-analog signals. The parameter values are based on measured performance and experience and are considered the minimum essential values necessary to provide acceptable service to the user. The parameters apply to all single channel and multichannel VF circuits regardless of transmission distance, transmission medium and multiplexing scheme employed for multichannel transmission, except where stated otherwise in the applicable subparagraph. The parameters apply to data modulation rates of up to 2400 Bd, with the exception of the insertion loss versus frequency characteristic (see 5.2.3.1.6.1) and the envelope delay distortion (see 5.2.3.1.6.2) which are subdivided into parameter values for modulation rates of up to 1200 Bd and up to 2400 Bd.

VF circuits engineered for telegraph and data service may also be used for acceptable analog voice and analog facsimile service. VF circuits that cannot meet applicable requirements for telegraph and data service may have to be conditioned (equalized) or equipped with regenerative repeaters (see 5.4.3.5) to provide acceptable service. Appendix F provides general guidance for the engineering of telegraph and data circuits, including equalization of circuits and regeneration of data signals.

5.2.3.1.1 Bit error rate (BER). As a Design Objective for the tactical subsystems type I, II and III (see Table I), the BER for telegraph and data service on a user-to-user basis should not exceed 1 erroneous bit in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

100,000 information bits at the applicable data signaling rate for 98 percent or more of the time that a user utilizes the subsystem. This requirement should be achieved by the tactical subsystem without using EDC measures. As an additional Design Objective, the user throughput should be equal to or greater than 50 percent during 98 percent of the time that a user utilizes the subsystem.

NOTE 1: The 98 percent time period, as specified in 5.2.3.1.1, may occur in a continuous uninterrupted interval or may occur accumulatively, but the time period does not include any equipment outages.

NOTE 2: The design objectives of 5.2.3.1.1 are applicable to system designs utilizing a mathematical model of error probability distributions. For other applications, such as acceptance tests, the unit of time (specified in 5.2.3.1.1 as a percentage of time) has to be determined, using the special requirements of the system under consideration as a basis.

NOTE 3: Automatic-repeat-request (ARQ) systems are generally sensitive to round trip delays. For example, block-by-block data transmission in AUTODIN over a satellite subsystem has throughputs appreciably less than 50 percent. Additionally, the throughput as a function of BER is also a function of the block size. These throughput requirements cannot be interpreted as requirements on transmission facilities only but rather must be interpreted as requirements on the entire system.

NOTE 4: For the tactical subsystems type I and type II (see Table I) it is, in general, difficult to control, maintain and guarantee to the user a specific BER performance since these subsystems normally operate in a highly maneuverable tactical environment. In such an environment, radio sites are frequently relocated and site selection is dictated usually more by tactical requirements and less by communications requirements. Additionally, the BER depends on many factors, such as modulation scheme, type of channel and transmission medium, and vehicle and lightning interferences. No conclusive data, justifying the establishment of a meaningful standard, are available at the present time. Tests of single channel VHF and UHF radio links and of multichannel LOS radio links have indicated that a BER on the order of 10^{-2} to 10^{-7} can be expected without utilizing error control methods. Data-grade circuits of the tactical subsystems type I and type II should be engineered in accordance with the design objectives of 5.2.3.1.1, even if the installation and operation of some of these circuits cannot be strictly controlled to guarantee a specific BER performance to the users.

5.2.3.1.2 Error control. Error control necessary to satisfy special requirements not covered in 5.2.3.1.1 shall be accomplished by ancillary devices or by input/output line units (see Note 1) designed for that purpose and shall not be included in the basic communications network design. In cases of special error rate requirements, the originator of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

such a requirement shall be responsible to supply the means, compatible with the communications system, to meet this special requirement.

NOTE 1: These ancillary devices or line units may be an integral part of the DTE.

NOTE 2: No standard or design objective for the type of error control technique to be used can be provided at the present time. (See 5.4.3.7.)

5.2.3.1.3 Character-count and bit-count integrity. Same as 5.2.2.2.2.

5.2.3.1.4 Transmitted quasi-analog signal level. (MIL-STD-188-100.) For the tactical subsystems type I and type III (see Table I), the transmitted quasi-analog signal level of telegraph and data equipment (modem, multiplexer, etc.) shall be adjustable at least from -18 dBm to +3 dBm, so as to provide -13 dBm0 (i.e., -13 dBm at a 0TLP) at the input terminals of a data trunk or switch. For multitone data signals, the level (v) of each data tone with reference to -13 dBm shall be derived from the formula v = -13 - 10 log t, measured in dBm, where t is the number of tones. For the tactical subsystem type II (see Table I), the transmitted quasi-analog signal level of telegraph and data equipment (modem, multiplexer, etc.,) shall be adjustable at least from -18 dBm to +3 dBm, so as to provide -6 dBm0 (i.e., -10 dBm at a -4TLP) at the input terminals of a data trunk or switch. For multitone data signals, the level (v) of each data tone with reference to -10 dBm shall be derived from the formula v = -10 - 10 log t, measured in dBm, where t is the number of tones.

NOTE 1: Compliance with 5.2.3.1.4 does not require the quasi-analog signal level to be continuously adjustable. The specified signal level may be obtained in incremental steps, continuously, or by a combination of both methods. The method of adjustment will be determined in applicable equipment specifications.

NOTE 2: The formulas -13 - 10 log t and -10 - 10 log t assume a random phase distribution for the data tones of a multitone modem. The multitone phases may not be distributed randomly if the multitone signals are derived from a common frequency source. In such a case, the probability of tones adding in phase increases and the composite multitone signal may have a higher level than given by the above formulas and could overload FDM equipment. (See 5.7.3.1.1.2.)

NOTE 3: The standard data level of -6 dBm0 applies also to TDM/PCM equipment even if this type of equipment is not subject to the same overload problems as FDM equipment is. A higher than standard data level will cause clipping of the peak amplitudes of the quasi-analog signals by the PCM equipment, resulting in unsatisfactory data transmission. (See 5.7.3.2.2.1 and Note of 5.7.3.2.6.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE 4: The different transmitted quasi-analog signal levels of the tactical subsystems type I or type III and the tactical subsystem type II require a special method (see 5.8.5) for interconnecting the VF channels of these subsystems.

5.2.3.1.5 Output parameters.

5.2.3.1.5.1 Received quasi-analog signal level. Telegraph and data equipment (modem, multiplexer, etc.) shall be capable of operating, without degradation of performance, with a received quasi-analog signal level ranging at least from -35 dBm to -5 dBm. The received quasi-analog signal level shall be measured at the input terminals of the DCE.

In the tactical subsystems type I and type III (see Table I), the received quasi-analog signal level:

a. Should be -22 dBm for switched and nonswitched (point-to-point) VF circuits used for both voice service and telegraph and data service.

b. Should be -13 dBm for nonswitched (point-to-point) VF circuits used only for telegraph and data service.

c. Is not standardized for switched VF circuits used only for telegraph and data service. (See 60.3.2.1 of Appendix F.)

In the tactical subsystem type II, the received quasi-analog signal level:

a. Should have a mean value of -13.5 dBm for switched and nonswitched (point-to-point) VF circuits used for both voice service and telegraph and data service.

b. Should be -10 dBm for nonswitched (point-to-point) VF circuits used only for telegraph and data service.

c. Is not standardized for switched VF circuits used only for telegraph and data service. (See 60.3.2.2 of Appendix F.)

5.2.3.1.5.2 Received noise power. (MIL-STD-188-100.) For acceptable telegraph and data service over FDM or metallic circuits, the non-impulse type of received noise power shall not exceed 50,000 pW0 (47.0 dBrn0) for 99 percent or more of the time (see Note 1) and shall not exceed 316,000 pW0 (55.0 dBrn0) for 1 percent or less of the time when referenced to a circuit OTLP. The received noise power shall be measured at the input terminals of telegraph and data equipment (demodulator, demultiplexer, etc.) with flat weighting over the frequency band occupied by the quasi-analog signals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE 1: The unit of time (to which the percentage of time in 5.2.3.1.5.2 and other subparagraphs applies) depends on the special requirements of the system under consideration and, therefore, cannot be given in a general form. (See also Note 1 of 5.2.3.1.1.)

NOTE 2: The received noise power for telegraph and data service stated in 5.2.3.1.5.2 has been based on a minimum SNR of 26 dB (see 5.2.3.1.5.3) and the lowest allowable level of -22 dBm of the quasi-analog signal in accordance with 5.2.3.1.5.1 (worst case assumption). For higher received signal levels the SNR will increase accordingly. (See Appendix F.)

NOTE 3: The standards of 5.2.3.1.5.2 apply to FDM transmission and metallic circuits only. For the transmission of quasi-analog signals through VF channels derived by PCM, the correlation of quantizing noise to BER has not been clearly established at the present time. In addition, the correlation of BER to thermal noise depends on the modulation technique employed in the conversion device (modem). In any case, the BER stated in 5.2.3.1.1 should be the dominating design objective for data transmission.

NOTE 4: The commonly used noise measuring sets will measure total rms noise power including the average impulse noise power. However, the contribution of the average impulse noise, as registered by these meters, is usually small enough to be neglected.

5.2.3.1.5.3 Signal-to-noise ratio (SNR). As a Design Objective for telegraph and data service over VF circuits, the rms-signal-to-rms-noise ratio should be at least 26 dB. The parameter should be measured at the quasi-analog input terminals of telegraph and data equipment (demodulator, demultiplexer, etc.) with flat weighting over the frequency band occupied by the quasi-analog signals.

NOTE: The SNR stated in 5.2.3.1.5.3 is a necessary but not sufficient requirement for acceptable telegraph and data service since signal discontinuities (see 5.2.3.1.7) may increase the BER for certain unpredictable periods of time.

5.2.3.1.5.4 Single tone interference. (MIL-STD-188-100.) For telegraph and data service over VF circuits, no interfering single-frequency tone shall exceed 30 dBrn (Design Objective: 24 dBrn). The parameter shall be measured at the quasi-analog input terminals of telegraph and data equipment (demodulator, demultiplexer, etc.) with flat weighting over the frequency band occupied by the quasi-analog signals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.2.3.1.6 Transfer function parameters.

5.2.3.1.6.1 Insertion loss versus frequency characteristics.

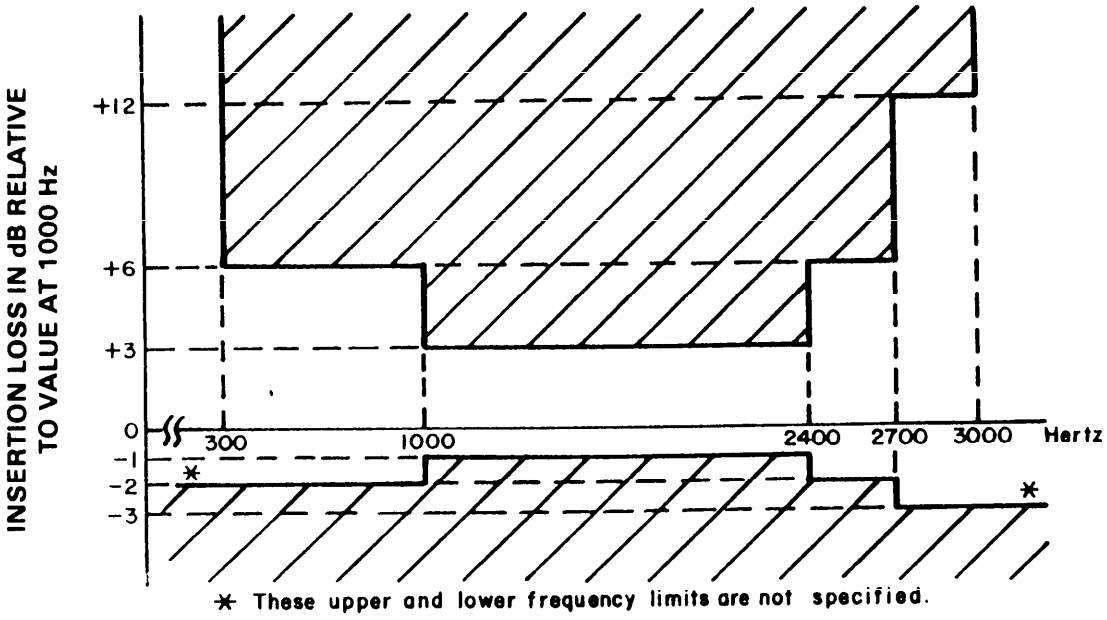
5.2.3.1.6.1.1 Telegraph and data service with modulation rates of up to 1200 Bd. As a Design Objective for telegraph and data service with modulation rates of up to 1200 Bd over VF circuits, the insertion loss versus frequency characteristic, referenced to 1000 Hz, should not exceed the values given in Table IV over the frequency bands stated. Loss is indicated by a (+) and gain by a (-) sign. (See Figure 7A.)

NOTE: The parameter values listed in Tables IV and VI are identical to the values of the D2 circuit parameters, and the parameter values listed in Tables V and VII are identical to the values of the D1 and S1 circuit parameters. The D1, D2 and S1 circuit parameters are part of the DCS technical schedule published in DCAC 300-175-9. The DCS technical schedule is based on corresponding parameter values used by commercial carriers. The parameter values of the DCS technical schedule are considered the best information available to specify VF circuit characteristics necessary for acceptable data transmission. However, experience has shown that VF circuits may provide acceptable data service when the measured parameters are close to, but not exactly within the limits of the values listed in Tables IV through VII. Therefore, the parameter values listed in these tables are given as design objectives and should not be used exclusively to determine the need for regenerating data signals and conditioning data circuits. (See Appendix F.)

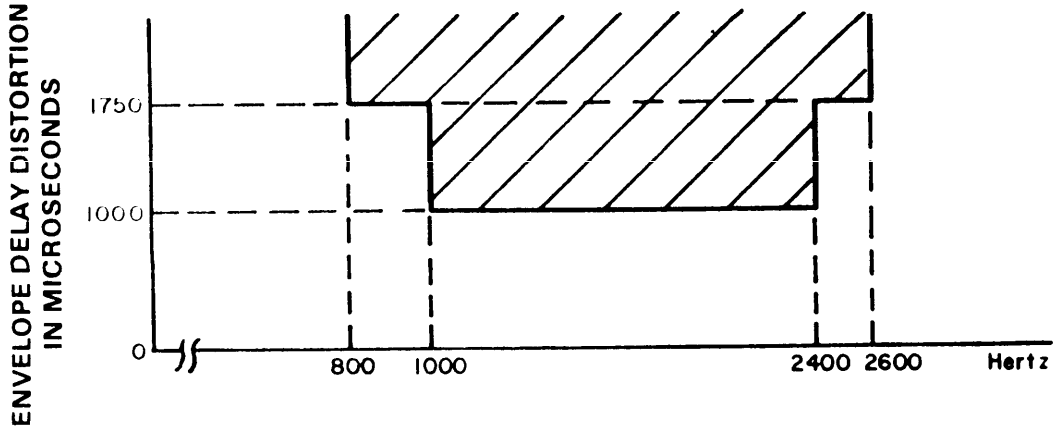
TABLE IV. Insertion loss versus frequency characteristic of VF circuits for telegraph and data service with modulation rates of up to 1200 Bd (user-to-user).

Frequency in Hz	Insertion Loss in dB (Referenced to 1000 Hz)
$f < 300$	≥ -2
$300 \leq f < 1000$	-2 to +6
$1000 \leq f < 2400$	-1 to +3
$2400 \leq f < 2700$	-2 to +6
$2700 \leq f < 3000$	-3 to +12
$3000 \leq f$	≥ -3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 7. VF circuit parameters for telegraph and data service with modulation rates of up to 1200 Bd (user-to-user).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.2.3.1.6.1.2 Telegraph and data service with modulation rates of up to 2400 Bd. As a Design Objective for telegraph and data service with modulation rates of up to 2400 Bd over VF circuits, the insertion loss versus frequency characteristic, referenced to 1000 Hz, should not exceed the values given in Table V over the frequency bands stated. Loss is indicated by a (+) and gain by a (-) sign. (See Figure 8A and Note of 5.2.3.1.6.1.1.)

TABLE V. Insertion loss versus frequency characteristic of VF circuits for telegraph and data service with modulation rates of up to 2400 Bd (user-to-user).

Frequency in Hz	Insertion Loss in dB (Referenced to 1000 Hz)
$f < 300$	≥ -2
$300 \leq f < 500$	-2 to +6
$500 \leq f < 2800$	-1 to +3
$2800 \leq f < 3000$	-2 to +6
$3000 \leq f$	≥ -2

5.2.3.1.6.2 Envelope delay distortion.

5.2.3.1.6.2.1 Telegraph and data service with modulation rates of up to 1200 Bd. As a Design Objective for telegraph and data service with modulation rates of up to 1200 Bd over VF circuits, the envelope delay distortion measured between 800 Hz and 2600 Hz (see 70.5 of Appendix G) should not exceed the values given in Table VI over the frequency bands stated. (See Figure 7B and Note of 5.2.3.1.6.1.1.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE VI. Envelope delay distortion of VF circuits for telegraph and data service with modulation rates of up to 1200 Bd (user-to-user).

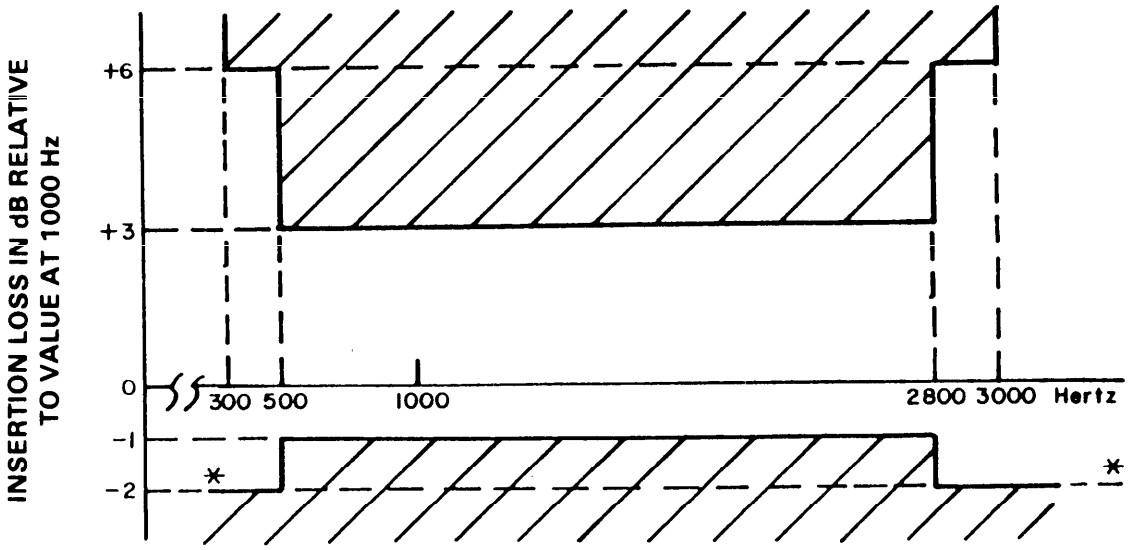
Frequency in Hz	Maximum Envelope Delay Distortion in microseconds
800 ≤ f < 1000	1750
1000 ≤ f < 2400	1000
2400 ≤ f ≤ 2600	1750

5.2.3.1.6.2.2 Telegraph and data service with modulation rates of up to 2400 Bd. As a Design Objective for telegraph and data service with modulation rates of up to 2400 Bd over VF circuits, the envelope delay distortion measured between 500 Hz and 2800 Hz (see 70.5 of Appendix G) should not exceed the values given in Table VII over the frequency bands stated. (See Figure 8B and Note of 5.2.3.1.6.1.1.)

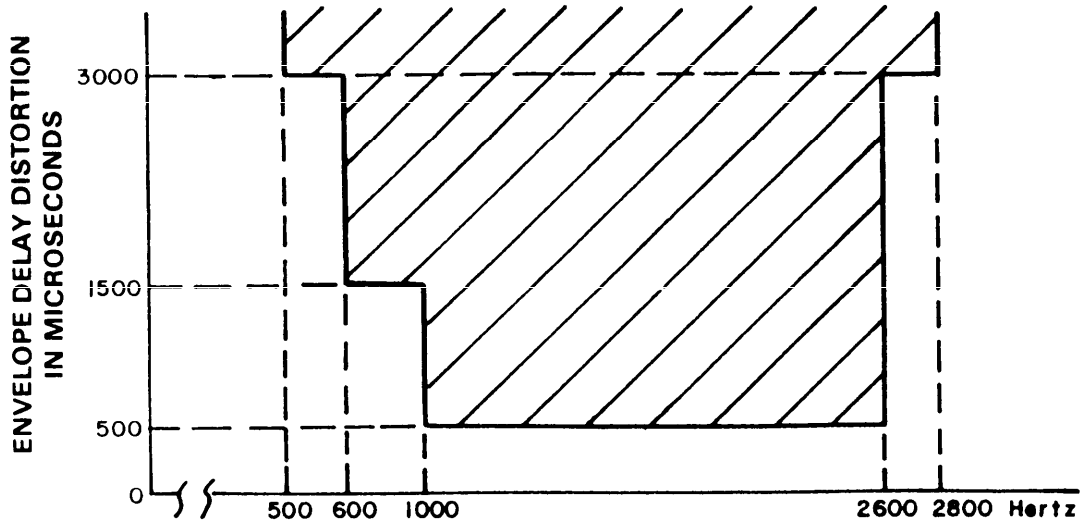
TABLE VII. Envelope delay distortion of VF circuits for telegraph and data service with modulation rates of up to 2400 Bd (user-to-user).

Frequency in Hz	Maximum Envelope Delay Distortion in microseconds
500 ≤ f < 600	3000
600 ≤ f < 1000	1500
1000 ≤ f < 2600	500
2600 ≤ f ≤ 2800	3000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 8. VF circuit parameters for telegraph and data service with modulation rates of up to 2400 Bd (user-to-user).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.2.3.1.6.3 Total harmonic distortion. For telegraph and data service over VF circuits, the total harmonic distortion produced by any single-frequency test signal between 300 Hz and 3000 Hz shall be at a level which is at least 30 dB below reference level (-30 dBm0), with a Design Objective of -40 dBm0. The total harmonic distortion shall be measured with a standard test signal in accordance with 5.6.6.1.

5.2.3.1.6.4 Intermodulation distortion. Under consideration.

5.2.3.1.6.5 Total distortion of digital signals. The maximum total distortion of binary signals introduced by a VF circuit due to any combination of causes (bias, fortuitous, cyclic or characteristic) shall not exceed ±25 percent of the unit interval, measured from data source to data sink.

5.2.3.1.6.6 Frequency displacement. Any single-frequency signal between 300 Hz and 3400 Hz transmitted over a VF circuit shall be reproduced at the input terminals of a telegraph or data modem (or a regeneration point) with a frequency error of not more than ±4 Hz.

NOTE: The standard of 5.2.3.1.6.6 is not applicable to speech plus half-duplex and to speech plus full-duplex operations. Frequency displacement for these types of operation is given in 5.4.1.5.1.3.

5.2.3.1.6.7 Net loss variations. (MIL-STD-188-100.) The net loss variations measured from a telegraph or data modulator output (or a regeneration point) to a telegraph or data demodulator input (or a regeneration point) shall not exceed ±5 dB over any 30 consecutive days and ±1.5 dB over any 15 consecutive minutes.

NOTE: The values given in 5.2.3.1.6.7 for the net loss variations do not account for the net loss variations that occur in unprotected field wire loops due to weather conditions. (See Note of 5.5.2.8.)

5.2.3.1.7 Signal discontinuities.

5.2.3.1.7.1 Impulse noise. (MIL-STD-188-100.) For telegraph and data service over VF circuits, the impulse noise shall not exceed 15 counts above a level of 71 dBrn0 over any 15 consecutive minutes. The impulse noise shall be measured at the quasi-analog input terminals of telegraph and data equipment (demodulator, demultiplexer, etc.) with flat weighting over the frequency bandwidth occupied by the quasi-analog signals. The test instrument shall be capable of counting rates of up to 7.5 counts per second.

NOTE: Impulse noise performance of a connection is closely related to the type of switching equipment and to the total site complex, in terms of relays, battery charging and filtering, etc. Electronic switching (solid state) subsystems are approximately 10 dB quieter than mechanical relay switchboards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.2.3.1.7.2 Signal level dropouts. As a Design Objective for telegraph and data service over VF circuits, the number of signal level dropouts equal to or greater than 12 dB and exceeding a period of 10 milliseconds (ms) should be less than two dropouts over any 15 consecutive minutes.

5.2.3.1.7.3 Signal level change. Under consideration.

5.2.3.1.7.4 Phase jitter. (MIL-STD-188-100.) The total peak-to-peak phase jitter, imparted to a test tone traversing a VF circuit at any frequency between 300 Hz and 3000 Hz, shall not exceed 15 degrees (Design Objective: 10 degrees).

5.2.3.1.7.5 Phase hits. (MIL-STD-188-100.) For telegraph and data service over VF circuits, the number of phase hits of greater than ±20 degrees shall not exceed 15 hits over any 15 consecutive minutes.

5.2.3.1.7.6 Amplitude (or gain) hits. Under consideration.

5.2.3.1.8 Equalization. Not standardized.

NOTE: Equalization of VF circuits has to be determined by circuit engineering on a case-by-case basis. (See Appendix F.)

5.2.3.1.9 Regeneration. Not standardized.

NOTE: Regeneration points for telegraph and data service over VF circuits have to be determined by circuit engineering on a case-by-case basis. (See Appendix F.)

5.2.3.2 Parameters for telegraph and data service over 48 kHz FDM group circuits (user-to-user). Under consideration.

NOTE: No detailed parameters for telegraph and data service over 48 kHz FDM group circuits are provided on a user-to-user basis, since loop characteristics for this service cannot be specified at the present time. (See 5.5.5.) Until these loop characteristics are established, the transmission channel parameters stated in 5.8.3 should be used as a general guidance for telegraph and data service over 48 kHz FDM group circuits. (See also 60.2 of Appendix F.)

5.2.3.3 Parameters for telegraph and data service over digital circuits (user-to-user).

5.2.3.3.1 Bit error rate (BER). As a Design Objective for the tactical subsystem type IV (see Table I), the BER for telegraph and data service on a user-to-user basis should not exceed 1 erroneous bit in 100,000 information bits for 98 percent or more of the time (see Notes 1 and 2 of 5.2.2.2.1) at the applicable data signaling rate, exclusive of equipment outages. This requirement should be achieved by using EDC techniques in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

the DTE or DCE. The EDC device should provide a BER improvement factor of not less than 100 in a random error environment of 1 erroneous bit in 1000 information bits.

5.2.3.3.2 Character-count and bit-count integrity. Same as 5.2.2.2.2.

5.2.4 Parameters for facsimile service (user-to-user). Parameters for facsimile service on a user-to-user basis are subdivided into parameters for analog facsimile service (see 5.2.4.1) and for digital facsimile service (see 5.2.4.2). The parameter values are based on measured performance and on experience and are considered the minimum essential values necessary to provide acceptable service to the user. The parameters apply to all single channel and multichannel VF circuits regardless of transmission distance, transmission medium and multiplexing scheme employed for multichannel transmission, except where stated otherwise in the applicable subparagraph.

5.2.4.1 Parameters for analog facsimile service (user-to-user).

5.2.4.1.1 Transmitted signal level. (MIL-STD-188-100.) The facsimile transmitter output corresponding to a high-contrast signal level shall be adjustable at least from -10 dBm to +10 dBm. Considering the loop loss, the transmitted level shall be -13 dBm at a OTLP (-13 dBmO) of the tactical subsystems type I and type III (see Table I) and -10 dBm at a -4TLP (-6 dBmO) of the tactical subsystem type II.

5.2.4.1.2 Received signal level. (MIL-STD-188-100.) Analog facsimile receiving equipment shall be capable of operating with a received high-contrast signal level ranging at least from -9 dBm to -36 dBm. The average received level should be approximately -20 dBm.

NOTE: The exact value of the received level of analog facsimile signals depends on the circuit configuration.

5.2.4.1.3 Received noise power. (MIL-STD-188-100.) The received noise power shall not exceed 25,000 pW (44.0 dBrn) (Design Objective: 38 dBrn) measured at the line terminals of a facsimile receiver with flat weighting over the frequency bandwidth from 400 Hz to 2800 Hz.

5.2.4.1.4 Insertion loss versus frequency characteristic. Same as for analog voice service given in 5.2.2.1.4.

5.2.4.1.5 Envelope delay distortion. Under consideration.

NOTE: Until parameter values are established, the envelope delay distortion for data-grade circuits given in 5.2.3.1.6.2 should be used as guidance for analog facsimile service.

5.2.4.1.6 Single tone interference. (MIL-STD-188-100.) No interfering single-frequency tone shall exceed 30 dBrn (Design Objective: 24 dBrn),

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

measured at the line terminals of a facsimile receiver with flat weighting over the frequency band from 400 Hz to 2800 Hz.

5.2.4.1.7 Frequency displacement. Not standardized.

NOTE: No standard or design objective is given for analog facsimile service since the frequency displacement generally encountered over VF circuits has a negligible effect on analog facsimile quality.

5.2.4.1.8 Net loss variations. (MIL-STD-188-100.) The net loss variations measured between the line terminals of facsimile transmitting and receiving equipment shall not exceed ±5 dB over any 30 consecutive days and ±1.5 dB over any 15 consecutive minutes.

NOTE: The values given in 5.2.4.1.8 for the net loss variations of a circuit do not account for the net loss variations that occur in unprotected field wire loops due to weather conditions. (See Note of 5.5.2.8.)

5.2.4.2 Parameters for digital facsimile service (user-to-user). Under consideration.

5.2.5 Parameters for tactical digital information links (TADIL). Message formats and related information for TADIL A, B and C are published in JCS PUB 10. Technical characteristics for all TADIL are being published in volumes of the MIL-STD-188-203 series. It is intended that the technical characteristics of other TADIL currently under development, such as the Joint Tactical Information Distribution System (JTIDS), will be included in the MIL-STD-188-203 series.

5.2.5.1 Parameters for TADIL A. The technical characteristics of TADIL A shall comply with the applicable requirements of the current edition of MIL-STD-188-203-1. Message formats and related information for TADIL A shall comply with the applicable requirements of the current edition of JCS PUB 10.

5.2.5.2 Parameters for TADIL B. Message formats and related information for TADIL B shall comply with the applicable requirements of the current edition of JCS PUB 10.

NOTE: The technical characteristics of TADIL B are being developed and will be published in MIL-STD-188-203-2. (See Appendix B.)

5.2.5.3 Parameters for TADIL C. Message formats and related information for TADIL C shall comply with the applicable requirements of the current edition of JCS PUB 10.

NOTE: The technical characteristics of TADIL C are being developed and will be published in MIL-STD-188-203-3. (See Appendix B.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.3 Terminal subsystems. The following subparagraphs deal with standards and design objectives for terminal subsystems, including interface characteristics of analog and digital equipment. The configuration of terminal subsystems will vary widely, ranging from a single telephone or teletypewriter to a very complex data terminal consisting of different types of equipment for receiving, controlling, processing, displaying and transmitting of data. It is not intended that all possible configurations of terminal subsystems be covered in this paragraph. Instead, emphasis is placed on those parameters considered essential to ensure interoperation between terminal and transmission subsystems, among different types of equipment within a terminal subsystem, and between users and the terminal equipment. An example of a data terminal subsystem in the form of a block diagram is depicted in Figure 9. Basic functions of equipment that may be located in a data terminal subsystem are described in 4.9.1.1 through 4.9.1.3. Requirements for other equipment that may be located in, and form part of, a data terminal subsystem are given in 5.4.3.

5.3.1 Analog terminal equipment. Standards and design objectives for tactical analog telephone sets and analog facsimile equipment contained in MIL-STD-188C are being updated and will be published in MIL-STD-188-260 (See Appendix B). Until MIL-STD-188-260 is approved and published, MIL-STD-188C will continue to apply to tactical analog terminal equipment. Standards and design objectives which are common to long haul and tactical analog terminal equipment are contained in paragraph 5.7 of MIL-STD-188-100.

5.3.1.1 Analog telephone sets. The technical characteristics of tactical analog telephone sets shall comply with the applicable requirements stated in paragraph 6.2 of MIL-STD-188C.

5.3.1.2 Analog facsimile equipment. The technical characteristics of tactical analog facsimile equipment shall comply with the applicable requirements stated in paragraph 6.3 of MIL-STD-188C.

5.3.2 Data terminal equipment (DTE). The technical characteristics of tactical DTE, including teletypewriter equipment, tape punches and tape readers, card punches and card readers, magnetic tape recorders and readers, optical character recognition equipment (OCRE), keyboard display terminals and digital telephone sets shall comply with the applicable requirements stated in 5.3.5 through 5.3.9.

NOTE: Standards and design objectives for tactical DTE, besides those stated in 5.3.5 through 5.3.9, are being developed and will be published in MIL-STD-188-260 (see Appendix B). Standards and design objectives which are common to long haul and tactical DTE are contained in paragraph 5.8 of MIL-STD-188-100.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

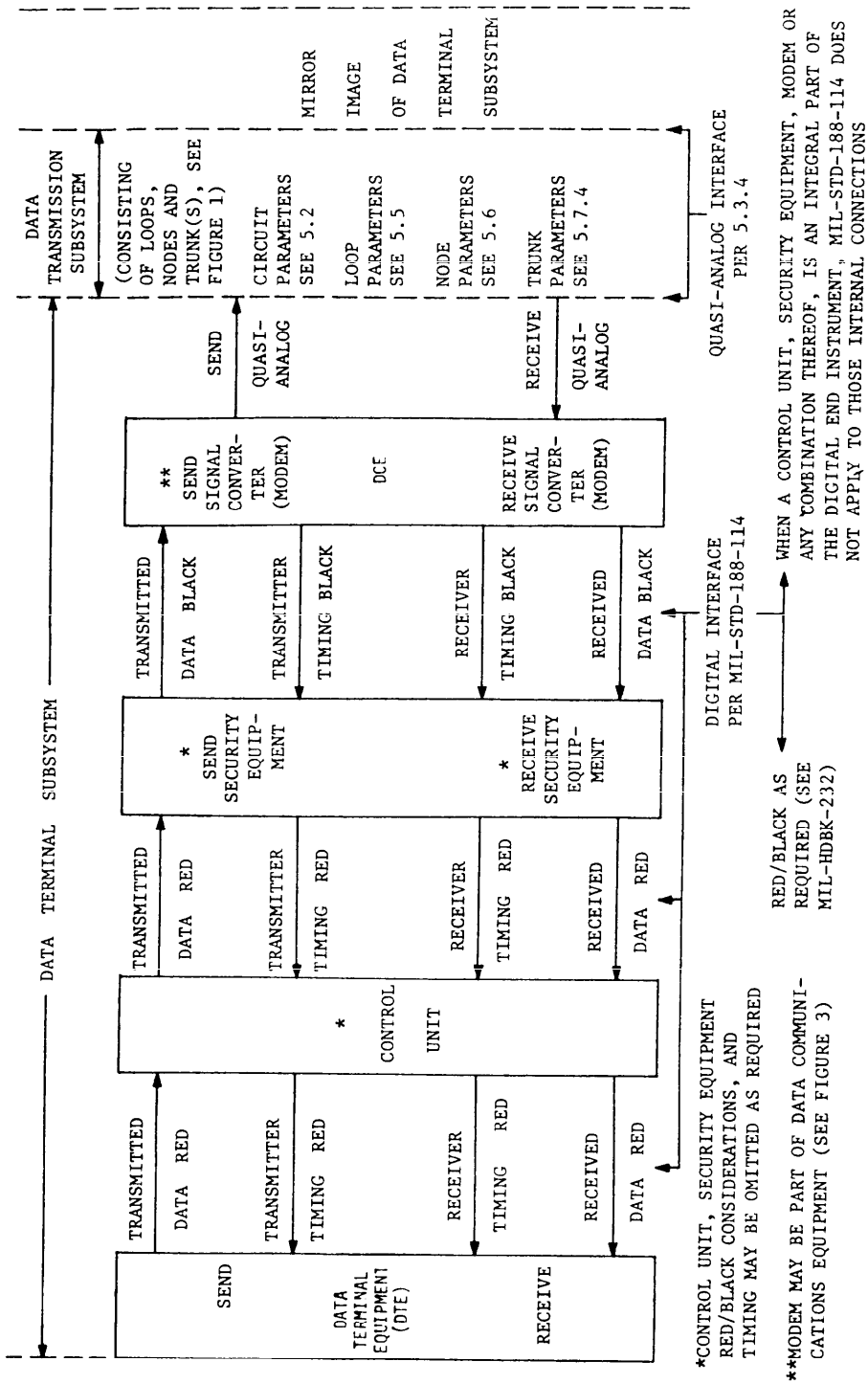


FIGURE 9. Example of block diagram of data terminal subsystem.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.3.2.1 Digital facsimile equipment. The technical characteristics of tactical digital facsimile equipment shall comply with the applicable requirements of the current edition of MIL-STD-188-161.

5.3.3 Analog interface characteristics.

5.3.3.1 Nominal 3 kHz VF channel interface.

5.3.3.1.1 Impedance. (MIL-STD-188-100.) The impedance of the transmitting and receiving terminals of a nominal 3 kHz VF channel interface for analog signals shall each be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band from 350 Hz to 3000 Hz. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

5.3.3.1.2 Transmitted analog signal level. The transmitted level of analog facsimile signals at a VF channel interface shall be adjusted to a value, depending on the transmission level point (TLP) at the interface, that results in the correct signal level given in 5.2.4.1.1.

NOTE: The transmitted speech volume at a VF channel interface is not adjusted and depends on the TLP at the interface and on the talker volume distribution (see 5.2.2.1.1).

5.3.3.1.3 Received analog signal level. Not standardized.

NOTE: The received level of analog signals at a VF channel interface depends on the TLP at this interface.

5.3.3.2 Nominal 4 kHz VF channel interface.

5.3.3.2.1 Impedance. (MIL-STD-188-100.) The impedance of the transmitting and receiving terminals of a nominal 4 kHz VF channel interface for analog signals shall each be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band from 300 Hz to 3400 Hz. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

5.3.3.2.2 Transmitted analog signal level. Same as 5.3.3.1.2.

5.3.3.2.3 Received analog signal level. Not standardized. (See Note of 5.3.3.1.3.)

5.3.4 Quasi-analog interface characteristics.

5.3.4.1 Nominal 3 kHz VF channel interface.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.3.4.1.1 Impedance. (MIL-STD-188-100.) The impedance of the transmitting and receiving terminals of a nominal 3 kHz VF channel interface for quasi-analog signals, other than for single channel radios, shall each be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band from 350 Hz to 3000 Hz. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

For modems used with radio equipment of single channel radio subsystems, the terminal impedance at the modulator output shall be 150 ohms, unbalanced to ground, with a minimum return loss of 20 dB against a 150-ohm resistance over the frequency band of interest. The terminal impedance at the demodulator input shall be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band of interest. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0). As a Design Objective, the terminal impedance at the modulator output should be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band of interest. (See Note.) The electrical symmetry should be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

NOTE: A terminal impedance balanced to ground is recommended for equipment (radios, data modems, etc.) operating in an environment that has a high EMI level, such as in aircraft and vehicles. Measurements have shown that an electrical noise rejection improvement of up to 20 dB can be achieved for balanced terminations, compared with unbalanced terminations.

5.3.4.1.2 Transmitted quasi-analog signal level. The signal level shall be adjusted to a value, depending on the TLP at the interface, that results in the correct signal level given in 5.2.3.1.4.

5.3.4.1.3 Received quasi-analog signal level. Not standardized.

NOTE: The received level of quasi-analog signals at a VF channel interface depends on the TLP at this interface. (See 5.2.3.1.5.1.)

5.3.4.2 Nominal 4 kHz VF channel interface.

5.3.4.2.1 Impedance. (MIL-STD-188-100.) The impedance of the transmitting and receiving terminals of a nominal 4 kHz VF channel interface for quasi-analog signals, other than for single channel radios, shall each be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band from 300 Hz to 3400 Hz. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

For modems used with radio equipment of single channel radio subsystems, the terminal impedance at the modulator output shall be 150 ohms, unbalanced to ground, with a minimum return loss of 20 dB against a 150-ohm resistance over the frequency band of interest. The terminal impedance at the demodulator input shall be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band of interest. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0). As a Design Objective, the terminal impedance at the modulator output should be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band of interest. (See Note.) The electrical symmetry should be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0). (See Note of 5.3.4.1.1.)

5.3.4.2.2 Transmitted quasi-analog signal level. Same as 5.3.4.1.2.

5.3.4.2.3 Received quasi-analog signal level. Not standardized. (See Note of 5.3.4.1.3.)

5.3.4.3 48 kHz FDM group channel interface.

5.3.4.3.1 Impedance. (MIL-STD-188-100.) The impedance of the transmitting and receiving terminals of a 48 kHz FDM group channel interface for quasi-analog signals shall each be 135 ohms, balanced to ground, with a minimum return loss of 20 dB (Design Objective: 26 dB) against a 135-ohm resistance over the frequency band of interest. For the purpose of interfacing with other equipment, a strapping option for 150-ohm impedance termination, balanced to ground, shall be provided. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

5.3.4.3.2 Transmitted quasi-analog signal level. The transmitted level of the modulated data signal shall be adjusted to a value, depending on the TLP at the interface, that results in the correct signal level given in 5.8.3.1.1.

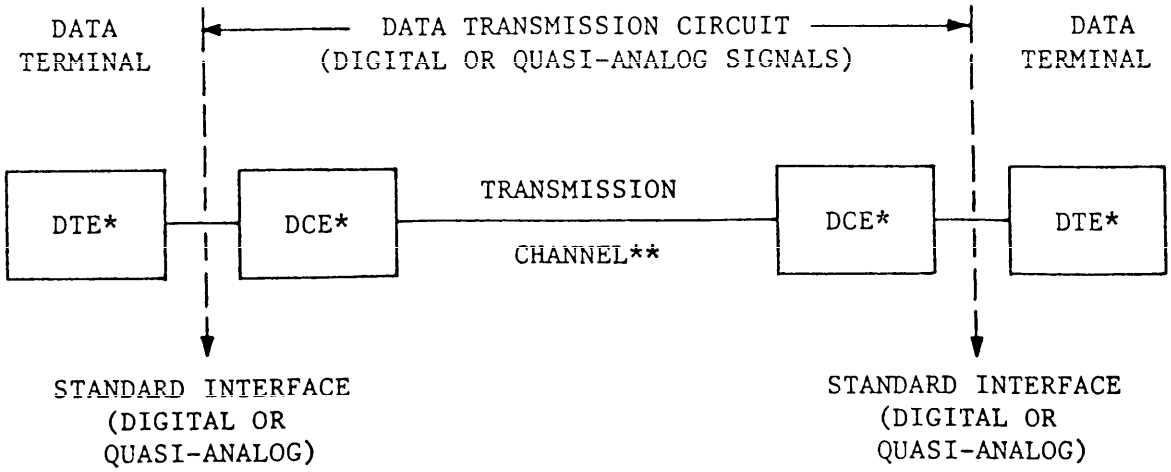
5.3.4.3.3 Received quasi-analog signal level. Not standardized.

NOTE: The received level of the modulated data signal at a 48 kHz FDM group channel interface depends on the TLP at this interface.

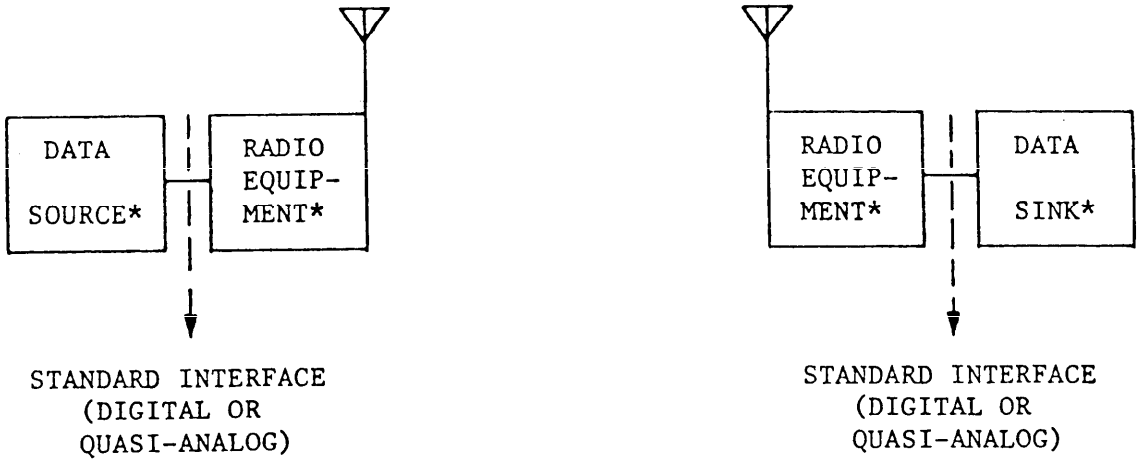
5.3.5 Telegraph and data interface characteristics.

5.3.5.1 Modulation and data signaling rates. (MIL-STD-188-100.) The modulation rates (expressed in Bd) and the data signaling rates (expressed in b/s) at the standard interfaces shown in Figure 10 shall be as listed

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



A. Block Diagram of Multichannel Circuit.



B. Block Diagram of Single Channel Radio Circuit.

* May include data adapters, modems, error control devices, control units and other equipment as required.

** May include nodes and multichannel transmission facilities. (See Figure 1.)

NOTE: Modulation rates and data signaling rates at the standard interface are given in 5.3.5.1.

FIGURE 10. Standard interfaces between data terminal equipment and data circuit terminating equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

below under items a through c. For transmission over nominal 3 kHz and nominal 4 kHz VF channels, only the modulation and data signaling rates specified under items a and b shall apply.

a. 50 Bd or b/s.

b. 75×2^{m} Bd or b/s, up to and including 9600 Bd or b/s, where m is a positive integer 0, 1, 2, ... 7.

c. The modulation rates (expressed in Bd) and the data signaling rates (expressed in b/s) above 9600 Bd or b/s are based on the formula 8000 x N and shall be 16 kBd or kb/s and 32 kBd or kb/s.

NOTE 1: The series of 25×2^{-m} contained in older MIL-STD-188 documents has been deleted.

NOTE 2: The modulation and data signaling rates of 5.3.5.1 for data transmission over VF channels comply with the preferred rates of FIPS PUB 22-1, with the following two exceptions:

a. The rate of 50 Bd or b/s is a DoD standard but is not a Federal standard.

b. The rate of 7200 Bd or b/s is a preferred rate in the Federal standard but has not been adopted as a DoD standard.

FIPS PUB 22-1 has also been identified as FED-STD-1013. Both Federal standards have adopted ANSI X3.1-1976 (a revision of X3.1-1968). ANSI X3.1-1976 specifies a series of signaling rates based on the formula 600 x N with a subset of preferred standard rates.

NOTE 3: It is not intended that new equipment must have the capability to operate with all modulation and data signaling rates specified in 5.3.5.1. Only those rates listed in 5.3.5.1 should be selected that are necessary to meet the operational requirements of the equipment under development.

NOTE 4: The data signaling rate is expressed in b/s; the modulation rate is expressed in Bd. Data signaling rates in b/s and modulation rates in Bd are the same if, and only if, all pulses have equal duration, all pulses occupy the complete unit interval and binary (rather than m-ary) signaling is used. (See also 4.9.1.5.)

5.3.5.2 Logic and signal sense for binary signals. (MIL-STD-188-100.) For information interchange between and among communications systems, automatic data processing (ADP) systems and associated equipment, the logic and signal sense for binary data and timing signals shall comply

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

with the applicable requirements stated in Table VIII. The signal voltage with respect to signal ground shall be negative to represent the significant condition of Mark, and positive to represent the significant condition of Space. For recording binary information on magnetic tape, the magnetic flux polarity shall comply with the applicable requirements of 5.4.1.8.3.2 through 5.4.1.8.3.4.

An alternative capability shall be provided to interoperate with equipment which conforms to other standards, such as older versions of MIL-STD-188C and MIL-STD-188-100, and which have polarities opposite to those polarities stated in Table VIII. (See Note 1.)

NOTE 1: In MIL-STD-188-114, the significant conditions of Mark and Space for telegraphy and data transmission and for timing have been changed from Mark represented by a positive voltage (positive Mark) and Space represented by a negative voltage (negative Space) as stated in MIL-STD-188-100, subparagraph 4.3.1.5, to negative Mark and positive Space. It is not intended that existing equipment and facilities be converted to the negative Mark/ positive Space conditions unless a major facility modification or rehabilitation is planned and such conversion could be accommodated during that work. However, a provision for negative and positive Mark must be accommodated in the acquisition of new equipment. This does not mean that negative Mark will be implemented immediately, simply because the equipment or facility is new. This guidance is only intended to mean that new equipment or facilities will be purchased with the positive and negative Mark capability and the facilities engineering or operating agency will be expected to make the changeover to negative Mark on a case-by-case basis, subject to operational and fiscal constraints.

NOTE 2: Two methods of recording binary signals on 12.7 millimeters (mm) or 0.5 inch (in) wide magnetic tape have been standardized by ANSI and adopted as FIPS (see 5.4.1.8.3). The first method is non-return-to-zero inverted (NRZI) recording (see Note 3) standardized for recording densities of 800 characters per inch (cpi) or approximately 32 characters per millimeter (cpmm), and 6250 cpi (approximately 246 cpmm). The second method is phase encoded recording (see Note 4) standardized for recording densities of 1600 cpi (approximately 63 cpmm).

NOTE 3: In NRZI recording, ANSI X3.22-1973 and X3.54-1976 define a "1" data bit as a reversal of flux polarity and a "0" data bit as no reversal of flux polarity.

NOTE 4: In phase encoded recording, ANSI X3.39-1973 defines a "1" data bit as a flux reversal to the polarity of the interblock gap, when reading in the forward direction; and a "0" data bit is defined as a flux reversal to the polarity opposite to that of the interblock gap, when reading in the forward direction. A flux reversal is recorded at the nominal midpoint between successive "1" bits or between successive "0" bits, to establish proper polarity. This flux reversal is called a phase flux reversal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE VIII. Logic and signal sense for binary signals.

APPLICATION	CONDITION	CONDITION
Voltage to Signal Ground	Negative (-)	Positive (+)
Conventional Term	Mark	Space
Binary Digit Value	One (1)	Zero (0)
Timing Signal State	Off	On
Perforated Tape (see 5.4.1.8.2)	Hole	No Hole
FSK Signal State	Lower Frequency	Higher Frequency
Neutral System Current	On	Off
Tone, Single-Frequency AM System	On	Off
Tone, Dual-Frequency AM System	Lower Frequency	Higher Frequency
Magnetic Tape (see 5.4.1.8.3)		
NRZI (see Note 3 of 5.3.5.2)	Change Polarity	No Change in Polarity
Phase Encoded	(see Note 4 of 5.3.5.2)	(see Note 4 of 5.3.5.2)

5.3.5.3 Electrical characteristics of digital interface. The electrical characteristics for binary signals at the standard interfaces (see Figure 10) shall comply with the applicable requirements of the current edition of MIL-STD-188-114.

NOTE 1: The low level digital interfaces (balanced and unbalanced configurations) contained in subparagraph 4.3.1.3 of MIL-STD-188-100 have been superseded by MIL-STD-188-114.

NOTE 2: The high level digital interface used in some existing facilities is nonstandard and must not be used for the design of new equipment, subsystems and systems. Information about the high level digital interface can be found in MIL-STD-188-100, subparagraph 4.3.1.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.3.5.4 Data signal connections. The connection for the exchange of information between a DTE or a terminal subsystem and a transmission subsystem shall be by serial transfer of bits in a single data channel.

NOTE 1: Timing, control and alarm functions may require additional connections between a DTE or a terminal subsystem and a transmission subsystem.

NOTE 2: Parallel data lines may be used to connect equipment within a terminal subsystem. The information transfer in this parallel connection is serial-by-character, parallel-by-bit.

5.3.5.5 Input/output signal characteristics. Telegraph and data equipment sources shall deliver, and sinks shall accept, unit interval signals.

5.3.5.6 AUTODIN interface characteristics. (MIL-STD-188-100.) Data terminals which interface with AUTODIN shall comply with the applicable interface and control criteria specified in the current edition of DCAC 370-D175-1 and with the applicable operating procedures specified in the current edition of JANAP 128.

5.3.5.7 Data interface characteristics with tactical NATO systems. Telegraph and data channels of US systems shall be interconnected with telegraph and data channels of a tactical system of another NATO member nation in accordance with the applicable requirements of the current edition of STANAG 5040. Tape relay procedures shall comply with the applicable requirements of ACP 127, NATO Supplement 3.

5.3.6 Clock equipment, control and timing. (MIL-STD-188-100.) A clock is the device which provides the time base for controlling the operation of digital equipment. An equipment clock provides for the peculiar needs of its equipment and in some cases may control the flow of data at its equipment interface. A master or station clock, regardless of its physical location, controls two or more equipments which are linked together as a subsystem. 5.3.6.1 through 5.3.6.4.5 are primarily concerned with master or station clocks.

5.3.6.1 Transmission modes of station clocks. (MIL-STD-188-100.) All new equipment, such as DTE, DCE, radios and multiplexers, requiring a stable clock or precise character interval control shall have provisions for operating from station clocks in any or all of the modes specified in 5.3.6.1.1 through 5.3.6.1.4.

NOTE 1: There are no exceptions permitted to the requirement of 5.3.6.1. The reasons for this requirement are:

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

a. The cost of adding an external clock option is minimal if it is done in the original equipment design as compared to the cost of interfacing asynchronous signals to synchronous transmission subsystems, error control devices, multiplexing and switching equipment, etc., that have no external clock option.

b. The extension of communications system timing to the terminal subsystem insures that terminal equipment with a need for bit-count integrity in its traffic will not be exposed to undetected time differential blanks that are encountered in those communications systems wherein isochronous/synchronous interfaces are not maintained.

NOTE 2: An example of the benefits derived from an external clock option is the ability of a DTE to operate with a wide range of data signaling rates without the need to change anything in the DTE. The DTE simply follows the external timing supplied across the interface.

NOTE 3: Questions have arisen in the past as to the specific intent of 5.3.6.1. A synchronous time division multiplexer, modem or DTE, for example, requires a stable clock; therefore, an external clock option is required for these devices in accordance with 5.3.6.1.1. A typical asynchronous transmit terminal requires precise character interval control within each character interval; therefore, the requirements of 5.3.6.1.2 must be met.

5.3.6.1.1 Bit synchronous mode. (MIL-STD-188-100.) In this mode, clock timing shall be delivered at twice the data modulation rate. A device shall release one data bit (see Note 1) within the duration of one clock cycle. Data bits shall be sent only when there is a clock cycle present.

NOTE 1: For the purpose of 5.3.6, data includes all information bits plus all bits added to the bit stream for whatever purpose they may serve in the system, such as error control and framing.

NOTE 2: During periods when the data source has no traffic to send, an idle pattern or all "ones" may be transmitted.

NOTE 3: Bit synchronous mode is a useful mode of operation. In this mode, a DTE transmits a character interval asynchronous format and the transmission is under the control of an external clock signal at twice the data modulation rate. This mode of operation has the advantages of character interval asynchronous operation coupled with the advantages of synchronous transmission and is often referred to as isochronous operation.

NOTE 4: The reason that the clock modulation rate is required at the DTE interface at twice the data modulation rate is to insure that the interface of a send DTE with a receive DTE may be electrically and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

technically as simple as possible. If an external clock signal is supplied at a modulation rate greater than twice the data modulation rate, it is more difficult to assure that the phase relationship between clock and data signals will be correct at the data sink. (See also 5.3.6.3.) An incorrect phase relationship between clock and data signals over a wire or cable connection is normally caused by different propagation delay times of a data signal and an external clock signal that has a much higher modulation rate than the data signal. (See also Note 3 of 5.3.6.2.3.)

5.3.6.1.2 Character interval synchronous mode. (MIL-STD-188-100.) In this mode, any character interval from 4 to 16 unit intervals per character interval shall be permitted.

NOTE 1: It is assumed that, having programmed a given facility for a particular character interval, no other character interval operation would be expected except by reprogramming. An example of such operation would be a 7-unit per character interval tape reader being stepped at 8 units per character interval. (See 5.3.6.1.3.)

NOTE 2: One problem synchronous operation can present is the need for the receive DTE to remain quiescent during periods when no traffic is being transmitted. If a particular bit sequence is chosen to cause the receiver to remain in an idle state, then this bit sequence cannot be used to represent information which is to be passed on to a user.

NOTE 3: Character interval synchronized equipment has an operational advantage over bit synchronous equipment in that the start and stop bits are outside of the information bit sequence and, therefore, the receive DTE is able to distinguish between an all Mark (1's) state on the receive input data line and an all Mark (1's) sequence in the information bits. (For example, an eight-level tape punched in all eight levels.) The start and stop intervals also serve as framing information which may be useful to the receive DTE.

5.3.6.1.3 Character interval control (stepping) mode. In this mode, the character interval of the step signal shall be greater than the free running character interval of the controlled device. (See Note 1.) The release of the controlled signal referenced to its step signal should be consistent for sequential step signals. (See Note 2.) This consistency should be within ±10 percent for electromechanical devices, and ±1 percent for electronic devices, of the duration of the unit interval of the output signal of the controlled device. (See Note 3.) When character interval control (stepping) is employed with electromechanical devices, the receiver shall be prepared to accept the next stop-start transition not more than 0.6 unit intervals into the stop time. (See Note 4.)

NOTE 1: For example, a 7-unit per character interval, 5-unit code, start-stop device would require a step signal with a character interval of at least 7.2 units. As an additional example, devices operating with a step signal of 7 units per character interval require the free running

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Page 75 is missing from the original legacy source document.

either balanced or unbalanced to ground, shall comply with the applicable

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE 2: The time period of 100,000 consecutive seconds, stated in 5.3.6.2.2, does not apply to tactical single channel radio equipment. The time period, expected to be less than 100,000 seconds for this type of equipment, is under consideration.

5.3.6.2.3 Modulation rate phase adjustment. (MIL-STD-188-100.) The following requirements shall apply to all digital transmission, switching, terminal and security equipment that have the means of passing on to another equipment or transmission link a received data signal in a synchronous mode of operation, or a timing signal recovered from a received data signal:

The equipment shall have the capability to shift the phase of the incoming data signal with relation to the clock signal over a period of at least three data unit intervals, i.e., a shift of at least 1.5 data unit intervals early and late from the theoretical center of the data unit interval at the applicable modulation rate. (See Note 1.) The objective of this phase shift is a minimum delay between the clock signal transitions and the resulting data signal transitions, but in no case shall this delay exceed 12.5 percent of the duration of the data unit interval. Once this delay is adjusted, the delay shall be consistent within ±1 percent of itself for each clock transition in accordance with the applicable requirements of 5.3.6.3.

NOTE 1: There may be significant changes in the relative phase of the incoming data signal when a communications link is derived from a satellite subsystem. Cyclic path delays of up to approximately 1200 microseconds over a 12-hour period are not uncommon in satellite links. Therefore, it may be necessary to shift the phase of the incoming data signal with relation to the clock signal significantly more than three data unit intervals, as stated in 5.3.6.2.3, in order to compensate for these cyclic path delay changes in satellite links. Such changes are related to the suborbital motion of the satellite and, therefore, the instantaneous rate of change is related to the orbital position of the satellite within a 12-hour period. The purpose of 5.3.6.2.3 is to insure that phase changes due to path changes are not passed on to associated equipment and propagated through a digital network.

NOTE 2: The purpose of 5.3.6.2.3 is to ensure that any phase change of a data signal, which may occur along a transmission path and is reflected in the recovered timing signal, must not be passed on to the next portion of a transmission subsystem. Therefore, 5.3.6.2.3 does not apply to those DTE (such as a receive printer) that do not have the means of passing on to associated equipment a received data signal or a recovered timing signal. However, those DTE must comply with the applicable requirement(s) of 5.3.6.3. On the other hand, a receive modem or a demultiplexer that recovers the timing signal from the received data signal must provide the means of shifting the phase of the data signal with reference to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

timing signal in accordance with 5.3.6.2.3. The phase of the recovered timing signal must be the phase of the local station clock and not the phase of the incoming data signal.

NOTE 3: At modulation rates higher than 2400 Bd, there may be a significant relative phase shift between clock and data signals which must be adjusted in accordance with 5.3.6.2.3. This relative phase shift may be caused by different path length over which the clock and data signals travel. These paths may consist of wire, cable or equipment interconnections. The magnitude of the relative phase shift between clock and data signals may be significant where either the modulation rate of an external clock signal is much higher than twice the data modulation rate (see also Note 4 of 5.3.6.1.1) or the clock signal modulation rate is twice the data signal modulation rate but the clock signal and the data signal are physically separated and routed through different type or length of connections. However, practical operating experience has shown that the shift in relative phase over typical inside plant telephone cables for two signals differing in their modulation rates by a factor of two is insignificant up to those higher modulation rates where the cable capacitance acts as the limiting factor by reducing the amplitudes of data and clock signals to useless levels. For most practical cases, typical multiple-pair paper or plastic insulated exchange-grade telephone cable can be expected to function at modulation rates up to about 50 kBd and distances up to about 5000 cable feet without concern for relative phase shifts between clock and data signals, noise or insufficiently low signal levels, if the electrical characteristics of the digital interface circuits for both clock and data signals comply with the applicable requirements of the current edition of MIL-STD-188-114. (See 5.3.5.3 and 5.3.6.2.4.)

NOTE 4: Compliance with 5.3.6.2.3 does not require the phase shift to be continuously adjustable over the specified periods of a data unit interval. The required phase shift may be accomplished in incremental steps, continuously, or by a combination of both methods. The decision as to whether this phase adjustment is to be manual, semiautomatic, or fully automatic is left to the equipment designer.

5.3.6.2.4 Clock output signal. (MIL-STD-188-100.) The output signal of a clock shall be an alternating symmetrically-shaped wave at the required clock modulation rate (see 5.3.6.2.1). The clock output signal, either balanced or unbalanced to ground, shall comply with the applicable voltage and waveshaping requirements of the current edition of MIL-STD-188-114. When the clock is quiescent, the clock output signal shall have negative polarity. (See Figure 14.)

5.3.6.2.5 Clock signal period. (MIL-STD-188-100.) A clock signal period or cycle shall consist of one half-cycle of positive polarity (sense) and one half-cycle of negative polarity (sense). The duty cycle shall be 50 percent, ±1 percent. (See Note.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: In the binary sense, each clock signal period or cycle is composed of two clock signal unit intervals, and it follows that a clock signaling rate of 50 Hz is a clock modulation rate of 100 Bd.

5.3.6.3 Phase relationship between clock and data signals. (MIL-STD-188-100.) The following requirements of 5.3.6.3.1 through 5.3.6.3.5, concerning the phase relationship between clock and data signals, shall apply to all digital devices operated at dc baseband which are interconnected by wire or cable or other equipment which provides in effect the same function as a wire or cable. (See Note 3 of 5.3.6.2.3.)

5.3.6.3.1 Direct clock control. (MIL-STD-188-100.) All data signal transitions emitted by a source under direct control of an external clock signal (see Note) shall occur on (be caused by) the negative to positive transitions of that clock signal. The delay between the clock signal transition and the resulting data signal transition should be a minimum, but in no case shall this delay exceed 12.5 percent of the duration of the data unit interval. For each equipment, once this delay is fixed in hardware, the delay shall be consistent within ±1 percent of itself for each clock signal transition. These delay limits shall apply directly at the data source interface. (See Figure 11.)

NOTE: Direct clock control means control of the data signal by a clock signal with a modulation rate of twice the data modulation rate. (See also 5.3.6.4.)

5.3.6.3.2 Indirect clock control. (MIL-STD-188-100.) For indirect clock control (see Note 1), the phase difference between an external clock signal and a data signal shall be consistent within ±1 percent of the data unit interval at the applicable modulation rate, regardless of the absolute value of the phase difference. (See Note 2.) In the case that the clock modulation rate of an external clock is divided, within an equipment, to a clock modulation rate of twice the data modulation rate, and then supplied as the equipment clock output, the data signal transitions shall coincide within ±1 percent of the data unit interval with the negative to positive transitions of the clock output signal. (See Figure 12.)

NOTE 1: Indirect clock control means control of the data signal by a clock signal with a standard modulation rate (see 5.3.6.2.1) greater than twice the data modulation rate.

NOTE 2: The absolute value of the phase difference between an external clock signal and a data signal is not specified for devices in which the external clock signal is related only indirectly to the data signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

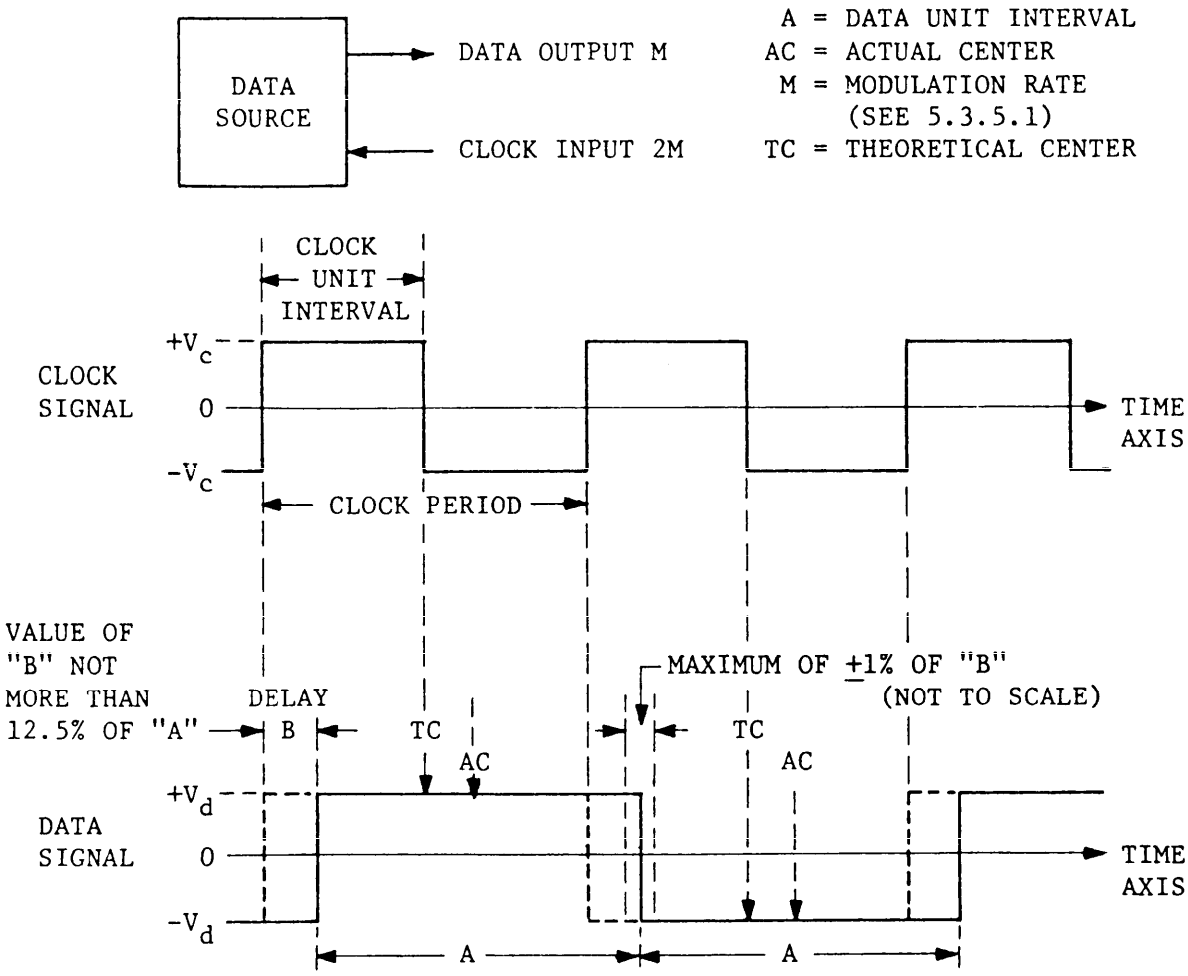


FIGURE 11. Phase relationship between clock and data signals for direct clock control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

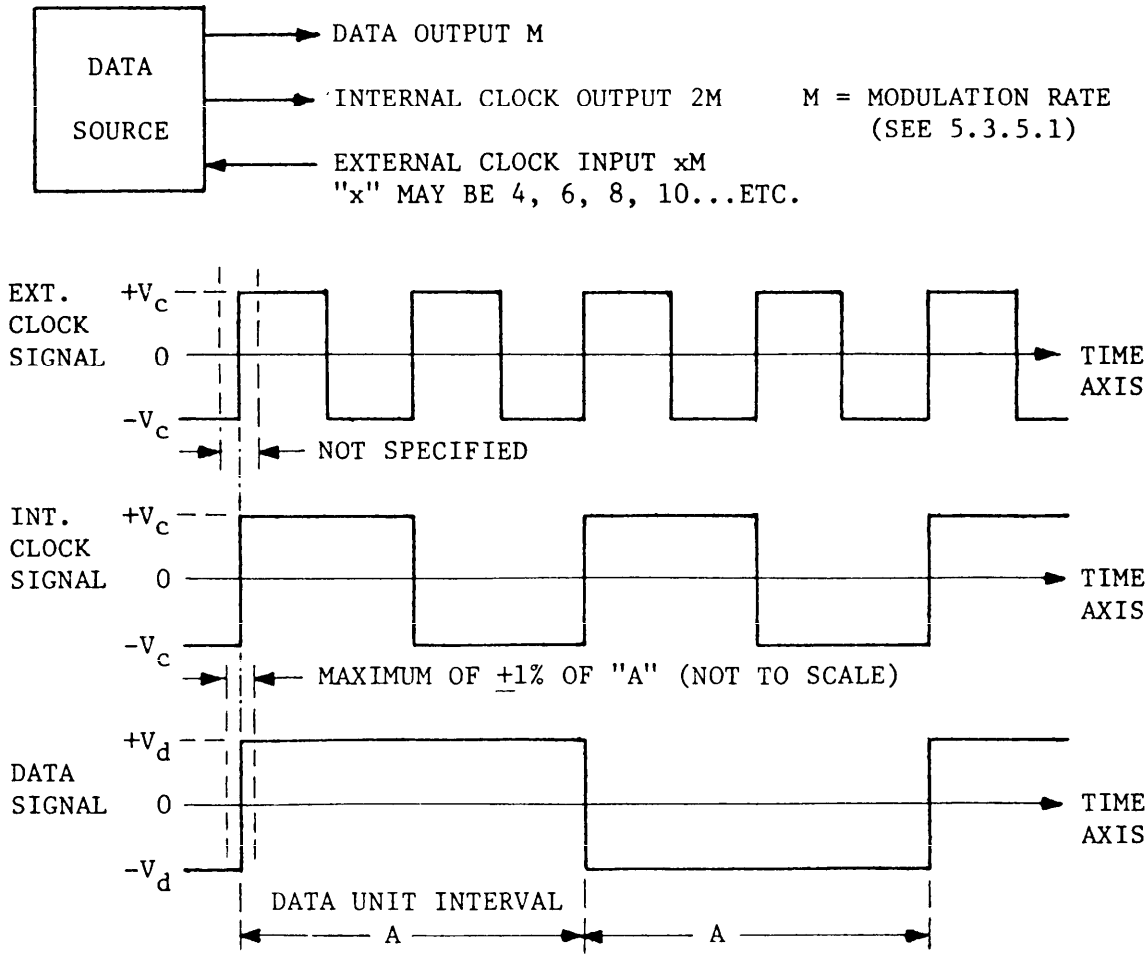


FIGURE 12. Phase relationship between clock and data signals for indirect clock control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.3.6.3.3 Sampling of data signals. (MIL-STD-188-100.) Sampling of data signals at a sink interface under the control of an external clock signal shall occur on (be caused by) positive to negative clock signal transitions. (See Figure 13.) For devices in which input data to a sink are sampled under the control of a clock signal with a modulation rate that is not directly related to the data modulation rate (see 5.3.6.4), the phase relationship of the data signal to the clock signal shall be maintained such that each data unit interval shall be sampled within ±1 percent of the theoretical center of the data unit interval. For start-stop devices utilizing internal, low stability sampling sources, incoming data signals shall be sampled within ±12.5 percent of the data unit interval, measured at the actual center of the data unit interval. (See Figure 11.) Any distortion caused by this sampling method shall not be passed on to an output interface. The data signal shall be regenerated before it is retransmitted.

5.3.6.3.4 Intermittent data transmission. (MIL-STD-188-100.) When a gated clock signal is used for controlling intermittent data transmission, the data signal shall not change state except when requested by a negative to positive clock signal transition. The quiescent state of the clock signal shall be at a negative voltage. The quiescent state of the data signal shall be that state resulting from the last negative to positive clock signal transition. (See Figure 14.)

5.3.6.3.5 Start-stop signals. (MIL-STD-188-100.) For synchronous devices transmitting start-stop signals and operating from an external clock signal with a modulation rate that is directly related to the data modulation rate (see 5.3.6.4), data signal transitions shall be coincident with negative to positive clock signal transitions within ±1 percent of the data unit interval. (See Note 1.) For start-stop devices utilizing internal, low stability sampling sources, incoming data signals shall be sampled within ±12.5 percent of the data unit interval, measured at the actual center of the data unit interval. (See Figure 11.) Any distortion caused by this sampling method (see Note 2) shall not be passed on to an output interface. The data signal shall be regenerated before it is retransmitted.

NOTE 1: This mode is also referred to as isochronous operation.

NOTE 2: This mode is also referred to as asynchronous operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

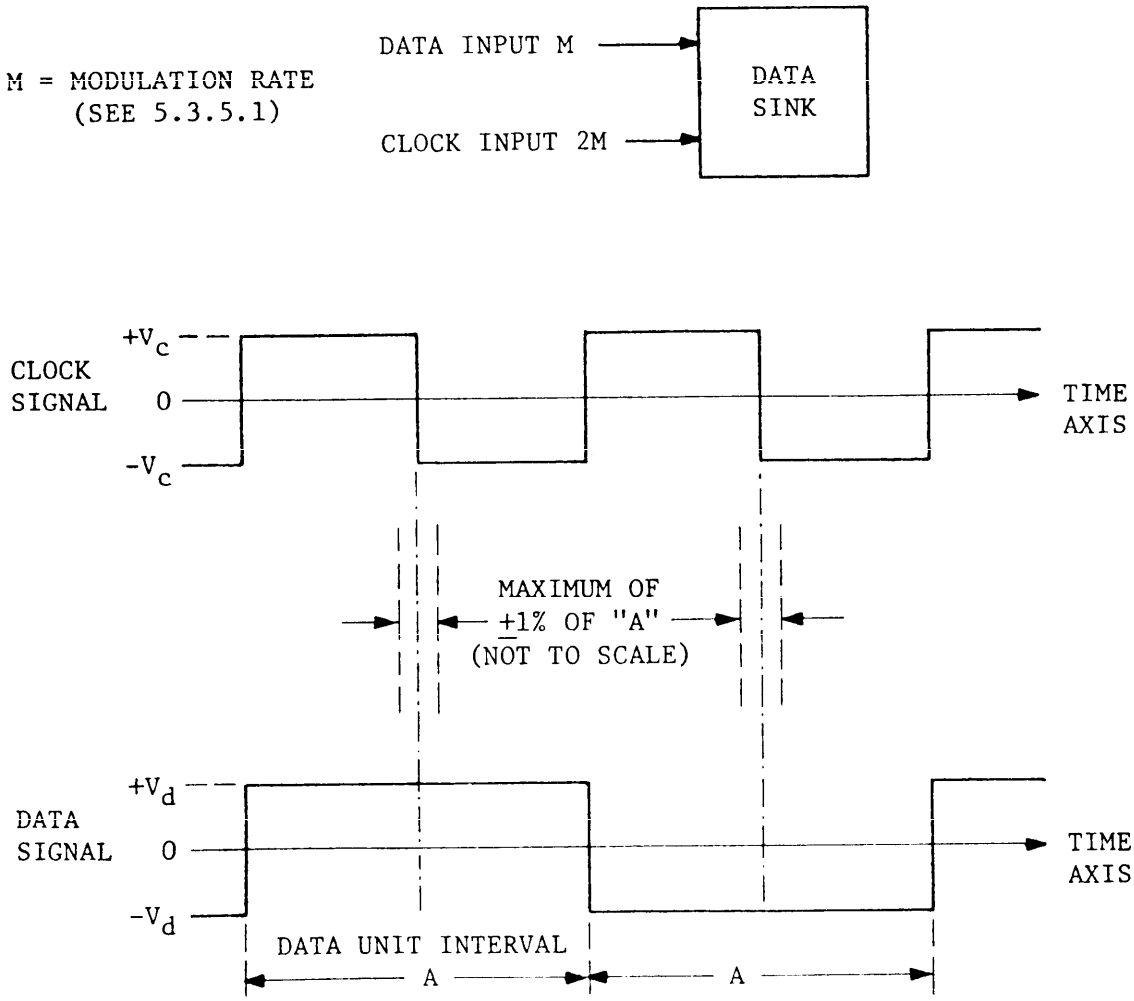
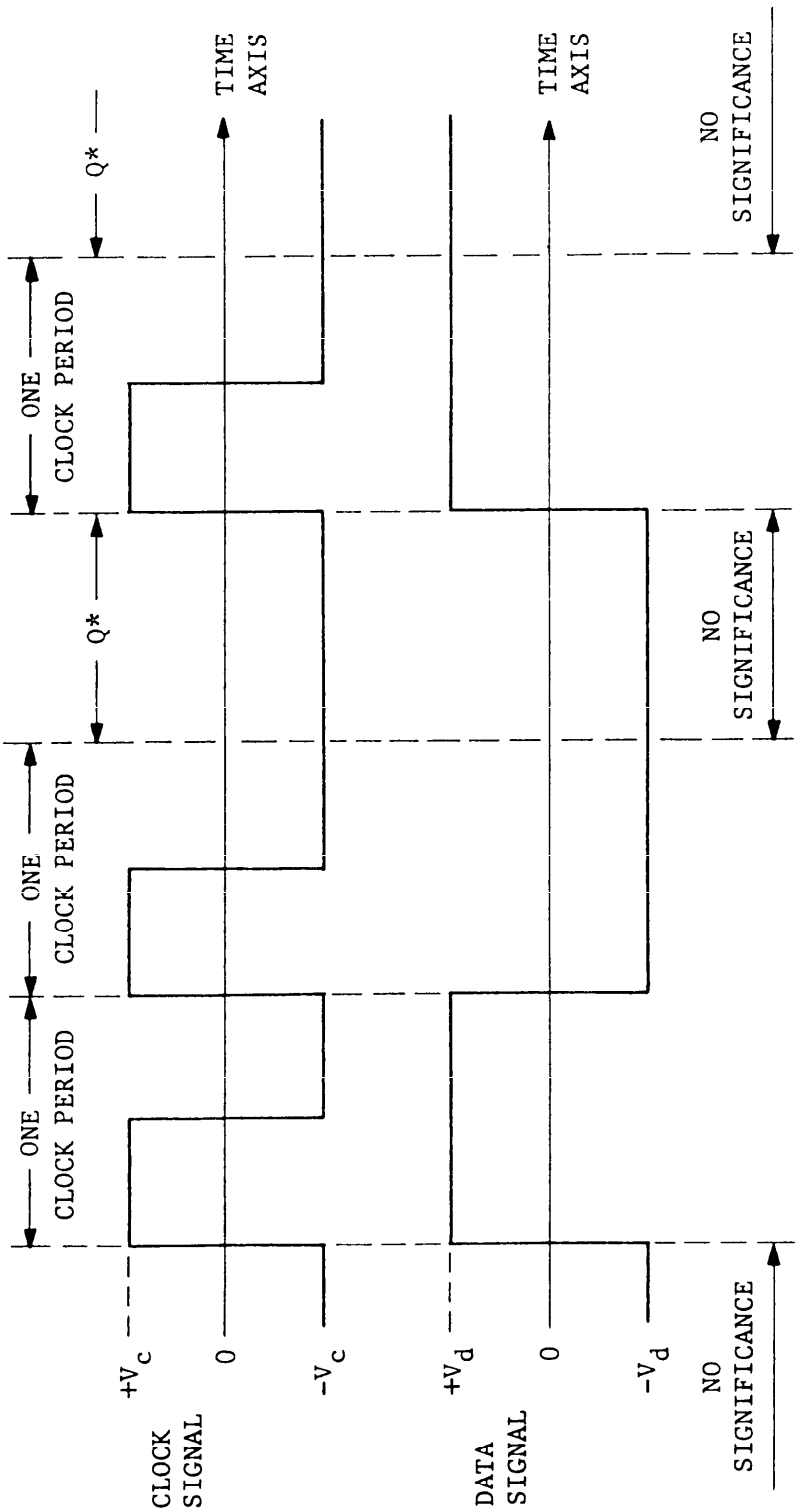


FIGURE 13. Phase relationship between clock and data signals for sampling of data (direct clock control).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



*Q is the quiescent state of the clock. (See 5.3.6.2.4.)

FIGURE 14. Phase relationship between clock and data signals for intermittent data transmission (gated clock).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.3.6.4 Standard arrangements for supplying clock signals. (MIL-STD-188-100.) There are five typical cases listed in 5.3.6.4.1 through 5.3.6.4.5 showing arrangements that shall be used for supplying clock signals to data sources and data sinks. One or more cases shall be selected to meet a specific application.

The following definitions shall apply:

a. Data modulation rate = M.

b. Clock modulation rate = 2M. The clock modulation rate is directly related to the data modulation rate. The 2M clock signal shall not be permitted to change state except at negative to positive transitions (see Figure 11) of a continuous square wave of the same modulation rate as the data modulation rate in order to preclude data intervals occurring at other than integral multiples of the data unit interval.

c. Clock modulation rate = xM, where x equals an even integer greater than 2. The clock modulation rate is indirectly related to the data modulation rate. An external clock signal, supplied at a modulation rate greater than twice the data modulation rate and divided within the equipment to provide a data modulation rate of M (see Figure 15), would be defined as some number (x) multiplied by the data modulation rate; e.g., 128M designates that the modulation rate of the external clock signal is 128 times the data modulation rate.

5.3.6.4.1 Case one. (MIL-STD-188-100.) (See Figure 15.) The source-to-sink data modulation rate shall be determined by external clock signal(s). (See Note.) For collocated source and sink driven by the same external clock signal at rate xM, x may equal 2. In this case, the clock modulation rate is directly related to the data modulation rate, and center sampling, within ±1 percent of the data unit interval, shall be provided. (See Figure 13.) For collocated source and sink driven by different clock signals, or driven by the same clock signal when x is an even integer greater than 2, or for electrically separated source and sink, phasing techniques (see 5.3.6.2.3) shall be employed which permit an initial adjustment of the sample point to within ±1 percent of the theoretical data unit interval measured from the center of the data unit intervals arriving at the sink.

NOTE: In some cases the external clock signal is supplied to the source and sink and in other cases there may be two separate external clock signals.

5.3.6.4.2 Case two. (MIL-STD-188-100.) (See Figure 16.) An external clock signal with a modulation rate of xM shall be supplied to the source, where x may be 2 or may be an even integer greater than 2. For x being

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

greater than 2, the 2M clock signal shall be divided down to 2M (see Note 1) to serve as a directly related clock signal that determines the data modulation rate. The 2M clock signal which may be gated (see Note 2) or nongated, is released along with the M data signal, and the clock signal shall act as an instruction to the sink that the data lead contains data that are to be sensed. The phase relationship between the data signal M and the clock signal 2M shall be maintained within ±1 percent at the source, regardless of whether a gated or a nongated clock signal is furnished.

NOTE 1: There is no specific phase relationship between xM and 2M or between xM and M defined or implied. There is a specific modulation rate relationship which is determined by the ratio xM/2M or xM/M.

NOTE 2: The method of gating the 2M clock signal is not standardized and will be determined in the applicable equipment specification.

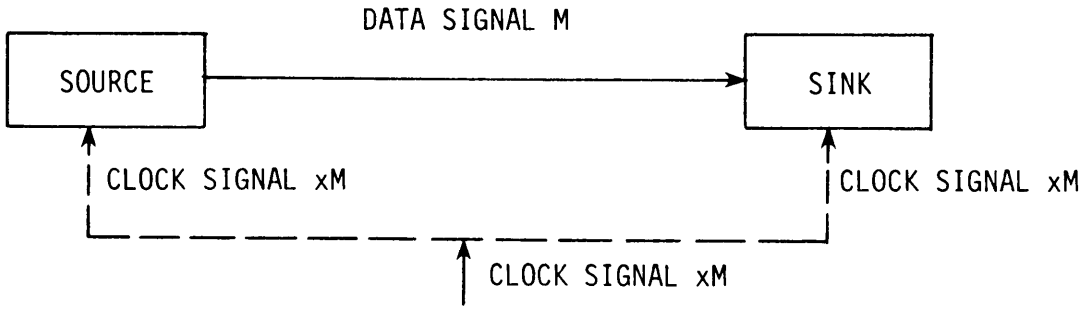


FIGURE 15. Standard arrangement case 1 for supplying clock signals.

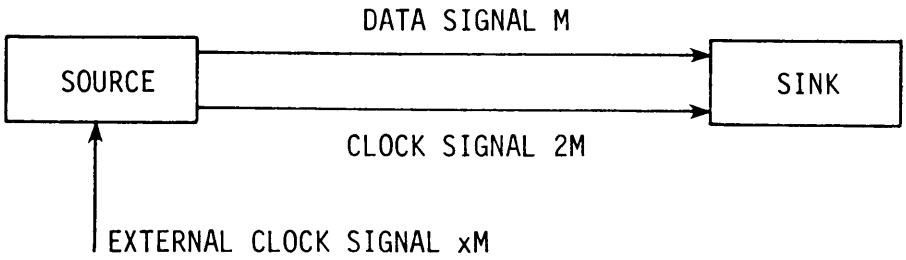


FIGURE 16. Standard arrangement case 2 for supplying clock signals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.3.6.4.3 Case three. (MIL-STD-188-100.) (See Figure 17.) Case three is similar to case two, except an external clock signal with a modulation rate of xM shall be supplied to the sink, where x may be 2 or may be an even integer greater than 2. For x being greater than 2, the xM clock signal shall be divided down to 2M (see Note 1 of 5.3.6.4.2) to serve as a directly related clock signal that determines the data modulation rate. The 2M clock signal, which may be gated (see Note 2 of 5.3.6.4.2) or nongated, shall act as an instruction to the source to release data. The phase relationship of M to 2M shall be such that data transitions shall occur only after negative to positive transitions of the 2M clock signal. The data line shall not again change state until the next negative to positive transition at the 2M clock lead occurs. The phase relationship between the data signal M and the clock signal 2M shall be maintained within ±1 percent at the source, regardless of whether a gated or a nongated clock signal is furnished.

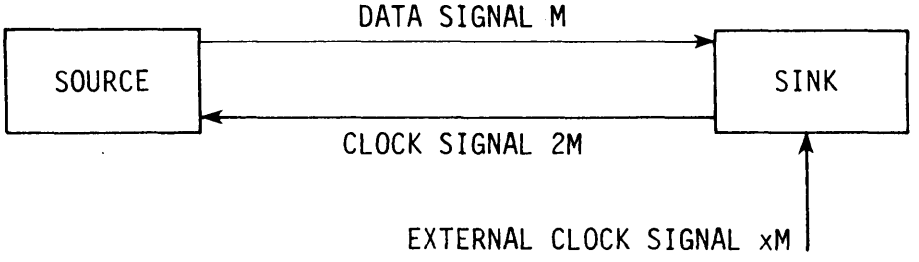


FIGURE 17. Standard arrangement case 3 for supplying clock signals.

5.3.6.4.4 Case four. (MIL-STD-188-100.) (See Figure 18.) An internal clock signal with a modulation rate of 2M shall be supplied from the source to the sink. The data modulation rate shall be completely under the control of the source. The state of the clock lead shall determine the sensing of the data lead at the sink. The 2M clock signal shall request (cause) changes in the state of the data only on negative to positive transitions of the clock signal. The receiver shall determine the state of the data lead only on the positive to negative transitions of the 2M clock signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

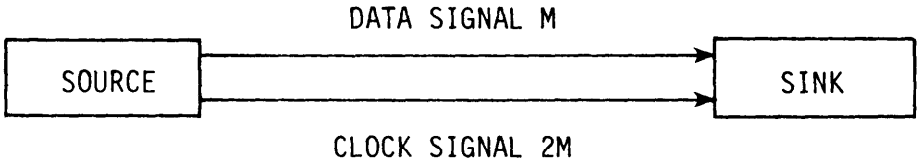


FIGURE 18. Standard arrangement case 4 for supplying clock signals.

5.3.6.4.5 Case five. (MIL-STD-188-100.) For sinks operating with start-stop signals and utilizing internal, low stability sampling sources or an external clock signal with a modulation rate of xM, incoming data shall be sampled within ±12.5 percent of the data unit interval, measured at the actual center of the data unit intervals (see Figure 11) arriving at the sink. Any distortion caused by this sampling method shall not be passed on to an output interface. The data signal shall be regenerated before it is retransmitted.

5.3.7 Codes for information interchange. (MIL-STD-188-100.) The code for information interchange shall be the USA Standard Code for Information Interchange (USASCII or ASCII) in accordance with the current edition of FIPS PUB 1-1.

NOTE: FIPS PUB 1-1 has adopted ANSI X3.4-1977.

5.3.7.1 Scope and application. (MIL-STD-188-100.) ASCII defines a basic or standard set of 128 characters (see 5.3.7.4), three subsets (see 5.3.7.7) and code extension techniques (see 5.3.7.8) that shall apply to teleprinters and other character conscious data equipment for the interchange of common user information among and between communications systems, information processing systems and associated equipment. ASCII shall not apply to such established specialized functions as telephone signaling and supervision, and A-D conversion techniques, such as PCM and CVSD modulation. ASCII shall not necessarily apply to equipment or subsystems (such as central processors and peripheral ADP equipment) used substantially full time as part of the control element in a system (such as a weapon control system) where that system is not itself primarily concerned with common user information interchange. The implementation and applicability of ASCII and related matters, such as waiver procedures, shall comply with the applicable requirements of the current edition of FIPS PUB 7. (See Note.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: The requirements of 5.3.7 and 5.3.7.1 are intended to mean that all character conscious data equipment for the interchange of common user information, such as teleprinters, must have the capability to use the basic ASCII character set (see 5.3.7.4) or one or more of the ASCII subsets (see 5.3.7.7). In addition to the ASCII capabilities, character conscious data equipment may also have the capability to use any other code, such as the five-unit code of ITA No. 2 (see 80.3 of Appendix H) if required for interoperability with older equipment still in the inventory; or the capability to use any other alphabet if required for interoperability with equipment of allied forces in foreign countries. The requirement for interoperating with NATO military and civil organizations is stated in 5.3.7.9.

5.3.7.2 Character substitutions. Character substitutions in the ASCII character sets, including subsets and code extensions with 7-bit or 8-bit code structures, shall not be permissible except by specific approval as described in Section 9 of FIPS PUB 7 and other applicable FIPS Publications.

5.3.7.3 Character parity. (MIL-STD-188-100.) This standard does not specify that a character parity bit shall or shall not be transmitted. As a Design Objective, all data equipment using ASCII should be designed to transmit or receive (or transmit and receive) a bit in the parity position to preclude the development of several types of equipment that are incompatible and limited in its use. If the bit in the parity position is not used as a parity bit, it should be transmitted as a Mark and ignored by the receiver. If the bit in the parity position is used as a parity bit, the transmitter and receiver shall be programmed or adjusted accordingly and the parity sense shall comply with the applicable requirements of 5.3.7.5.4 and 5.3.7.5.5. Any equipment designed not to transmit or receive parity bits shall be used only in limited and closed networks where parity is not needed and transmission efficiency is paramount over compatibility considerations.

5.3.7.4 Basic ASCII character set. (MIL-STD-188-100.) The basic or standard ASCII character set shall comply with the current edition of FIPS PUB 1 (see Note of 5.3.7) and shall consist of a 7-bit coded character set comprised of 128 characters as shown in Table IX.

NOTE: The basic or standard ASCII character set defines a set of 128 characters, consisting of 95 graphic characters (upper and lower case alphabet, decimal digits, space, punctuation and special symbols), 32 characters used for control functions and a "Delete" (DEL) character. The control characters listed in Table IX are described in 80.4.1 of Appendix H.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE IX. Basic ASCII character set.

b_7 b_6 b_5 → / Bits b_4 b_3 b_2 b_1	0 0 0	0 0 1	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
COLUMN → / ROW ↓	0	1	2	3	4	5	6	7
0 0 0 0	0	NUL	DLE	SP	0	@	P	`	p
0 0 0 1	1	SOH	DC1	!	1	A	Q	a	q
0 0 1 0	2	STX	DC2	"	2	B	R	b	r
0 0 1 1	3	ETX	DC3	#	3	C	S	c	s
0 1 0 0	4	EOT	DC4	$	4	D	T	d	t
0 1 0 1	5	ENQ	NAK	%	5	E	U	e	u
0 1 1 0	6	ACK	SYN	&	6	F	V	f	v
0 1 1 1	7	BEL	ETB	'	7	G	W	g	w
1 0 0 0	8	BS	CAN	(	8	H	X	h	x
1 0 0 1	9	HT	EM	)	9	I	Y	i	y
1 0 1 0	10	LF	SUB	*	:	J	Z	j	z
1 0 1 1	11	VT	ESC	+	;	K	[	k	{
1 1 0 0	12	FF	FS	,	<	L	\	l	|
1 1 0 1	13	CR	GS	-	=	M	]	m	}
1 1 1 0	14	SO	RS	.	>	N	^	n	~
1 1 1 1	15	SI	US	/	?	O	_	o	DEL

NOTE 1: Reprinted from American National Standard Code for Information Interchange, X3.4-1977, with permission of the copyright holder, the American National Standards Institute.

NOTE 2: See Tables X and XI for the legend.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T07:28Z
Check the source to verify that this is the current version before use.

5.3.7.4.1 Character representation. (MIL-STD-188-100.) The representation of a 7-bit ASCII character shall comply with the applicable requirements of the current edition of FIPS PUB 1 (see Note of 5.3.7) and shall be with b_7 as the high-order bit and b_1 as the low-order bit, as shown in the following example for the character "K" (see Table IX):

b_7	b_6	b_5	b_4	b_3	b_2	b_1
1	0	0	1	0	1	1

NOTE: The code table position for the character "K" may also be represented by the notation "column 4, row 11" or alternatively as "4/11." The decimal equivalent of the binary number formed by bits b_7, b_6 and b_5, collectively, forms the column number, and the decimal equivalent of the binary number formed by bits b_4, b_3, b_2 and b_1, collectively, forms the row number.

5.3.7.4.2 Control characters. (MIL-STD-188-100.) ASCII control characters shall comply with the applicable requirements of the current edition of FIPS PUB 1 (see Note of 5.3.7) and shall be as shown in Table X.

5.3.7.4.3 Graphic representation of control characters. The graphic representation of the ASCII control characters shall comply with the applicable requirements of the current edition of FIPS PUB 36.

NOTE: FIPS PUB 36 has adopted ANSI X3.32-1973, which specifies graphical representation for the 32 control functions of columns 0 and 1 (see Table IX), as well as the normally nonprinting characters Space and Delete. Two forms of graphical representation for each of the 34 characters are provided: A pictorial symbol and a 2-letter alphanumeric code. These representations are intended for use in the display of control characters on devices, such as paper tape punches, diagnostic printers and cathode-ray tubes, where the graphic representation of these normally nonprinting characters is required.

5.3.7.4.4 Graphic characters. (MIL-STD-188-100.) ASCII graphic characters shall comply with the applicable requirements of the current edition of FIPS PUB 1 (see Note of 5.3.7) and shall be as shown in Table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE X. Control characters of basic ASCII character set.

Column/ Row	Symbol	Name	Type (see Note)	Column/ Row	Symbol	Name	Type (see Note)
0/0	NUL	Null		1/2	DC2	Device Control 2	
0/1	SOH	Start of Heading	CC	1/3	DC3	Device Control 3	
0/2	STX	Start of Text	CC	1/4	DC4	Device Control 4 (Stop)	
0/3	ETX	End of Text	CC				
0/4	EOT	End of Transmission	CC	1/5	NAK	Negative Acknowledge	CC
0/5	ENQ	Enquiry	CC				
0/6	ACK	Acknowledge	CC	1/6	SYN	Synchronous Idle	CC
0/7	BEL	Bell (audible or attention signal)		1/7	ETB	End of Transmission Block	CC
0/8	BS	Backspace	FE	1/8	CAN	Cancel	
0/9	HT	Horizontal Tabulation (punched card skip)	FE	1/9	EM	End of Medium	
0/10	LF	Line Feed	FE	1/10	SUB	Substitute	
0/11	VT	Vertical Tabulation	FE	1/11	ESC	Escape	
0/12	FF	Form Feed	FE	1/12	FS	File Separator	IS
0/13	CR	Carriage Return	FE	1/13	GS	Group Separator	IS
0/14	SO	Shift Out		1/14	RS	Record Separator	IS
0/15	SI	Shift In		1/15	US	Unit Separator	IS
1/0	DLE	Data Link Escape	CC	7/15	DEL	Delete (in the strict sense, DEL is not a control character)	
1/1	DC1	Device Control 1					

LEGEND: CC = Communication Control
FE = Format Effector
IS = Information Separator

Note: For a description of these control characters see 80.4.1 of Appendix H.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XI. Graphic characters of basic ASCII character set.

Column/ Row	Symbol	Name	Column/ Row	Symbol	Name
2/0	SP	Space (Normally Non-Printing)	3/11	;	Semicolon
2/1	!	Exclamation Point	3/12	<	Less Than
2/2	"	Quotation Marks (Diaeresis, see Note 1)	3/13	=	Equals
			3/14	>	Greater Than
2/3	#	Number Sign (see Notes 2 and 3)	3/15	?	Question Mark
			4/0	@	Commercial At (see Note 2)
2/4	$	Dollar Sign	5/11	[	Opening Bracket (see Note 2)
2/5	%	Percent	5/12	\	Reverse Slant (see Note 2)
2/6	&	Ampersand	5/13	]	Closing Bracket (see Note 2)
2/7	'	Apostrophe (Closing Single Quotation Mark; Acute Accent; see Note 1)	5/14	^	Circumflex (see Notes 1 and 2)
			5/15	_	Underline
			6/0	`	Grave Accent (see Notes 1 and 2) (Opening Single Quotation Mark)
2/8	(	Opening Parenthesis			
2/9	)	Closing Parenthesis	7/11	{	Opening Brace (see Note 2)
2/10	*	Asterisk	7/12	|	Vertical Line (see Notes 2 and 4)
2/11	+	Plus			
2/12	,	Comma (Cedilla, see Note 1)	7/13	}	Closing Brace (see Note 2)
2/13	-	Hyphen (Minus)	7/14	~	Overline (see Note 2) (Tilde; General Accent; see Note 1)
2/14	.	Period (Decimal Point)			
2/15	/	Slant			
3/10	:	Colon			

Note 1: The use of the symbols in 2/2, 2/7, 2/12, 5/14, 6/0 and 7/14 as diacritical marks is described in Appendix A, paragraph A5.2 of ANSI X3.4-1977.

Note 2: These characters should not be used in international interchange without determining that there is agreement between sender and receiver. (See 5.3.7.9 and Appendix B5 of ANSI X3.4-1977.)

Note 3: In applications where there is no requirement for the symbol #, the symbol £ may be used in position 2/3.

Note 4: The Vertical Line character is a solid line extended so that a continuous vertical line can be printed by using the appropriate control characters.

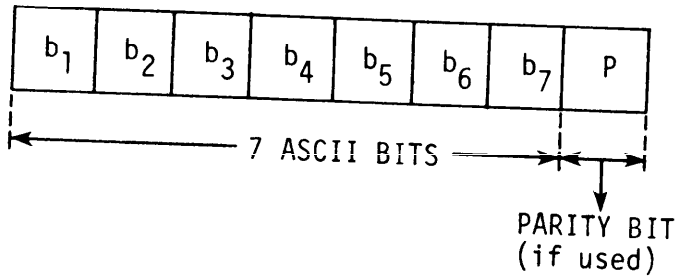
Source: https://assist.dla.mil -- Downloaded: 2026/10-26T07:28Z
Check the source to verify that this is the current version before use.

5.3.7.5 Serial-by-bit data communications. (See 5.3.5.4.)

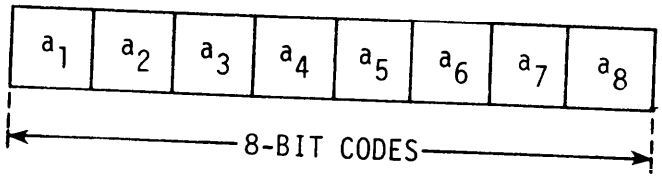
5.3.7.5.1 Bit sequencing. In serial-by-bit, serial-by-character data communications, the bit sequence for an ASCII character shall comply with the applicable requirements of the current edition of FIPS PUB 16 (see Note) and shall be as follows:

For the 7-bit basic ASCII character set (see Table IX), the bit sequence shall be least significant bit first to most significant bit last, b_1 through b_7 (see Figure 19A). For the 8-bit expanded ASCII character set (see 5.3.7.8), the bit sequence shall be least significant bit first to most significant bit last, a_1 through a_8 (see Figure 19B).

NOTE: FIPS PUB 16-1 has also been identified as FED-STD-1010. Both Federal standards have adopted ANSI X3.15-1976.



A. 7-Bit Basic ASCII.



B. 8-Bit Expanded ASCII.

NOTE: The order of transmission is from left to right.

FIGURE 19. ASCII character structure for synchronous serial-by-bit data communications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.3.7.5.2 Character structure for synchronous data communications. In synchronous serial-by-bit, serial-by-character data communications, the character structure shall comply with the applicable requirements of the current edition of FIPS PUB 17 (see Note) and shall be as follows:

As shown in Figure 19A, the character structure of the 7-bit basic ASCII character set (see Table IX) shall consist of seven ASCII bits, b_1 through b_7, plus one character parity bit P (see 5.3.7.3). As shown in Figure 19B, the character structure of the 8-bit expanded ASCII character set (see 5.3.7.8) shall consist of eight bits, a_1 through a_8. All bits shall have equal time intervals.

NOTE: FIPS PUB 17-1 has also been identified as FED-STD-1011. Both Federal standards have adopted ANSI X3.16-1976.

5.3.7.5.3 Character structure for asynchronous data communications. In asynchronous serial-by-bit, serial-by-character data communications, the character structure shall comply with the applicable requirements of the current edition of FIPS PUB 17 (see Note of 5.3.7.5.2) and shall be as follows:

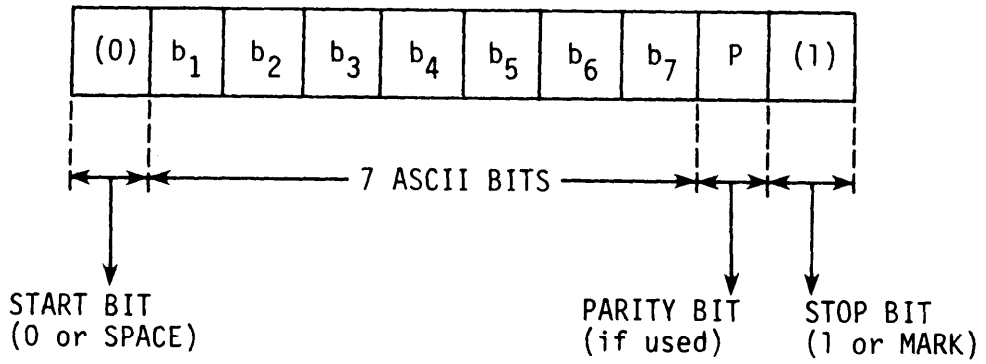
As shown in Figure 20A, the character structure of the 7-bit basic ASCII character set (see Table IX) shall consist of one Space (0) start element, seven ASCII bits b_1 through b_7, one character parity bit P (see 5.3.7.3) and one Mark (1) stop element. As shown in Figure 20B, the character structure of the 8-bit expanded ASCII character set (see 5.3.7.8) shall consist of one Space (0) start element, eight bits a_1 through a_8 and one Mark (1) stop element. All signal elements shall have equal time intervals. (See Note 1.) The intercharacter interval (see Note 2) shall have the same sense as the stop element, but may be of any length.

NOTE 1: Some configurations of communication facilities cannot operate satisfactorily with the stop element as specified in 5.3.7.5.3. Where this is the case, a stop element consisting of two time intervals will be necessary. This exception to the character structure is intended to provide relief where character regenerators are employed (as in long haul, multi-station networks). The use of a stop element consisting of two time intervals will require prior agreement between sender and receiver.

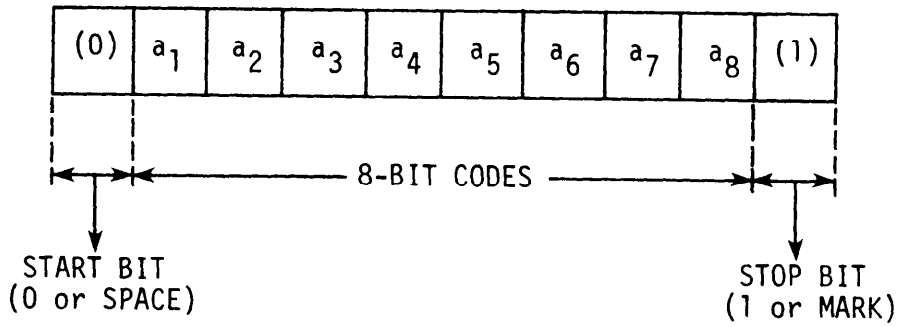
NOTE 2: The term intercharacter interval is defined in FED-STD-1037.

NOTE 3: Receiving equipment must be capable of operating with any intercharacter interval. In the case of transmission with a single-unit stop element, the receiving equipment must be capable of responding in less than one unit time interval in accordance with 5.3.6.1.3.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the currentversion before use.



A. 7-Bit Basic ASCII.



B. 8-Bit Expanded ASCII.

NOTE: The order of transmission is from left to right.

FIGURE 20. ASCII character structure for asynchronous serial-by-bit data communications.

5.3.7.5.4 Parity sense for synchronous data communications. In synchronous serial-by-bit, serial-by-character data communications, the sense of the character parity of the 7-bit basic ASCII character set (see Table IX) shall comply with the applicable requirements of the current edition of FIPS PUB 17 (see Note of 5.3.7.5.2). The sense of character parity

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

shall be odd over the eight bits (seven ASCII bits and one character parity bit); that is, an odd number of Mark (1) bits per character.

NOTE 1: There is no parity bit in the 8-bit expanded ASCII character set.

NOTE 2: The use of character parity is not required. (See 5.3.7.3.)

NOTE 3: AUTODIN does not comply with FIPS PUB 17. In AUTODIN, odd parity is used for transmitting message characters in both the synchronous and asynchronous modes. Even parity is used for transmitting control characters for channel control and coordination in the synchronous mode.

5.3.7.5.5 Parity sense for asynchronous data communications. In asynchronous serial-by-bit, serial-by-character data communications, the sense of the character parity of the 7-bit basic ASCII character set (see Table IX) shall comply with the applicable requirements of the current edition of FIPS PUB 17 (see Note of 5.3.7.5.2). The sense of character parity shall be even over the eight bits (seven ASCII bits and one character parity bit); that is, an even number of Mark (1) bits per character. (See Notes 1, 2 and 3 of 5.3.7.5.4.)

5.3.7.6 Parallel-by-bit data communications. (See 5.3.5.4.)

5.3.7.6.1 Bit-to-channel relationship. In parallel-by-bit, serial-by-character data communications, the bit-to-channel relationship for the 7-bit basic ASCII character set (see Table IX) and the 8-bit expanded ASCII character set (see 5.3.7.8) shall comply with the applicable requirements of the current edition of FIPS PUB 18 (see Note) and shall be as shown in Table XII.

NOTE: FIPS PUB 18-1 has also been identified as FED-STD-1012. Both Federal standards have adopted ANSI X3.25-1976.

TABLE XII. ASCII bit-to-channel relationship.

Channel	1	2	3	4	5	6	7	8
7-Bit Basic ASCII	b_1	b_2	b_3	b_4	b_5	b_6	b_7	P
8-Bit Expanded ASCII	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.3.7.6.2 Character structure. In parallel-by-bit, serial-by-character data communications, the character structure shall comply with the applicable requirements of the current edition of FIPS PUB 18 (see Note of 5.3.7.6.1) and shall be as follows:

For the 7-bit basic ASCII character set (see Table IX), the character structure shall consist of seven ASCII bits, b_1 through b_7, plus one character parity bit P (see 5.3.7.3). For the 8-bit expanded ASCII character set (see 5.3.7.8), the character structure shall consist of eight bits, a_1 through a_8.

5.3.7.6.3 Character parity sense. In parallel-by-bit, serial-by-character data communications, the sense of the character parity of the 7-bit basic ASCII character set (see Table IX) shall comply with the applicable requirements of the current edition of FIPS PUB 18 (see Note of 5.3.7.6.1) and shall be as follows:

For a transmission system where the character timing is not separately signaled, the sense of the character parity shall be odd; that is, the parity bit for each character shall be such that there are an odd number of Mark (1) bits in the 8-bit character. For a transmission system where the character timing information is provided by means of a separate timing channel, the sense of the character parity shall be even; that is, the parity bit for each character shall be such that there are an even number of Mark (1) bits in the 8-bit character.

NOTE 1: There is no parity bit in the 8-bit expanded ASCII character set.

NOTE 2: The use of character parity is not required. (See 5.3.7.3.)

NOTE 3: AUTODIN does not comply with FIPS PUB 18. (See Note 3 of 5.3.7.5.4.)

5.3.7.7 ASCII subsets. (MIL-STD-188-100.) Any ASCII subset shall comply with the applicable requirements of the current edition of FIPS PUB 15. If the full 128-character ASCII set (see Table IX) cannot be used, then the largest possible subset specified in 5.3.7.7.1 through 5.3.7.7.3 shall be used and the ASCII collating sequence shall be observed.

NOTE: Three ASCII subsets have been standardized in FIPS PUB 15: the 95-character graphic subset, the 64-character graphic subset and the 16-character graphic numeric subset.

5.3.7.7.1 95-Character ASCII graphic subset. (MIL-STD-188-100.) This subset shall comply with Section 1 of FIPS PUB 15 and shall consist of all characters in columns 2, 3, 4, 5, 6 and 7 of the basic ASCII character set, except the character Delete (DEL) in position 7/15. (See Table XIII.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XIII. 95-Character ASCII graphic subset.

Bits b7 b6 b5 / b4 b3 b2 b1	COLUMN / ROW	0 0 0	0 0 1	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
		0	1	2	3	4	5	6	7
0 0 0 0	0			SP	0	@	P	`	p
0 0 0 1	1			!	1	A	Q	a	q
0 0 1 0	2			"	2	B	R	b	r
0 0 1 1	3			#	3	C	S	c	s
0 1 0 0	4			$	4	D	T	d	t
0 1 0 1	5			%	5	E	U	e	u
0 1 1 0	6			&	6	F	V	f	v
0 1 1 1	7			'	7	G	W	g	w
1 0 0 0	8			(	8	H	X	h	x
1 0 0 1	9			)	9	I	Y	i	y
1 0 1 0	10			*	:	J	Z	j	z
1 0 1 1	11			+	;	K	[	k	{
1 1 0 0	12			,	<	L	\	l	|
1 1 0 1	13			-	=	M	]	m	}
1 1 1 0	14			.	>	N	^	n	~
1 1 1 1	15			/	?	O	_	o	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: The 95-character ASCII graphic subset is derived from the 128-character basic ASCII set (see Table IX) and is intended to be used in those systems or applications whose needs are adequately met by a 95-character set.

5.3.7.7.2 64-Character ASCII graphic subset. (MIL-STD-188-100.) This subset shall comply with Section 2 of FIPS PUB 15 and shall consist of all characters in columns 2, 3, 4 and 5 of the basic ASCII character set. (See Table XIV.)

NOTE: The 64-character ASCII graphic subset is derived from the 128-character basic ASCII set (see Table IX) and is intended to be used in those systems or applications whose needs are adequately met by a 64-character set.

5.3.7.7.3 16-Character ASCII graphic numeric subset. (MIL-STD-188-100.) This subset shall comply with Section 3 of FIPS PUB 15 and shall consist of the ten numerals from the top ten positions of column 3 and six symbols (asterisk, plus, comma, hyphen (minus), period (decimal point) and slant) from the bottom six positions of column 2 of the basic ASCII character set. (See Table XV.)

NOTE: The 16-character ASCII graphic numeric subset is derived from the 128-character basic ASCII set (see Table IX) and is intended to be used in those systems or applications whose needs are adequately met by a 16-character set. Systems and applications employing this limited 16-character set, when receiving information via standard media (such as perforated tape or magnetic tape) or communications from systems employing the full 128-character set, the 95-character subset or the 64-character subset, should ignore all characters outside of the 16-character subset.

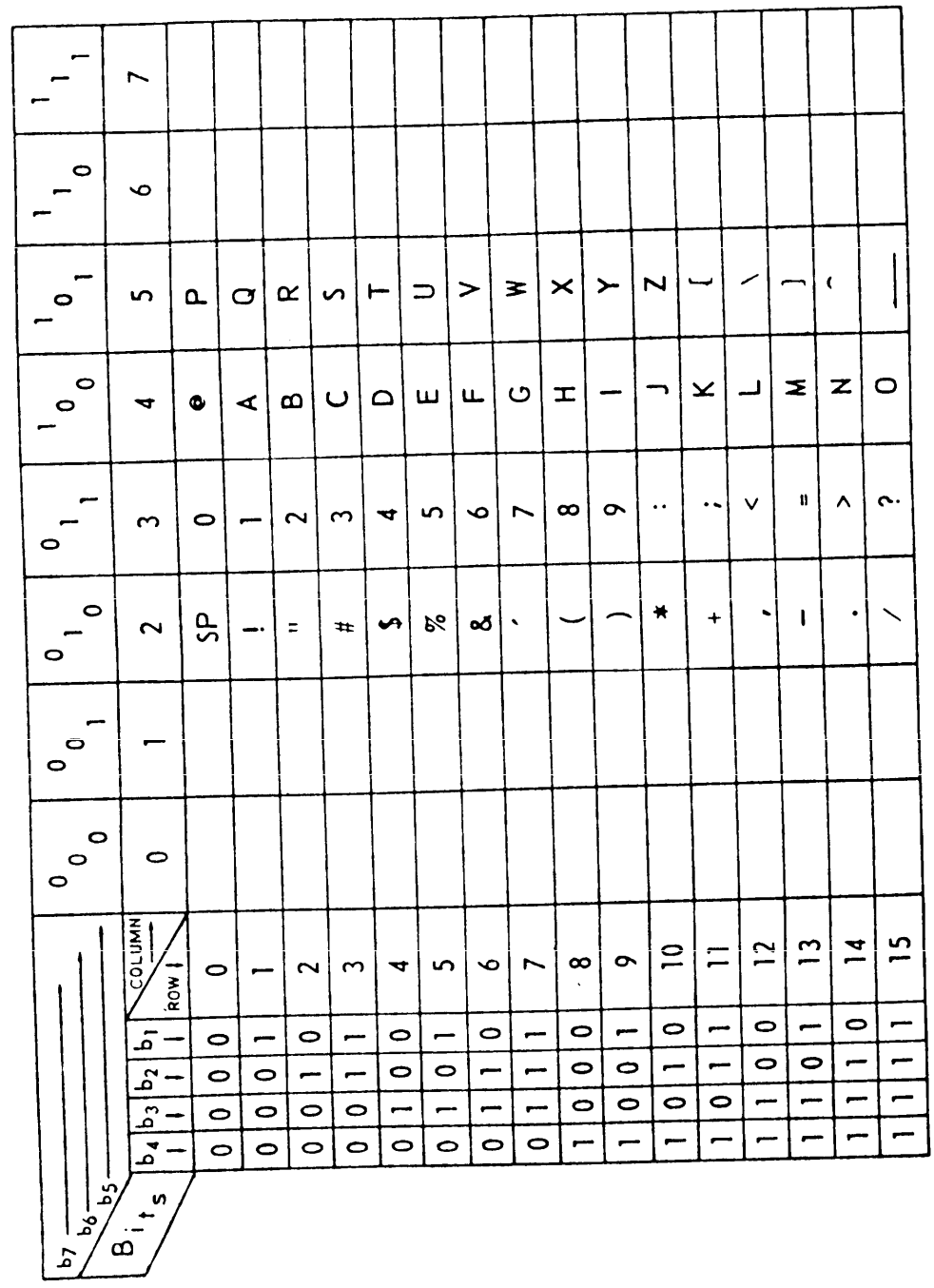
5.3.7.8 Code extension techniques. Code extension techniques for use with the basic ASCII character set (see Table IX) shall comply with the current edition of FIPS PUB 35.

NOTE 1: FIPS PUB 35 has adopted ANSI X3.41-1974 which specifies methods of extending the 7-bit code of ASCII (see Note 2), remaining in a 7-bit environment or increasing to an 8-bit environment, building upon the structure of ASCII to describe various means of extending the control and graphic sets of the code. ANSI X3.41 also describes techniques for constructing expanded codes (see Note 3) related to ASCII (so as to allow application-dependent usage without preventing the interchangeability of their data) and describes 8-bit codes for general information interchange in which ASCII is a subset.

NOTE 2: Extended sets use alternate assignments of the 128 binary patterns of the basic ASCII character set (see Table IX). This may be accomplished by use of the ASCII control characters Shift Out (SO), Shift In (SI) and Escape (ESC). Such use may be planned by agencies having

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the currentversion before use.

TABLE XIV. 64-Character ASCII graphic subset.



Bits b7 b6 b5 / b4 b3 b2 b1	COLUMN / ROW	0 0 0	0 0 1	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
		0	1	2	3	4	5	6	7
0 0 0 0	0			SP	0	@	P		
0 0 0 1	1			!	1	A	Q		
0 0 1 0	2			"	2	B	R		
0 0 1 1	3			#	3	C	S		
0 1 0 0	4			$	4	D	T		
0 1 0 1	5			%	5	E	U		
0 1 1 0	6			&	6	F	V		
0 1 1 1	7			'	7	G	W		
1 0 0 0	8			(	8	H	X		
1 0 0 1	9			)	9	I	Y		
1 0 1 0	10			*	:	J	Z		
1 0 1 1	11			+	;	K	[		
1 1 0 0	12			,	<	L	\		
1 1 0 1	13			-	=	M	]		
1 1 1 0	14			.	>	N	^		
1 1 1 1	15			/	?	O	_		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XV. 16-Character ASCII graphic numeric subset.

Bits b7 b6 b5 / b4 b3 b2 b1	COLUMN / ROW	0 0 0	0 0 1	0 1 0	0 1 1	1 0 0	1 0 1	1 1 0	1 1 1
		0	1	2	3	4	5	6	7
0 0 0 0	0				0				
0 0 0 1	1				1				
0 0 1 0	2				2				
0 0 1 1	3				3				
0 1 0 0	4				4				
0 1 0 1	5				5				
0 1 1 0	6				6				
0 1 1 1	7				7				
1 0 0 0	8				8				
1 0 0 1	9				9				
1 0 1 0	10			*					
1 0 1 1	11			+					
1 1 0 0	12			,					
1 1 0 1	13			-					
1 1 1 0	14			.					
1 1 1 1	15			/					

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

special requirements, but must not be finalized until approved by the National Bureau of Standards (NBS) and entered in the FIPS Register. (See Sections 8 and 9 of FIPS PUB 7.)

NOTE 3: Expanded sets use 256 eight-bit character codes made possible by the availability of eight information channels on standard magnetic tape and other media with a capacity for eight data bits. The character set of the expanded code must include the character set of the basic ASCII character set (see Table IX). Use of expanded sets must be approved by NBS and entered in the FIPS Register. (See Sections 8 and 9 of FIPS PUB 7.)

5.3.7.9 NATO 7-bit code for information interchange. The interchange of coded character-oriented information among information processing systems, communications systems and associated equipment within NATO military and civil organizations shall comply with the applicable requirements of the current edition of STANAG 5036.

NOTE: STANAG 5036 applies to information interchange among fixed stations, ships, vehicles and aircraft. The NATO 7-bit code, derived from the International Alphabet No. 5 (see 80.3 of Appendix H), is compatible with ASCII.

5.3.8 Data Modems. The technical characteristics of data modems shall comply with the applicable requirements of the current edition of MIL-STD-188-110.

5.3.9 Keyboards. Keyboards shall comply with the applicable requirements of the current edition of MIL-STD-1280.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.4 Transmission subsystems. This paragraph deals with standards and design objectives of various types of subsystems for the transmission of user information, control signals and supervisory signals in analog, digital and quasi-analog form over wire, cable and radio links.

Additional subjects in the area of transmission are covered under loop subsystems in 5.5, trunk subsystems in 5.7, and transmission channel subsystems in 5.8.

5.4.1 Telegraph and data transmission subsystems.

5.4.1.1 Bit error rate (BER). Same as 5.2.3.1.1.

5.4.1.2 Interface characteristics. Same as 5.3.4 through 5.3.5.7.

5.4.1.3 Clock equipment, control and timing. Same as 5.3.6 through 5.3.6.4.5.

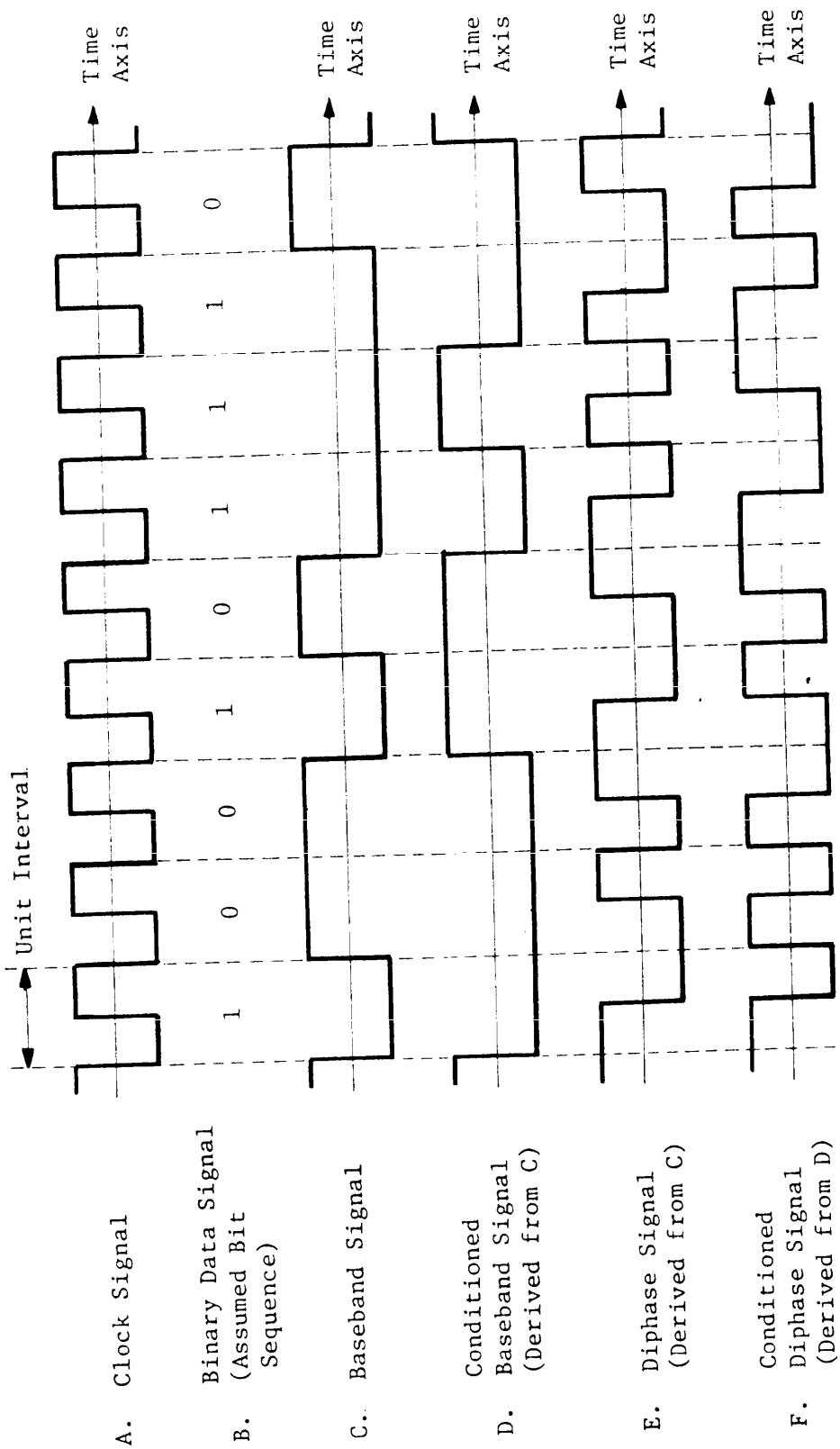
5.4.1.4 Signal conditioning, diphase and conditioned diphase modulation. Signal conditioning and diphase modulation are two independent means of processing a serial binary digital signal to facilitate data transmission over wire, cable or radio links. Telegraph and data transmission subsystems employing binary digital signals with conditioning or diphase modulation, or both, shall comply with the applicable standards of 5.3.5.1 through 5.3.5.7 and 5.4.1.4.1 through 5.4.1.4.3.

5.4.1.4.1 Signal conditioning. Conditioning of binary data signals shall be accomplished by using the following rules in accordance with Figure 21D: The significant condition (see 5.3.5.2) of Space (0) is transmitted by repeating the same level as in the immediately preceding unit interval (regardless of which level exists in the preceding unit interval); the significant condition of Mark (1) is transmitted by reversing the level at the beginning of the unit interval from the level in the immediately preceding unit interval (regardless of which level exists in the preceding unit interval).

NOTE: The rules used for conditioning of binary data signals are known as differential coding (see 50.3 of Appendix E) where bits are transmitted in terms of "changes" and "no changes" rather than amplitude levels. A conditioned signal allows correct bit recovery at the data sink regardless of any polarity reversals that may occur in associated transmission links, such as radio links. (See 5.4.6.1.)

5.4.1.4.2 Diphase modulation. Diphase modulation of binary data signals shall be accomplished by using the following rules in accordance with Figure 21E: The significant condition (see 5.3.5.2) of Space (0) is transmitted by a level change from negative to positive levels at the middle of the unit interval. The significant condition of Mark (1) is transmitted by a level change from positive to negative levels at the

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



NOTE: Additional standards for signal conditioning, diphase and conditioned diphase modulation are given in 5.3.5.

FIGURE 21. Signal conditioning, diphase and conditioned diphase modulation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

middle of the unit interval, resulting in a phase shift of 180 degrees with reference to the phase of the significant condition of Space (0).

NOTE 1: The purpose of diphase modulation is to eliminate the dc component in the frequency spectrum of a baseband data signal and to enhance the recovery of timing from the data signal by ensuring that every unit interval has at least one signal transition (in the middle of the unit interval).

NOTE 2: The waveform shown in Figure 21E is also known as Manchester II data code. (See 50.2.1.2 of Appendix E.)

5.4.1.4.3 Conditioned diphase modulation. Conditioned diphase modulation of binary data signals shall be accomplished by using the following rules in accordance with Figure 21F: The significant condition (see 5.3.5.2) of Space (0) is transmitted by repeating the same phase as in the immediately preceding unit interval, regardless of which phase exists in the preceding unit interval. The significant condition of Mark (1) is transmitted by changing the phase 180 degrees with reference to the phase in the immediately preceding unit interval, regardless of which phase exists in the preceding unit interval.

NOTE 1: Conditioned diphase modulation may be accomplished, as shown in Figure 21, by diphase modulation of a conditioned binary baseband signal in accordance with the rules given in 5.4.1.4.1 and 5.4.1.4.2.

NOTE 2: If a conditioned diphase modulated signal is demodulated by comparing the signal phases in two adjacent unit intervals, there is a one-bit error extension inherent in this demodulation process.

5.4.1.5 Speech plus half-duplex operation. This type of operation uses a half-duplex circuit for the simultaneous transmission of speech and telegraphy. The telegraph channel uses part of the frequency bandwidth available for the telephone channel which causes a degradation in the speech quality.

5.4.1.5.1 Telephone channel characteristics. Trunks shall not be used for speech plus half-duplex operation since this type of operation degrades the speech quality and also precludes the use of these trunks for higher speed data transmission, such as 1200 Bd or 2400 Bd.

5.4.1.5.1.1 Insertion loss versus frequency characteristic. The spread of the insertion loss versus frequency characteristic shall not be greater than 3 dB between 300 Hz and 1000 Hz and between 1600 Hz and 3400 Hz.

5.4.1.5.1.2 Single harmonic distortion. The single harmonic distortion, produced by any single-frequency test signal between 300 Hz and 1000 Hz and between 1600 Hz and 3400 Hz, shall be at a level which is at least 25 dB below reference level (-25 dBm0). The single harmonic distortion shall be measured with a standard test signal in accordance with 5.6.6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.4.1.5.1.3 Frequency displacement. The frequency displacement of any voice frequency due to the difference in carrier frequencies, measured at each end of a speech plus half-duplex link, shall not exceed ±2 Hz.

5.4.1.5.2 Data modems. A modem with standardized characteristics in accordance with MIL-STD-188-110 shall be used for speech plus half-duplex operation for modulation rates of 150 Bd or less.

5.4.1.5.3 Telegraph channel characteristics.

5.4.1.5.3.1 Insertion loss versus frequency characteristic. The spread of the insertion loss versus frequency characteristic shall not be greater than 4 dB between 1200 Hz and 1350 Hz.

5.4.1.5.3.2 Crosstalk coupling loss. The crosstalk coupling loss from the telephone channel into the telegraph channel shall be equal to or greater than 80 dB for any frequency between 300 Hz and 3400 Hz, when measured at or referenced to equal TLP.

5.4.1.6 Speech plus full-duplex operation. Speech plus full-duplex operation shall follow the same standards as those given in 5.4.1.5.1 through 5.4.1.5.3.2 for speech plus half-duplex operation.

5.4.1.7 Monitoring, signal connections and path characteristics.

5.4.1.7.1 Monitoring. (MIL-STD-188-100.) Digital transmission shall be monitored by an essentially nonreactive tap with a minimum dc resistance of 10,000 ohms (Design Objective: 47,000 ohms to 68,000 ohms) for polar operation, and an essentially nonreactive series insertion with a maximum dc resistance of 200 ohms for neutral operation.

5.4.1.7.2 Signal paths. (MIL-STD-188-100.) The signal path for digital signals shall be either two-wire with metallic ground or four-wire full metallic for a single full-duplex circuit.

NOTE: The use of earth return is not recommended. It is to be noted that a distinction is implied between ground and earth return. Use of the first term is meant to imply that a single metallic path either insulated or not insulated from the earth is employed for the return path of one or more dc circuits; whereas, use of the term earth return is meant to imply that the signal current literally flows through the earth on a non-metallic basis.

5.4.1.7.3 Data signal connections. Same as 5.3.5.4.

5.4.1.7.4 Transmission path performance. The quality of data transmission shall be expressed by the following formulas:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

a. Path Error Rate (in percent) =

$$\frac{\text{number of incorrectly received characters or bits}}{\text{number of characters or bits sent}} \times 100$$

b. Path Utilization (in percent) =

$$\frac{\text{number of characters or bits sent}}{\text{theoretical number of characters or bits that could be sent with EDC systems}} \times 100$$

c. Path Efficiency (in percent) =

$$\frac{\text{number of correctly received characters or bits}}{\text{total number of characters or bits sent (including repeated characters or bits, control characters, etc.)}} \times 100$$

5.4.1.8 Recording media.

5.4.1.8.1 Teletypewriter paper.

5.4.1.8.1.1 Teletypewriter roll paper. (MIL-STD-188-100.) Teletypewriter roll paper used with friction feed operation shall be 215.9 mm (8.5 in) wide and the roll shall have a diameter of nominally 127 mm (5 in) in accordance with the current edition of Federal Specification UU-P-547.

5.4.1.8.1.2 Teletypewriter fanfold paper. (MIL-STD-188-100.) Teletypewriter fanfold paper used with sprocket feed operation shall be 215.9 mm (8.5 in) wide in accordance with the current edition of MIL-P-40023.

5.4.1.8.2 Perforated tape. The requirements of 5.4.1.8.2.1 through 5.4.1.8.2.3 apply to paper tape, and tape made of materials other than paper, used for recording information by a perforated tape code. (See 5.4.1.8.2.3.)

5.4.1.8.2.1 Physical characteristics of unpunched tape. Under consideration.

5.4.1.8.2.2 Physical dimensions of perforated tape. (MIL-STD-188-100.) For 25.4 mm (1 in) wide perforated tape and the related 8-channel tape punching and reading equipment, the physical dimensions, including the

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

size and location of the perforations used for recording information, shall comply with the applicable requirements of the current edition of FIPS PUB 26.

NOTE 1: FIPS PUB 26 has adopted ANSI X3.18-1967.

NOTE 2: Federal Specification UU-T-120B contains the physical dimensions of 11/16 in and 7/8 in wide perforated tape which is still used with older equipment.

5.4.1.8.2.3 Perforated tape code for information interchange. (MIL-STD-188-100.) For information interchange between and among communications systems, ADP systems and associated equipment, the representation of the ASCII (see 5.3.7) on perforated tape shall comply with the applicable requirements of the current edition of FIPS PUB 2.

NOTE: FIPS PUB 2 has adopted ANSI X3.6-1965.

5.4.1.8.3 Magnetic tape.

5.4.1.8.3.1 Physical characteristics of unrecorded magnetic tape. Under consideration.

5.4.1.8.3.2 Recorded magnetic tape for information interchange (800 cpi,NRZI). For representing the ASCII (see 5.3.7) on magnetic tape at the recording density of nominally 800 cpi (approximately 32 cpmm) using NRZI, the recording characteristics and the data format shall comply with the applicable requirements of FIPS PUB 3-1 which has adopted, with one exception (see Note), ANSI X3.22-1973. FIPS PUB 3-1 shall apply to 9-track 12.7 mm (0.5 in) wide magnetic tape used for information interchange between and among communications systems, ADP systems and associated equipment.

NOTE: The exception specified in FIPS PUB 3-1 changes the text of paragraph 5.4.3 of ANSI X3.22-1973 to read as follows: "Bit Z shall be zero or treated as a bit of higher order than the ASCII bits." This is interpreted to mean that bit Z will be a zero when recording the ASCII characters but may be other than zero when recording dense numeric, binary, or extended ASCII code (see 5.3.7.8) representation.

5.4.1.8.3.3 Recorded magnetic tape for information interchange (1600 cpi, phase encoding). For representing the ASCII (see 5.3.7) on magnetic tape at the recording density of nominally 1600 cpi (approximately 63 cpmm) using phase encoding, the recording characteristics and the data format shall comply with the applicable requirements of FIPS PUB 25 which has adopted, with one exception (see Note), ANSI X3.39-1973. FIPS PUB 25 shall apply to 9-track 12.7 mm (0.5 in) wide magnetic tape used for information interchange between and among communications systems, ADP systems and associated equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: The exception specified in FIPS PUB 25 changes the text of paragraph 5.4.3 of ANSI X3.39-1973 to read as follows: "Bit Z shall be zero or treated as a bit of higher order than the ASCII bits." See Note of 5.4.1.8.3.2 for an interpretation of this change.

5.4.1.8.3.4 Recorded magnetic tape for information interchange (6250 cpi, group-coded recording). For representing the ASCII (see 5.3.7) on magnetic tape at the recording density of nominally 6250 cpi (approximately 246 cpmm) using NRZI with group-coded recording, the recording characteristics and the data format shall comply with the applicable requirements of FIPS PUB 50 which has adopted, with one exception (see Note), ANSI X3.54-1976. FIPS PUB 50 shall apply to 9-track 12.7 mm (0.5 in) wide magnetic tape used for information interchange between and among communications systems, ADP systems and associated equipment.

NOTE: The exception specified in FIPS PUB 50 changes the text of paragraph 5.4.4 of ANSI X3.54-1976 to read as follows: "Bit Z shall be zero or treated as a bit of higher order than the ASCII bits." See Note of 5.4.1.8.3.2 for an interpretation of this change.

5.4.2 FDM telegraph subsystems. Several telegraph channels may be frequency division multiplexed into a nominal 3 kHz or 4 kHz VF channel. This type of operation is also called voice frequency carrier telegraph (VFCT). Two FDM telegraph subsystems are standardized: One subsystem with 12 telegraph channels and the other one with 16 telegraph channels, to be used with conditioned or unconditioned nominal 4 kHz VF channels.

5.4.2.1 Twelve-channel FDM telegraph subsystem.

5.4.2.1.1 Digital interface characteristics. The electrical characteristics of the digital interface at the input of a telegraph multiplexer and at the output of a telegraph demultiplexer shall comply with the applicable requirements of the current edition of MIL-STD-188-114. (See 5.3.5.3.)

5.4.2.1.2 Quasi-analog interface characteristics.

5.4.2.1.2.1 Impedance. Same as 5.3.4.2.1.

5.4.2.1.2.2 Multiplexer output level. Same as 5.2.3.1.4.

5.4.2.1.2.3 Demultiplexer input level. Same as 5.2.3.1.5.1.

5.4.2.1.3 Telegraph distortion. (MIL-STD-188-100.) The total telegraph distortion due to any combination of causes (bias, fortuitous, cyclic, or characteristic) as measured in the receiving loop shall not exceed ±4 percent (Design Objective: ±2 percent) of the unit interval. This total distortion shall apply to all modulation rates for which the equipment was designed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: The total telegraph distortion is measured in the receiving loops, with the multiplexer transmit terminals and the demultiplexer receive terminals connected back-to-back, transmitting and receiving loops properly terminated, all transmit tones activated at proper levels, and random undistorted signals keyed simultaneously into each transmitting loop at the modulation rates for which the equipment was designed.

5.4.2.1.4 Crosstalk coupling loss. (MIL-STD-188-100.) The crosstalk coupling loss between telegraph channels, with the multiplexer and demultiplexer connected back-to-back, shall be equal to or greater than 55 dB for any frequency between 300 Hz and 3400 Hz, when measured at or referenced to equal TLP.

5.4.2.1.5 Center frequencies. The nominal center frequencies of the telegraph channels shall be as shown in Table XVI.

TABLE XVI. Frequency characteristics for 12-channel FDM telegraph subsystem.

Channel Designation	Mark Frequency (Hz)	Center Frequency (Hz)	Space Frequency (Hz)	Center Frequency Spacing	Frequency Shift (Hz)
1	382.5	425	467.5		±42.5
				170	
2	552.5	595	637.5		±42.5
				170	
3	722.5	765	807.5		±42.5
				170	
4	892.5	935	977.5		±42.5
				170	
5	1062.5	1105	1147.5		±42.5
				170	
6	1232.5	1275	1317.5		±42.5
				170	
7	1402.5	1445	1487.5		±42.5
				170	
8	1572.5	1615	1657.5		±42.5
				340	
9	1870.0	1955	2040.0		±85.0
				425	
10	2295.0	2380	2465.0		±85.0
				425	
11	2720.0	2805	2890.0		±85.0
				425	
12	3145.0	3230	3315.0		±85.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.4.2.1.6 Signal type and signal sense. (MIL-STD-188-100.) Frequency shift modulation shall be employed. The Mark signal shall be at the center frequency minus the frequency shift, and the Space signal shall be at the center frequency plus the frequency shift. (See Tables VIII and XVI.)

5.4.2.1.7 Frequency tolerance. (MIL-STD-188-100.) The tolerance of the Mark and Space frequencies, as generated, shall not exceed ±4 Hz.

5.4.2.1.8 Frequency shift. The frequency shift in telegraph channels 1 through 8 (see Table XVI) shall be 42.5 Hz below and above the center frequencies. The frequency shift in telegraph channels 9 through 12 (see Table XVI) shall be 85.0 Hz below and above the center frequencies.

5.4.2.2 Sixteen-channel FDM telegraph subsystem.

5.4.2.2.1 Digital interface characteristics. Same as 5.4.2.1.1.

5.4.2.2.2 Quasi-analog interface characteristics. Same as 5.4.2.1.2.

5.4.2.2.3 Telegraph distortion. Same as 5.4.2.1.3.

5.4.2.2.4 Crosstalk coupling loss. Same as 5.4.2.1.4.

5.4.2.2.5 Center frequencies. (MIL-STD-188-100.) The nominal center frequencies of the telegraph channels shall be as shown in Table XVII.

5.4.2.2.6 Signal type and signal sense. (MIL-STD-188-100.) Frequency shift modulation shall be employed. The Mark signal shall be at the center frequency minus the frequency shift, and the Space signal shall be at the center frequency plus the frequency shift. (See Tables VIII and XVII.)

5.4.2.2.7 Frequency tolerance. Same as 5.4.2.1.7.

5.4.2.2.8 Frequency shift. (MIL-STD-188-100.) The frequency shift in all telegraph channels shall be 42.5 Hz below and above the center frequencies. (See Table XVII.)

5.4.3 Telegraph and data equipment.

5.4.3.1 Common parameters. All telegraph and data equipment shall comply with the applicable interface requirements given in 5.3.5.1 through 5.3.5.7 and the applicable requirements for clock equipment, control and timing given in 5.3.6.1 through 5.3.6.4.5.

5.4.3.2 Data modems. The technical characteristics of data modems shall comply with the applicable requirements of the current edition of MIL-STD-188-110.

Source: https://assist.dla.mil/ Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XVII. Frequency characteristics for 16-channel FDM telegraph subsystem.

Channel Designation	Mark Frequency (Hz)	Center Frequency (Hz)	Space Frequency (Hz)
1	382.5	425	467.5
2	552.5	595	637.5
3	722.5	765	807.5
4	892.5	935	977.5
5	1062.5	1105	1147.5
6	1232.5	1275	1317.5
7	1402.5	1445	1487.5
8	1572.5	1615	1657.5
9	1742.5	1785	1827.5
10	1912.5	1955	1997.5
11	2082.5	2125	2167.5
12	2252.5	2295	2337.5
13	2422.5	2465	2507.5
14	2592.5	2635	2677.5
15	2762.5	2805	2847.5
16	2932.5	2975	3017.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.4.3.3 Telegraph multiplexer equipment. Same as 5.4.2.

5.4.3.4 Modulation rate converters.

5.4.3.4.1 General. A modulation rate conversion process may take place in different types of equipment. The inherent characteristics of a modulation rate converter, regenerative repeater, distortion monitor and delay line can take on a close resemblance that may be usefully exploited in combining these functional areas. However, the equipment referred to in the following subparagraphs are devices whose basic purpose is to change the relationship of the input/output modulation rate. Modulation rate converters find application in such areas as interconnecting 50 Bd with 75 Bd teleprinter equipment. It is to be understood that in connecting a higher to a lower modulation rate device, provision must be made to control the average information rate of the faster device in order not to exceed the information rate of the slower device; e.g., a 100 words per minute (wpm) to 60 wpm conversion process operates at an average rate of 60 wpm, a 100-card per minute to 10-card per minute conversion process operates at an average rate of 10 cards per minute, etc. In addition to the requirements stated in 5.4.3.1, the following parameters and explanatory information given in 5.4.3.4.2.1 and 5.4.3.4.2.2 also apply to modulation rate converters.

5.4.3.4.2 Clocking methods.

5.4.3.4.2.1 Start-stop input. A modulation rate converter shall become quiescent and ready to commence sampling the character following the one just sampled, at some point after the beginning of the stop interval and in no case in excess of half the duration of the unit interval at the applicable modulation rate.

NOTE: For example, if the start-stop character interval is 10 unit intervals (1 unit start element, 8 units of information elements, and 1 unit stop element), the input logic will be prepared to correctly accept the following character at any time between 9.5 and 10 unit intervals from the stop-start transition of that character interval.

5.4.3.4.2.2 Start-stop output. As a Design Objective, the output character interval should be equal to or greater than the input character interval.

NOTE 1: The design objective of 5.4.3.4.2.2 will eliminate the need for various amounts of storage capabilities.

NOTE 2: There are two situations, depending on whether the modulation rate is increased or decreased by the modulation rate converter:

a. The modulation rate is increased. In this case, the output modulation rate is greater than the input modulation rate. Since the infor-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

mation rate is controlled by the character rate of the input signal, each output character must consist of a character of the same character structure as the input signal, but with the stop element extended until the time duration is the same as that of the input character. In terms of the output modulation rate, the output character interval must contain at least as many unit intervals as the input character, multiplied by the ratio of the output modulation rate to the input modulation rate.

EXAMPLE: Given an input signal with a character interval of 7 units and a modulation rate of 75 Bd, and an output signal to be delivered at a rate of 150 Bd, then the output character interval must be at least:

$$\frac{7 \times 150}{75} = 14 \text{ units}$$

b. The modulation rate is decreased. In this case, the output modulation rate is less than the input modulation rate. The character rate of the input signals must be reduced to that of the output circuit, or lower, so as not to outrun the capability of the modulation rate converter. A typical way to accomplish this is to use stepped operation. (See 5.3.6.1.3.) In stepped operation, the modulation rate converter or some external device generates step pulses that cause the input circuit to deliver only the next character. The step pulses are spaced sufficiently apart to allow the converter to deliver the output signal with the correct character interval.

EXAMPLE: Given an input signal with a character interval of 11 units and a modulation rate of 150 Bd, and an output signal to be delivered at a rate of 50 Bd, then the stepping pulses must be spaced by an interval, in terms of the input modulation rate, of at least:

$$\frac{11 \times 150}{50} = 33 \text{ units}$$

5.4.3.5 Regenerative repeaters. Regenerative repeaters reshape and retime digital signals. It is generally accepted that retiming characteristics make a regenerative repeater distinct from a nonregenerative repeater which reshapes but does not retime the signal. There are two basic classes of regenerative repeaters, synchronous repeaters and start-stop repeaters.

In addition to the requirements stated in 5.4.3.1, the following parameters and explanatory information given in 5.4.3.5.1.1 through 5.4.3.5.4.2 also apply to regenerative repeaters.

5.4.3.5.1 Synchronous repeaters.

5.4.3.5.1.1 Modulation rate coverage. Not standardized.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: The range of modulation rates accepted by the repeater depends on particular requirements. A range of at least 50 Bd to 32 kBd is desirable.

5.4.3.5.1.2 Propagation delay time. As a Design Objective, the internal propagation delay time of the repeater should be greater than one-half and less than two unit intervals at the applicable modulation rate.

NOTE: The design objective of 5.4.3.5.1.2 is intended to ensure correct center sampling operation.

5.4.3.5.1.3 Remote control. Not standardized.

5.4.3.5.1.4 Distortion. A synchronous repeater shall be capable of operating, without degradation of performance, with input signals having a maximum distortion for the logic sense of Mark and Space as shown in Table XVIII.

NOTE: The distortion values of Table XVIII are applicable to either externally or internally clocked repeaters. The modulation rates of the input signal and the clock signal must be within 1 percent of each other in accordance with the applicable requirements of 5.3.6.3.

TABLE XVIII. Input signal distortion for synchronous and start-stop regenerative repeaters.

Parameters	Maximum Distortion (in percent) of Input Signals	
	Mark	Space
Bias Distortion	49	49
Cyclic Distortion	24.5	24.5
Fortuitous Distortion	49	49

5.4.3.5.1.5 Alarm. Whenever the repeater input signal exceeds the distortion limits of 5.4.3.5.1.4, the output signal shall be forced to assume a Mark condition. Means should be incorporated in the repeater to

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

cause an alarm, either local or remote, whenever the repeater input signal exceeds the distortion limits of 5.4.3.5.1.4 for a period longer than 10 seconds.

5.4.3.5.2 Synchronous repeaters controlled by station clocks.

5.4.3.5.2.1 Phase adjustment of data to clock signals. Synchronous repeaters controlled by station or master clocks (external timing) shall provide the means to shift the phase of the incoming data signal with relation to the clock signal in accordance with 5.3.6.2.3.

NOTE: The adjustment of the phase relationship between input data signals and clock signals must not be reflected in a similar shift in the phase of the repeater output data signal. Rather, the output data signal must maintain the phase of the station clock supplied to the regenerative repeater. (See 5.3.6.2.3.)

5.4.3.5.2.2 Clock modulation rate. The clock signal shall be supplied to the repeater with a modulation rate that is two times the applicable data modulation rate in accordance with 5.3.6.2.1.

5.4.3.5.3 Synchronous repeaters not controlled by station clocks.

5.4.3.5.3.1 Timing control. As a Design Objective, the timing control signals of synchronous regenerative repeaters not controlled by station or master clocks should be derived by an integrating process so that the repeater follows the average data modulation rate rather than attempt to enforce its own rate.

NOTE: Such timing control is analogous to the automatic frequency control employed in radio receiver equipment.

5.4.3.5.4 Start-stop regenerative repeaters. In start-stop regenerative repeaters it is expected that an equipment propagation delay time equal to the number of bits normally stored, plus one-half the duration of the unit interval will be encountered. For example, a propagation delay time of 10.5-unit intervals would be considered reasonable for start-stop signaling consisting of 1-unit interval start element, 8-unit intervals of information elements, and 1-unit interval stop element. Normally, start-stop regenerative repeaters cannot regenerate distortion encountered at the stop element. Therefore, the distortion of the stop element may appear at the output of the regenerative repeater. There are exceptions to this rule. For example, synchronous transmission equipment with start-stop, input/output buffers regenerate the stop element, due to the storage and timing functions necessary for synchronous operations. The requirements of 5.4.3.5.4.1 and 5.4.3.5.4.2 do not apply to these types of start-stop repeaters which regenerate the stop element.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.4.3.5.4.1 Distortion. A start-stop regenerative repeater shall be capable of operating, without degradation of performance, with input signals having a maximum distortion for the logic sense of Mark and Space as shown in Table XVIII.

NOTE: When measuring the maximum distortion, the modulation rates of the input signal and the clock signal must be within 1 percent of each other in accordance with the applicable requirements of 5.3.6.3.

5.4.3.5.4.2 Alarm. Same as 5.4.3.5.1.5.

5.4.3.6 Telegraph code converters. The code conversion process shall provide that all permutations of a telegraph code being converted can be processed either from the basic (unconverted) code to the converted code or from the converted code to the basic code. The type of code conversion technique to be used is not standardized. However, all code converters shall comply with the requirements stated in 5.4.3.1, regardless of the type of code conversion technique employed.

NOTE 1: Telegraph code converters generally are an inherent part of, or at least closely associated with, other equipment, such as the input or output of TDM equipment. Unfortunately, a degree of confusion exists in differentiating between telegraph code converters and devices more properly called alphabet translators (see 5.4.3.8). Telegraph code converters are those devices whose basic function is to accept a digital signal in one telegraph code and convert it to another telegraph code. In certain cases, such as the conversion from start-stop asynchronous telegraph equipment using the 7-unit ITA No. 2 to synchronous equipment, a telegraph code conversion process may consist only of discarding the start and stop elements and adding a sixth element to indicate the start and stop condition. The code conversion process in this example does not manipulate the five information elements. Rather, these elements are passed to the output in the same physical relationship to each other as they were presented to the converter input.

NOTE 2: A device may contain a code conversion process and an alphabet translation process, e.g., a device converting the International Morse Code to ITA No. 2.

5.4.3.7 Error control devices. The type of error control technique to be used is not standardized. (See 5.2.3.1.2.) However, all error control devices shall comply with the requirements stated in 5.4.3.1, regardless of the type of error control technique employed. Error control devices to be used in tactical communications systems shall be designed to protect against the particular type of interference encountered in these systems, such as impulse noise bursts, multipath effects, deep fading and selective fading, and shall be compatible with the characteristics of the communications systems.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

Page 118 is missing from the original legacy source document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.4.4 Radio subsystems. The radio subsystems are treated from two different points of view in this document. Parameters for radio subsystems are listed according to functional employment in 5.4.4.2 and according to frequency bands in 5.4.4.3.

5.4.4.1 Radio regulations. The use of the frequency spectrum is regulated by international agreements embodied in the Radio Regulations, published by the General Secretariat of the ITU, Geneva, Switzerland, and modified periodically by a World Administrative Radio Conference (WARC). These radio regulations are further qualified at the national level through Federal Government agencies, such as the Interdepartment Radio Advisory Committee (IRAC), and through military agencies, such as the JCS and the Military Communications-Electronics Board (MCEB). Military frequency planning, including joint functional frequency allocation tables, is established as a joint action area under the MCEB.

For subsystem and equipment design, the choice and performance of the equipment, as well as frequencies and emissions of any radio subsystem, shall satisfy the provisions of those radio regulations. Adequate familiarity with these regulations is, therefore, required of designers and users of radio subsystems. Final approval of frequency bands, operating modes and equipment characteristics within the DoD rests with the MCEB.

Appendix J contains information based on the ITU Radio Regulations concerning the classification and designation of emissions and necessary bandwidths for radio transmission.

5.4.4.2 Parameters for radio subsystems according to functional employment. Radio transmitters and receivers are employed in three general classes of tactical radio subsystems: Single channel point-to-point subsystems, single channel net radio subsystems and multichannel subsystems. A radio subsystem may be used as a loop, link, trunk or circuit, provided the characteristics of the radio equipment meet the applicable standards for such a connection.

5.4.4.2.1 Single channel point-to-point radio subsystems. This general class of tactical radio subsystems provides for the transmission and reception of voice, teletype, data, facsimile or any other type of signals for which the subsystem is suitable. Simplex or duplex operation, or both, may be employed.

5.4.4.2.2 Single channel net radio subsystems. This general class of tactical radio subsystems includes radio equipment used in net operations for the transmission and reception of voice, teletype, data, facsimile or any other type of signals for which the equipment is suitable. These subsystems usually operate on the same frequency for transmitting and receiving on a push-to-talk or push-to-operate (half-duplex) basis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.4.4.2.3 Telegraph and data transmission over single channel radio subsystems. In the past, single channel radio equipment has been designed primarily for voice and low speed telegraph transmission. Recent requirements for higher speed telegraph and data transmission necessitates increased attention to those radio characteristics essential for satisfactory transmission of telegraph and data signals. The standards and design objectives in 5.4.4.2.3.1 through 5.4.4.2.3.3 apply to all single channel radio equipment that has a requirement for telegraph and data transmission regardless of RF frequency bands covered.

More detailed standards and design objectives for single channel radio equipment are being developed and will be published in MIL-STD-188-141, MIL-STD-188-242, MIL-STD-188-243 and MIL-STD-188-146. (See Appendix B.)

5.4.4.2.3.1 Protection of signal sense. All single channel radio equipment shall be designed in such a manner that neither a radio transmitter nor a radio receiver inverts the logic and signal sense of the binary signals specified in 5.3.5.2.

5.4.4.2.3.2 Receiver and transmitter control time delays. Current and expected future requirements in the areas of communications security and higher speed data transmission (e.g., 16 kb/s for digitized voice) make it necessary to design single channel radio equipment with relatively short delay times for the control functions to change from the receive to the transmit mode and vice versa. The design objectives of 5.4.4.2.3.2.1 through 5.4.4.2.3.2.6 have been established to meet these requirements. For a definition of the terms used in these subparagraphs see FED-STD-1037.

5.4.4.2.3.2.1 Receiver attack time delay. As a Design Objective, the receiver attack time delay should not exceed 40 ms.

5.4.4.2.3.2.2 Receiver release time delay. As a Design Objective, the receiver release time delay should not exceed 40 ms.

5.4.4.2.3.2.3 Receive-after-transmit time delay. As a Design Objective, the receive-after-transmit time delay should not exceed 60 ms.

5.4.4.2.3.2.4 Transmitter attack time delay. As a Design Objective, the transmitter attack time delay should not exceed 25 ms.

5.4.4.2.3.2.5 Transmitter release time delay. As a Design Objective, the transmitter release time delay should not exceed 10 ms.

5.4.4.2.3.2.6 Transmit-after-receive time delay. As a Design Objective, the transmit-after-receive time delay should not exceed 70 ms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.4.4.2.3.3 Data modems. The technical characteristics of data modems used for data transmission over single channel radio subsystems shall comply with the applicable requirements of the current edition of MIL-STD-188-110.

5.4.4.2.4 Multichannel radio subsystems. This general class of tactical radio subsystems includes radio equipment used for the transmission and reception of multiplexed voice and data signals. This class is normally subdivided into two general types of subsystems: Radio equipment with four or less VF channels, such as HF SSB equipment; and radio equipment with more than four multiplexed channels, such as LOS microwave radio equipment.

5.4.4.2.4.1 HF radio subsystems. Same as 5.4.4.3.5.

5.4.4.2.4.2 Tropospheric scatter radio subsystems. Under consideration.

NOTE: Common long haul/tactical standards and design objectives for tropospheric scatter multichannel radio subsystems are being developed and will be published in MIL-STD-188-144. (See Appendix B.)

5.4.4.2.4.3 LOS microwave radio subsystems. Under consideration.

NOTE: Common long haul/tactical standards and design objectives for LOS microwave multichannel radio subsystems are being developed and will be published in MIL-STD-188-145. (See Appendix B.)

5.4.4.2.4.4 LOS millimeter wave radio equipment. Same as 5.4.4.3.9.

5.4.4.2.4.5 Radio orderwire for multichannel equipment. Orderwires are required to provide the minimum acceptable transmission quality for reliable maintenance of radio terminals and repeaters. Communication channels of multichannel equipment may in some cases be used for orderwire purposes. If a separate orderwire is provided, the standards and design objectives in 5.4.4.2.4.5.1 through 5.4.4.2.4.5.4 apply. In any case, radio orderwires must be specially engineered to meet the radio subsystem requirements. This is particularly important where orderwires are required to be extended over long distances from the radio terminal or where a large number of radio relay repeaters are required between terminals.

5.4.4.2.4.5.1 Input impedance. For a local orderwire at the radio set, the input impedance shall be 150 ohms, unbalanced to ground, with a minimum return loss of 20 dB against a 150-ohm resistance over the frequency band of interest. For a remote orderwire at the radio set, the input impedance shall be 600 ohms, balanced to ground, with a minimum return loss of 20 dB (Design Objective: 26 dB) against a 600-ohm resistance over the frequency band of interest.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.4.4.2.4.5.2 Output impedance. For both local or remote orderwires at the radio set, the output impedance shall be 600 ohms, balanced to ground, with a minimum return loss of 20 dB (Design Objective: 26 dB) against a 600-ohm resistance over the frequency band of interest.

5.4.4.2.4.5.3 Voice input level. As a Design Objective, the nominal input level of a remote orderwire should be -10 dBm based on an assumed mean speech volume of -10 VU with a standard deviation (sigma) of 5 VU in accordance with 5.2.2.1.1.

NOTE: The nominal input level of a local orderwire (when a microphone is connected directly to the radio set) is not standardized and depends on the mean power output of the microphone associated with the radio set.

5.4.4.2.4.5.4 Voice output level. The radio receiver output level delivered to the headset or handset for local orderwire operation or delivered to a remote orderwire shall be continuously adjustable from -20 dBm to 0 dBm. The radio receiver output level delivered to a loudspeaker for local orderwire operation shall be continuously adjustable from 10 dBm to 30 dBm.

5.4.4.2.5 Satellite subsystems. Under consideration.

NOTE: Common long haul/tactical standards and design objectives for satellite radio subsystems are being developed and will be published in MIL-STD-188-146. (See Appendix B.)

5.4.4.2.6 Manpack and handheld radio sets. Under consideration.

NOTE: Radio sets which are carried by one individual in field operations have severe constraints in their design from the standpoint of size, weight, ruggedness, reliability and ease of operation by non-technical personnel. Standards and design objectives for manpack and handheld radio equipment are being developed and will be published in MIL-STD-188-141, MIL-STD-188-242 and MIL-STD-188-243. (See Appendix B.)

5.4.4.3 Parameters for radio subsystems according to frequency bands.

5.4.4.3.1 ELF band (below 300 Hz) and ILF band (300 Hz to 3 kHz). Under consideration.

NOTE: Standards and design objectives for radio equipment operating in the ELF and ILF bands may be developed at a later time and will then be published in MIL-STD-188-140.

5.4.4.3.2 VLF band (3 kHz to 30 kHz). The technical characteristics of tactical radio equipment operating in the VLF band shall comply with the applicable requirements stated in paragraph 5.3 of MIL-STD-188-140.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: The VLF band is used primarily for broadcast in the maritime and aeronautical mobile service and for limited long distance communications by tactical forces. The usage of the VLF band is restricted to lower speed data transmission due to the relatively small information bandwidth inherent in the VLF band.

5.4.4.3.3 LF band (30 kHz to 300 kHz). The technical characteristics of tactical radio equipment operating in the LF band shall comply with the applicable requirements stated in paragraph 5.4 of MIL-STD-188-140.

NOTE: The LF band is used for a limited amount of tactical communications. This usage is restricted to lower speed data transmission due to the relatively small information bandwidth inherent in the LF band.

5.4.4.3.4 MF band (300 kHz to 3000 kHz). The technical characteristics of tactical radio equipment operating in the MF band shall comply with the applicable requirements stated in subparagraph 4.5.6 of MIL-STD-188C.

NOTE: The MF band is primarily utilized for broadcasting purposes. Assigned bandwidths may be up to 12 kHz. Military HF communication equipment generally covers the 2000 kHz to 3000 kHz portion of the MF band. In this event, the standards of the HF band apply to the extended frequency coverage. Standards and design objectives for tactical MF radio equipment currently contained in MIL-STD-188C are being updated and will be published in MIL-STD-188-141. (See Appendix B.)

5.4.4.3.5 HF band (3000 kHz to 30 MHz). The technical characteristics of tactical radio equipment operating in the HF band shall comply with the applicable requirements stated in subparagraph 4.5.7 of MIL-STD-188C. Additionally, the technical characteristics of HF combat net radio equipment shall comply with the applicable requirements of the current edition of QSTAG-263A.

NOTE: Standards and design objectives for tactical HF radio equipment currently contained in MIL-STD-188C are being updated and will be published in MIL-STD-188-141. (See Appendix B.)

5.4.4.3.6 VHF band (30 MHz to 300 MHz). The technical characteristics of tactical radio equipment operating in the VHF band shall comply with the applicable requirements stated in subparagraph 4.5.8 of MIL-STD-188C. Additionally, the technical characteristics of VHF combat net radio equipment shall comply with the applicable requirements of the current edition of QSTAG-263B.

NOTE: Standards and design objectives for tactical single channel VHF radio equipment currently contained in MIL-STD-188C are being updated and will be published in MIL-STD-188-242. Common long haul/tactical standards and design objectives for multichannel radio equipment and subsystems

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

operating in part of the VHF band are being developed and will be published in MIL-STD-188-145. (See Appendix B.)

5.4.4.3.7 UHF band (300 MHz to 3000 MHz). The technical characteristics of tactical radio equipment operating in the UHF band shall comply with the applicable requirements stated in subparagraph 4.5.9 of MIL-STD-188C. Additionally, the technical characteristics of UHF combat net radio equipment shall comply with the applicable requirements of the current edition of QSTAG-263C.

NOTE: The UHF band is used for single channel, tropospheric scatter, LOS microwave and satellite communications. Standards and design objectives for these different types of tactical radio equipment currently contained in MIL-STD-188C are being updated and will be published in MIL-STD-188-243, MIL-STD-188-144, MIL-STD-188-145 and MIL-STD-188-146, respectively. (See Appendix B.)

5.4.4.3.8 SHF band (3000 MHz to 30 GHz). The technical characteristics of tactical radio equipment operating in the SHF band shall comply with the applicable requirements stated in subparagraph 4.5.10 of MIL-STD-188C.

NOTE: The SHF band is used for tropospheric scatter, LOS microwave and satellite communications. Standards and design objectives for these different types of tactical radio equipment currently contained in MIL-STD-188C are being updated and will be published in MIL-STD-188-144, MIL-STD-188-145 and MIL-STD-188-146, respectively. (See Appendix B.)

5.4.4.3.9 Extremely high frequency (EHF) band (30 GHz to 300 GHz). All equipment designed for operation in the EHF band shall be compatible with the existing standards for communications equipment and shall be capable of interoperation with voice and data circuits or multiplexed carrier equipment designed in accordance with the applicable standards for lower frequency bands.

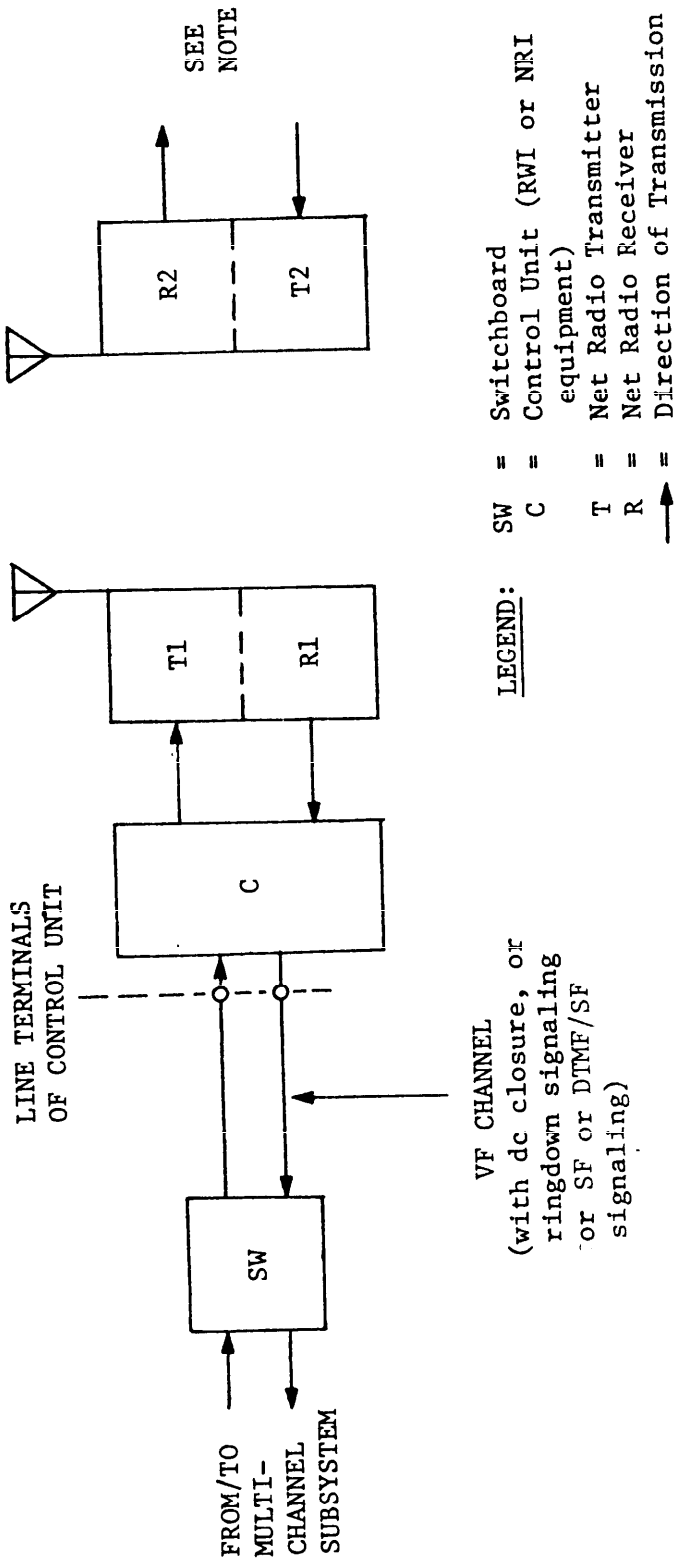
NOTE: Standards and design objectives for tactical and long haul EHF radio equipment will be developed and published in MIL-STD-188-147. (See Appendix B.)

5.4.5 Interconnection of single channel radio subsystems and telephone facilities. In tactical communications systems, single channel radio sets may be interconnected with local telephones or switchboards associated with multichannel trunking subsystems in order to provide communications to mobile, airborne and stationary users. Special control units, also called radio and wire integration (RWI) or, more specifically, net radio interface (NRI) equipment, are used to integrate these radio and telephone facilities. (See Figure 22.) The control units, either in the form of a remote control applique or designed as an integral part of the radio set incorporating equivalent features, may provide ringing and supervision signals in both directions of transmission which eliminates the need for

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

Page 125 is missing from the original legacy source document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



NOTE: T2/R2 is normally a net radio without a control unit, which may be added to provide extended service.

FIGURE 22. Block diagram of net radio interface.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Radio receiver R1, upon receiving a carrier from radio transmitter T2, shall signal control unit C and the control unit shall respond by alerting a local operator (see Note 2), or by providing a dc closure across the line terminals of the control unit, or by performing both functions. The control unit shall also provide, either manually or automatically, a VF channel path from its line terminals to the output terminals of the demodulator of receiver R1.

NOTE 1: The control time delays for keying-on and keying-off receivers and transmitters are given in 5.4.4.2.3.2.1 through 5.4.4.2.3.2.6

NOTE 2: The means by which a control unit alerts a local radio operator are not standardized and will be determined in applicable equipment specifications.

5.4.5.2.5 Ringdown signaling and dc closure control. For operating with ringdown supervisory signals, the control unit shall comply with the applicable requirements of 5.4.5.2.4 and, additionally, shall be capable of receiving a ringdown signal with a frequency of 20 Hz, ±5 Hz, and an amplitude between 15 volts rms and 110 volts rms. The ringdown signal generator of the control unit shall be capable of delivering a ringdown signal with a frequency of 20 Hz, ±5 Hz, and a level of not less than 36 volts rms into a resistive load of 1500 ohms, ±10 percent. The open circuit voltage of the ringdown signal generator shall not exceed 60 volts rms. The control unit shall be capable, if required, to modulate the carrier of the associated radio transmitter in accordance with 5.4.5.2.6.

NOTE: A ringdown signal is used to alert local switchboard or control unit operators.

5.4.5.2.6 Modulating signal. A control unit shall be capable, if required, of responding to a ringdown supervisory signal by providing to the associated radio transmitter a modulating signal with a frequency of 1600 Hz, ±30 Hz, and a level of -8 dBm, ±2 dB. Additionally, the control unit shall be capable of responding in accordance with 5.4.5.2.5 to a modulating signal with a frequency of 1600 Hz, ±30 Hz, and a level between -15 dBm and +10 dBm from the associated radio receiver. NOTE: The modulating signal of 1600 Hz is required if ringdown supervisory signals are to be transmitted by the single channel radio subsystem. A 20 Hz ringdown supervisory signal cannot be transmitted by the single channel radio subsystem because this signal is outside of the radio baseband frequency characteristic.

5.4.5.2.7 Dual-tone multifrequency (DTMF) signaling control. (See Figure 22.) When a telephone operating with DTMF supervisory signals is connected via a manual, semiautomatic or automatic switchboard (see Note) to the line terminals of control unit C, the reception of DTMF digit "1" (see Table XIX) from the telephone through the switchboard shall cause the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XIX. DTMF and SF signals for NRI control unit.

Function	Frequency* in Hz	Single Tone Level in dBm	Duration	Remarks
Digit 1 2 3 4 5 6 7 8 9 0	697/1209 697/1336 697/1477 770/1209 770/1336 770/1477 852/1209 852/1336 852/1477 941/1336	-7, ±2	Not standardized	Operator originated
Seize	2250	-7, ±2	Not standardized. (Normally operator controlled)	Held until seize acknowledge is heard
Seize Acknowledge	570	-7, ±2	Continuous up to 250 ms, ±20%, after seize stops	
Ringback Tone	425	-14, ±2	Not standardized. (Until operator answers)	2 s on, 4 s off, modulated at 10 Hz rate with 50 ms on, 50 ms off
Conference	941/1477	-7, ±2	Not standardized	Operator originated
Recall	941/1209	-7, ±2	Not standardized	Operator originated
Release	2600	-7, ±2	Continuous up to 10 s, ±20%	Operator originated, cutoff by release acknowledge
Release Acknowledge	570	-7, ±2	Cutoff 250 ms after release ends	

*For tolerances see 5.4.5.2.7.1 and 5.4.5.2.7.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

control unit to key-on the carrier of transmitter T1 (see Note 1 of 5.4.5.2.4). The control unit shall also provide a VF channel path from its line terminals to the input terminals of the modulator of transmitter T1. The reception of DTMF digit "3" (see Table XIX) from the telephone through the switchboard shall cause the control unit to return the radio T1/R1 to the receive mode (see Note 1 of 5.4.5.2.4), and the control unit shall provide a VF channel path from its line terminals to the output terminals of the demodulator of receiver R1. The control unit shall perform the above functions upon receiving DTMF digits 1 or 3 for a duration of at least 150 ms, ±25 ms, but the transmit relay shall not be activated until the DTMF signal ends. It shall also be possible to key-on and key-off manually the transmitter carrier at a control unit. Radio receiver R1, upon receiving a carrier from radio transmitter T2, shall signal control unit C and the control unit shall respond by alerting a local operator (see Note 2 of 5.4.5.2.4), or by providing a seize signal across the line terminals of the control unit, or by performing both functions. The control unit shall also provide, either manually or automatically, a VF channel path from its line terminals to the output terminals of the demodulator of receiver R1.

NOTE: For DTMF signaling control, the switchboard does not need the capability of generating DTMF supervisory signals but it must be capable of passing DTMF supervisory signals from connected telephone sets to the control unit.

5.4.5.2.7.1 Signaling and supervisory signals. A control unit shall be capable of generating DTMF and single-frequency (SF) signals with frequency tolerances that are within ±1.3 percent of the frequencies and with single tone levels as listed in Table XIX. A control unit shall be capable of responding to received DTMF and SF signals in accordance with 5.4.5.2.7.2.

5.4.5.2.7.2 Tone detectors. The tone detectors of a control unit shall respond to DTMF and SF signals with frequency tolerances that are within ±2 percent of the frequencies listed in Table XIX and with single tone levels between -5 dBm and -30 dBm. The tone detectors shall not respond to DTMF and SF signals that have frequency tolerances outside of ±5 percent (Design Objective: ±3 percent) of the frequencies listed in Table XIX.

5.4.5.3 Digital NRI. Under consideration.

5.4.6 Wire and cable transmission subsystems. Wire and cable transmission subsystems shall be engineered to comply with the applicable requirements for the type of subsystem employed, as stated in 5.4.6.1 through 5.4.6.4.

5.4.6.1 Wire and cable loops. Wire and cable loops shall comply with the applicable requirements of 5.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.4.6.2 Wire and cable trunks. Wire and cable trunks shall comply with the applicable requirements of 5.7.4, with the exception of crosstalk coupling loss given in 5.4.6.2.1.

5.4.6.2.1 Crosstalk coupling loss. In wire and cable trunks where transmit and receive channels are in close proximity (e.g., multipair cable), the near-end crosstalk coupling loss shall be equal to or greater than 25 dB for any frequency between 300 Hz and 3400 Hz, when measured at or referenced to equal TLP.

5.4.6.3 Digital transmission over wires and cables. These transmission subsystems shall comply with the applicable requirements of 5.3.5.1 through 5.3.5.7.

5.4.6.4 Signal conditioning, diphase and conditioned diphase modulation. Same as 5.4.1.4.

5.4.7 Fiber optic communications subsystems. Under consideration.

NOTE: Common long haul/tactical standards for fiber optic communications subsystems are being developed and will be published in MIL-STD-188-111. (See Appendix B.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.5 Loop subsystems.

5.5.1 General. Loops are the connections between terminal equipment or terminal subsystems and nodes. A loop may be identified as either an analog loop or a digital loop. Analog loops are used for analog terminal equipment, and for DTE when the data modem is located in the terminal subsystem. Digital loops are used for DTE when digital transmission is employed in a circuit or the data modem is located in the nodal subsystem. Loops may range from very short field wire loops to very long loaded or nonloaded cable loops. A loop may also be a radio extension loop or an access loop connecting a four-wire node via a two-wire switch or a private branch exchange (PBX) with terminal equipment. The characteristics of the loops will vary widely and are predictable only when the specific makeup of a loop is known. Each loop should be considered as a separate problem and each loop must be individually engineered to ensure adequate user-to-user communications service. In particular, loops for telegraph and data service have to be carefully engineered to achieve a minimum of SNR degradation at the data sink. (See 60.3 of Appendix F.)

5.5.2 Parameters of wire or cable loops for analog and quasi-analog signals.

5.5.2.1 Insertion loss of wire or cable loops for tactical subsystems type I and type III.

5.5.2.1.1 Loops for voice service and telegraph and data service. The following requirements apply to wire or cable loops used only for voice service or used for both voice service and telegraph and data service (employing quasi-analog signals) provided over switched and nonswitched (point-to-point) circuits of the tactical subsystems type I and type III (see Table I.) As a Design Objective, the insertion loss of a single two-wire loop (see Note 1) should be between 4 dB and 6 dB with a mean loss of 5 dB; the insertion loss of a single four-wire loop should be between 8 dB and 10 dB with a mean loss of 9 dB. The loop insertion loss shall be measured with a test signal in accordance with 5.5.2.9. Loop loss pads shall be provided as part of the nodal equipment. When terminal equipment is connected to a node, a means shall be provided to adjust for the loop length falling within the four distance ranges of approximately 0 - 300 meters (m) or approximately 0 - 1000 feet (ft), 300 m - 1500 m (approximately 1000 ft - 5000 ft), 1500 m - 3000 m (approximately 5000 ft - 10,000 ft), and greater than 3000 m (approximately 10,000 ft).

NOTE 1: Two-wire loops are not recommended for telegraph and data service employing quasi-analog signals.

NOTE 2: The loop insertion loss given in 5.5.2.1.1 does not include the two-wire/four-wire hybrid transformer insertion loss nor any two-wire switchboard insertion loss, if applicable, since a hybrid transformer or a switchboard is considered part of the node. (See 4.3.4.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE 3: The loop insertion loss for the tactical subsystems type I and type III has been chosen so that the requirements of -28 VU in accordance with 5.2.2.1.2 can be met with a mean transmitted speech volume of -10 VU (see 5.2.2.1.1) providing the insertion loss of the transmission channel (see Figure 1) is 0 dB.

5.5.2.1.2 Loops for telegraph and data service. The following requirements apply to wire or cable loops used only for telegraph and data service (employing quasi-analog signals) provided over switched and nonswitched (point-to-point) circuits of the tactical subsystems type I and type III (see Table I). As a Design Objective, the loop insertion loss should be as small as possible and no adjustments should be made to obtain a specific value of the loop insertion loss. (See 60.3.1.1 of Appendix F.)

5.5.2.2 Insertion loss of wire or cable loops for tactical subsystem type II. The insertion loss of a single wire or cable loop (two-wire or four-wire) of the tactical subsystem type II (see Table I), measured at 1000 Hz, ±25 Hz, has been found to approximate an exponential distribution with a mean loss of 3.5 dB and a standard deviation (sigma) of 3.5 dB. (See Figure 23.) Wire or cable loops used for telegraph and data service employing quasi-analog signals should be engineered to obtain the smallest possible loop insertion loss. (See 60.3.1.2 of Appendix F.)

NOTE: The insertion loss of wire or cable loops employed in the tactical subsystem type II is not controlled and no adjustments are made to keep the loop insertion loss within prescribed limits. A statistical distribution of the loop insertion loss is assumed for the purpose of system engineering and circuit design. This statistical distribution has been based on field wire loops installed in typical configurations of the tactical subsystem type II.

5.5.2.3 Insertion loss of wire or cable loops for tactical subsystem type IV. Under consideration.

5.5.2.4 Insertion loss versus frequency characteristics of wire or cable loops.

5.5.2.4.1 Loops for analog voice and analog facsimile service. For a single wire or cable loop (two-wire or four-wire), the insertion loss versus frequency characteristic, referenced to 1000 Hz, shall not exceed the values given in Table XX over the frequency bands stated. (See Figure 24A.) This requirement shall apply to loops for analog voice and analog facsimile service of the tactical subsystems type I, II and III. (See Table I.)

NOTE 1: In Figures 24A, 25A and 26A, the loop insertion loss versus frequency characteristic is referenced to the loop insertion loss at 1000 Hz. Therefore, relative loop insertion loss (with respect to the loss at

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

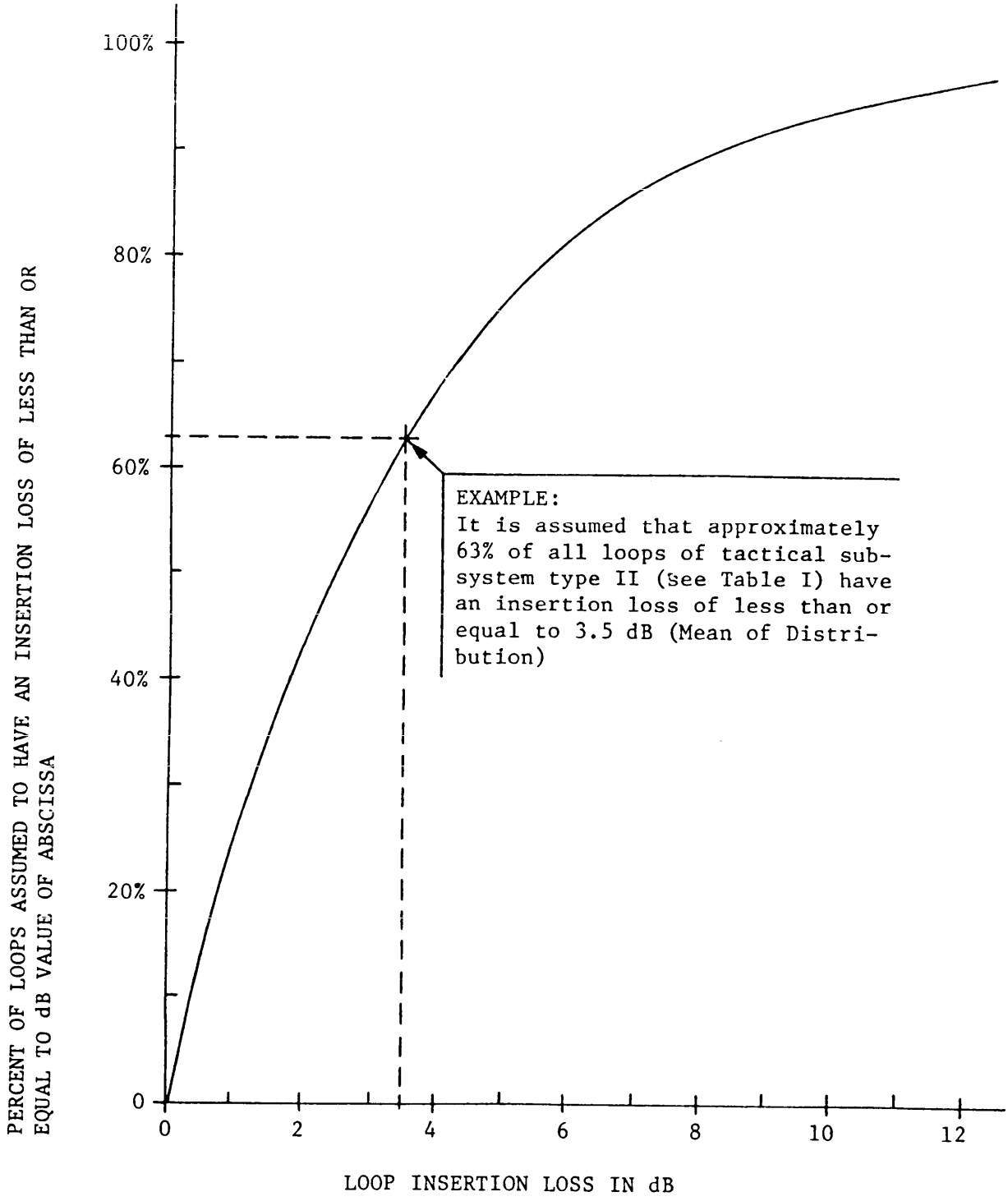
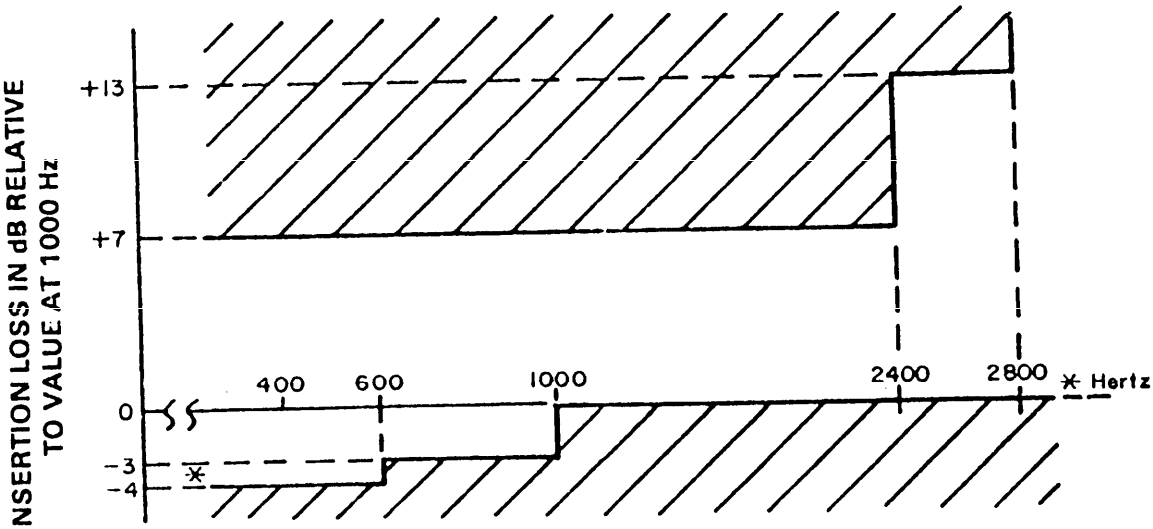


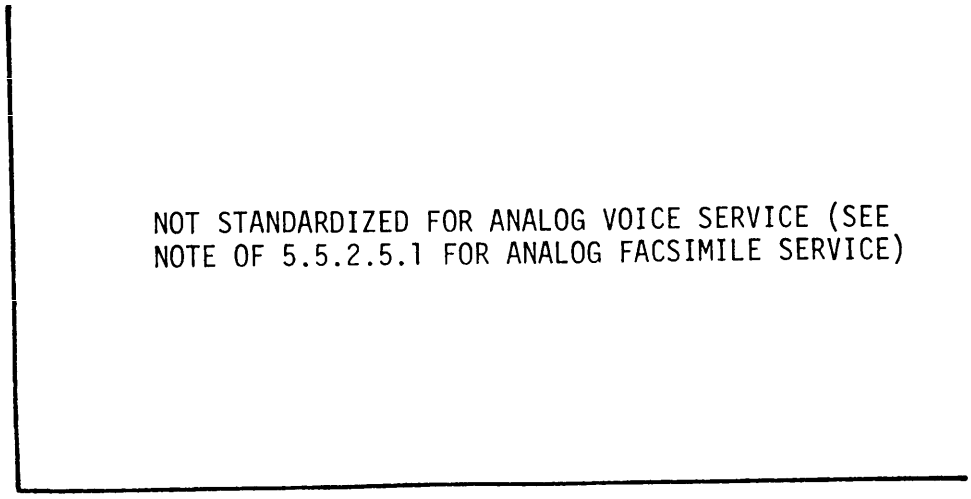
FIGURE 23. Assumed insertion loss distribution for wire or cable loops of tactical subsystem type II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



* These upper and lower frequency limits are not specified

A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 24. Parameters of wire or cable loops for analog voice and analog facsimile service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

1000 Hz) is indicated by a (+) sign and relative loop insertion gain is indicated by a (-) sign.

NOTE 2: It should be noted that an unconditioned loop is unconditioned only in terms of data conditioning. (See 60.3.3 of Appendix F.) Such unconditioned loops may be "conditioned" for analog service by the use of loading coils or other devices.

TABLE XX. Insertion loss versus frequency characteristic of wire or cable loops for analog voice and analog facsimile service.

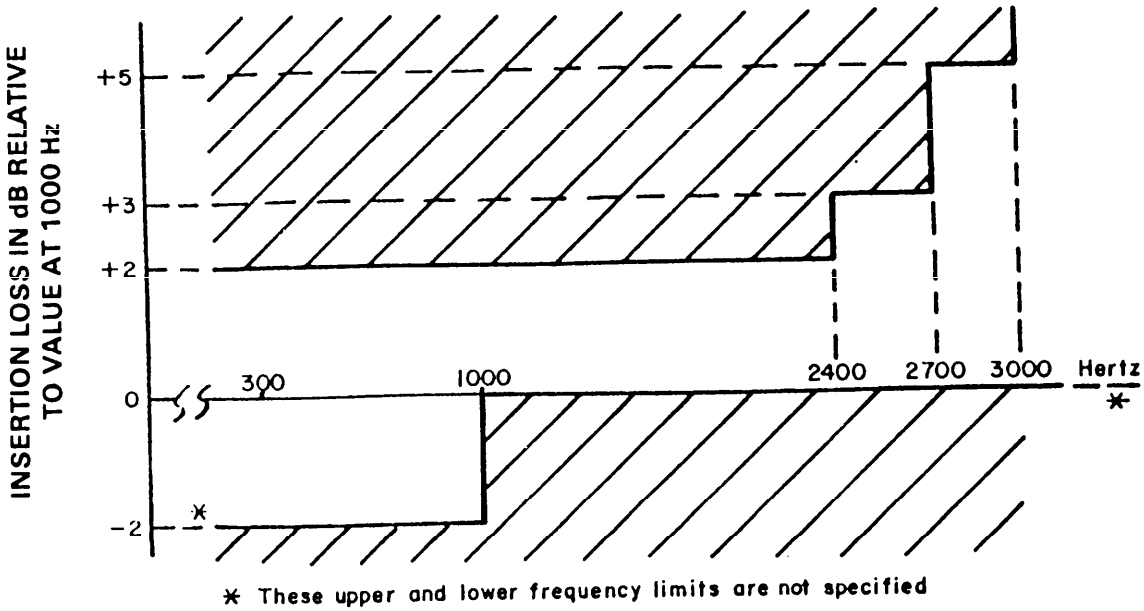
Frequency in Hz	Insertion Loss in dB (Referenced to 1000 Hz)
$f < 400$	≥ -4
$400 \leq f < 600$	-4 to +7
$600 \leq f < 1000$	-3 to +7
$1000 \leq f < 2400$	0 to +7
$2400 \leq f < 2800$	0 to +13
$2800 \leq f$	≥ 0

5.5.2.4.2 Loops for telegraph and data service with modulation rates of up to 1200 Bd. For a single wire or cable loop (two-wire or four-wire), the insertion loss versus frequency characteristic, referenced to 1000 Hz, shall not exceed the values given in Table XXI over the frequency bands stated. (See Figure 25A.) This requirement shall apply to loops of the tactical subsystems type I, II and III (see Table I) when the loops are used for the transmission of quasi-analog signals with modulation rates of up to 1200 Bd. (See Notes 1 and 2 of 5.5.2.4.1.)

NOTE 1: The standards of 5.5.2.4.2 and 5.5.2.4.3 have been established for the transmission of quasi-analog signals. Standards and design objectives on loop characteristics for the transmission of digital signals are under consideration. (See 5.5.6.)

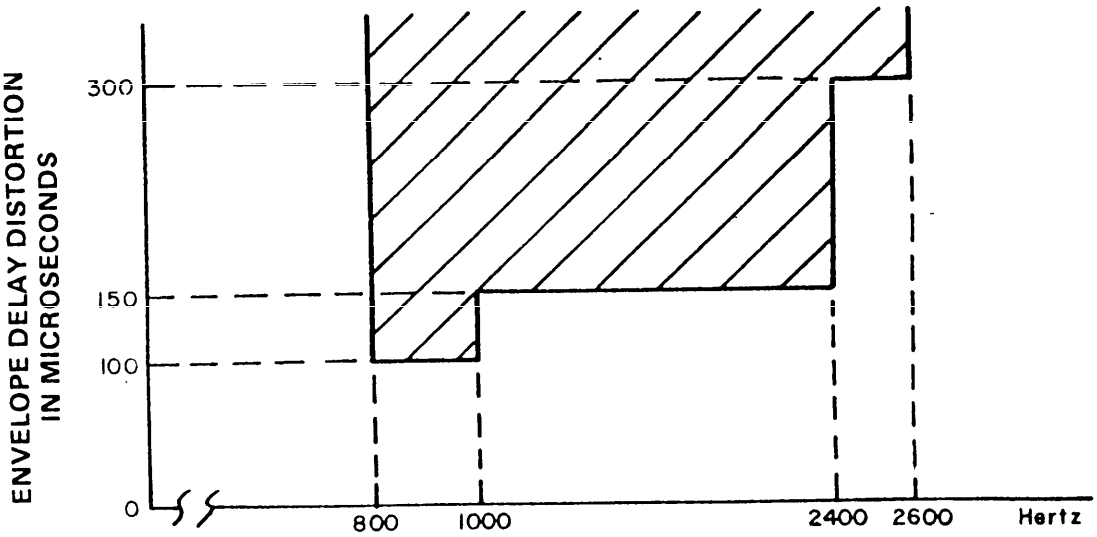
NOTE 2: The insertion loss versus frequency characteristics listed in Tables XXI and XXII are based on a worst case criteria. It is not

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



* These upper and lower frequency limits are not specified

A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 25. Parameters of wire or cable loops for telegraph and data service with modulation rates of up to 1200 Bd.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

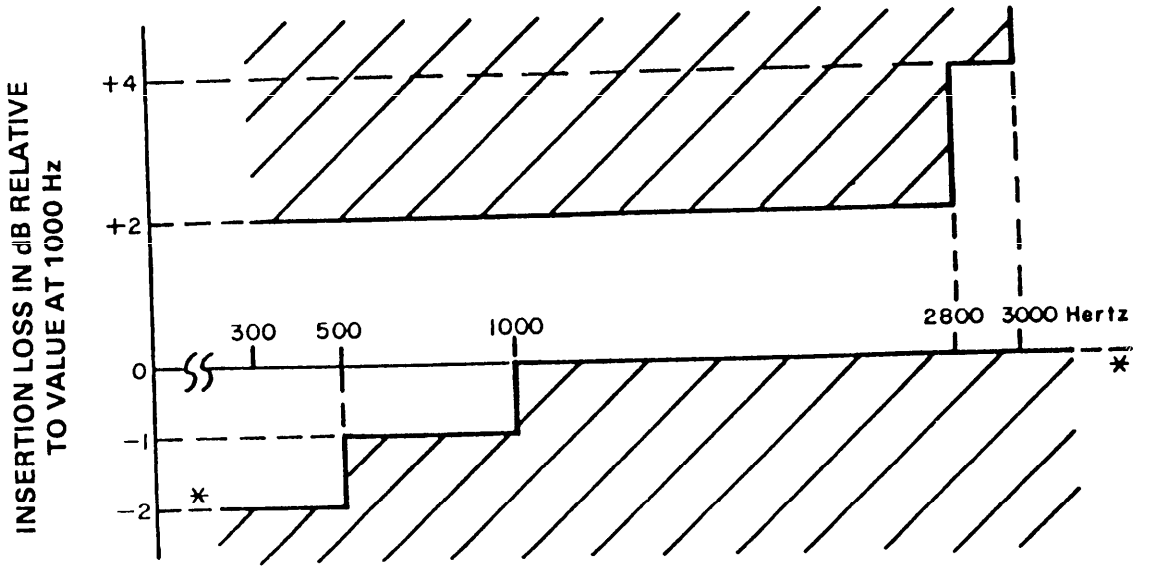
intended that this loop parameter together with those of tandem trunks "add up" (see 5.9.3.3) to the total allowable insertion loss versus frequency characteristic of VF circuits for telegraph and data service stated in 5.2.3.1.6.1. Circuit engineering which may include the use of necessary equalization (or conditioning) and regeneration (see 60.3.3 and 60.3.4 of Appendix F) must take this situation into account.

TABLE XXI. Insertion loss versus frequency characteristic of wire or cable loops for telegraph and data service with modulation rates of up to 1200 Bd.

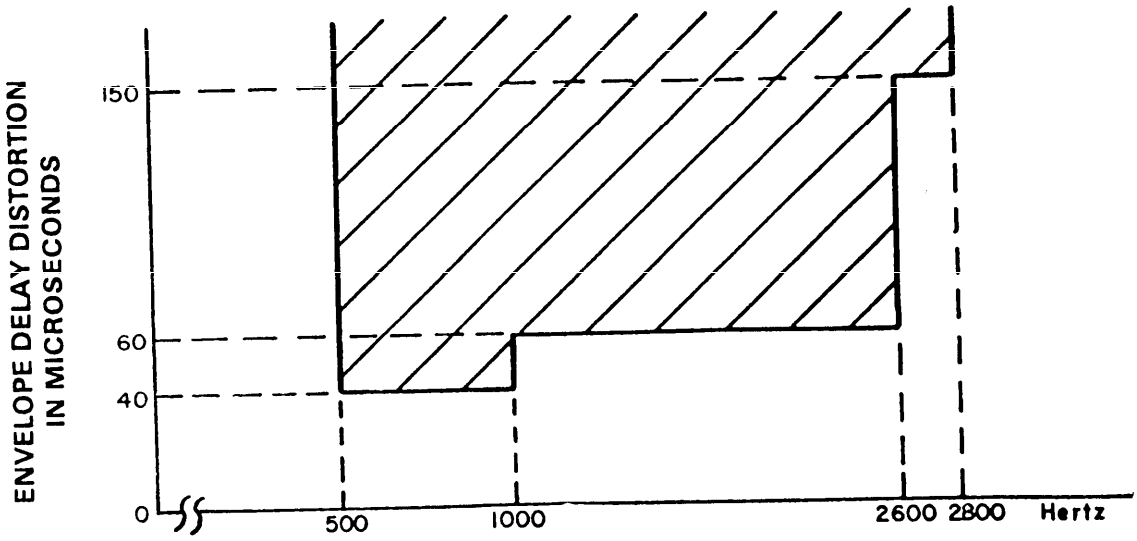
Frequency in Hz	Insertion Loss in dB (Referenced to 1000 Hz)
$f < 300$	≥ -2
$300 \leq f < 1000$	-2 to +2
$1000 \leq f < 2400$	0 to +2
$2400 \leq f < 2700$	0 to +3
$2700 \leq f < 3000$	0 to +5
$3000 \leq f$	≥ 0

5.5.2.4.3 Loops for telegraph and data service with modulation rates of up to 2400 Bd. For a single wire or cable loop (two-wire or four-wire), the insertion loss versus frequency characteristic, referenced to 1000 Hz, shall not exceed the values given in Table XXII over the frequency bands stated. (See Figure 26A.) This requirement shall apply to loops of the tactical subsystems type I, II and III (see Table I) when the loops are used for the transmission of quasi-analog signals with modulation rates of up to 2400 Bd. (See Notes 1 and 2 of 5.5.2.4.1 and Notes 1 and 2 of 5.5.2.4.2.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 26. Parameters of wire or cable loops for telegraph and data service with modulation rates of up to 2400 Bd.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XXII. Insertion loss versus frequency characteristic of wire or cable loops for telegraph and data service with modulation rates of up to 2400 Bd.

Frequency in Hz	Insertion Loss in dB (Referenced to 1000 Hz)
$f < 300$	≥ -2
$300 \leq f < 500$	-2 to +2
$500 \leq f < 1000$	-1 to +2
$1000 \leq f < 2800$	0 to +2
$2800 \leq f < 3000$	0 to +4
$3000 \leq f$	≥ 0

5.5.2.5 Envelope delay distortion of wire or cable loops.

5.5.2.5.1 Loops for analog voice and analog facsimile service. The envelope delay distortion of a loop is not standardized for analog voice service (see Note of 5.2.2.1.5) and is under consideration for analog facsimile service.

NOTE: Envelope delay distortion may play a significant role for analog facsimile transmission and the transmission of quasi-analog signals requiring the use of loops for telegraph and data service. The envelope delay distortion of loops for telegraph and data service is given in 5.5.2.5.2 and 5.5.2.5.3. Until parameter values are established for the envelope delay distortion of loops for facsimile service, the envelope delay distortion of loops for telegraph and data service given in 5.5.2.5.2 and 5.5.2.5.3 should be used as guidance for analog facsimile service.

5.5.2.5.2 Loops for telegraph and data service with modulation rates of up to 1200 Bd. As a Design Objective for a single wire or cable loop (two-wire or four-wire), the envelope delay distortion measured between 800 Hz and 2600 Hz (see 70.5 of Appendix G), should not exceed the values given in Table XXIII over the frequency bands stated. (See Figure 25B.) This requirement applies to loops of the tactical subsystems type I, II and III (see Table I) when the loops are used for the transmission of quasi-analog signals with modulation rates of up to 1200 Bd.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE 1: The design objectives of 5.5.2.5.2 and 5.5.2.5.3 have been established for the transmission of quasi-analog signals. Standards and design objectives on loop characteristics for the transmission of digital signals are under consideration. (See 5.5.6.)

NOTE 2: The parameter values listed in Table XXIII have been based on measured performance of installed loops. These values have been selected to meet the envelope delay distortion of a telegraph and data circuit given in 5.2.3.1.6.2.1 for circuits of the tactical subsystems type II and type III engineered in accordance with the configurations shown in Figures 54, 55, 57 and 58. Experience has shown that loops may provide acceptable telegraph and data service with modulation rates of up to 1200 Bd when the envelope delay distortion of the loop is outside of the limits given in Table XXIII. Therefore, the envelope delay distortion values listed in Table XXIII should not be used exclusively to determine the need for equalization (or conditioning) data loops and regenerating data signals (see 60.3.3 and 60.3.4 of Appendix F), rather, the characteristics of the entire circuit from data source to data sink should be considered.

TABLE XXIII. Envelope delay distortion of wire or cable loops for telegraph and data service with modulation rates of up to 1200 Bd.

Frequency in Hz	Maximum Envelope Delay Distortion in microseconds
$800 \leq f < 1000$	100
$1000 \leq f < 2400$	150
$2400 \leq f \leq 2600$	300

5.5.2.5.3 Loops for telegraph and data service with modulation rates of up to 2400 Bd. As a Design Objective for a single wire or cable loop (two-wire or four-wire), the envelope delay distortion measured between 500 Hz and 2800 Hz (see 70.5 of Appendix G), should not exceed the values given in Table XXIV over the frequency bands stated. (See Figure 26B.) This requirement applies to loops of the tactical subsystems type I, II and III (see Table I) when the loops are used for the transmission of quasi-analog signals with modulation rates of up to 2400 Bd. (See Note 1 of 5.5.2.5.2.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: The parameter values listed in Table XXIV have been based on measured performance of installed loops. Experience has shown that loops may provide acceptable telegraph and data service with modulation rates of up to 2400 Bd when the envelope delay distortion of the loop is outside of the limits given in Table XXIV. Therefore, the envelope delay distortion values listed in Table XXIV should not be used exclusively to determine the need for equalization (or conditioning) data loops and regenerating data signals (see 60.3.3 and 60.3.4 of Appendix F), rather, the characteristics of the entire circuit from data source to data sink should be considered.

TABLE XXIV. Envelope delay distortion of wire or cable loops for telegraph and data service with modulation rates of up to 2400 Bd.

Frequency in Hz	Maximum Envelope Delay Distortion in microseconds
$500 \leq f < 1000$	40
$1000 \leq f < 2600$	60
$2600 \leq f \leq 2800$	150

5.5.2.6 Noise power. As a Design Objective, the noise power of a single wire or cable loop (two-wire or four-wire) should not exceed 6300 pWp (38.0 dBrnc) for analog voice service and 6300 pW (38.0 dBrn) for analog facsimile service and telegraph and data service. The noise power shall be measured at the loop output terminals with the loop disconnected from the terminal equipment and the nodal equipment, and both input and output terminals of the loop properly terminated.

NOTE 1: The noise power of a loop is not referenced to a circuit OTLP but stated as an absolute value. The contribution of the loop noise to the total circuit noise is considered to be independent of the TLP assigned to the loop when connected to terminal and nodal equipment to form part of a circuit. (See Figures 50 through 60 and 120.3 of Appendix L.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE 2: Noise power with C-message weighting is used for loops employed for analog voice transmission, whereas noise power with flat weighting over the frequency band of interest is used for loops employed for analog facsimile transmission and telegraph and data transmission.

5.5.2.7 Impedance. Not standardized.

NOTE: The impedance of wire or cable loops varies widely with frequency, temperature, moisture and length. The nominal value of the impedance is taken to be between 600 ohms and 900 ohms, balanced to ground.

5.5.2.8 Net loss variations. (MIL-STD-188-100.) The net loss variations of a single wire or cable loop shall not exceed ±1.5 dB over any 30 consecutive days and ±0.5 dB over any 15 consecutive minutes.

NOTE: The values given in 5.5.2.8 do not account for net loss variations of unprotected loops, such as field wire loops generally installed in the tactical subsystems type I and II (see Table I). Net losses of unprotected loops may vary up to 0.7 dB/km (1.3 dB/nmi) between wet and dry weather conditions.

5.5.2.9 Test signal characteristics. The signal used for transmission testing on loops shall have a frequency of 1000 Hz, ±25 Hz, and shall be at a level of 0 dBm0, e.g., -4 dBm at a -4TLP. (See Figures 50 and 54.)

NOTE: A test tone signal (see 5.6.6.2) with a level of -10 dBm0 may be used as an option providing it does not cause operational problems.

5.5.3 Parameters for two-wire radio loops. Two-wire radio loops shall be operated in one of the following two ways: (1) Two-wire at both ends on a push-to-talk basis (one-way at a time) employing NRI equipment (see 5.4.5), or (2) four-wire at the subscriber end and two-wire at the switchboard end.

5.5.3.1 Loop insertion loss. The insertion loss of a two-wire radio loop measured at 1000 Hz, ±25 Hz, shall be 0 dB, ±2 dB.

5.5.3.2 Noise power. As a Design Objective, the noise power of a single two-wire radio loop should not exceed 25,000 pWp (44.0 dBrnc) for analog voice service, and 25,000 pW (44.0 dBrn) flat weighted over the frequency band of interest for analog facsimile service and telegraph and data service. The noise power shall be measured at the loop output terminals with the loop disconnected from the terminal equipment and the nodal equipment and both input and output terminals of the loop properly terminated. (See Notes 1 and 2 of 5.5.2.6.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.5.3.3 Terminal impedance. The terminal impedance of two-wire radio loops shall be 600 ohms, balanced to ground, with a minimum return loss of 20 dB (Design Objective: 26 dB) against a 600-ohm resistance over the frequency band of interest. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

5.5.3.4 Test signal characteristics. Same as 5.5.2.9.

5.5.4 Parameters for four-wire radio loops. Four-wire radio loops shall comply with the same standards as two-wire radio loops, stated in 5.5.3.1 through 5.5.3.4.

5.5.5 Parameters of wire or cable loops for data service over 48 kHz FDM group circuits. Under consideration.

5.5.6 Parameters of wire or cable loops for digital signals. Under consideration.

NOTE: The characteristics of loops used to transmit digital signals will vary widely among tactical systems. In addition to the different types of loops, the wire or cable parameters (impedance, dc resistance, capacitance and length) are not standardized and will be determined by application engineering. General guidance is provided in MIL-STD-188-114 concerning wire and cable characteristics for digital interface circuits.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.6 Nodal subsystems.

5.6.1 General. The configuration of a tactical nodal subsystem will vary widely depending on the various functions this subsystem has to perform (see also 4.3.4). It is not intended that all possible configurations of a nodal subsystem be covered in this paragraph. Instead, emphasis is placed on those parameters that are considered essential to ensure interoperation between loop and trunk subsystems, and among different types of equipment within a nodal subsystem.

5.6.2 Switching equipment.

5.6.2.1 Analog automatic circuit switches. Analog automatic circuit switches shall be designed for switching nominal 4 kHz trunks and loops on a four-wire full-duplex basis and shall comply with the applicable requirements of 5.6.2.1.1 through 5.6.2.1.7.

NOTE: It is recognized that some switches have been designed with transmission characteristics that permit the switching of channels with a bandwidth greater than 4 kHz. These wideband characteristics are considered as optional and are not standardized in this document.

5.6.2.1.1 Interconnection requirements. Conversion equipment shall be provided in tactical automatic or semiautomatic circuit switches that have requirements to interconnect with other types of switching equipment as listed in 5.6.2.1.1.1 through 5.6.2.1.1.3.

5.6.2.1.1.1 Interconnection with manual switchboards. Automatic or semiautomatic circuit switches that interconnect with manual switchboards shall comply with the applicable requirements of 5.6.2.3.1 through 5.6.2.3.4.

5.6.2.1.1.2 Interconnection with tactical NATO circuit switches. Circuit switches that interconnect with tactical NATO circuit switches shall comply with the applicable requirements of the current editions of STANAG 5018 and STANAG 5040. (See Note 1 of 5.1.1.4.)

5.6.2.1.1.3 Interconnection with long haul AUTOVON switching centers. Tactical automatic or semiautomatic circuit switches that interconnect with long haul AUTOVON switching centers shall comply with the applicable requirements of the current edition of DCAC 370-V175-6.

5.6.2.1.2 Transmission characteristics.

5.6.2.1.2.1 Insertion loss. For any VF channel connection through an analog circuit switch, the insertion loss shall be 0 dB, ±0.5 dB, when measured with a standard test signal in accordance with 5.6.6.1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.6.2.1.2.2 Insertion loss versus frequency characteristic. For any VF channel connection through an analog circuit switch, the insertion loss versus frequency characteristic, referenced to 1000 Hz, shall be within the limits of ±1 dB (Design Objective: ±0.5 dB) between 300 Hz and 3400 Hz.

5.6.2.1.2.3 Envelope delay distortion. For any VF channel connection through an analog circuit switch, the envelope delay distortion shall not exceed 25 microseconds between 500 Hz and 3200 Hz. (See 70.5 of Appendix G.)

5.6.2.1.2.4 Total harmonic distortion. For any VF channel connection through an analog circuit switch, the total harmonic distortion produced by any single-frequency test signal between 300 Hz and 3400 Hz shall be at a level which is at least 35 dB below reference level (-35 dBm0) with a Design Objective of 40 dB (-40 dBm0). The total harmonic distortion shall be measured with a standard test signal in accordance with 5.6.6.1.

5.6.2.1.2.5 Noise power. The idle channel noise power contributed by an analog circuit switch shall not exceed 400 pWp (26.0 dBrnc) for voice transmission and 630 pW (28.0 dBrn) for quasi-analog data transmission. The impulse noise level, due to switching or any other switch function not associated with the connection being measured, shall not exceed 0.004 volts (4 millivolts) peak amplitude. The idle channel and impulse noise shall be measured at the output terminals of any idle VF channel connection with input and output terminals properly terminated and the insertion loss of the idle channel adjusted in accordance with 5.6.2.1.2.1.

5.6.2.1.2.6 Limiting. An analog circuit switch shall not overload and start limiting when a signal with a frequency between 300 Hz and 3400 Hz is transmitted through any VF channel connection at a level of at least 8 dB above the switch transmission level (see 5.6.2.1.2.9).

5.6.2.1.2.7 Terminal impedance. The impedance of the input and output terminals of any VF channel connection through an analog circuit switch shall each be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band from 300 Hz to 3400 Hz. The electrical symmetry shall be sufficient to suppress longitudinal currents in each direction of transmission to a level which is at least 40 dB below reference level (-40 dBm0).

5.6.2.1.2.8 Crosstalk coupling loss. The crosstalk coupling loss between any two VF channel connections through an analog circuit switch shall be equal to or greater than 60 dB (Design Objective: 70 dB) for any frequency between 300 Hz and 3400 Hz, measured at or referenced to equal TLP.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

5.6.2.1.2.9 Transmission levels. The transmission levels at the input and output terminals of any VF channel connection through an analog circuit switch shall be as shown in Table XXV.

TABLE XXV. Transmission levels for analog circuit switches.

Tactical Subsystem (See Table I)	Type of Connection	Transmission Level (in dBm)
Type I	2-Wire	0
	4-Wire	0
Type II	2-Wire	0*
	4-Wire	-4
Type III	2-Wire	0
	4-Wire	0

* In the tactical subsystem type II, the test signal level of a circuit containing two-wire switches is adjusted so that only the first two-wire switch in the direction of transmission will have a level of 0 dBm. All other two-wire switches will have transmission levels lower than 0 dBm due to the insertion loss of 3 dB of each trunk connected to a two-wire output termination. (See Figure 45.)

5.6.2.1.3 Control and signaling supervision procedures. Control and signaling supervision procedures between and among analog automatic or semiautomatic circuit switches shall be as follows:

a. The seizure signal shall be an untimed signal transmitted in the forward direction (calling to called switch).

b. The called switch, after proper detection and identification of the seizure signal, shall initiate instructions for connecting an available idle register receiver to the trunk requesting service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

c. When the register is ready to receive address and routing information, a DTMF restart signal (R, start sending) shall be sent to the calling switch.

d. The calling switch, upon detection of the R signal, shall cease transmitting the seizure signal to the called switch.

e. The called switch shall continue to transmit the R signal until the calling switch starts sending the first DTMF digit signal in the address sequence.

f. When the called switch receives this digit signal, the switch shall confirm its receipt by sending a confirmation signal (see 5.6.2.1.4) to the calling switch.

g. The calling switch, upon detection and verification of the confirmation signal (see 5.6.2.1.4), shall proceed to send the interdigit signal (X).

h. The called switch, upon receipt of the X signal, shall return a proceed signal (P) to the calling switch as an instruction to send the next digit in the address sequence.

i. This procedure between the calling and called switches shall be repeated until the last digit in the Primary Area Switch Locator (PRSL) office code (see 5.6.3) is sent and confirmed.

j. When the register of the called switch receives the next interdigit signal, after the office code, the called switch shall examine the received office code digits.

k. If the called switch determines that the received office code is not its own and the call has to be transferred to another (third) switch, then the process shall continue with step m. If the called switch determines that the received office code is its own, then the address sending shall continue until the receiving register detects that the last digit is received.

l. The last X signal sent forward shall stop the sending of the last confirmation signal in the reverse direction. The register in the calling switch shall be released and the called switch shall proceed to process the call to the subscriber corresponding to the received address. This process completes the procedure between two switches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

m. If the called switch has determined that the call has to be transferred to another switch, then the called switch shall seize an idle trunk and proceed to send the trunk seizure signal to the next (third) switch.

n. Once this seizure is accomplished, in the same manner as it was accomplished between the calling and the first called switch, a register receiver in the third switch shall be connected to the calling trunk.

o. An R signal shall then be sent in the reverse direction from the third to the second switch.

p. When this R signal is received in the second switch, the register in this switch shall be released and a connection shall be established between the two trunks (first and third switch).

q. The R signal, which is still being sent from the third switch, shall now be received in the first switch and this shall be interpreted as a start R signal to restart sending the called address.

r. The first switch shall start anew to send the first digit in the address sequence and the entire procedure shall be repeated between the first and the third switch, instead of the first and the second switch.

s. When the office code and the X signal is sent, the digits shall now be examined, but this time by the third switch. This procedure can be repeated for establishing tandem trunk connections through more than three switches.

t. The last X signal sent forward shall stop the sending of the last confirmation signal in the reverse direction. The register in the calling switch shall be released and the called switch shall proceed to process the call to the subscriber corresponding to the received address. This process completes the procedure among three or more switches.

5.6.2.1.4 Confirmation signaling. The following requirement shall apply to analog automatic or semiautomatic circuit switches. For each digit or precedence signal sent over a trunk by the calling switch, a confirmation signal shall be returned from the called switch to the calling switch. The confirmation signals returned shall consist of the frequencies shown in Table XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: For example, if digit 1 with frequencies of 697/1209 Hz (see Table XXVII) is sent, the confirmation signal returned will have frequencies of 941/1633 Hz.

TABLE XXVI. Confirmation signaling for analog automatic circuit switches.

Frequency Group	Frequency (in Hz) of Digit or Precedence Signal Sent	Frequency (in Hz) of Confirmation Signal Returned
Low Group	697 770 852 941	941 852 770 697
High Group	1209 1336 1477 1633	1633 1477 1336 1209

5.6.2.1.5 Signaling and supervisory signals. Signaling and supervisory signals of analog automatic circuit switches shall be DTMF and SF signals with frequency tolerances that are within ±1.3 percent of the frequencies and with single tone levels as listed in Table XXVII. For DTMF digits 1 through 0, the total time (see Note) shall be nominally 200 ms but no DTMF digit shall be transmitted for longer than 1 s.

NOTE: The total time starts at the beginning of transmitting one digit until the beginning of transmitting the next digit and includes confirmation signaling and interdigit, proceed or restart signals, if applicable.

5.6.2.1.6 Tone detectors. The tone detectors of analog automatic circuit switches shall respond to DTMF and SF signals with frequency tolerances that are within ±1.5 percent of the frequencies listed in Table XXVII and with single tone levels between 0 dBm and -25 dBm. The tone detectors shall not respond to DTMF and SF signals that have frequency tolerances outside of ±5 percent (Design Objective: ±3 percent) of the frequencies listed in Table XXVII.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XXVII. Signaling and supervisory signals for analog automatic circuit switches.

Function	Frequency (Hz)	Single Tone Level (dBm)	Duration	Remarks
Digits 1 2 3 4 5 6 7 8 9 0	697/1209 697/1336 697/1477 770/1209 770/1336 770/1477 852/1209 852/1336 852/1477 941/1336	-7, ±2	See 5.6.2.1.5	
Seize	2250	-7, ±2	250 ms to 1.5 s	Cutoff by seize acknowledge
Seize Acknowledge (Loops)	570	-7, ±2	Continuous up to 1.5 s	Until seize stops plus 250 ms
Seize Acknowledge (Trunks) Restart Signal (R)	852/1209 (Digit 7)	-7, ±2	Continuous up to 1.5 s	Until seize stops plus 250 ms
Dial Tone	425	-14, ±2	Continuous up to 10 s	Cutoff by first digit
Ring Loop (Normal)	570	-4, ±2	3 min, ±0.5 min cutoff by ring trip tone	50 ms on, 50 ms off, 2 s on, 4 s off
Ring Loop (Priority)	570/425 (FSK)	-4, ±2	3 min, ±0.5 min cutoff by ring trip tone	50 ms 570 Hz 50 ms 425 Hz 1 s on, 1 s off
Loop Busy Tone	425	-14, ±2	10 s, ±3 s	250 ms on 250 ms off

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XXVII. (CONTINUED) Signaling and supervisory signals for analog automatic circuit switches.

Function	Frequency (Hz)	Single Tone Level (dBm)	Duration	Remarks
Trunk Busy Tone	425	-14, ±2	10 s, ±3 s	250 ms on 250 ms off
Ringback Tone	570	-14, ±2	Up to 3 min, ±0.5 min	Stops when called telephone goes off hook
Ring Trip Tone	570	-14, ±2	250 ms to 2 s	Until ring trip stops
Released Party Preempt Tone	570	-14, ±2	1.25 s, ±0.25 s	
Preempt Tone	570/425 (FSK)	-14, ±2	1.25 s, ±0.25 s	50 ms 570 Hz 50 ms 425 Hz
Precedence Levels FO F I P	697/1633 770/1633 852/1633 941/1633	-7, ±2	See 5.6.2.1.5	
Release	2600	-7, ±2	Continuous up to 3 s	Cutoff by release acknowledge
Release Acknowledge (Trunks)	2600	-7, ±2	250 ms to 3 s	Cutoff by the end of release
Recall Response	570	-14, ±2	Until recall stops	
Proceed Signal	941/1209	-7, ±2	See 5.6.2.1.5	Cutoff by digit signal
Interdigit Signal	941/1633	-7, ±2	See 5.6.2.1.5	Cutoff by proceed or restart signals

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.6.2.1.7 Release signal protection (voice guard). The design of circuit switches shall be such that release signals are properly recognized by the switch in single or tandem connections. Only a specified number of connections (see Note) shall be falsely released as a result of voice simulations.

NOTE: The number of calls falsely released and other details regarding acceptance testing of circuit switches will be determined in applicable equipment specifications based on validated requirements.

5.6.2.2 Analog semiautomatic circuit switches. Analog semiautomatic (cordless) circuit switches shall be designed for switching nominal 4 kHz trunks and loops on a four-wire full-duplex basis and shall comply with the applicable requirements of 5.6.2.2.1 through 5.6.2.2.7.

5.6.2.2.1 Interconnection requirements.
Same as 5.6.2.1.1 through 5.6.2.1.1.3.

5.6.2.2.2 Transmission characteristics.
Same as 5.6.2.1.2.1 through 5.6.2.1.2.9.

5.6.2.2.3 Control and signaling supervision procedures.
Same as 5.6.2.1.3.

5.6.2.2.4 Confirmation signaling. Same as 5.6.2.1.4.

5.6.2.2.5 Signaling and supervisory signals. Signaling and supervisory signals of analog semiautomatic (cordless) circuit switches shall be DTMF and SF signals with frequency tolerances that are within ±1.5 percent of the frequencies and with single tone levels as listed in Table XXVIII. For DTMF digits 1 through 0, the total time (see Note of 5.6.2.1.5) shall be nominally 200 ms but no DTMF digit shall be transmitted for longer than 1 s.

5.6.2.2.6 Tone detectors. The tone detectors of analog semiautomatic (cordless) circuit switches shall respond to DTMF and SF signals with frequency tolerances that are within ±2 percent of the frequencies listed in Table XXVIII and with single tone levels between -5 dBm and -25 dBm. The tone detectors shall not respond to DTMF and SF signals that have frequency tolerances outside of ±5 percent (Design Objective: ±3 percent) of the frequencies listed in Table XXVIII.

5.6.2.2.7 Release signal protection (voice guard). Same as 5.6.2.1.7.

5.6.2.3 Analog manual switchboards.

5.6.2.3.1 Interconnection with tactical NATO circuit switches. Analog manual switchboards that interconnect with a tactical manual switchboard or the manual position of an automatic circuit switch of another NATO member nation shall comply with the applicable requirements of the current edition of STANAG 5018.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XXVIII. Signaling and supervisory signals for analog semiautomatic circuit switches.

Function	Frequency (Hz)	Single Tone Level (dBm)	Duration	Remarks
Digits 1 2 3 4 5 6 7 8 9 0	697/1209 697/1336 697/1477 770/1209 770/1336 770/1477 852/1209 852/1336 852/1477 941/1336	-14, ±2	Operator controlled	Operator originated
Seize	2250	-14, ±1	Minimum of 250 ms	Cutoff by seize acknowledge
Seize Acknowledge	570	-14, ±1	Continuous until seize stops	
Dial Tone (to automatic switch)	425	-14, ±1	Continuous up to 10 s	Cutoff by first digit
Ring Loop (Normal)	570	-4, ±2	3 min, ±0.5 min cutoff by ring trip tone	50 ms on, 50 ms off, 2 s on, 4 s off
Ring Loop (Priority)	570/425 (FSK)	-4, ±2	3 min, ±0.5 min cutoff by ring trip tone	50 ms 570 Hz 50 ms 425 Hz 1 s on, 1 s off
Ringback Tone	570	-14, ±1	Up to 3 min, ±0.5 min	Stops when called telephone goes off hook
Ring Trip Tone	570	-14, ±1	Minimum of 250 ms	Cutoff by end of ring trip
Preempt Tone	1000	-14, ±1	Operator controlled or timed	0.25 s on 0.25 s off

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XXVIII. (CONTINUED) Signaling and supervisory signals for analog semiautomatic circuit switches.

Function	Frequency (Hz)	Single Tone Level (dBm)	Duration	Remarks
Precedence Levels FO F I P	697/1633 770/1633 852/1633 941/1633	-14, ±1	Operator controlled	
All Links Busy	1000	-14, ±1	Subscriber controlled	0.25 s on 0.25 s off
Recall/Line Priority	941/1209	-14, ±1	Operator controlled	
Recall Response	570	-14, ±1	Until recall stops	
Release	2600	-14, ±1	Continuous, 3 s to 10 s	Cutoff by release acknowledge
Release Acknowledge	570	-14, ±1	250 ms to 10 s	Cutoff by end of release

5.6.2.3.2 Trunk signaling.

5.6.2.3.2.1 Wire or cable trunks. Manual switchboards connected to wire or cable trunks shall be capable of delivering a ringdown signal with a frequency of 20 Hz, ±5 Hz, and a level of not less than 36 volts rms into a resistive load of 1500 ohms, ±10 percent. The open circuit voltage of the ringdown signal generator shall not exceed 60 volts rms. The duration of the ringdown signal shall be determined by manual operation of a key or an equivalent device. The ringdown signal receiver in the switchboard shall be capable of operating with a received signal level between 15 volts rms and 110 volts rms and a frequency of 20 Hz, ±5 Hz.

NOTE: Interoperability with older ringdown signaling equipment still in the inventory requires an upper limit of 110 volts rms of the received signal level.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

Page 155 is missing from the original legacy source document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE 2: Starting in 1978 some NATO member nations will use a deducible constant-length telephone directory numbering plan in accordance with STANAG 5046 in tactical multichannel systems employing automatic circuit switches.

TABLE XXIX. Telephone directory numbering plan.

Digit Position	Identification	Range of Numbers
PR	Primary Area	70-98*
SL	Switch Locator	00-99
IXX	3-Digit Subscriber Identification and Abbreviated Dialing	100-699
JXX	Manual Switchboard Identification	700-999
	Local Operator	0

*See 5.6.3.3 for restriction of AUTOVON connections.

5.6.3.1 Switch identification. Every automatic and semiautomatic nodal switch shall be identified by a code of the form PRSL and every manual switchboard shall be identified by a code of the form JXX in accordance with Table XXIX.

5.6.3.2 Routing. Automatic switching equipment shall have appropriate programs to route a long distance call and to establish a connection on a primary or an alternate route by using the 7-digit directory number of the form PRSLIXX in accordance with 5.6.3.

5.6.3.3 Primary area (PR) codes. The range of numbers used as PR codes shall be from 70 through 98 inclusive. PR codes 70, 71, 80, 81, 90 and 91 shall not be used for automatic interconnections between tactical subscribers and the AUTOVON. (See Note.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: The AUTOVON follows the commercial practice where the second digit of a 3-digit area code is always "0" or "1" which is used by a switch as an indication that a 10-digit number (3-digit area code plus 7-digit subscriber number) is being dialed. Consequently, a "0" or "1" cannot be assigned as the second digit of any 7-digit subscriber number.

5.6.3.4 Switchboard locator (SL) codes. The range of numbers used as SL codes shall be from 00 through 99 inclusive.

5.6.3.5 Subscriber number codes. The range of numbers used as subscriber number codes shall be from 100 through 699 inclusive. Subscriber number code 0 shall be reserved for the local operator.

5.6.3.6 Abbreviated dialing codes. The range of numbers used as abbreviated dialing codes shall be from 100 to 600 inclusive.

5.6.3.7 Manual switchboard codes. The range of numbers used as manual switchboard codes (JXX) shall be from 700 through 999 inclusive.

5.6.3.8 Fixed directory codes. As a Design Objective for automatic circuit switches, a capability for fixed directory codes should be provided to assign fixed directory numbers to those subscribers or units that have frequent relocation requirements.

5.6.4 Control facilities. Under consideration.

5.6.5 Patch and test facilities. Under consideration.

5.6.6 Analog signal characteristics for testing, measuring and monitoring.

5.6.6.1 Standard test signal. (MIL-STD-188-100.) The level of the standard test signal shall be 0 dBm at a OTLP, or 0 dBm0, with a frequency of 1000 Hz, ±25 Hz. A test signal frequency of 1004 Hz is preferred for PCM transmission. (See Notes 1 and 2 of 5.7.3.2.2.3.)

NOTE 1: The standard test signal is generally used for testing the peak power transmission capability and for measuring the harmonic distortion of a link, trunk or transmission channel. In the tactical subsystem type II (see Table I), the standard test signal is also used for level alignment of links in tandem provided that the circuit to be aligned does not include links of the tactical subsystems type I or type III or a long haul system. The standard test signal should not be used in the tactical subsystems type I and type III for level alignment of links in tandem since the test signal may overload FDM channels. (See Note 2 of 5.6.6.2.)

NOTE 2: The standard test signal (with a level of 0 dBm0) must not be transmitted across a VF channel interface between the tactical subsystem type II and the tactical subsystems type I or III, or across a VF channel

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

interface between the tactical subsystem type II and a long haul system in either direction. Any TLP of the tactical subsystem type II has to be considered internal to this subsystem and must not be related to a TLP of either the tactical subsystems type I and type III or a long haul system in terms of signal levels. For interfacing VF channels of the tactical subsystem type II with VF channels of the tactical subsystems type I or type III see 5.8.5. For interfacing VF channels of a tactical subsystem with VF channels of a long haul system see 5.1.1.5.

5.6.6.2 Standard test tone. (MIL-STD-188-100). In the tactical subsystems type I and type III (see Table I), the level of the standard test tone shall be -10 dBm0, that is -10 dBm at a OTLP of these subsystems, with a frequency of 1000 Hz, ±25 Hz. In the tactical subsystem type II, the level of the standard test tone shall be -3 dBm0, that is -7 dBm at a -4TLP of that subsystem or -3 dBm at a OTLP of that subsystem, with a frequency of 1000 Hz, ±25 Hz. (See Note 3.) A test tone frequency of 1004 Hz is preferred for PCM transmission. (See Notes 1 and 2 of 5.7.3.2.2.3.)

NOTE 1: The difference in test tone levels between the tactical subsystem type II and the tactical subsystems type I and type III (or a long haul system) is caused by different traffic signal levels (voice and quasi-analog signals) at the respective OTLP of the subsystem under consideration. The different traffic signal levels are based on different overload characteristics of the communications equipment employed in these subsystems. Therefore, any TLP of the tactical subsystem type II has to be considered internal to this subsystem and must not be related to a TLP of either the tactical subsystems type I and type III or a long haul system in terms of signal levels. For interfacing VF channels of the tactical subsystem type II with VF channels of the tactical subsystems type I or type III see 5.8.5. For interfacing VF channels of a tactical subsystem with VF channels of a long haul system see 5.1.1.5.

NOTE 2: The standard test tone is generally used for level alignment of VF channels of single links and of links in tandem in the tactical subsystems type I and type III and in a long haul system. It has been found necessary to use, for link alignments, the much lower level of the standard test tone (as compared to the standard test signal stated in 5.6.6.1) in order to prevent overloading of those multichannel wideband transmission subsystems that use FDM or radio equipment, or both, designed for voice service with an activity factor as low as 25 percent (see 5.7.3.1.1). This type of equipment is normally employed in tactical subsystems type I and III and in long haul systems.

NOTE 3: In the tactical subsystem type II, the standard test tone should be used only for level alignment of VF channels of those links that interconnect, and form part of a circuit, with the tactical subsystems

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

type I or III or a long haul system. Details of this level alignment across a VF channel interface between the tactical subsystems type I or III and the tactical subsystem type II are provided in 5.8.5 and across a VF channel interface between a tactical subsystem and a long haul system are provided in MIL-STD-188-100. (See 5.1.1.5.)

5.6.6.3 Monitoring test tone. (MIL-STD-188-100). In the tactical subsystems type I and III (see Table I), the level of the monitoring test tone shall be -15 dBm0, that is -15 dBm at a OTLP of these subsystems, with a frequency of 1000 Hz, ±25 Hz.

NOTE 1: The monitoring test tone is used for testing links, trunks or transmission channels of the tactical subsystems type I and III (or a long haul system) when the test period exceeds approximately 4 hours, e.g., during net loss variation measurements. The monitoring test tone is used because it is comparable to the long term loading of the FDM and radio equipment normally employed in these subsystems. (See 5.7.3.1.1.)

NOTE 2: The monitoring test tone is not required in the tactical subsystem type II since the overload characteristic of the communications equipment employed in this subsystem allows monitoring with the standard test signal (see 5.6.6.1). If the monitoring test tone is transmitted across an interface between the tactical subsystems type I or III and the tactical subsystem type II, it will appear at a level of -12 dBm at a -4TLP (-8 dBm0) of the tactical subsystem type II. (See Note 4 of 5.8.5.1.)

5.6.6.4 Levels for intermodulation distortion (IMD) measurements. (MIL-STD-188-100). The level of the composite two-frequency test signal for IMD measurements of links, trunks or transmission channels shall be -3 dBm0, that is -3 dBm at a OTLP of the subsystem under consideration, and each single-frequency signal level shall be -6 dBm0.

NOTE: The level of -3 dBm0 of the composite signal results in a comparable peak loading of either voice signals or quasi-analog signals. (See 5.7.3.1.1.)

5.6.6.5 Summary of test levels. The various test levels described in 5.6.6.1 through 5.6.6.4 for the tactical subsystems type I, II and III (see Table I) are shown for ready reference in Table XXX.

5.6.7 Digital signal characteristics for testing, measuring and monitoring. Under consideration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XXX. Summary of test levels.

Parameters	For Tactical Subsystem (see Table I)	
	Type I and III Levels in dBm0	Type II Levels in dBm0*
Standard Test Signal	0	0
Standard Test Tone	-10	-3
Monitoring Test Tone	-15	-8**
IMD Measurements:		
Composite Two-Frequency Signal	-3	-3
Single-Frequency Signal	-6	-6

* Referenced to a OTLP of the tactical subsystem type II. This OTLP has to be considered internal to the subsystem and must not be related to a OTLP of either the tactical subsystem type I or the tactical subsystem type III (or a long haul system). (See also 5.8.5.)

** Normally not used. (See Note 2 of 5.6.6.3.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7 Trunk subsystems.

5.7.1 General. Trunks are the connections between nodal subsystems and may consist of one link or of two or more links in tandem. In general, these links consist of multiplexing equipment and wideband radio facilities utilizing LOS microwave, tropospheric scatter or satellite transmission. A connection between nodes is normally installed as a trunk group (see 4.3.3) and is shared by many users of a switched communications system.

5.7.2 Multiplexing equipment. Multiplexing equipment currently employed in trunk subsystems uses either FDM or TDM/PCM techniques. The modulation link parameters given in 5.7.3 and the trunk parameters given in 5.7.4 are based on the characteristics of current FDM and TDM/PCM equipment. The next generation of multiplexers will be TDM equipment using CVSD as the A-D conversion techniques. Common long haul/tactical standards and design objectives for TDM/CVSD equipment will be developed and published in MIL-STD-188-102 (see Appendix B) when sufficient test and evaluation data are available.

5.7.3 Modulation links. The following subparagraphs deal with modulation links which can be considered "building block" configurations of transmission and modulation. Historically, these configurations were developed for voice service and as such they did not deal with technical control configurations nor with switched subsystems or networks. However, with the advent of data service and switched network applications, equipment for additional functions, such as conditioning, regeneration, switching and technical control, will have to be introduced to the basic link configuration to ensure satisfactory service.

The parameters in 5.7.3.1 and 5.7.3.2 apply to single links which are properly terminated and generally operated in tandem connections. A single link in this context comprises a connection between two points where the signal is modulated/multiplexed and demodulated/demultiplexed to/from the nominal frequency bandwidth of the link under consideration. Two different modulation links are considered: Nominal 4 kHz VF links derived from FDM or TDM/PCM and 48 kHz FDM group links. A link will be understood to include only four-wire, or equivalent four-wire, portions of a circuit and will not include two-wire/four-wire hybrid transformers. (See 4.3.2.)

For the purpose of system engineering and circuit design, 5.7.3.1 and 5.7.3.2 specify the characteristics of links employing existing FDM and TDM/PCM equipment. Current equipment design is based on digital transmission employing CVSD as the A-D conversion techniques. Standards and design objectives for links employing TDM/CVSD equipment will be developed when sufficient test and evaluation data are available. (See 5.7.2.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.1 Parameters for FDM modulation links. The basic configuration of an FDM modulation link with typical channel groupings and frequency bands is shown in Figure 27. The link shown in Figure 27 has no through-group or through-supergroup equipment and is used in tactical subsystems type I and III (see Table I), sometimes without the supergroup translation equipment. The FDM VF link begins at the left of Figure 27 at VF band (V), is modulated in successive stages to basic group frequency band (G) and basic supergroup frequency band (S) and then, as required, to either baseband frequency (B) for radio transmission or to line frequency band (L) for metallic line transmission. The transmission medium may consist of LOS radio relay, tropospheric scatter or cable (land or submarine). At the receiving end of the link, corresponding demodulation stages return the received signal to VF band (V). The FDM group link begins at the left of Figure 27 at group frequency band (G), is modulated and demodulated in a similar manner and terminates at (G) on the right. Figure 27 shows only one direction of transmission; the other direction is identical.

Figure 28 depicts a block diagram of FDM equipment based on the configuration of Figure 27 and shows test signal levels and impedances at the different distribution frames.

Another typical configuration of an FDM modulation link is shown in Figure 29. This link contains additional multiplex translations between the originating and the terminating high frequency distribution frames (HFDF) shown in Figure 27. The translations are accomplished by employing through-group and through-supergroup equipment, normally located at technical control facilities (TCF). The link shown in Figure 29 is generally used in the tactical subsystem type III. (See Table I.)

5.7.3.1.1 FDM channel loading. The total power applied to the input of FDM equipment depends on the vector sum of the instantaneous voltages in the separate channels and is a function of both the phase and amplitude of each signal. More detailed information on FDM channel loading is contained in Appendix K.

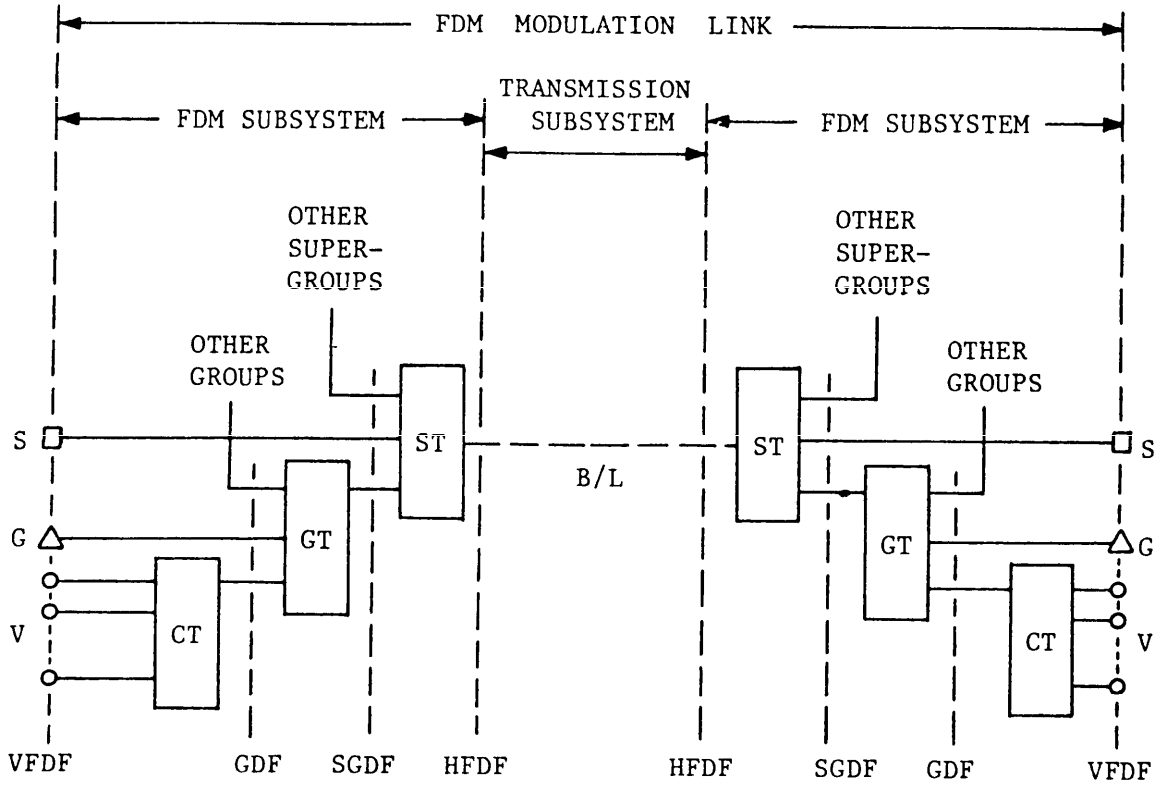
5.7.3.1.1.1 Voice loading (analog signals). (MIL-STD-188-100.) The mean rms power (P_m) of the statistically distributed voice signals that an FDM subsystem must be capable of transmitting depends on the number of channels involved and shall be calculated from one of the following two formulas. (See Figure 30.)

$P_m = -1 + 4 \log N$, ($12 \leq N < 240$ channels) in dBm0, or

$P_m = -15 + 10 \log N$, ($N \geq 240$ channels) in dBm0.

NOTE: The above formulas include a small margin for channel loading caused by signaling tones, pilot signals and carrier leak.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

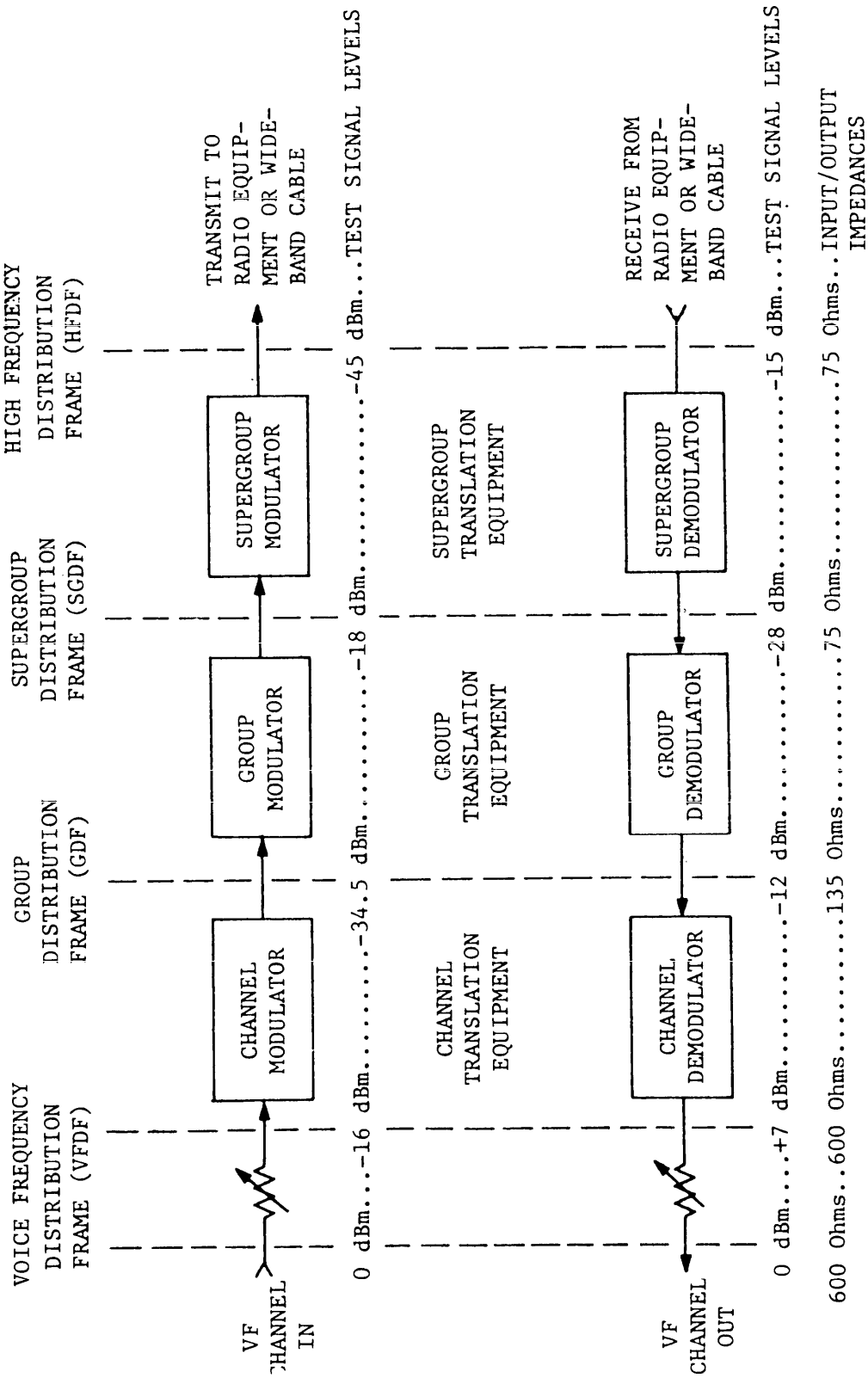


LEGEND:

B	=	Baseband Frequency for Radio Transmission
CT	=	Channel Translation Equipment
G	=	Basic Group Frequency Band, 60 kHz - 108 kHz
G to G	=	FDM Group Link
GDF	=	Group Distribution Frame
GT	=	Group Translation Equipment
HFDF	=	High Frequency Distribution Frame
L	=	Line Frequency Band for Cable Transmission
S	=	Basic Supergroup Frequency Band, 312 kHz - 552 kHz
S to S	=	FDM Supergroup Link
SGDF	=	Supergroup Distribution Frame
ST	=	Supergroup Translation Equipment
V	=	4-Wire, Nominal 4 kHz VF Channel
V to V	=	FDM VF Link
VFDF	=	VF Distribution Frame
○	=	Access Point, Nominal 4 kHz VF Channel
△	=	Access Point, 48 kHz FDM Group Channel
□	=	Access Point, 240 kHz FDM Supergroup Channel

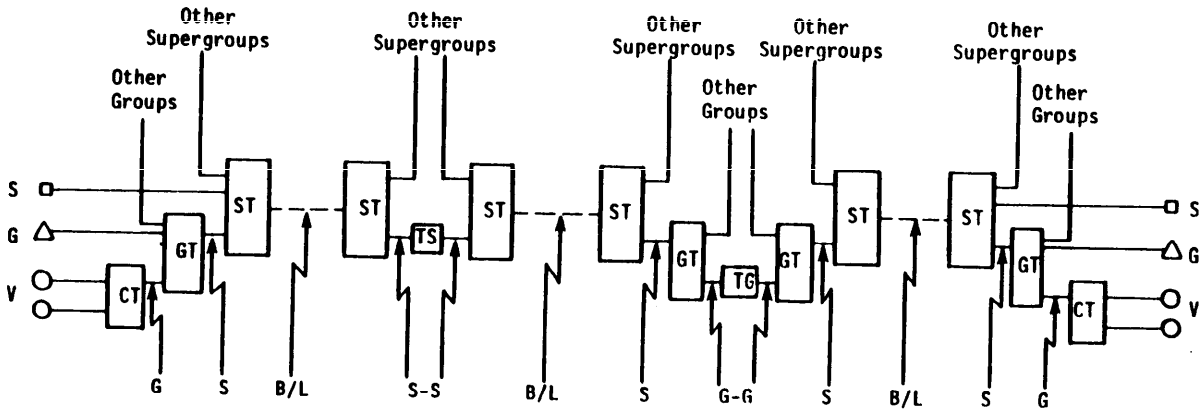
FIGURE 27. Basic configuration of FDM modulation link without through-equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



• FIGURE 28. Block diagram of FDM equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

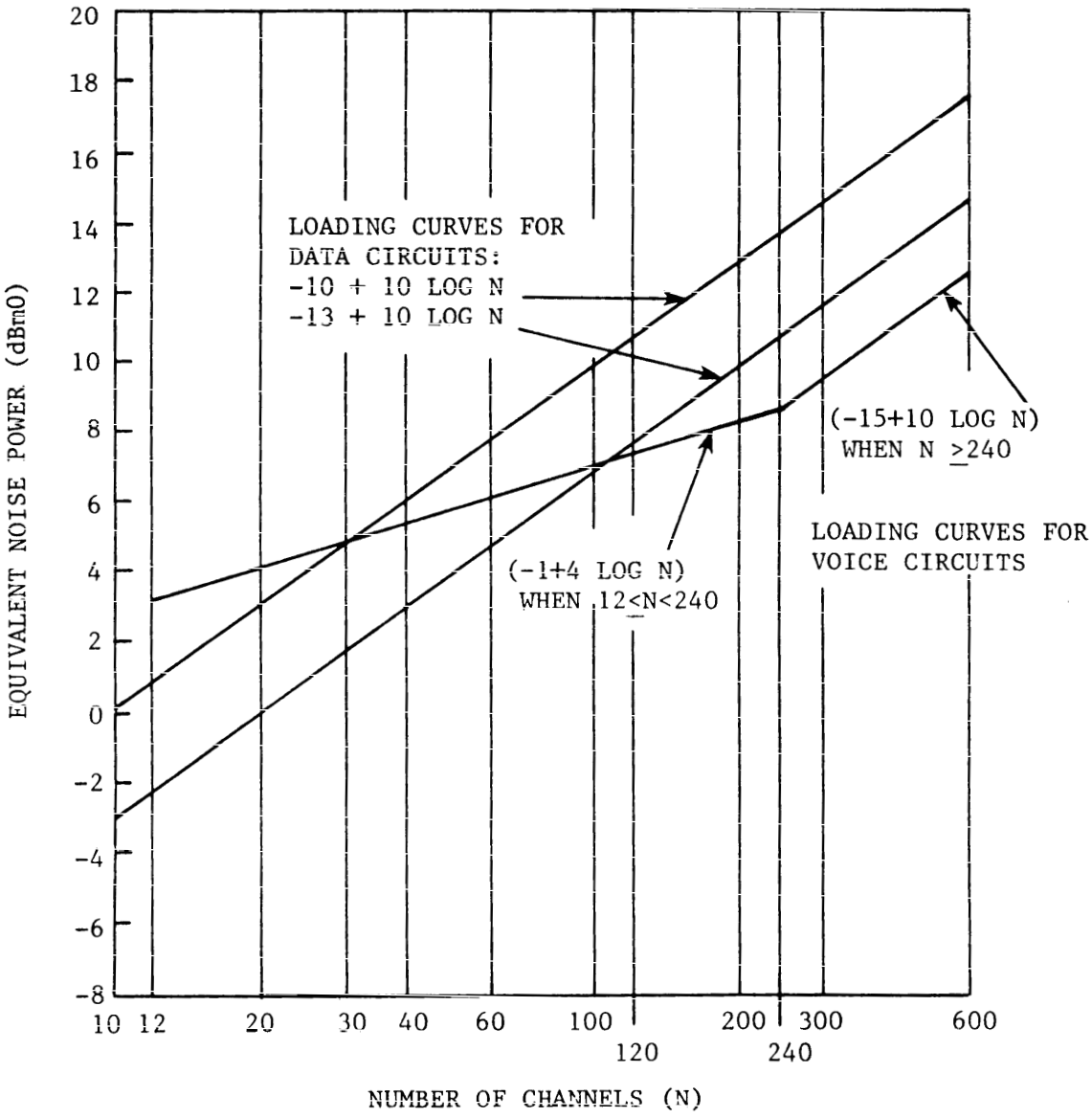


LEGEND:

B	= Baseband Frequency for Radio Transmission
CT	= Channel Translation Equipment
G	= Basic Group Frequency Band, 60 kHz - 108 kHz
G to G	= FDM Group Link
GT	= Group Translation Equipment
L	= Line Frequency Band for Cable Transmission
S	= Basic Supergroup Frequency Band, 312 kHz - 552 kHz
S to S	= FDM Supergroup Link
ST	= Supergroup Translation Equipment
TG	= Through-Group Equipment
TS	= Through-Supergroup Equipment
V	= 4-Wire, Nominal 4 kHz VF Channel
V to V	= FDM VF Link
○	= Access Point, Nominal 4 kHz VF Channel
△	= Access Point, 48 kHz FDM Group Channel
□	= Access Point, 240 kHz FDM Supergroup Channel

FIGURE 29. Basic configuration of FDM modulation link with through-equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



The power of a Gaussian noise test signal for simulating traffic loads of FDM equipment depends on the number of channels and on the type of traffic. Lower values of signal power for simulating voice traffic result from lower activity factors as compared to data traffic. (See Appendix K.)

FIGURE 30. FDM channel loading curves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.1.1.2 Data loading (quasi-analog signals). (MIL-STD-188-100.) The mean rms power (P_m) of quasi-analog signals that an FDM subsystem must be capable of transmitting depends on the power in the separate data channels and on the number of channels involved. This mean rms power shall be calculated from the following formula. (See Figure 30.)

$$P_m = P_c + 10 \log N$$

where P_m = rms power of the multichannel signal in dBm0
P_c = rms power of the input telegraph or data signal in dBm0
N = number of channels carrying telegraph or data signals.

5.7.3.1.2 Test signal levels. (MIL-STD-188-100.) The test signal levels for FDM modulation links shall be as shown in Table XXXI. Figure 31 summarizes test and traffic levels for VF links of the tactical subsystems type I and III. (See Table I.)

5.7.3.1.3 Noise power parameters. For the tactical subsystem type I (see Table I), the total noise power (consisting of multiplex and transmission media noise power) of an FDM VF link is allocated on a per link basis. For the tactical subsystem type III, the total noise power of an FDM modulation link is divided into multiplex noise power (consisting of idle channel and loaded channel noise power) and transmission media noise power, and allocated separately for an FDM VF link and an FDM group link. Noise power values for the transmission of analog voice signals over VF links are psophometrically weighted and expressed in pWp or dBrnc. Noise power values for the transmission of quasi-analog signals over FDM VF links and for the transmission of analog and quasi-analog signals over 48 kHz FDM group links are flat weighted over the frequency band of interest and expressed in pW or dBrn.

5.7.3.1.3.1 Total noise power for FDM VF links of tactical subsystem type I. For a single FDM VF link employing LOS radio transmission of the tactical subsystem type I (see Table I), the total noise power (multiplex plus transmission media noise power) shall not exceed 5000 pWp (37.0 dBrnc) for voice transmission and 8000 pW (39.0 dBrn) for the transmission of quasi-analog signals. For a single FDM VF link employing tropospheric scatter radio transmission of the tactical subsystem type I, the total noise power (multiplex plus transmission media noise power) shall not exceed 16,000 pWp (42.0 dBrnc) for voice transmission and 25,000 pW (44.0 dBrn) for the transmission of quasi-analog signals. The noise power values shall be measured at the output terminals of the link, properly terminated, and the link insertion loss adjusted to 0 dB, ±0.5 dB. The noise power values shall be obtained during 99 percent or more of the time (see Note 1 of 5.2.3.1.5.2).

NOTE: Fade margins for LOS and tropospheric scatter radio links are given in 5.7.3.1.3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XXXI. Test signal levels for FDM modulation links.

Test Points	Tactical Subsystem (See Table I)	Test Signal Levels (dBm)*
Channel Modulator In (Input to channel translation equipment)	Type I	0
	Type III	-16
Channel Demodulator Out (Output from channel translation equipment)	Type I	0
	Type III	+7
Transmit Group Distribution Frame	Types I and III	-34.5
Receive Group Distribution Frame	Types I and III	-12, ±0.5
Transmit Supergroup Distribution Frame	Types I and III	-18
Receive Supergroup Distribution Frame	Types I and III	-28, ±0.5
Transmit High Frequency Distribution Frame	Types I and III	-45
Receive High Frequency Distribution Frame	Types I and III	-15, ±0.5

* These values equate to a 1000 Hz, ±25 Hz, test signal level of 0 dBm (1 mW across 600 ohms) at the OTLP (0 dBm0) (see 5.6.6.1), and should not be confused with the standard test tone of -10 dBm0 for the tactical subsystems type I and III (or a long haul system) and of -3 dBm0 for the tactical subsystem type II (see 5.6.6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

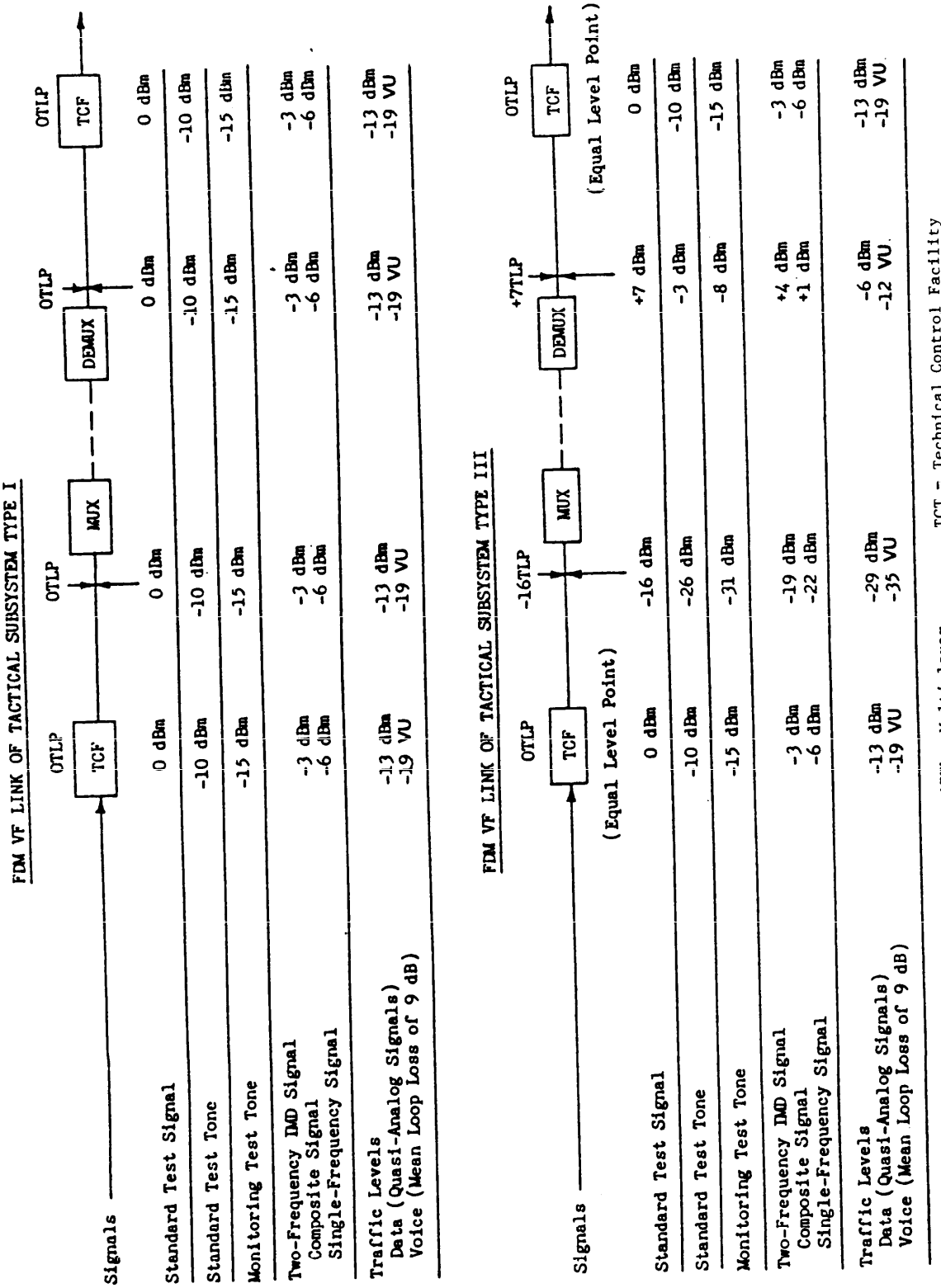


FIGURE 31. Interface and test levels for FDM VF links.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.1.3.2 Multiplex noise power for FDM equipment and VF links of tactical subsystem type III. For multiplexing equipment employed in FDM VF links of the tactical subsystem type III (see Table I), the multiplex idle channel noise power and the loaded channel noise power shall not exceed the values given in Table XXXII, when measured at or referenced to a OTLP of the link.

NOTE: The multiplex idle channel noise power and loaded channel noise power values for the configurations of FDM VF links shown in Figures 27 and 29 are listed in Table XXXII. Multiplex noise power values for other link configurations can be obtained by selecting the applicable equipment parameters listed in Table XXXII and using the summation method of 5.9.3.2.

TABLE XXXII. Multiplex noise power for FDM equipment and VF links of tactical subsystem type III.

Configuration	FDM Noise Power Allocation			
	Idle Channel Noise Power		Loaded Channel Noise Power	
	pWp0	pW0	pWp0	pW0
Pair, Multiplexer Channel Translation Equipment	10	16	31	50
Pair, Multiplexer Group Translation Equipment	40	64	50	80
Pair, Multiplexer Supergroup Translation Equipment	25	40	50	80
Multiplexer Through-Group Equipment	0	0	10	16
Multiplexer Through-Supergroup Equipment	25	40	50	80
Single FDM VF Link Without Through-Equipment (See Figure 27)	75*	120*	131*	210*
Single FDM VF Link With Through-Equipment (See Figure 29)	190*	304*	341*	546*

* These noise power values have been obtained by applying the summation method given in 5.9.3.2 to the link configurations shown in Figures 27 and 29.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.1.3.3 Multiplex noise power for FDM equipment and group links of tactical subsystem type III. For multiplexing equipment employed in FDM group links of the tactical subsystem type III (see Table I), the multiplex idle channel noise power and the loaded channel noise power shall not exceed the values given in Table XXXIII, when measured at or referenced to a OTLP of the link.

NOTE: The multiplex idle channel noise power and loaded channel noise power values for the configurations of FDM group links shown in Figures 27 and 29 are listed in Table XXXIII. Multiplex noise power values for other link configurations can be obtained by selecting the applicable equipment parameters listed in Table XXXIII and using the summation method of 5.9.3.2.

TABLE XXXIII. Multiplex noise power for FDM equipment and group links of tactical subsystem type III.

Configuration	FDM Noise Power Allocation	
	Idle Channel Noise Power (pW0)	Loaded Channel Noise Power (pW0)
Pair, Multiplexer Group Translation Equipment	1,140	1,425
Pair, Multiplexer Supergroup Translation Equipment	710	1,425
Multiplexer Through-Group Equipment	0	284
Multiplexer Through-Supergroup Equipment	710	1,425
Single FDM Group Link Without Through-Equipment (See Figure 27)	1,850*	2,850*
Single FDM Group Link With Through-Equipment (See Figure 29)	5,120*	8,834*

* These noise power values have been obtained by applying the summation method given in 5.9.3.2 to the link configurations shown in Figures 27 and 29.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.1.3.4 Transmission media noise power. For FDM radio links and radio relay links employing tropospheric scatter transmission of the tactical subsystem type III (see Table I), the median value of the transmission medium noise power for the worst hour of a year shall not exceed the applicable value given in Table XXXIV, when measured at or referenced to a OTLP of the link. For FDM radio links and radio relay links employing LOS transmission of the tactical subsystem type III, the short term median value of the transmission medium noise power shall not exceed the applicable value given in Table XXXIV when measured at or referenced to a OTLP of the link.

NOTE 1: For FDM radio links or radio relay links, the allocation of noise power due to the transmission media depends on the type of medium, on the length of the radio link or radio relay links in tandem, and on the type of repeater, i.e., baseband, intermediate frequency (IF) or rf repeater.

NOTE 2: The noise power allocations in Table XXXIV for nominal link lengths above 50 km (27 nmi) include the noise power contributed by IF repeaters (not baseband repeaters) that may be required for LOS transmission distances greater than 50 km.

NOTE 3: Multiplex idle channel and loaded channel noise power values are not included in the noise power allocations of Table XXXIV. (See 5.7.3.1.3.2 and 5.7.3.1.3.3.)

5.7.3.1.3.5 Fade margin for FDM radio links. For LOS and tropospheric scatter radio links of the tactical subsystems type I and III (see Table I), the fade margin shall be such as to ensure that the total noise power or the transmission media noise power of these links shall not exceed the limits given in 5.7.3.1.3.1 or 5.7.3.1.3.4, as applicable, more than 1 percent of the time. (See Note 1 of 5.2.3.1.5.2.)

5.7.3.1.4 Link input/output impedances.

5.7.3.1.4.1 FDM VF link. (MIL-STD-188-100.) For a nominal 4 kHz channel of an FDM modulation link, the input and output impedances shall each be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band from 300 Hz to 3400 Hz. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

5.7.3.1.4.2 FDM group link. (MIL-STD-188-100.) For a 48 kHz channel of an FDM modulation link, the input and output impedances shall each be 135 ohms, balanced to ground, with a minimum return loss of 20 dB (Design Objective: 26 dB) against a 135-ohm resistance over the frequency band of interest. The electrical symmetry shall be sufficient to suppress

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0). For the purpose of interfacing with other equipment, a strapping option for 150-ohm impedance termination, balanced to ground, should be provided.

5.7.3.1.4.3 Multiplex baseband. (MIL-STD-188-100.) For the multiplex baseband of an FDM modulation link, the input and output impedances shall each be 75 ohms, unbalanced to ground, with a minimum return loss of 20 dB (Design Objective: 26 dB) against a 75-ohm resistance over the frequency band of interest.

TABLE XXXIV. Transmission media noise power for FDM modulation links of tactical subsystem type III.

Nominal 4 kHz FDM VF Link			
Nominal Link Length	Maximum Noise Power for Voice Service	Maximum Noise Power for Telegraph and Data Service	Repeater
L ≤ 50 km (27 nmi)	140 pWp0	220 pW0	None
	180 pWp0	285 pW0	One IF Repeater
L > 50 km (27 nmi)	3.6 pWp0/km (6.7 pWp0/nmi)	5.7 pW0/km (10.6 pW0/nmi)	IF Repeater(s)* as required
48 kHz FDM Group Link			
Nominal Link Length	Maximum Noise Power for Voice Service and Telegraph and Data Service		Repeater
L ≤ 50 km (27 nmi)	3600 pW0		None
	4600 pW0		One IF Repeater
L > 50 km (27 nmi)	92 pW0/km (170 pW0/nmi)		IF Repeater(s)* as required

*See Note 2 of 5.7.3.1.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.1.5 Insertion loss versus frequency characteristics.

5.7.3.1.5.1 FDM VF link of tactical subsystem type I. For a single FDM VF link of the tactical subsystem type I (see Table I), the insertion loss versus frequency characteristic, referenced to 1000 Hz, shall not exceed the values given in Table XXXV over the frequency bands stated. Loss is indicated by a (+) and gain by a (-) sign. (See Figure 32A.)

TABLE XXXV. Insertion loss versus frequency characteristic for FDM VF links of tactical subsystem type I.

Frequency in Hz	Insertion Loss in dB (Referenced to 1000 Hz)
$f < 300$	≥ -3
$300 \leq f < 400$	-3 to +1
$400 \leq f < 3000$	-2 to +1
$3000 \leq f < 3400$	-3 to +1
$3400 \leq f$	≥ -3

5.7.3.1.5.2 FDM equipment and VF links of tactical subsystem type III. (MIL-STD-188-100.) For multiplexing and radio equipment employed in FDM VF links of the tactical subsystem type III (see Table I), the insertion loss versus frequency characteristic, referenced to 1000 Hz, shall not exceed the values given in Table XXXVI over the frequency bands stated. Loss is indicated by a (+) and gain by a (-) sign. (See Figures 33A and 34A.)

NOTE: The insertion loss versus frequency characteristic for the configurations of FDM VF links shown in Figures 27 and 29 are listed in Table XXXVI. The insertion loss versus frequency characteristic for other link configurations can be obtained by selecting the applicable equipment parameters listed in Table XXXVI and using the summation method of 5.9.3.3.

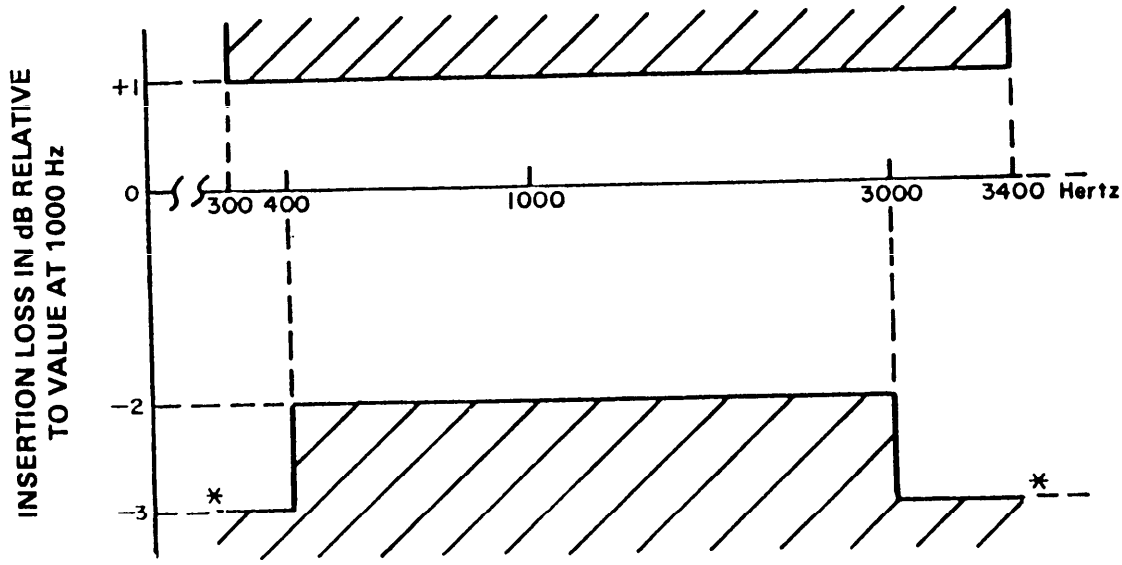
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XXXVI. Insertion loss versus frequency characteristics for FDM equipment and VF links of tactical subsystem type III.

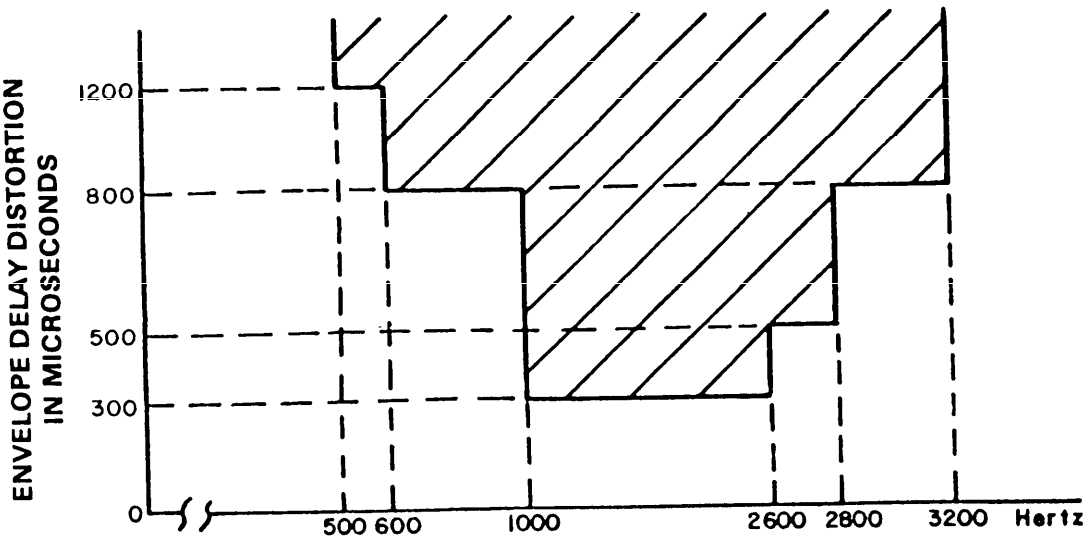
Configuration	Insertion Loss in dB, Referenced to 1000 Hz, Versus Frequency in Hz		
	$300 \leq f < 400$ $3000 \leq f < 3400$	$400 \leq f < 3000$	$f < 300$ $f \geq 3400$
Pair, Multiplexer Channel Translation Equipment	-0.3 to +1.5	±0.3	$\geq$ -0.3
Pair, Multiplexer Group Translation Equipment	±0.3	±0.3	$\geq$ -0.3
Pair, Multiplexer Supergroup Translation Equipment	±0.5	±0.5	$\geq$ -0.5
Multiplexer Through-Group Equipment	±0.5	±0.5	$\geq$ -0.5
Multiplexer Through-Supergroup Equipment	±0.5	±0.5	$\geq$ -0.5
Radio Equipment (back-to-back)	±0.5	±0.5	±0.5
Single FDM VF Link Without Through-Equipment (See Figures 27 and 33A)	-0.8 to +1.7*	±0.8*	$\geq$ -0.8*
Single FDM VF Link With Through-Equipment (See Figures 29 and 34A)	-1.5 to +2.1*	±1.5*	$\geq$ -1.5*

* These values have been obtained by applying the summation method given in 5.9.3.3 to the link configurations shown in Figures 27 and 29 and rounding off the results within 0.1 dB.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



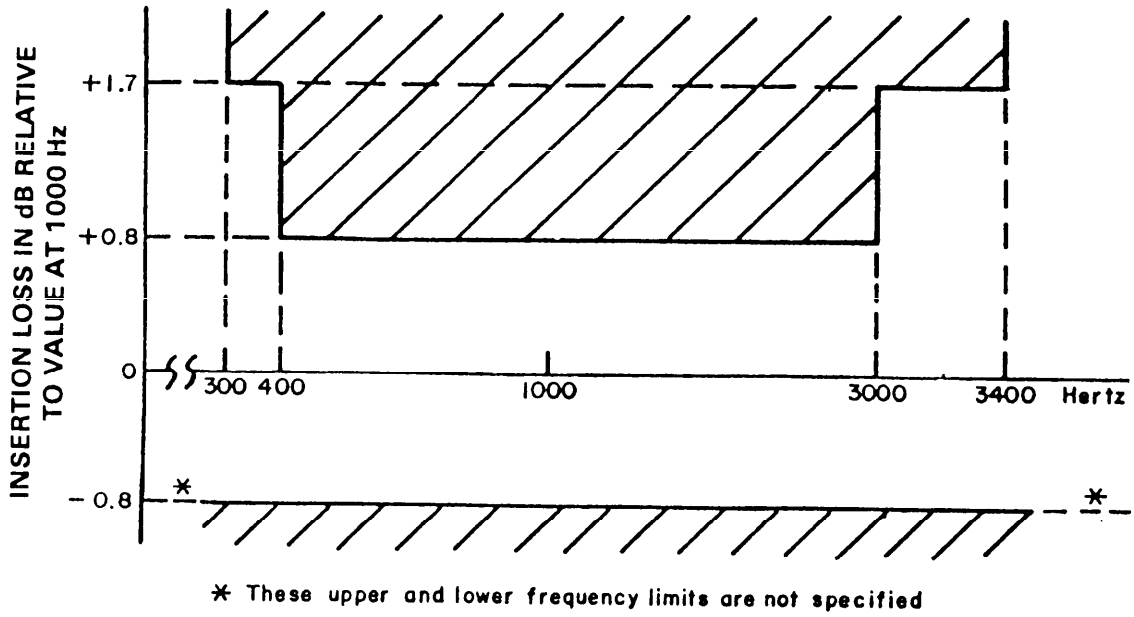
A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

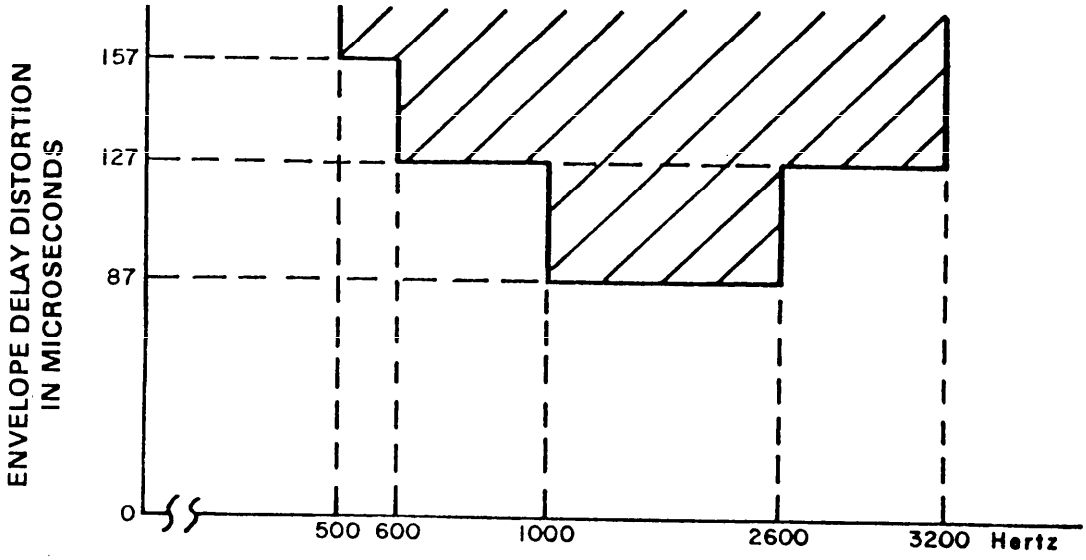
FIGURE 32. Parameters for FDM VF links of tactical subsystem type I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



* These upper and lower frequency limits are not specified

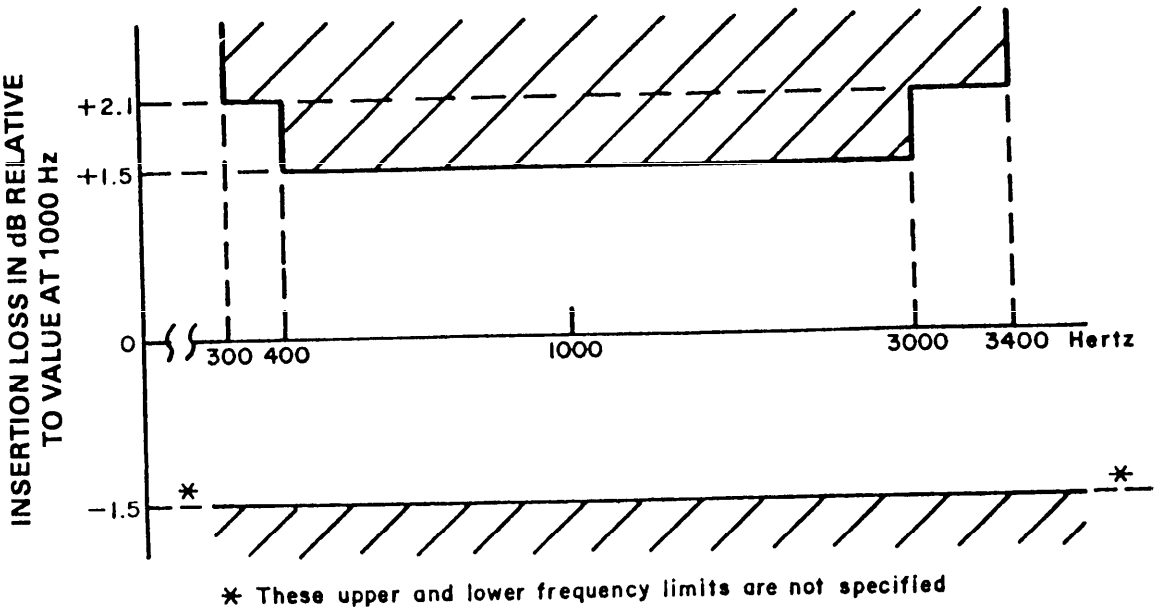
A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



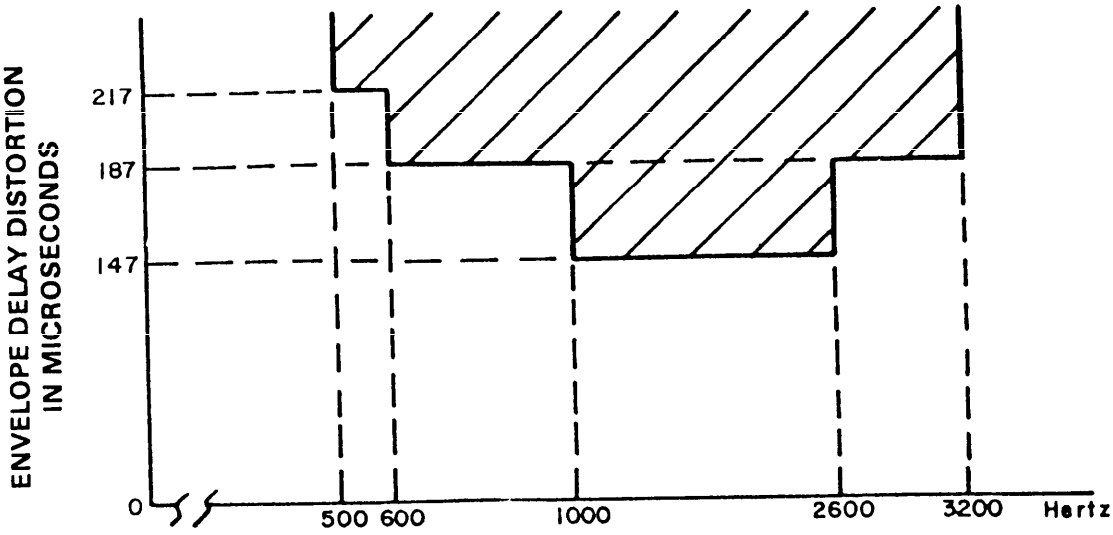
B. ENVELOPE DELAY DISTORTION

FIGURE 33. Parameters for FDM VF link without through-equipment of tactical subsystem type III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 34. Parameters for FDM VF link with through-equipment of tactical subsystem type III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.1.5.3 FDM equipment and group links of tactical subsystem type III. (MIL-STD-188-100.) For multiplexing and radio equipment employed in FDM group links of the tactical subsystem type III (see Table I), the insertion loss versus frequency characteristic shall not exceed the values given in Table XXXVII over the frequency bands stated. Loss is indicated by a (+) and gain by a (-) sign.

NOTE: The insertion loss versus frequency characteristic for the configurations of FDM group links shown in Figures 27 and 29 are listed in Table XXXVII. The insertion loss versus frequency characteristic for other link configurations can be obtained by selecting the applicable equipment parameters listed in Table XXXVII and using the summation method of 5.9.3.3.

TABLE XXXVII. Insertion loss versus frequency characteristics for FDM equipment and group links of tactical subsystem type III.

Configuration	Maximum Insertion Loss in dB from 60.6 kHz to 107.7 kHz
Pair, Multiplexer Group Translation Equipment	±0.6
Pair, Multiplexer Supergroup Translation Equipment	±2.0
Multiplexer Through-Group Equipment	±1.0
Multiplexer Through-Supergroup Equipment	±1.4
Radio Equipment (back-to-back)	±0.5
Single FDM Group Link Without Through-Equipment (See Figure 27)	±2.1*
Single FDM Group Link With Through-Equipment (See Figure 29)	±3.9*

* These values have been obtained by applying the summation method given in 5.9.3.3 to the link configurations shown in Figures 27 and 29 and rounding off the results within 0.1 dB.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.1.6 Envelope delay distortion.

5.7.3.1.6.1 FDM VF link of tactical subsystem type I. For a single FDM VF link of the tactical subsystem type I (see Table I), the envelope delay distortion measured between 500 Hz and 3200 Hz (see 70.5 of Appendix G), shall not exceed the values given in Table XXXVIII over the frequency bands stated. (See Figure 32B.)

TABLE XXXVIII. Envelope delay distortion for FDM VF links of tactical subsystem type I.

Frequency in Hz	Maximum Envelope Delay Distortion in microseconds
$500 \leq f < 600$	1200
$600 \leq f < 1000$	800
$1000 \leq f < 2600$	300
$2600 \leq f < 2800$	500
$2800 \leq f \leq 3200$	800

5.7.3.1.6.2 FDM equipment and VF links of tactical subsystem type III. (MIL-STD-188-100.) For multiplexing equipment employed in FDM VF links of the tactical subsystem type III (see Table I), the envelope delay distortion measured between 500 Hz and 3200 Hz (see 70.5 of Appendix G), shall not exceed the values given in Table XXXIX over the frequency bands stated. (See Figures 33B and 34B.)

NOTE 1: The envelope delay distortion for the link configurations shown in Figures 27 and 29 are listed in Table XXXIX for the FDM VF links and in Table XL for the FDM group links. The envelope delay distortion for other link configurations can be obtained by selecting the applicable equipment parameters listed in Tables XXXIX and XL and using the summation method of 5.9.3.4.

NOTE 2: The contributions to the envelope delay distortion of FDM VF links and FDM group links by wideband radio transmission subsystems are considered negligible for stable transmission media.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XXXIX. Envelope delay distortion for FDM equipment and VF links of tactical subsystem type III.

Configuration	Maximum Envelope Delay Distortion in microseconds versus Frequency in Hz		
	$500 \leq f < 600$	$600 \leq f < 1000$ $2600 \leq f \leq 3200$	$1000 \leq f < 2600$
Pair, Multiplexer Channel Translation Equipment	150	120	80
Pair, Multiplexer Group Translation Equipment	4 (in any 4 kHz channel slot, 60 kHz - 108 kHz band)		
Pair, Multiplexer Supergroup Translation Equipment	3 (in any 4 kHz channel slot, 60 kHz - 108 kHz band) (See Note 1)		
Multiplexer Through-Group Equipment	20	20	20
Multiplexer Through-Supergroup Equipment	30	30	30
Single FDM VF Link Without Through-Equipment (See Figures 27 and 33B)	157 (See Notes 1 and 2)	127 (See Notes 1 and 2)	87 (See Notes 1 and 2)
Single FDM VF Link With Through-Equipment (See Figures 29 and 34B)	217 (See Notes 1 and 2)	187 (See Notes 1 and 2)	147 (See Notes 1 and 2)

NOTE 1: Except channels 1 and 2 of group 1 of supergroup 1, and channels 11 and 12 of group 5 of supergroup 3.

NOTE 2: These values have been obtained by applying the summation method given in 5.9.3.4 to the link configurations shown in Figures 27 and 29.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.1.6.3 FDM equipment and group links of tactical subsystem type III. (MIL-STD-188-100.) For multiplexing equipment employed in FDM group links of the tactical subsystem type III (see Table I), the envelope delay distortion measured between 64 kHz and 104 kHz (see 70.5 of Appendix G), shall not exceed the values given in Table XL over the frequency bands stated. (See Notes 1 and 2 of 5.7.3.1.6.2.)

TABLE XL. Envelope delay distortion for FDM equipment and group links of tactical subsystem type III.

Configuration	Maximum Envelope Delay Distortion in microseconds versus Frequency in kHz	
	$64 \leq f < 68$ $100 \leq f \leq 104$	$68 \leq f < 100$
Pair, Multiplexer Group Translation Equipment	12	12
Pair, Multiplexer Supergroup Translation Equipment	10 (See Note 1) 56 (See Note 2)	10 (See Note 1) 20 (See Note 2)
Multiplexer Through-Group Equipment	120	40
Multiplexer Through-Supergroup Equipment	70	70
Single FDM Group Link Without Through-Equipment (See Figure 27)	22 (See Notes 1 and 3) 68 (See Notes 2 and 3)	22 (See Notes 1 and 3) 32 (See Notes 2 and 3)
Single FDM Group Link With Through-Equipment (See Figure 29)	244 (See Notes 1 and 3) 382 (See Notes 2 and 3)	164 (See Notes 1 and 3) 194 (See Notes 2 and 3)

NOTE 1: Only for supergroups 2, and 4 through 10.

NOTE 2: For supergroups 1 and 3, except group 1 of supergroup 1 and group 5 of supergroup 3.

NOTE 3: These values have been obtained by applying the summation method given in 5.9.3.4 to the link configurations shown in Figures 27 and 29.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

Page 183 is missing from the original legacy source document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XLI. Net loss variations for FDM equipment and modulation links of tactical subsystem type III.

Configuration	Net Loss Variation in dB
Pair, Multiplexer Channel Translation Equipment	±0.5
Pair, Multiplexer Group Translation Equipment	±0.5
Pair, Multiplexer Supergroup Translation Equipment	±0.5
Multiplexer Through-Group Equipment	±0.2
Multiplexer Through-Supergroup Equipment	±0.2
Radio Equipment (back-to-back)	±0.5
Single FDM VF Link Without Through-Equipment (See Figure 27)	±1.0*
Single FDM Group Link Without Through-Equipment (See Figure 27)	±0.9*
Single FDM VF Link With Through-Equipment (See Figure 29)	±1.5*
Single FDM Group Link With Through-Equipment (See Figure 29)	±1.4*

* These values have been obtained by applying the summation method given in 5.9.3.5 to the link configurations shown in Figures 27 and 29 and rounding off the results within 0.1 dB.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

5.7.3.2 Parameters for TDM/PCM VF link. Basic features of TDM and PCM have been outlined in 4.5.3 and 4.6.2.1, respectively. Digital transmission is gaining wide acceptance not only in commercial, but also in military communications systems, due to the advantages of transmitting and receiving digital signals rather than analog signals. Based on current tactical system planning, existing TDM/PCM equipment will be replaced with TDM/CVSD equipment for the next generation of switched multichannel communications systems. (See 5.7.2.)

A block diagram of a TDM/PCM VF link of the tactical subsystem type II (see Table I) is shown in Figure 35. The link begins with a group of VF channels at the multiplexer input. Each VF channel has a nominal bandwidth of 4 kHz and is capable of transmitting analog voice signals or data signals in quasi-analog form (see Note of 5.7.3.2.6). The amplitudes of the analog or quasi-analog signal carried in each VF channel are periodically sampled in time, converted into a TDM/PAM signal, and then coded into 6-bit PCM code words. This PCM signal, together with framing and synchronization pulses, appears at the multiplexer output where it is fed into a cable interface converter or a radio for transmission. At the receiving end the PCM pulse stream is decoded, demultiplexed, and the PAM samples are demodulated to analog baseband.

The tactical subsystem type II provides up to 96 VF channels over a cable or radio link. The characteristics of these VF channels are compatible with the characteristics of FDM VF channels given in 5.7.3.1, allowing tandem connections of TDM/PCM VF channels with FDM VF channels provided the proper level adjustment is made in accordance with 5.8.5.1.

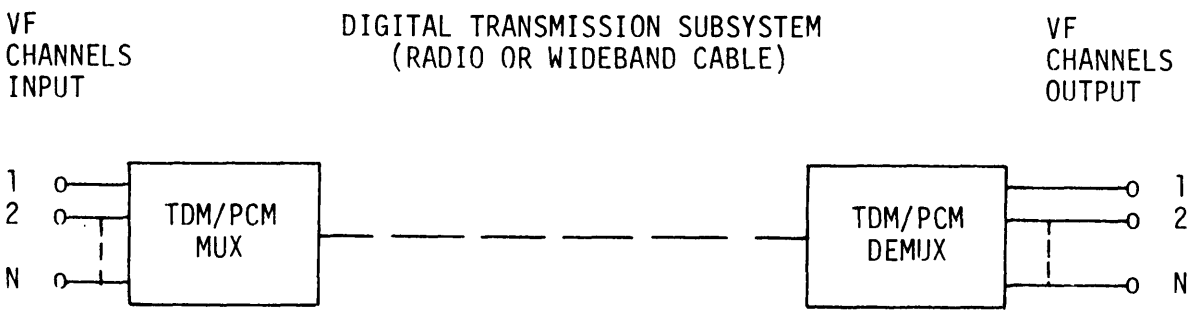


FIGURE 35. Block diagram of TDM/PCM VF link of tactical subsystem type II.

5.7.3.2.1 Characteristics of TDM/PCM signal. The transfer characteristics of a TDM link transmitting digitized voice signals are primarily determined by the A-D/D-A conversion process employed in the link. 5.7.3.2.1.1 through 5.7.3.2.1.4 provide basic characteristics of the 6-bit PCM signal used in multiplexers currently in the inventory and employed in the tactical subsystems type II (see Table I).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.2.1.1 Sampling rate. (MIL-STD-188-100.) For a TDM/PCM VF link of the tactical subsystem type II (see Table I), each nominal 4 kHz channel shall be sampled at a nominal rate of 8000 samples per second.

5.7.3.2.1.2 Number of digits per sample. For a TDM/PCM VF link of the tactical subsystem type II (see Table I), each sample from a nominal 4 kHz channel shall be encoded into a 6-bit binary code word with the most significant bit transmitted first.

5.7.3.2.1.3 Compandor characteristics. An instantaneous compandor shall be used in the multiplexer of a TDM/PCM VF link of the tactical subsystem type II (see Table I). The compressor shall have a 3-segment, straight-line, transfer characteristic approximating a μ companding law with a degree of compression of $\mu = 100$. The expandor shall have the complementary transfer characteristic of the compressor. The compression ratio, defined as the initial slope to average slope, shall be 20 dB, ±1 dB.

NOTE: The transfer characteristic of the compandor is derived by the μ-law equation:

$$V_o = V_{i\ max} \frac{\ln (1+\mu V_i / V_{i\ max})}{\ln (1+\mu)} \qquad \text{for } V_i > 0$$

where: V_o = compandor output voltage

V_i = compandor input voltage

$V_{i\ max}$ = maximum compandor input voltage

μ = degree of compression

ln = natural logarithm (to base e)

5.7.3.2.1.4 Frame synchronization. Frame synchronization information shall be added to the PCM output of the multiplexer of a TDM/PCM VF link of the tactical subsystem type II (see Table I). This information shall consist of one frame bit substituted for the least significant bit of the binary code word in the last channel of a twelve-channel group. The frame bit shall have the same shape and amplitude characteristics as the remaining code word bits.

5.7.3.2.2 Link input/output parameters.

5.7.3.2.2.1 TDM/PCM channel loading. For TDM/PCM VF links of the tactical subsystem type II (see Table I), the level of quasi-analog signals, if used (see Note of 5.7.3.2.6), shall be adjusted in accordance with 5.2.3.1.4 for satisfactory telegraph and data transmission.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

NOTE 1: The level of analog voice signals at the input terminals of a TDM/PCM VF link is not standardized. This level depends on the talker volume, the loop loss and the circuit configuration. Typical levels of voice signals for the tactical subsystem type II are shown in Figures 53, 54 and 60.

NOTE 2: TDM/PCM VF links are not subject to the same multichannel overloading problems as FDM VF links. Overloading individual VF channels of TDM/PCM links with analog voice signals or quasi-analog data signals will result in clipping of the peak amplitudes of these signals but will not affect other VF channels within the same TDM channel group. Clipping of analog voice signals generally can be tolerated since its effect on intelligibility is negligible. However, clipping of quasi-analog data signals will result in distortion which may cause excessive error rates.

5.7.3.2.2.2 Test signal levels. The test signal levels at the input and output terminals of a TDM/PCM VF link of the tactical subsystem type II (see Table I) shall be 0 dBm0, e.g., -4 dBm at a -4TLP or +1 dBm at a +1TLP.

NOTE: The TLP at the input and output terminals of a TDM/PCM VF link depend on the circuit configuration and the type of service provided. (See 5.7.4.6 and Figures 53, 54, 55 and 60.)

5.7.3.2.2.3 Test signal frequency. (MIL-STD-188-100.) For transmission testing on TDM/PCM VF links of the tactical subsystem type II (see Table I), the test signal shall have a frequency of 1000 Hz, ±25 Hz, preferably 1004 Hz.

NOTE 1: A test signal frequency of 1004 Hz is used in long haul and commercial systems for testing TDM/PCM VF links.

NOTE 2: PCM signal processing can generate unique forms of nonlinear distortion which have no direct counterpart in analog transmission. The sources of nonlinear PCM distortion are sampling, quantizing, mistracking of the instantaneous compandor, and filtering methods used to limit the demultiplexed PAM samples to voice bandwidth signals. Test signal frequencies which are exactly rational fractions of (subharmonically related to) the 8000 Hz PCM sampling rate, such as 800 Hz, 1000 Hz and 1600 Hz, can cause nonlinear PCM distortion and, therefore, should not be used. This nonlinear PCM distortion results in fluctuating meter readings which can be avoided by changing the 1000 Hz test frequency by a few hertz.

5.7.3.2.2.4 Link input/output impedances. (MIL-STD-188-100.) For a TDM/PCM link of the tactical subsystem type II (see Table I), the input and output impedances shall each be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band from 300 Hz to 3400 Hz. The electrical symmetry shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

5.7.3.2.3 Noise power parameters.

5.7.3.2.3.1 Idle channel noise power. For a single TDM/PCM VF link of the tactical subsystem type II (see Table I), the noise power measured at the output terminals of any idle VF channel, properly terminated, shall not exceed the values given in Table XLII, when the link insertion gain is adjusted as stated in Table XLII. The noise power for voice transmission shall be measured with psophometric weighting; the noise power for telegraph and data transmission with quasi-analog signals (see Note of 5.7.3.2.6) shall be measured with flat weighting over the frequency band occupied by the quasi-analog signals.

TABLE XLII. Idle channel noise power for TDM/PCM VF links of tactical subsystem type II.

Type of Transmission	Maximum Idle Channel Noise Power with Link Insertion Gain Adjusted to	
	0 dB, ±0.5 dB	5 dB, ±0.5 dB
Voice Transmission	2000 pWp (33.0 dBrnc)	6300 pWp (38.0 dBrnc)
Telegraph and Data Transmission	3150 pW (35.0 dBrn)	10,000 pW (40.0 dBrn)

5.7.3.2.3.2 Quantizing noise ratio. For a single TDM/PCM link of the tactical subsystem type II (see Table I), a sine wave signal with a frequency of 1000 Hz, ±25 Hz, preferably 1004 Hz (see Notes 1 and 2 of 5.7.3.2.3), at the input of any VF channel shall produce at the output of the same VF channel a minimum test signal-to-quantizing noise ratio as shown in Table XLIII. The quantizing noise shall be measured with flat weighting over the frequency band from 300 Hz to 3400 Hz and with the link insertion gain adjusted to 0 dB, ±0.5 dB.

5.7.3.2.3.3 Equivalent PCM noise power. As a Design Objective for analog voice transmission over a single TDM/PCM VF link of the tactical subsystem type II (see Table I), the equivalent PCM noise power should not exceed 5,000 pWp (37.0 dBrnc) when measured with the link insertion gain

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

adjusted to 0 dB, ±0.5 dB, and should not exceed 16,000 pWp (42.0 dBrnc) when measured with the link insertion gain adjusted to 5 dB, ±0.5 dB.

NOTE: For a definition of equivalent PCM noise see FED-STD-1037.

TABLE XLIII. Quantizing noise ratio for TDM/PCM VF links of tactical subsystem type II.

Input Signal (in dBm)	Minimum Test Signal-to-Quantizing Noise Ratio (in dB)
-4	30
-5 to -20	17
-21 to -30	15
-31 to -40	14

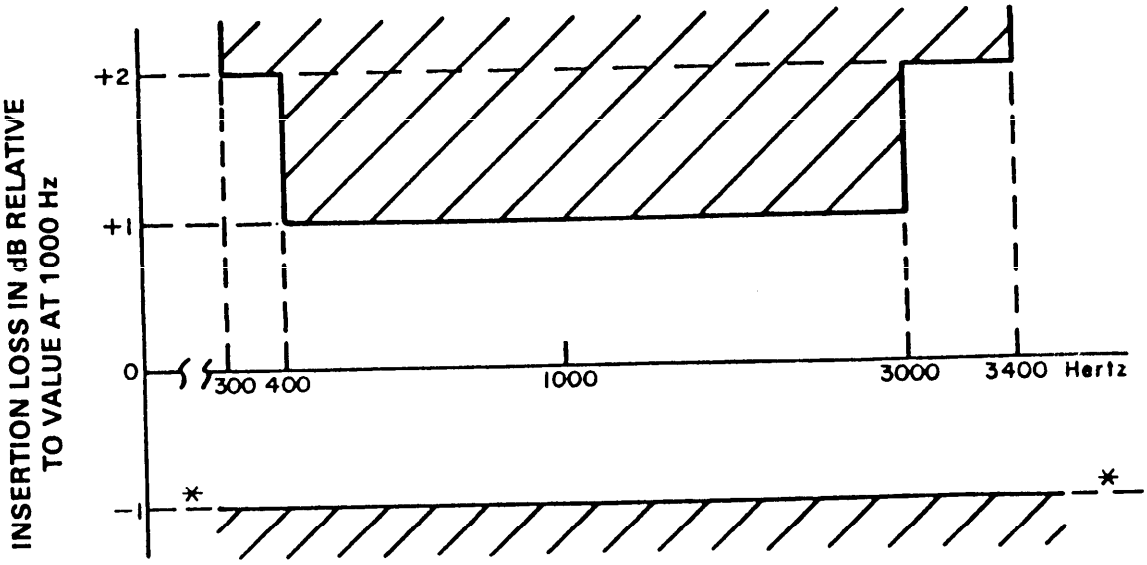
5.7.3.2.4 Transfer function parameters.

5.7.3.2.4.1 Insertion loss versus frequency characteristic. For a single TDM/PCM VF link of the tactical subsystem type II (see Table I), the insertion loss versus frequency characteristic, referenced to 1000 Hz, shall not exceed the values given in Table XLIV over the frequency bands stated. Loss is indicated by a (+) and gain by a (-) sign. (See Figure 36A.)

TABLE XLIV. Insertion loss versus frequency characteristic for TDM/PCM VF links of tactical subsystem type II.

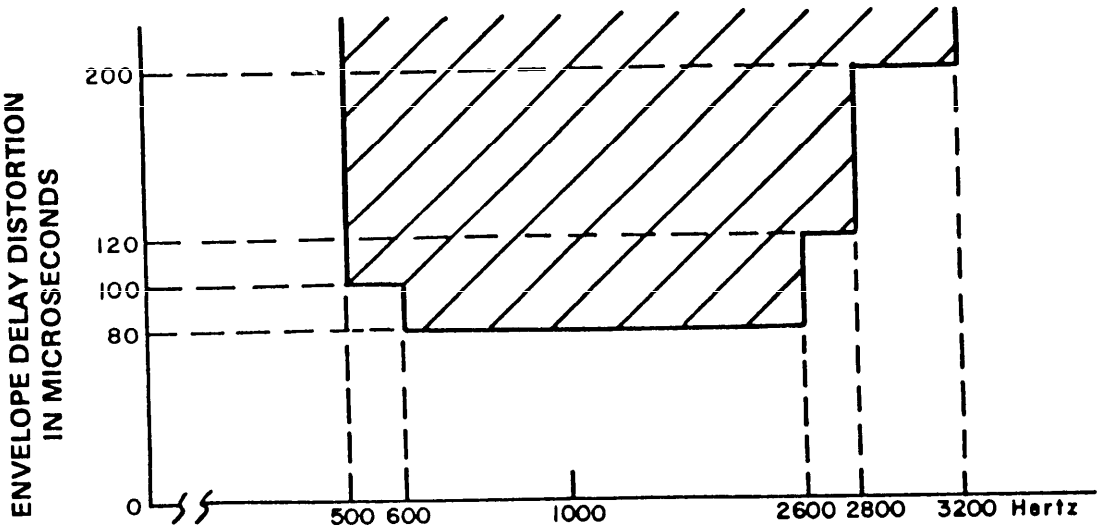
Frequency in Hz	Insertion Loss in dB (Referenced to 1000 Hz)
$f < 300$	≥ -1
$300 \leq f < 400$	-1 to +2
$400 \leq f < 3000$	-1 to +1
$3000 \leq f < 3400$	-1 to +2
$3400 \leq f$	≥ -1

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



* These upper and lower frequency limits are not specified

A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 36. Parameters for TDM/PCM VF links of tactical subsystem type II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.2.4.2 Envelope delay distortion. For a single TDM/PCM VF link of the tactical subsystem type II (see Table I), the envelope delay distortion measured between 500 Hz and 3200 Hz (see 70.5 of Appendix G), shall not exceed the values given in Table XLV over the frequency bands stated. (See Figure 36B.)

TABLE XLV. Envelope delay distortion for TDM/PCM VF links of tactical subsystem type II.

Frequency in Hz	Maximum Envelope Delay Distortion in microseconds
$500 \leq f < 600$	100
$600 \leq f < 2600$	80
$2600 \leq f < 2800$	120
$2800 \leq f \leq 3200$	200

5.7.3.2.4.3 Total harmonic distortion. For a single TDM/PCM VF link of the tactical subsystem type II (see Table I), the total harmonic distortion, produced by any single-frequency test signal between 300 Hz and 3400 Hz, shall be at a level which is at least 30 dB below reference level (-30 dBm0). The total harmonic distortion shall be measured with a standard test signal in accordance with 5.6.6.1.

5.7.3.2.4.4 Intermodulation distortion (IMD). For a single TDM/PCM VF link of the tactical subsystem type II (see Table I), the individual IMD products, produced by any two equal level single-frequency signals between 300 Hz and 3000 Hz, shall be at a level which is at least 30 dB below reference level (-30 dBm0). The signal level for IMD measurements shall comply with the requirements of 5.6.6.4.

NOTE: The frequencies of the two equal level signals should be selected so that at least the third order harmonic products fall within the specified frequency band.

5.7.3.2.4.5 Frequency displacement. For a single TDM/PCM VF link of the tactical subsystem type II (see Table I), any single-frequency signal between 300 Hz and 3400 Hz transmitted over this link shall be reproduced at the link output with a frequency error of not more than ±1 Hz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.3.2.4.6 Net loss variations. For a single TDM/PCM VF link of the tactical subsystem type II (see Table I), the net loss variations shall not exceed ±1.5 dB over any 30 consecutive days and ±0.5 dB over any 15 consecutive minutes.

5.7.3.2.4.7 Input/output linearity. As a Design Objective for a single TDM/PCM VF link of the tactical subsystem type II (see Table I), the link input versus output level characteristic should be linear within ±0.5 dB for input signal levels from -4 dBm to -40 dBm. This measurement should be made with a test frequency of 1000 Hz, ±25 Hz, preferably 1004 Hz. (See Notes 1 and 2 of 5.7.3.2.2.3.)

5.7.3.2.5 Crosstalk. For a single TDM/PCM VF link of the tactical subsystem type II (see Table I), the near-end and the far-end test signal-to-noise plus crosstalk ratio shall be equal to or greater than 50 dB for any frequency between 300 Hz and 3400 Hz, when measured at or referenced to equal TLP. This parameter shall be measured with flat weighting in any idle VF channel, properly terminated, by applying a standard test signal with a level of 0 dBm0, e.g., a level of -4 dBm at a -4TLP, to any other VF channel input of the TDM/PCM VF link.

5.7.3.2.6 Data transmission over TDM/PCM links. Under consideration.

NOTE: Two types of equipment, a time division digital multiplexer (TDDM) and a high speed serial data buffer (HSSDB), have been developed. The TDDM is capable of accepting up to 24 digital telegraph or data signals and combining these signals into a 32 kb/s bit stream which is fed into the HSSDB. For digital signals, the HSSDB automatically bypasses the A-D/D-A converters in the TDM/PCM equipment and provides direct access into the digital bit stream of a TDM/PCM link. As this equipment becomes available, it is recommended that digital telegraph and data signals not be converted into quasi-analog signals and transmitted through a TDM/PCM VF channel but, rather, be transmitted as digital signals through the TDM link. A better utilization of available channel capacity and better signal quality can be achieved if a digital signal is transmitted through the tactical subsystem type II without converting this signal into a quasi-analog signal.

5.7.4 Trunk parameters. The parameters in 5.7.4.1 through 5.7.4.8 apply to all trunks consisting of nominal 4 kHz VF channels of the tactical subsystems type I, II and III (see Table I), regardless of the specific configuration selected for the engineering and installation of these trunks, except where stated otherwise in the applicable subparagraph. A trade-off may be necessary regarding through-equipment connections for FDM trunks consisting of several modulation links in tandem in order to achieve the required performance. Figure 29 shows a typical configuration of an FDM modulation link with through-equipment connections.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

5.7.4.1 Trunk input parameters.

5.7.4.1.1 FDM channel loading. Same as 5.7.3.1.1.

5.7.4.1.2 TDM/PCM channel loading. Same as 5.7.3.2.2.1.

5.7.4.1.3 Speech volume. Not standardized.

NOTE: The speech volume at a trunk input depends on such factors as the talker volume distribution (see 5.2.2.1.1) and the loop insertion loss (see 5.5.2). Typical values of speech volumes that should be achieved at a trunk input are shown in Figures 50, 51, 53, 54, 56, 57, 59 and 60 for different types of trunks.

5.7.4.1.4 Quasi-analog signal level. Same as 5.2.3.1.4.

5.7.4.1.5 Standard test signal. Same as 5.6.6.1.

5.7.4.1.6 Standard test tone. Same as 5.6.6.2.

5.7.4.1.7 Monitoring test tone. Same as 5.6.6.3.

5.7.4.1.8 Levels for IMD measurements. Same as 5.6.6.4.

5.7.4.2 Trunk output parameters.

5.7.4.2.1 Speech volume. Not standardized.

NOTE: The speech volume at a trunk output depends on such factors as the talker volume distribution (see 5.2.2.1.1) and the loop insertion loss (see 5.5.2). Typical values of speech volumes that should be achieved at a trunk output are shown in Figures 50, 51, 53, 54, 56, 57, 59 and 60 for different types of trunks.

5.7.4.2.2 Quasi-analog signal level. The quasi-analog signal level at a trunk output shall be -13 dBm0 for trunks of the tactical subsystems type I and III (see Table I) and -6 dBm0 for trunks of the tactical subsystem type II in accordance with 5.2.3.1.4. The TLP at the trunk output shall be adjusted to a value as stated in 5.7.4.3.6.

NOTE: For the transmission of data signals over TDM/PCM trunks of the tactical subsystem type II see Note of 5.7.3.2.6.

5.7.4.2.3 Noise power for FDM trunks of tactical subsystem type I. For a nominal 4 kHz VF channel of an FDM trunk consisting of one LOS radio link of the tactical subsystem type I (see Table I), the total noise power (multiplex plus transmission media noise power) shall not exceed 5000 pWp (37.0 dBrnc) for voice transmission and 8000 pW (39.0 dBrn) for the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

transmission of quasi-analog signals. For a nominal 4 kHz VF channel of an FDM trunk consisting of one tropospheric scatter radio link of the tactical subsystem type I, the total noise power (multiplex plus transmission media noise power) shall not exceed 16,000 pWp (42.0 dBrnc) for voice transmission and 25,000 pW (44.0 dBrn) for the transmission of quasi-analog signals. The noise power values shall be measured at the output terminals of the trunk, properly terminated, and the trunk insertion loss adjusted to 0 dB, ±0.5 dB. The noise power values shall be obtained during 99 percent or more of the time (see Note 1 of 5.2.3.1.5.2). Noise power values for telegraph and data transmission, expressed in pW or dBrn, shall be measured with flat weighting over the frequency band occupied by the quasi-analog signals.

NOTE: The maximum noise power values for FDM trunks of the tactical subsystem type I are the same as for FDM VF links given in 5.7.3.1.3.1.

5.7.4.2.4 Noise power for TDM/PCM trunks of tactical subsystem type II. The following requirements shall apply to TDM/PCM trunks of the tactical subsystem type II (see Table I), regardless of whether an LOS radio link, a tropospheric scatter radio link, a satellite link or a cable link is employed in the transmission subsystem of the trunk. For an idle nominal 4 kHz VF channel of a TDM/PCM trunk, the noise power shall not exceed 2000 pWp (33.0 dBrnc) for voice transmission and 3150 pW (35.0 dBrn) for the transmission of quasi-analog signals (see Note of 5.7.3.2.6). Noise power values for telegraph and data transmission, expressed in pW or dBrn, shall be measured with flat weighting over the frequency band occupied by the quasi-analog signals. As a Design Objective for analog voice transmission over a nominal 4 kHz VF channel of a TDM/PCM trunk, the equivalent PCM noise power (see Note of 5.7.3.2.3.3) should not exceed 5000 pWp (37.0 dBrnc). The noise power values shall be measured at the output terminals of the trunk, properly terminated, and the trunk insertion loss adjusted to 0 dB, ±0.5 dB.

NOTE: The maximum values of idle channel noise power and equivalent PCM noise power for TDM/PCM trunks of the tactical subsystem type II are the same as for TDM/PCM VF links given in 5.7.3.2.3.1 and 5.7.3.2.3.3, respectively.

5.7.4.2.5 Noise power for FDM trunks of tactical subsystem type III. The following requirements shall apply to FDM trunks, consisting of one or more LOS radio links or one or more tropospheric scatter radio links, or combinations of both, of the tactical subsystem type III (see Table I). For a nominal 4 kHz VF channel of an FDM trunk, the total noise power (multiplex plus transmission media noise power) shall not exceed 2500 pWp0 (34.0 dBrnc0) for voice transmission and 4000 pW0 (36.0 dBrn0) for the transmission of quasi-analog signals. These noise power values shall be measured at or referenced to a OTLP of the trunk under consideration and shall be obtained during 99 percent or more of the time (see Note 1 of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.2.3.1.5.2). Noise power values for data transmission, expressed in pW0 or dBrn0, shall be measured with flat weighting over the frequency band occupied by the quasi-analog signals.

NOTE: The maximum noise power values for FDM trunks of the tactical subsystem type III have been based on the noise power values for FDM VF links given in 5.7.3.1.3.2 and 5.7.3.1.3.4. The noise power values stated in 5.7.4.2.5 allow configurations of FDM trunks with modulation links different from those shown in Figures 27 and 29.

5.7.4.2.6 Noise power of trunks derived from SSB HF radio links. Under consideration.

NOTE: Trunks with nominal 3 kHz channels derived from SSB HF radio links have very limited use in tactical systems. These trunks cannot meet the standards for general purpose trunks derived from wideband radio links, such as LOS or tropospheric scatter radio links. Standards for SSB HF radio links contained in MIL-STD-188C are being updated and will be published in MIL-STD-188-141. (See Appendix B.)

5.7.4.2.7 Noise power of wire and cable trunks. Wire and cable trunks shall meet the same noise power standards as trunks derived from wideband radio links, such as LOS or tropospheric scatter radio links.

5.7.4.2.8 Single tone interference. Under consideration.

5.7.4.3 Transfer function parameters.

5.7.4.3.1 Insertion loss versus frequency characteristics.

5.7.4.3.1.1 Trunks of tactical subsystem type I. The following requirement shall apply to FDM trunks, consisting of one LOS radio link or one tropospheric scatter radio link, of the tactical subsystem type I (see Table I). For a nominal 4 kHz VF channel of an FDM trunk, the insertion loss versus frequency characteristic, referenced to 1000 Hz, shall not exceed the values given in Table XLVI over the frequency bands stated. Loss is indicated by a (+) and gain by a (-) sign. (See Figure 37A.)

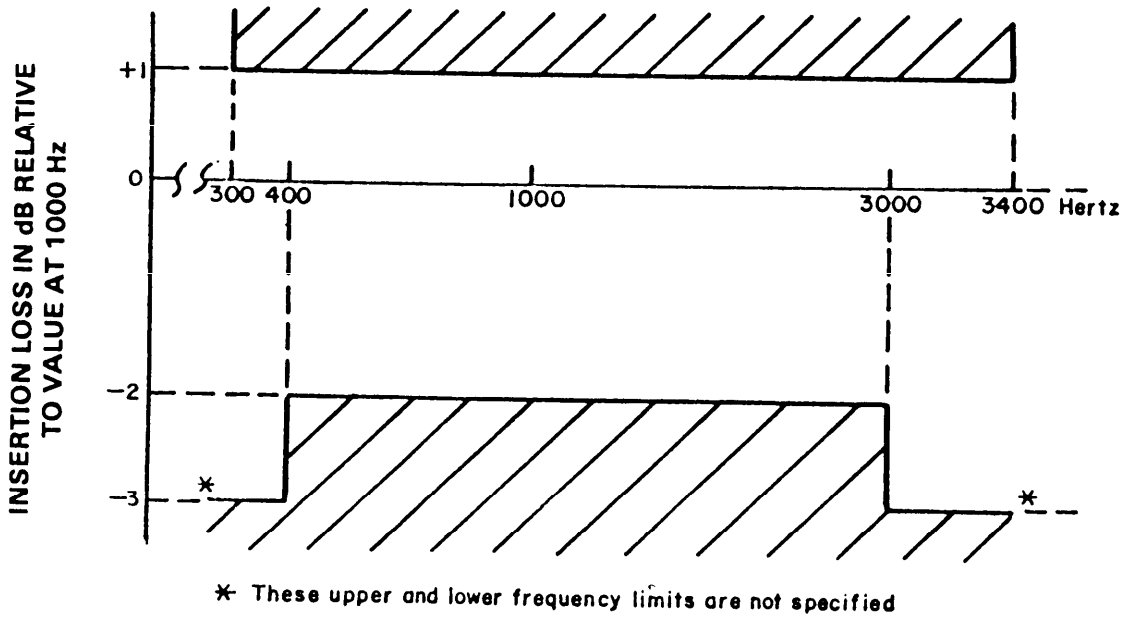
NOTE: The insertion loss versus frequency characteristic for FDM trunks of the tactical subsystem type I is the same as for FDM VF links given in 5.7.3.1.5.1.

5.7.4.3.1.2 Trunks of tactical subsystem type II. The following requirement shall apply to TDM/PCM trunks of the tactical subsystem type II (see Table I), regardless of whether an LOS radio link, a tropospheric scatter radio link, a satellite link or a cable link is employed in the transmission subsystem of the trunk. For a nominal 4 kHz VF channel of a TDM/PCM trunk, the insertion loss versus frequency characteristic,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

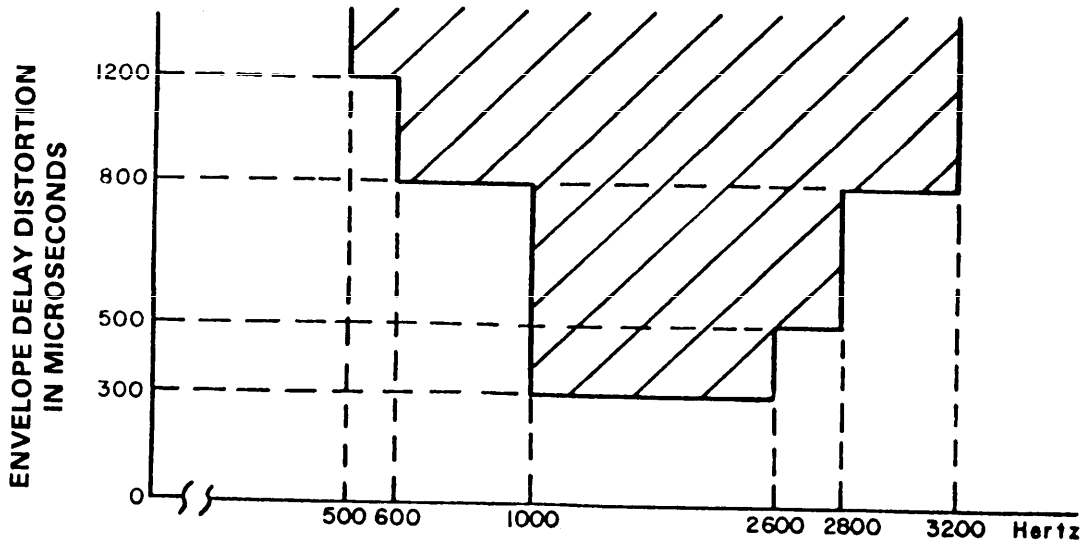
Page 196 is missing from the original legacy source document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



* These upper and lower frequency limits are not specified

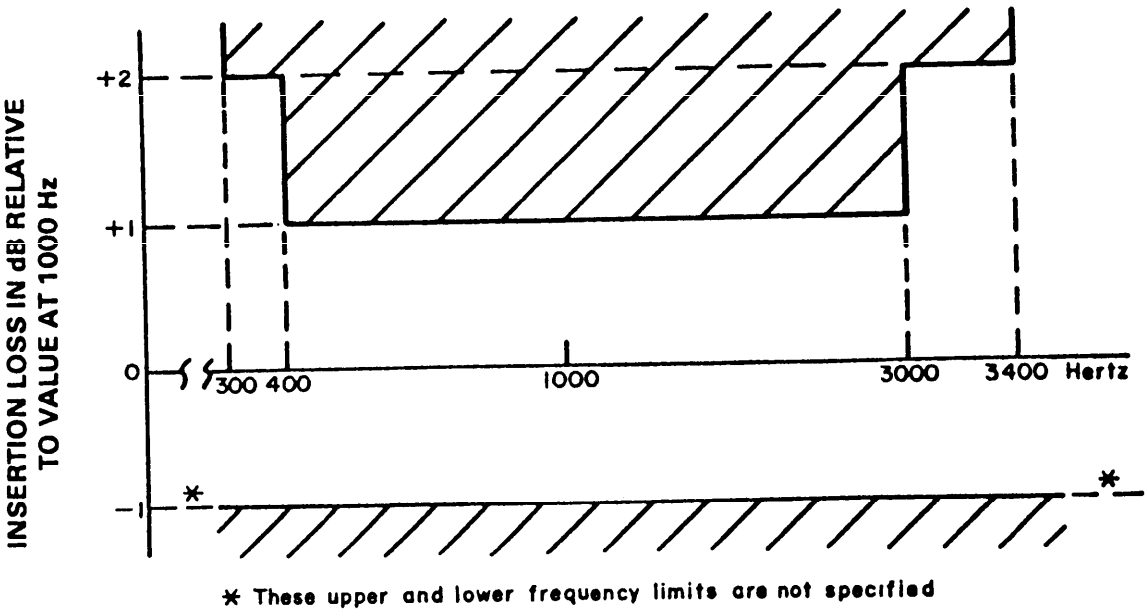
A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



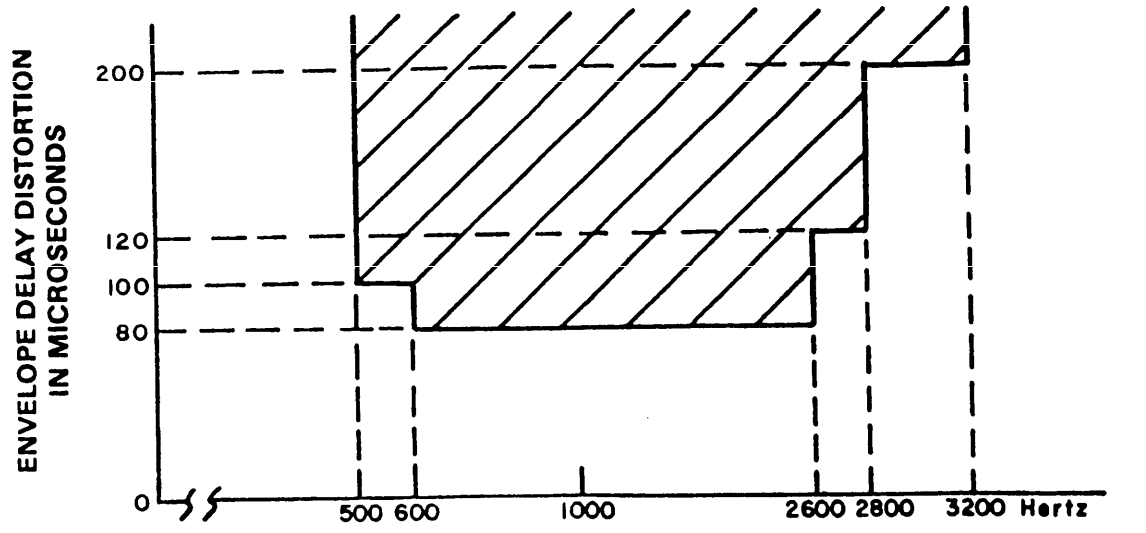
B. ENVELOPE DELAY DISTORTION

FIGURE 37. Parameters for FDM trunks of tactical subsystem type I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 38. Parameters for TDM/PCM trunks of tactical subsystem type II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.4.3.1.3 Trunks of tactical subsystem type III. The following requirement shall apply to FDM trunks, consisting of one or more LOS radio links or one or more tropospheric scatter radio links, or combinations of both, of the tactical subsystem type III (see Table I). For a nominal 4 kHz VF channel of an FDM trunk, the insertion loss versus frequency characteristic, referenced to 1000 Hz, shall not exceed the values given in Table XLVIII over the frequency bands stated. Loss is indicated by a (+) and gain by a (-) sign. (See Figure 39A.)

NOTE: The insertion loss versus frequency characteristic for FDM trunks of the tactical subsystem type III has been based on the corresponding parameters for FDM equipment and VF links given in 5.7.3.1.5.2.

TABLE XLVIII. Insertion loss versus frequency characteristic for FDM trunks of tactical subsystem type III.

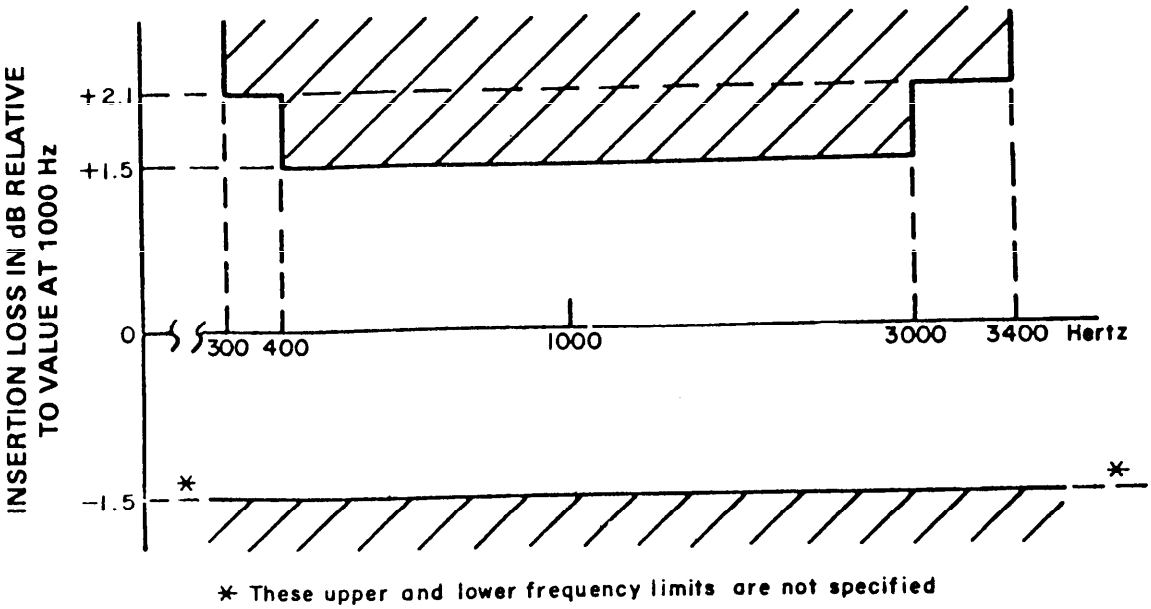
Frequency in Hz	Insertion Loss in dB (Referenced to 1000 Hz)
$f < 300$	≥ -1.5
$300 \leq f < 400$	-1.5 to +2.1
$400 \leq f < 3000$	-1.5 to +1.5
$3000 \leq f < 3400$	-1.5 to +2.1
$3400 \leq f$	≥ -1.5

5.7.4.3.2 Envelope delay distortion.

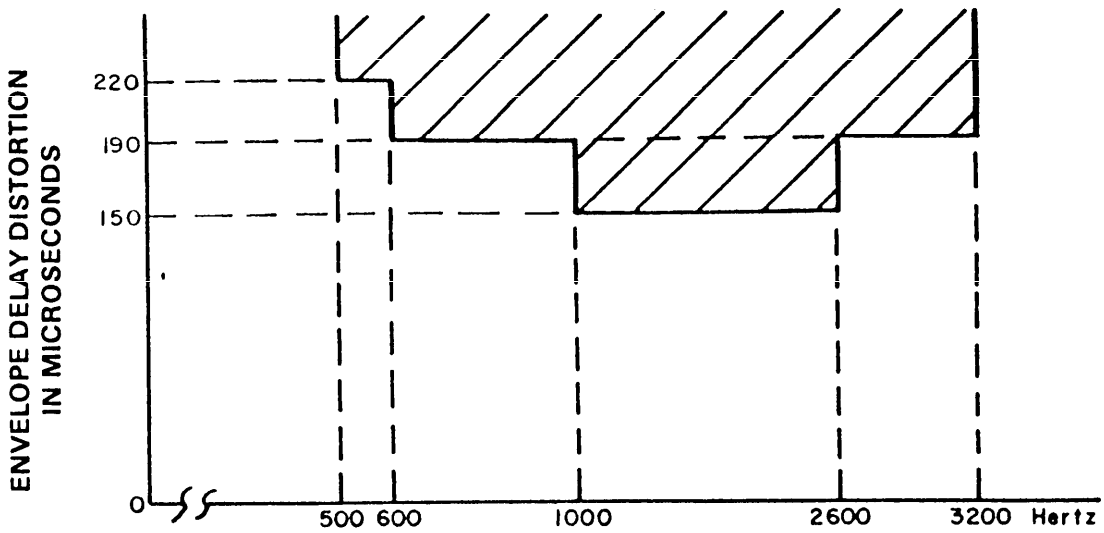
5.7.4.3.2.1 Trunks of tactical subsystem type I. The following requirement shall apply to FDM trunks, consisting of one LOS radio link or one tropospheric scatter radio link, of the tactical subsystem type I (see Table I). For a nominal 4 kHz VF channel of an FDM trunk, the envelope delay distortion measured between 500 Hz and 3200 Hz (see 70.5 of Appendix G), shall not exceed the values given in Table XLIX over the frequency bands stated. (See Figure 37B.)

NOTE: The envelope delay distortion for FDM trunks of the tactical subsystem type I is the same as for FDM VF links given in 5.7.3.1.6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 39. Parameters for FDM trunks of tactical subsystem type III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE XLIX. Envelope delay distortion for FDM trunks of tactical subsystem type I.

Frequency in Hz	Maximum Envelope Delay Distortion in microseconds
$500 \leq f < 600$	1200
$600 \leq f < 1000$	800
$1000 \leq f < 2600$	300
$2600 \leq f < 2800$	500
$2800 \leq f \leq 3200$	800

5.7.4.3.2.2 Trunks of tactical subsystem type II. The following requirement shall apply to TDM/PCM trunks of the tactical subsystem type II (see Table I), regardless of whether an LOS radio link, a tropospheric scatter radio link, a satellite link or a cable link is employed in the transmission subsystem of the trunk. For a nominal 4 kHz VF channel of a TDM/PCM trunk, the envelope delay distortion measured between 500 Hz and 3200 Hz (see 70.5 of Appendix G), shall not exceed the values given in Table L over the frequency bands stated. (See Figure 38B.)

NOTE: The envelope delay distortion for TDM/PCM trunks of the tactical subsystem type II is the same as for TDM/PCM VF links given in 5.7.3.2.4.2.

5.7.4.3.2.3 Trunks of tactical subsystem type III. The following requirement shall apply to FDM trunks, consisting of one or more LOS radio links or one or more tropospheric scatter radio links, or combinations of both, of the tactical subsystem type III (see Table I). For a nominal 4 kHz VF channel of an FDM trunk, the envelope delay distortion measured between 500 Hz and 3200 Hz (see 70.5 of Appendix G), shall not exceed the values given in Table LI over the frequency bands stated. (See Figure 39B.)

NOTE: The envelope delay distortion for FDM trunks of the tactical subsystem type III has been based on the corresponding parameters for FDM equipment and VF links given in 5.7.3.1.6.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE L. Envelope delay distortion for TDM/PCM trunks of tactical subsystem type II.

Frequency in Hz	Maximum Envelope Delay Distortion in microseconds
$500 \leq f < 600$	100
$600 \leq f < 2600$	80
$2600 \leq f < 2800$	120
$2800 \leq f \leq 3200$	200

TABLE LI. Envelope delay distortion for FDM trunks of tactical subsystem type III.

Frequency in Hz	Maximum Envelope Delay Distortion in microseconds
$500 \leq f < 600$	220
$600 \leq f < 1000$	190
$1000 \leq f < 2600$	150
$2600 \leq f \leq 3200$	190

5.7.4.3.3 Total harmonic distortion. For FDM trunks of the tactical subsystems type I and III (see Table I), the total harmonic distortion, produced by any single-frequency test signal between 300 Hz and 3400 Hz, shall be at a level which is at least 38 dB below reference level (-38 dBm0). The total harmonic distortion shall be measured with a standard test signal in accordance with 5.6.6.1.

NOTE: The standard of 5.7.4.3.3 cannot be met by trunks comprised of TDM/PCM VF links of the tactical subsystem type II. (See 5.7.3.2.4.3.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.4.3.4 Intermodulation distortion (IMD). The individual IMD products of a trunk produced by any two equal level single-frequency signals between 300 Hz and 3000 Hz, shall be at a level which is at least 30 dB below reference level (-30 dBm0). The signal level for IMD measurements shall comply with the requirements of 5.6.6.4.

NOTE: The frequencies of the two equal level signals should be selected so that at least the third order harmonic products fall within the specified frequency band.

5.7.4.3.5 Frequency displacement. For FDM trunks of the tactical subsystem type I and TDM/PCM trunks of the tactical subsystem type II (see Table I), any single-frequency signal between 300 Hz and 3400 Hz transmitted over these trunks shall be reproduced at the trunk output with a frequency error of not more than ±1 Hz. For FDM trunks of the tactical subsystem type III (see Table I), any single-frequency signal between 300 Hz and 3400 Hz transmitted over this trunk shall be reproduced at the trunk output with a frequency error of not more than ±0.1 Hz.

5.7.4.3.6 Insertion gain. For voice service or telegraph and data service, or both, provided over switched and nonswitched (point-to-point) circuits of the tactical subsystems type I and type III (see Table I), the insertion gain of trunks, regardless of two-wire or four-wire terminations (see Note 1), shall be 0 dB, ±0.5 dB, measured at 1000 Hz, ±25 Hz, with the following exception: For telegraph and data service provided over nonswitched (point-to-point) circuits, the last trunk in the direction of transmission connected to a four-wire receiving loop (see Note 1) should have an insertion gain (in dB) that equals the insertion loss (in dB) of the receiving loop, so that a circuit OTLP is obtained at the input terminals of the data sink. (See Figures 52 and 58 and 60.3.2.1 of Appendix F.)

For voice service or telegraph and data service, or both, provided over switched and nonswitched (point-to-point) circuits of the tactical subsystem type II (see Table I), the insertion gain of trunks connected to a two-wire termination at the trunk output (see Note 2) shall be 5 dB, ±0.5 dB, and the insertion gain of trunks connected to a four-wire termination at the trunk output shall be 0 dB, ±0.5 dB, measured at 1000 Hz, ±25 Hz (see 5.7.3.2.2.3), with the following exception: For telegraph and data service provided over nonswitched (point-to-point) circuits, the last trunk in the direction of transmission connected to a four-wire receiving loop (see Note 1) should have an insertion gain (in dB) that equals the insertion loss (in dB) of the receiving loop, so that a circuit -4TLP is obtained at the input terminals of the data sink. (See Figure 55 and 60.3.2.2 of Appendix F.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE 1: Two-wire loops are not recommended for telegraph and data service employing quasi-analog signals.

NOTE 2: The insertion gain of 5 dB applies to trunks of the tactical subsystem type II connected to a two-wire output termination for both a two-wire or a four-wire termination at the trunk input. In the case of a two-wire termination at the trunk input, the trunk insertion gain of 5 dB will result in an overall insertion loss of 3 dB, measured from the two-wire input to the two-wire output terminations. (See Figure 45.) The insertion loss of 3 dB will assure a sufficient singing margin for stable operation of such a two-wire connection, even in the absence of an idle line termination (the trunk output is open or not terminated). In the case of a four-wire termination at the trunk input, the trunk insertion gain of 5 dB will result in an overall insertion gain of 1 dB, measured from the four-wire input to the two-wire output terminations, and will compensate for the hybrid transformer insertion loss assumed to be 4 dB. (See Figure 46.)

5.7.4.3.7 Limiting. The limiting characteristic of trunks, a smooth curve, shall be such that input signal levels inserted at or referenced to a OTLP at the trunk input are compressed at the trunk output and do not exceed the output signal levels as shown in Table LII.

TABLE LII. Limiting characteristic for trunks.

Input Signal Level (dBm0)	Maximum Output Signal Level (dBm0)	Compression
-10	-10	None
0	0	None
+3	+3	None
+5	+4	1 dB
+8	+5	3 dB

5.7.4.3.8 Net loss variations. The net loss variations of a trunk shall not exceed ±1.5 dB over any 30 consecutive days and ±0.5 dB over any 15 consecutive minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.7.4.4 Crosstalk coupling loss. Between any two trunks of a trunk group, the crosstalk coupling loss shall be equal to or greater than 55 dB (Design Objective: 65 dB) for any frequency between 300 Hz and 3400 Hz, when measured at or referenced to equal TLP. The near-end crosstalk coupling loss of a trunk shall be equal to or greater than 50 dB for any frequency between 300 Hz and 3400 Hz, when measured at or referenced to equal TLP.

NOTE: The requirements of 5.7.4.4 do not apply to trunks derived from wire and cable transmission subsystems. The near-end crosstalk coupling loss for wire and cable trunks is given in 5.4.7.2.1.

5.7.4.5 Trunk input/output impedances. (MIL-STD-188-100.) The input and output impedances of a trunk shall each be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band from 300 Hz to 3400 Hz. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

5.7.4.6 Transmission level points (TLP) and test signal levels. For voice service or telegraph and data service, or both, provided over switched and nonswitched (point-to-point) circuits of the tactical subsystems type I and type III (see Table I), the input and output terminals of an FDM trunk that forms part of a circuit and is connected to a two-wire or a four-wire termination shall each have a OTLP with a test signal level of 0 dBm (0 dBm0) at these points, with the following exception: For telegraph and data service provided over nonswitched (point-to-point) circuits, the output terminals of the last trunk in the direction of transmission connected to a four-wire receiving loop (see Note 1 of 5.7.4.3.6) should have a TLP which is higher than the circuit OTLP by an amount equal to the insertion loss of the receiving loop, so that a circuit OTLP is obtained at the input terminals of the data sink. (See Figures 52 and 58 and 60.3.2.1 of Appendix F.)

For voice service or telegraph and data service, or both, provided over switched and nonswitched (point-to-point) circuits of the tactical subsystem type II (see Table I), the input terminals of a TDM/PCM trunk that forms part of a circuit and is connected to a two-wire or a four-wire termination shall each have a -4TLP with a test signal level of -4 dBm (0 dBm0) at this point (see Note); the output terminals of a TDM/PCM trunk that forms part of a circuit and is connected to a two-wire termination shall have a +1TLP with a test signal level of +1 dBm (0 dBm0) at this point (see Note 2 of 5.7.4.3.6); the output terminals of a TDM/PCM trunk that forms part of a circuit and is connected to a four-wire termination shall have a -4TLP with a test signal level of -4 dBm (0 dBm0) at this point. The requirements stated above apply with the following exception: For telegraph and data service provided over nonswitched (point-to-point) circuits, the output terminals of the last trunk in the direction of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

transmission connected to a four-wire receiving loop (see Note 1 of 5.7.4.3.6) should have a TLP which is higher than the circuit -4TLP by an amount equal to the insertion loss of the receiving loop, so that a circuit -4TLP is obtained at the input terminals of the data sink. (See Figure 55 and 60.3.2.2 of Appendix F.)

NOTE: Current multiplexing equipment (TDM/PCM and FDM) used in the tactical subsystem type II has been designed with a -4TLP at the four-wire VF input and output terminals to obtain a OTLP at the connection between the output of a two-wire switchboard and the input of a two-wire/four-wire hybrid transformer connected to the trunk input. (See Figure 53.)

5.7.4.7 Signal discontinuities. Under consideration.

5.7.4.8 Grade of service for trunk groups. The size of trunk groups (number of trunks between nodes) shall be determined on the basis of three lost (or blocked) calls per 100 attempted calls during the busy hour of a switched multichannel network.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.8 Transmission channel subsystems. This paragraph deals with standards and design objectives of transmission channels between the originating node and the terminating node of a circuit. The transmission channels do not include the loops to and from the terminal equipment or terminal subsystem. (See Figure 1.) The standards and design objectives are expressed in the form of technical parameters, such as received noise power or harmonic distortion, which apply to all transmission channels regardless of transmission distance, transmission medium and multiplexing scheme used (FDM, TDM or a mix of FDM and TDM), except where stated otherwise in the applicable subparagraph.

In general, a transmission channel will be a nominal 4 kHz VF channel (see 5.8.2). There may be also, within the communications network, a nominal 3 kHz VF transmission channel (see 5.8.1), consisting in its basic configuration of nominal 4 kHz links in tandem with one or more nominal 3 kHz links derived from submarine cable or HF radio subsystems. Additionally, a nominal 48 kHz group transmission channel (see 5.8.3) may be provided through the use of FDM equipment normally employed in the tactical subsystem type III (see Table I).

5.8.1 Parameters for nominal 3 kHz VF transmission channels. Under consideration.

NOTE: These transmission channels are derived from submarine cable subsystems or from HF radio links. Submarine cables are normally not employed in tactical communications systems. Therefore, standards for submarine cable transmission are not included in this document. Standards for nominal 3 kHz VF transmission channels derived from HF radio links are being developed and will be published in MIL-STD-188-141. (See Appendix B.)

5.8.2 Parameters for nominal 4 kHz VF transmission channels.

5.8.2.1 Transmission channel input parameters.

5.8.2.1.1 Speech volume. Not standardized.

NOTE: The speech volume at a transmission channel input depends on the talker volume distribution (see 5.2.2.1.1) and the loop insertion loss (see 5.5.2). Typical values of speech volumes that should be achieved at a transmission channel input are shown in Figures 50, 51, 53, 54, 56, 57, 59 and 60.

5.8.2.1.2 Quasi-analog signal level. Same as 5.2.3.1.4.

5.8.2.1.3 Standard test signal. Same as 5.6.6.1.

5.8.2.1.4 Standard test tone. Same as 5.6.6.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.8.2.1.5 Monitoring test tone. Same as 5.6.6.3.

5.8.2.1.6 Levels for IMD measurements. Same as 5.6.6.4.

5.8.2.2 Transmission channel output parameters.

5.8.2.2.1 Speech volume. Not standardized.

NOTE: The speech volume at a transmission channel output depends on the talker volume distribution (see 5.2.2.1.1), the loop insertion loss (see 5.5.2) and the insertion loss of the transmission channel. Typical values of speech volumes that should be achieved at a transmission channel output are shown in Figures 50, 51, 53, 54, 56, 57, 59 and 60.

5.8.2.2.2 Quasi-analog signal level. The quasi-analog signal level at the output of a nominal 4 kHz transmission channel (see Figure 1) shall be -13 dBm0 for FDM transmission channels of the tactical subsystems type I and type III (see Table I), and -6 dBm0 for TDM/PCM transmission channels of the tactical subsystem type II in accordance with 5.2.3.1.4. The TLP at the transmission channel output shall be adjusted to a value as stated in 5.7.4.3.6.

NOTE: For the transmission of digital signals over TDM/PCM transmission channels see Note of 5.7.3.2.6.

5.8.2.2.3 Noise power for FDM transmission channels. For voice transmission over a nominal 4 kHz FDM transmission channel (see Figure 1) of the tactical subsystems type I and type III (see Table I), the total transmission channel noise power (multiplex plus transmission media noise power) shall not exceed 40,000 pWp0 (46.0 dBrnc0), when measured at or referenced to a OTLP of the transmission channel. (See Notes 1 and 2.) For the transmission of quasi-analog signals over a nominal 4 kHz FDM transmission channel of the tactical subsystems type I and type III, the non-impulse type of the total transmission channel noise power (multiplex plus transmission media noise power) shall not exceed 63,000 pW0 (48.0 dBrn0) during 99 percent or more of the time (see Note 3) and shall not exceed 400,000 pW0 (56.0 dBrn0) for 1 percent or less of the time, when measured at or referenced to a OTLP of the transmission channel. The total transmission channel noise power shall be measured with flat weighting over the frequency band from 300 Hz to 3400 Hz. (See Notes 4 and 5.)

NOTE 1: The requirements of 5.8.2.2.3 apply to all configurations of a transmission channel regardless of the type of trunks employed (LOS, tropospheric scatter or wire and cable).

NOTE 2: For the tactical subsystem type I, the maximum noise power of 40,000 pWp0 for voice service will be met by a transmission channel covering the probable maximum transmission distance of 300 km (see 4.1.1) and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

consisting of up to six LOS trunks or up to two tropospheric scatter trunks in tandem, or a combination of both, each trunk having the maximum allowable noise power according to 5.7.4.2.3. For the tactical subsystem type III, the maximum noise power of 40,000 pWp0 for voice service allows the engineering of a transmission channel with trunk configurations different from those shown in Figures 27 and 29 to cover the probable maximum transmission distance of 1800 km (see 4.1.2). Additionally, the maximum noise power of 40,000 pWp0 for voice service over a nominal 4 kHz transmission channel will assure that the requirement of 5.2.2.1.3 for the received noise power at the listener telephone set can be met.

NOTE 3: The unit of time (to which the percentage of time applies) depends on the special requirements of the system under consideration and, therefore, cannot be given in a general form. (See also Note 1 of 5.2.3.1.5.2.)

NOTE 4: The commonly used noise measuring sets will measure total rms noise power including the average impulse noise power. However, the contribution of the average impulse noise, as registered by these meters, is small enough to be neglected.

NOTE 5: The requirement of 5.8.2.2.3, stating a maximum FDM channel noise power of 63,000 pW0 for the transmission of quasi-analog signals, has been based on a minimum SNR of 26 dB at the data sink in accordance with 5.2.3.1.5.3, assuming that the spectrum of the quasi-analog signal occupies the entire nominal 4 kHz bandwidth of the transmission channel (worst case assumption) and that the loop noise is in accordance with 5.5.2.6.

5.8.2.2.4 Equivalent PCM noise power for TDM/PCM transmission channels. The following requirement applies to TDM/PCM transmission channels (see Figure 1) regardless of whether LOS radio links, tropospheric scatter radio links, satellite links or cable links are employed in the trunks of the transmission channel. As a Design Objective for voice transmission over a nominal 4 kHz TDM/PCM transmission channel of a tactical subsystem type II (see Table I), the equivalent PCM noise power shall not exceed 31,500 pWp (45.0 dBrnc), when measured at or referenced to a -4TLP of this transmission channel.

NOTE 1: For a definition of equivalent PCM noise see FED-STD-1037.

NOTE 2: The design objective of 5.8.2.2.4, stating a maximum equivalent PCM noise power of 31,500 pWp for a transmission channel, has been based on six TDM/PCM trunks in tandem, each trunk having the maximum allowable equivalent PCM noise power of 5000 pWp according to 5.7.4.2.4. For the tactical subsystem type II, the maximum equivalent PCM noise power of 31,500 pWp for a transmission channel will result in an acceptable noise power value measured at the line terminals of a listener telephone set in

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the currentversion before use.

accordance with 5.2.2.1.3, provided the loop noise is in accordance with 5.5.2.6.

5.8.2.2.5 Idle channel noise power for TDM/PCM transmission channels. The following requirements shall apply to TDM/PCM transmission channels (see Figure 1) regardless of whether LOS radio links, tropospheric scatter radio links, satellite links or cable links are employed in the trunks of the transmission channel. For voice transmission over a nominal 4 kHz TDM/PCM transmission channel of the tactical subsystem type II (see Table I), the idle channel noise power shall not exceed 16,000 pWp (42.0 dBrnc), when measured at or referenced to a -4TLP of the transmission channel. For transmission of quasi-analog signals over a nominal 4 kHz TDM/PCM transmission channel of the tactical subsystem type II, the idle channel noise power shall not exceed 25,000 pW (44.0 dBrn) flat-weighted over the frequency band from 300 Hz to 3400 Hz, when measured at or referenced to a -4TLP of the transmission channel.

NOTE 1: The standards of 5.8.2.2.5, stating a maximum idle channel noise power of 16,000 pWp for voice transmission and 25,000 pW for the transmission of quasi-analog signals, have been based on six TDM/PCM trunks in tandem, each trunk having the maximum idle channel noise power of 2000 pWp or 3150 pW according to 5.7.4.2.4, and include the noise power contributed by all nodes of a transmission channel.

NOTE 2: The maximum idle channel noise power of 25,000 pW stated in 5.8.2.2.5 is a necessary but not sufficient requirement for satisfactory telegraph and data service since signal discontinuities (see 5.2.3.1.7) may increase the BER for certain unpredictable periods of time. For future applications, the transmission of quasi-analog signals over TDM/PCM transmission channels is not recommended. (See Note of 5.7.3.2.6.)

5.8.2.2.6 Single tone interference. (MIL-STD-188-100.) No interfering tone of a single frequency shall exceed 30 dBrnc0, measured at or referenced to a 0TLP of a nominal 4 kHz transmission channel (see Figure 1).

5.8.2.3 Transmission channel input/output impedances. (MIL-STD-188-100.) For a nominal 4 kHz transmission channel (see Figure 1), the input and output impedances shall each be 600 ohms, balanced to ground, with a minimum return loss of 26 dB against a 600-ohm resistance over the frequency band from 300 Hz to 3400 Hz. The electrical symmetry shall be sufficient to suppress longitudinal currents to a level which is at least 40 dB below reference level (-40 dBm0).

5.8.2.4 Insertion loss versus frequency characteristic. (MIL-STD-188-100.) For a nominal 4 kHz transmission channel (see Figure 1), the insertion loss versus frequency characteristic, referenced to 1000 Hz, shall not exceed the values given in Table LIII over the frequency bands stated. Loss is indicated by a (+) and gain by a (-) sign. (See Figure 40A.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the currentversion before use.

NOTE: Transmission channels meeting the requirements of 5.8.2.4 may need regeneration or conditioning, or both, to meet the additional requirements of 5.2.3.1.6.1 for acceptable telegraph and data service on a user-to-user basis.

TABLE LIII. Insertion loss versus frequency characteristic for nominal 4 kHz transmission channels.

Frequency in Hz	Insertion Loss in dB (Referenced to 1000 Hz)
$f < 300$	≥ -4
$300 \leq f < 400$	-4 to +8
$400 \leq f < 3000$	-4 to +4
$3000 \leq f < 3400$	-4 to +8
$3400 \leq f$	≥ -4

5.8.2.5 Distortion parameters.

5.8.2.5.1 Envelope delay distortion. (MIL-STD-188-100.) For telegraph and data service over a nominal 4 kHz transmission channel (see Figure 1), the envelope delay distortion measured between 500 Hz and 3200 Hz (see 70.5 of Appendix G), shall not exceed the values given in Table LIV over the frequency bands stated. (See Figure 40B.)

NOTE: Transmission channels meeting the requirements of 5.8.2.5.1 may need regeneration or conditioning, or both, to meet the additional requirements of 5.2.3.1.6.2 for acceptable telegraph and data service on a user-to-user basis.

5.8.2.5.2 Total harmonic distortion. As a Design Objective for a nominal 4 kHz FDM transmission channel (see Figure 1) of the tactical subsystems type I and III (see Table I), the total harmonic distortion, produced by any single-frequency test signal between 300 Hz and 3400 Hz, should be at a level which is at least 30 dB below reference level (-30 dBm0). The total harmonic distortion shall be measured with a standard test signal in accordance with 5.6.6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: The design objective of 5.8.2.5.2 cannot be met by transmission channels comprised of up to six TDM/PCM trunks with multiplexer equipment currently in the inventory. (See 5.7.3.2.4.3.)

TABLE LIV. Envelope delay distortion for nominal 4 kHz transmission channels.

Frequency in Hz	Maximum Envelope Delay Distortion in microseconds
$500 \leq f < 1000$	2500
$1000 \leq f < 2600$	1750
$2600 \leq f \leq 3200$	2500

5.8.2.5.3 Intermodulation distortion (IMD). Under consideration.

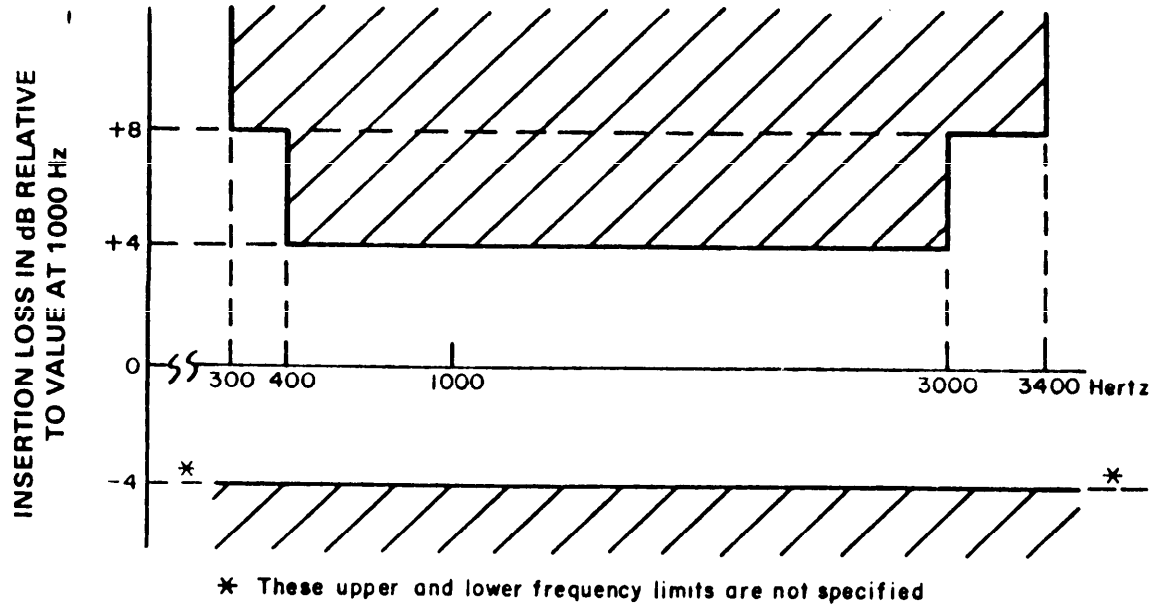
5.8.2.6 Signal discontinuities. Under consideration.

NOTE: At the present time, the statistical characteristics of the various signal discontinuities are not known well enough to establish parameter values for the signal discontinuities of a transmission channel based on measurements or on prorating and summation methods. However, experience has shown that a single link or a loop will normally be the major source of signal discontinuities of a transmission channel or a circuit. These signal discontinuities may often be traceable to excessive traffic levels, to propagation anomalies of a single link or to maintenance procedures. It is recommended that the signal discontinuity parameters of a circuit given in 5.2.3.1.7 be used as a general guidance for the signal discontinuities of a transmission channel.

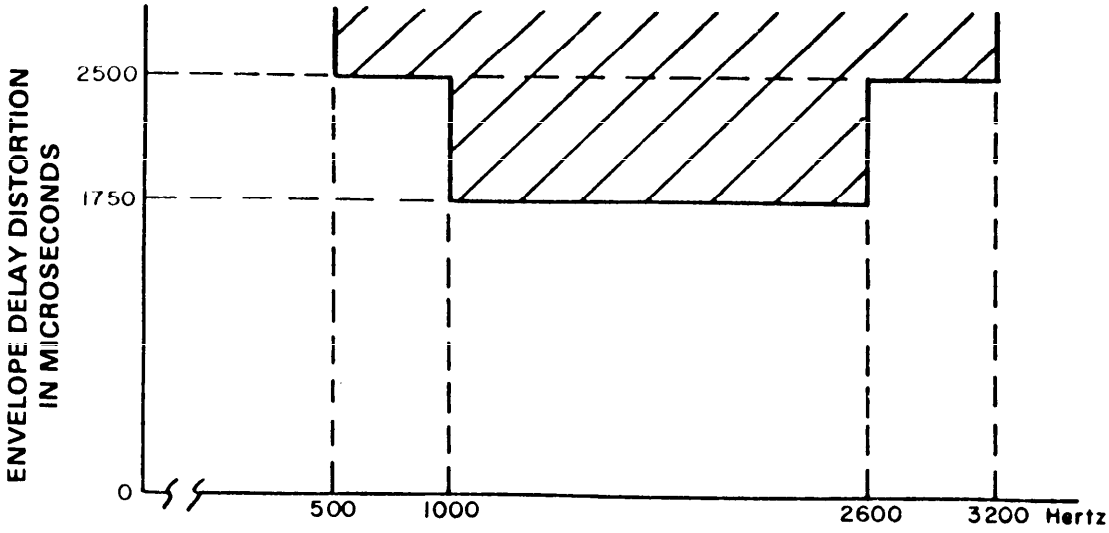
5.8.2.7 Frequency displacement. For a nominal 4 kHz transmission channel (see Figure 1), any single-frequency signal between 300 Hz and 3400 Hz transmitted over this channel shall be reproduced at the channel output with a frequency error of not more than ±4 Hz.

5.8.2.8 Net loss variations. (MIL-STD-188-100.) For a nominal 4 kHz transmission channel (see Figure 1), the net loss variations shall not exceed ±4.5 dB over any 30 consecutive days and ±1.2 dB over any 15 consecutive minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 40. Parameters for nominal 4 kHz transmission channels.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

5.8.3 Parameters for 48 kHz FDM group transmission channels. The basic FDM group channel is derived by frequency division multiplexing of twelve nominal 4 kHz VF channels and by translation into a basic FDM group frequency band, resulting in a bandwidth of 48 kHz. This basic FDM group channel can be utilized for higher speed data transmission employing quasi-analog signals. The FDM group channel normally terminates at each end of the GDF or equivalent points. (See Figure 27.) Loop parameters for data service over 48 kHz FDM group circuits have not yet been established (see 5.5.5), but loop characteristics must be carefully considered and loops must be individually engineered to ensure satisfactory user-to-user service. General guidance for the engineering of telegraph and data circuits is provided in Appendix F. The following subparagraphs of 5.8.3 apply to 48 kHz FDM group transmission channels (see Figure 1) used for the transmission of quasi-analog data signals. This data service is generally provided by the tactical subsystem type III (see Table I).

5.8.3.1 Transmission channel input parameters.

5.8.3.1.1 Quasi-analog signal level. (MIL-STD-188-100.) For the transmission of quasi-analog data signals over a 48 kHz FDM group transmission channel (see Figure 1), the modulated data signal at the input terminals (GDF or equivalent points) of the transmission channel shall not exceed a level which is 5 dB below the test signal level of 0 dBm0, e.g., the level of the modulated data signal shall be equal to or less than -39.5 dBm at a -34.5TLP. Any single-frequency component of the modulated data signal shall not exceed a level which is 13 dB below the test signal level (-13 dBm0).

5.8.3.1.2 Test signal level. (MIL-STD-188-100.) The level of the test signal at the input terminals (GDF or equivalent points) of a 48 kHz FDM group transmission channel (see Figure 1) shall be -34.5 dBm. (See 5.7.3.1.2.)

5.8.3.2 Transmission channel output parameters.

5.8.3.2.1 Quasi-analog signal level. Not standardized.

5.8.3.2.2 Test signal level. (MIL-STD-188-100.) The level of the test signal at the output terminals (GDF or equivalent points) of a 48 kHz FDM group transmission channel (see Figure 1) shall be -12 dBm, ±0.5 dB. (See 5.7.3.1.2.)

5.8.3.2.3 Noise power. (MIL-STD-188-100.) For the transmission of quasi-analog data signals over a 48 kHz FDM group transmission channel (see Figure 1), the long term median noise power of the transmission channel shall not exceed 60 dBrn0 (-30 dBm0) flat weighted over the 48 kHz bandwidth, when measured at or referenced to a OTLP of the transmission channel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.8.3.2.4 Impulse noise power. (MIL-STD-188-100.) For the transmission of quasi-analog data signals over a 48 kHz FDM group transmission channel (see Figure 1), the unweighted impulse noise power measured at the output terminals (GDF or equivalent points) of the transmission channel shall not exceed 45 counts over any 30 consecutive minutes above a level which is 5 dB below the total signal power level.

5.8.3.3 Transmission channel input/output impedances. Same as 5.7.3.1.4.2.

5.8.3.4 Insertion loss versus frequency characteristic. (MIL-STD-188-100.) For the transmission of quasi-analog data signals over an unconditioned 48 kHz FDM group transmission channel (see Figure 1), the insertion loss versus frequency characteristic, referenced to 83 kHz, shall be within the limits of ±8 dB between 60.6 kHz and 107.7 kHz. (See Figure 41A.)

NOTE: Conditioning of the transmission channel, if required, will be provided externally to the FDM group equipment on a case-by-case basis.

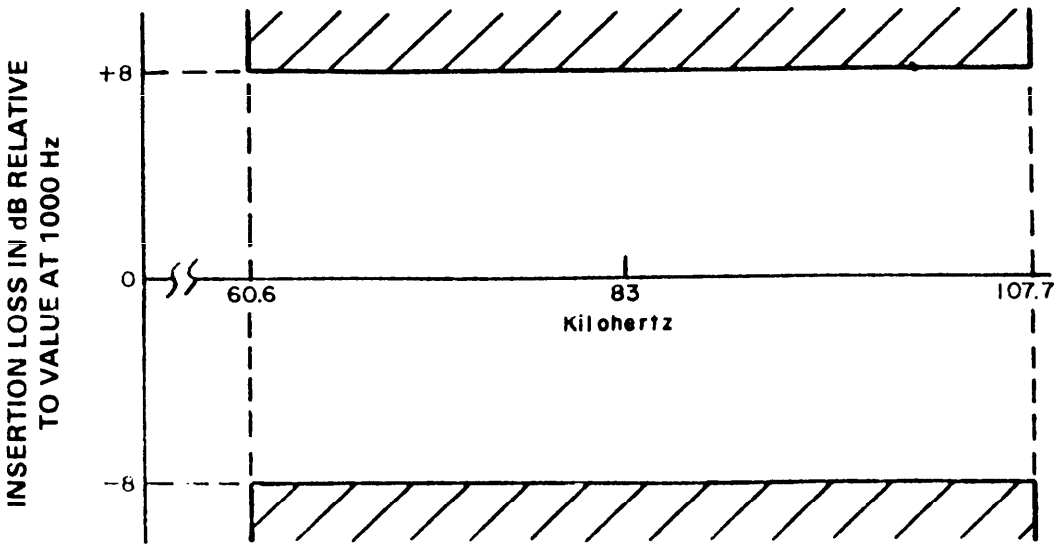
5.8.3.5 Envelope delay distortion. (MIL-STD-188-100.) For the transmission of quasi-analog data signals over an unconditioned 48 kHz FDM group transmission channel (see Figure 1), the envelope delay distortion measured between 64 kHz and 104 kHz (see 70.5 of Appendix G), shall not exceed the values given in Table LV over the frequency bands stated. (See Figure 41B.)

NOTE: Delay equalization of the transmission channel, if required, will be provided externally to the FDM group equipment as part of the FDM group data modem on a case-by-case basis.

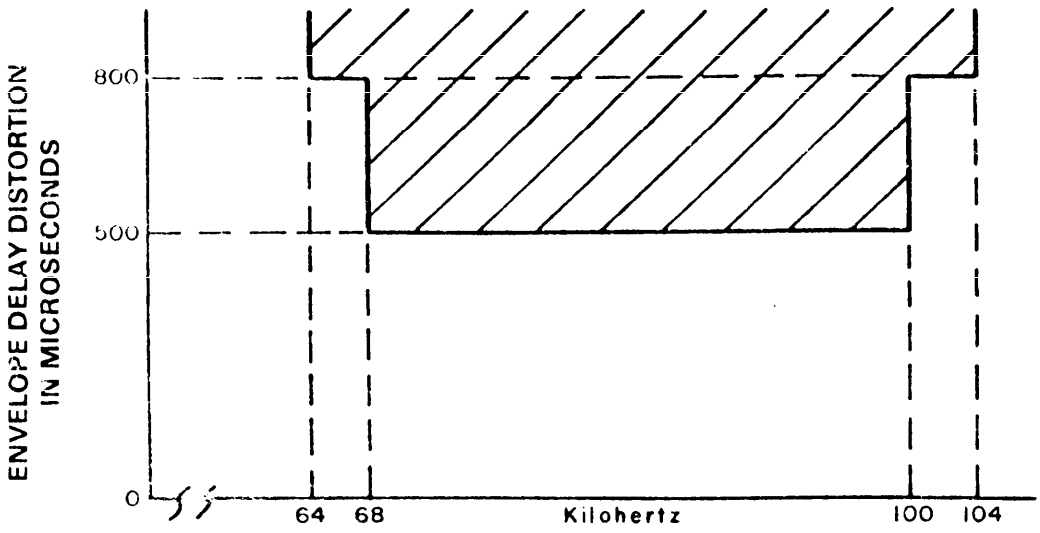
TABLE LV. Envelope delay distortion for 48 KHz FDM group transmission channels.

Frequency in kHz	Maximum Envelope Delay Distortion in microseconds
$64 \leq f < 68$	800
$68 \leq f < 100$	500
$100 \leq f \leq 104$	800

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



A. INSERTION LOSS VERSUS FREQUENCY CHARACTERISTIC



B. ENVELOPE DELAY DISTORTION

FIGURE 41. Parameters for unconditioned 48 kHz FDM group transmission channels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.8.3.6 Net loss variations. (MIL-STD-188-100.) For the transmission of quasi-analog data signals over a 48 kHz FDM group transmission channel (see Figure 1), the net loss variations of the transmission channel shall not exceed ±2.5 dB over any 30 consecutive days, measured between GDF or equivalent points.

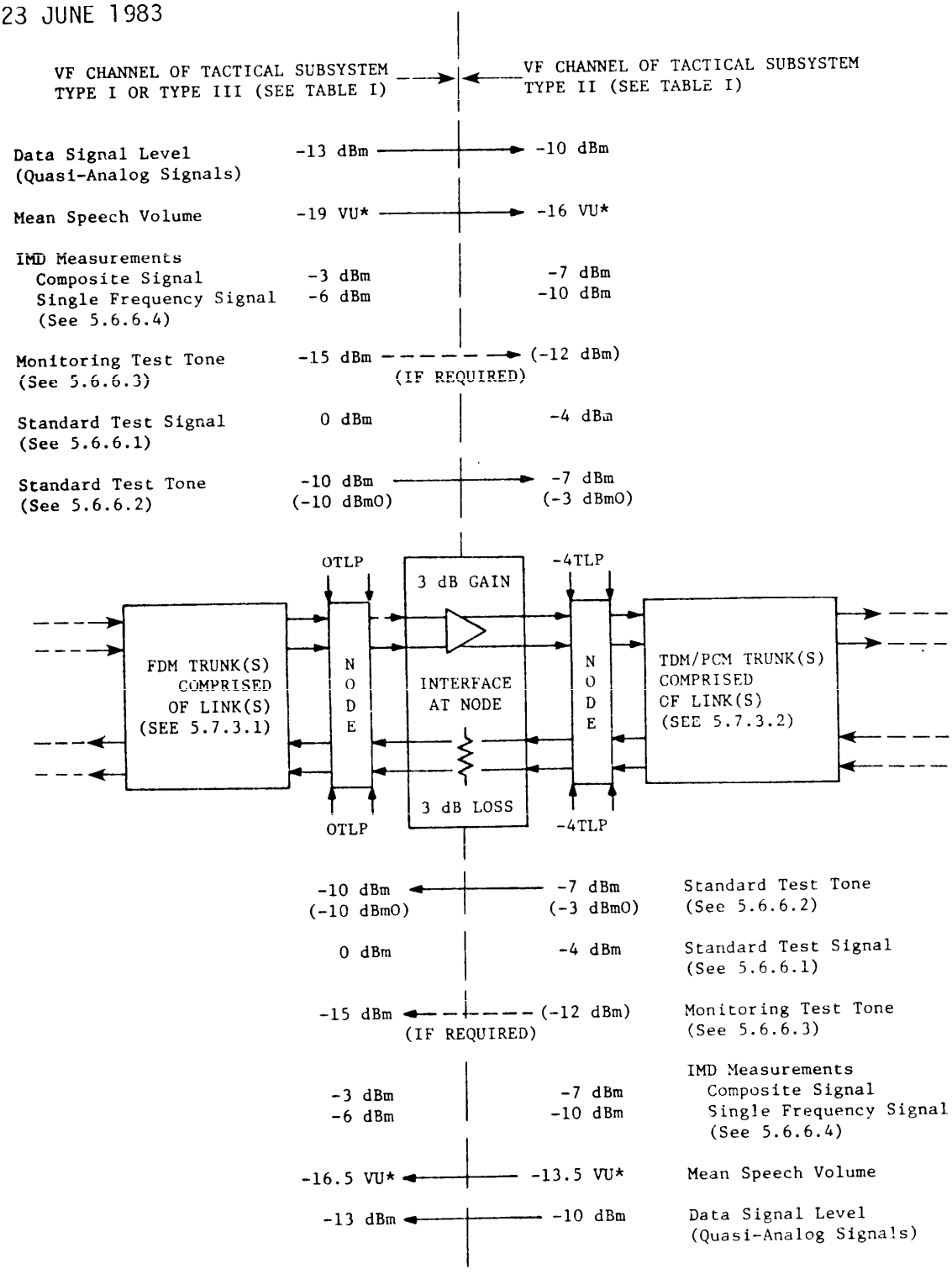
5.8.4 Parameters for digital transmission channels. Under consideration.

5.8.5 Methods for interconnecting VF channels of tactical subsystems type I, II, III and IV. Different requirements for communications subsystems have led to the development of equipment, such as multiplexers and wideband radios, with significantly different signal levels (voice, data and test tone) and overload characteristics. Therefore, the applicable method, specified in 5.8.5.1 through 5.8.5.3, shall be used to interconnect VF channels of the tactical subsystems type I, II, III and IV (see Table I).

5.8.5.1 Special method for interconnecting VF channels of tactical subsystems type I or III with type II. (See Figure 42.) VF channels of the tactical subsystems type I or type III (see Table I) shall be interconnected with VF channels of the tactical subsystem type II on the basis of the standard test tone levels, specified in 5.6.6.2 for the subsystems under consideration, and not on the basis of the TLP of these subsystems. This requirement shall apply regardless of the type of service, such as voice service or telegraph and data service, and of the type of loop employed, i.e., two-wire loops, four-wire loops or a two-wire and a four-wire loop at either end of the connection. The TLP shown in Figure 42 shall be considered only as internal reference for each subsystem, and shall not be transferred across the interface between the tactical subsystems type I or type III and the tactical subsystem type II. For this reason, the standard test signal and the signal for IMD measurements, as shown in Figure 42, shall not be transmitted across this interface. A standard test tone shall be transmitted across the interface to adjust the transmission levels to the specified values.

NOTE 1: The reason for the requirements of 5.8.5.1 is that the TLP must be considered only as internal references for each subsystem and cannot be transferred across a subsystem interface without a level translation. This necessary level translation is achieved by the standard test tone specified at different levels for the subsystems under consideration (see 5.6.6.2). In other words, if a VF channel of the tactical subsystems type I or type III and a VF channel of the tactical subsystem type II would be interconnected based on their different TLP at the interface, a serious level mismatch would result with totally unacceptable traffic signal levels in both subsystems. It should be emphasized that in Figure 42 the standard test tone level of -10 dBm0 is referenced to a OTLP of the tactical subsystems type I or type III, whereas the standard test tone

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the current version before use.



*Nominal values of speech volume, depending on talker volume distribution and loop insertion loss.

FIGURE 42. Signal level and interface diagram for connecting VF channels of tactical subsystems type I or III with type II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

level of -3 dBm0 (-7 dBm at a -4TLP) is referenced to a OTLP of the tactical subsystem type II and not to a OTLP of the other subsystems, even though a OTLP may be physically unavailable in the configuration of the tactical subsystem type II.

NOTE 2: The following items have been considered for the development of an interface standard between the tactical subsystems type I or type III and the tactical subsystem type II:

a. The level difference across the interface is caused by current equipment inventory which cannot be changed significantly in the foreseeable future due to economic factors, such as equipment life cycles.

b. Any solution of the interface problem should have a minimal impact on existing procedures and practices used in all subsystems to be interfaced.

c. The introduction of a new definition into the tactical subsystem type II, namely the standard test tone of -3 dBm0 (see 5.6.6.2), is considered less confusing to communications personnel than a change of established procedures and methods employed within a system, such as the TLP concept.

d. The method of interfacing on the basis of the standard test tone will affect only channels traversing the tactical subsystems type I or type III and the tactical subsystem type II and will not change established test or alignment procedures within any of these subsystems.

NOTE 3: The special method for interconnecting VF channels in accordance with 5.8.5.1 results in a gain of 3 dB at the interface point for signals traversing from the tactical subsystems type I or type III to the tactical subsystem type II, and in an attenuation of 3 dB at the interface point for signals traversing in the opposite direction, as shown in Figure 42.

NOTE 4: The monitoring test tone is not used in the tactical subsystem type II (see 5.6.6.3). The monitoring test tone may be transmitted across an interface between the tactical subsystems type I or type III and type II, if required, and would appear at a level of -12 dBm at a -4TLP of the tactical subsystem type II. (See Figure 42.)

5.8.5.2 Method for interconnecting VF channels of tactical subsystems type I with type III. (See Figure 43.) VF channels of the tactical subsystem type I (see Table I) shall be interconnected with VF channels of the tactical subsystem type III on the basis of the standard test tone level specified in 5.6.6.2 as -10 dBm0 for both types of subsystems. This requirement shall apply regardless of the type of service, such as voice

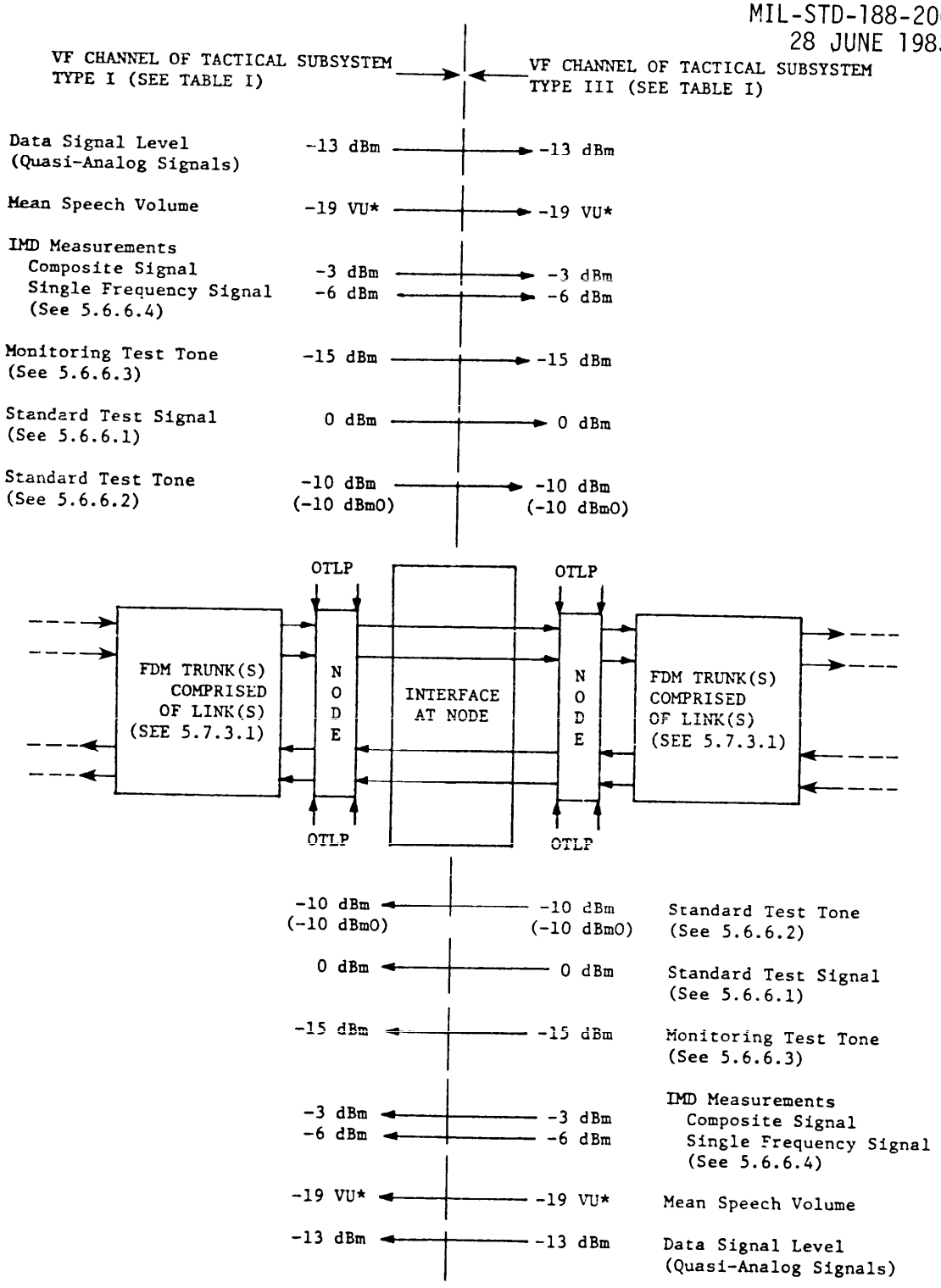
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

service or telegraph and data service, and of the type of loop employed, i.e., two-wire loops, four-wire loops or a two-wire and a four-wire loop at either end of the connection.

NOTE: The method stated in 5.8.5.2 for interconnecting VF channels of the tactical subsystems type I with type III requires no level translation at the interface point between the two subsystems, as opposed to the special method stated in 5.8.5.1. Consequently, the TLP of each subsystem can be transferred across the interface and the method of interconnecting VF channels of the tactical subsystems type I with type III on the basis of the test tone levels is identical with the older method of interconnecting VF channels of two subsystems on the basis of the TLP at the interface. Therefore, all signals including the standard test signal, the standard test tone, the monitoring test tone and the signal for IMD measurements may be transmitted across the interface as shown in Figure 43.

5.8.5.3 Method for interconnecting channels of tactical subsystems type I, II or III with type IV. Under consideration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



*Nominal values of speech volume, depending on talker volume distribution and loop insertion loss.

FIGURE 43. Signal level and interface diagram for connecting VF channels of tactical subsystems type I with type III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.9 Hypothetical reference circuits.

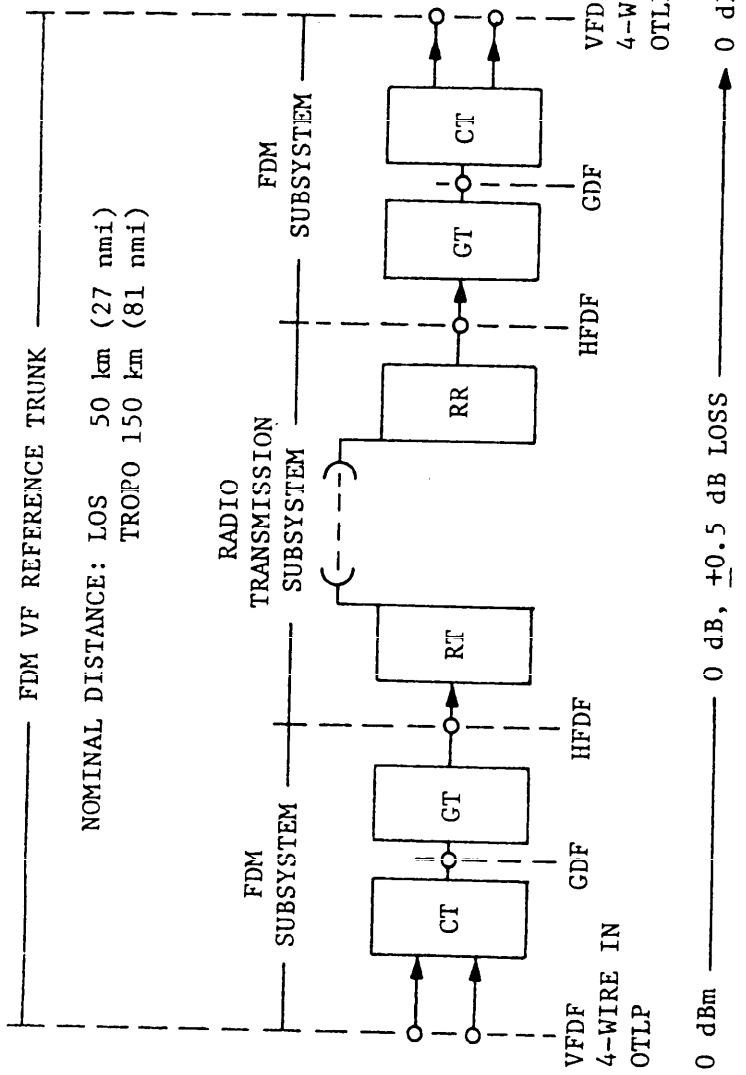
5.9.1 General. The following subparagraphs deal with configurations of reference trunks (see 5.9.2), assumed transfer characteristics of tandem connections (see 5.9.3), and examples of hypothetical reference circuits (see 5.9.4). The parameter values established in the following subparagraphs are not considered to be mandatory system standards as stated in 1.4 and defined in FED-STD-1037. Rather, these parameter values are intended to show typical performance characteristics that can be expected from circuits engineered in accordance with the requirements contained in this document with respect to FDM and TDM/PCM multichannel transmission. More detailed information about hypothetical reference circuits is contained in Appendix L.

5.9.2 Reference trunks. Reference trunks may be considered as the building blocks of hypothetical reference circuits. Reference trunks connect two nodes of a hypothetical reference circuit and may consist of one modulation link (see 5.7.3) or of two or more modulation links in tandem. The modulation links may be used to transmit voice or telegraph and data signals and may consist of a multiplexing subsystem employing FDM or TDM/PCM, and a radio transmission subsystem or a cable transmission subsystem with or without repeaters. The radio transmission subsystem may be comprised of LOS, tropospheric scatter or satellite links.

Reference trunks are identified by a specific configuration, such as the type of multiplexing and transmission subsystem employed; the nominal distance between transmitting and receiving terminals; and a set of parameter values, such as test signal levels and maximum noise power levels. These parameter values are derived from the corresponding parameter values for modulation links specified in 5.7.3 and for trunks specified in 5.7.4. Reference trunks include only four-wire, or equivalent four-wire, portions of a connection and do not include two-wire terminations. (See 4.3.3.)

5.9.2.1 Reference trunks of tactical subsystem type I. The reference trunks of the tactical subsystem type I (see Table I) consist of an FDM subsystem and a radio transmission subsystem providing nominal 4 kHz VF channels for the transmission of analog voice or quasi-analog telegraph and data signals. The radio transmission subsystem may consist of one LOS radio link covering a nominal transmission distance of 50 km (27 nmi) or one tropospheric scatter radio link covering a nominal transmission distance of 150 km (81 nmi). Figure 44 shows a block diagram of FDM VF reference trunks of the tactical subsystem type I. Figure 44 shows only one direction of transmission; the other direction is identical. Table LVI lists the parameter values which describe the characteristics of these reference trunks. The reference trunk parameters are identical to the corresponding parameters for the FDM trunks of the tactical subsystem type I given in 5.7.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



LEGEND:

CT	= Channel Translation Equipment	RR	= Radio Receiver
GDF	= Group Distribution Frame	RT	= Radio Transmitter
GT	= Group Translation Equipment	TLP	= Transmission Level Point
HFDF	= HF Distribution Frame	VFDF	= VF Distribution Frame
⟶	= Direction of Transmission		

FIGURE 44. FDM VF reference trunks of tactical subsystem type I.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

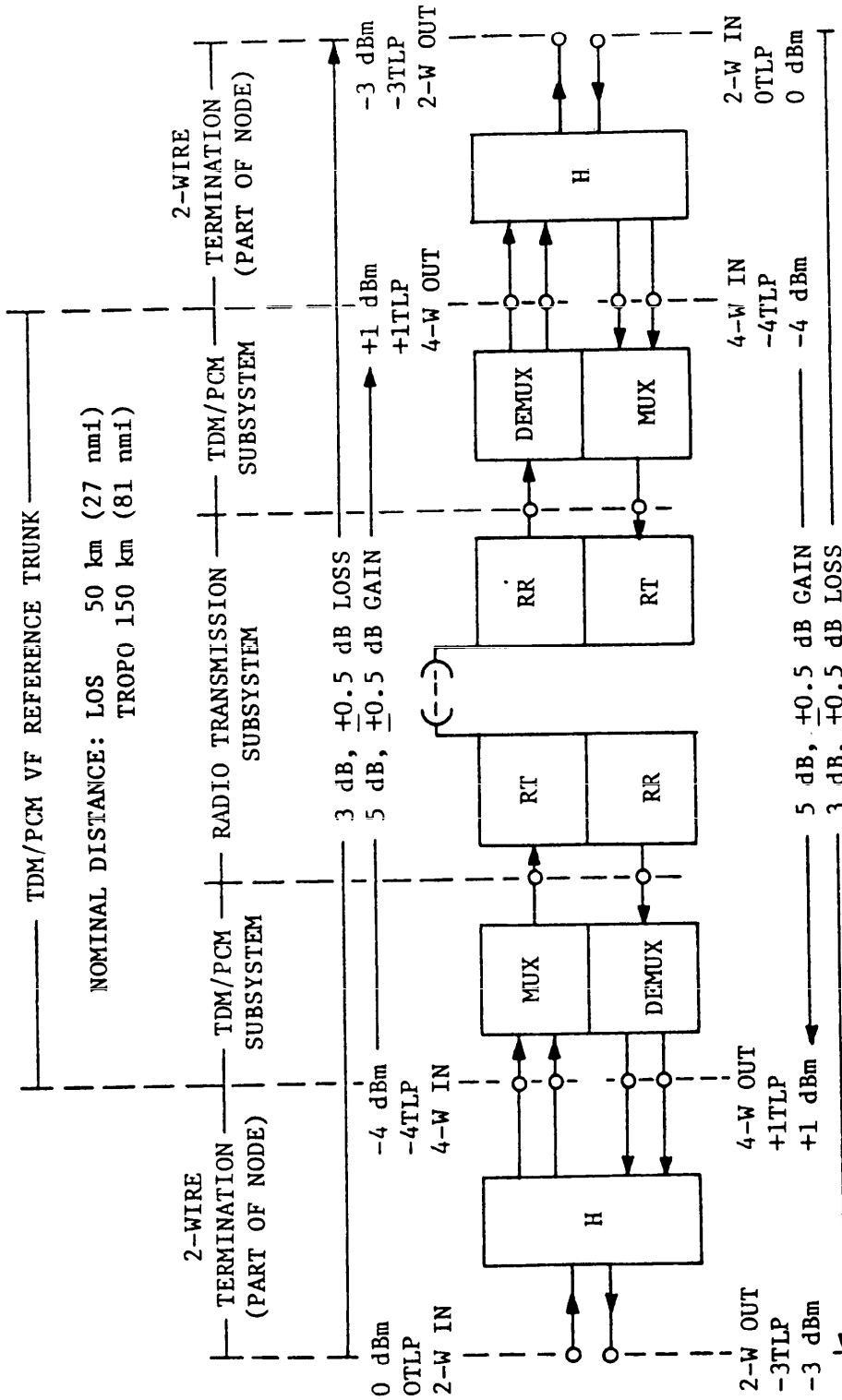
TABLE LVI. Parameters for FDM VF reference trunks of tactical subsystem type I.

Reference Trunk Parameters	Parameter Values (Applicable to Figure 44)	
	LOS	Tropospheric Scatter
Nominal Transmission Distance	50 km (27 nmi)	150 km (81 nmi)
Maximum Noise Power (See 5.7.4.2.3) for Voice Service	5,000 pWp (37.0 dBrnc)	16,000 pWp (42.0 dBrnc)
Maximum Noise Power (See 5.7.4.2.3) for Telegraph and Data Service	8,000 pW (39.0 dBrn)	25,000 pW (44.0 dBrn)
Insertion Loss Versus Frequency Characteristic (See 5.7.4.3.1.1)	f < 300 Hz: ≥ -3 dB 300 Hz ≤ f < 400 Hz: -3 dB to +1 dB 400 Hz ≤ f < 3000 Hz: -2 dB to +1 dB 3000 Hz ≤ f < 3400 Hz: -3 dB to +1 dB 3400 Hz ≤ f ≥ -3 dB	
Maximum Envelope Delay Distortion (See 5.7.4.3.2.1)	500 Hz ≤ f < 600 Hz: 1200 µs 600 Hz ≤ f < 1000 Hz: 800 µs 1000 Hz ≤ f < 2600 Hz: 300 µs 2600 Hz ≤ f < 2800 Hz: 500 µs 2800 Hz ≤ f ≤ 3200 Hz: 800 µs	
Insertion Gain (See 5.7.4.3.6)	0 dB, ±0.5 dB	
Net Loss Variations (See 5.7.4.3.8) Over any 30 consecutive days Over any 15 consecutive min.	±1.5 dB ±0.5 dB	
Test Signal Level and TLP (See 5.7.4.6)	4-Wire VF Input : 0 dBm at OTLP 4-Wire VF Output: 0 dBm at OTLP	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.9.2.2 Reference trunks of tactical subsystem type II. The reference trunks of the tactical subsystem type II (see Table I) consist of a TDM/PCM subsystem and a radio transmission subsystem providing nominal 4 kHz VF channels for the transmission of analog voice or quasi-analog telegraph and data signals. The radio transmission subsystem may consist of one LOS radio link covering a nominal transmission distance of 50 km (27 nmi) or one tropospheric scatter radio link covering a nominal transmission distance of 150 km (81 nmi). The LOS or tropospheric scatter radio link may be replaced with a satellite link or a cable link (with or without repeaters) covering different nominal transmission distances without significantly changing the performance of the reference trunk provided the digital TDM/PCM signal remains above threshold (see 4.6.2.1). Figures 45 through 47 depict block diagrams of TDM/PCM VF reference trunks of the tactical subsystem type II showing the different test signal levels and TLP for two-wire and four-wire terminations in accordance with 5.7.4.6. The two-wire/four-wire hybrid transformers shown in Figures 45 and 46 have an assumed insertion loss of nominal 4 dB and are used as an example to illustrate a two-wire termination for the reference trunks. It should be noted that a two-wire termination at the output of a VF trunk of the tactical subsystem type II requires a 5 dB gain of the trunk in accordance with 5.7.4.6. Table LVII lists the parameter values which describe the performance of the TDM/PCM VF reference trunks. There is no difference in the electrical parameter values listed in Table LVII between LOS and tropospheric scatter radio links since the transfer characteristics of a TDM/PCM VF channel is primarily determined by the A-D/D-A conversion process in the multiplexer/demultiplexer equipment and not by the radio transmission subsystem.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

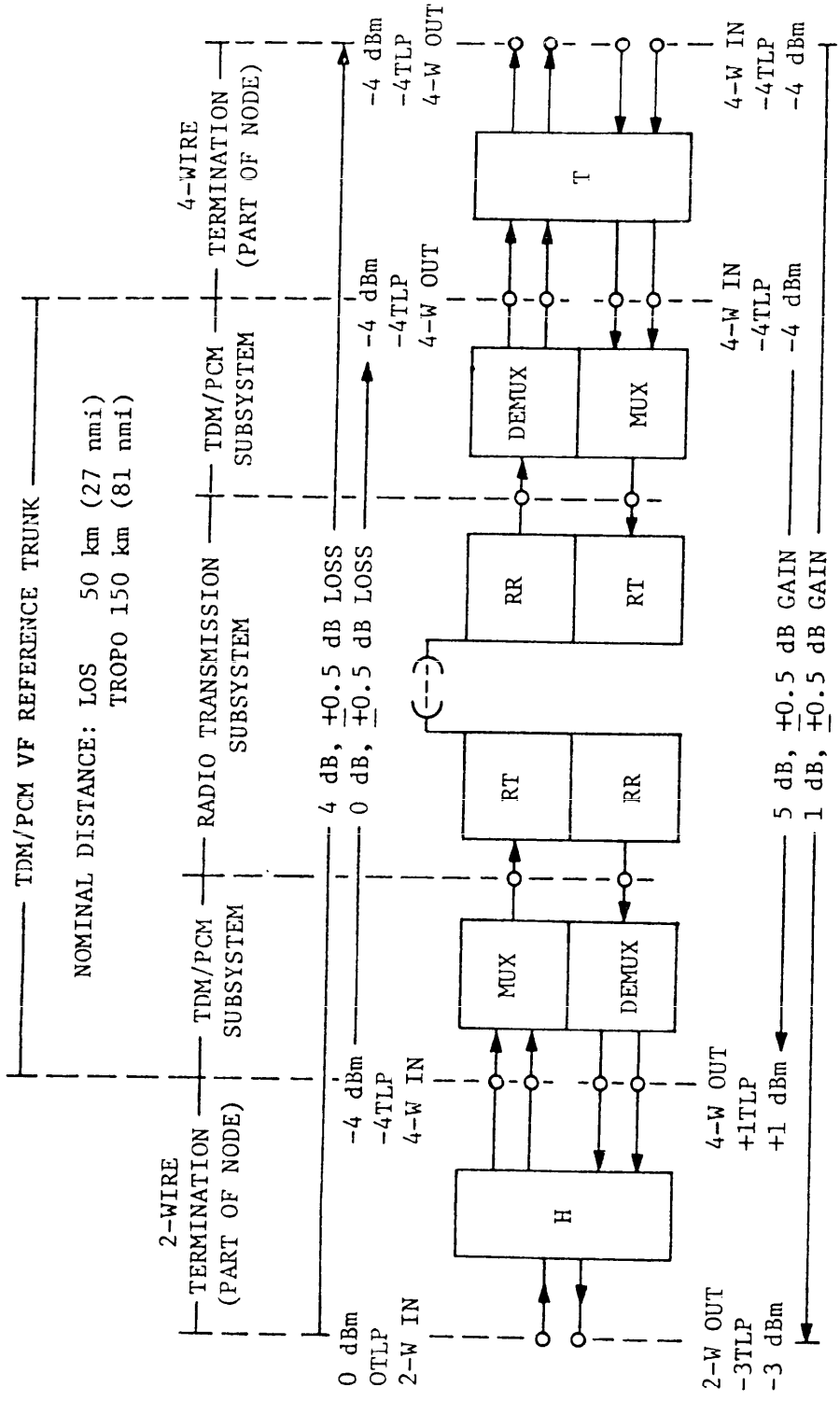


NOTE: For TLP and test signal levels see 5.7.4.6.

LEGEND: DEMUX = Demultiplexer
H = 2-Wire/4-Wire Hybrid Transformer
MUX = Multiplexer
RR = Radio Receiver
RT = Radio Transmitter
TLP = Transmission Level Point
→ = Direction of Transmission

FIGURE 45. TDM/PCM VF reference trunks with two-wire terminations of tactical subsystem type II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

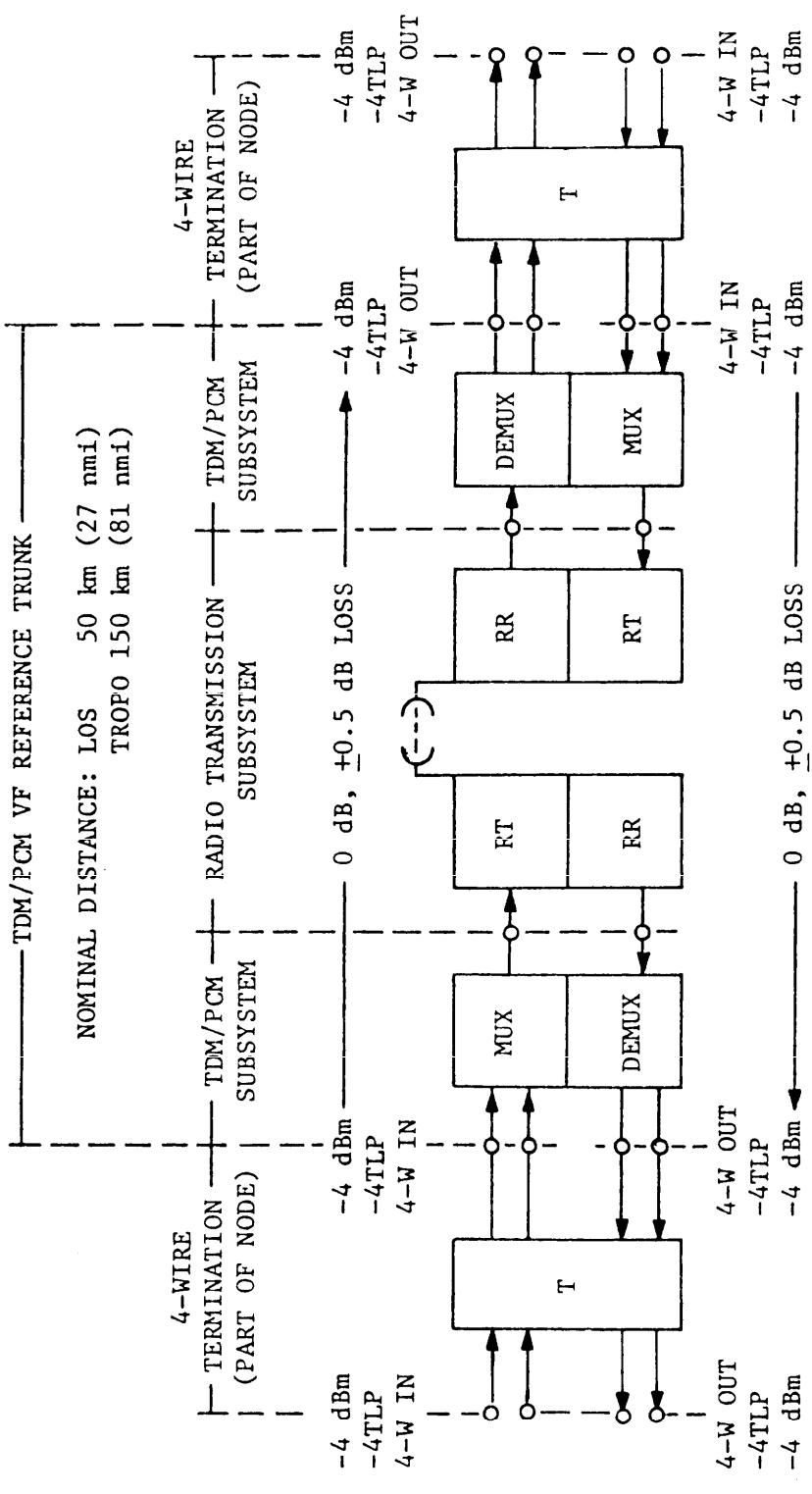


NOTE: For TLP and test signal levels see 5.7.4.6.

LEGEND: DEMUX = Demultiplexer
H = 2-Wire/4-Wire Hybrid Transformer
MUX = Multiplexer
RR = Radio Receiver
RT = Radio Transmitter
T = 4-Wire Termination, e.g. Switch
TLP = Transmission Level Point
→ = Direction of Transmission

FIGURE 46. TDM/PCM VF reference trunks with two-wire/four-wire terminations of tactical subsystem type II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



NOTE: For TLP and test signal levels see 5.7.4.6.

LEGEND: DEMUX = Demultiplexer
MUX = Multiplexer
RR = Radio Receiver
RT = Radio Transmitter
T = 4-Wire Termination, e.g. Switch
TLP = Transmission Level Point
→ = Direction of Transmission

FIGURE 47. TDM/PCM VF reference trunks with four-wire terminations of tactical subsystem type II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LVII. Parameters for TDM/PCM VF reference trunks of tactical subsystem type II.

Reference Trunk Parameters	Parameter Values (Applicable to Figures 45 through 47)
Nominal Transmission Distance	LOS: 50 km (27 nmi) Tropospheric Scatter: 150 km (81 nmi)
Maximum Idle Channel Noise Power (See 5.7.4.2.4)	
for Voice Service	2,000 pWp (33.0 dBrnc)
for Telegraph and Data Service	3,150 pW (35.0 dBrn)
Equivalent PCM Noise Power (See 5.7.4.2.4)	5,000 pWp (37.0 dBrnc)
Insertion Loss Versus Frequency Characteristic (See 5.7.4.3.1.2)	f < 300 Hz: > -1 dB 300 Hz ≤ f < 400 Hz: -1 dB to +2 dB 400 Hz ≤ f < 3000 Hz: -1 dB to +1 dB 3000 Hz ≤ f < 3400 Hz: -1 dB to +2 dB 3400 Hz ≤ f: ≥ -1 dB
Maximum Envelope Delay Distortion (See 5.7.4.3.2.2)	500 Hz ≤ f < 600 Hz: 100 µs 600 Hz ≤ f < 2600 Hz: 80 µs 2600 Hz ≤ f < 2800 Hz: 120 µs 2800 Hz ≤ f ≤ 3200 Hz: 200 µs
Insertion Gain (See 5.7.4.3.6)	2-Wire Output Termination: 5 dB, ±0.5 dB 4-Wire Output Termination: 0 dB, ±0.5 dB
Net Loss Variations (See 5.7.4.3.8) Over any 30 consecutive days 15 consecutive min.	 ±1.5 dB ±0.5 dB
Test Signal Level and TLP (See 5.7.3.2.2.2 and 5.7.4.6)	4-Wire VF Input: -4 dBm at -4TLP 4-Wire VF Output with 2-Wire Termination: +1 dBm at +1TLP with 4-Wire Termination: -4 dBm at -4TLP

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.9.2.3 Reference trunks of tactical subsystem type III. The reference trunks of the tactical subsystem type III (see Table I) consist of an FDM subsystem and a radio transmission subsystem providing nominal 4 kHz VF channels for the transmission of analog voice or quasi-analog telegraph and data signals. The nominal transmission distance of a reference trunk of the tactical subsystem type III has been taken as 300 km (162 nmi) which may be achieved by six LOS radio links in tandem, each LOS radio link having a nominal transmission distance of 50 km (27 nmi), or by one tropospheric scatter radio link covering a nominal transmission distance of 200 km (108 nmi) in tandem with two LOS radio links of 50 km (27 nmi) each. Figure 48 shows the FDM VF reference trunk consisting of six LOS radio links in tandem. Figure 49 shows the FDM VF reference trunk consisting of one tropospheric scatter radio link connected at each end with one LOS radio link. Figures 48 and 49 show only one direction of transmission; the other direction is identical. Table LVIII lists the parameter values which describe the characteristics of these reference trunks. The parameter values listed in Table LVIII have been calculated by selecting the applicable values contained in Tables XXXII, XXXIV, XXXVI, XXXIX and XLI, and applying the applicable summation methods given in 5.9.3 to the configuration of the reference trunk under consideration. The reference trunk shown in Figure 48 has an FDM subsystem that consists of one pair of channel translation equipment, one pair of group translation equipment and one pair of supergroup translation equipment, and a radio transmission subsystem consisting of six pairs of radio equipment. The reference trunk shown in Figure 49 has the same FDM subsystem configuration as the one shown in Figure 48 but the radio transmission subsystem consists of three pairs of radio equipment.

5.9.2.4 Reference trunks of tactical subsystem type IV. Under consideration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

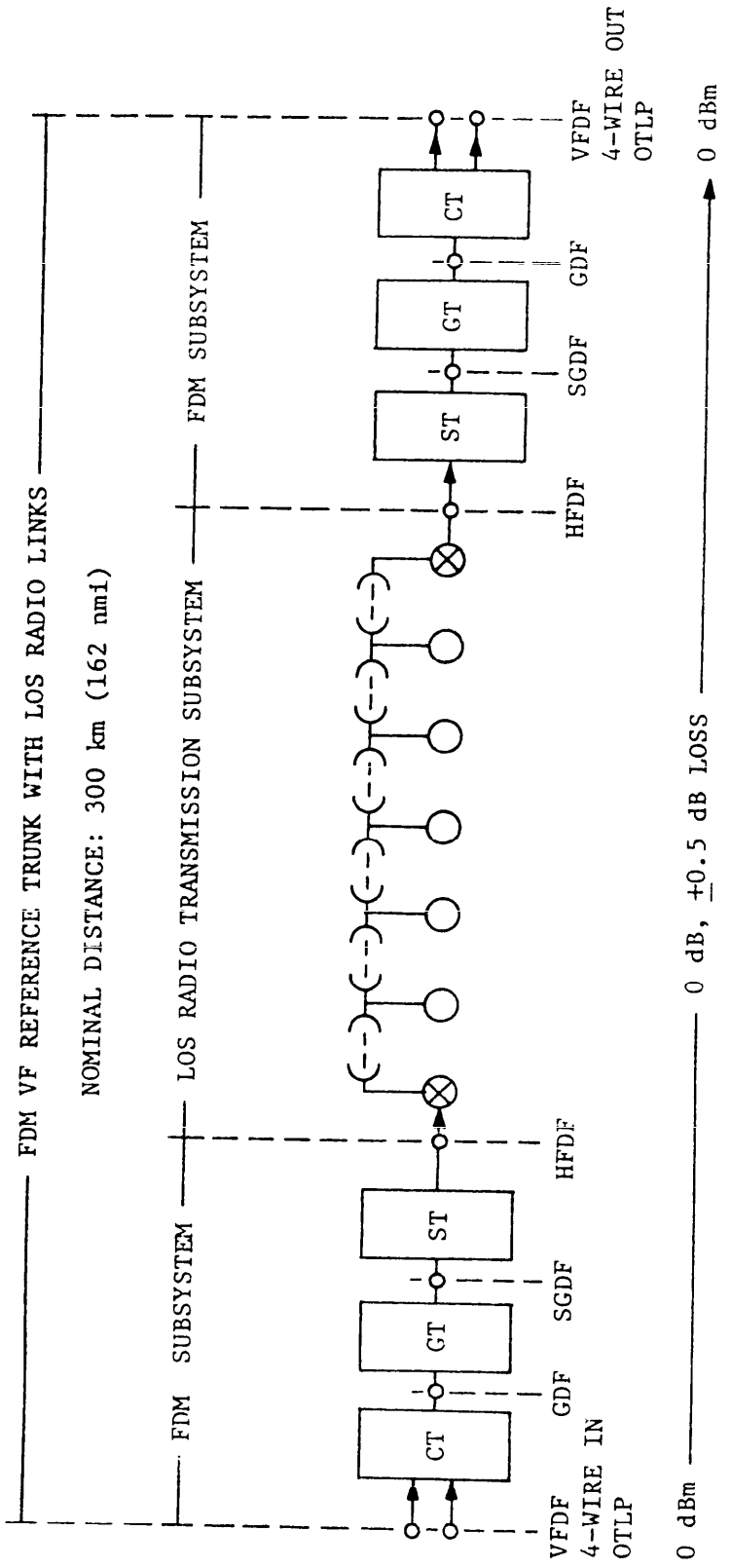


FIGURE 48. FDM VF reference trunk with LOS radio links of tactical subsystem type III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

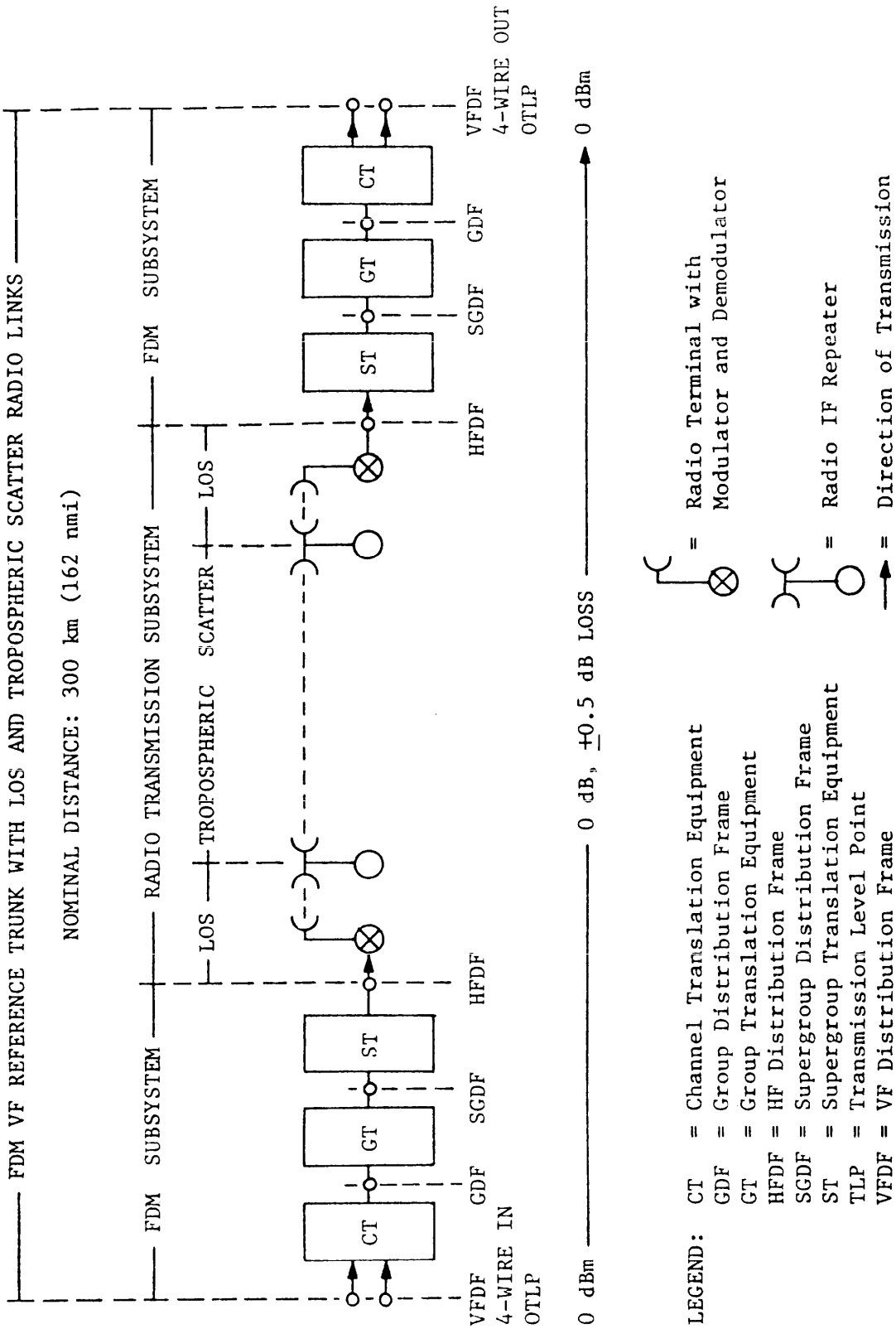


FIGURE 49. FDM VF reference trunk with LOS and tropospheric scatter radio links of tactical subsystem type III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LVIII. Parameters for FDM VF reference trunks of tactical subsystem type III.

Reference Trunk Parameters		Parameter Values for Figure 48	Parameter Values for Figure 49
Nominal Transmission Distance		300 km (162 nmi)	300 km (162 nmi)
Maximum Noise Power for Voice Service	Idle Channel Noise	75 pWp0	
	Loaded Channel Noise (See 5.7.3.1.3.2)	131 pWp0	
	Transmission Medium Noise (see 5.7.3.1.3.4)	1080 pWp0	
	Total Maximum Trunk Noise	1286 pWp0 (31.1 dBrnc0)	
for Telegraph and Data Service	Idle Channel Noise	120 pW0	
	Loaded Channel Noise (See 5.7.3.1.3.2)	210 pW0	
	Transmission Medium Noise (See 5.7.3.1.3.4)	1710 pW0	
	Total Maximum Trunk Noise	2040 pW0 (33.1 dBrn0)	
Insertion Loss Versus Frequency Characteristic (in dB)* (See 5.7.3.1.5.2)	f < 300 Hz	> -1.4	> -1.1
	300 Hz ≤ f < 400 Hz	-1.4 to +2.0	-1.1 to +1.8
	400 Hz ≤ f < 3000 Hz	-1.4 to +1.4	-1.1 to +1.1
	3000 Hz ≤ f < 3400 Hz	-1.4 to +2.0	-1.1 to +1.8
	3400 Hz ≤ f	≥ -1.4	≥ -1.1
Maximum Envelope Delay Distortion (in microseconds) (See 5.7.3.1.6.2)	500 Hz ≤ f < 600 Hz	157	157
	600 Hz ≤ f < 1000 Hz	127	127
	1000 Hz ≤ f < 2600 Hz	87	87
	2600 Hz ≤ f ≤ 3200 Hz	127	127
Insertion Gain (in dB) (See 5.7.4.3.6)		0, ±0.5	0, ±0.5
Net Loss Variation (in dB)* (See 5.7.3.1.9.2) Over any 30 consecutive days		±1.5	±1.2
Test Signal Level and TLP (See 5.7.4.6)		4-Wire VF Input (VFDF): 0 dBm at OTLP 4-Wire VF Output (VFDF): 0 dBm at OTLP	

*dB values rounded off within 0.1 dB.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.9.3 Transfer characteristics of tandem connections.

5.9.3.1 General. Methods for the summation of transfer characteristics of individual loops, links, or trunks to obtain the corresponding tandem characteristic are given in 5.9.3.2 through 5.9.3.5. These summation methods are applicable to analog service (voice and facsimile) on a user-to-user basis by considering loop and transmission channel characteristics in tandem. These methods are also applicable to telegraph and data service from the transmitting quasi-analog modem output to the receiving quasi-analog modem input, provided there is no regeneration employed between the modems. If regeneration is employed, then each regeneration point acts as data sink/data source combination and the summation methods will only be applicable from regeneration point to regeneration point and not on a circuit basis from the transmitting DTE to the receiving DTE.

NOTE 1: In general, the methods of 5.9.3.2 through 5.9.3.5 should only be used if the parameters contributed by the different portions of a tandem connection are specified over identical frequency ranges. However, if the parameters are specified over different frequency ranges, then the smaller frequency range will govern the overall frequency response and the summation rule should only be applied to the smaller frequency range. For example, when the insertion loss versus frequency characteristic of loops and trunks are combined to obtain the reference circuit characteristics of 5.9.4.2 through 5.9.4.7, the summation rules have been applied with respect to the frequency bands specified for the loops.

NOTE 2: The results obtained by the summation methods of 5.9.3.2 through 5.9.3.5 should not be interpreted as deterministic results intended to replace measured performance of installed and operational circuits but, rather, should be interpreted as a first approximation for transfer functions of hypothetical reference circuits and as guidance to system and circuit engineering. Results obtained by these summation methods have to be confirmed by actual measurements of installed and operational circuits or portions of circuits in order to determine whether the applicable standards have been met.

5.9.3.2 Noise power. (MIL-STD-188-100.) The total noise power in picowatts of a tandem connection (circuit, trunk, etc.) is assumed to be the sum of the noise powers in picowatts contributed by each portion of the tandem connection (loop, link, trunk, etc.), provided the noise powers are referenced to the same TLP (see Note 3); represent the same type of noise, e.g., all-year median noise power of LOS radio links; and the same noise weighting is used.

NOTE 1: Noise powers expressed in units of dB cannot be added by the method of 5.9.3.2, rather, one of the methods given in 120.2.3 of Appendix L must be used.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE 2: Worst-hour median noise of tropospheric scatter radio links in tandem cannot be added by the method of 5.9.3.2 since the worst-hour noise of these links usually does not occur at the same time. Adding worst-hour noise power values results in extremely pessimistic and unrealistic noise power predictions of tropospheric scatter radio links in tandem.

NOTE 3: If the noise power values are not referenced to the same TLP, then they have to be converted to a common TLP before they can be summed. It is accepted practice in circuit engineering to reference each noise power value, contributed by the different portions of a circuit, to the OTLP of the circuit under consideration and thus arrive at the total circuit noise power in units of pWp0, pW0, dBrnc0 or dBrn0. Noise power values may be referenced to a circuit OTLP even if a OTLP does not exist or is physically not accessible. (See Figure 54.) The total noise power, referenced to the circuit OTLP, can be transferred to any other TLP of the same circuit by considering the transmission level difference between the circuit OTLP and the TLP of interest. (See 120.2.4 of Appendix L.)

NOTE 4: When FDM and TDM/PCM links are connected in tandem for analog voice service, the equivalent PCM noise power (see 5.7.3.2.3.3) is used to determine the contributions by TDM/PCM links to the total circuit noise power. (See Figure 60 and Table LXIII.)

5.9.3.3 Insertion loss versus frequency characteristic. (MIL-STD-188-100.) The lower (or upper) limit of the insertion loss versus frequency characteristic (in dB) over the frequency band of interest (see Note 1 of 5.9.3.1) of a tandem connection (circuit, trunk, etc.) is assumed to be obtained by summing the squares of the lower (or upper) limit of the insertion loss versus frequency characteristic (in dB) contributed by each portion of the tandem connection (loop, link, trunk, etc.) and taking the square root of this sum (root-sum-square law).

NOTE: The summation method of 5.9.3.3 has been based on a statistical distribution of the insertion loss versus frequency characteristic of communications equipment and wire or cable connections, considering that in this document, upper and lower limits (worst case criteria) are specified for this characteristic. In the case that the insertion loss (in dB) of each portion of a tandem connection is measured at selected frequencies, then the overall insertion loss of the tandem connection is obtained by adding the dB values of each portion of the tandem connection at the selected frequencies. However, using the upper and lower limits (in dB) of the insertion loss versus frequency characteristic, specified in this document for different portions of a tandem connection, and adding these dB values would result in too pessimistic and unrealistic predictions of the insertion loss versus frequency characteristic of a hypothetical connection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.9.3.4 Envelope delay distortion. (MIL-STD-188-100.) The envelope delay distortion over the frequency band of interest (see Note 1 of 5.9.3.1) of a tandem connection (circuit, trunk, etc.) is assumed to be the sum of the envelope delay distortion contributed by each portion of the tandem connection (loop, link, trunk, etc.). (See 70.5 of Appendix G.)

5.9.3.5 Net loss variation. (MIL-STD-188-100.) The net loss variation (in dB) of a tandem connection (circuit, trunk, etc.) is assumed to be obtained by summing the squares of the net loss variations (in dB) contributed by each portion of the tandem connection (loop, link, trunk, etc.) and taking the square root of this sum (root-sum-square law).

5.9.4 Examples of hypothetical reference circuits.

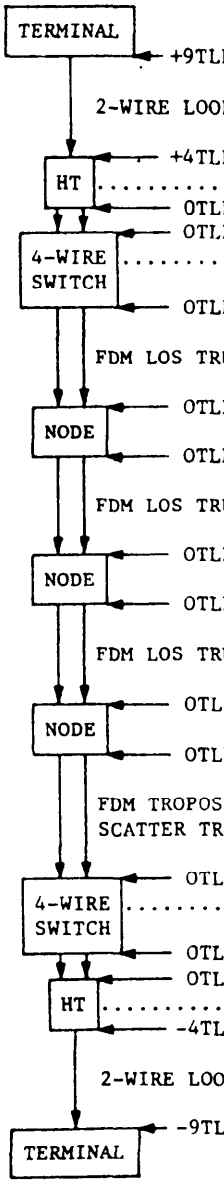
5.9.4.1 General. Hypothetical reference circuits consist of combinations of different types of reference trunks (see 5.9.2) connected in tandem with wire or cable loops (see 5.5.2) which have characteristics either for analog voice service or for telegraph and data service. The reference circuit examples of 5.9.4.2 through 5.9.4.7 illustrate the application of standards defined in this document for analog FDM and digital TDM/PCM multichannel transmission. Only a limited number of hypothetical reference circuits have been selected since it is impractical to cover all possible combinations of reference circuit configurations for the tactical subsystems type I, II and III. (See Table I.) The configurations of the reference circuits have been restricted to circuits consisting of up to six reference trunks connected with two-wire or four-wire loops. Based on experience, a circuit consisting of six tandem trunks connecting seven nodes should satisfy the maximum transmission distance of a switched or nonswitched (point-to-point) VF circuit. Therefore, the reference circuit examples shown in Figures 50 through 60 are considered to be representative of the probable maximum length of typical installed circuits in the tactical subsystem under consideration. More detailed information about tactical reference circuits is contained in Appendix L.

5.9.4.2 Hypothetical reference circuits of tactical subsystem type I. The nominal transmission distance of a hypothetical reference circuit of the tactical subsystem type I (see Table I) has been taken as 300 km (162 nmi), not including the length of the loops. This distance is considered to be the probable maximum transmission distance over which switched and nonswitched (point-to-point) communications are required in the tactical subsystem type I. Three different examples of VF reference circuits of the tactical subsystem type I are shown in Figures 50, 51 and 52. These figures show the configurations of the reference circuits, circuit TLP, signal levels, noise power levels contributed by the different parts of the circuit to the total circuit noise, and total circuit noise power referenced to selected TLP. The noise power contributed by each loop to the total circuit noise has been taken as the maximum allowable noise

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Page 237 is missing from the original legacy source document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



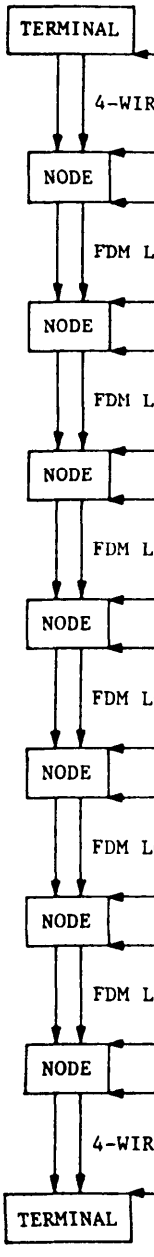
	MEAN SPEECH VOLUME	TEST SIGNAL LEVEL	NOISE POWER CONTRIBUTED BY PARTS OF CIRCUIT (pWp)	(dBrnc)	REFERENCED TO	REFERENCED TO OTLP (pWpO)	(dBrncO)
+9TLP	-10 VU	+9 dBm (0 dBmO)					
2-WIRE LOOP			6,300	38.0	+4TLP	2,510	34.0
+4TLP	-15 VU	+4 dBm					
HT			0				
OTLP	-19 VU	0 dBm					
OTLP	-19 VU	0 dBm					
4-WIRE SWITCH			400	26.0	OTLP	400	26.0
OTLP	-19 VU	0 dBm					
FDM LOS TRUNK #1 (FIG. 44)			5,000	37.0	OTLP	5,000	37.0
OTLP	-19 VU	0 dBm					
NODE			400	26.0	OTLP	400	26.0
OTLP	-19 VU	0 dBm					
FDM LOS TRUNK #2 (FIG. 44)			5,000	37.0	OTLP	5,000	37.0
OTLP	-19 VU	0 dBm					
NODE			400	26.0	OTLP	400	26.0
OTLP	-19 VU	0 dBm					
FDM LOS TRUNK #3 (FIG. 44)			5,000	37.0	OTLP	5,000	37.0
OTLP	-19 VU	0 dBm					
NODE			400	26.0	OTLP	400	26.0
OTLP	-19 VU	0 dBm					
FDM TROPOSPHERIC SCATTER TRUNK #4 (FIG. 44)			16,000	42.0	OTLP	16,000	42.0
OTLP	-19 VU	0 dBm					
4-WIRE SWITCH			400	26.0	OTLP	400	26.0
OTLP	-19 VU	0 dBm					
OTLP	-19 VU	0 dBm					
HT			0				
-4TLP	-23 VU	-4 dBm					
2-WIRE LOOP			6,300	38.0	-9TLP	50,040	47.0
-9TLP	-28 VU	-9 dBm					
TERMINAL							

HT = TWO-WIRE/FOUR-WIRE HYBRID TRANSFORMER WITH NOMINAL 4 dB OF INSERTION LOSS.

TOTAL CIRCUIT NOISE POWER		
	AT OTLP:	85,550 pWpO (49.3 dBrncO)
	AT -4TLP:	34,060 pWp (45.3 dBrnc)
	AT -9TLP:	10,770 pWp (40.3 dBrnc)

FIGURE 50. Hypothetical 300 km (162 nmi) switched VF reference circuit for voice service of tactical subsystem type I.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



	MEAN SPEECH VOLUME	DATA SIGNAL LEVEL	TEST SIGNAL LEVEL	NOISE POWER REFERENCED TO OTLP: CONTRIBUTED BY PARTS OF CIRCUIT VOICE (pWpO)	CONTRIBUTED BY PARTS OF CIRCUIT DATA (pWO)	ACCUMULATED AT POINTS INDICATED VOICE (pWpO)	ACCUMULATED AT POINTS INDICATED DATA (pWO)
TERMINAL							
+9TLP	-10 VU	-4 dBm (-13 dBmO)	+9 dBm (0 dBmO)				
4-WIRE LOOP				6,300	6,300		
OTLP	-19 VU	-13 dBm (-13 dBmO)	0 dBm (0 dBmO)			6,300	6,300
NODE				400	630		
OTLP						6,700	6,930
FDM LOS TRUNK #1 (FIG. 44)				5,000	8,000		
OTLP	-19 VU	-13 dBm	0 dBm			11,700	14,930
NODE				400	630		
OTLP						12,100	15,560
FDM LOS TRUNK #2 (FIG. 44)				5,000	8,000		
OTLP	-19 VU	-13 dBm	0 dBm			17,100	23,560
NODE				400	630		
OTLP						17,500	24,190
FDM LOS TRUNK #3 (FIG. 44)				5,000	8,000		
OTLP	-19 VU	-13 dBm	0 dBm			22,500	32,190
NODE				400	630		
OTLP						22,900	32,820
FDM LOS TRUNK #4 (FIG. 44)				5,000	8,000		
OTLP	-19 VU	-13 dBm	0 dBm			27,900	40,820
NODE				400	630		
OTLP						28,300	41,450
FDM LOS TRUNK #5 (FIG. 44)				5,000	8,000		
OTLP	-19 VU	-13 dBm	0 dBm			33,300	49,450
NODE				400	630		
OTLP						33,700	50,080
FDM LOS TRUNK #6 (FIG. 44)				5,000	8,000		
OTLP	-19 VU	-13 dBm	0 dBm			38,700	58,080
NODE				400	630		
OTLP	-19 VU	-13 dBm	0 dBm			39,100	58,710
4-WIRE LOOP				6,300*	6,300*		
-9TLP	-28 VU	-22 dBm	-9 dBm			11,220*	13,690*
TERMINAL							

TOTAL CIRCUIT NOISE POWER	AT OTLP:	89,140 pWpO (49.5 dBrncO)	108,750 pWO (50.4 dBrnO)
	AT -9TLP:	11,220 pWp (40.5 dBrnc)	13,690 pW (41.4 dBrn)

SNR = -22 dBm - (41.4 dBrn -90) = 26.6 dB

*THESE NOISE POWER VALUES ARE REFERENCED TO THE -9TLP AT THE RECEIVING TERMINAL AND NOT TO A OTLP.

FIGURE 51. Hypothetical 300 km (162 nmi) switched VF reference circuit for voice service and telegraph and data service of tactical subsystem type I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Page 240 is missing from the original legacy source document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

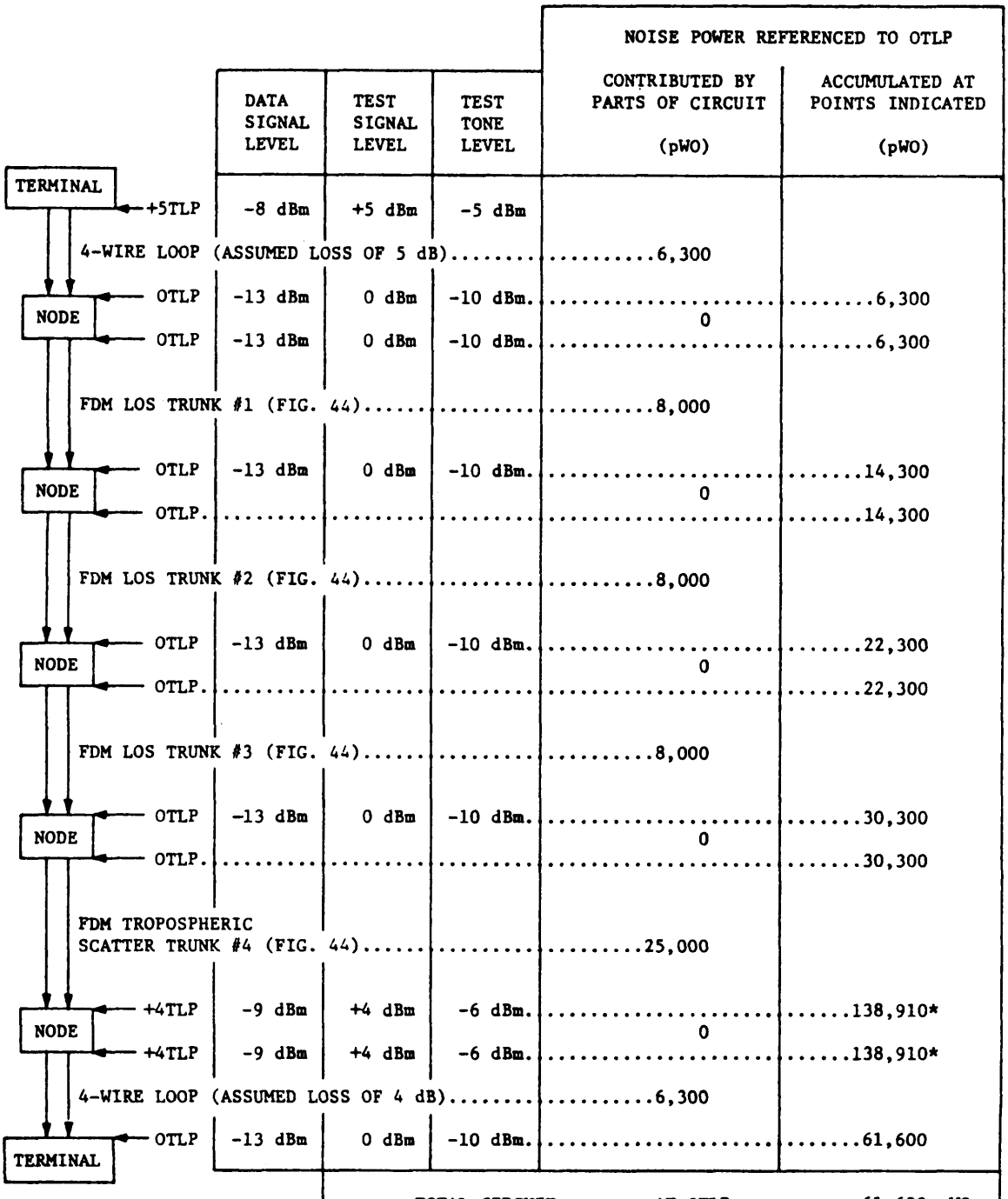


FIGURE 52. Hypothetical 300 km (162 nmi) nonswitched (point-to-point) VF reference circuit for telegraph and data service of tactical subsystem type I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LIX. Parameters for hypothetical 300 km (162 nmi) VF reference circuits of tactical subsystem type I.

Reference Circuit Parameters		Parameter Values for Figure 50	Figure 51	Figure 52
Nominal Transmission Distance		300 km (162 nmi) (without Loops)		
Total Circuit Noise Power for Voice Service	referenced to Circuit OTLP	85,550 pWp0 (49.3 dBrnc0)	89,140 pWp0 (49.5 dBrnc0)	not applicable
	referenced to TLP at Receiving Terminal	10,770 pWp (40.3 dBrnc)	11,220 pWp (40.5 dBrnc)	not applicable
for Telegraph and Data Service	referenced to Circuit CTLP	not applicable	108,750 pW0 (50.4 dBrn0)	61,600 pW0 (47.9 dBrn0)
	referenced to TLP at Receiving Terminal	not applicable	13,960 pW (41.4 dBrn)	61,600 pW0 (47.9 dBrn0)
Quasi-Analog Data SNR		not appl.	26.6 dB	29.1 dB
Insertion Loss Versus Frequency Characteristics (in dB, rounded off within 0.1 dB) with Loops for Voice Service	$f < 400$ Hz	≥ -7.3	not applicable	not applicable
	$400 \text{ Hz} \leq f < 600$ Hz	-7.3 to +10.3		
	$600 \text{ Hz} \leq f < 1000$ Hz	-6.2 to +10.3		
	$1000 \text{ Hz} \leq f < 2400$ Hz	-4.6 to +10.3		
	$2400 \text{ Hz} \leq f < 2800$ Hz	-4.6 to +18.6		
	$2800 \text{ Hz} \leq f$	≥ -4.6		
with Loops for Telegraph and Data Service up to 1200 Bd	$f < 300$ Hz	not applicable	≥ -8.3	≥ -6.6
	$300 \text{ Hz} \leq f < 400$ Hz		-8.3 to +4.6	-6.6 to +3.5
	$400 \text{ Hz} \leq f < 1000$ Hz		-6.2 to +4.6	-4.9 to +3.5
	$1000 \text{ Hz} \leq f < 2400$ Hz		-5.6 to +4.6	-4.0 to +3.5
	$2400 \text{ Hz} \leq f < 2700$ Hz		-5.6 to +5.6	-4.0 to +4.7
	$2700 \text{ Hz} \leq f < 3000$ Hz		-5.6 to +7.9	-4.0 to +7.4
	$3000 \text{ Hz} \leq f$		≥ -5.6	≥ -4.0
Maximum Envelope Delay Distortion (in microseconds) with Loops for Voice Service		Not standardized (See 5.2.2.1.5)		
with Loops for Telegraph and Data Service up to 1200 Bd	$800 \text{ Hz} \leq f < 1000$ Hz	not applicable	5175	3400
	$1000 \text{ Hz} \leq f < 2400$ Hz		2275	1500
	$2400 \text{ Hz} \leq f \leq 2600$ Hz		2575	1800
Mean of Insertion Loss (in dB)		18	18	5
Net Loss Variations (in dB, rounded off within 0.1 dB)	Over any 30 consecutive days	± 3.7	± 4.2	± 3.7
	Over any 15 consecutive min.	± 1.2	± 1.4	± 1.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Table LIX indicate that the reference circuits shown in Figures 50 and 51 meet all requirements for voice service stated in 5.2.2.1. However, the reference circuits shown in Figures 51 and 52 do not meet the requirements for telegraph and data service stated in 5.2.3.1.5.2 for the received noise power, in 5.2.3.1.6.1.1 for the insertion loss versus frequency characteristic and in 5.2.3.1.6.2.1 for the envelope delay distortion. Therefore, these circuits may require regeneration or equalization, or both, to provide acceptable telegraph and data service.

5.9.4.3 Hypothetical reference circuits of tactical subsystem type II. The nominal transmission distance of a hypothetical reference circuit of the tactical subsystem type II (see Table I) has been taken as 300 km (162 nmi), not including the length of the loops. This distance is considered to be the probable maximum transmission distance over which switched and nonswitched (point-to-point) communications are required in the tactical subsystem type II. Three different examples of VF reference circuits of the tactical subsystem type II are shown in Figures 53, 54 and 55. These figures show the configurations of the reference circuits, circuit TLP, signal levels, noise power levels contributed by the different parts of the circuit to the total circuit noise, and total circuit noise power referenced to selected TLP. The noise power contributed by each loop to the total circuit noise has been taken as the maximum allowable noise power of 6300 pWp for analog voice service and 6300 pW for telegraph and data service, measured at the output terminals of the loop in accordance with 5.5.2.6. The insertion loss of the loops shown in Figures 53 and 54 has been taken as the mean loss of 3.5 dB in accordance with 5.5.2.2. The reference trunks have a configuration as shown in Figures 45, 46 or 47 with characteristics as listed in Table LVII. Figures 53, 54 and 55 show only one direction of transmission; the other direction is identical.

Figure 53 shows an example of a hypothetical 300 km (162 nmi) switched VF reference circuit for voice service consisting of three TDM/PCM LOS reference trunks and one tropospheric scatter reference trunk, five switches and two-wire loops extending to the telephone sets. Collocated with the switches may also be other equipment, such as equipment for patching, signal processing and technical control. The noise power contributed by each switch to the total circuit noise has been taken as the maximum allowable noise power in accordance with 5.6.2.1.2.5. The two-wire loops are connected to two-wire switches which are connected via two-wire/four-wire hybrid transformers with the four-wire trunks. The hybrid transformers together with the two-wire switches are part of the originating and terminating nodes. The noise power levels contributed by the loops, switches and trunks to the total circuit noise are listed with reference to the associated circuit TLP at the output of each loop, switch and trunk. All noise power contributions are then added at each TLP, starting with the OTLP at the output of the transmitting loop and considering the level changes as indicated by the applicable TLP, to obtain the total circuit noise power referenced to the -6.5TLP at the receiving terminal which is transferred to the circuit OTLP. The received

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

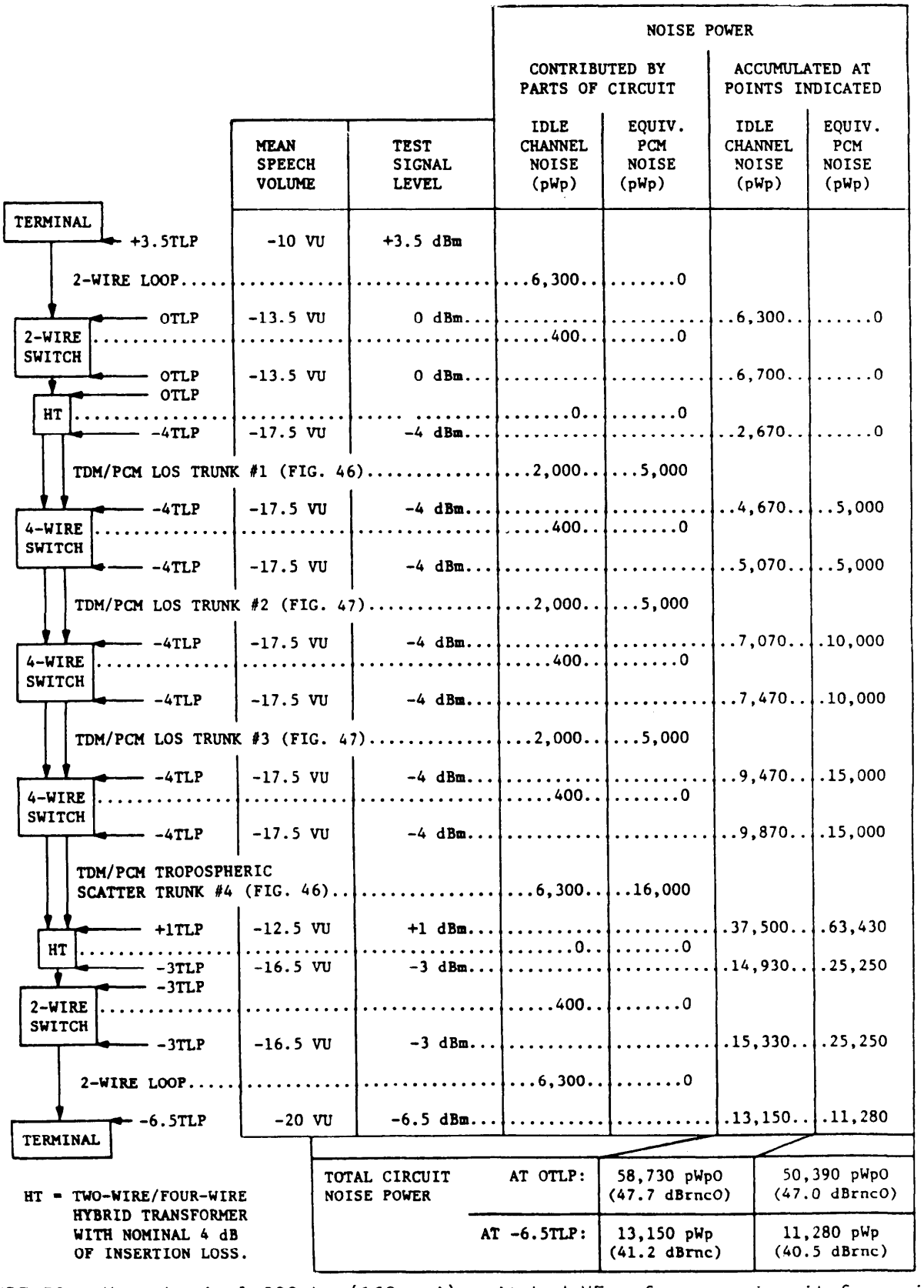


FIGURE 53. Hypothetical 300 km (162 nmi) switched VF reference circuit for voice service of tactical subsystem type II.

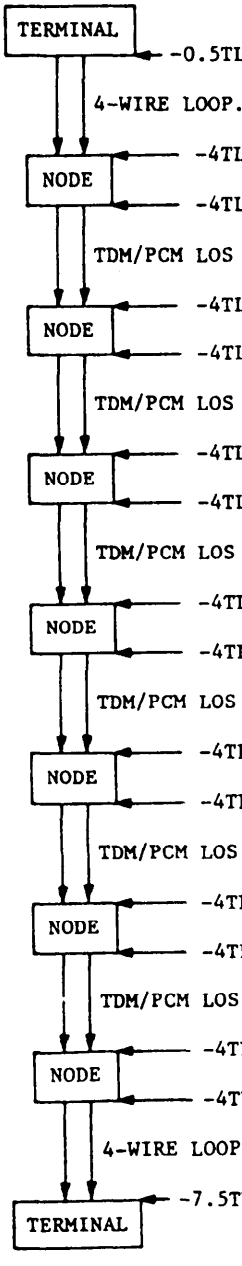
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

PCM idle channel noise power of 13,150 pWp (41.2 dBrnc) and the equivalent PCM noise power of 11,280 pWp (40.5 dBrnc) are within the limits of 5.2.2.1.3. The transmitted mean speech volume of -10 VU is in accordance with 5.2.2.1.1, whereas the received speech volume of -20 VU is higher (see Note 2 of 5.2.2.1.2) than the nominal value of -28 VU stated in 5.2.2.1.2.

Figure 54 shows an example of a hypothetical 300 km (162 nmi) switched VF reference circuit for voice service and telegraph and data service consisting of six TDM/PCM LOS reference trunks, seven nodes and four-wire loops extending to the terminal equipment or terminal subsystems. The nodes contain a four-wire analog switch and may also contain other equipment, such as equipment for patching, signal processing and technical control (see 4.3.4). The noise power contributed by each node to the total circuit noise has been taken as the maximum allowable noise power of an analog switch in accordance with 5.6.2.1.2.5. The noise power levels contributed by the loops, nodes and trunks to the total circuit noise are listed with reference to the associated circuit TLP at the output of each loop, node and trunk. The associated circuit TLP are -4TLP with the exception of the circuit TLP at the output of the receiving loop which has a -7.5TLP. All noise power contributions are then added, considering 3.5 dB of attenuation of the receiving loop, to obtain the total circuit noise power referenced to the -7.5TLP at the receiving terminal. The received PCM idle channel noise power of 15,730 pWp (42.0 dBrnc) for voice service and the equivalent PCM noise power of 13,400 pWp (41.3 dBrnc) are within the limits given in 5.2.2.1.3. The transmitted mean speech volume of -10 VU is in accordance with 5.2.2.1.1, whereas the received speech volume of -17 VU is higher (see Note 2 of 5.2.2.1.2) than the nominal value of -28 VU stated in 5.2.2.1.2. The received noise power of 19,530 pW (42.9 dBrn) for telegraph and data service consists only of the idle channel noise power and does not include quantizing noise power generated by the A-D/D-A conversion process of the TDM/PCM equipment. (See Note 2 of 5.2.3.1.5.2.) Also, the quasi-analog SNR of 33.6 dB is based only on idle channel noise and does not include any quantizing noise. The data signal level of -10 dBm at each -4TLP (-6 dBm0) is in accordance with 5.2.3.1.4.

Figure 55 shows an example of a hypothetical 300 km (162 nmi) non-switched (point-to-oint) VF reference circuit for telegraph and data service consisting of three TDM/PCM LOS reference trunks and one tropospheric scatter reference trunk, five nodes and four-wire loops extending to the terminal equipment or terminal subsystems. The nodes may contain equipment for such functions as patching, signal processing and technical control (see 4.3.4). The noise power contributed by each node to the total circuit noise has been taken as zero since the nodes do not contain switches. The noise power levels contributed by the loops and trunks to the total circuit noise are listed with reference to the associated circuit TLP at the output of each loop and trunk. The associated circuit TLP are -4TLP with the exception of the circuit TLP at the output of the TDM/PCM tropospheric scatter reference trunk which has a 0TLP in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



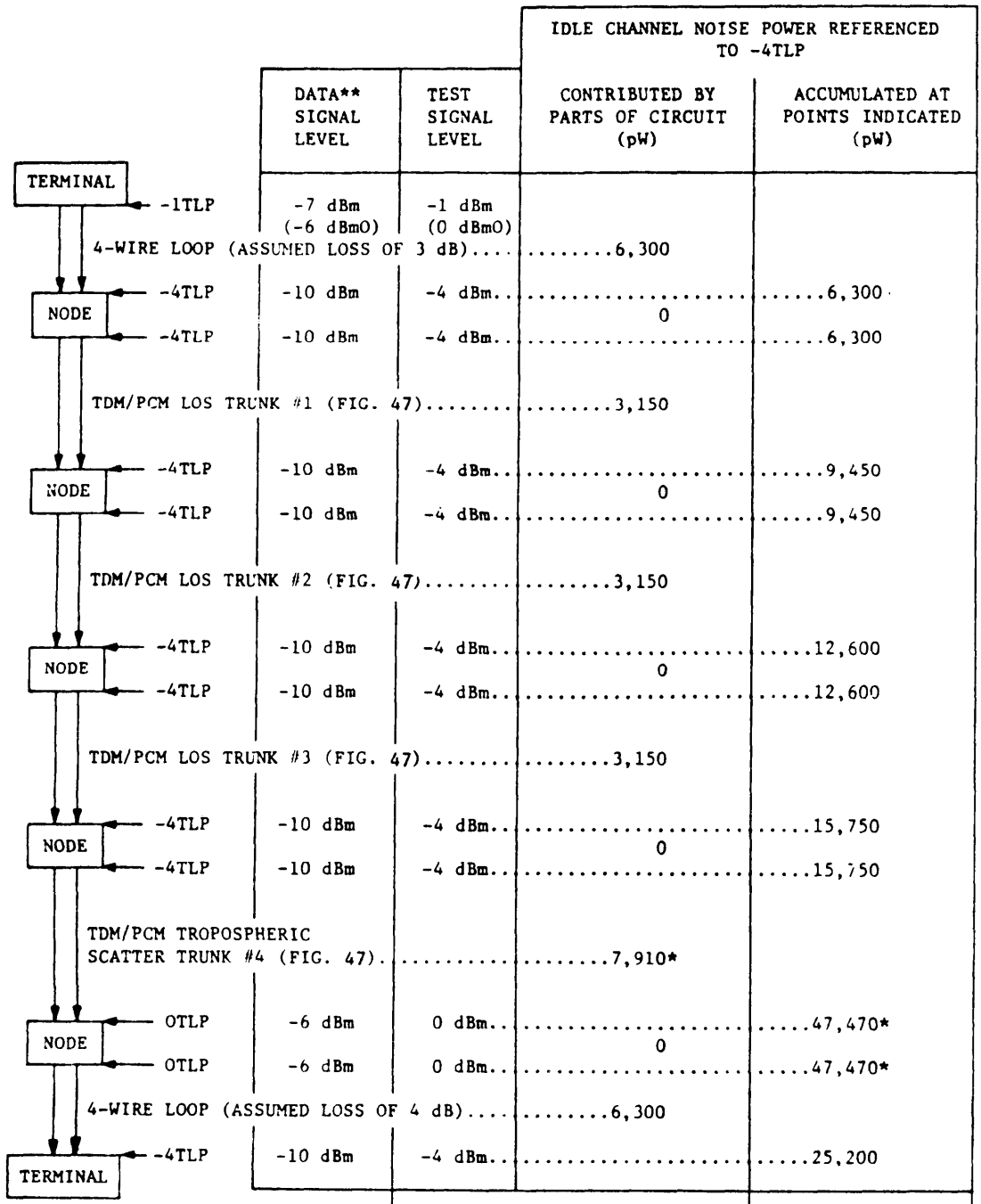
	MEAN SPEECH VOLUME	DATA** SIGNAL LEVEL	TEST SIGNAL LEVEL	NOISE POWER REFERENCED TO -4TLP CONTRIBUTED BY PART OF CIRCUIT: IDLE CHANNEL NOISE POWER VOICE (pWp)	IDLE CHANNEL NOISE POWER DATA** (pW)	EQUIVALENT PCM NOISE POWER (pWp)
-0.5TLP	-10 VU	-6.5 dBm (-6 dBmO)	-0.5 dBm (0 dBmO)			
4-WIRE LOOP...	.....	.....	.....	.....6,300.....	..6,300.....	0
-4TLP	-13.5 VU	-10 dBm	-4 dBm	400	630	0
-4TLP	-13.5 VU	-10 dBm	-4 dBm			
TDM/PCM LOS TRUNK #1 (FIG. 47)....	.....	.....	.....	.....2,000.....	..3,150.....	...5,000
-4TLP	-13.5 VU	-10 dBm	-4 dBm	400	630	0
-4TLP	-13.5 VU	-10 dBm	-4 dBm			
TDM/PCM LOS TRUNK #2 (FIG. 47)....	.....	.....	.....	.....2,000.....	..3,150.....	...5,000
-4TLP	-13.5 VU	-10 dBm	-4 dBm	400	630	0
-4TLP	-13.5 VU	-10 dBm	-4 dBm			
TDM/PCM LOS TRUNK #3 (FIG. 47)....	.....	.....	.....	.....2,000.....	..3,150.....	...5,000
-4TLP	-13.5 VU	-10 dBm	-4 dBm	400	630	0
-4TLP	-13.5 VU	-10 dBm	-4 dBm			
TDM/PCM LOS TRUNK #4 (FIG. 47)....	.....	.....	.....	.....2,000.....	..3,150.....	...5,000
-4TLP	-13.5 VU	-10 dBm	-4 dBm	400	630	0
-4TLP	-13.5 VU	-10 dBm	-4 dBm			
TDM/PCM LOS TRUNK #5 (FIG. 47)....	.....	.....	.....	.....2,000.....	..3,150.....	...5,000
-4TLP	-13.5 VU	-10 dBm	-4 dBm	400	630	0
-4TLP	-13.5 VU	-10 dBm	-4 dBm			
TDM/PCM LOS TRUNK #6 (FIG. 47)....	.....	.....	.....	.....2,000.....	..3,150.....	...5,000
-4TLP	-13.5 VU	-10 dBm	-4 dBm	400	630	0
-4TLP	-13.5 VU	-10 dBm	-4 dBm			
4-WIRE LOOP...	.....	.....	.....	.....6,300*.....	..6,300*....	0
-7.5TLP	-17 VU	-13.5 dBm	-7.5 dBm			
TOTAL CIRCUIT NOISE POWER			AT OTLP***:	88,430 pWpO (49.5 dBrncO)	109,800 pWO (50.4 dBrnO)	75,360 pWpO (48.8 dBrncO)
			AT -7.5TLP:	15,730 pWp (42.0 dBrnc)	19,530 pW (42.9 dBrn)	13,400 pWp (41.3 dBrnc)
SNR = -13.5 dBm - (42.9 dBrn - 90) = 33.6 dB						

*THESE NOISE POWER VALUES ARE REFERENCED TO -7.5TLP AT RECEIVING TERMINAL AND NOT TO A -4TLP.

**SEE NOTE OF 5.7.3.2.6 FOR DATA TRANSMISSION OVER TDM/PCM VF CHANNELS.
***NOISE POWER MAY BE REFERENCED TO A OTLP EVEN IF IT DOES NOT EXIST (SEE NOTE 3 OF 5.9.3.2).

FIGURE 54. Hypothetical 300 km (162 nmi) switched VF reference circuit for voice service and telegraph and data service of tactical subsystem type II.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



**SEE NOTE OF 5.7.3.2.6 FOR DATA TRANSMISSION OVER TDM/PCM VF CHANNELS.

FIGURE 55. Hypothetical 300 km (162 nmi) nonswitched (point-to-point) VF reference circuit for telegraph and data service of tactical subsystem type II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

accordance with 5.7.4.3.6. The noise power contributions are then added at each TLP, starting with the -4TLP at the output of the transmitting loop and considering 4 dB of amplification of the TDM/PCM tropospheric scatter reference trunk and 4 dB of insertion loss of the receiving loop, to obtain the total circuit noise power referenced to the -4TLP at the receiving terminal. The received noise power of 25,200 pW (44.0 dBrn) consists only of the idle channel noise power and does not include quantizing noise power generated by the A-D/D-A conversion process of the TDM/PCM equipment. (See Note 2 of 5.2.3.1.5.2.) Also, the quasi-analog SNR of 36 dB is based only on idle channel noise and does not include any quantizing noise. The data signal level of -10 dBm at each -4TLP (-6 dBm0) is in accordance with 5.2.3.1.4.

Table LX lists the parameter values for the reference circuits shown in Figures 53, 54 and 55. The parameter values have been obtained by applying the summation methods of 5.9.3 to the applicable reference trunk parameters listed in Table LVII, the applicable loop parameters given in 5.5.2 and the applicable switch transmission characteristics given in 5.6.2.1.2. Only switch transmission characteristics have been considered for calculating the contributions of the nodes to the reference circuit parameters of Figures 53 and 54. The contribution of the nodes shown in Figure 55 to the reference circuit parameters have been taken as zero since these nodes do not contain switches. The parameter values listed in Table LX indicate that the reference circuits shown in Figures 53 and 54 meet all requirements for voice service stated in 5.2.2.1, but the reference circuits shown in Figures 54 and 55 do not meet all requirements for telegraph and data service stated in 5.2.3.1. The reference circuit shown in Figure 54 does not meet the insertion loss versus frequency characteristic for modulation rates of up to 1200 Bd and up to 2400 Bd and does not meet, but the parameter values listed in Table LX are considered sufficiently close to, the envelope delay distortion for modulation rates of up to 2400 Bd. The reference circuit shown in Figure 55 does not meet, but the parameter values listed in Table LX are considered sufficiently close to, the insertion loss versus frequency characteristics for modulation rates of up to 1200 Bd and up to 2400 Bd. Therefore, the reference circuits shown in Figures 54 and 55 may require equalization (see 5.2.3.1.8) to provide acceptable telegraph and data service. Digital equipment has been developed to transmit and receive over TDM/PCM links digital data signals with a wide range of data signaling rates. (See Note of 5.7.3.2.6.) This equipment will eliminate the need for data modems and also the need for equalization of TDM/PCM VF channels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LX. Parameters for hypothetical 300 km (162 nmi) VF reference circuits of tactical subsystem type II.

Reference Circuit Parameters		Parameter Values for Figure 53	Figure 54	Figure 55
Nominal Transmission Distance		300 km (162 nmi) (without Loops)		
Total Idle Channel Noise Power for Voice Service	referenced to Circuit OTLP	58,730 pWp0 (47.7 dBrnc0)	88,430 pWp0 (49.5 dBrnc0)	not applicable
	referenced to TLP at Receiving Terminal	13,150 pWp (41.2 dBrnc)	15,730 pWp (42.0 dBrnc)	not applicable
for Telegraph and Data Service	referenced to Circuit OTLP	not applicable	109,800 pW0 (50.4 dBrn0)	63,300 pW0 (48.0 dBrn0)
	referenced to TLP at Receiving Terminal	not applicable	19,530 pW (42.9 dBrn)	25,200 pW (44.0 dBrn)
Quasi-Analog Data SNR		not appl.	33.6 dB	36 dB
Total Equivalent PCM Noise Power	referenced to Circuit OTLP	50,390 pWp0 (47.0 dBrnc0)	75,360 pWp0 (48.8 dBrnc0)	not applicable
	referenced to TLP at Receiving Terminal	11,280 pWp (40.5 dBrnc)	13,400 pWp (41.3 dBrnc)	not applicable
Insertion Loss Versus Frequency Characteristic (in dB, rounded off within 0.1 dB) with Loops for Voice Service	$f < 400$ Hz	≥ -6.4	not applicable	not applicable
	400 Hz $\leq f <$ 600 Hz	-6.4 to +10.3		
	600 Hz $\leq f <$ 1000 Hz	-5.2 to +10.3		
	1000 Hz $\leq f <$ 2400 Hz	-3.0 to +10.3		
	2400 Hz $\leq f <$ 2800 Hz	-3.0 to +18.6		
	2800 Hz $\leq f$	≥ -3.0		
with Loops for Telegraph and Data Service up to 1200 Bd	$f < 300$ Hz	not applicable	≥ -4.6	≥ -3.5
	300 Hz $\leq f <$ 400 Hz		-4.6 to +6.2	-3.5 to +4.9
	400 Hz $\leq f <$ 1000 Hz		-4.6 to +4.6	-3.5 to +3.5
	1000 Hz $\leq f <$ 2400 Hz		-3.6 to +4.6	-2.0 to +3.5
	2400 Hz $\leq f <$ 2700 Hz		-3.6 to +5.6	-2.0 to +4.7
	2700 Hz $\leq f <$ 3000 Hz		-3.6 to +7.9	-2.0 to +7.3
	3000 Hz $\leq f$		≥ -3.6	≥ -2.0
with Loops for Telegraph and Data Service up to 2400 Bd	$f < 300$ Hz	not applicable	≥ -4.6	≥ -3.5
	300 Hz $\leq f <$ 400 Hz		-4.6 to +6.2	-3.5 to +4.9
	400 Hz $\leq f <$ 500 Hz		-4.6 to +4.6	-3.5 to +3.5
	500 Hz $\leq f <$ 1000 Hz		-3.9 to +4.6	-2.4 to +3.5
	1000 Hz $\leq f <$ 2800 Hz		-3.6 to +4.6	-2.0 to +3.5
	2800 Hz $\leq f <$ 3000 Hz		-3.6 to +6.7	-2.0 to +6.0
	3000 Hz $\leq f$		≥ -3.6	≥ -2.0
Maximum Envelope Delay Distortion (in microseconds) with Loops for Voice Service		Not standardized (see 5.2.2.1.5)		
with Loops for Tel. and Data up to 1200 Bd	800 Hz $\leq f <$ 1000 Hz	not applicable	855	520
	1000 Hz $\leq f <$ 2400 Hz		955	620
	2400 Hz $\leq f \leq$ 2600 Hz		1255	920
with Loops for Telegraph and Data Service up to 2400 Bd	500 Hz $\leq f <$ 600 Hz	not applicable	855	480
	600 Hz $\leq f <$ 1000 Hz		735	400
	1000 Hz $\leq f <$ 2600 Hz		775	440
	2600 Hz $\leq f \leq$ 2800 Hz		1195	780
Mean of Insertion Loss (in dB)		10	7	3
Net Loss Variations (in dB, rounded off within 0.1 dB) Over any 30 consec. days		$\pm$3.7	$\pm$4.2	$\pm$3.7
Over any 15 consec. min.		$\pm$1.2	$\pm$1.4	$\pm$1.2

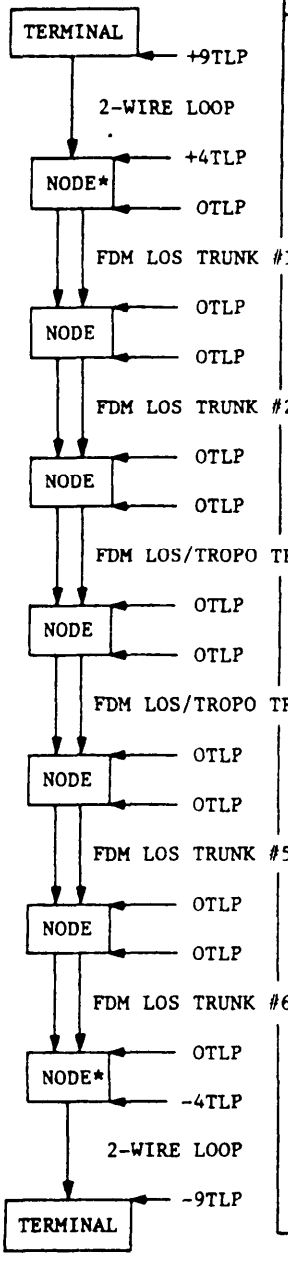
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.9.4.4 Hypothetical reference circuits of tactical subsystem type III. The nominal transmission distance of a hypothetical reference circuit of the tactical subsystem type III (see Table I) has been taken as 1800 km (972 nmi), not including the length of the loops. This distance is considered to be the probable maximum transmission distance over which switched and nonswitched (point-to-point) communications are required in the tactical subsystem type III. Three different examples of VF reference circuits of the tactical subsystem type III are shown in Figures 56, 57 and 58. These figures show the configurations of the reference circuits, circuit TLP, signal levels, noise power levels contributed by the different parts of the circuit to the total circuit noise, and total circuit noise power referenced to selected TLP. The noise power contributed by each loop to the total circuit noise has been taken as the maximum allowable noise power of 6300 pWp for analog voice service and 6300 pW for telegraph and data service, measured at the output terminals of the loop in accordance with 5.5.2.6. The reference trunks have a configuration as shown in Figures 48 or 49 with characteristics as listed in Table LVIII. Figures 56, 57 and 58 show only one direction of transmission; the other direction is identical.

Figure 56 shows an example of a hypothetical 1800 km (972 nmi) switched VF reference circuit for voice service consisting of four FDM LOS reference trunks and two FDM reference trunks with LOS and tropospheric scatter radio links, seven nodes and two-wire loops extending to the telephone sets. The nodes contain a four-wire analog switch and may also contain other equipment, such as equipment for patching, signal processing and technical control (see 4.3.4). The noise power contributed by each node to the total circuit noise has been taken as the maximum allowable noise power of an analog switch in accordance with 5.6.2.1.2.5. The two-wire loops are assumed to have a mean insertion loss of 5 dB in accordance with 5.5.2.1.1. The loops are connected to two-wire/four-wire hybrid transformers which are part of the originating and terminating nodes. The noise power levels contributed by the loops, nodes and trunks to the total circuit noise are listed with reference to the associated circuit TLP at the output of each loop, node and trunk and then transferred, where applicable, to a circuit OTLP. All noise power contributions referenced to OTLP are then added to obtain the total circuit noise power referenced to OTLP which is transferred to the -9TLP at the receiving terminal. The received noise power of 7940 pWp (39.0 dBrnc) is within the limit given in 5.2.2.1.3. The transmitted mean speech volume of -10 VU and the received mean speech volume of -28 VU are in accordance with 5.2.2.1.1 and 5.2.2.1.2, respectively.

Figure 57 shows an example of a hypothetical 1800 km (972 nmi) switched VF reference circuit for voice service and telegraph and data service consisting of six FDM LOS reference trunks, seven nodes and four-wire loops extending to the terminal equipment or terminal subsystems. The nodes contain a four-wire analog switch and may also contain other equipment, such as equipment for patching, signal processing and technical

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



	MEAN SPEECH VOLUME	TEST SIGNAL LEVEL	NOISE POWER CONTRIBUTED BY PARTS OF CIRCUIT (pWp)	(dBrnc)	REFERENCED TO	REFERENCED TO OTLP (pWp0)	(dBrnc0)
+9TLP	-10 VU	+9 dBm (0 dBm0)					
2-WIRE LOOP			6,300	38.0	+4TLP	2,510	34.0
+4TLP	-15 VU	+4 dBm					
OTLP	-19 VU	0 dBm	400	26.0	OTLP	400	26.0
FDM LOS TRUNK #1 (FIG. 48)			1,286	31.1	OTLP	1,286	31.1
OTLP	-19 VU	0 dBm					
OTLP	-19 VU	0 dBm	400	26.0	OTLP	400	26.0
FDM LOS TRUNK #2 (FIG. 48)			1,286	31.1	OTLP	1,286	31.1
OTLP	-19 VU	0 dBm					
OTLP	-19 VU	0 dBm	400	26.0	OTLP	400	26.0
FDM LOS/TROPO TRUNK #3 (FIG. 49)			1,286	31.1	OTLP	1,286	31.1
OTLP	-19 VU	0 dBm					
OTLP	-19 VU	0 dBm	400	26.0	OTLP	400	26.0
FDM LOS/TROPO TRUNK #4 (FIG. 49)			1,286	31.1	OTLP	1,286	31.1
OTLP	-19 VU	0 dBm					
OTLP	-19 VU	0 dBm	400	26.0	OTLP	400	26.0
FDM LOS TRUNK #5 (FIG. 48)			1,286	31.1	OTLP	1,286	31.1
OTLP	-19 VU	0 dBm					
OTLP	-19 VU	0 dBm	400	26.0	OTLP	400	26.0
FDM LOS TRUNK #6 (FIG. 48)			1,286	31.1	OTLP	1,286	31.1
OTLP	-19 VU	0 dBm					
-4TLP	-23 VU	-4 dBm	400	26.0	OTLP	400	26.0
2-WIRE LOOP			6,300	38.0	-9TLP	50,040	47.0
-9TLP	-28 VU	-9 dBm					
TOTAL CIRCUIT NOISE POWER					AT OTLP:	63,066 pWp0 (48.0 dBrnc0)	
					AT -9TLP:	7,940 pWp (39.0 dBrnc)	

*THESE NODES CONTAIN A 2-WIRE/4-WIRE HYBRID TRANSFORMER WITH AN INSERTION LOSS OF NOMINAL 4 dB.

FIGURE 56. Hypothetical 1800 km (972 nmi) switched VF reference circuit for voice service of tactical subsystem type III.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

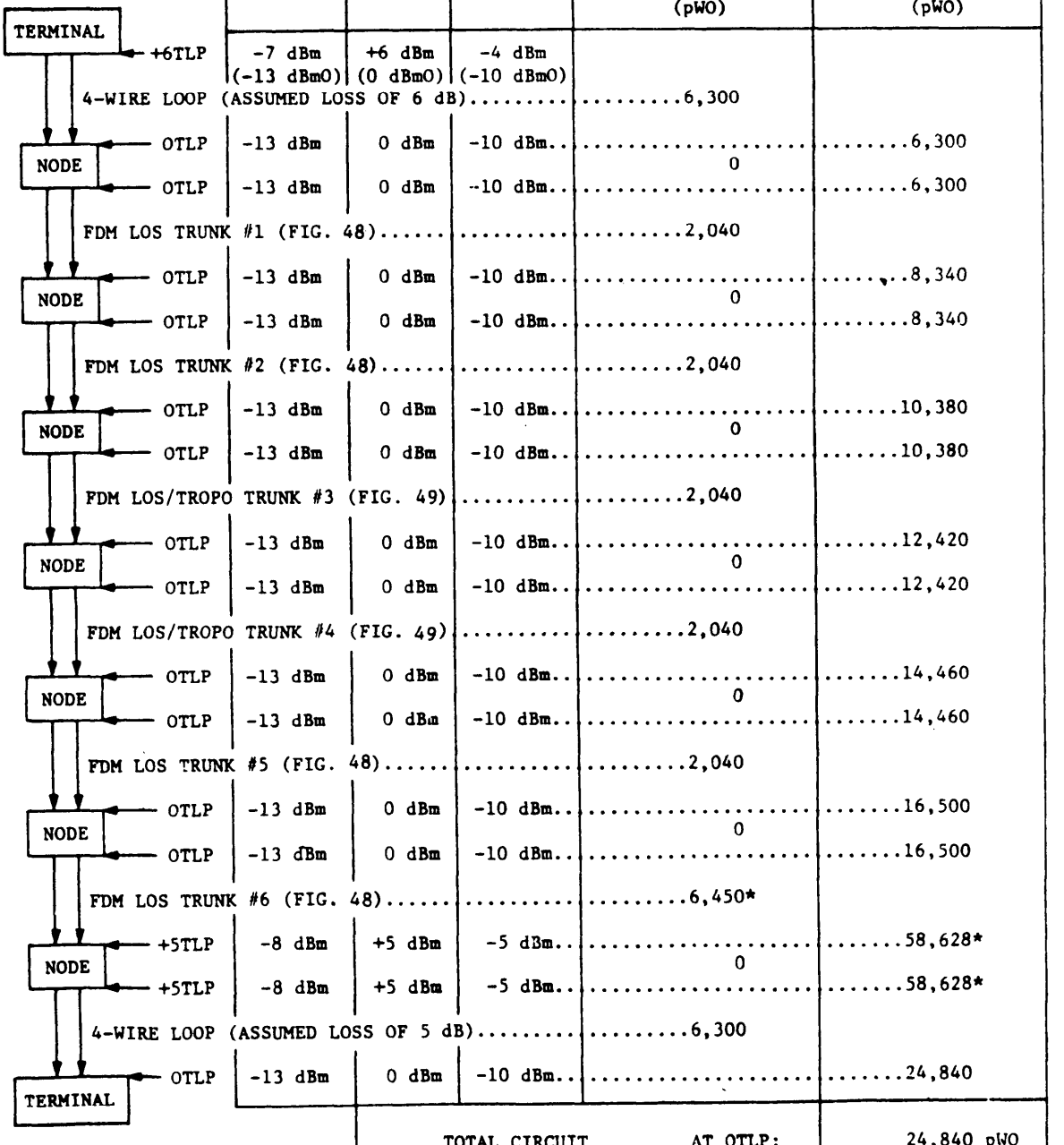
Page 252 is missing from the original legacy source document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

control (see 4.3.4). The noise power contributed by each node to the total circuit noise has been taken as the maximum allowable noise power of an analog switch in accordance with 5.6.2.1.2.5. The four-wire loops are assumed to have a mean insertion loss of 9 dB in accordance with 5.5.2.1.1. The noise power levels contributed by the loops, nodes and trunks to the total circuit noise are listed with reference to the associated circuit TLP at the output of each loop, node and trunk. The associated circuit TLP are OTLP with the exception of the circuit TLP at the output of the receiving loop which has a -9TLP. All noise power contributions are then added at each TLP, starting with the OTLP at the output of the transmitting loop and considering 9 dB of attenuation of the receiving loop, to obtain the total circuit noise power referenced to the -9TLP at the receiving terminal. The received noise power of 8417 pWp (39.3 dBrnc) for voice service is within the limit given in 5.2.2.1.3. The received noise power of 73,000 pW0 (48.6 dBrn0) for telegraph and data service is 1.6 dB higher than the limit given in 5.2.3.1.5.2. Therefore, the quasi-analog data signal may require regeneration for acceptable telegraph and data service even if the SNR of 28.4 dB meets the requirement of 5.2.3.1.5.3. The transmitted mean speech volume of -10 VU and the received mean speech volume of -28 VU are in accordance with 5.2.2.1.1 and 5.2.2.1.2, respectively. The data signal level of -13 dBm at each OTLP (-13 dBm0) and the received data signal level of -22 dBm are in accordance with 5.2.3.1.4 and 5.2.3.1.5.1, respectively.

Figure 58 shows an example of a hypothetical 1800 km (972 nmi) non-switched (point-to-point) VF reference circuit for telegraph and data service consisting of four FDM LOS reference trunks and two FDM reference trunks with LOS and tropospheric scatter radio links, seven nodes and four-wire loops extending to the terminal equipment or terminal subsystems. The nodes may contain equipment for such functions as patching, signal processing and technical control (see 4.3.4). The noise power contributed by each node to the total circuit noise has been taken as zero since the nodes do not contain switches. It is assumed that one four-wire loop has a minimum insertion loss of 6 dB, the other four-wire loop has a minimum insertion loss of 5 dB, and that no adjustments are made to obtain a specific insertion loss value in accordance with 5.5.2.1.2. The noise power levels contributed by the loops and trunks to the total circuit noise are listed with reference to the associated circuit TLP at the output of each loop and trunk. The associated circuit TLP are OTLP with the exception of the circuit TLP at the output of the FDM LOS reference trunk #6 which has a +5TLP in accordance with 5.7.4.3.6. The noise power contributions are then added at each TLP, starting with the OTLP at the output of the transmitting loop and considering 5 dB of amplification of the FDM LOS reference trunk #6, to obtain the total circuit noise power referenced to the OTLP at the receiving terminal. The received noise power of 24,840 pW0 (44.0 dBrn0) is within the limit given in 5.2.3.1.5.2. The data signal level of -13 dBm at each OTLP (-13 dBm0) and the received data signal level of -13 dBm0 are in accordance with

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.



	DATA SIGNAL LEVEL	TEST SIGNAL LEVEL	TEST TONE LEVEL	NOISE POWER REFERENCED TO OTLP: CONTRIBUTED BY PARTS OF CIRCUIT (pWO)	NOISE POWER REFERENCED TO OTLP: ACCUMULATED AT POINTS INDICATED (pWO)
+6TLP	-7 dBm (-13 dBm0)	+6 dBm (0 dBm0)	-4 dBm (-10 dBm0)		
4-WIRE LOOP (ASSUMED LOSS OF 6 dB)				6,300	
OTLP	-13 dBm	0 dBm	-10 dBm		6,300
				0	
OTLP	-13 dBm	0 dBm	-10 dBm		6,300
FDM LOS TRUNK #1 (FIG. 48)				2,040	
OTLP	-13 dBm	0 dBm	-10 dBm		8,340
				0	
OTLP	-13 dBm	0 dBm	-10 dBm		8,340
FDM LOS TRUNK #2 (FIG. 48)				2,040	
OTLP	-13 dBm	0 dBm	-10 dBm		10,380
				0	
OTLP	-13 dBm	0 dBm	-10 dBm		10,380
FDM LOS/TROPO TRUNK #3 (FIG. 49)				2,040	
OTLP	-13 dBm	0 dBm	-10 dBm		12,420
				0	
OTLP	-13 dBm	0 dBm	-10 dBm		12,420
FDM LOS/TROPO TRUNK #4 (FIG. 49)				2,040	
OTLP	-13 dBm	0 dBm	-10 dBm		14,460
				0	
OTLP	-13 dBm	0 dBm	-10 dBm		14,460
FDM LOS TRUNK #5 (FIG. 48)				2,040	
OTLP	-13 dBm	0 dBm	-10 dBm		16,500
				0	
OTLP	-13 dBm	0 dBm	-10 dBm		16,500
FDM LOS TRUNK #6 (FIG. 48)				6,450*	
+5TLP	-8 dBm	+5 dBm	-5 dBm		58,628*
				0	
+5TLP	-8 dBm	+5 dBm	-5 dBm		58,628*
4-WIRE LOOP (ASSUMED LOSS OF 5 dB)				6,300	
OTLP	-13 dBm	0 dBm	-10 dBm		24,840

TOTAL CIRCUIT NOISE POWER	AT OTLP:	24,840 pWO (44.0 dBrnO)
	AT +5TLP:	78,550 pW (49.0 dBrn)
SNR = -13 dBm - (44 dBrn - 90) = 33 dB		

*THESE NOISE POWER VALUES ARE REFERENCED TO A +5TLP AT THE TERMINATING NODE AND NOT TO A OTLP

FIGURE 58. Hypothetical 1800 km (972 nmi) nonswitched (point-to-point) VF reference circuit for telegraph and data service of tactical subsystem type III.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.2.3.1.4 and 5.2.3.1.5.1, respectively. The SNR of 33 dB is within the limit given in 5.2.3.1.5.3.

Table LXI lists the parameter values for the reference circuits shown in Figures 56, 57 and 58. The parameter values have been obtained by applying the summation methods of 5.9.3 to the applicable reference trunk parameters listed in Table LVIII, the applicable loop parameters given in 5.5.2 and the applicable switch transmission characteristics given in 5.6.2.1.2. Only switch transmission characteristics have been considered for calculating the contributions of the nodes to the reference circuit parameters of Figures 56 and 57. The contribution of the nodes shown in Figure 58 to the reference circuit parameters have been taken as zero since these nodes do not contain switches. The parameter values listed in Table LXI indicate that the reference circuits shown in Figures 56 and 57 meet all requirements for voice service stated in 5.2.2.1, but the reference circuits shown in Figures 57 and 58 do not meet all requirements for telegraph and data service stated in 5.2.3.1. The reference circuits shown in Figures 57 and 58 do not meet the insertion loss versus frequency characteristics for modulation rates of up to 1200 Bd and up to 2400 Bd and do not meet, but the parameter values listed in Table LXI are considered sufficiently close to, the envelope delay distortion for modulation rates of up to 2400 Bd. Additionally, the reference circuit shown in Figure 57 does not meet the received noise power for telegraph and data service stated in 5.2.3.1.5.2. Therefore, these circuits may require regeneration or equalization, or both, to provide acceptable telegraph and data service.

5.9.4.5 Hypothetical reference circuits of tactical subsystem type IV. Under consideration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

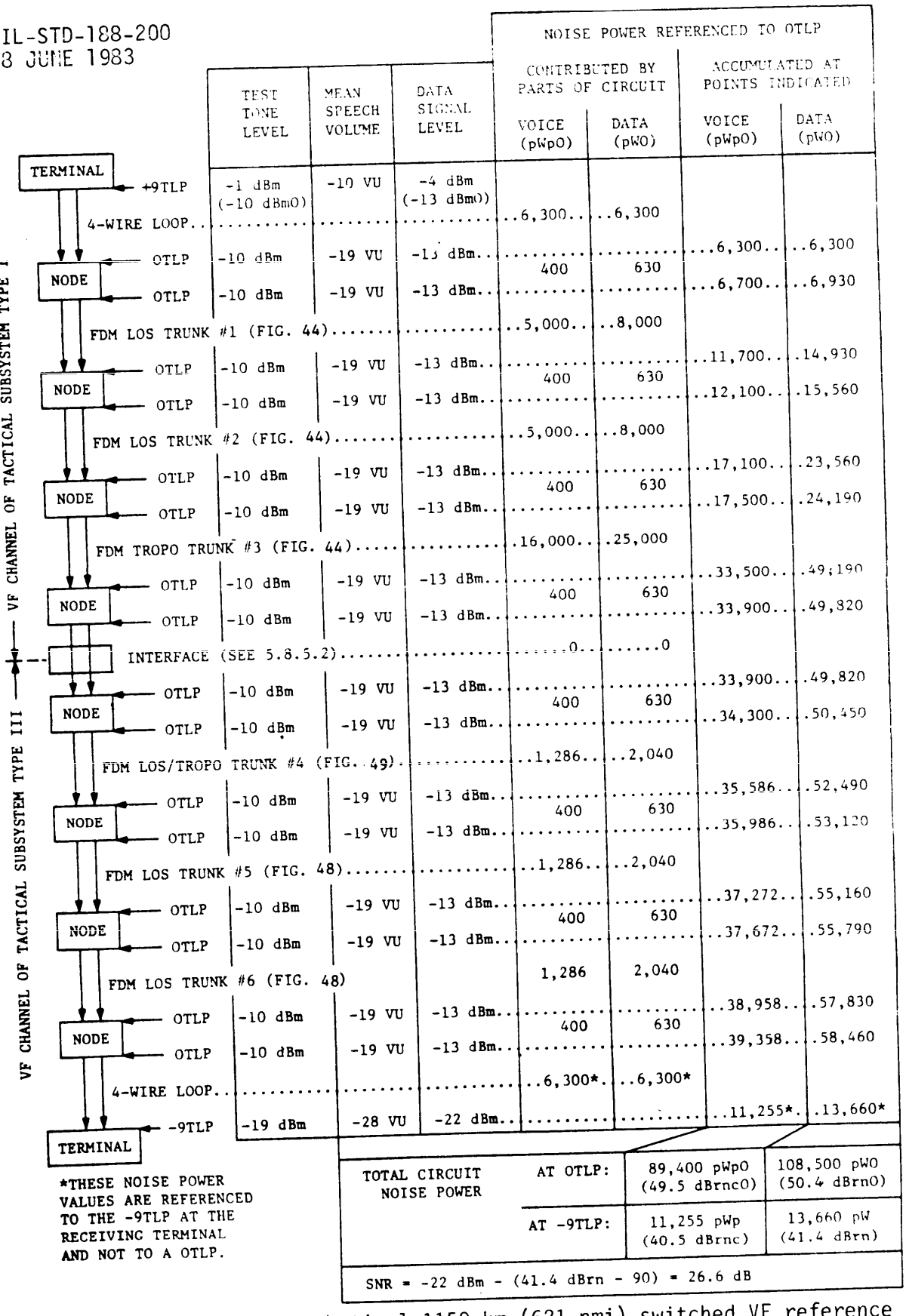
TABLE LXI. Parameters for hypothetical 1800 km (972 nmi) VF reference circuits of tactical subsystem type III.

Reference Circuit Parameters		Parameter Values for Figure 56	Figure 57	Figure 58
Nominal Transmission Distance		1800 km (972 nmi) (without Loops)		
Total Circuit Noise Power for Voice Service	referenced to Circuit OTLP	63,066 pWp0 (48.0 dBrnc0)	66,860 pWp0 (48.3 dBrnc0)	not applicable
	referenced to TLP at Receiving Terminal	7,940 pWp (39.0 dBrnc)	8,417 pWp (39.3 dBrnc)	not applicable
for Telegraph and Data Service	referenced to Circuit OTLP	not applicable	73,000 pW0 (48.6 dBrn0)	24,840 pW0 (44.0 dBrn0)
	referenced to TLP at Receiving Terminal	not applicable	9,190 pW (39.6 dBrn)	24,840 pW0 (44.0 dBrn0)
Quasi-Analog Data SNR		not appl.	28.4 dB	33 dB
Insertion Loss Versus Frequency Characteristics (in dB, rounded off within 0.1 dB) with Loops for Voice Service	$f < 400$ Hz	≥ -7.0	not applicable	not applicable
	$400 \text{ Hz} \leq f < 600$ Hz	-7.0 to +10.7		
	$600 \text{ Hz} \leq f < 1000$ Hz	-5.9 to +10.7		
	$1000 \text{ Hz} \leq f < 2400$ Hz	-4.2 to +10.7		
	$2400 \text{ Hz} \leq f < 2800$ Hz	-4.2 to +18.8		
	$2800 \text{ Hz} \leq f$	≥ -4.2		
with Loops for Telegraph and Data Service up to 1200 Bd	$f < 300$ Hz	not applicable	≥ -5.2	≥ -4.3
	$300 \text{ Hz} \leq f < 400$ Hz		-5.2 to +6.2	-4.3 to +5.5
	$400 \text{ Hz} \leq f < 1000$ Hz		-5.2 to +5.2	-4.3 to +4.3
	$1000 \text{ Hz} \leq f < 2400$ Hz		-4.3 to +5.2	-3.2 to +4.3
	$2400 \text{ Hz} \leq f < 2700$ Hz		-4.3 to +6.1	-3.2 to +5.3
	$2700 \text{ Hz} \leq f < 3000$ Hz		-4.3 to +8.3	-3.2 to +7.8
	$3000 \text{ Hz} \leq f$		≥ -4.3	≥ -3.2
with Loops for Telegraph and Data Service up to 2400 Bd	$f < 300$ Hz	not applicable	≥ -5.2	≥ -4.3
	$300 \text{ Hz} \leq f < 400$ Hz		-5.2 to +6.2	-4.3 to +5.5
	$400 \text{ Hz} \leq f < 500$ Hz		-5.2 to +5.2	-4.3 to +4.3
	$500 \text{ Hz} \leq f < 1000$ Hz		-4.6 to +5.2	-3.5 to +4.3
	$1000 \text{ Hz} \leq f < 2800$ Hz		-4.3 to +5.2	-3.2 to +4.3
	$2800 \text{ Hz} \leq f < 3000$ Hz		-4.3 to +7.1	-3.2 to +6.5
	$3000 \text{ Hz} \leq f$		≥ -4.3	≥ -3.2
Maximum Envelope Delay Distortion (in microseconds) with Loops for Voice Service		Not standardized (See 5.2.2.1.5)		
with Loops for Tel. and Data up to 1200 Bd	$800 \text{ Hz} \leq f < 1000$ Hz	not applicable	1137	962
	$1000 \text{ Hz} \leq f < 2400$ Hz		997	822
	$2400 \text{ Hz} \leq f \leq 2600$ Hz		1297	1122
with Loops for Telegraph and Data Service up to 2400 Bd	$500 \text{ Hz} \leq f < 600$ Hz	not applicable	1197	1022
	$600 \text{ Hz} \leq f < 1000$ Hz		1017	842
	$1000 \text{ Hz} \leq f < 2600$ Hz		817	642
	$2600 \text{ Hz} \leq f \leq 2800$ Hz		1237	1062
Mean of Insertion Loss (in dB)		18	18	6
Net Loss Variation (in dB, rounded off within 0.1 dB) Over any 30 consecutive days		± 4.0	± 4.2	± 4.0

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.9.4.6 Common hypothetical reference circuit of tactical subsystems type I and type III. Figure 59 shows an example of a hypothetical switched VF reference circuit for voice service and telegraph and data service connecting a VF channel of the tactical subsystem type I (see Table I) with a VF channel of the tactical subsystem type III. The method of interconnecting the VF channels of the two tactical subsystems is based on the standard test tone level stated in 5.8.5.2. The reference circuit shown in Figure 59 consists of a four-wire loop, two FDM LOS reference trunks and one FDM tropospheric scatter reference trunk of the tactical subsystem type I, connected in tandem with one FDM LOS/tropospheric scatter reference trunk, two FDM LOS reference trunks and a four-wire loop of the tactical subsystem type III. Figure 59 shows only one direction of transmission; the other direction is identical. The nominal transmission distance of the reference circuit is 1150 km (621 nmi), not including the length of the loops. Figure 59 shows the configuration of the reference circuit, circuit TLP, signal levels, noise power levels contributed by the different parts of the circuit to the total circuit noise, and total circuit noise power for voice service and telegraph and data service referenced to a circuit OTLP and to the -9TLP at the receiving terminal. The noise power contributed by each loop to the total circuit noise has been taken as the maximum allowable noise power of 6300 pWp for analog voice service and 6300 pW for telegraph and data service, measured at the output terminals of the loop in accordance with 5.5.2.6. The nodes contain a four-wire analog switch and may also contain other equipment, such as equipment for patching, signal processing and technical control (see 4.3.4). The noise power contributed by each node to the circuit noise has been taken as the maximum allowable noise power of an analog switch in accordance with 5.6.2.1.2.5. The four-wire loops are assumed to have a mean insertion loss of 9 dB in accordance with 5.2.2.1.1. The noise power levels contributed by the loops, nodes and trunks to the total circuit noise are listed in Figure 59 with reference to the associated circuit TLP at the output of each loop, node and trunk. The associated circuit TLP are OTLP with the exception of the circuit TLP at the output of the receiving loop which has a -9TLP. The noise power contributions are then added at each TLP, starting with the OTLP at the output of the transmitting loop and considering 9 dB of attenuation of the receiving loop, to obtain the total circuit noise power referenced to the -9TLP at the receiving terminal. The received noise power of 11,255 pWp (40.5 dBrnc) for voice service is within the limit given in 5.2.2.1.3. The received noise power of 108,500 pW0 (50.4 dBrn0) for telegraph and data service is 3.4 dB higher than the limit given in 5.2.3.1.5.2. Therefore, the quasi-analog data signals may require regeneration for acceptable telegraph and data service even if the SNR of 26.6 dB meets the requirements of 5.2.3.1.5.3. The transmitted mean speech volume of -10 VU and the received mean speech volume of -28 VU are in accordance with 5.2.2.1.1 and 5.2.2.1.2, respectively. The data signal level of -13 dBm at each OTLP (-13 dBm0) and the received data signal level of -22 dBm are in accordance with 5.2.3.1.4 and 5.2.3.1.5.1, respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



	TEST TONE LEVEL	MEAN SPEECH VOLUME	DATA SIGNAL LEVEL	NOISE POWER REFERENCED TO OTLP: CONTRIBUTED BY PARTS OF CIRCUIT VOICE (pWpO)	CONTRIBUTED BY PARTS OF CIRCUIT DATA (pWO)	ACCUMULATED AT POINTS INDICATED VOICE (pWpO)	ACCUMULATED AT POINTS INDICATED DATA (pWO)
TERMINAL +9TLP	-1 dBm (-10 dBmO)	-10 VU	-4 dBm (-13 dBmO)				
4-WIRE LOOP				6,300	6,300		
NODE OTLP	-10 dBm	-19 VU	-13 dBm			6,300	6,300
NODE OTLP	-10 dBm	-19 VU	-13 dBm	400	630	6,700	6,930
FDM LOS TRUNK #1 (FIG. 44)				5,000	8,000		
NODE OTLP	-10 dBm	-19 VU	-13 dBm			11,700	14,930
NODE OTLP	-10 dBm	-19 VU	-13 dBm	400	630	12,100	15,560
FDM LOS TRUNK #2 (FIG. 44)				5,000	8,000		
NODE OTLP	-10 dBm	-19 VU	-13 dBm			17,100	23,560
NODE OTLP	-10 dBm	-19 VU	-13 dBm	400	630	17,500	24,190
FDM TROPO TRUNK #3 (FIG. 44)				16,000	25,000		
NODE OTLP	-10 dBm	-19 VU	-13 dBm			33,500	49,190
NODE OTLP	-10 dBm	-19 VU	-13 dBm	400	630	33,900	49,820
INTERFACE (SEE 5.8.5.2)				0	0		
NODE OTLP	-10 dBm	-19 VU	-13 dBm			33,900	49,820
NODE OTLP	-10 dBm	-19 VU	-13 dBm	400	630	34,300	50,450
FDM LOS/TROPO TRUNK #4 (FIG. 49)				1,286	2,040		
NODE OTLP	-10 dBm	-19 VU	-13 dBm			35,586	52,490
NODE OTLP	-10 dBm	-19 VU	-13 dBm	400	630	35,986	53,120
FDM LOS TRUNK #5 (FIG. 48)				1,286	2,040		
NODE OTLP	-10 dBm	-19 VU	-13 dBm			37,272	55,160
NODE OTLP	-10 dBm	-19 VU	-13 dBm	400	630	37,672	55,790
FDM LOS TRUNK #6 (FIG. 48)				1,286	2,040		
NODE OTLP	-10 dBm	-19 VU	-13 dBm			38,958	57,830
NODE OTLP	-10 dBm	-19 VU	-13 dBm	400	630	39,358	58,460
4-WIRE LOOP				6,300*	6,300*		
-9TLP TERMINAL	-19 dBm	-28 VU	-22 dBm			11,255*	13,660*

TOTAL CIRCUIT NOISE POWER		
AT OTLP:	89,400 pWpO (49.5 dBrncO)	108,500 pWO (50.4 dBrnO)
AT -9TLP:	11,255 pWp (40.5 dBrnc)	13,660 pW (41.4 dBrn)

SNR = -22 dBm - (41.4 dBrn - 90) = 26.6 dB

FIGURE 59. Common hypothetical 1150 km (621 nmi) switched VF reference circuit for voice service and telegraph and data service of tactical subsystems type I and type III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Table LXII lists the parameter values for the reference circuit shown in Figure 59. The parameter values have been obtained by applying the summation methods of 5.9.3 to the applicable reference trunk parameters listed in Tables LVI and LVIII, the applicable loop parameters given in 5.5.2 and the applicable switch transmission characteristics given in 5.6.2.1.2. Only switch transmission characteristics have been considered for calculating the contributions of the nodes to the reference circuit parameters. The parameter values listed in Table LXII show that the reference circuit of Figure 59 meets the requirements for analog voice service given in 5.2.2.1. However, the reference circuit does not meet the insertion loss versus frequency characteristic given in 5.2.3.1.6.1 and the envelope delay distortion given in 5.2.3.1.6.2. Therefore, the circuit may require equalization or regeneration, or both, to provide acceptable telegraph and data service.

5.9.4.7 Common hypothetical reference circuit of tactical subsystems type II and type III. Figure 60 shows an example of a hypothetical switched VF reference circuit for voice service connecting a VF channel of the tactical subsystem type II (see Table I) with a VF channel of the tactical subsystem type III. The method of interconnecting the VF channels of the two tactical subsystems is based on the standard test tone level stated in 5.8.5.1. The reference circuit shown in Figure 60 consists of a two-wire loop, two TDM/PCM LOS reference trunks and one TDM/PCM tropospheric scatter reference trunk of the tactical subsystem type II, connected in tandem with one FDM LOS/tropospheric scatter reference trunk, two FDM LOS reference trunks and a four-wire loop of the tactical subsystem type III. The two-wire loop is assumed to have a mean insertion loss of 3.5 dB in accordance with 5.5.2.2. The four-wire loop is assumed to have a mean insertion loss of 9 dB in accordance with 5.5.2.1.1. Figure 60 shows the configuration of the reference circuit and a level diagram of the mean speech volume for both directions of transmission. The nominal transmission distance of the reference circuit is 1150 km (621 nmi), not including the length of the loops. The nodes shown in Figure 60 may contain switching equipment and may also contain other equipment, such as equipment for patching, signal processing and technical control (see 4.3.4). The transmitted mean speech volume of -10 VU is in accordance with 5.2.2.1.1. The received mean speech volume of -29.5 VU at the telephone set of the tactical subsystem type III is within the limits given in 5.2.2.1.2, whereas the received mean speech volume of -18.5 VU at the telephone set of the tactical subsystem type II is higher (see Note 2 of 5.2.2.1.2) than the nominal value of -28 VU stated in 5.2.2.1.2.

Table LXIII lists the noise power values contributed by the different components to the total circuit noise and shows how the total circuit noise has been calculated, considering the 3 dB level difference at the interface between the two tactical subsystems. The noise power contributed by each loop to the total circuit noise has been taken as the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXII. Parameters for common hypothetical VF reference circuit of tactical subsystems type I and type III.

Reference Circuit Parameters		Parameter Values for Figure 59
Nominal Transmission Distance (without Loops)		1150 km (621 nmi)
Total Circuit Noise Power for Voice Service	referenced to Circuit OTLP	89,400 pWp0 (49.5 dBrnc0)
	referenced to TLP at Receiving Terminal	11,255 pWp (40.5 dBrnc)
for Telegraph and Data Service	referenced to Circuit OTLP	108,500 pW0 (50.4 dBrn0)
	referenced to TLP at Receiving Terminal	13,660 pW (41.4 dBrn)
Quasi-Analog Data SNR		26.6 dB
Insertion Loss Versus Frequency Characteristic (in dB)* with Loops for Voice Service	f < 400 Hz	> -7.6
	400 Hz ≤ f < 600 Hz	-7.6 to +10.7
	600 Hz ≤ f < 1000 Hz	-6.6 to +10.7
	1000 Hz ≤ f < 2400 Hz	-5.0 to +10.7
	2400 Hz ≤ f < 2800 Hz	-5.0 to +18.8
	2800 Hz ≤ f	≥ -5.0
with Loops for Telegraph and Data Service up to 1200 Bd	f < 300 Hz	> -6.9
	300 Hz ≤ f < 400 Hz	-6.9 to +5.5
	400 Hz ≤ f < 1000 Hz	-5.8 to +4.9
	1000 Hz ≤ f < 2400 Hz	-5.0 to +4.9
	2400 Hz ≤ f < 2700 Hz	-5.0 to +5.8
	2700 Hz ≤ f < 3000 Hz	-5.0 to +8.1
	3000 Hz ≤ f	≥ -5.0
Maximum Envelope Delay Distortion (in microseconds) with Loops for Voice Service		Not standardized (See 5.2.2.1.5)
with Loops for Tel. and Data Service up to 1200 Bd	800 Hz ≤ f < 1000 Hz	3181
	1000 Hz ≤ f < 2400 Hz	1661
	2400 Hz ≤ f ≤ 2600 Hz	1961
Mean of Insertion Loss (in dB)		18
Net Loss Variation (in dB)* Over any 30 consecutive days		±4.1

*dB values rounded off within 0.1 dB.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

maximum allowable noise power of 6300 pWp for analog voice service, measured at the output terminals of the loop in accordance with 5.5.2.6. The noise power contributed by each node to the total circuit noise has been taken as the maximum allowable noise power of an analog switch in accordance with 5.6.2.1.2.5. For voice service, the equivalent PCM noise of each TDM/PCM trunk has been combined with the idle channel noise of each node, the total trunk noise of each FDM trunk and the loop noise, to obtain the total circuit noise at the listener telephone set. (See Note 4 of 5.9.3.2.)

Table LXIV lists the parameter values for the reference circuit shown in Figure 60. The parameter values have been obtained by applying the summation methods of 5.9.3 to the applicable reference trunk parameters listed in Tables LVII and LVIII, the applicable loop parameters given in 5.5.2 and the applicable switch transmission characteristics given in 5.6.2.1.2. Only switch transmission characteristics have been considered for calculating the contributions of the nodes to the reference circuit parameters. The parameter values listed in Table LXIV show that the reference circuit of Figure 60 meets the requirements for analog voice service given in 5.2.2.1. Considering the two directions of transmission shown in Figure 60, the reference circuit is unsymmetrical with respect to circuit mean insertion losses and received mean speech volumes at each terminal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

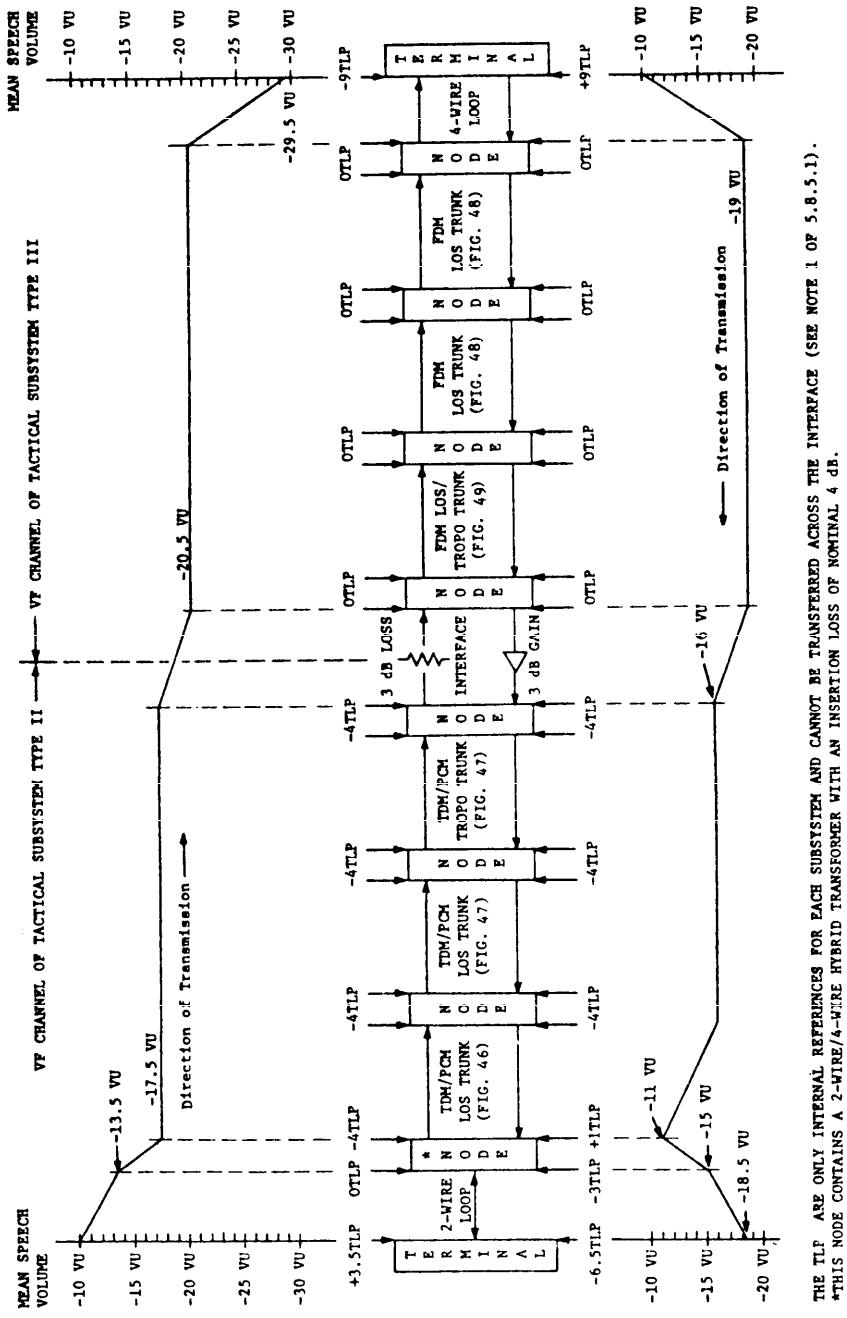


FIGURE 60. Common hypothetical 1150 km (621 nmi) switched VF reference circuit for voice service of tactical subsystems type II and type III

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXIII. Calculations of circuit noise power for reference circuit of Figure 60.

A. Direction of transmission from terminal of tactical subsystem type II to terminal of tactical subsystem type III.		
Item	Maximum Noise Power for Voice Service	Remarks
1A	2,500 pWp*	Loop Noise. (See 5.5.2.6.)
2A	400 pWp*	Idle channel noise of 1st node (See 5.6.2.1.2.5.)
3A	5,000 pWp*	Equivalent PCM noise of 1st TDM/PCM trunk. (See Table LVII.)
4A	400 pWp*	Same as item 2A for 2nd node.
5A	5,000 pWp*	Same as item 3A for 2nd TDM/PCM trunk.
6A	400 pWp*	Same as item 2A for 3rd node.
7A	5,000 pWp*	Same as item 3A for 3rd TDM/PCM trunk.
8A	400 pWp*	Same as item 2A for 4th node.
9A	19,100 pWp* (42.8 dBrnc*)	Total noise contributed by tactical subsystem type II and referenced to -4TLP of this subsystem. (Sum of items 1A through 8A.)
10A	9,570 pWp0 (39.8 dBrnc0)	Noise power of item 9A transfered across interface (with attenuation of 3 dB) and referenced to OTLP of tactical subsystem type III.
11A	400 pWp0	Same as item 2A for 5th node.
12A	1,286 pWp0	Total noise of 1st FDM trunk. (See Table LVIII.)
13A	400 pWp0	Same as item 2A for 6th node.
14A	1,286 pWp0	Same as item 12A for 2nd FDM trunk.
15A	400 pWp0	Same as item 2A for 7th node.
16A	1,286 pWp0	Same as item 12A for 3rd FDM trunk.
17A	400 pWp0	Same as item 2A for 8th node.
18A	50,000 pWp0	Loop noise. (See 5.5.2.6.)
19A	55,458 pWp0 (47.4 dBrnc0)	Total noise contributed by tactical subsystem type III and referenced to OTLP of this subsystem. (Sum of items 11A through 18A.)
20A	65,028 pWp0 (48.1 dBrn0)	Total circuit noise referenced to OTLP of tactical subsystem type III. (Sum of items 10A and 19A.)
21A	8,190 pWp (39.1 dBrnc)	Total circuit noise referenced to -9TLP at receiving terminal of tactical subsystem type III.

*Indicates that noise power is referenced to -4TLP of tactical subsystem type II which has a 3 dB higher level than OTLP of tactical subsystem type III. (See 5.8.5.1.)

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the currentversion before use.

TABLE LXIII. (CONTINUED) Calculations of circuit noise power for reference circuit of Figure 60.

B. Direction of transmission from terminal of tactical subsystem type III to terminal of tactical subsystem type II.		
Item	Maximum Noise Power for Voice Service	Remarks
1B	6,300 pWp0	Loop noise. (See 5.5.2.6.)
2B	400 pWp0	Idle channel noise of 1st node. (See 5.6.2.1.2.5.)
3B	1,286 pWp0	Total noise of 1st FDM trunk. (See Table LVIII.)
4B	400 pWp0	Same as item 2B for 2nd node.
5B	1,286 pWp0	Same as item 3B for 2nd FDM trunk.
6B	400 pWp0	Same as item 2B for 3rd node.
7B	1,286 pWp0	Same as item 3B for 3rd FDM trunk.
8B	400 pWp0	Same as item 2B for 4th node.
9B	11,758 pWp0 (40.7 dBrnc0)	Total noise contributed by tactical subsystem type III and referenced to OTLP of this subsystem. (Sum of items 1B through 8B.)
10B	23,460 pWp* (43.7 dBrnc*)	Noise power of item 9B transferred across interface (with gain of 3 dB) and referenced to -4TLP of tactical subsystem type II.
11B	400 pWp*	Same as item 2B for 5th node.
12B	5,000 pWp*	Same as item 3A for 1st TDM/PCM trunk.
13B	400 pWp*	Same as item 2B for 6th node.
14B	5,000 pWp*	Same as item 3A for 2nd TDM/PCM trunk.
15B	400 pWp*	Same as item 2B for 7th node.
16B	5,000 pWp*	Same as item 3A for 3rd TDM/PCM trunk.
17B	400 pWp*	Same as item 2B for 8th node.
18B	11,200 pWp*	Loop noise. (See 5.5.2.6.)
19B	27,800 pWp* (44.4 dBrnc*)	Total noise contributed by tactical subsystem type II and referenced to -4TLP of this subsystem. (Sum of items 11B through 18B.)
20B	51,260 pWp* (47.1 dBrnc*)	Total circuit noise referenced to -4TLP of tactical subsystem type II. (Sum of items 10B and 19B.)
21B	28,830 pWp (44.6 dBrnc)	Total circuit noise referenced to -6.5TLP at receiving terminal of tactical subsystem type II.

*Indicates that noise power is referenced to -4TLP of tactical subsystem type II which has a 3 dB higher level than OTLP of tactical subsystem type III. (See 5.8.5.1.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXIV. Parameters for common hypothetical VF reference circuit of tactical subsystems type II and type III.

Reference Circuit Parameters		Parameter Values for Figure 60
Nominal Transmission Distance (without Loops)		1150 km (621 nmi)
Total Circuit Noise Power for Voice Service	at Receiving Terminal of Tactical Subsystem Type II	28,830 pWp (44.6 dBrnc)
	at Receiving Terminal of Tactical Subsystem Type III	8,190 pWp (39.1 dBrnc)
Insertion Loss Versus Frequency Characteristic (in dB)* with Loops for Voice Service	f < 400 Hz	≥ -6.9
	400 Hz ≤ f < 600 Hz	-6.9 to +10.7
	600 Hz ≤ f < 1000 Hz	-5.8 to +10.7
	1000 Hz ≤ f < 2400 Hz	-4.0 to +10.7
	2400 Hz ≤ f < 2800 Hz	-4.0 to +18.8
	2800 Hz ≤ f	≥ -4.0
Maximum Envelope Delay Distortion with Loops for Voice Service		Not standardized (see 5.2.2.1.5)
Mean of Insertion Loss for Direction of Transmission (see Figure 60)	from left to right:	19.5 dB
	from right to left:	8.5 dB
Net Loss Variation (in dB)* Over any 30 consecutive days		±4.1

*dB values rounded off within 0.1 dB.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CUSTODIANS:

ARMY CR
NAVY - EC
AIR FORCE - 90

PREPARING ACTIVITY:

ARMY - CR

REVIEW ACTIVITIES:

ARMY - SC, CR, AV, MI, AD, TE, CU
NAVY - AS, EC, SH, OM, NC
AIR FORCE - 02, 11, 13, 17, 71, 80, 90
DCA - DC
NSA - NS
TRI-TAC - TT

USER ACTIVITIES:

ARMY
NAVY
AIR FORCE

OTHER INTERESTS:

JCS
DODECAC
NCS

PROJECT NO.:

TCTS-2000

INTERNATIONAL INTERESTS:

NATO
ABCA Armies

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX A

MEMORANDUM FROM THE ASSISTANT SECRETARY OF DEFENSE FOR COMMUNICATIONS, COMMAND, CONTROL AND INTELLIGENCE, 10 MAY 1977, SUBJECT: MANDATORY USE OF MILITARY STANDARDS IN THE 188 SERIES.

This Appendix contains information related to MIL-STD-188-200. Appendix A is a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

ASSISTANT SECRETARY OF DEFENSE
WASHINGTON, D. C. 20301

COMMUNICATIONS, COMMAND,
CONTROL, AND INTELLIGENCE

10 MAY 1977

MEMORANDUM FOR Assistant Secretary of the Army (I&L)
Assistant Secretary of the Navy (I&L)
Assistant Secretary of the Air Force (I&L)
Commandant of the Marine Corps
Director, Defense Communications Agency
Director, National Security Agency

SUBJECT: Mandatory use of military standards in the 188 Series

On January 3, 1972, the Assistant Secretary of Defense (I&L) found it necessary to make a significant change in the DoD Standardization Manual 4120.3M because of recurring misapplications of military standards in general. The essence of the change is that military standards as a general rule are now cited as "approved for use" rather than "mandatory for use" in the Department of Defense.

This deference to the judgment of the designing and procuring agencies is clearly appropriate to standards dealing with process, component ruggedness and reliability, paint finishes, and the like. It is clearly not appropriate to standards such as those in the MILSTD 188 series which address telecommunication design parameters. These influence the functional integrity of telecommunication systems and their ability to efficiently interoperate with other functionally similar Government and commercial systems. Therefore, relevant military standards in the 188 series will continue to be mandatory for use within the Department of Defense.

To minimize the probability of misapplication of these standards, it is incumbent upon the developers of the MILSTD 188 series to insure that each standard is not only essential but of uniformly high quality, clear and concise as to application, and wherever possible compatible with existing or proposed national, international and Federal telecommunication standards. It is also incumbent upon the users of these standards to cite in their procurement specifications only those standards which are clearly necessary to the proper functioning of the device or systems over its projected lifetime.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

This direction is in consonance with the recommendations made by the Director, Defense Materiel Specifications and Standards Office, in his letter of March 4, 1977, which was addressed to the U.S. Army Electronics Command.

Gerald P. Dinneen

cc:
OASD(MRA&L), DMSSO
Co-Chairman, JSC, ECOM (DRSEL-CE-CS)
Co-Chairman, JSC, DCEC (R110)
OJCS (J-3), Telecom Div

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX B

LIST OF TACTICAL AND COMMON LONG HAUL/TACTICAL STANDARDIZATION PROJECTS OF THE MIL-STD-188 SERIES.

This Appendix contains general information related to MIL-STD-188-200. Appendix B is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

20 List of tactical and common long haul/tactical standardization projects of the MIL-STD-188 series.

20.1 List of tactical standardization projects.

MIL-STD-188-203-2, Subsystem Design and Engineering Standards for Tactical Digital Information Link (TADIL) B.
Preparing Activity: Army.
Estimated Completion Date: 3rd Quarter of 1983.

MIL-STD-188-203-3, Subsystem Design and Engineering Standards for Tactical Digital Information Link (TADIL) C.
Preparing Activity: Navy.
Estimated Completion Date: 2nd Quarter of 1983.

MIL-STD-188-242, Design and Engineering Standards for Tactical Single Channel VHF Radio Equipment.
Preparing Activity: Army.
Estimated Completion Date: 1st Quarter of 1984.

MIL-STD-188-243, Design and Engineering Standards for Tactical Single Channel UHF Radio Equipment.
Preparing Activity: Air Force.
Estimated Completion Date: 3rd Quarter of 1984.

MIL-STD-188-256, Design and Engineering Standards for Digital Signaling and Supervision of Tactical Communications Systems.
Preparing Activity: Army
Estimated Completion Date: 4th Quarter of 1983.

MIL-STD-188-260, Design and Engineering Standards for Tactical Analog and Digital Terminal Equipment.
Preparing Activity: Army.
Estimated Completion Date: 4th Quarter of 1983.

20.2 List of common long haul/tactical standardization projects.

MIL-STD-188-102, Equipment Technical Design Standards for Common Long Haul/Tactical Time Division Multiplexers.

NOTE: This project will be initiated when sufficient data for developing a standard is available.

MIL-STD-188-111, Subsystem Design and Engineering Standards for Common Long Haul/Tactical Fiber Optics Communications.
Preparing Activity: Navy.
Estimated Completion Date: 4th Quarter of 1983.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

MIL-STD-188-113, Common Long Haul/Tactical Standards for Analog/Digital Conversion Techniques.
Preparing Activity: Army.
Estimated Completion Date: 1st Quarter of 1984.

MIL-STD-188-115, Subsystem Design and Engineering Standards for Common Long Haul/Tactical Communications Timing and Synchronization.
Preparing Activity: Joint Tactical Communications Office (TRI-TAC).
Estimated Completion Date: 2nd Quarter of 1984.

MIL-STD-188-141, Subsystem Design and Engineering Standards for Common Long Haul/Tactical Radio Communications in the Medium Frequency (MF) and High Frequency (HF) Bands.
Preparing Activity: Army.
Estimated Completion Date: 3rd Quarter of 1983.

MIL-STD-188-144, Subsystem Design and Engineering Standards for Common Long Haul/Tactical Tropospheric Scatter Radio Communications.
Preparing Activity: Air Force.
Estimated Completion Date: To be determined.

MIL-STD-188-145, Subsystem Design and Engineering Standards for Common Long Haul/Tactical LOS Microwave Radio Communications.
Preparing Activity: Army.
Estimated Completion Date: 2nd Quarter of 1984.

MIL-STD-188-146, Subsystem Design and Engineering Standards for Common Long Haul/Tactical Satellite Communications.
Preparing Activity: Army.
Estimated Completion Date: To be determined.

MIL-STD-188-154, Subsystem Design and Engineering Standards for Common Long Haul/Tactical Systems and Technical Controls.
Preparing Activity: Air Force.
Estimated Completion Date: 4th Quarter of 1984.

MIL-STD-188-190, Methods of Measurement for Common Long Haul/Tactical Communications.
Preparing Activity: Air Force.
Estimated Completion Date: 3rd Quarter of 1984.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX C

LIST OF ABBREVIATIONS AND ACRONYMS USED IN MIL-STD-188-200.

This Appendix contains general information in support of MIL-STD-188-200. Appendix C is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

30 List of abbreviations and acronyms used in MIL-STD-188-200.

ABCA	American, British, Canadian, Australian (Armies)
ACP	Allied Communication Publication(s)
A-D	analog-to-digital
ADP	automatic data processing
AM	amplitude modulation
AMI	alternate mark inversion
ANSI	American National Standards Institute
ARQ	automatic-repeat-request
ASCII	American standard code for information interchange
AUTODIN	automatic digital network (of the DCS)
AUTOVON	automatic voice network (of the DCS)
Bd	baud(s)
BER	bit error rate
bit	binary digit
BITE	built-in test equipment
b/s	bit(s) per second
CCIR	International Radio Consultative Committee (part of the ITU)
CCITT	International Telegraph and Telephone Consultative Committee (part of the ITU)
CMI	coded mark inversion
COMSEC	communications security
cpi	character(s) per inch
cpmm	character(s) per millimeter
CVSD	continuously variable slope delta
D-A	digital-to-analog
dB	decibel(s)
dBa	dB, adjusted
dBa0	dBa, referenced to OTLP
dBm	dB, referred to 1 mW
dBm0	dBm, referenced to OTLP
dBrn	dB above reference noise
dBrn(144)	dBrn, 144-weighted

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

dBrnc	dBrn, C-message weighted
dBrnc0	dBrnc, referenced to OTLP
dBrn0	dBrn, referenced to OTLP
dc	direct current
DCA	Defense Communications Agency
DCAC	DCA circular
DCE	data circuit terminating equipment
DCS	Defense Communications System
DoD	Department of Defense
DoDD	DoD directive
DPSK	differential phase shift keying (or keyed)
DTE	data terminal equipment
DTMF	dual-tone multifrequency
ECCM	electronic counter-countermeasure
EDC	error detection and correction
EHF	extremely high frequency, (30 GHz to 300 GHz)
ELF	extremely low frequency, (below 300 Hz)
EMC	electromagnetic compatibility
EMI	electromagnetic interference
EMP	electromagnetic pulse
EMSEC	emanations security
EW	electronic warfare
FDM	frequency division multiplex(ing)
FED-STD	Federal Standard(s)
FIPS	Federal Information Processing Standard(s)
FIPS PUB	Federal Information Processing Standard Publication(s)
FSK	frequency shift keying (or keyed)
ft	foot or feet
GDF	group distribution frame(s)
GHz	gigahertz, (1 GHz = 1000 MHz)
HF	high frequency, (3000 kHz to 30 MHz)
HFDF	high frequency distribution frame(s)
Hz	hertz, (1 Hz = 1 cycle per second)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

IEEE	Institute of Electrical and Electronics Engineers
IF	intermediate frequency
IFF	identification friend or foe
ILD	injection laser diode
ILF	infra low frequency, (300 Hz to 3000 Hz)
IMD	intermodulation distortion
in	inch(es)
IRAC	Interdepartment Radio Advisory Committee
ISB	independent sideband
ISO	International Standardization Organization
ITA	international telegraph alphabet
ITU	International Telecommunication Union
JANAP	Joint Army-Navy-Air Force publication(s)
JCS	Joint Chiefs of Staff
JCS PUB	JCS publication(s)
JTIDS	joint tactical information distribution system
kBd	kilobaud(s), (1 kBd = 1000 Bd)
kb/s	kilobit(s) per second, (1 kb/s = 1000 b/s)
kHz	kilohertz, (1 kHz = 1000 Hz)
km	kilometer(s), (1 km = 1000 m)
LED	light emitting diode
LF	low frequency, (30 kHz to 300 kHz)
ln	natural logarithm (to base of e = 2.718...)
log	logarithm (to base of 10)
LOS	line-of-sight
LPC	linear predictive coding
m	meter(s), (1 m = 1000 mm)
m-ary	other than binary; e.g., ternary, quarternary, etc.
Mb/s	megabit(s) per second, (1 Mb/s = 1000 kb/s)
MCEB	Military Communications-Electronics Board
MF	medium frequency, (300 kHz to 3000 kHz)
MHz	megahertz, (1 MHz = 1000 kHz)
MIL-HDBK	military handbook
MIL-STD	military standard
min	minute(s)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

mm	millimeter(s), (1 mm = 1/1000 m)
Modem	modulator/demodulator
ms	millisecond(s), (1 ms = 1/1000 s)
mW	milliwatt(s), (1 mW = 1/1000 W)
MTBF	mean time between failures
MTBPM	mean time between preventive maintenance
MTTR	mean time to repair
NACSEM	national COMSEC/EMSEC information memorandum
NACSIM	national COMSEC information memorandum
NATO	North Atlantic Treaty Organization
NBS	National Bureau of Standards
NICS	NATO integrated communications system
NMCS	national military command system
nmi	nautical mile(s), (1 nmi = 1852 m)
Np	neper(s)
NRI	net radio interface
NRZ	non-return-to-zero (of digital waveform)
NRZI	non-return-to-zero inverted (See Notes 2 and 3 of 5.3.5.2)
OCRE	optical character recognition equipment
PAM	pulse amplitude modulation (or modulator)
PBX	private branch exchange
PCM	pulse code modulation
PE	phase encoded
PRSL	primary area switch locator
PSK	phase shift keying (or keyed)
pW	picowatt(s), (1 pW = 10^{-12} W)
pW0	pW, referenced to OTLP
pWp	pW, psophometrically weighted
pWp0	pWp, referenced to OTLP
QPSK	quadrature phase shift keying (or keyed)
QSTAG	quadripartite standardization agreement(s) (of the Armies of the United States, United Kingdom, Australia and the Canadian Forces; ABCA Armies)
rf	radio frequency

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

RFI	radio frequency interference
rms	root-mean-square
RWI	radio and wire integration
RZ	return-to-zero (of digital waveform)
s	second(s), (1 s = 1000 ms)
SF	single-frequency
S-F	store-and-forward
SGDF	supergroup distribution frame(s)
SHF	super high frequency, (3000 MHz to 30 GHz)
SNR	signal-to-noise ratio
SSB	single sideband
STANAG	standardization agreement(s) (of the NATO)
TADIL	tactical digital information link(s)
TCF	technical control facility
TDM	time division multiplex(ing)
TEMPEST	(see 4.2.1)
THF	tremendously high frequency, (300 GHz to 3000 GHz)
TLP	transmission level point(s)
TROPO	tropospheric scatter
UHF	ultra high frequency, (300 MHz to 3000 MHz)
USASCII	USA standard code for information interchange (same as ASCII)
VF	voice frequency
VFCT	voice frequency carrier telegraph
VFDF	voice frequency distribution frame(s)
VHF	very high frequency, (30 MHz to 300 MHz)
VLF	very low frequency, (3000 Hz to 30 kHz)
VU	volume unit(s)
W	watt(s)
WARC	World Administrative Radio Conference
wpm	word(s) per minute
OTLP	zero transmission level point(s)
µs	microsecond(s), (1 µs = 1/1000 ms)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

where P_{rms} = rms power of the multichannel signal in dBm0

APPENDIX D

MODULATION CHARACTERISTICS FOR CONTINUOUSLY VARIABLE SLOPE DELTA (CVSD) AND LINEAR PREDICTIVE CODING (LPC).

This Appendix contains information related to MIL-STD-188-200. Appendix D is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph	Title	Page
40	Digital voice conversion techniques	D-284
40.1	CVSD modulation characteristics	D-284
40.1.1	General description of CVSD converter	D-284
40.1.2	Common CVSD encoder/decoder characteristics	D-286
40.1.2.1	Data signaling rates	D-286
40.1.2.2	Clock characteristics	D-286
40.1.2.3	Three-bit shift register	D-286
40.1.2.4	Overload logic	D-286
40.1.2.5	Syllabic filter	D-286
40.1.2.6	Pulse amplitude modulator (PAM)	D-287
40.1.2.7	Reconstruction integrator	D-287
40.1.3	CVSD encoder characteristics	D-287
40.1.3.1	Comparator	D-287
40.1.3.2	Input filter	D-287
40.1.4	CVSD decoder characteristics	D-287
40.1.4.1	Output filter	D-287
40.1.4.2	Reference digital patterns	D-288
40.1.4.3	Decoder performance	D-288
40.1.5	CVSD converter characteristics	D-288
40.1.5.1	Test signal levels	D-288
40.1.5.2	Idle channel noise	D-288

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

CONTENTS (CONTINUED)

Paragraph	Title	Page
40.1.5.3	Quantizing noise ratio	D-288
40.1.5.4	Input/output linearity	D-288
40.2	Linear predictive coding (LPC) modulation characteristics	D-290
40.2.1	LPC encoder process	D-290
40.2.2	LPC decoder process	D-292

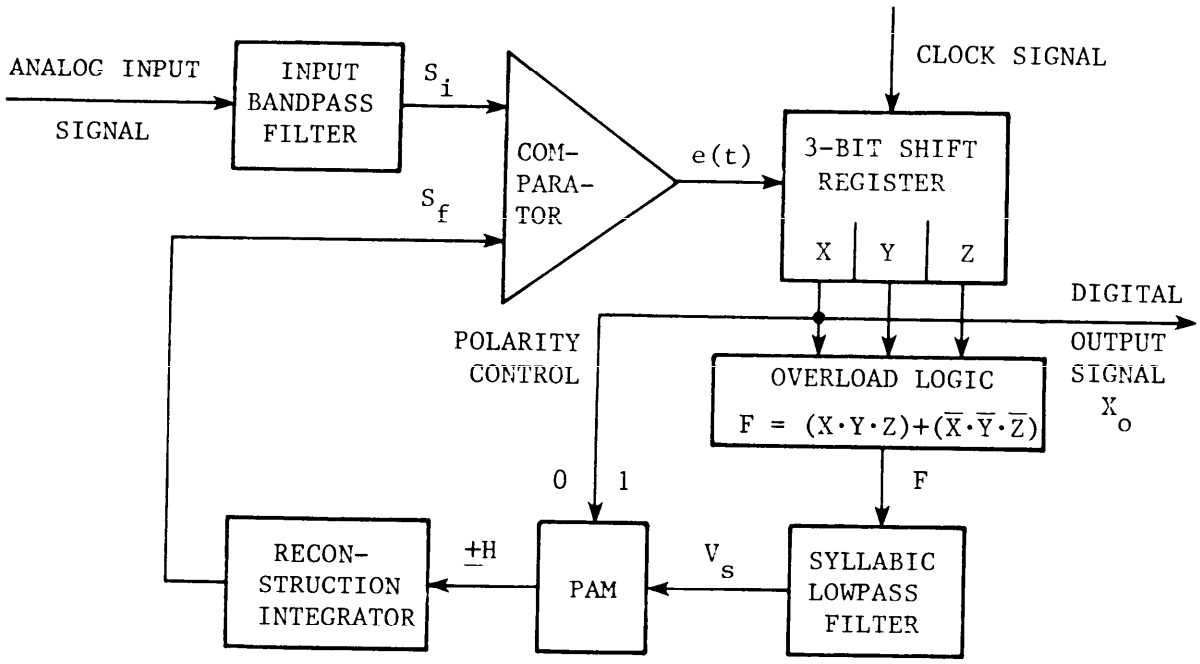
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

40 Digital voice conversion techniques.

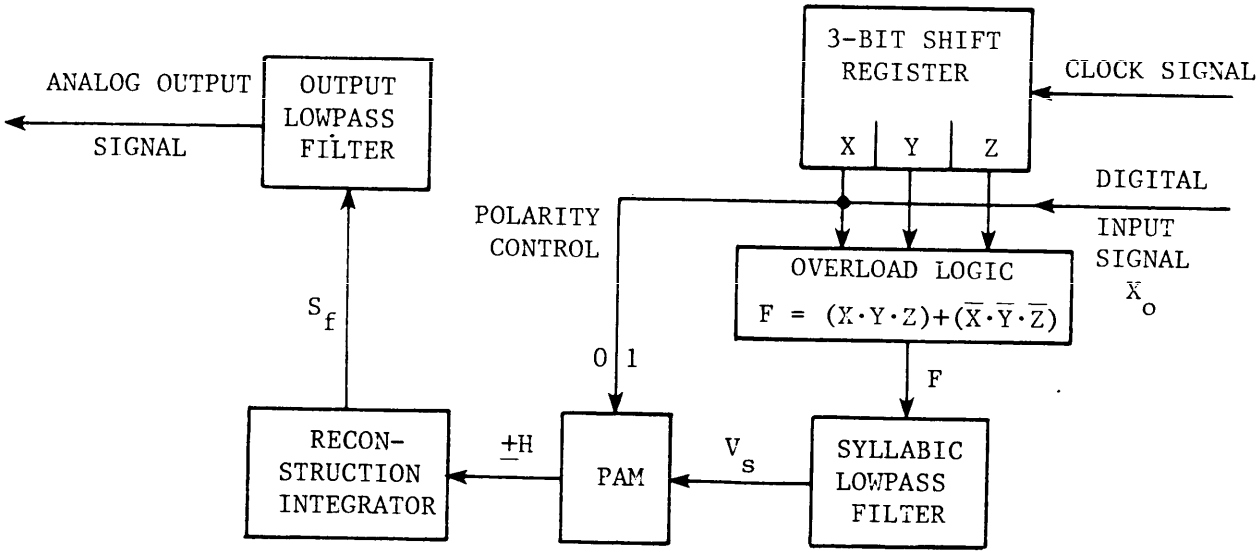
40.1 CVSD modulation characteristics. The A-D/D-A conversion technique that will be used in digital wideband equipment of the tactical subsystem type IV (see Table I) is CVSD modulation. CVSD modulation is a non-linear, sampled data, feedback system which accepts a band-limited analog signal and encodes it into binary form for transmission through a digital channel. At the receiver, the binary signal is decoded into a close approximation of the original analog signal. The CVSD converter consists of an encoder and a decoder as shown in Figure 61 and described in 40.1.1 through 40.1.4.

40.1.1 General description of CVSD converter. (See Figure 61.) The CVSD converter consists basically of an encoder and a decoder. The analog input signal of the CVSD encoder is band-limited by the input bandpass filter. The CVSD encoder compares the band-limited analog input signal S_i with the analog feedback approximation signal S_f, which is generated at the reconstruction integrator output. The digital output signal X_o of the encoder is the output of the first register in the "run-of-three" counter. The digital output signal is transmitted at the clock (sample) rate and will equal "1" if $S_i \geq S_f$ at the instant of sampling. For this value of X_o, the pulse amplitude modulator (PAM) applies a positive feedback pulse (+H) to the reconstruction integrator; otherwise, a negative pulse (-H) is applied. This function is accomplished by the polarity control signal, which is equal to the digital encoder output X_o. The amplitude of the feedback pulse is derived by means of a three-bit shift register , logic sensing for overload, and a syllabic lowpass filter. When a string of three consecutive ONES or ZEROS appears at the output X_o, a discrete voltage level F is applied to the syllabic filter and the feedback pulse amplitude (+H) increases until the overload string is broken. In such an event, ground potential is fed to the filter by the overload logic, forcing a decrease in the amplitude of the slope voltage V_s. The encoder and decoder have identical characteristics, except for the comparator and filter functions. The CVSD decoder consists of the shift register, overload logic, syllabic filter, PAM and reconstruction integrator used in the encoder, followed by a 4 kHz lowpass filter. The decoder performs the inverse function of the encoder and regenerates speech by passing the output signal S_f of the reconstruction integrator through the lowpass filter. Other characteristics optimize the CVSD modulation technique for voice signals. These characteristics include: Changes in the slope of the analog input signal determine the step size changes of the digital output signal; the feedback loop is adaptive to the extent that the loop

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



A. CVSD Encoder.



B. CVSD Decoder.

FIGURE 61. Block diagram of CVSD encoder/decoder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

provides continuous or smoothly incremental changes in step size; companding is performed at a syllabic rate, to extend the dynamic range of the analog input signal; and the reconstruction integrator of the CVSD converter is of the exponential (leaky) type to reduce the effects of digital errors.

40.1.2 Common CVSD encoder/decoder characteristics.

40.1.2.1 Data signaling rates. The digital output of the encoder and the digital input of the decoder operate at 16 kb/s or 32 kb/s.

40.1.2.2 Clock characteristics. The clock of the CVSD converter will comply with the applicable requirements of 5.3.6.2.

40.1.2.3 Three-bit shift register. (See Figure 61.) The three-bit shift register acts as a sampler which clocks the digital error signal e(t) at the specified data signaling rate and stores the current and two previous samples of the error signal. The output signal X_o is a binary signal having the same polarity as the input signal e(t) at the time of the clock signal. The output signal X_o is also the digital output of the encoder and is referred to as the baseband signal. Further processing for transmission, such as conditioned diphase modulation (see 5.4.1.4.3), may be applied to the baseband signal. It is required that the inverse of any such processing be accomplished and the baseband signal restored before the CVSD decoding process is attempted.

40.1.2.4 Overload logic. (See Figure 61.) The overload logic operates on the output of the three-bit shift register (X, Y, Z) using the run-of-three coincidence algorithm, so that the algorithm output is $F = (X \cdot Y \cdot Z) + (\overline{X} \cdot \overline{Y} \cdot \overline{Z})$. The output signal F is a binary signal at the clock signaling rate and is TRUE for one clock period following the detection of three like bits, and FALSE at all other times.

40.1.2.5 Syllabic filter. (See Figure 61.) The syllabic filter acts as a lowpass filter, with a time constant of 5 ms, ±1 ms, for the output signal F from the overload logic. The output of the syllabic filter, shown as the slope voltage V_s in Figure 61, is the modulating input to the PAM. The step function response of the syllabic filter is related to the syllabic rate of speech, independent of the sampling rate, and is exponential in nature. When F is TRUE, a charging curve is applicable. When F is FALSE, a discharging curve is applicable. The compression ratio, defined as $V_{s\ max}$ divided by $V_{s\ min}$ has a nominal value of 15:1 with an allowable maximum value of 20:1 and an allowable minimum value of 12:1. The maximum slope voltage $V_{s\ max}$ is measured at the output of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

syllabic filter when the digital output of the encoder, or the digital input of the decoder, is a 50 percent run-of-threes digital pattern. The minimum slope voltage $V_{s\ min}$ is measured at the output of the syllabic filter when the digital output of the encoder, or the digital input of the decoder, is an idle pattern that does not contain any run-of-threes digital pattern.

40.1.2.6 Pulse amplitude modulator (PAM). (See Figure 61.) The PAM operates with two input signals, the signal V_s from the syllabic filter and the digital signal X_0 from the three-bit shift register. The signal V_s determines the amplitude of the PAM output signal and the signal X_0 is the polarity control that determines the direction, plus or minus, of the PAM output signal ±H. The name CVSD of the converter is derived from the way the PAM output signal varies almost continuously.

40.1.2.7 Reconstruction integrator. (See Figure 61.) The reconstruction integrator should have a time constant between 0.75 ms and 1.0 ms. The integrator operates on the output signal of the PAM to produce an analog feedback signal approximation S_f of the analog input signal S_i.

40.1.3 CVSD encoder characteristics.

40.1.3.1 Comparator. (See Figure 61A.) The comparator compares the band-limited analog input signal S_i with the output signal S_f of the reconstruction integrator. This comparison produces the digital error signal e(t). The transfer characteristic of the comparator is such that the difference between the two input signals (S_i-S_f) causes the output signal e(t) to be driven to saturation in the direction of the sign of that difference.

40.1.3.2 Input filter. (See Figure 61A.) The input filter is a bandpass filter whose frequency response should be 3 dB down at approximately 50 Hz and approximately 4 kHz. The frequency response should have an asymptotic rolloff from approximately 50 Hz with a minimum slope of 6 dB/octave, and from approximately 4 kHz with a minimum slope of 12 dB/octave. The input filter should be a second-order or higher-order filter. The same input filter characteristic is used for encoder digital output signals of either 16 kb/s or 32 kb/s.

40.1.4 CVSD decoder characteristics.

40.1.4.1 Output filter. (See Figure 61B.) The output filter is a lowpass filter whose frequency response should be 3 dB down at approximately 4 kHz. The frequency response should have an asymptotic rolloff from approximately 4 kHz with a minimum slope of 40 dB/octave.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

The stopband rejection should be a minimum of 45 dB. The same output filter characteristic is used for encoder digital output signals of either 16 kb/s or 32 kb/s.

40.1.4.2 Reference digital patterns. The following repeating digital patterns represent the input signal, at the data signaling rates indicated, to be applied to the digital input of the decoder for verifying the CVSD decoder performance specified in 40.1.4.3.

32 kb/s: 1111111011010101000010000001001010101111 0

16 kb/s: 1111101101000001001 0

40.1.4.3 Decoder performance. (See Figure 62.) With the applicable reference digital pattern (see 40.1.4.2) applied to the digital input of the decoder (interface point C), the analog output signal should be 800 Hz, ±1.3 percent at a level of -17 dBm, ±1 dB, when measured at the interface point D terminated at 600 ohms.

40.1.5 CVSD converter characteristics. The performance characteristics of the CVSD converter, stated in 40.1.5.1 through 40.1.5.4, apply to one CVSD conversion process obtained by connecting an encoder with a decoder back-to-back with interface points as shown in Figure 62. No bit errors should be introduced between interface points B and C of Figure 62.

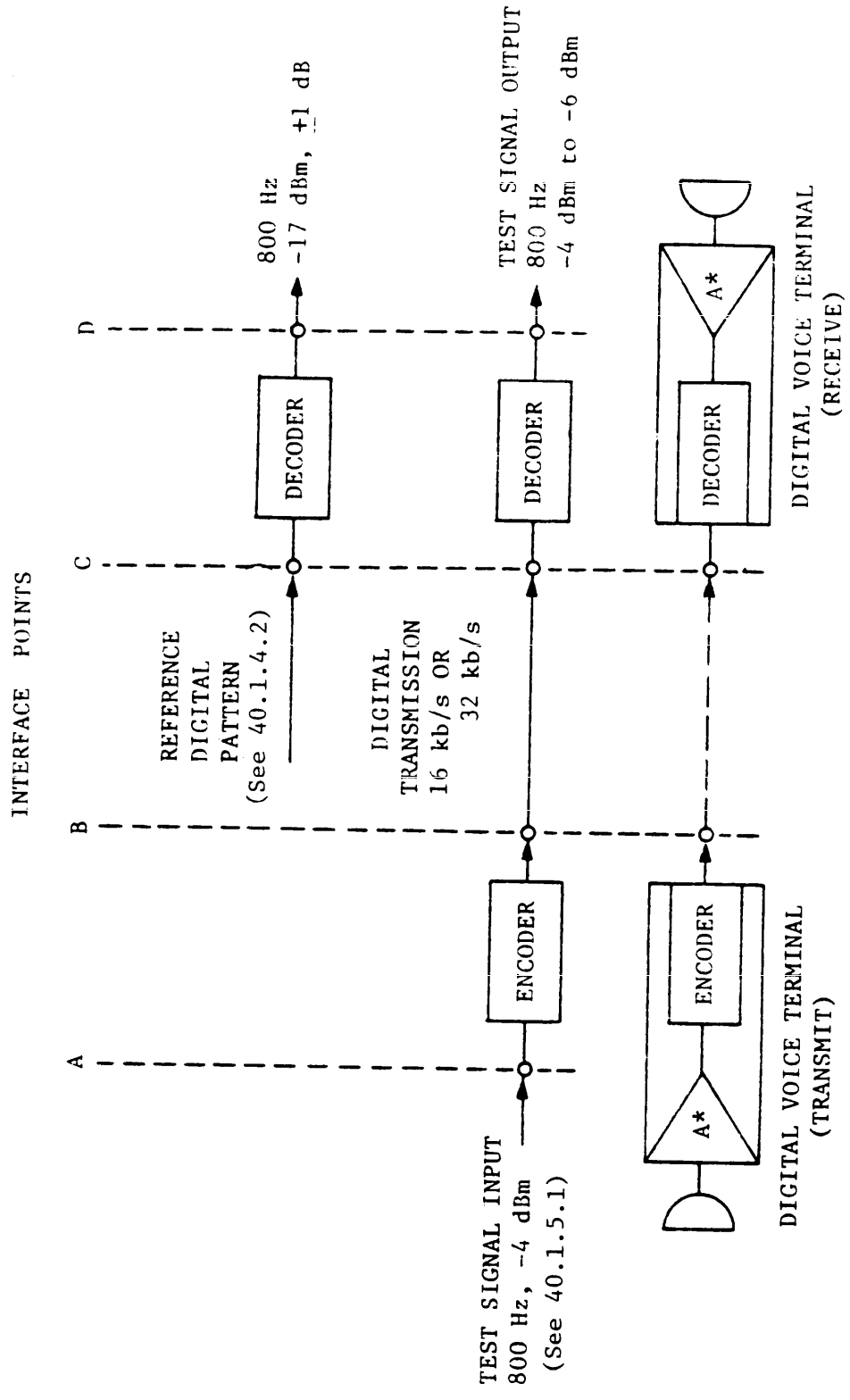
40.1.5.1 Test signal levels. (See Figure 62.) When a test signal with a level of -4 dBm and a nominal frequency of 800 Hz is connected to the encoder input at interface point A, the test signal level at the decoder output at interface point D should be between -4 dBm and -6 dBm. For this measurement, both the encoder input and the decoder output are terminated in 600 ohms.

40.1.5.2 Idle channel noise. The idle channel noise should not exceed 2000 pWp (33.0 dBrnc) when measured at the CVSD decoder output with both the encoder input and the decoder output terminated in 600 ohms.

40.1.5.3 Quantizing noise ratio. The CVSD converter should produce a minimum test signal-to-quantizing plus idle channel noise ratio over the input signal range as shown in Table LXV. The noise ratio should be measured with flat weighting at the decoder output, with a nominal 800 Hz test signal at the encoder input and with both the encoder input and the decoder output terminated in 600 ohms.

40.1.5.4 Input/output linearity. The input versus output level characteristic of the CVSD converter should be linear within ±0.5 dB for a range of input signals from +2 dBm to -38 dBm. This measurement should be made with a nominal 800 Hz test signal.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



*Attenuation or amplification as required to satisfy end-to-end signal levels.

FIGURE 62. Signal level and interface diagram for CVSD encoder/decoder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXV. Quantizing noise ratio for CVSD converter.

Input Signal in dBm	Minimum Test Signal-to-Quantizing Plus Idle Channel Noise Ratio in dB	
	32 kb/s	16 kb/s
+2 to -3	22	15
-4 to -8	25	14
-9 to -13	26	14
-14 to -18	24	14
-19 to -23	22	14
-24 to -28	21	14
-29 to -33	19	13
-34 to -39	16	13

40.2 Linear predictive coding (LPC) modulation characteristics. The A-D/D-A conversion technique to be used in narrowband digital voice equipment will be LPC. Figure 63 shows a block diagram of an LPC encoder/decoder. The encoder converts the analog speech input signal into a digital output signal; the decoder performs the inverse operation of the encoder. The digital output signal of the encoder may be used for transmission over a narrowband channel, or the digital signal may be converted to a quasi-analog signal and transmitted over a VF channel.

40.2.1 LPC encoder process. The LPC method of digital voice coding is by analysis. The encoder consists of a low-pass filter, A/D converter, analyzer and multiplexer. The A/D converter changes the speech signal into a digital format suitable for processing by the analyzer. The analyzer is a digital processor which extracts vocal source information, such as the fundamental frequency or pitch, the voiced/unvoiced (V/UV) decision, and an amplitude information (A) which is a measure of the speech energy from the input speech signal. (A) is used to set the gain at the decoder. The analyzer also contains an N-th order linear predictor

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

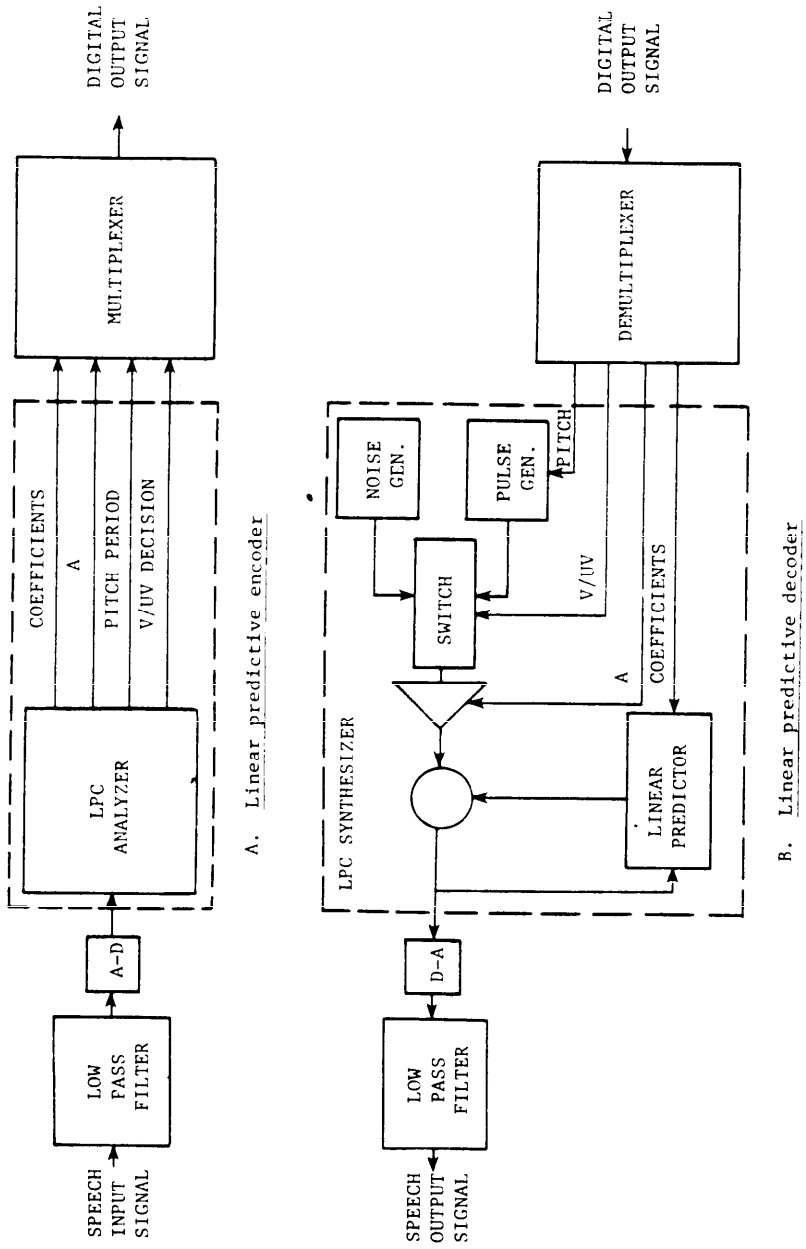


FIGURE 63. Block diagram of linear predictive encoder/decoder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

(non-recursive filter) whose coefficients are optimized to produce a minimum mean-squared difference between the actual and predicted values of the speech signal over finite analysis intervals. The output of the analyzer consists of time quantized parameters. These parameters are predictor coefficients, rms value of speech energy, pitch and the V/UV decision. The multiplexer formats the speech signal parameters into successive frames to form a serial binary digital output signal.

40.2.2 LPC decoder process. The linear predictive decoding of digital voice is performed by a demultiplexer, synthesizer, D/A converter and low-pass output filter. The demultiplexer performs the inverse operation of the encoder multiplexer. The synthesizer processes an excitation signal through an optimum linear predictor which is the identical inverse of the linear predictor at the analyzer. The coefficients of the predictor correspond to those generated by the analyzer. Depending on the voicing decision, a series of white noise samples or pitch pulses comprise the excitation signal. The spacing of the pitch pulses corresponds to pitch intervals in the original speech signal. Gain (A) is applied to the excitation signal at the synthesizer in order that the synthetic speech energy corresponds to the energy in the original speech signal. The D/A converter performs an inverse operation of the A/D converter. The low-pass filter removes high-frequency quantization noise components from the reconstructed speech output signal.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX E

CODING AND MODULATION TECHNIQUES FOR BINARY DIGITAL SIGNALS.

This Appendix contains general information in support of MIL-STD-188-200. Appendix E is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph	Title	Page
50	Coding and modulation techniques for binary digital signals	E-295
50.1	General	E-295
50.2	Binary state coding techniques	E-296
50.2.1	State symmetric coding	E-296
50.2.1.1	Level coding	E-296
50.2.1.2	Phase coding	E-298
50.2.1.3	Pulse duration coding	E-298
50.2.2	State unsymmetric coding	E-299
50.2.2.1	Unbalanced coding	E-299
50.2.2.2	Balanced coding	E-299
50.2.2.3	Conditioned coding	E-301
50.3	Binary differential coding techniques	E-301
50.3.1	Differential symmetric coding	E-301
50.3.1.1	Level coding	E-302
50.3.1.2	Phase coding	E-304
50.3.1.3	Pulse duration coding	E-304
50.3.2	Differential unsymmetric coding	E-305
50.3.2.1	Unbalanced coding	E-305
50.3.2.2	Balanced coding	E-305
50.3.2.3	Conditioned coding	E-306

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

50 Coding and modulation techniques for binary digital signals.

50.1 General. Digital symbols may be represented in many different ways by electrical signals to facilitate data transmission. All these methods for representing (or coding) digital symbols assign electrical parameter values to the digital symbols. In binary coding these digital symbols are restricted to two states, i.e., Space (or 0) and Mark (or 1). The electrical parameters used to code digital symbols are levels (or amplitudes), transitions between different levels, phases (normally 0 degree and 180 degrees for binary coding) and pulse durations (or frequencies), or a combination of these parameters. There are a variety of coding techniques for different areas of application and no particular technique has been found to be optimum for all applications, considering such factors as implementing the coding technique in hardware, type of transmission facility employed, decoding methods at the data sink and timing and synchronization requirements.

One method to classify digital signals would be to distinguish between conditioned and unconditioned signals. A signal that has been conditioned has the property that the binary information contained in the signal remains unchanged even though the polarity of the received signal may have been reversed by the transmission process. Therefore, conditioned signals are useful for radio transmission where polarity reversals may occur. (See 5.4.4.2.3.1.) If a conditioned diphase modulated signal is demodulated by comparing the signal phases in two adjacent unit intervals, there is a one bit error extension inherent in this demodulation process. Digital signals may also be divided into synchronous and asynchronous signals. Synchronous and asynchronous signals should not be confused with synchronous transmission and asynchronous transmission. (See FED-STD-1037 for definitions.) A digital signal is said to be synchronous if, and only if, all pulses have equal durations, all pulses occupy the entire unit interval and there is no time separation between the pulses. A synchronous signal, such as the polar NRZ signal shown in Figure 64A1, requires timing coordination between the data source (generator) and the data sink (receiver), otherwise the receiver would be unable to separate a sequence of Space (0) or Mark (1) bits. An asynchronous signal, such as the polar RZ signal shown in Figure 64A2, has a definite separation between pulses and does not require a timing signal for correct decoding at the data sink. Signals having a definite separation between pulses are also called self-clocking signals since it is possible to extract timing information from the received data bit stream. The separation between pulses may be obtained by RZ coding, as shown in Figure 64A2, or by other coding techniques, such as those resulting in transitions between every unit interval, as shown in Figures 64C1 and 64C2.

In this appendix an attempt has been made to establish a new classification for the different types of coding techniques and to show

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

examples of waveforms resulting from this classification. The examples have been restricted to binary and pseudo-ternary waveforms of digital signals. Waveforms of other signals, such as quasi-analog FSK or PSK signals, are not included. The waveforms shown in Figures 64 through 67 are considered to be typical examples and are not intended to cover all possible variations of digital coding techniques.

50.2 Binary state coding techniques. In state coding of binary signals the value of the electrical parameter assigned to a binary symbol depends only on the binary symbol to be coded in the unit interval under consideration. Binary state coding techniques are subdivided into state symmetric coding (see 50.2.1) where both binary symbols are used in the coding process, and state unsymmetric coding (see 50.2.2) where only one binary symbol is used in the coding process.

50.2.1 State symmetric coding. In state symmetric coding of binary signals both binary symbols are used in the coding process and every unit interval of the electrical signal contains energy. One binary symbol is represented by one value of the selected electrical parameter (level, phase or transition, or pulse duration) and the other binary symbol is represented by the other value of the same electrical parameter. Normally, state symmetric coding techniques employ polar waveforms with positive and negative voltage levels having the same absolute amplitude and, therefore, are balanced with respect to the zero voltage axis. State symmetric coding techniques are subdivided into level coding, phase coding and pulse duration coding.

50.2.1.1 Level coding. In state symmetric level coding of binary signals the binary symbols are represented by two different levels of an electrical signal. 5.3.5.2 requires that the binary symbol of Mark (1) be represented by a negative level and the binary symbol of Space (0) be represented by a positive level. In the corresponding waveform, shown in Figure 64A1, the levels occupy the entire unit interval and do not return to zero level. Therefore, this waveform is a synchronous signal which requires timing coordination between the data source and the data sink. The waveform shown in Figure 64A1 is also an unconditioned signal known as polar or balanced non-return-to-zero (NRZ) signal. A polar NRZ signal allows for a simple detection method of the binary symbols at the data sink since the threshold for determining the binary symbols of Mark (1) and Space (0) is below and above the zero voltage axis, respectively.

In state symmetric level coding the positive and negative levels do not have to occupy the entire unit interval but may return to zero level within each unit interval. Figure 64A2 shows an example of a state symmetric level coded signal where the positive and negative levels representing Space (0) and Mark (1), respectively, are not occupying the entire unit interval but return to zero level in the second half-unit interval. This waveform, known as polar return-to-zero (RZ) signal, has a definite

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

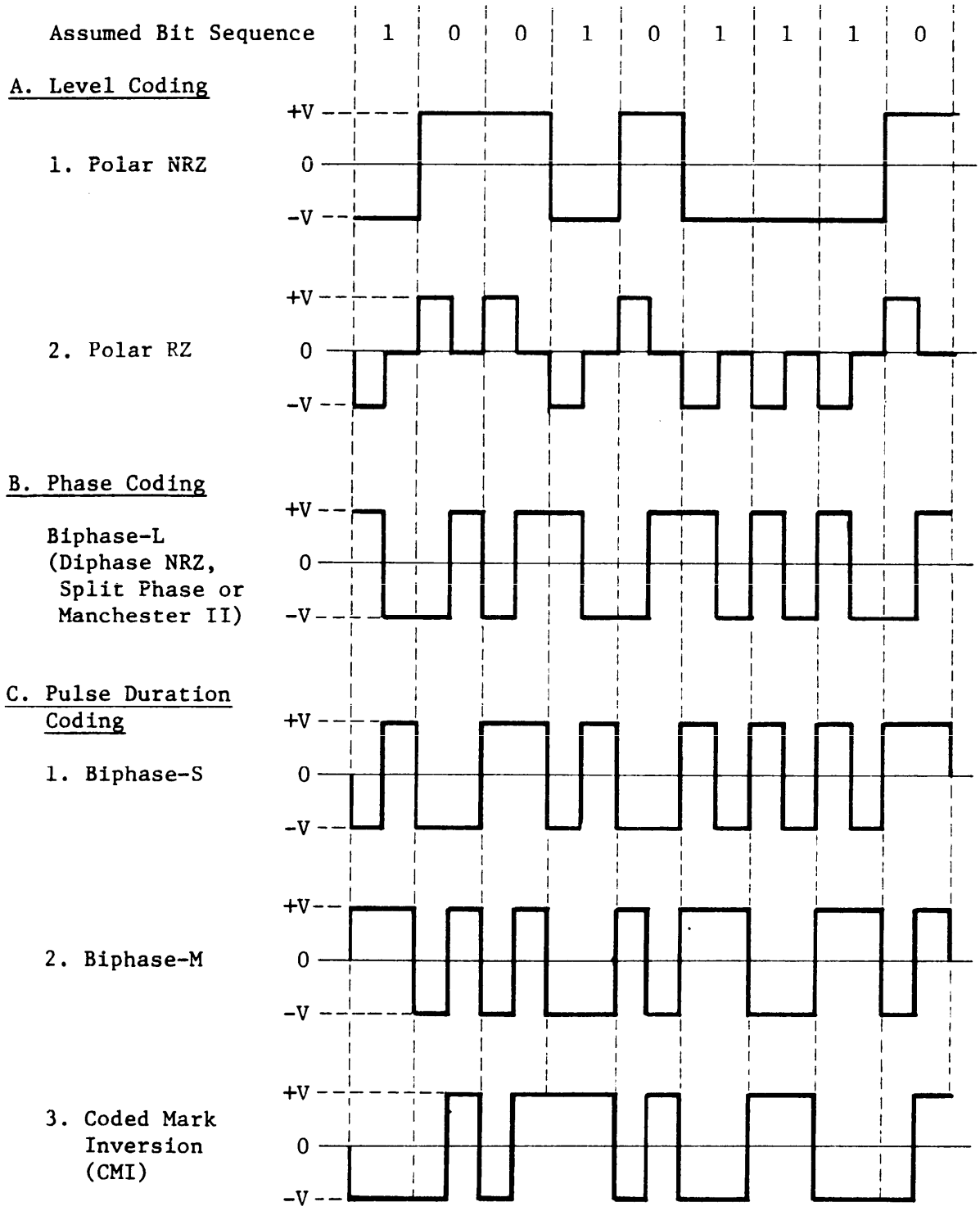


FIGURE 64. Examples of binary state symmetric coded waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

separation between levels representing the digital symbols. A polar RZ signal is a pseudo-ternary asynchronous and self-clocking signal.

50.2.1.2 Phase coding. In state symmetric phase coding of binary signals the binary symbols are represented by two different phases of an electrical signal, normally 0 degree and 180 degrees. Figure 64B shows an example of a state symmetric phase coded waveform where the binary symbol of Mark (1) is represented by one phase value and the binary symbol of Space (0) is represented by the other phase value which is shifted 180 degrees with respect to the Mark (1) phase. This waveform, also called biphase-level (Biphase-L), split-phase or diphase NRZ signal, is self-clocking and identical to the diphase modulated signal standardized in 5.4.1.4.2. The signal has a transition in the middle of each unit interval. The transition is positive-to-negative going for the binary symbol of Mark (1) and negative-to-positive going for the binary symbol of Space (0). The waveform shown in Figure 64B is also known as the Manchester II waveform or the Manchester II data code which is standardized in MIL-STD-1553 as the data code for a TDM data bus on aircraft. In state symmetric phase coding it is not necessary to change the phase at the beginning of every unit interval, as shown in Figure 64B. Phases may also be changed at the middle or the end of every unit interval. The resulting waveforms are then shifted accordingly along the time axis with respect to the waveform shown in Figure 64B.

50.2.1.3 Pulse duration coding. In state symmetric pulse duration (or frequency) coding of binary signals the binary symbols are represented by two different pulse durations of an electrical signal. The two pulse durations are normally equal to one unit interval and one half of a unit interval. Figure 64C1 shows an example of a state symmetric pulse duration coded waveform where the binary symbol of Space (0) is represented by a pulse duration equal to the unit interval and the binary symbol of Mark (1) is represented by a pulse duration equal to one half of the unit interval. This waveform, also called biphase-space (Biphase-S), has a transition in the middle of every unit interval for the binary symbol of Mark (1) and no transition in the middle of a unit interval for the binary symbol of Space (0). Reversing the coding rule, so that the binary symbol of Mark (1) is represented by a pulse duration equal to the unit interval and the binary symbol of Space (0) is represented by a pulse duration equal to one half of the unit interval, results in a waveform shown in Figure 64C2. This waveform is known as biphase-mark (Biphase-M). Both waveforms shown in Figures 64C1 and 64C2 have transitions between every unit interval and, therefore, are self-clocking. These waveforms also have the property of being conditioned signals.

A modification of the Biphase-M signal, so that the binary symbol of Space (0) is represented always by a negative-to-positive going transition in the middle of the unit interval, results in the coded mark inversion

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

(CMI) signal shown in Figure 64C3. The CMI signal is standardized in CCITT Recommendation G.703, published in Volume III, Fascicle III.3, of the Yellow Books, Geneva, 1981. The CMI signal is a binary NRZ signal in which the binary symbol of Space (0) is coded so that both negative and positive amplitude levels are attained consecutively, each for one half of a unit interval, and the binary symbol of Mark (1) is coded so that either the positive or the negative amplitude levels are attained alternately for one full unit interval.

50.2.2 State unsymmetric coding. In state unsymmetric coding of binary signals only one binary symbol is used in the coding process. The absence of coding represents the other binary symbol. Consequently, not every unit interval of the electrical signal contains energy. The resulting waveforms may be balanced or unbalanced with respect to the zero voltage axis. State unsymmetric coding techniques are subdivided into unbalanced coding, balanced coding and conditioned coding.

50.2.2.1 Unbalanced coding. In state unsymmetric unbalanced coding of binary signals only one binary symbol is represented by an electrical parameter so that the resulting waveform is unbalanced with respect to the zero voltage axis. Figure 65A1 shows an example of a state unsymmetric unbalanced coded signal where the binary symbol of Space (0) is represented by a positive voltage occupying the entire unit interval and the binary symbol of Mark (1) is represented by zero voltage or the absence of coding. The waveform shown in Figure 65A1 is a synchronous signal known as unipolar NRZ, NRZ-level (NRZ-L), NRZ-change or unbalanced NRZ signal.

The pulses of a state unsymmetric unbalanced coded signal do not have to occupy the entire unit interval but may occupy only part of the unit interval and return to zero voltage for the remainder of the unit interval. Figure 65A2 shows an example of a waveform where the binary symbol of Space (0) is represented by a positive voltage occupying the first half of the unit interval and returning to zero voltage in the second half of the unit interval. The binary symbol of Mark (1) is represented by zero voltage or the absence of coding. The waveform shown in Figure 65A2 is an asynchronous signal known as unipolar RZ or RZ-level (RZ-L) signal. Unipolar RZ signals, as well as the unipolar NRZ signal shown in Figure 65A1, contain a dc component that could be difficult to transmit due to bandwidth limitations of the transmission facility.

50.2.2.2 Balanced coding. In state unsymmetric balanced coding of binary signals only one binary symbol is represented by an electrical parameter so that the resulting waveform is balanced with respect to the zero voltage axis. Figure 65B shows an example of a state unsymmetric balanced coded waveform where the binary symbol of Mark (1) is represented by a pulse with a positive and negative voltage level, each polarity occupying one half of the unit interval, and the binary symbol of Space (0)

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

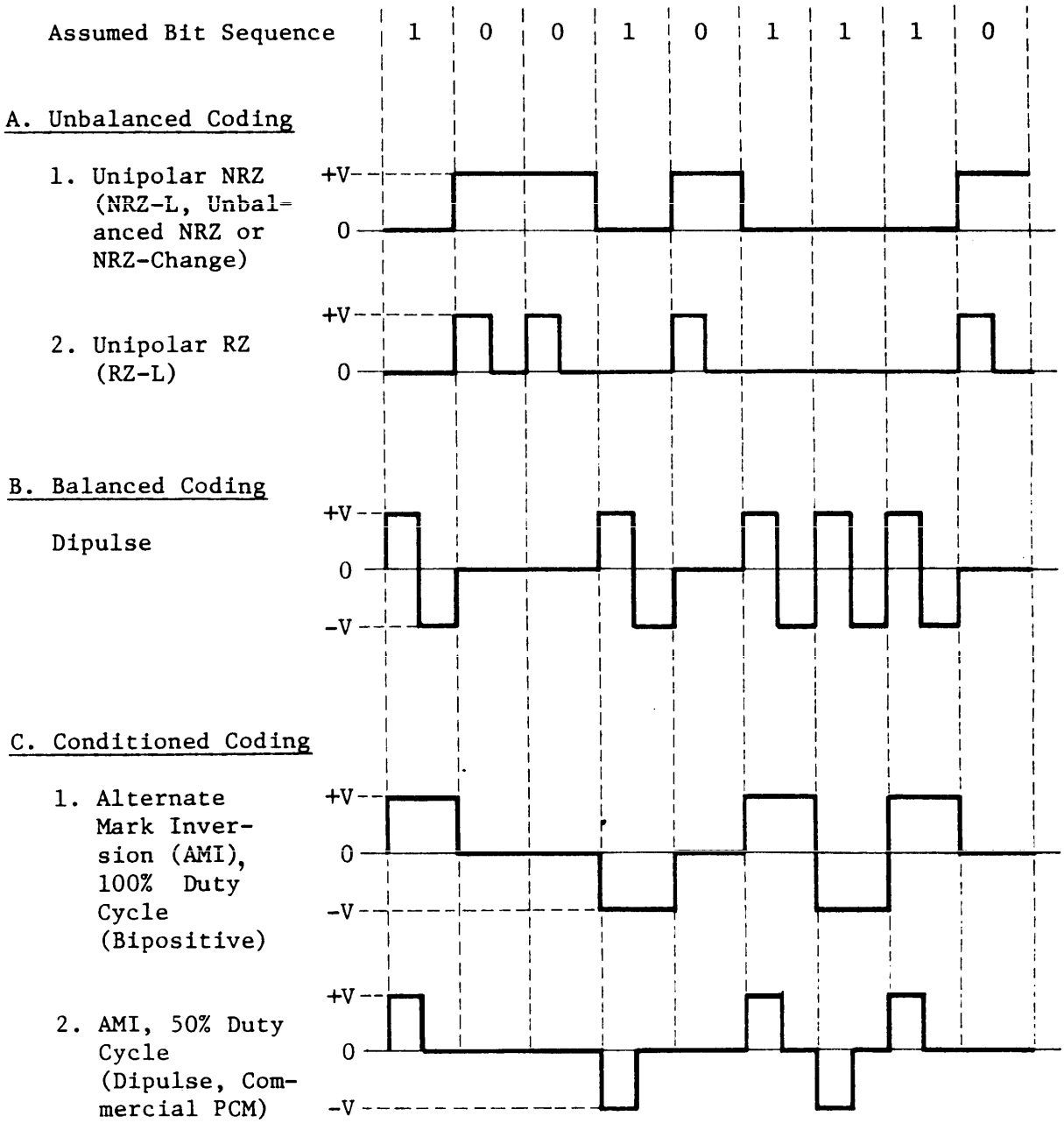


FIGURE 65. Examples of binary state unsymmetric coded waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

is represented by zero voltage or the absence of coding. This waveform, also called a dipulse signal, is a pseudo-ternary signal that contains no dc component.

50.2.2.3 Conditioned coding. In state unsymmetric conditioned coding of binary signals only one binary symbol is represented by an electrical parameter so that the coding results in a conditioned signal. The conditioning is achieved by alternately assigning each of the two different values of the same electrical parameter to one binary symbol. The other binary symbol is represented by the absence of coding. An example of a state unsymmetric conditioned coded waveform is shown in Figure 65C1 where the binary symbol of Mark (1) is represented by alternating positive and negative voltages having the same absolute value and occupying the entire unit interval. This waveform is a pseudo-ternary signal known as bipositive or alternate mark inversion (AMI) signal. Another form of an AMI signal with a 50 percent duty cycle, where the alternating positive and negative Mark (1) pulses occupy only one half of the unit interval, is shown in Figure 65C2. This AMI signal is used in commercial TDM/PCM systems. Both forms of AMI signals shown in Figures 65C1 and 65C2 are pseudo-ternary signals and are also known as bipolar signals. The AMI signal shown in Figure 65C1 is standardized in CCITT Recommendation G.703, published in Volume III, Fascicle III.3, of the Yellow Books, Geneva, 1981.

50.3 Binary differential coding techniques. In differential coding techniques of binary signals the value of the electrical parameter assigned to a binary symbol depends not only on the binary symbol to be coded in the unit interval under consideration but also on the value of the electrical parameter representing the binary symbol in the immediately preceding unit interval. Differential coding techniques are subdivided into differential symmetric coding (see 50.3.1) that result in NRZ signals and differential unsymmetric coding (see 50.3.2) that result in RZ signals. A differential coding process is based essentially on a "change" and "no change (repeat)" coding rule of the electrical parameter values selected to represent the digital symbols and, therefore, results in waveforms which are inherently conditioned signals. It should be noted that all differential coding techniques result in conditioned signals but not all conditioned signals are differentially coded. For example, state symmetric pulse duration coding (see 50.2.1.3) results in a conditioned signal which is not differentially coded.

50.3.1 Differential symmetric coding. In differential symmetric coding of binary signals two values of the same electrical parameter are used in the coding process and every unit interval of the electrical signal contains energy. The resulting waveforms are NRZ signals that are balanced with respect to the zero voltage axis. Instead of coding the two electrical parameter values so that one value is assigned only to one binary symbol as in state symmetric coding (see 50.2.1), in differential

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

symmetric coding either one of the two electrical parameter values may be assigned to a binary symbol depending on the particular coding rule. One binary symbol is represented by repeating the same electrical parameter value as in the immediately preceding unit interval. The other binary symbol is represented by changing the electrical parameter value from the value in the immediately preceding unit interval to the other selected parameter value. Symmetric differential coding techniques are subdivided into level coding, phase coding and pulse duration coding.

50.3.1.1 Level coding. In differential symmetric level coding of binary signals each binary symbol is represented by either one of two different levels of an electrical signal. Normally, the two levels are positive and negative voltages with the same absolute value, resulting in NRZ signals that are balanced with respect to the zero voltage axis. The level that is assigned to a binary symbol in a unit interval depends not only on the binary symbol in this unit interval but also on the level representing the binary symbol in the immediately preceding unit interval. Figure 66A1 shows an example of a differential symmetric level coded signal where the binary symbol of Space (0) is represented by repeating the same level as in the immediately preceding unit interval, regardless of which level exists in the preceding unit interval. The binary symbol of Mark (1) is represented by changing the level at the beginning of the unit interval from the level in the immediately preceding unit interval to the other selected level, regardless of which level exists in the preceding unit interval. This waveform is also called an NRZ-mark (NRZ-M) signal or a conditioned signal which is standardized in 5.4.1.4.1.

Another differential symmetric level coded waveform is obtained by reversing the coding rule for the NRZ-M signal so that the binary symbol of Space (0) is represented by changing the level at the beginning of the unit interval and the binary symbol of Mark (1) is represented by repeating the same level as in the immediately preceding unit interval. This waveform, known as an NRZ-Space (NRZ-S) signal, is shown in Figure 66A2.

The level change which represents a binary symbol in differential symmetric level coding does not have to occur at the beginning of the unit interval, but may occur at the middle or the end of the unit interval under consideration. A waveform that employs level changes at the middle or the end of a unit interval is shown in Figure 66A3. This waveform, known as delay modulation or Miller code, is a variation of the NRZ-M signal shown in Figure 66A1. In the Miller code the binary symbol of Mark (1) is represented by repeating in the first half-unit interval the same level as in the immediately preceding half-unit interval and changing the level at the middle of the unit interval for the second half-unit interval. The binary symbol of Space (0) is represented by repeating in the full unit interval the same level as in the immediately preceding half-unit interval, if the Space (0) symbol to be coded is preceded by a Mark (1) symbol, and by changing the level at the end of the unit

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

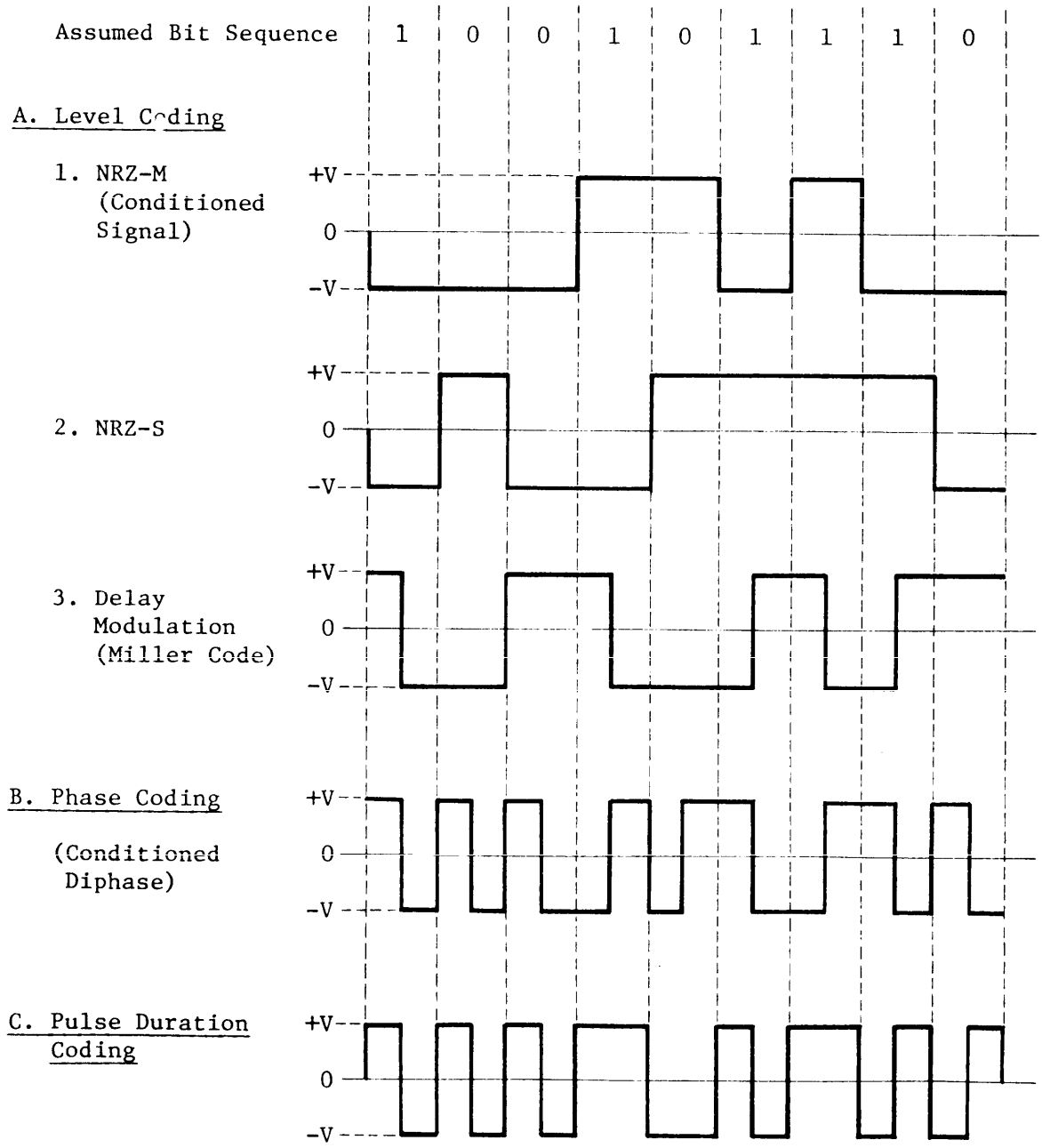


FIGURE 66. Examples of binary differential symmetric coded waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

interval, if the Space (0) symbol is followed by another Space (0) symbol. This coding rule separates a string of continuous Space (0) symbols by transitions between these symbols, resulting in a self-clocking waveform. It should be noted that the Miller code results in a conditioned and self-clocking signal which has a pulse duration equal to or greater than an entire unit interval. Therefore, this signal occupies less bandwidth for the same data signaling rate than other conditioned and self-clocking signals, such as those shown in Figures 66B and 66C.

50.3.1.2 Phase coding. In differential symmetric phase coding of binary signals each binary symbol is represented by either one of two different phases of an electrical signal, normally 0 degree and 180 degrees. The phase that is assigned to a binary symbol in a unit interval depends not only on the binary symbol in this unit interval but also on the phase representing the binary symbol in the immediately preceding unit interval. Figure 66B shows an example of a differential symmetric phase coded waveform where the binary symbol of Space (0) is represented by repeating the same phase as in the immediately preceding unit interval, regardless of which phase exists in the preceding unit interval. The binary symbol of Mark (1) is represented by changing the phase 180 degrees with reference to the phase in the immediately preceding unit interval, regardless of which phase exists in the preceding unit interval. This waveform is identical to the conditioned diphase signal standardized in 5.4.1.4.3. The waveform has transitions at the middle of every unit interval and, therefore, is self-clocking. In differential symmetric phase coding it is not necessary to change the phase at the beginning of every unit interval as shown in Figure 66B. Phases may also be changed at the middle or the end of every unit interval. The resulting waveforms are then shifted accordingly along the time axis with respect to the waveform shown in Figure 66B.

50.3.1.3 Pulse duration coding. In differential symmetric pulse duration coding of binary signals each binary symbol is represented by either one of two different pulse durations of an electrical signal. The two pulse durations are normally equal to one unit interval and one half of a unit interval. The pulse duration assigned to a binary symbol in a unit interval depends not only on the binary symbol in this unit interval but also on the pulse duration representing the binary symbol in the immediately preceding unit interval. Figure 66C shows an example of a differential symmetric pulse duration coded waveform where the binary symbol of Space (0) is represented by repeating the same pulse duration as in the immediately preceding unit interval, regardless of which pulse duration exists in the preceding unit interval. The binary symbol of Mark (1) is represented by changing the pulse duration at the beginning of the unit interval from the pulse duration in the immediately preceding unit interval to the other selected pulse duration, regardless of which pulse duration exists in the preceding unit interval. The waveform shown in Figure 66C has a transition between every unit interval and, therefore, is a self-clocking signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

50.3.2 Differential unsymmetric coding. In differential unsymmetric coding of binary signals the two electrical parameter values that represent the binary symbols are the presence and the absence of a pulse. This coding rule results in RZ signals since not every unit interval of the electrical signal contains energy. The coding of a binary symbol depends not only on the binary symbol to be coded in the unit interval under consideration but also on the electrical parameter value representing the binary symbol in the immediately preceding unit interval. The resulting waveforms may be balanced or unbalanced with respect to the zero voltage axis. Differential unsymmetric coding techniques are subdivided into unbalanced coding, balanced coding and conditioned coding. Differential unsymmetric coding techniques have been included in this appendix to complete the new classification even if these coding techniques have not found the same application as the other coding techniques shown in Figures 64 through 66.

50.3.2.1 Unbalanced coding. In differential unsymmetric unbalanced coding of binary signals the binary symbols are represented by the presence or absence of a pulse so that the resulting waveform is unbalanced with respect to the zero voltage axis. Figure 67A shows an example of a differential unsymmetric unbalanced coded signal where one electrical parameter value consists of a positive voltage during the entire unit interval (presence of a pulse) and the other parameter value consists of zero voltage during the entire unit interval (absence of a pulse). The binary symbol of Mark (1) is represented by repeating the same electrical parameter value as in the immediately preceding unit interval, regardless of which parameter value exists in the preceding unit interval. The binary symbol of Space (0) is represented by changing the electrical parameter value from the parameter value in the immediately preceding unit interval to the other selected parameter value, regardless of which parameter value exists in the preceding unit interval.

50.3.2.2 Balanced coding. In differential unsymmetric balanced coding of binary signals the binary symbols are represented by the presence or absence of a pulse so that the resulting waveform is balanced with respect to the zero voltage axis. Figure 67B shows an example of a differential unsymmetric balanced coded signal where one electrical parameter value consists of a pulse having a positive voltage during the first half of the unit interval and a negative voltage during the second half of the unit interval; the other parameter value consists of zero voltage during the entire unit interval. The binary symbol of Mark (1) is represented by repeating the same electrical parameter value as in the immediately preceding unit interval, regardless of which parameter value exists in the preceding unit interval. The binary symbol of Space (0) is represented by changing the electrical parameter value from the parameter value in the immediately preceding unit interval to the other selected parameter value, regardless of which parameter value exists in the preceding unit interval. The waveform shown in Figure 67B is a pseudo-ternary asynchronous signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

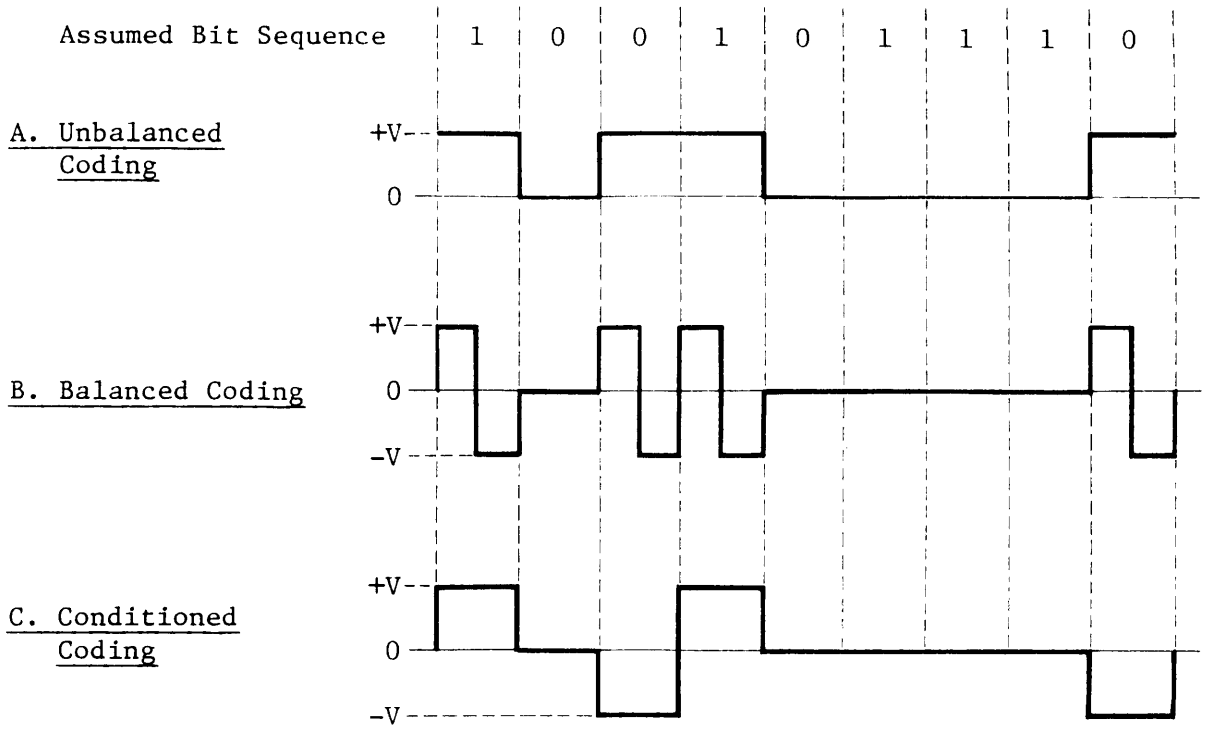


FIGURE 67. Examples of binary differential unsymmetric coded waveforms.

50.3.2.3 Conditioned coding. In differential unsymmetric conditioned coding of binary signals the binary symbols are represented by the presence or absence of pulses that alternate in polarity. Figure 67C shows an example of a differential unsymmetric conditioned signal where one electrical parameter value consists of a pulse with either a positive or a negative level during the entire unit interval and the other parameter value consists of zero voltage during the entire unit interval. The binary symbol of Mark (1) is represented by repeating the same electrical parameter value as in the immediately preceding unit interval, regardless of which parameter value exists in the preceding unit interval. The binary symbol of Space (0) is represented by changing the electrical parameter value from the parameter value in the immediately preceding unit interval to the other selected parameter value, regardless of which parameter value exists in the preceding unit interval. Additionally, every other pulse of the resulting sequence of pulses is reversed in polarity resulting in a waveform with alternating pulse polarities as shown in Figure 67C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX F

GENERAL GUIDANCE FOR THE ENGINEERING OF TELEGRAPH AND DATA CIRCUITS.

This Appendix contains tutorial information in support of MIL-STD-188-200. Appendix F is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph	Title	Page
60	General guidance for the engineering of telegraph and data circuits	F-309
60.1	Application	F-309
60.2	General	F-309
60.3	Telegraph and data transmission over multichannel VF circuits	F-310
60.3.1	Loop insertion loss	F-310
60.3.1.1	Loop insertion loss for tactical subsystems type I and type III	F-310
60.3.1.2	Loop insertion loss for tactical subsystem type II	F-310
60.3.2	Received quasi-analog signal level	F-311
60.3.2.1	Received quasi-analog signal level for tactical subsystems type I and type III	F-311
60.3.2.2	Received quasi-analog signal level for tactical subsystem type II	F-312
60.3.3	Equalization	F-313
60.3.4	Regeneration	F-313
60.4	Telegraph and data transmission over 48 kHz FDM group circuits	F-314

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

60 General guidance for the engineering of telegraph and data circuits.

60.1 Application. The information provided in this appendix applies to multichannel FDM and TDM/PCM VF circuits and FDM group circuits used for the transmission of quasi-analog telegraph and data signals. General requirements for telegraph and data transmission over single channel radio subsystems are given in 5.4.6. For the transmission of digital telegraph and data signals over TDM circuits of the tactical subsystem type II see Note of 5.7.3.2.6.

60.2 General. Telegraph and data transmission is influenced by many factors, such as irregularities of the transmission medium and internal and external interferences. Most of these factors are difficult to predict and to control. Equipment and facility design has traditionally been based on voice transmission and low speed telegraph transmission using quasi-analog signals over VF channels. These channels were later used for higher speed data transmission, resulting often in unsatisfactory performance due to such phenomena as impulse noise, level dropouts, and other signal discontinuities which were found to be much more disturbing to data transmission than to voice transmission.

In general, experience has shown that signal discontinuities, such as phase and amplitude hits, impulse noise and dropouts, as well as excessive envelope delay distortion and low SNR at the data sink, are the major causes of high bit error rates. In many cases, unsatisfactory telegraph and data transmission is caused by power supply problems, e.g., transients or unstable supply voltages, or by problems related to control and timing signals (see 5.3.6).

For VF and FDM group circuits, two principal methods have evolved to improve telegraph and data service: Equalization of channel characteristics and regeneration of digital signals. General guidance for the engineering of VF circuits and FDM group circuits is provided in 60.3 and 60.4, respectively, based on experience gained in using these circuits for telegraph and data transmission with quasi-analog signals.

Circuit designers are expected to make the necessary trade-off among all factors influencing the performance of switched and nonswitched (point-to-point) telegraph and data circuits, based on the information provided in this appendix, and to obtain the optimum solution to insure acceptable and reliable telegraph and data service. Factors that should be considered in the trade-off are: Reducing loop insertion loss and loop noise, obtaining the correct received quasi-analog signal level at a data sink with or without additional amplifiers between the output of a transmission channel (see Figure 1) and the input of a data loop, regeneration of data signals and equalization of data circuits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

60.3 Telegraph and data transmission over multichannel VF circuits. Telegraph and data service over these circuits may be provided by the use of controlled loop characteristics and by the use of TDM/PCM VF channels (see Note of 5.7.3.2.6) and those FDM VF channels which are best suited for telegraph and data service employing quasi-analog signals. FDM VF channels 1 and 2 of group 1 of supergroup 1, and FDM VF channels 11 and 12 of group 5 of supergroup 3 are not recommended for this purpose due to unsatisfactory channel characteristics.

60.3.1 Loop insertion loss. The main objectives of engineering a loop for quasi-analog telegraph and data signals are to keep the noise generated in the loop as low as possible and the quasi-analog signal level sufficiently high to obtain an optimum SNR at the data sink. Reducing the insertion loss of a wire or cable loop to the lowest possible value will facilitate a sufficiently high level of the received quasi-analog data signals and will obtain the best SNR at the data sink for a given loop noise power level. Possible methods for reducing the loop insertion loss or the loop noise, or both, are rerouting, using a lower gauge wire or cable, shielding and better ground connections.

60.3.1.1 Loop insertion loss for tactical subsystems type I and type III. Wire or cable loops used for switched or nonswitched (point-to-point) telegraph and data service in the tactical subsystems type I and type III (see Table I) are not adjusted for a nominal value of the loop insertion loss, rather, the loop insertion loss should be as low as possible. Engineering wire or cable loops for telegraph and data service to achieve the minimum insertion loss is different from engineering wire or cable loops for voice service (see 5.5.2.1.1) where loop loss pads are used in the tactical subsystems type I and type III for adjusting the loop insertion loss to obtain a received speech volume of nominal -28 VU in accordance with 5.2.2.1.2. In those cases, where a wire or cable loop is used for both voice service and telegraph and data service in switched and nonswitched (point-to-point) circuits of the tactical subsystems type I and type III, the requirements of 5.5.2.1.1 for voice service normally govern the adjustment of the loop insertion loss, provided that a satisfactory SNR for the telegraph and data signals can be achieved at the data sink. Otherwise, a compromise among the parameter values of 5.2.2.1.1, 5.2.2.1.2 and 5.5.2.1.1 may have to be found to increase the received quasi-analog signal level by reducing the loop insertion loss, and a mean received speech volume higher than nominal -28 VU may have to be tolerated.

60.3.1.2 Loop insertion loss for tactical subsystem type II. In the tactical subsystem type II (see Table I), the insertion loss of wire or cable loops is not adjusted to a specific value, rather, a statistical distribution of the loop insertion loss is assumed for the purpose of system engineering and circuit design. (See 5.5.2.2.) The insertion loss of wire or cable loops, based on the distribution shown in Figure 23, will assure a satisfactory received speech volume at the listener telephone

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

set. However, the insertion loss distribution will not assure a sufficiently high level of received quasi-analog telegraph and data signals at the data sink. Therefore, an effort should be made to keep the insertion loss of wire or cable loops used for telegraph and data transmission as low as possible to obtain a sufficiently high level of quasi-analog telegraph and data signals received at the data sink.

60.3.2 Received quasi-analog signal level. The received quasi-analog signal level at the output of a wire or cable loop used for telegraph and data service should be sufficiently high to obtain a high SNR at the data sink and a low BER of the received telegraph or data signal. A sufficiently high level of the received quasi-analog signal is particularly important for noisy wire or cable loops where the high signal level is often a more economical method than using signal regeneration to maintain a satisfactory SNR at all points of a telegraph and data circuit from data source to data sink. In nonswitched (point-to-point) VF circuits used for telegraph and data service, a sufficiently high level of the quasi-analog signals received at a data sink is obtained by increasing the signal level at the output of a transmission channel (see Figure 1) to compensate for the insertion loss of the data loop connected to the output of the transmission channel. The higher quasi-analog signal level at the output of the transmission channel and the input of the receiving loop may increase the crosstalk into other loops, particularly in multipair cable loops. However, this possible increase of the crosstalk is considered tolerable since the quasi-analog signal level at the input of the receiving loop is in most cases very close to the level of the quasi-analog signal at the output of the data source and the input of the transmitting loop. (See Figures 52, 55 and 58.)

60.3.2.1 Received quasi-analog signal level for tactical subsystems type I and type III. In nonswitched (point-to-point) VF circuits used only for telegraph and data service of the tactical subsystems type I and type III (see Table I), a OTLP is assigned to the input terminals of a data sink, resulting in a quasi-analog signal level of -13 dBm (-13 dBm0) at this point. (See Figures 52 and 58.) In order to obtain the quasi-analog signal level of -13 dBm at the data sink, the output of a transmission channel (see Figure 1) must have a TLP which is higher than the OTLP at the data sink by an amount equal to the loop insertion loss. Most types of multiplexing equipment used in the tactical subsystems type I and type III have adjustable level controls at the VF channel output of the demultiplexer that allows a signal level adjustment of up to +7 dBm. This range of level adjustment should be sufficient to compensate for the insertion loss of the majority of data loops normally encountered in these tactical subsystems.

In switched VF circuits used only for telegraph and data service of the tactical subsystems type I and type III, the demultiplexer output level must be adjusted in accordance with 5.6.2.1.2.9, otherwise the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

switching equipment would be overloaded. Consequently, the demultiplexer level adjustment cannot be used to compensate for the insertion loss of a data loop connected through a switch to the VF channel output of the demultiplexer. This restriction makes it important to keep the loop insertion loss and the loop noise as low as possible to obtain the highest possible quasi-analog SNR at the data sink. In those cases where it is infeasible or ineffective to reduce the loop insertion loss or the loop noise, a VF amplifier may be used, connected between the VF channel output of the switch and the input of the loop, to increase the received quasi-analog signal level at the data sink to the specified level.

For switched and nonswitched (point-to-point) VF circuits used for both voice service and telegraph and data service in the tactical subsystems type I and III, the requirements of 5.2.2.1 for voice service normally determine the adjustment of the insertion loss of wire or cable loops. With a mean insertion loss of the loops as stated in 5.5.2.1.1, the received quasi-analog signal level of the telegraph and data signals is -22 dBm at the -9TLP of the data sink. (See Figures 51 and 57.) In those cases, where a satisfactory SNR for the quasi-analog telegraph and data signals cannot be achieved at the data sink with the received quasi-analog signal level of -22 dBm, a compromise among the parameter values of 5.2.2.1.1, 5.2.2.1.2 and 5.5.2.1.1 may have to be found to increase the received quasi-analog signal level by reducing the loop insertion loss, and a mean received speech volume higher than nominal -28 VU may have to be tolerated.

60.3.2.2 Received quasi-analog signal level for tactical subsystem type II. In nonswitched (point-to-point) VF circuits used only for telegraph and data service of the tactical subsystem type II (see Table I), a -4TLP is assigned to the input terminals of a data sink, resulting in a quasi-analog signal level of -10 dBm (-6 dBm0) at this point. (See Figure 55.) In order to obtain the quasi-analog signal level of -10 dBm at the data sink, the output of a transmission channel (see Figure 1) must have a TLP which is higher than the -4TLP at the data sink by an amount equal to the loop insertion loss. The multiplexing equipment used in the tactical subsystem type II has an adjustable level control at the VF channel output of the demultiplexer that allows a signal level adjustment of up to +5 dBm. This range of level adjustment should be sufficient to compensate for the insertion loss of the majority of data loops normally encountered in the tactical subsystem type II.

In switched VF circuits used only for telegraph and data service or used for both voice service and telegraph and data service, the demultiplexer output level must be adjusted in accordance with 5.6.2.1.2.9, otherwise the switching equipment would be overloaded. Consequently, the demultiplexer level adjustment cannot be used to compensate for the insertion loss of a data loop connected through a switch to the VF channel output of the demultiplexer. This restriction makes it important to keep

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

the loop insertion loss and the loop noise as low as possible to obtain the highest possible quasi-analog SNR at the data sink. The specific value of the received quasi-analog signal level at the data sink is not standardized, since this value depends on the actual insertion loss of the installed loop. In the tactical subsystem type II, the insertion loss of wire or cable loops is not adjusted to a specific value, rather, a statistical distribution of the loop insertion loss is assumed. (See 5.5.2.2.) A mean received quasi-analog signal level of -13.5 dBm is stated in 5.2.3.1.5.1, based on an assumed mean loop insertion loss of 3.5 dB and on a quasi-analog signal level of -6 dBm0 in accordance with 5.2.3.1.4. These values have also been used in the hypothetical reference circuit shown in Figure 54.

60.3.3 Equalization. Equalization (or conditioning) equipment is used to improve the envelope delay distortion and the insertion loss versus frequency characteristic of VF channels employed for the transmission of quasi-analog signals. Four-wire TDM/PCM VF circuits of the tactical subsystem type II and FDM VF circuits of the tactical subsystem type III (see Table I) engineered in accordance with the configurations shown in 5.9.4 will have transmission characteristics that are generally suitable for telegraph and data transmission in quasi-analog form with modulation rates up to and including 1200 Bd. Consequently, these four-wire circuits usually do not require special conditioning or equalization devices, provided that the data circuits meet, or are close to the values given in 5.2.3.1.6.1 and 5.2.3.1.6.2. Equalization devices are normally required for modulation rates higher than 1200 Bd. Several types of higher speed modems for VF circuits have been designed with equalizers which are an integral part of the modem. This technique has been standardized in documents, such as FED-STD-1005 and FED-STD-1006.

Standards on the use of equalization devices for VF circuits cannot be provided due to the wide variety of channel characteristics of operational circuits. As a basic rule, equalization should be considered if a circuit cannot meet all applicable parameters of 5.2.3.1. In any case, the employment of equalization devices has to be determined by circuit engineering on an individual basis.

60.3.4 Regeneration. Regeneration of digital signals is used to retime and reshape these signals and to remove undesired signal components, such as noise or jitter. As general guidance, regeneration of data signals transmitted over VF circuits may be necessary:

a. For modulation rates greater than or equal to 1200 Bd.

b. After more than six FDM trunks in tandem of the tactical subsystem type I (see Table I) to keep the circuit noise within allowable limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

c. At the interface point between a loop and a transmission channel (see Figure 1) when relatively long wire or multipair cable loops are used; and

d. At the interface point between a VF channel of a tactical subsystem connected with a VF channel of a long haul system (see MIL-STD-188-100, subparagraphs 4.3.2 and 4.3.3).

It should be noted that each regeneration point will act as a data sink/data source combination and the transfer function parameters of 5.2.3.1.6 will only be applicable to those portions of a circuit extending from regeneration point to regeneration point and not to the entire circuit from user to user. (See 5.9.3.1.)

Standards on the use of regeneration for VF circuits cannot be provided due to the wide variety of channel characteristics of operational circuits. As a basic rule, regeneration should be considered if a circuit cannot meet all applicable parameters of 5.2.3.1. In any case, the employment of regeneration devices has to be determined by circuit engineering on an individual basis.

60.4 Telegraph and data transmission over 48 kHz FDM group circuits. The basic FDM group channel is derived by frequency division multiplexing of 12 VF channels and by translation into a basic group frequency band resulting in a bandwidth of 48 kHz. This basic group channel can be utilized for higher speed telegraph and data transmission with quasi-analog signals. Such channels are provided over those FDM groups which are best suited for higher speed telegraph and data service. FDM group A (12 kHz to 60 kHz), FDM group 1 of supergroup 1, and FDM group 5 of supergroup 3 are not recommended for this purpose due to unsatisfactory channel characteristics.

The nominal bandwidth of an FDM group B channel extends from 60 kHz to 108 kHz; however, the standard group pilot for system regulation and alarm appears at 104.08 kHz, which effectively prevents the use of the 104 kHz to 108 kHz portion of the group B bandwidth for higher speed telegraph and data service. The 104 kHz to 108 kHz portion can be used for a VF channel or a lower speed data channel for coordination and control of the higher speed data channel. The remaining bandwidth is suitable for higher speed data transmission. In systems having pilots at frequencies other than 104.08 kHz, such pilots must be disabled and the filters associated with these pilots must be removed from the FDM group channel before this channel may be used for higher speed telegraph and data service.

Standards on the use of equalization and regeneration devices for FDM group circuits cannot be provided due to the wide variety of channel characteristics of operational circuits. Experience has shown that it may be necessary to regenerate data signals after three data trunks in tandem to provide satisfactory telegraph and data service.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX G

PHASE AND ENVELOPE DELAYS, AND DELAY DISTORTIONS.

This Appendix contains tutorial information in support of MIL-STD-188-200. Appendix G is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph	Title	Page
70	Phase and envelope delays, and delay distortions	G-317
70.1	Phase delay	G-317
70.2	Delay distortion	G-317
70.3	Envelope delay	G-318
70.4	Absolute envelope delay and relative envelope delay	G-320
70.5	Envelope delay distortion	G-320

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

70 Phase and envelope delays, and delay distortions. The following explanatory information is provided for a better understanding of the terms: Phase delay, delay distortion, absolute envelope delay, relative envelope delay and envelope delay distortion.

70.1 Phase delay. The time delay between the phases of a single-frequency signal at the input and the output of a device is called the phase delay of the device under consideration. The device, such as an equipment, a network or a circuit, may include passive or active components. The time delay between the phases of the single-frequency input and output signals may be determined from the phase shift characteristic as shown in Figure 68. The phase delay (T_p) at a specific frequency is calculated as:

$$T_p = \frac{b_1 \text{ (in radians)}}{\omega_1 \text{ (in radian/s)}} \qquad \text{where } \omega_1 = 2\pi f_1$$

or

$$T_p = \frac{b_2 \text{ (in radians)}}{\omega_2 \text{ (in radian/s)}} \qquad \text{where } \omega_2 = 2\pi f_2$$

It should be noted that phase shift, the difference between two phase angles, is measured in units of radians, whereas phase delay is measured in units of time, as expressed in the above formulas.

70.2 Delay distortion. Normally, the phase delay versus frequency characteristic of a device is not a straight line but a nonlinear curve, as shown in Figure 68. Such a nonlinear characteristic causes distortion of a waveform passing through the device when the waveform consists of more than one frequency component. This type of distortion is called delay distortion. Delay distortion of a multifrequency waveform is caused by the different arrival times of each frequency component at the output of a device due to a nonlinear phase delay versus frequency characteristic of the device. Delay distortion (T_d) of a device is calculated as:

$$T_d = \frac{b_2}{\omega_2} - \frac{b_1}{\omega_1}, \qquad \text{where } \omega_1 = 2\pi f_1 \text{ and } \omega_2 = 2\pi f_2$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Delay distortion (T_d) is the difference in phase delay or propagation time at the two frequencies f_1 and f_2. Data transmission is more impaired by phase irregularities in the transmission path than voice transmission since the human ear is relatively insensitive to the phase distortion of a voice signal.

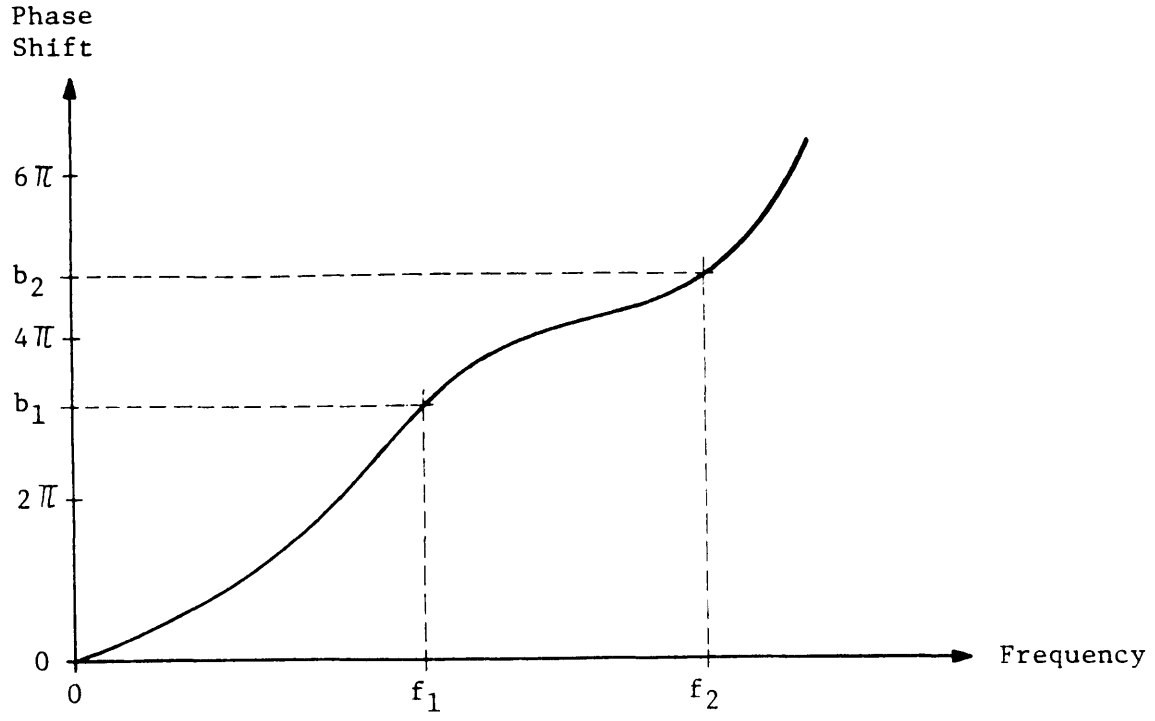


FIGURE 68. Phase delay versus frequency characteristic.

70.3 Envelope delay. The envelope delay of a device at a specified frequency is the slope of the phase delay (see 70.1) of the device at this frequency. Mathematically, the envelope delay is the derivative with respect to frequency of the phase delay versus frequency characteristic shown in Figure 68. Sometimes it is rather difficult to measure the phase delay, particularly of long transmission channels, due to the problem of establishing a phase reference at the channel output with respect to the phase of the test signal at the channel input. Additionally, phase delay is normally not too important to data transmission since the delay distortion gives a better indication of the transmission impairments of data signals than the phase delay. One method of determining how much a nonlinear phase delay characteristic affects the transmission quality of data signals is to transmit a narrowband AM signal through a channel and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

to evaluate at the channel output the phases of the upper and lower sidebands with respect to the phase of the carrier signal. The result of such a measurement is called the envelope delay since the envelope of the narrowband AM signal indicates the amount of delay contributed by the channel under consideration. The main advantage of using a narrowband AM signal is that it does not require a phase reference at the channel output with respect to the phase of the test signal at the channel input. An envelope delay versus frequency characteristic of a channel can be obtained by varying the carrier frequency of the narrowband AM signal over the passband of the channel. Test equipment for envelope delay measurements is commercially available. This equipment uses principles and modulating frequencies that have been standardized by telephone companies and the ITU.

Figure 69 depicts a typical envelope delay versus frequency characteristic of an FDM VF channel. The curve shows the increase in envelope delay at both ends of the channel passband due to the very selective channel filters employed in FDM.

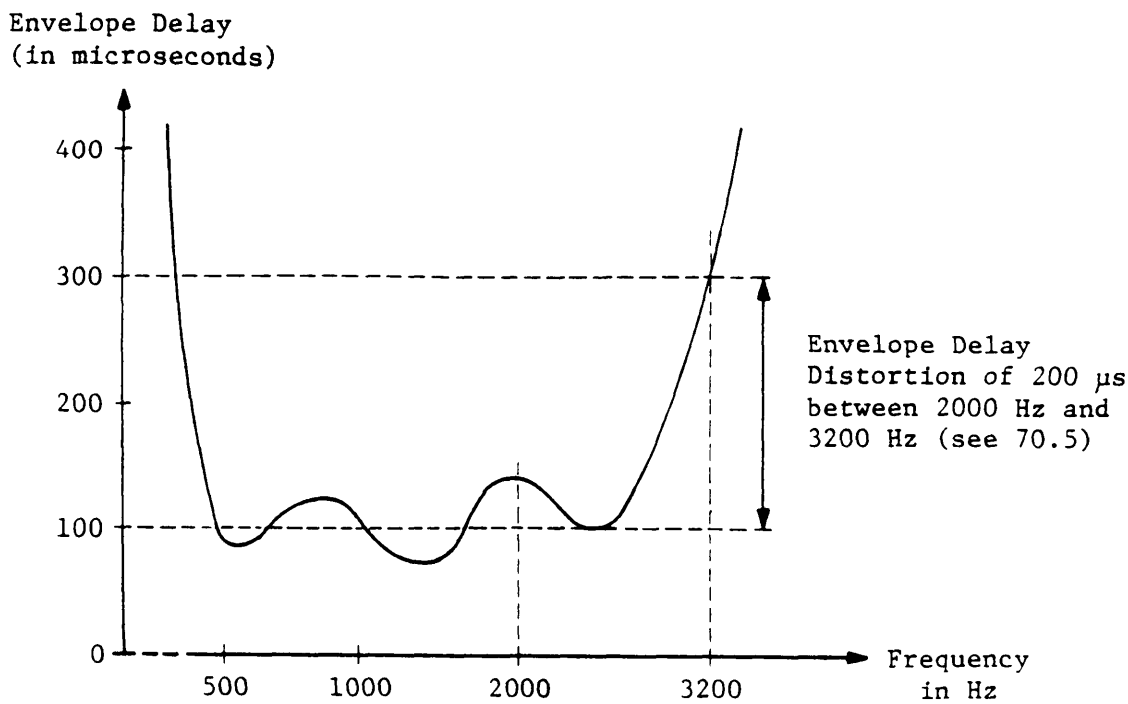


FIGURE 69. Envelope delay versus frequency characteristic.

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the current version before use.

70.4 Absolute envelope delay and relative envelope delay. Absolute envelope delay is the amount of delay encountered by a narrowband AM signal with a specified carrier frequency between the input and the output terminals of a device or a transmission path, as explained in 70.3 and shown in Figure 69. Relative envelope delay is the difference between the envelope delay at a specified frequency and the envelope delay at another frequency which has been selected as a reference. The envelope delay at the reference frequency is considered to be zero. The envelope delay at all other frequencies will then be either more (positive) or less (negative) than, or the same as, the envelope delay at the reference frequency. Figure 70 shows an envelope delay versus frequency characteristic of a VF channel with the scale for absolute envelope delay depicted on the left hand side and the scale for the relative envelope delay with respect to 1000 Hz on the right hand side of the figure.

70.5 Envelope delay distortion. Envelope delay distortion, often confused with delay distortion (see 70.2) or envelope delay (see 70.3), is the difference between the maximum and the minimum values of the envelope delay in a frequency band between two specified frequencies. Envelope delay distortion may be obtained from the envelope delay versus frequency characteristic as shown in Figure 69. It should be noted that envelope delay distortion is not directly related to delay distortion.

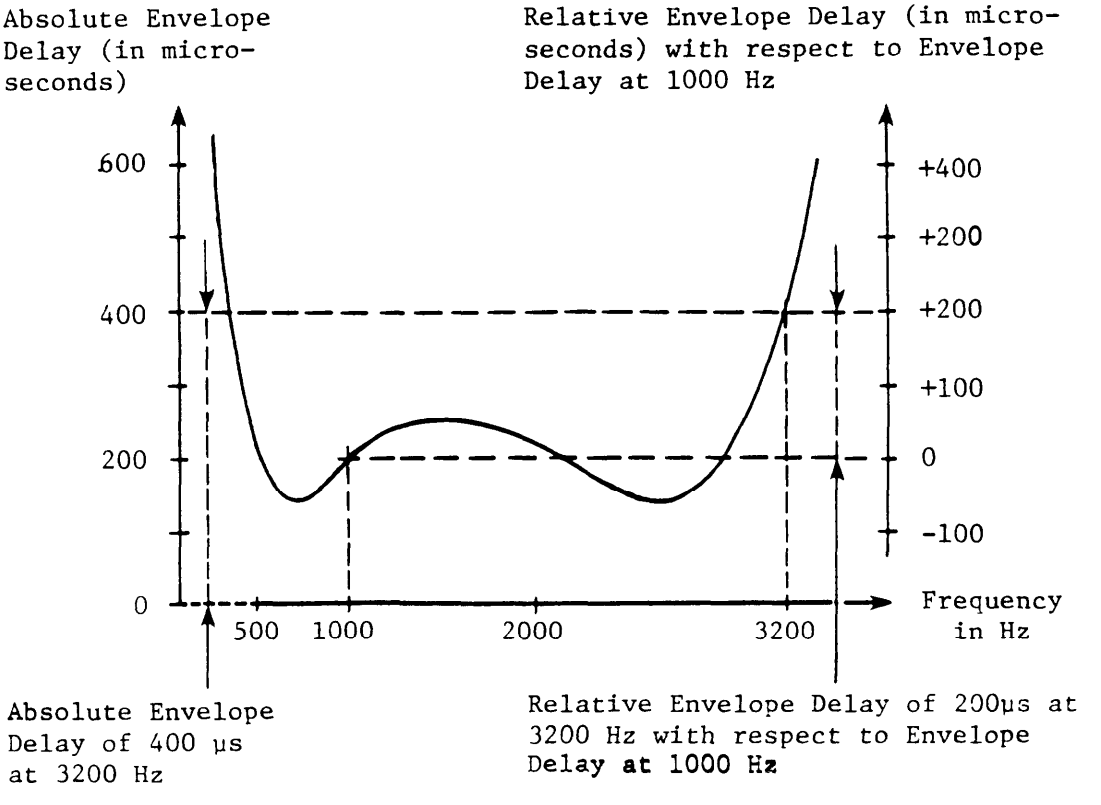


FIGURE 70. Absolute and relative envelope delay versus frequency characteristic.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the currentversion before use.

APPENDIX H

CODES FOR INFORMATION INTERCHANGE.

This Appendix contains information related to MIL-STD-188-200. Appendix H is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph	Title	Page
80	Codes for information interchange	H-324
80.1	General	H-324
80.2	Baudot code	H-324
80.3	International alphabets	H-324
80.4	American standard code for information interchange (ASCII)	H-326
80.4.1	Description of ASCII control characters	H-326
80.4.1.1	Communication control (CC) characters	H-326
80.4.1.1.1	Start of heading (SOH)	H-326
80.4.1.1.2	Start of text (STX)	H-326
80.4.1.1.3	End of text (ETX)	H-326
80.4.1.1.4	End of transmission (EOT)	H-326
80.4.1.1.5	Enquiry (ENQ)	H-327
80.4.1.1.6	Acknowledge (ACK)	H-327
80.4.1.1.7	Data link escape (DLE)	H-327
80.4.1.1.8	Negative acknowledge (NAK)	H-327
80.4.1.1.9	Synchronous idle (SYN)	H-327
80.4.1.1.10	End of transmission block (ETB)	H-327
80.4.1.2	Format effector (FE) characters	H-327
80.4.1.2.1	Backspace (BS)	H-327
80.4.1.2.2	Horizontal tabulation (HT)	H-327
80.4.1.2.3	Vertical tabulation (VT)	H-327
80.4.1.2.4	Line feed (LF)	H-327

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Paragraph	Title	Page
80.4.1.2.5	Form feed (FF)	H-328
80.4.1.2.6	Carriage return (CR)	H-328
80.4.1.2.7	Space (SP)	H-328
80.4.1.3	Information separator (IS) characters	H-328
80.4.1.4	Other control characters	H-328
80.4.1.4.1	Null (NUL)	H-328
80.4.1.4.2	Bell (BEL)	H-328
80.4.1.4.3	Shift out (SO)	H-328
80.4.1.4.4	Shift in (SI)	H-328
80.4.1.4.5	Device control (DC)	H-328
80.4.1.4.6	Cancel (CAN)	H-329
80.4.1.4.7	End of medium (EM)	H-329
80.4.1.4.8	Substitute (SUB)	H-329
80.4.1.4.9	Escape (ESC)	H-329
80.4.1.5	Delete (DEL) character	H-329
80.4.1.6	Space (SP) character	H-329

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

80 Codes for information interchange.

80.1 General. Information exchange employing teletypewriters is a means of record traffic communications between users. Special telegraph codes have been developed over the years for converting alphanumeric characters into electrical signals to facilitate the exchange of record traffic. Two of the oldest telegraph codes are the Morse code and the Baudot code (see 80.2). The terms "Mark" and "Space" originated in the earlier days of telegraphy when binary electrical signals were first recorded on paper tape and these terms are still used today. (See Table VIII.)

80.2 Baudot code. The Baudot code, invented by Emile Baudot in 1874, was a five-unit code for synchronous transmission used mainly with TDM systems. The code was used with a five-key keyboard, constructed similar to a piano keyboard, where the operator used three fingers of the right hand and two fingers of the left hand to generate the 32 possible combinations of the five-unit code. Today, the Baudot code is only of historical interest. However, the term "Baudot code" is still used erroneously to denote the ITA No. 2 which is a five-unit start/stop code for asynchonous transmission (see 80.3) and distinctly different from the Baudot code. Many teletypewriters still in use today operate with the American version of the ITA No. 2 (see Table LXVI). The use of the term "Baudot" in connection with codes, machines, signaling, etc., should be restricted to the historical meaning of this term and should not be confused with the ITA No. 2 or any other international alphabet.

80.3 International alphabets. Several international alphabets for telegraph and data transmission have been developed and standardized by the International Standardization Organization (ISO) and the International Telegraph and Telephone Consultative Committee (CCITT). The International Telegraph Alphabet (ITA) No. 1, a five-unit code for synchronous transmission, had been based on the Baudot code and is not used anymore.

The ITA No. 2 is a five-unit start-stop code for asynchronous transmission and is standardized in CCITT Recommendation F.1, published in Volume II, Fascicle II.4, of the Yellow Books, Geneva, 1981. The American version of ITA No. 2 is shown in Table LXVI for information only since this code is still used by many military and commercial teletypewriters. It should be emphasized that ITA No. 2 is not a DoD standard and, therefore, not a mandatory requirement for new equipment. (See 5.3.7.1.) The ITA No. 2 is often erroneously called the Baudot code. (See 80.2.)

The ITA No. 3 is a seven-unit code for synchronous transmission with an EDC capability using ARQ procedures and is intended to be used on radio-telegraph circuits. The ITA No. 3 is standardized in CCITT Recommendation S.13, published in Volume VII, Fascicle VII.2, of the Yellow Books, Geneva, 1981 and CCIR Recommendation 342-2, published in Volume III of the Green Books, Geneva 1978.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXVI. American version of ITA No. 2.

CHARACTERS			CODE SIGNALS							
LOWER (LETTER) CASE	UPPER (FIGURE) CASE		S = SPACE M = MARK							ITA No. 2 (SEE NOTE)
	COMMUNI-CATIONS	WEATHER SYMBOLS	START	1	2	3	4	5	STOP	
A	-	↑	S	M	M	S	S	S	M	
B	?	⊕	S	M	S	S	M	M	M	
C	:	○	S	S	M	M	M	S	M	
D	$	↗	S	M	S	S	M	S	M	WHO ARE YOU
E	3	3	S	M	S	S	S	S	M	
F	!	→	S	M	S	M	M	S	M	UNASSIGNED
G	&	↘	S	S	M	S	M	M	M	UNASSIGNED
H	STOP OR #	↓	S	S	S	M	S	M	M	UNASSIGNED
I	8	8	S	S	M	M	S	S	M	
J	'	↙	S	M	M	S	M	S	M	AUDIBLE SIGNAL
K	(	←	S	M	M	M	M	S	M	
L	)	↖	S	S	M	S	S	M	M	
M	.	.	S	S	S	M	M	M	M	
N	,	⊖	S	S	S	M	M	S	M	
O	9	9	S	S	S	S	M	M	M	
P	0	0	S	S	M	M	S	M	M	
Q	1	1	S	M	M	M	S	M	M	
R	4	4	S	S	M	S	M	S	M	
S	BELL	BELL	S	M	S	M	S	S	M	APOSTROPHE
T	5	5	S	S	S	S	S	M	M	
U	7	7	S	M	M	M	S	S	M	
V	;	⊘	S	S	M	M	M	M	M	=
W	2	2	S	M	M	S	S	M	M	
X	/	/	S	M	S	M	M	M	M	
Y	6	6	S	M	S	M	S	M	M	
Z	"	+	S	M	S	S	S	M	M	+
BLANK		-	S	S	S	S	S	S	M	UNASSIGNED
SPACE			S	S	S	M	S	S	M	
CARRIAGE RETURN			S	S	S	S	M	S	M	
LINE FEED			S	S	M	S	S	S	M	
FIGURE SHIFT			S	M	M	S	M	M	M	
LETTER SHIFT			S	M	M	M	M	M	M	

NOTE: The column on the right side lists only those upper (figure) case characters of ITA No. 2 which differ from the American version. All lower (letter) case characters of ITA No. 2 are identical to the American version.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

The ITA No. 4 is a six-unit code for synchronous transmission over two or three telegraph channels derived by TDM from one VF channel. The ITA No. 4 is standardized in CCITT Recommendation R.44, published in Volume VII, Fascicle VII.1, of the Yellow Books, Geneva, 1981.

The International Alphabet No. 5 is a seven-unit code to be used for the general exchange of information among telegraph systems, ADP systems and associated equipment. The American version of the International Alphabet No. 5 is the ASCII. (See 5.3.7.) The International Alphabet No. 5 is standardized in CCITT Recommendation V.3, published in Volume VIII, Fascicle VIII.1, of the Yellow Books, Geneva, 1981.

80.4 American standard code for information interchange (ASCII). The ASCII, as defined in FIPS PUB 1 and shown in Table IX, is the American version of the International Alphabet No. 5. The ASCII is the Federal and DoD standard for teleprinters and other character conscious data equipment for the interchange of common user information for synchronous and asynchronous transmission.

80.4.1 Description of ASCII control characters. Most of the control characters of the basic ASCII character set, listed in Tables IX and X, are classified as one of three types: Communication control characters, format effector characters and information separator characters. All ASCII control characters and the graphic character of Space (SP) are described in 80.4.1.1 through 80.4.1.6.

80.4.1.1 Communication control (CC) characters. These are functional characters intended to control or facilitate transmission of information over communications networks.

80.4.1.1.1 Start of heading (SOH). SOH is a communication control character used at the beginning of the sequence of characters which constitute a machine-sensible address or routing information. Such a sequence is referred to as the "heading". An STX character (see 80.4.1.1.2) has the effect of terminating a heading.

80.4.1.1.2 Start of text (STX). STX is a communication control character which precedes a sequence of characters that is to be treated as an entity and entirely transmitted through to the ultimate destination. Such a sequence is referred to as "text". STX may be used to terminate a sequence of characters started by SOH. (See 80.4.1.1.1.)

80.4.1.1.3 End of text (ETX). ETX is a communication control character used to terminate a sequence of characters started with STX (see 80.4.1.1.2) and transmitted as an entity.

80.4.1.1.4 End of transmission (EOT). EOT is a communication control character used to indicate the conclusion of a transmission, which may have contained one or more texts and any associated headings.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

80.4.1.1.5 Enquiry (ENQ). ENQ is a communication control character used in data communications systems as a request for a response from a remote station. ENQ may be used as a "Who Are You" (WRU) to obtain identification, or may be used to obtain station status, or both.

80.4.1.1.6 Acknowledge (ACK). ACK is a communication control character transmitted by a receiver as an affirmative response to a sender.

80.4.1.1.7 Data link escape (DLE). DLE is a communication control character which will change the meaning of a limited number of contiguously following characters. DLE is used exclusively to provide supplementary controls in data communication networks. DLE is usually terminated by an SI character (see 80.4.1.4.4).

80.4.1.1.8 Negative acknowledge (NAK). NAK is a communication control character transmitted by a receiver as a negative response to the sender.

80.4.1.1.9 Synchronous idle (SYN). SYN is a communication control character used by a synchronous transmission system in the absence of any other character to provide a signal from which synchronism may be achieved or retained.

80.4.1.1.10 End of transmission block (ETB). ETB is a communication control character used to indicate the end of a block of data for communication purposes. ETB is used for grouping data where the block structure is not necessarily related to the processing format.

80.4.1.2 Format effector (FE) characters. These are functional characters which control the layout or positioning of information in printing or display devices.

80.4.1.2.1 Backspace (BS). BS is a format effector which controls the movement of the printing position one printing space backward on the same printing line. (Applicable also to display devices.)

80.4.1.2.2 Horizontal tabulation (HT). HT is a format effector which controls the movement of the printing position to the next in a series of predetermined positions along the printing line. (Applicable also to display devices and the skip function on punched cards.)

80.4.1.2.3 Vertical tabulation (VT). VT is a format effector which controls the movement of the printing position to the next in a series of predetermined printing lines. (Applicable also to display devices.)

80.4.1.2.4 Line feed (LF). LF is a format effector which controls the movement of the printing position to the next printing line. (Applicable also to display devices.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

NOTE: In commercial applications, LF may also have the meaning "New Line (NL)", a format effector which controls the movement of the printing point to the first printing position on the next printing line. Use of this convention requires agreement between sender and receiver.

80.4.1.2.5 Form feed (FF). FF is a format effector which controls the movement of the printing position to the first predetermined printing line on the next form or page. (Applicable also to display devices.)

80.4.1.2.6 Carriage return (CR). CR is a format effector which controls the movement of the printing position to the first printing position on the same printing line. (Applicable also to display devices.)

80.4.1.2.7 Space (SP). See 80.4.1.6.

80.4.1.3 Information separator (IS) characters. These are characters used to separate and qualify information in a logical sense. There is a group of four such characters, which are to be used in a hierarchical order. These characters are: File separator (FS), group separator (GS), record separator (RS) and unit separator (US). These information separators may be used within data in optional fashion, except that their hierarchical relationship is as follows: FS is the most inclusive, then GS, then RS, and US is least inclusive. The content and length of a file, group, record or unit are not specified.

80.4.1.4 Other control characters.

80.4.1.4.1 Null (NUL). NUL is the all-zeros character which may serve to accomplish time fill and media fill.

80.4.1.4.2 Bell (BEL). BEL is a control character which may be used when there is a need to call for human attention. BEL may control alarm or attention devices.

80.4.1.4.3 Shift out (SO). SO is a control character indicating that the code combinations which follow are to be interpreted as outside of the character set of the standard code table until an SI character (see 80.4.1.4.4.) is reached.

80.4.1.4.4 Shift in (SI). SI is a control character indicating that the code combinations which follow are to be interpreted according to the standard code table.

80.4.1.4.5 Device control (DC). There are four control characters, labeled DCI, DC2, DC3 and DC4, which are used to control ancillary devices associated with data processing or telecommunication systems, more especially switching devices "on" or "off". If a single "stop" control is

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

required to interrupt or turn off ancillary devices, DC4 is the preferred assignment.

80.4.1.4.6 Cancel (CAN). CAN is a control character used to indicate that the data with which CAN is sent is in error or is to be disregarded.

80.4.1.4.7 End of medium (EM). EM is a control character associated with the sent data which may be used to identify the physical end of the medium, or the end of the used, or wanted, portion of information recorded on a medium. The position of this character does not necessarily correspond to the physical end of the medium.

80.4.1.4.8 Substitute (SUB). SUB is a control character that may be substituted for a character which is determined to be invalid or in error.

80.4.1.4.9 Escape (ESC). ESC is a control character intended to provide code extension (supplementary characters) in general information interchange. ESC itself is a prefix affecting the interpretation of a limited number of contiguously following characters. ESC is usually terminated by an SI character (see 80.4.1.4.4).

80.4.1.5 Delete (DEL) character. This character is used primarily to "erase" or "obliterate" erroneous or unwanted characters in perforated tape. In the strict sense, DEL is not a control character.

80.4.1.6 Space (SP) character. SP is a normally nonprinting graphic character used to separate words. SP is also a format effector which controls the movement of the printing position, one printing position forward. (Applicable also to display devices.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX I

TIMING AND SYNCHRONIZATION OF COMMUNICATIONS NETWORKS.

This Appendix contains information related to MIL-STD-188-200. Appendix I is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph	Title	Page
90	Timing and synchronization of communications networks	I-333
90.1	General	I-333
90.2	Timing and synchronization techniques	I-333
90.2.1	Independent atomic clock technique	I-334
90.2.1.1	Network timing and synchronization	I-334
90.2.1.2	Advantages and disadvantages of independent atomic clock technique	I-335
90.2.2	Master-slave technique	I-335
90.2.2.1	Network timing and synchronization	I-335
90.2.2.2	Advantages and disadvantages of master-slave technique	I-336
90.2.3	Frequency averaging technique	I-336
90.2.3.1	Network timing and synchronization	I-336
90.2.3.2	Advantages and disadvantages of frequency averaging technique	I-337
90.2.4	Bit stuffing technique	I-337
90.2.4.1	Network timing and synchronization	I-337
90.2.4.2	Advantages and disadvantages of bit stuffing technique	I-338

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

90 Timing and synchronization of communications networks.

90.1 General. Network timing and synchronization are critical design areas since the tactical communications objectives reflect a need for automated digital systems which are specifically designed to accommodate both voice and data traffic and to provide communications security on an end-to-end basis. For satisfactory system operation, stringent requirements must be satisfied with regard to the timing variations which are allowable both within and among the digital bit streams handled by networks. Several different synchronous and asynchronous transmission concepts can be employed to provide network synchronization. Independent stable clocks and bit stuffing represent two possible asynchronous transmission concepts. Master-slave and frequency averaging represent two possible synchronous transmission concepts. The basic requirement concerning network synchronization relates to the need for a system to maintain character-count and bit-count integrity in order to provide end-to-end communications security and an acceptable data transmission capability.

In a digital system, all switching operations within a node are timed at a local clock rate so that processor operation is synchronous. Since the incoming bit streams from other nodes may differ slightly in the actual transmission rate, it is necessary to regenerate the received signals to avoid losing information. Data is read into the retiming unit at the incoming rate and transferred out at the rate determined by the local clock. If the incoming rates and the local rate are the same, then the retiming unit is needed only for the proper phasing of the sampling times. Since the rates of all links are probably not identical, data can accumulate in the stores associated with the retiming units. From a practical viewpoint, the link rates may differ instantaneously due to propagation variations even if the links have the same long term average value. Consequently, buffers would be required in any case to compensate for large transient disturbances. When buffers are employed, it is necessary to reset the stores when overflow or depletion occurs. To minimize the operational impact, the buffers should be sized so that no more than one reset is required during any one 24-hour period.

90.2 Timing and synchronization techniques. Four basic and qualitatively different network timing and synchronization techniques considered useable in a digital system are as follows:

a. Independent atomic clocks
b. Master-slave
c. Frequency averaging
d. Bit stuffing

The independent atomic clock approach appears to be well suited to the tactical application because of its operational simplicity, jamming immunity and network flexibility which result from not depending on external timing signals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

90.2.1 Independent atomic clock technique.

90.2.1.1 Network timing and synchronization. With the independent clock approach, network timing is established by employing a frequency standard at each node to provide a stable local timing reference. This technique is inherently asynchronous since the frequency standards will vary slightly from node to node. The differences between the basic bit rates of the incoming trunk groups and that of the local timing source are accommodated through employment of buffers which are used to retime the incoming trunk traffic. To minimize the differences among timing sources, atomic clocks are used to implement the independent clock approach.

Atomic clocks are generally categorized in three groups based upon the nature of their construction; namely, atomic beam tubes, resonance cells and masers. Atomic clocks operate on the principle that the transition between two hyperfine energy states in an atom will emit or absorb energy at an extremely accurate wavelength. The energy transition of the atom is caused by the application of an electromagnetic field at a frequency which is the same as the transition frequency of the atom. The energy in these transitions is used to control the frequency of crystal oscillators via feedback control circuits.

The following atomic clocks have been developed and are considered to be operational:

a. Cesium beam clock.
b. Rubidium resonance cell clock.
c. Hydrogen maser clock.

The cesium beam and the rubidium resonance cell offer the greatest promise for application in an independent clock network timing system. The hydrogen maser clock was rejected as a candidate frequency standard for tactical applications due to its high cost, poor reliability, large support requirements and size/weight disadvantages.

Cesium beam clocks, which can operate at a theoretically predictable frequency with a high degree of accuracy (1×10^{-11}), are primary standards and no other reference is required for calibration. Cesium beam clocks are used wherever highly precise and accurate frequency and time standards are required. These clocks are currently employed as the basis for the United States frequency standard.

Rubidium clocks are incapable of operating at a theoretically predictable frequency, since there is a significant frequency offset from the natural transition frequency. This frequency offset is attributed mainly to collisions of the rubidium atoms within the resonance cell. Since the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

frequency offset may vary depending on the physical and electrical characteristics of the clock, rubidium clocks must be calibrated periodically with a known frequency standard; therefore, these clocks are referred to as secondary frequency standards.

The basic elements of the independent clock network timing approach include a modem, an m-bit buffer and an atomic frequency standard. The modem is used to regenerate the incoming digital bit stream. The atomic clock is used to provide a stable local clock rate. The buffer (m-bit capacity) is used to absorb differences between the frequency of the local clock at the receiving node and the frequency of the incoming digital bit stream.

The buffer is an asynchronous first-in, first-out shift register. With asynchronous operation, data can be entered and withdrawn at different rates. Since the output rate of the buffer is controlled by the local clock, the incoming digital bit stream is retimed in accordance with the frequency of the local clock.

The apparent basic clock frequency of the incoming digital bit stream can differ from that of the frequency of the local nodal clock due to the inherent differences in the clock frequencies at the two nodes and the special properties of the internodal transmission links. These factors, as well as the link transmission rate and the allowable buffer reset period, dictate the capacity requirements for the buffer.

90.2.1.2 Advantages and disadvantages of independent atomic clock technique. The independent atomic clock technique provides frequency stability which is not affected by system operation or by any external signal source. The stability is, therefore, immune to jamming signals. The approach is not susceptible to transient disturbances in the network frequency. Buffers must be employed on each trunk to accommodate timing differences between the sending and receiving nodes and to accommodate frequency variations caused by the transmission media. The buffers must be periodically reset due to long-term accumulated phase differences. A major advantage of the approach is that the effects of malfunctions in links and nodes are not propagated throughout the network. Cesium beam clocks require only routine operational checks. Rubidium resonance cell clocks must be recalibrated periodically using a primary frequency standard.

90.2.2 Master-slave technique.

90.2.2.1 Network timing and synchronization. With the master-slave technique, network synchronization is established by transmitting a reference timing signal from a master node to all network nodes that are slaved to that node. Ideally, the slaved nodes then operate at the same frequency and a synchronous system would be established.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

The basic elements of a master-slave configuration are a data modem, an m-bit buffer and a phase-locked loop. The modem is used to regenerate the incoming digital bit stream. The buffer is required for phasing and to compensate for propagation delay variations that may exist in the transmission media between the master and the slaved equipment. Normally a one or two-bit buffer is adequate for this purpose. The buffer is located at the master node. The local clock rate at each node is derived from a phase-locked loop which is slaved to the stable frequency standard located at the master node.

To achieve network synchronization, the slaved node will track the reference timing signal generated by the master node. The slaved equipment will derive this reference (master) timing from the received data signal which contains the reference timing as an integral part of its digital waveform. The clock, derived from the incoming signal at the slave node, is used to regenerate and transfer received data. The transmit timing source at the slave node is also phase-locked to the reference timing recovered by the receiver.

90.2.2.2 Advantages and disadvantages of master-slave technique. The primary advantage of a master-slave technique is its low cost relative to other techniques. The approach also provides an inherent time division multiple access (TDMA) capability and facilitates the implementation of coherent detection. The need for large buffers is also eliminated. The two major disadvantages of a master-slave technique are its susceptibility to jamming and its vulnerability due to the potential loss or failure of the master node or the links used to transmit the reference timing signal.

90.2.3 Frequency averaging technique.

90.2.3.1 Network timing and synchronization. Frequency averaging is a network synchronization technique in which a local clock at each node adjusts the clock frequency to the average frequency of the digital bit streams received from the nodes to which that node is connected. Using this scheme, all clocks in the network are assigned equal weight in determining the ultimate network frequency, since there is no reference clock. The frequency averaging technique involves the establishment of a reference phase at each node by averaging the phases of the local clock and all incoming trunk group bit streams. Frequency averaging is a synchronous transmission technique since ideally all nodal frequencies should have the same average value. Buffer storage is required on each link to compensate for dynamic variations in propagation delay and to accommodate the differences between the local clock rate and that of the distant node.

To implement frequency averaging, each internodal link terminated at a node would be equipped with its own phase-locked loop which is designed to track the incoming bit stream to within a small fraction of a bit phase

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

error. The local clock is used for regeneration of incoming bit streams, frame synchronization, and readout of bits to the retiming buffer store. Buffer storage is necessary to accommodate the accumulation of input bits when the input rate is slightly above the local clock rate, or to supply bits when the input rate is less than the local clock rate.

The regenerated input bit stream is shifted through the buffer at the input clock rate for that link. The rate of extracting the data is the local clock rate, and the position for readout of the bit stream from the shift register is controlled by continuously monitoring the phase changes between the input clock rate and the local clock rate. When the transmission link is initially established, the point at which bits are readout from the retiming buffer is set at the center of the register thereby allowing the buffer to compensate for positive or negative phase changes.

90.2.3.2 Advantages and disadvantages of frequency averaging technique. The major advantage of the frequency averaging technique is the relatively small size of equipment required to implement this technique. Implementation costs would fall between the higher cost of the independent stable clock technique and the lower cost of the master-slave technique. Although the frequency averaging technique is considered to represent a plausible method for network synchronization, the technique does possess certain major disadvantages. Loss of a link, or jamming of a link in the network will introduce a transient disturbance which will be propagated throughout the network. Further, there are design limitations which require parameters such as gain, propagation delay and bandwidth to be maintained within certain bounds. Excessive gain-delay products and improper choice of loop bandwidths can produce instabilities in the network. Parameter bounds are a function of network size and it is difficult to establish these bounds for a large network and also difficult to maintain these bounds for a tactical network which may frequently change the size and configuration.

90.2.4 Bit stuffing technique.

90.2.4.1 Network timing and synchronization. Bit stuffing is a synchronization technique which has been found to be effective in point-to-point communication links. However, the technique has distinct disadvantages when employed in a network configuration.

With bit stuffing, dummy bits are added to all synchronous inputs received at a node which have rates slightly lower than the outgoing rate of the node to make the outgoing rate equal to the rate of the local clock. The additional bits or stuff bits are inserted at prearranged locations in the bit stream so that the stuff bits can be removed at some later time. The location of each stuff bit is signaled via a separate channel. Since correct removal of all stuff bits is essential, an effective error control must be provided for the signaling channel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

To provide synchronization on a network basis, bit deletion is also required to accommodate input rates which are received at rates slightly higher than the rate of the local clock. The deleted bits, along with information to identify the exact locations at which the bits should be inserted, must be transmitted in an error protected manner to the distant node via the separate signaling channel.

90.2.4.2 Advantages and disadvantages of bit stuffing technique. A distinct disadvantage of the bit stuffing technique is the amount of processing required. At the transmitter end, processing is required to determine where each stuff bit should occur and to signal the locations of the stuff bit to the receiving end. At the receiving end, processing is required to read the location of the stuff bits. The technique is inherently complex for a network configuration. The technique is also expensive to implement and the cost depends on the volume of traffic in the network. Another disadvantage of this technique is that jitter will be introduced. The effect of jitter build-up in a complex network is not easily analyzed and may result in uncontrolled loss of synchronization.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX J

SPECTRUM DESIGNATION, EMISSIONS AND NECESSARY BANDWIDTHS.

This Appendix contains information in support of MIL-STD-188-200. Paragraph 100.1 of this Appendix contains a mandatory requirement. The remainder of Appendix J is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph	Title	Page
100	Spectrum designation, emissions and necessary bandwidths	J-341
100.1	Spectrum designation	J-341
100.2	Designation of emissions	J-342
100.3	Necessary bandwidth	J-342
100.4	Classification	J-342
100.4.1	Basic characteristics	J-342
100.4.1.1	First symbol - type of modulation of the main carrier	J-343
100.4.1.2	Second symbol - nature of signal(s) modulating the main carrier	J-344
100.4.1.3	Third symbol - type of information to be transmitted	J-344
100.4.2	Optional characteristics	J-345
100.4.2.1	Fourth symbol - details of signal(s)	J-345
100.4.2.2	Fifth symbol - nature of multiplexing	J-346
100.5	Classification symbols	J-346

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

100 Spectrum designation, emissions and necessary bandwidths. The content of this appendix has been based on FED-STD-1037 and the 1979 edition of the ITU Radio Regulations.

100.1 Spectrum designation. The radio spectrum is subdivided into frequency bands which shall be designated in accordance with FED-STD-1037 and as shown in Table LXVII. Frequencies shall be expressed:

a. In kilohertz (kHz) up to and including 3000 kHz;

b. In megahertz (MHz) thereafter up to and including 3000 MHz;

c. In gigahertz (GHz) thereafter up to and including 3000 GHz.

Where adherence to these provisions would introduce difficulties, for example, in connection with the notification and registration of frequencies, the lists of frequencies and related matters, reasonable departures may be made.

TABLE LXVII. Spectrum designation of frequency.

FREQUENCY RANGE (Lower Limit Exclusive, Upper Limit Inclusive)	US DESIGNATION	ITU DESIGNATION
Below 300 Hz	ELF (Extremely Low Frequency)	-
300 Hz - 3 kHz	ILF (Infra Low Frequency)	-
3 kHz - 30 kHz	VLF (Very Low Frequency)	4
30 kHz - 300 kHz	LF (Low Frequency)	5
300 kHz - 3000 kHz	MF (Medium Frequency)	6
3000 kHz - 30 MHz	HF (High Frequency)	7
30 MHz - 300 MHz	VHF (Very High Frequency)	8
300 MHz - 3000 MHz	UHF (Ultra High Frequency)	9
3000 MHz - 30 GHz	SHF (Super High Frequency)	10
30 GHz - 300 GHz	EHF (Extremely High Frequency)	11
300 GHz - 3000 GHz	THF (Tremendously High Frequency)	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

100.2 Designation of emissions. The ITU at the 1979 WARC adopted an international standard method of forming radio emission designators. This method is contained in the ITU Radio Regulations and is effective as of 1 January 1982; it supersedes old designators which are shown in this appendix for reference only. Emissions will be designated according to their necessary bandwidth and their classification symbol as prescribed by the ITU and outlined in this appendix.

100.3 Necessary bandwidth. For a given class of emission the necessary bandwidth is the width of the frequency band which is just sufficient to ensure the transmission of information at the rate and with the quality required under specified conditions. When used, the necessary bandwidth is determined by computation in accordance with recommendations of the International Radio Consultative Committee (CCIR), or by measurement. However, the necessary bandwidth so determined is not the only characteristic of an emission to be considered in evaluating the interference that may be caused by that emission.

a. The necessary bandwidth is added just before the classification symbols and is expressed by three numerals and one letter. The letter occupies the position of the decimal point and represents the unit of bandwidth. The first character will be neither zero nor K, M or G.

b. Necessary bandwidths are expressed as follows:

(1) between .001 and 999 Hz, expressed in Hz (letter H)
(2) between 1.00 and 999 kHz, expressed in kHz (letter K)
(3) between 1.00 and 999 MHz, expressed in MHz (letter M)
(4) between 1.00 and 999 GHz, expressed in GHz (letter G)

Examples:

0.002	Hz	= H002	6.0	kHz	= 6K00	1.25	MHz	= 1M25
0.1	Hz	= H100	12.5	kHz	= 12K5	2.00	MHz	= 2M00
25.3	Hz	= 25H3	180.4	kHz	= 180K	10.00	MHz	= 10M0
400.0	Hz	= 400H	180.5	kHz	= 181K	5.65	GHz	= 5G65

100.4 Classification. Emissions are classified and symbolized according to their characteristics. Optional characteristics may be used. Normally, only the basic characteristics are required.

100.4.1 Basic characteristics. The basic characteristics are:

First symbol - type of modulation of the main carrier.
Second symbol - nature of signal(s) modulating the main carrier.
Third symbol - type of information to be transmitted.

NOTE: Modulation used only for short periods and for incidental purposes (such as, for identification or calling) may be ignored provided that the necessary bandwidth as indicated is not thereby increased.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

100.4.1.1 First symbol - type of modulation of the main carrier.

a.	Emission of an unmodulated carrier	N
b.	Emission in which the main carrier is amplitude-modulated (including cases where sub-carriers are angle-modulated)	
	(1) Double-sideband	A
	(2) Single-sideband, full carrier	H
	(3) Single-sideband, reduced or variable level carrier	R
	(4) Single-sideband, suppressed carrier	J
	(5) Independent sidebands	B
	(6) Vestigial sideband	C
c.	Emission in which the main carrier is angle-modulated	
	(1) Frequency modulation	F
	(2) Phase modulation	G
d.	Emission in which the main carrier is amplitude and angle-modulated either simultaneously or in a pre-established sequence	D
e.	Emission of pulses (Emissions, where the main carrier is directly modulated by a signal which has been coded into quantized form (e.g., PCM), should be designated under 100.4.1.1b or 100.4.1.1c)	
	(1) Sequence of unmodulated pulses	P
	(2) A sequence of pulses	
	(a) modulated in amplitude	K
	(b) modulated in width/duration	L
	(c) modulated in position/phase	M
	(d) in which the carrier is angle-modulated during the period of the phase	Q

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

(e)	which is a combination of the foregoing or is produced by other means	V
(f)	cases not covered above, in which an emission consists of the main carrier modulated, either simultaneously or in a pre-established sequence, in a combination of two or more of the following modes: amplitude, angle or pulse	W
(g)	Cases not otherwise covered	X

100.4.1.2 Second symbol - nature of signal(s) modulating the main carrier.

a.	No modulating signal	Ø
b.	A single channel containing quantized or digital information without the use of a modulating subcarrier (This excludes TDM)	1
c.	A single channel containing quantized or digital information with the use of a modulating subcarrier (This excludes TDM)	2
d.	A single channel containing analog information	3
e.	Two or more channels containing quantized or digital information	7
f.	Two or more channels containing analog information	8
g.	Composite system with one or more channels containing quantized or digital information together with one or more channels containing analog information	9
h.	Cases not otherwise covered	X

100.4.1.3 Third symbol - type of information to be transmitted. In this context the word "information" does not include information of a constant unvarying nature, such as provided by standard frequency emissions, continuous wave and pulse radars, etc.

a.	No information transmitted	N
b.	Telegraphy - for aural reception	A
c.	Telegraphy - for automatic reception	B

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

d.	Facsimile	C
e.	Data transmission, telemetry, telecommand	D
f.	Telephony (including sound broadcasting)	E
g.	Television (video)	F
h.	Combination of the above	W
i.	Cases not otherwise covered	X

100.4.2 Optional characteristics. The optional additional characteristics, not normally required, are:

Fourth symbol - details of signal(s)
Fifth symbol - nature of multiplexing

100.4.2.1 Fourth symbol - details of signal(s).

a.	Two-condition code with elements of differing numbers or durations, or numbers and durations	A
b.	Two-condition code with elements of the same number and duration without error-correction	B
c.	Two-condition code with elements of the same number and duration with error-correction	C
d.	Four-condition code in which each condition represents a signal element (of one or more bits)	D
e.	Multi-condition code in which each condition represents a signal element (of one or more bits)	E
f.	Multi-condition code in which each condition or combination of conditions represents a character	F
g.	Sound of broadcasting quality (monophonic)	G
h.	Sound of broadcasting quality (stereophonic or quadraphonic)	H
i.	Sound of commercial quality (excluding categories given in subparagraphs j and k below)	J
j.	Sound of commercial quality with the use of frequency inversion or band-splitting	K

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

k.	Sound of commercial quality with separate frequency-modulated signals to control the level of demodulated signal	L
l.	Monochrome	M
m.	Color	N
n.	Combination of the above	W
o.	Cases not otherwise covered	X

100.4.2.2 Fifth symbol - nature of multiplexing.

a.	None	N
b.	Code division multiplex (this includes bandwidth expansion techniques)	C
c.	FDM	F
d.	TDM	T
e.	Combination of FDM and TDM	W
f.	Other types of multiplexing	X

100.5 Classification symbols. Classification symbols for amplitude modulated emissions, frequency and phase modulated emissions, pulse modulated emissions and other modulated emissions are shown in Tables LXVIII through LXX, respectively. These tables also show old symbols in parenthesis for reference.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXVIII. Amplitude modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
NØN (AØ)	N	Emission of an unmodulated carrier	Ø	No modulating signal	N	No information transmitted
A1A (A1)	A	Double-sideband	1	A single channel containing quantized or digital information without the use of a modulating sub-carrier	A	Telegraphy - for aural reception
A2A (A2)	A	Double-sideband	2	A single channel containing quantized or digital information with the use of a modulating sub-carrier	A	Telegraphy - for aural reception
A2D (A2)	A	Double-sideband	2	A single channel containing quantized or digital information with the use of a modulating sub-carrier	D	Data transmission, telemetry, telecommand

Source: https://assist.dla.mil - Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXVIII. (CONTINUED) Amplitude modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
R2B (A2A)	R	Single-sideband, reduced or variable level carrier	2	A single channel containing quantized or digital information with the use of a modulating subcarrier	B	Telegraphy - for automatic reception
B2B (A2B)	B	Independent sidebands	2	A single channel containing quantized or digital information with the use of a modulating subcarrier	B	Telegraphy - for automatic reception
H2B (A2H)	H	Single-sideband, full carrier	2	A single channel containing quantized or digital information with the use of a modulating subcarrier	B	Telegraphy - for automatic reception

Source: https://assist.dla.mil -- Downloaded: 202640-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXVIII. (CONTINUED) Amplitude modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
J2B (A2J)	J	Single-sideband, suppressed carrier	2	A single channel containing quantized or digital information with the use of a modulating subcarrier	B	Telegraphy - for automatic reception
A3E (A3)	A	Double-sideband	3	A single channel containing analog information	E	Telephony (including sound broadcasting)
R3E (A3A)	R	Single-sideband, reduced or variable carrier level	3	A single channel containing analog information	E	Telephony (including sound broadcasting)
B8E (A3B)	B	Independent sidebands	8	Two or more channels containing analog information	E	Telephony (including sound broadcasting)
H3E (A3H)	H	Single-sideband, full carrier	3	A single channel containing analog information	E	Telephony (including sound broadcasting)
J3E (A3J)	J	Single-sideband, suppressed carrier	3	A single channel containing analog information	E	Telephony (including sound broadcasting)

Source: https://assist.dla.mil -- Downloaded: 2026/10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXVIII. (CONTINUED) Amplitude modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
A3C (A4)	A	Double-sideband	3	A single channel containing analog information	C	Facsimile
R3C (A4A)	R	Single-sideband, reduced or variable level carrier	3	A single channel containing analog information	C	Facsimile
J3C (A4J)	J	Single-sideband, suppressed carrier	3	A single channel containing analog information	C	Facsimile
A3F (A5)	A	Double-sideband	3	A single channel containing analog information	F	Television (video)
C3F (A5C)	C	Vestigial sideband	3	A single channel containing analog information	F	Television (video)
A7B (A6)	A	Double-sideband	7	Two or more channels containing quantized or digital information	B	Telegraphy - for automatic reception

Source: https://assist.dla.mil -- Downloaded: 2026/10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXVIII. (CONTINUED) Amplitude modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
A2B (A7)	A	Double-sideband	2	A single channel containing quantized or digital information with the use of a modulating subcarrier	B	Telegraphy - for automatic reception
B7B (A7B)	B	Independent sidebands	7	Two or more channels containing quantized or digital information	B	Telegraphy - for automatic reception
J7B (A7J)	J	Single-sideband, suppressed carrier	7	Two or more channels containing quantized or digital information	B	Telegraphy - for automatic reception
J2B (A7J)	J	Single-sideband, suppressed carrier	2	A single channel containing quantized or digital information with the use of a modulating subcarrier	B	Telegraphy - for automatic reception

Source: https://assist.dla.mil -- Downloaded: 2026/10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXVIII. (CONTINUED) Amplitude modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
B7D (A9B)	B	Independent sidebands	7	Two or more channels containing quantized or digital information	D	Data transmission, telemetry, telecommand
B9W (A9B)	B	Independent sidebands	9	Composite system with one or more channels containing quantized or digital information, together with one or more channels containing analog information	W	Combination of telephony and data transmission
J7B (A9B)	J	Single-sideband, suppressed carrier	7	Two or more channels containing quantized or digital information	B	Telegraphy - for automatic reception
AXX	A	Double-sideband	X	Cases not otherwise covered	X	Cases not otherwise covered

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXIX. Frequency and phase modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
FØN (FØ)	F	Frequency modulation	Ø	No modulating signal	N	No information transmitted
F7B (F1)	F	Frequency modulation	7	Two or more channels containing quantized or digital information	B	Telegraphy - for automatic reception
F1B (F1)	F	Frequency modulation	1	A single channel containing quantized or digital information without the use of a modulating sub-carrier	B	Telegraphy - for automatic reception
F2A (F2)	F	Frequency modulation	2	A single channel containing quantized or digital information with the use of a modulating sub-carrier	A	Telegraphy - for aural reception

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXIX. (CONTINUED) Frequency and phase modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
F2D (F2)	F	Frequency modulation	2	A single channel containing quantized or digital information with the use of a modulating sub-carrier	D	Data Transmission, telemetry, telecommand
F1E (F3)	F	Frequency modulation	1	A single channel containing quantized or digital information without the use of a modulating sub-carrier	E	Telephony (including sound broadcasting)
F3E (F3)	F	Frequency modulation	3	A single channel containing analog information	E	Telephony (including sound broadcasting)
F3C (F4)	F	Frequency modulation	3	A single channel containing analog information	C	Facsimile

Source: https://assist.dla.mil -- Downloaded: 2026/0-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXIX. (CONTINUED) Frequency and phase modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
F3F (F5)	F	Frequency modulation	3	A single channel containing analog information	F	Television (video)
FXX (F9)	F	Frequency modulation	X	Cases not otherwise covered	X	Cases not otherwise covered
G7W	G	Phase Modulation	7	Two or more channels containing quantized or digital information	W	Combination of telephony and data transmission

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXX. Pulse modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
PØN (PØ)	P	A sequence of unmodulated pulses	Ø	No modulating signal	N	No information transmitted
P1B (P1)	P	A sequence of unmodulated pulses	1	A single channel containing quantized or digital information without the use of a modulating sub-carrier	B	Telegraphy - for automatic reception
K1A (P1D)	K	A sequence of pulses modulated in amplitude	1	A single channel containing quantized or digital information without the use of a modulating sub-carrier	A	Telegraphy - for aural reception
P2D (P2)	P	A sequence of unmodulated pulses	2	A single channel containing quantized or digital information with the use of a modulating sub-carrier	D	Data transmission, telemetry, telecommand

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXX. (CONTINUED) Pulse modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
L2A (P2E)	L	A sequence of pulses modulated in width/duration	2	A single channel containing quantized or digital information with the use of a modulating sub-carrier	A	Telegraphy - for aural reception
M2A (P2F)	M	A sequence of pulses modulated in position/phase	2	A single channel containing quantized or digital information with the use of a modulating sub-carrier	A	Telegraphy - for aural reception
K3E (P3)	K	A sequence of pulses modulated in amplitude	3	A single channel containing analog information	E	Telephony (including sound broadcasting)
M3E (P3F)	M	A sequence of pulses modulated in position/phase	3	A single channel containing analog information	E	Telephony (including sound broadcasting)
P7B (P6)	P	A sequence of unmodulated pulses	7	Two or more channels containing quantized or digital information	B	Telegraphy - for automatic reception

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXX. (CONTINUED) Pulse modulated emission designators.

New Symbol (Old Symbol)	First Symbol	Type of Modulation of the Main Carrier	Second Symbol	Nature of the Signal(s) Modulating the Main Carrier	Third Symbol	Type of Information to be Transmitted
KXX (P9)	K	A sequence of pulses modulated in amplitude	X	Cases not otherwise covered	X	Cases not otherwise covered
MXX (P9)	M	A sequence of pulses modulated in position/ phase	X	Cases not otherwise covered	X	Cases not otherwise covered
PXX (P9)	P	A sequence of un-modulated pulses	X	Cases not otherwise covered	X	Cases not otherwise covered
WXX	W	Cases not covered under amplitude-modulated, angle modulated or pulse modulated emissions in which an emission consists of the main carrier modulated, either simultaneously or in a pre-established sequence, in a combination of two or more of the following modes: amplitude, angle, pulse	X	Cases not otherwise covered	X	Cases not otherwise covered
XXX	X	Cases not otherwise covered	X	Cases not otherwise covered	X	Cases not otherwise covered

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX K

FDM CHANNEL LOADING.

This Appendix contains tutorial information in support of MIL-STD-188-200. Appendix K is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

110 FDM channel loading.

110.1 Voice loading (analog signals). The total signal power (or loading) applied to the input of multichannel FDM equipment is the sum of the signal powers in the individual channels. There are two factors in multichannel voice transmission that tend to reduce and stabilize the total loading as the number of channels increases. These factors are:

a. The numbers and distribution of channels actively transmitting voice; and

b. The speech volumes in the individual channels.

In determining the FDM channel loading of a multichannel voice system, considerable advantage may be taken of the statistical distribution of the voice signals. For example, the activity factor for a channel, or the percentage of time during the busiest traffic hour a channel might be active (i.e., voice is transmitted in the channel) in multichannel systems, is about 25 percent. For smaller groups of channels this percentage may be larger, but it is highly unlikely that any increase in group size would change it appreciably. Utilization of this activity factor results in a decrease of the mean power of a voice channel from -10 dBm0 (short term average of an active channel) to -16 dBm0 (long term average of an active channel).

The volumes of speech in active voice channels will also affect the total load. Speech volumes are not constant in a channel but vary considerably, depending on the characteristics of the talker's speech and the loudness of the voice. The changes in the numbers of active channels and in speech volumes concern only the maximum rms load of multichannel FDM equipment, causing more and more gradual variations as the number of channels increases. However, it is the peak input voltage applied to the multichannel FDM equipment, and not the rms input voltage, that determines whether the equipment will overload. This peak input voltage is the vector sum of the instantaneous voltages at the input of the individual channels and is a function of both the phase and amplitude of the voice signal in each channel.

In consideration of the above statistical aspects of voice and the associated probabilities, loading formulas for multichannel FDM equipment, as established by Holbrook and Dixon, are recommended by the CCIR. These formulas give the mean absolute power of the distributed speech signals that the FDM subsystem must be capable of carrying. The signal power depends on the number of FDM channels involved, and this signal power is calculated from one of the two formulas given in 5.7.3.1.1.1.

110.2 Data loading (quasi-analog signals). Telegraph and data signals are of relatively constant amplitude in contrast to the wide variations of voice signals. Thus, the average power of a telegraph and data signal is

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

continuous and imposes a greater loading on the multichannel FDM equipment than that of a voice signal which varies considerably in its amplitude and is present only part of the time. The loading imposed by telegraph and data signals is calculated from the formula given in 5.7.3.1.1.2.

In contemporary switched multichannel systems, the combination of voice, teletype and data loading (with a data loading of -10 dBm0 mean power per channel) produces a resultant composite loading level that is reasonably compatible with the specific overload characteristics of FDM equipment based on the criteria established by Holbrook and Dixon and recommended by the CCIR.

When all or nearly all of the FDM channels carry data, telegraph or facsimile signals, the resultant loading based on -10 + 10 log N (in dBm0), where N is the number of channels, is approximately 3.2 times higher than the loading recommended by the CCIR. Thus, if there were a need for 300 data channels, and only voice-type FDM and radio equipment (not designed for continuous data loading of -10 + 10 log N) is available, then the total loading must be the same as if a 960-channel voice-loaded FDM subsystem were deployed. In actual system practice, where both types of FDM and radio equipment (designed only for voice service, or for voice and data service) may be found, the data loading must be reduced from -10 dBm0 to -13 dBm0 mean power per channel in order to prevent overloading the FDM subsystem.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX L

HYPOTHETICAL REFERENCE CIRCUITS AND RELATED INFORMATION.

This Appendix contains general information in support of MIL-STD-188-200. Appendix L is not a mandatory part of this standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph	Title	Page
120	Hypothetical reference circuits and related information	L-365
120.1	General	L-365
120.2	Signal level and noise power calculations	L-365
120.2.1	Noise power units and noise weighting curves	L-365
120.2.2	Conversion formulas for noise power units	L-369
120.2.3	Adding dB values	L-369
120.2.4	TLP conversion formula	L-373
120.2.5	Neper and dB conversion formulas	L-373
120.3	Noise power calculations for hypothetical reference circuits	L-373

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

120 Hypothetical reference circuits and related information.

120.1 General. A hypothetical reference circuit is a circuit of specified configuration and length, and a defined transmission characteristic which is based on such factors as communications requirements, user satisfaction, the measured performance of equipment in the inventory, and installation and operation procedures and experience. The reference circuit configurations, such as the number of links, trunks and nodes in tandem with associated transmission equipment, are chosen so that each configuration can be considered representative of typical installed and operational circuits of the network or subsystem under consideration. The nominal length of a reference circuit normally represents the probable maximum distance over which communications are required in the network or subsystem under consideration.

A hypothetical reference circuit is used: (a) As a reference for the performance of planned or installed and operational circuits; (b) as guidance for planning and engineering of circuits and networks; (c) to prorate transmission parameters and allocate these parameters to different portions of a circuit and to associated equipment; and (d) as a basis to derive interface standards and subsystem and equipment standards.

In an installed and operational communications system, it is normally required that various circuits be employed which differ in their length and in other parameters from the reference circuits described in 5.9.4.2 through 5.9.4.7. It is impractible to standardize the performance of every link, trunk or circuit that may have to be engineered and installed. The purpose of standardizing certain transmission parameters on a user-to-user basis and to indicate what can be achieved by defining hypothetical reference circuits, is to establish a grade of system design which will ensure that actual links, trunks and circuits based on such a design will perform satisfactorily as parts of an overall subsystem or system.

Designers and circuit engineers are expected to make their own assumptions and decide on such factors as length of repeater sections, amount and effect of fading, and provisions to minimize or control circuit noise so that a similar performance is obtained from actual circuits as from corresponding hypothetical reference circuits given in 5.9.4.2 through 5.9.4.7.

120.2 Signal level and noise power calculations.

120.2.1 Noise power units and noise weighting curves. Several noise power units have been established over the years to measure the quality of voice signals transmitted over VF circuits. These noise power units have been based on different noise weighting curves (see Figure 71) which have been developed to describe the performance of certain types of telephone sets. Measurements made with the Western Electric Company type 144 handset resulted in a 144 noise weighting curve and in noise power units expressed

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T07:28Z
Check the source to verify that this is the currentversion before use.

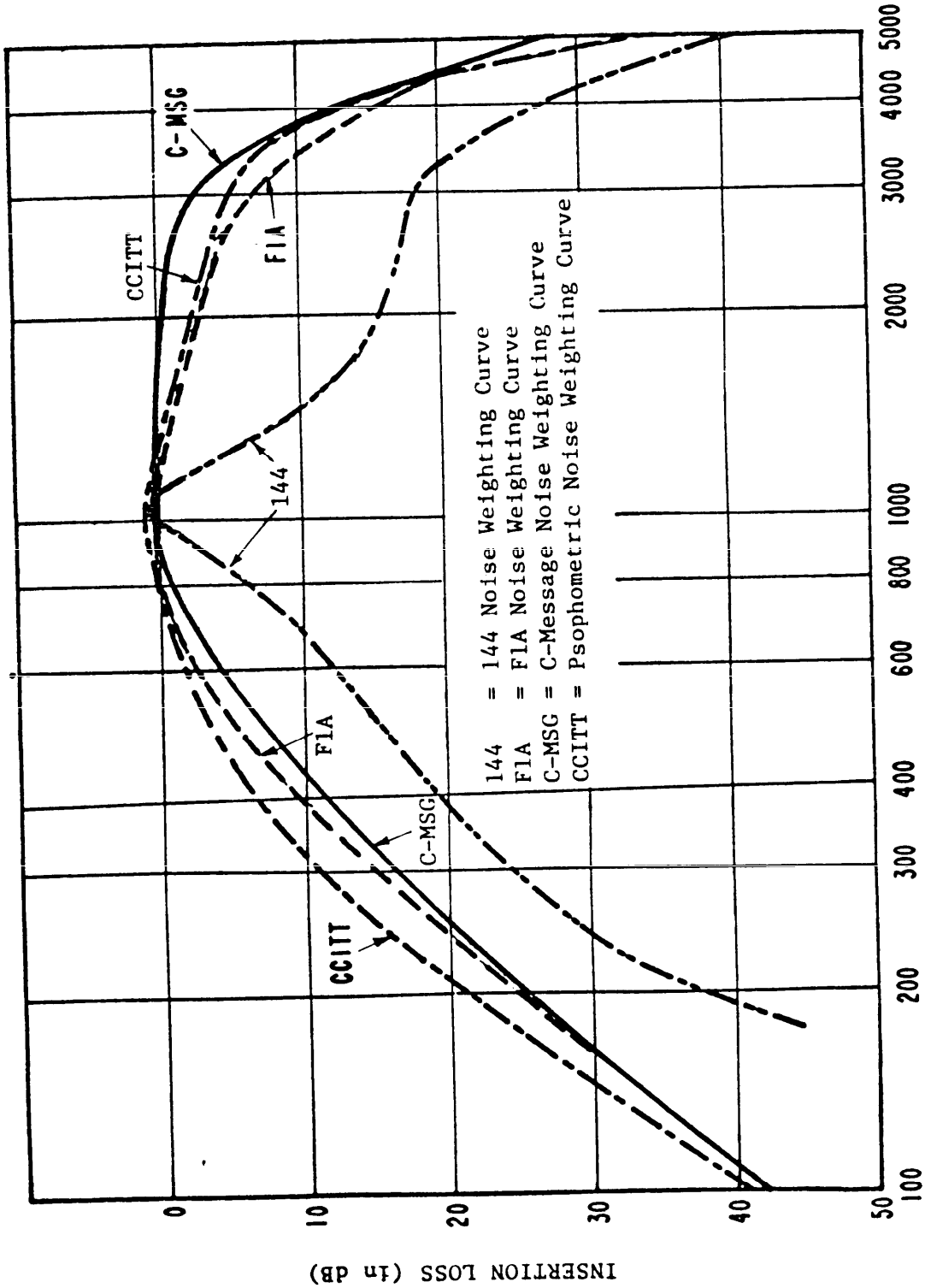


FIGURE 71. Noise weighting curves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

in dBrn(144). The unit of dBrn(144) has been referenced to -90 dBm in order to obtain positive values for all noise power levels likely to be encountered in VF circuits. A Western Electric type 2A or equivalent noise measuring set uses the 144 noise weighting curve and is calibrated in dBrn(144) units. A sinusoidal 1000 Hz signal with a power level of 0 dBm (1 mW) will read +90 dBrn(144) on a type 2A or equivalent noise measuring set.

Later, an improved telephone set, the type 302, was developed which used a type F1 transmitter and type HA1 receiver. The type 302 telephone set, which has a better frequency response than the older type 144, required a new noise weighting curve. This new weighting curve, called F1A weighting, resulted in noise power readings of about 5 dB higher than the older 144 noise weighting curve. Therefore, a new noise power unit, called dB adjusted (dBa), was introduced and referenced to -85 dBm to be compatible with the older dBrn(144) unit. A Western Electric type 2B or equivalent noise measuring set which uses the F1A noise weighting curve is calibrated in dBa units. A sinusoidal 1000 Hz signal with a power level of 0 dBm (1mW) will read +85 dBa on a type 2B or equivalent noise measuring set with the F1A noise weighting curve. Noise measurements made in dBa units using either the 144 or F1A noise weighting curves represent the same amount of noise power and have the same interfering effect on a telephone circuit. Also, dBa and dBrn(144) readings using the 144 noise weighting curve are numerically equal since the difference between dBa and dBrn(144) units has been eliminated by the different reference levels for these noise power units.

Further improvements in telephone set performance resulted in the 500-type telephone set with a new noise weighting curve called C-message weighting. Measurements made with C-message weighting are expressed in units of dBrnc which is referenced to -90 dBm. It should be emphasized that dBrnc is not identical to dBrn(144), although both units have the same reference level of -90 dBm, since the 144 and the C-message weighting curves are different. (See Figure 71.) Another weighting curve called psophometric weighting, which is similar to the C-message weighting curve, has been standardized by the CCITT and is published in Recommendation P.53, contained in Volume V of the Yellow Books, Geneva, 1981. Measurements made with a psophometric weighting curve are expressed in units of pWp or dBmp. The psophometric unit of pWp is not referenced to a reference level, such as -85 dBm for dBa or -90 dBm for dBrn(144) and dBrnc, rather this unit represents absolute measurements of noise power.

Figure 71 shows the insertion loss versus frequency characteristics of the 144, F1A, C-message and psophometric noise weighting networks. The characteristics of the 144, F1A and C-message noise weighting networks have a reference frequency of 1000 Hz, whereas the characteristic of the psophometric noise weighting network has a reference frequency of 800 Hz. At the reference frequency the insertion loss of a noise weighting network is 0 dB. Table LXXI summarizes the output power levels of the four noise weighting networks when a sinusoidal signal or bandlimited white noise is applied to the input of the network. The insertion loss of the four

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXXI. Characteristics of noise weighting networks.

Noise Weighting Curve (See Figure 71)	Unit(s) of Measurement	Reference Frequency (0 dB of attenuation) of Noise Weighting Curve	The following readings are obtained at the network output if the type of signal shown with a power of 0 dBm (1 mW) is applied to the input of the network having a noise weighting curve as listed in left column:		
			Single Frequency		White Noise in 300 Hz to 3400 Hz Frequency Band
			800 Hz	1000 Hz	
144	dBrn(144)	1000 Hz	84 dBrn(144)	90 dBrn(144) (0 dBm)	81.5 dBrn(144) (82 dBrn(144))*
F1A	dBa	1000 Hz	84 dBa	85 dBa (0 dBm)	81.5 dBa (82 dBa)*
C-Message	dBrnc	1000 Hz	89 dBrnc	90 dBrnc (0 dBm)	88.5 dBrnc (88 dBrnc)*
Psophometric	dBmp, pWp	800 Hz	0 dBmp (0 dBm)	1 dBmp	-2.5 dBmp (-2 dBmp)*
Flat (No Weighting)	dBm, dBrn, pW, mW	not applicable	0 dBm	0 dBm	0 dBm

*Rounded-off values used in other publications.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T07:28Z
Check the source to verify that this is the current version before use.

noise weighting networks is different for different input frequencies and noise power spectra since the shape and reference frequency of the corresponding noise weighting curves are different. Also, a flat weighting has been included in Table LXXI as well as a column listing the noise power units corresponding to the different noise weighting networks.

120.2.2 Conversion formulas for noise power units. The differences in shape and reference frequency among the noise weighting curves shown in Figure 71 make it difficult to derive exact conversion factors, valid for all types of signals and noise power spectra, to convert measurement results made on a VF circuit with one type of noise weighting network to equivalent numbers for other types of noise weighting networks. However, for practical comparison purposes the approximations shown in Table LXXII are generally used and are considered applicable to flat (non-impulse) type of noise power spectra normally encountered in VF circuits. The conversion factors listed in Table LXXII are based on the values listed in Table LXXI and may differ by up to 1 dB as compared with values published elsewhere, depending on the amount of rounding off used. For example, the CCITT recommends the following rounded off conversion factors: 0 dBa = 6 dBrnc = -84 dBmp.

For converting from x units of pW or pWp to y units of dBrn or dBrnc the following conversion formulas have been used in MIL-STD-188-200:

y (in dBrn) = 10 log x (in pW)

y (in dBrnc) = 10 log x (in pWp)

Example: Convert -51 dBm of noise power measured with flat weighting (from about 300 Hz to 3400 Hz) at the output of a VF circuit to a noise power value expressed in dBrnc that would have been measured in the same VF circuit with a C-message noise weighting network. According to Table LXXII, 0 dBm equals approximately 88.5 dBrnc, therefore, -51 dBm is about the same noise power as 88.5 - 51 = 37.5 dBrnc.

120.2.3 Adding dB values. The addition of two parameter values expressed in dB corresponds to the multiplication of logarithmic power ratios, according to the definition of the dB. The parameter may be a single frequency test signal level expressed in dBm or a noise power expressed in dBrn, dBrnc, etc. If noise powers expressed in dB units, such as dBm or dBrnc, are to be added then both units must be the same and both units must represent white (or Gaussian) type of noise, otherwise the following addition rules are not applicable.

One method of adding two dB values is to convert each dB value to units of pW or W, adding these units and reconverting the result to a

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

value expressed in dB. Another method of adding two dB values consists of calculating the absolute difference between the two dB values and determining a number (in dB) that depends on the absolute difference. This number (between 0 and 3 dB) is added to the larger of the two dB values to obtain the sum of the two dB values. Table LXXIII lists the absolute differences between two dB values and the corresponding number (in dB) that has to be added to the larger of the two dB values. Figure 72 depicts a curve for adding two dB values based on Table LXXIII.

TABLE LXXII. Conversion formulas for noise power units.

0 dBa	=	-81.5	dBm		
	=	-84	dBmp		
	=	0	dBrn(144)		
	=	8.5	dBrn		
	=	7	dBrnc		
	=	7.1	pW		
	=	4	pWp		
0 dBrn(144)	=	0	dBa		
	=	-81.5	dBm		
	=	-84	dBmp		
	=	8.5	dBrn		
	=	7	dBrnc		
	=	7.1	pW		
	=	4	pWp		
0 dBm	=	81.5	dBa		
	=	-2.5	dBmp		
	=	81.5	dBrn(144)		
	=	90	dBrn		
	=	88.5	dBrnc		
	=	1×10^9	pW		
	=	0.56×10^9	pWp		
0 dBrn	=	-8.5	dBa		
	=	-90	dBm		
	=	-92.5	dBmp		
	=	-8.5	dBrn(144)		
	=	-1.5	dBrnc		
	=	1	pW		
	=	0.56	pWp		
0 dBmp	=	84	dBa		
	=	2.5	dBm		
	=	84	dBrn(144)		
	=	92.5	dBrn		
	=	91	dBrnc		
	=	1.78×10^9	pW		
	=	1×10^9	pWp		
0 dBrnc	=	-7	dBa		
	=	-88.5	dBm		
	=	-91	dBmp		
	=	-7	dBrn(144)		
	=	1.5	dBrn		
	=	1.4	pW		
	=	0.8	pWp		

NOTE: The conversion formulas listed in Table LXXII are applicable only to white (Gaussian) noise and include the effect of the different shapes and reference frequencies of the noise weighting curves (see Figure 71) which are based on the characteristics of different types of telephone sets.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE LXXIII. Addition factors for calculating the sum of two dB values.

Absolute Difference (in dB) of two dB Values	Number (in dB) to be added to the larger of the two dB Values
0	3.0
1	2.5
2	2.1
3	1.8
4	1.5
5	1.2
6	1.0
8	0.6
10	0.4
12	0.3
14	0.2
16	0.1

NOTE: If the absolute difference of the two dB values to be added is greater than 16 dB, the number (in dB) to be added to the larger of the two dB values is less than 0.1 dB which may be neglected for most applications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Example: Calculate the sum of 36.5 dBrnc and 41.5 dBrnc of noise powers. The absolute difference between the two noise power values is 5 dB. Based on Table LXXIII and Figure 72, 1.2 dB has to be added to 41.5 dBrnc, the larger of the two noise power values, to obtain the sum of the two dB values. Therefore, 36.5 dBrnc + 41.5 dBrnc = 41.5 dBrnc + 1.2 dB = 42.7 dBrnc.

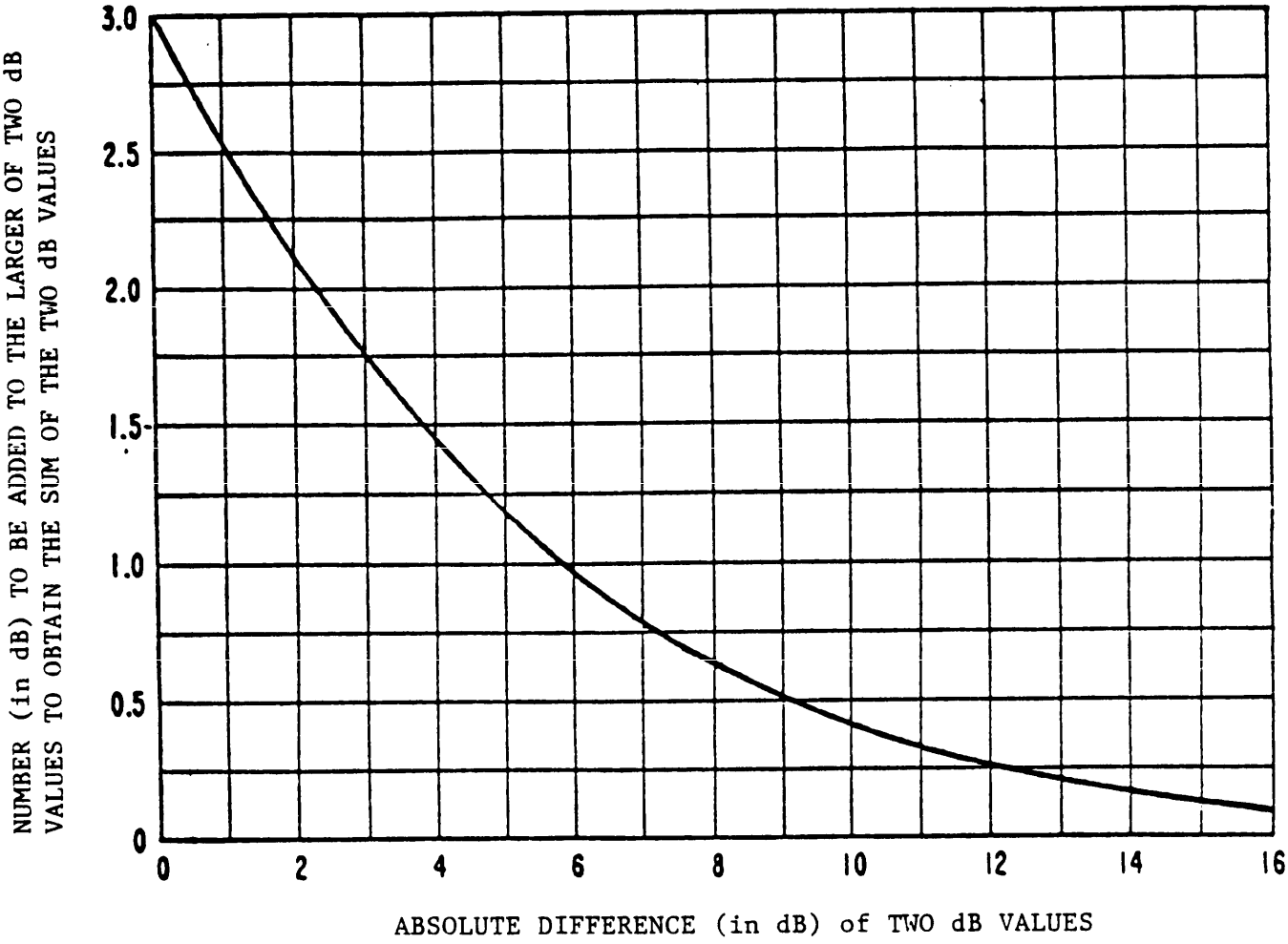


FIGURE 72. Sum of two dB values.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

120.2.4 TLP conversion formula. For converting signal levels or noise powers referenced to a OTLP and expressed in units of dBmO, dBrnO, etc., to the same unit referenced to a particular TLP (xTLP) other than a OTLP, the following formula applies:

$$\left[\begin{array}{l}\text{Signal Level or Noise Power, expressed in units of dBm, dBrn, etc., at xTLP}\end{array}\right] = \left[\begin{array}{l}\text{Signal Level or Noise Power, expressed in units of dBmO, dBrnO, etc., at OTLP}\end{array}\right] + \text{xTLP}$$

Example: Convert a signal level of -13 dBmO referenced to a OTLP to the equivalent level referenced to a -16TLP. According to the conversion formula given in 120.2.4, the signal level referenced to the -16TLP equals -13 dBmO + (-16) = -29 dBm.

120.2.5 Neper and dB conversion formulas. In European countries as well as in CCIR and CCITT documents, the unit of Neper (Np) is also used besides the unit of dB. The Np is based on the natural logarithm (with the base of e = 2.718...), whereas the dB is based on the decimal logarithm (with the base of 10). For converting from x units of Np to y units of dB, the following conversion formulas apply:

$$x \text{ (in Np)} = 8.686\ y \text{ (in dB)}$$

$$y \text{ (in dB)} = 0.115\ x \text{ (in Np)}.$$

120.3 Noise power calculations for hypothetical reference circuits. Noise power values of the various parts of a circuit, such as loops, switches, links and trunks, are normally standardized as an absolute value in terms of an upper limit of the noise power level and are not referenced to a particular circuit TLP, such as a circuit OTLP. The reason for stating an absolute noise power value is that, in general, it is not known beforehand where the various parts of a circuit are located in terms of TLP and how these parts are interconnected to establish the overall connection from the transmitting terminal to the receiving terminal. Only when the layout of a circuit has been established can the TLP be assigned to the various parts of the circuit and the total noise power of the circuit be calculated. For example, if the noise power of a loop has been measured as 6300 pWp in accordance with 5.5.2.6, this noise power value will be contributed (at the output terminals of the loop) to the total circuit noise power regardless of the particular circuit TLP assigned to the output terminals of the loop. (See Figures 51 and 52.) However, the noise power contributed by this loop and referenced to a circuit OTLP depends on the level difference between the TLP at the output terminals of the loop and the circuit OTLP according to the formula given in 120.2.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

In this document two methods have been used to calculate the total noise power of a hypothetical reference circuit. One method consists of referencing all noise power values contributed by the various parts of a circuit to a circuit OTLP and adding the noise power values at this OTLP in accordance with 5.9.3.2 to obtain the total circuit noise power (referenced to this OTLP). The total circuit noise power referenced to the circuit OTLP can then be transferred to any other TLP of interest, such as the TLP at the receiving terminal, by using the formula given in 120.2.4. This method has been used for the reference circuit examples shown in Figures 50 and 56.

The other method of calculating the total noise power of a hypothetical reference circuit consists of adding all noise power values, contributed by the various parts of the circuit, at the associated TLP of the circuit in accordance with 5.9.3.2. Starting at the transmitting terminal, the noise power values are accumulated at the circuit TLP along the direction of transmission, considering any level changes as indicated by the specific value of each TLP, until the total noise power has been calculated at the receiving terminal. This method has been used for the reference circuit examples shown in Figures 51, 52, 53, 55, 57, 58 and 59.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

INDEX

Key words are listed with reference to page numbers. A letter in front of a page number indicates an appendix; for example: "Abbreviations C-275" indicates that Abbreviations can be found in Appendix C on page number 275.

Abbreviated dialing codes 157
Abbreviations 12, C-275
Absolute envelope delay G-320
Acknowledge H-327
ACP 8, 10, 72, 118, 125, C-276
ACP 125; 10, 125
ACP 127; 10, 72, 118
ACP 127, NATO SUPP-3; 8, 72
Acronyms 12, C-275
Activity factor 166, K-360
A-D conversion 25-27, 31, 87, 161
  185, D-281 to D-292
Adding dB values L-369
ADP 87, C-276
Adjustment,
  loop loss 131, 132, F-310
  signal level 51, 52, F-311,
    F-312
  TLP 193
Agreements,
  NATO standardization 8
  quadripartite standardization 8
Alarm 115, 117
Alignment, level 157, 158
Allied communication publications
  10
Allocations, noise power 170-173
Alphabets, international 102,
  H-324
Alphabet translaters 118
Alternate mark inversion (AMI)
  C-276, E-300, E-301
AM C-276, J-343
American national standards 10
American standard code 87-102,
  C-276, H-326
AMI signal C-276, E-300, E-301
Amplitude frequency response (see:
  insertion loss versus frequency
  characteristic)
Amplitude hits 60
Amplitude modulation C-276, J-343
Analog circuit switches 144-155
Analog-digital conversion 25-27,
  31, 87, 161, 185, D-281 to D-292
Analog facsimile 63
Analog facsimile service 61
Analog interface 65
Analog pulse modulation 25, J-343
Analog signals 23
Analog telephone 63
Analog terminal equipment 63
Analog voice service 42-48
ANSI 10, C-276
ANSI X3.1; 10, 69
     X3.4; 10, 89
     X3.6; 10, 108
     X3.15; 10, 93
     X3.16; 10, 94
     X3.18; 10, 108
     X3.22; 10, 70, 108
     X3.25; 10, 96
     X3.32; 10, 90
     X3.39; 10, 70, 108, 109
     X3.41; 10, 99
     X3.54; 11, 70, 109
Application 2
ARQ 50, C-276
ARQ system 50
ASCII 6, 10, 87-102, C-276,
  H-326 to H-329
ASCII control characters 90, 91,
  H-326 to H-329
ASCII graphic characters 90, 92
ASCII subsets 97-102
Asynchronous mode 35, 75
Asynchronous operation 81
Asynchronous signal E-295, E-299,
  E-305
Atomic clock I-333 to I-335
Attack time delays 120
AUTODIN 9, 50, 72, 96, 97, C-276
AUTODIN interface 72
Automatic circuit switches 144-152,
  155
AUTOVON 144, 155-157, C-276

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

AUTOVON interconnection 144

Backspace H-327
Balanced coding E-299, E-305
Bandwidth, necessary J-342
Basic ASCII 88, 89
Baudot code H-324
Bd C-276
Bell H-328
BER 48, 49, 50, 53, 60, C-276
Bias distortion 59, 109, 115
Binary coding E-293 to E-306
Binary digit value 71
Binary signals, logic and signal sense 69-71
Binary waveforms E-293 to E-306
Biphase-L; E-297, E-298
Biphase-M; E-297, E-298
Biphase-S; E-297, E-298
Bit C-276
Bit-by-bit asynchronous mode 75
Bit-count integrity 17, 48, 51, 61, 73, I-333
BITE 19, C-276
Bit error rate (see: BER)
Bit sequencing 93
Bit stuffing I-333, I-337, I-338
Bit synchronous mode 73
Bit-to-channel relationship 96
Bonding 20, 41
b/s; C-276
Buffer I-333
Built-in test equipment 19, C-276
Busy tone 150, 151, 154

Cable transmission subsystems 129
Calculations, noise power L-365 to L-374
Cancel H-328
Card punches 63
Card readers 63
Carriage return H-328
Carrier signal J-343
Carrier suppression J-343
Carrier telegraph subsystems 109-111
CCIR C-276, H-324, J-342, K-361, L-373
CCITT C-276, E-299, E-301, H-324 H-326, L-369, L-373
Center frequencies 110, 111
Center sampling 79, 81, 84-87, 115
Channel 23
Channel interconnections 217-221
Channel interface characteristics 65-67
Channel translation equipment, FDM 162-165
Channel,
  digital transmission 217
  nominal 3 kHz 65, 109, 207
  nominal 4 kHz 65, 109, 207
  single, net radio 119
  telegraph 106, 109, 111
  telephone 105
  transmission 21, 23, 207, 214
  voice frequency 65, 109, 207
  48 kHz FDM group 67, 214
Character-count integrity 17, 48, 51, 61, I-333
Character interval control mode 74, 75
Character interval synchronous mode 74
Characteristic distortion 59
Characteristics,
  insertion loss versus frequency 44-46, 54-56, 58, 61, 105, 106, 125, 132, 134-139, 145, 174-179, 189, 190, 195-200, 210, 211, 213, 215, 216, 224, 229, 233, 235, 242, 249, 256, 260, 265, L-367
  of noise weighting networks L-368
Character parity 88
Character recognition, optical 63
Character representation 90
Character set 87-102, H-321 to H-329
Characters,
  control 90, 91, H-326 to H-329
  format effector H-327
  graphic 90, 92
  information separator H-328
Character structure (of ASCII) 93-97
Character substitutions 88
Circuit 21, 23
Circuit configuration 21

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Circuit parameters 42-62
Circuit switches 144-155
Circular, DCA 9, 44, 54, 72, 144
Classification of emissions J-342
Climatic conditions 20, 41
Clipping 187
Clock characteristics 75
Clock control,
  direct 78, 79, 82, 84-87
  indirect 78, 80, 84-87
Clock/data phase relationship 78-83
Clock equipment 72
Clocking methods 113, 114
Clock, master or station 72
Clock modulation rates 75, 116
Clock output signal 77
Clock signal cycle 77, 78
Clock signal,
  gated 81, 83
  quiescent 77, 81
  supplying of 84-87
Clock signal period 77, 78
Clock stability 75, 76
C-message weighting L-366 to L-369
CMI signal C-276, E-297 to E-299
Code, Baudot H-324
Code converters 117
Code extension techniques 99, 102
Code for perforated tape 108
Code,
  Manchester 105, E-298
  Miller E-302, E-303, E-304
  Morse H-324
Codes,
  abbreviated dialing 157
  fixed directory 157
Codes for information interchange
  87, 102, H-489
Codes,
  manual switchboard 157
  primary area 156
  subscriber number 157
  switchboard locator 157
Code word, PCM 186
Coding E-293 to E-306
Coding,
  balanced E-299, E-300, E-305
  conditioned E-300 to E-303, E-306
  differential E-301 to E-306
  level E-296, E-297, E-302 to
    E-304
Coding,
  unbalanced E-299, E-305
  unsymmetric E-299, E-305
Commonality 1
Common battery signaling 155
Common long haul/tactical
  standards 37
Communication control characters
  90, 91, H-326, H-327
Communications security 17, I-333
Communications service, parameters
  for 42-46
Communications system 1-3, 13,
  30-34, 37, 118
Companding law, PCM 186
Compandor, PCM 186
Compliance (with standards) 1-3,
  A-268
Composers, message 118
Compressor, PCM 186
Compromising emanations 18, 38
Conditioned coding E-300 to E-303,
  E-306
Conditioned diphase signal
  103-105, E-295
Conditioned diphase modulation
  103-105, E-295
Conditioned signal 103, 104,
  E-295, E-300, E-301, E-306
Conditioning 135
Conditioning, signal 103, 104,
  E-295, E-300, E-301, E-306
Condition, significant 70
Conference call 128
Configuration,
  circuit 21
  communications system 31
  data communications system 31,
    33-35
Confirmation signal 147, 148, 149,
  152
Connection,
  data signal 72
  VF channel 37, 38, 217-221
Control and timing 72
Control characters 90, 91, H-326
  to H-329
Control facilities 157, 169
Control supervision procedures
  146-148
Control time delays 120
Control unit 64

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Conversion, A-D 25-27, 31, 87, 161, 185, D-281 to D-292
Conversion formula,
noise power units L-369
TLP L-373
Neper to dB L-373
Converter,
modulation rate 113
telegraph code 117
cpi 70, 108, 109, C-276
cpmm 70, 108, 109, C-276
Cordless switches 152
Crosstalk 47, 192, F-311
Crosstalk coupling loss 106, 110, 111, 130, 145, 205
CVSD 27, 48, 49, 87, 161, 185, C-276, D-281
CVSD decoder D-285, D-287, D-288
CVSD encoder D-285, D-287
Cyclic distortion 59

D-A C-276, D-284
Data block H-327
Data communications equipment 31, 33
Data communications,
parallel-by-bit 72, 96
serial-by-bit 72, 93
Data communications system, configuration of 31
Data equipment 111
Data interface 67, 68
Data interface, NATO 72
Data link escape H-327
Data loading 167, K-360
Data modems 102, 106, 111, 121
Data processing system 118
Data service 49
Data signal connection 72
Data signaling rates 67, 69
Data signal level 51, 52, 65-67 109, 193, 207, 208, 214, F-311, F-312
Data terminal 63, 64
Data terminal equipment 31, 33 63, 64
Data transmission,
intermittent 81, 83
quality 106, 107
TDM/PCM 192
dB C-276
dBa C-276, L-367
dBm C-276
dBmp L-367, L-369
dBrn C-276
dBrn (144) C-276, L-367
dBrnc C-277, L-367
dc C-277
DCA iii, C-277
DCAC iii, 9, 54, 72, 144, C-277
DCAC 300-175-9; 9, 44, 54
DCAC 370-D175-1; 9, 72
DCAC 370-V175-6; 9, 144
DCA Circulars (see: DCAC)
dc closure control 125, 127
DCE 31, 33, 34, C-277
DCS iii, 44, 54, C-277
DCS technical schedule 44, 54
Defense Communications Agency iii, C-277
Defense Communications System (see DCS)
Definitions 12
Delay distortion G-315, G-317
Delay, envelope 47, 55-58, 61, 136, 138-141, 145, 176-178, 180-182, 190, 191, 197-202, 211-213, 215, 216, 224, 229, 233, 236, 242, 249, 256, 260, 265, G-319, G-320
Delay modulation E-302, E-303
Delay, phase G-317
Delay, time 47, G-317
Delay times 120
Delete character H-329
Delta modulation 26, D-281
Description of tactical communications 13
Designation of emissions J-342
Design objectives 3, 37
Design of tactical equipment 16, 17
Design requirements 38
Device control characters H-328
Dial tone 150, 153
Differential coding 103, E-301, E-303, E-305, E-306
Digital circuits, parameters for 48, 60
Digital facsimile equipment 65
Digital facsimile service 62
Digital information links 62

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Digital interface 71
Digital interface adapters 33
Digital pulse modulation 25, J-343
J-356 to J-358
Digital signals 23, 24
Digital switches 155
Digital telephone 63
Digital test signals 159
Digital transmission channel 217
Digitized voice service 42, 48, 49
Diphase modulation 103-105, E-303,
E-304
Diphase NRZ E-297, E-298
Diphase signal 103-105, E-297,
E-298, E-303, E-304
Dipulse E-300, E-301
Direct clock control 78, 79
Directives, DoD 1, 3, 9, 38, 40
Directory codes, fixed 158
Directory, telephone 155
Directory, NATO telephone 156
Discontinuities, signal 59, 60,
206, 212, F-309
Displacement, frequency 48, 59, 62,
106, 183, 191, 203, 212
Display terminals 63
Distortion 115, 117
Distortion,
bias 59, 109, 115
characteristics 59, 109
cyclic 59, 109, 115
delay G-317, G-318
envelope delay 47, 55-58, 61,
136, 138-141, 145, 176-178,
180-182, 190, 191, 197-202,
211-213, 215, 216, 224, 229,
233, 236, 242, 249, 256, 260,
265, G-319, G-320
fortuitous 59, 109, 115
intermodulation 59, 183, 191,
203
PCM 187
single harmonic 105
telegraph 109, 111
total harmonic 59, 125, 145, 183,
191, 202
total, of digital signals 59
Distribution frames, FDM 162-164,
168
Distribution,
loop loss 131, 132
Distribution,
talker volume 42, 43
Dixon K-360, K-361
Documents,
issue of 5
referenced 5
DoDD C-277
DoD Directives 1, 3, 9, 38, 40
DoDD 4120.21; 1, 9, 38
DoDD 4630.5; 3, 9
DoDD C-4600.3; 9, 40
DoD publications 9
DoD standards 7, 8
DPSK C-277
Dropouts, signal level 60
DTE 31-33, 63, 131, C-277
DTMF 127-130, 149, 152, C-277
Dual-tone multifrequency
signaling 127

Earth return 106
ECCM 18, 39, 40, C-277
Echo 47, 48
EDC 48, 60, 61, 107, C-277
Efficiency, path 107
EHF 124, C-277, J-341
Electrical balance (see:
Electrical symmetry)
Electrical characteristics of
digital interface 71
Electrical measurements 20, 41
Electrical symmetry 65-67, 125,
143, 145, 172, 187, 205, 210
Electromagnetic compatibility 18
Electronic counter-countermeasures
18, 39, 40, C-277
Electronic warfare 18, 39, 40,
C-277
ELF 28, 122, C-277, J-341
EMC 18, 38, 39, C-277
EMI 39, C-277
Emissions,
classification of 119, J-342 to
J-346
designation of 119, J-341, J-342
examples of J-347 to J-358
EMP 40, C-277
End of medium H-329
End of text H-326
End of transmission H-326

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

Engineering guidance for data
circuits F-307 to F-314
Enquiry character H-327
Envelope delay G-318 to G-320
Envelope delay distortion
47, 55-58, 61, 136, 138-141, 145,
176-178, 180-182, 190, 191,
197-202, 211-213, 215, 216, 224,
229, 233, 236, 242, 249, 256,
260, 265, G-319, G-320
Environmental test methods 20, 41
Equalization 49, 60, F-313
Equipment clock 72
Equipment, telegraph and data
111-118
Equivalent PCM noise 44, 188, 189,
209
Error control 50, 51
Error control devices 117, 118
Error correcting systems 118
Error detecting and feedback
systems 118
Error detecting systems 118
Error detection and correction 48
Error rate 48-50, 107
Escape character 99, H-329
EW 18, 39, 40, C-277
Examples of
emissions J-347 to J-358
reference circuits 236-265
Expanded codes 99, 102
Expandor, PCM 186
Exponential distribution (of loop
loss) 132
Extended character sets 99, 102
External clock 72, 73
Extremely low frequency 28, 122,
C-277, J-341

F1A noise weighting L-366 to L-368
Facsimile,
analog 63
digital 65
Facsimile service 61, 62
Factor, activity 166, K-360
Fade margin 172
Fanfold paper 107
FDM C-277
FDM channel loading 162, 167,
K-360, K-361
FDM equipment 161-165
FDM group A; F-314
FDM group B; F-314
FDM group channel interface 67
FDM group circuits 60, F-314
FDM group link 162-165
FDM group pilot F-314
FDM group transmission channel 214
FDM loading curves 166
FDM modulation links 161, 162
FDM noise power 167, 170-173
FDM subsystems 24, 163-165
FDM supergroup link 162-165
FDM telegraph subsystems 109-111
FDM translation equipment 162-165
FDM VF circuits 42, 49
FDM VF links 162-165, 169
FDM VF reference circuits 236-265
FDM VF reference trunks 222-233
Federal information processing
standards 6, 69, 87, 88, 90, 93,
95-97, 99, 102, 108, 109, H-326
Federal specifications 5, 107, 108
Federal standards 5, 6
Federal telecommunications
standards 5, 12, 21, 69, 94, 96,
118, 120, 222, E-295, F-313,
J-341
FED-STD C-277
FED-STD-1005; 5, F-313
-1006; 5, F-313
-1010; 5, 93
-1011; 5, 94
-1012; 5, 96
-1013; 5, 69
-1037; 5, 12, 21, 44, 94,
118, 120, 222, E-295,
J-341
Fiber optics 27, 30, 130, B-272
Fiber optic communications
subsystems 30, 130
File separator H-328
FIPS C-277
FIPS PUB 6, C-277
FIPS PUB 1-1; 6, 87, 90, H-326
PUB 2; 6, 108
PUB 3-1; 6, 108
PUB 7; 6, 87, 88, 102
PUB 15; 6, 97, 99
PUB 16-1; 6, 93
PUB 17-1; 6, 94-96
PUB 18-1; 6, 97

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

FIPS PUB 22-1; 6, 69
PUB 25; 6, 109
PUB 26; 6, 108
PUB 35; 6, 99
PUB 36; 6, 90
PUB 50; 6, 109
Fixed directory codes 157
Format effector 91, H-327
Form feed H-328
Fortuitous distortion 59, 109, 115
Frame synchronization 186
Frequencies, center 110, 112
Frequency averaging I-333, I-336, I-337
Frequency bands 28, 29, 119, 122-124, J-341
Frequency characteristics, insertion loss versus 44-46, 54-56, 58, 61, 105, 106, 125, 132, 134-139, 145, 174-179, 189, 190, 195-200, 210, 211, 213, 215, 216, 224, 229, 233, 235, 242, 249, 256, 260, 265, L-367
Frequency displacement 48, 59, 62, 106, 183, 191, 212
Frequency division multiplexing 24
Frequency modulation J-343, J-353 to J-355
Frequency response (see: Insertion loss versus frequency characteristic)
Frequency shift 111
Frequency spectrum 28, 29, 119, 122-124, J-341
Frequency standard I-334, I-335
Frequency, test signal 187
Frequency tolerance 111
FSK C-277, E-296
FSK signal state 71
Functions of communications system 30
Functions of transmission facilities 20

Gain hits 60
Gated clock signal 81, 83
Gating 75
GDF 163, 164, C-277
General design considerations 17
General design requirements 38
GHz C-277
Grade of service 206
Graphic characters 90, 92
Graphic numeric subset 99, 101
Graphic subsets 97-101
Grounding 20, 41
Ground return 106
Group-coded recording 109
Group distribution frame 163, 164 168, C-277
Group separator H-328
Group translation equipment 163-165

Harmonic distortion 59, 105, 125, 145, 191, 202, 211
Harmonic distortion, measurement of 157
Heading H-326
HF 28, 121, 123, C-277, J-341
HFDF 163, 164, C-277
HF distribution frame 163, 164
HF radio links 207
HF radio subsystems 28, 207
High-contrast signal level (facsimile) 61
High level digital interface 71
Hits, amplitude (or gain) 60
Hits, phase 60
Holbrook K-360, K-361
Horizontal tabulation H-327
Human engineering 16, 17, 18, 40
Hypothetical reference circuits 222, 236-265, L-365, L-373
Hz C-277

Idle channel noise 44, 145, 167, 170-172, 188, 210
Idle channel noise, PCM 44, 188, 210
Idle line termination 204
Idle pattern 73
Idle register receiver 146
IF C-278
IFF C-278
ILD C-278
ILF 28, 122, C-278, J-341
IMD 59, 183, C-278
IMD measurements 159
Impedance 65, 66, 67, 109, 121, 122, 125, 142, 143, 145, 172, 173, 187, 205, 210, 215

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Impulse noise 53, 59, 145, 215
Indirect clock control 78, 80
Information distribution system 62
Information interchange 69, 87, 102, H-321
Information interchange, codes for 87, 102, 108
Information processing standards 6
Information separator 91, H-328
Infra low frequency 28, 122, C-278, J-341
Input/output linearity 192
Input/output signal characteristics 72
Insertion gain/loss 131-132, 144, 155, 203
Insertion loss versus frequency characteristic 44-46, 54-56, 58, 61, 105, 106, 125, 132, 134-139, 145, 174-179, 189, 190, 195-200, 210, 211, 213, 215, 216, 224, 229, 233, 235, 242, 249, 256, 260, 265, L-367
Installation 16
Integrity, character and bit count 17, 48, 51, 61
Interblock gap 70
Intercharacter interval 94
Interconnection of radio/telephone facilities 124
Interconnection of VF channels 217-221
Interconnection requirements 37, 144, 217
Interconnection of subsystems 16, 37, 217
Interconnection with long haul systems 38
Interconnection with NATO systems 34, 144, 152
Interdepartment Radio Advisory Committee 119, C-278
Interdigit signal 147, 149, 151
Interface adapters 33
Interface,
analog 65
AUTODIN 72
data, NATO 72
digital 71, 109, 111
FDM group channel 67
Interface,
quasi-analog 65, 109, 111
Interfaces, standard 68
Interface, VF channel 65-67
Interference, single tone 47, 53 61, 195, 210
Intermittent data transmission 81, 83
Intermodulation distortion 59, 183, 191, 203, 212
International alphabets 102, H-324
International Radio Consultative Committee (see: CCIR)
International Regulations 11
International Standardization Organization H-324
International Telecommunication Union (see: ITU)
International Telegraph and Telephone Consultative Committee (See: CCITT)
Interoperability 1, 2
IRAC 119, C-278
ISB C-278
ISO H-324
Isochronous operation 78, 81
Isochronous signal 33
Issue of documents 5
ITA C-278, H-324
ITA No. 2., American version H-325
ITU 11, 119, C-278, J-341

JANAP 128; 9, 72, 118, C-278
JCS 119, C-278
JCS PUB 10; 9, 62, C-278
Jitter, phase 60
Joint publications 9
Joint Tactical Information Distribution System (see: JTIDS)
JTIDS 62

kBd C-278
kb/s C-278
Keyboards 102
Keyboard display terminals 63
kHz C-278
km C-278

LED C-278
Level adjustment F-311, F-312
Level alignment 157, 158

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Level coding E-296, E-302
Levels, transmission 146
LF 28, 123, C-278, H-327, J-341
Limiting 145, 204
Linearity, input/output 192
Line feed H-327
Link 22
Links,
  FDM group 162
  FDM VF 162
  modulation 162-192
  tactical digital information 62
  TDM/PCM VF 185, 192
Loaded channel noise power 167, 170
Loading coils 135
Loading curves 166
Loading, FDM 165
Loading, TDM/PCM 186, 187
Logic and signal sense 69
Long haul systems, interconnection with 38
Longitudinal currents 65, 66, 67, 125, 143, 145, 172, 173, 188, 205, 210
Loop 20
Loop insertion loss 131, 132, 142 F-310, F-311
Loop loss pads 131
Loop subsystems 131-143
Loops, radio 142, 143
Loops, wire or cable 131-142
LOS C-278
LOS microwave subsystem 121
LOS millimeter wave 121
Low Frequency (see: LF)
Low level digital interface 71
LPC 27, C-278, D-291, D-292

Magnetic flux polarity 70
Magnetic tape 108, 109
Magnetic tape reader 63
Magnetic tape recorder 63
Magnetic tape recording 70
Maintainability 19, 40, 41
Manchester II code 105, E-298
Mandatory standards 3, A-267
Manpack radios 122
Manual switchboards 154
Manual switchboard codes 157
Mark condition 70, H-324
m-ary 35, C-278
Master clock 72-87
Master-slave technique I-333, I-335
Mb/s C-278
MCEB 119, C-278
Mean time between failures 19
Mean time between losses 49
Mean time between preventive maintenance 19
Mean time to repair 19
Measurements
  IMD 159, C-278
  net loss variations 159
Media, recording 107
Medium frequency (see: MF)
Message composer 118
Message switches 155
Metallic lines 27
MF 28, 123, C-278, J-341
MHz C-278
Microwaves 28
MIL-HDBK 8, 9, 40, 64, C-278
MIL-HDBK-232; 8, 38, 64
  -235; 9, 39
  -237; 9, 39
  -241; 9, 39
  -253; 9, 39
Military Communications-Electronics Board 119
Military Handbooks (see: MIL-HDBK)
Military Specifications 5
Military standards 7
Miller code E-302, E-303, E-304
Millimeter wave radio 121
MIL-P-40023; 5, 107
MIL-STD C-278
MIL-STD-188; iii
MIL-STD-188C; iii, 7, 63, 70, 123, 124, 195
MIL-STD-188-100; iii, 7, 63, 70, 71
MIL-STD-188-102; 161, B-272
MIL-STD-188-110; 7, 102
MIL-STD-188-111; 130, B-272
MIL-STD-188-113; 49, B-273
MIL-STD-188-114; 7, 36, 64, 70, 71, 77, 109, 143
MIL-STD-188-115; B-273
MIL-STD-188-120; 12
MIL-STD-188-124; 7, 41
MIL-STD-188-140; 7, 122, 123

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

MIL-STD-188-141; 122, 123, 195, 207, B-273
MIL-STD-188-144; 124, B-273
MIL-STD-188-145; 124, B-273
MIL-STD-188-146; 122, 124, B-273
MIL-STD-188-147; 124
MIL-STD-188-154; B-273
MIL-STD-188-161; 7, 65
MIL-STD-188-190; 41, B-273
MIL-STD-188-203 series 7, 62, B-272
MIL-STD-188-242; 122, 123, B-272
MIL-STD-188-243; 122, 124, B-272
MIL-STD-188-260; 63, B-272
MIL-STD-210; 7, 41
-449; 7, 39
-461; 7, 38, 39
-462; 7, 39
-463, 7, 12
-470; 7, 41
-471; 7, 41
-781; 7, 40
-785; 8, 40
-810; 8, 41
-1280; 8, 102
-1472; 8, 40
-1553; E-298
min C-278
Mission bit stream 24
mm C-279
Mode,
asynchronous 35
bit-by-bit asynchronous 75
bit synchronous 73
character interval control 74
character interval synchronous 74
start-stop 35
stepping 74
synchronous 33
transmission, of clock 72
Modem 102, 106, 111, 121, C-279
Modes of operation 33
Modulation signal 127
Modulation,
amplitude J-343, J-347 to J-352
analog pulse 25
CVSD 27, 48, 49, 87, 161, 185, C-276, D-281
delay E-302
delta 26
digital pulse 25
Modulation,
frequency J-343, J-353 to J-355
Modulation links 161-192
Modulation,
phase J-343, J-353 to J-355
pulse J-343, J-356 to J-358
pulse code (see: PCM)
Modulation rate converter 113
Modulation rate phase adjustment 76
Modulation rates 67, 69
Modulation techniques 25-27, E-295
Modulation, types of J-343
Monitoring 106
Monitoring test tone 159
Morse code H-324
ms C-279
MTBF 19, C-279
MTBPM 19, C-279
MTTR 19, C-279
Multichannel radio subsystems 121
Multichannel subsystems 13-15
Multichannel trunking system, basic configuration of 21
Multiplexing equipment 161
Multiplexing,
frequency division 24
space division 24
time division 24
Multiplexing subsystems 24
Multiplex noise power 167-172
Multitone data signal level 51
mW C-279

NACSIM 5100 9, 38
NACSEM 5201 9, 38
National military command system iii
NATO C-279
NATO code for information interchange 102
NATO interface 72, 144, 152
NATO standards 8
NATO switches, interconnection with 144, 152
NATO systems, interconnection with 38
NATO telephone directory 156
NBS C-279
Necessary bandwidth J-339, J-342

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Negative acknowledge H-327
Neper L-373
Net loss variations 48, 59, 62, 142, 183, 192, 204, 212, 217, 236
Net loss variation measurements 159
Net radio interface 31, 124, 125
Net radio subsystems 119
Network 23
Network timing I-331, I-333
Network synchronization I-331, I-333
Neutral system current 71
New line character H-328
NICS 8, C-279
NMCS iii, C-279
nmi C-279
Nodal subsystems 144
Node 22
Noise,
  equivalent PCM 44, 188, 209
  idle channel 145, 170, 188
  impulse 53, 59, 170, 215
Noise measuring set L-366
Noise, PCM idle channel 44, 188, 210
Noise power 44, 52, 57, 59, 141, 142, 145, 167, 170, 171, 172, 188, 193-195, 208, 209, 214, 215
Noise power calculations L-373
Noise power,
  of loops 141, 142
  of switches 145
  of trunks 194, 195
  received 44, 52
  summation of 234
Noise power parameters 167
Noise power units L-365, L-367, L-369
Noise, quantizing 188
Noise rejection improvement 66
Noise weighting curves L-363, L-368
Noise weighting networks L-368
Non-multiplexed subsystems 24
Nonregenerative repeaters 114
Np C-279, L-373
NRI 31, 124, 125, C-279
NRZ C-279, E-295 to E-306
NRZ-change E-299
NRZI 70, 71, 108, 109, C-279
NRZ-L E-299, E-300
NRZ-M E-302, E-303
NRZ-S E-302, E-303
Null character H-328
Numbering plan, telephone 155, 156
Nyquist criteria 25, 26

Objectives, design 3, 37
Objectives (of this document) 2, 3
OCRE 63, C-279
Off-hook 155
Office code 147, 148
On-hook 155
Operation,
  asynchronous 81
  isochronous 81
  modes of 33
  tactical equipment 16, 17
Optical character recognition equipment (see: OCRE)
Orderwire, radio 121, 122
Overloading 51, 187

PAM 15, 16, 25, 185, C-279, D-284
Paper, teletypewriter 107
Parallel-by-bit data communications 72, 96
Parity, character 88
Parity sense 95, 96, 97
Patch facilities 157
Path efficiency 107
Path error rate 107
Path utilization 107
PBX 131, C-279
PCM 25, 26, 87, 185, C-279
PCM distortion 187
PCM, equivalent noise 44, 188, 189, 209
PCM idle channel noise 44, 188, 210
PCM quantizing noise 26, 188
Perforated tape 71, 107, 108
Perforated tape code 108
Phase adjustments 76, 77, 116
Phase coding E-297, E-298, E-303, E-304
Phase delay G-317, G-318
Phase encoded recording 70, 71, 108
Phase flux reversals 70
Phase hits 60
Phase jitter 60
Phase modulation J-343, J-353 to J-355

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Phase relationship 74, 78
Phase shift, relative 76, 77
Pilot, FDM group F-314
Polarity, signaling 69-71
Polar NRZ E-296, E-297
Polar RZ E-296, E-297
Polar waveforms E-296
Precedence 17
Precedence tones 151, 154
Preemption 17
Preempt tone 151, 153
Primary area codes 156
Private branch exchange 131, C-279
Proceed signal 147, 149, 151
Projects, standardization B-272, B-273
Promulgation sheet ii
Propagation delay time 115
Protection of signal sense 120
Pseudo-ternary waveforms E-296, E-298, E-301
Psophometric weighting L-367
PSK C-279, E-296
Publications,
  Allied Communication 8, 10
  Chiefs of Staff 9
  DoD 7, 9
  Federal Information Processing 6
  Joint 9
  NATO 8
Pulse duration coding E-297, E-298, E-303, E-304
Pulse modulation J-343, J-356 to J-358
Pulse modulation,
  amplitude 25, 26
  analog 25
  digital 25
Punches, tape 63
Purpose (of this document) 1
pW C-279
pWp C-279, L-367

QPSK C-279
QSTAG 8, 123, 124, C-279
QSTAG-263A; 8, 123
      -263B; 8, 123
      -263C; 8, 124
Quadripartite standardization agreement 8, 123, 124, C-279
Quality of data transmission 106, 107
Quantizing noise 26, 188, D-288
Quasi-analog interface 65, 66, 109
Quasi-analog signals 24
Quasi-analog signal levels 51, 52, 66, 67, 109, 207, 208, 214, F-311, F-312
Quiescent state of clock signal 81

Radio frequency interference 18
Radio loops 142, 143
Radio orderwire 121
Radio regulations 111, 119, J-341
Radio relay subsystems 28
Radio, single channel 15, 119, 120, 124
Radiotelephone procedures 125
Radio transmission subsystems 119
Radio wire integration 16, 31, 124
Rate converter 113
Rating, subjective 42, 44
Reader,
  card 63
  tape 63
Recall tone 128, 151, 154
Receive-after-transmit time delay 120
Received noise power,
  analog facsimile 61
  data 52, 53
  voice 44
Received signal level 52, 61, 65-67, F-311, F-312
Received speech volume 42, 44
Receiver attack time delay 120
Receiver release time delay 120
Recorded magnetic tape 108, 109
Recorder, tape 63
Recording, magnetic tape 70, 71
Recording media 107
Recording,
  NRZI 70, 71, 108, 109
  phase encoded 70, 71, 108
Record traffic 31, H-324
Record separator H-328
Reference circuits 222, 236-265
Referenced documents 5
Reference trunks 222-233
Regeneration 23, 24, 26, 49, 60, F-313

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Regenerative repeater 114
Register 146-148
Relative envelope delay G-320
Relative phase shift 77
Release acknowledge 128, 151, 154
Release signal 128, 151, 152, 154
Release signal protection 152
Release time delays 120
Release tone 128, 151, 154
Reliability 19, 40
Repeaters 114-117
Requirements,
  general design 38
  tactical systems 37
Restart signal 147, 149, 150
Return loss 65-67, 121, 122, 125, 143, 145, 172, 173, 187, 205, 210
Return path 106
rf 28, C-279
rf emissions J-342
RFI 18, C-280
rf spectrum,
  designation J-341
  measurements of 39
Ringback signal 128, 151, 153
Ringdown signaling 127, 154, 155
Ring loop signal 150, 153
Ring trip signal 151, 153
rms C-280
Root-sum-square law 235, 236
Roll paper 107
Routing 156
RWI 16, 31, 124, C-280
RZ C-280, E-295 to E-301, E-305

s C-280
Sample, PCM 186
Sampling 81, 115
Sampling rate 186
Satellite links,
  echo over 47
  path delay of 76
Satellite subsystems 122
Schedule, DCA technical 44, 54
Scope (of this document) 1
Security, communications 17
Seize 128, 129, 150, 153
Seize acknowledge 128, 150, 153
Seizure signal 146-158
Self-clocking signals E-295, E-298, E-304
Semiautomatic circuit switches 152
Serial-by-bit data communications 72, 93-96
Service,
  communications 42
  facsimile 61, 62
  grade of 206
  telegraph and data 49
  voice 42
S-F C-280
SF C-280
SF signals 128, 129, 149, 152
SGDF 163, 164, C-280
SHF 29, 124, C-280
SHF radio subsystems 29
Shielding 20, 41
Shift, frequency 111
Shift-in character H-328
Shift-out character H-328
Signal,
  analog 23
  asynchronous E-295
  busy 150, 151, 154
Signal characteristics 72
Signal,
  clock output 77
  conditioned 103, E-301
Signal conditioning 103
Signal, confirmation 147, 148, 149, 152
Signal connection 72
Signal conversion 35
Signal, digital 23
Signal discontinuities 59, 206, 212, F-309
Signal, DTMF 127-129, 147, 149-154
Signal element 24
Signaling,
  common battery 155
  dc 125-127
  dual-tone multifrequency 127-129
  loop 155
Signaling rates, data 67, 69
Signaling, ringdown 127, 155
Signaling supervision procedures 146-148, 152
Signaling/supervisory 150, 151, 153, 154
Signaling, trunk 154, 155
Signal,
  interdigit 147, 148, 151
  isochronous 35
Signal level change 60

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Signal level dropouts 60
Signal level,
  quasi-analog 51, 52, 65-67, 109, 111, 193, 207, 214
  received 52, 65-67
  transmitted 51, 65-67
Signal levels 205
Signal paths 106
Signal,
  preempt 151, 153
  proceed 147, 149, 151
  quasi-analog 24
  recall 151
Signal regeneration 23, 24
Signal,
  release 128, 151, 152, 154
  restart 147-150
  ringback 128, 151, 153
  ringdown 127, 154, 155
  ring loop 150, 153
  ring trip 151, 153
  self-clocking E-295, E-298 E-304
Signal sense 111, 120
Signal sense, protection of 120
Signal,
  seizure 147, 148, 153
  SF 128, 129, 149, 152
  standard test 157
  step 74
  supervisory 127, 129, 149-151, 153, 154
  synchronous E-295, E-299
Signals, type of 23
Signal-to-noise ratio 53, C-280
Signal type 111
Signal, unconditioned E-295
Significant condition 70
Single channel radio subsystems 15, 119, 120, 124
Single harmonic distortion 105
Single tone interference 47, 53, 61, 195, 210
Slewing 75
SNR 53, C-280
Space character H-328, H-329
Space condition 70, 71, H-324
Space division multiplexing 24
Specifications,
  federal 5
  military 5
Spectrum designation J-341
Speech plus full-duplex 106
Speech plus half-duplex 105
Speech volume 42, 122, 193, 207, 208, K-360
Speech volume,
  received 42
  transmitted 42
Split phase E-297
SSB C-280
Stability, clock modulation rate 75
STANAG C-280
STANAG 5018; 8, 38, 144, 152
       5036; 8, 102
       5040; 8, 38, 72, 144
       5046; 8, 156
Standard arrangement for supplying clock signals 84-87
Standardization agreements,
  NATO 8
  Quadripartite 8
Standardization projects B-271
Standards,
  American National 10
  DoD 7
  Federal 5
  Federal Information Processing 6
  Federal Telecommunications 5
  mandatory 3, A-267
  military 7
  NATO 8
  system 3, 37
Standard test signal 157
Standard test tone 158
Start of heading H-326
Start of text H-326
Start-stop device 74
           input 113
           mode 35
           output 113
           repeater 114, 116
           signals 81
Station clock 72
Stepping mode 74, 114
Step signal 74
Stop element 75
Subjective rating 42, 44, 45
Submarine cable 207
Subscriber number codes 157

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Subsets of ASCII 97-99
Substitute character H-329
Subsystems,
  carrier telegraph 109
  data terminal 63, 64
  FDM 24, 163
  FDM telegraph 109-111
  fiber optic communications 30, 130
  HF radio 28, 207
  interconnection of 16
  loop 131-143
  LOS microwave 121
  multichannel 13-15
  multichannel radio 121
  multiplexing 24
  net radio 119
  nodal 144-160
  non-multiplexed 24
  radio 28, 119, 122
  radio relay 28
  satellite 122
  SHF radio 29
  single channel radio 15, 119, 120, 124
  switched 13-15
  tactical 13
  TDM 24
  telegraph and data transmission 103-109, 120
  terminal 34, 63-102
  transmission 103, 163
  transmission channel 207-221
  tropospheric scatter 121
  trunk 161-206
  UHF radio 29
  VHF radio 29
  wire and cable transmission 129
Summation methods 234-236
Supergroup distribution frame 163, 164, 168
Supergroup translation equipment 163-165
Supervisory signals 127, 129, 149-154
Supplying clock signals 84-87
Survivability 19, 41
Switchboard locator codes 157
Switchboards, manual 152
Switch control and signaling 146
Switched subsystems 13-15
Switches,
  analog automatic 144-152
  circuit 144-155
  cordless 152-154
  digital circuit 155
  message 155
  semiautomatic 152-154
Switch identification code 156
Switching equipment 144-155
Switch transmission levels 146
Symmetry, electrical 65-67, 125, 143, 145, 172, 187, 205, 210
Synchronism 75
Synchronization,
  frame 186
  network I-331 to I-338
Synchronous idle character H-327
Synchronous mode 33, 74
  operation 74
  repeater 114-116
  signal E-295
System,
  communications 1-3, 13, 30-34, 37, 118
  data processing 118
Systems, requirements for 37
System standards 3, 37
System, tactical information distribution 62

Tabulation H-327
Tactical Communications,
  general description of 13
  requirements for 37
Tactical communications equipment 16
Tactical digital information links 62, C-280
Tactical equipment design, installation and operation 16
Tactical information distribution system 62
Tactical subsystems 13-15
Tactical subsystems, interconnection of 37
Tactical subsystem type I 13, 14
  type II 13, 14, 185
  type III 14, 15
  type IV 14, 15, D-284

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Tactical system 13
Tactical systems,
  basic functions of 30
  configuration of 31
  interconnection with long haul
    systems 38
TADIL 62, C-280
Talker volume distribution 42, 43
Tandem connections, summation
  methods for 234-236
Tape code 108
Tape,
  magnetic 108, 109
  perforated 107, 108
Tape punches 63
Tape readers 63
Tape recorders 63
TDM C-280
TDM/PCM signal characteristics
  185
TDM/PCM VF links 185
TDM/PCM VF reference circuits
  243-249
TDM/PCM VF reference trunks
  243-249
Technical schedule, DCS 44, 54
Telecommunications standards 5
Telegraph and data circuits,
  engineering of F-307 to F-314
Telegraph and data
  equipment 111
  interface 67-72
  service 49-61
  transmission over FDM group
    circuits F-314
  transmission subsystems 103, 109,
    120
Telegraph
  code converter 117
  distortion 109, 111
  subsystems, FDM 109, 111
Telephone,
  analog 63
  digital 63
Telephone directory 155
Telephone sets, types of L-365
Teletypewriter 63
Teletypewriter paper 107
TLP 205, C-280
TLP conversion formula L-373
Tempest 18, 38, C-280
Terminal equipment, analog 63
Terminal impedances 65, 66, 67,
  143, 145
Terminals, keyboard display 63
Terminal subsystems 34, 63-102
Termination, idle line 204
Terms, definitions of 12
Test facilities 157
Test levels, summary of 159, 160
Test methods,
  electrical 20
  environmental 20
Test signal frequency 187
Test signal levels 205, 214
Test signals 142, 157, 159, 168,
  169, 187
Test tone 142, 158
Text H-326
THF C-280, J-341
Through-group equipment 162, 165
Throughput 50
Through-supergroup equipment 162,
  165
Time delay 47, 120, G-317
Time differential blanks 73
Time division multiplexing 24
Timing control 116
Timing, network I-331
Timing requirements 72
Timing signal state 71
Tolerance, frequency 111
Tone,
  busy 150, 151, 154
  conference 128
  dial 150, 153
  monitoring test 159
  precedence 151, 154
  preempt 151, 153
  recall 128, 151, 154
  release 128, 151, 154
  release acknowledge 128, 151,
    154
  ringback 128, 150, 153
  ring trip 151, 153
  seize 128, 150, 153
  standard test 158
Tone detectors 129, 149, 152
Total distortion of digital
  signals 59

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Total harmonic distortion 59, 125
145, 191, 202, 211
Traffic
record H-324
types of 17
Transfer function parameters for
data circuits 54-59
Translaters, alphabet 118
Translation equipment, FDM 163-165
Transmission channel 21, 23
Transmission channels,
digital 217
FDM group 214
Transmission channel subsystems
207-223
Transmission
facilities 20-23
level points 205
levels 146
media 27
medium noise power 172
modes of clocks 72
path performance 106, 107
Transmission subsystems 103-130
Transmission subsystems, wire and
cable 129, 130
Transmit-after-receive time delay
120
Transmitted
signal level 51, 61, 65-67
speech volume 42
Transmitter attack time delay 120
Transmitter release time delay 120
Tropo C-280
Tropospheric scatter subsystems 121
Trunk 23
Trunking system, basic
configuration of 21
Trunk group 22
parameters 192
signaling 154
subsystems 161-206
Trunks,
grade of service 206
reference 222-233
wire and cable 129, 130
Type 144 handset L-365
302 handset L-367
500 handset L-367
Types of modulation J-343
signals 23
Types of telephone sets L-365
traffic 17
transmission 27, J-343

UHF 28, 29, 124, C-280, J-341
UHF radio subsystems 29
Unbalanced coding E-299, E-300,
E-305, E-306
Unbalanced NRZ E-299, E-300
Unconditioned signal E-295
Unipolar NRZ E-299, E-300
RZ E-299, E-300
Unit interval signals 72
Unit separator H-328
Unpunched tape 107
Unrecorded magnetic tape 108
Unsymmetric coding E-299, E-300,
E-305, E-306
USASCII 87, C-280
USA standard code 87
User-to-user, parameters for 42-62
Utilization, path 107
UU-P-547; 5, 107
UU-T-120; 5, 107

Variations, net loss 48, 59,
62, 142, 183, 192, 204, 212,
217, 236
Vertical tabulation H-327
Very low frequency (see: VLF)
VF C-280
VF channel interface 65-67
VF circuits, parameters for 42-48,
49-62
VFCT 109, C-280
VFDF 163, C-280
VF distribution frame 163, 164
VF transmission channels 207-213
VHF 28, 29, 123, 124, C-280,
J-341
VHF radio subsystems 29
VLF 28, 122, 123, C-280, J-341
Voice frequency carrier telegraph
109
Voice guard 152
Voice levels 122
Voice loading 162, K-360
Voice service,
analog 42-48
digitized 48-49
Voice simulations 152

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Volume,
  speech 42, 122, 193, 207, 208, K-360
  talker 42, 43
  unit 42
VU C-280

W C-280
WARC 119, C-280
Weighting curves, noise L-366
Weighting networks, noise L-368
Wire and cable trunks 130
Wire or cable loops 131-143
Wire transmission subsystems 129
World Administrative Radio Conference (see: WARC)
Worst-hour noise 235
wpm C-280

Zero transmission level point C-280
OTLP C-280, L-373

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Communications- Elevtronics Command
ATTN: DRSEL-ED-TO
Fort Monmouth, NJ 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STL-188-200	System Design & Engineering STLS for Tactical Communications

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.